CCIE Enterprise Infrastructure v1.0 Real Labs

Deploy Module

CCIE企业基础设施v1.0真实实验室部署模块

Lab Workbook Policy

1. We highly discourage sharing of the workbook hence the workbooks are mapped to Laptop/Desktop MAC address. If one tries to open the workbook on other desktop or laptop than the registered MAC address; account will get locked and we will not unlock it for any reasons.

2. The workbook does not have print access; kindly do not request to enable to print access. However you will have perpetual access to the workbook which you have purchased.

3. One will be provided with free updates up to 120 days from the date of purchase, post that one need to renew his/her account to access the latest update. However one will continue to have access to their existing workbooks. If you pass the lab within 120 days, you are not eligible for further updates.

4. If one wish to renew their subscription/account, you need to renew within 120 days or before the account gets expired. Post 120 days one can renew their account however the renewal will be considered has a new purchase. Hence we encourage one to renew within 120 days of the purchase.

5. The renewal cost is 999 USD if one pay within 120 days, if one fail to renew then the cost will be equivalent of a new purchase. (The renewal price can be changed at any time, without informing the client)

6. Every workbook is uniquely identified for each user with hidden words. If one shares his/her workbooks with others, and if the system detects the share, the account will be banned and we will not entertain any explanation of any sort.

7. For any queries regarding Questions/Solutions, you can contact us on email: support@chinesedumps.com or skype @ chinesexams@gmail.com. Response time to any of the queries is 24 hours.

8. We do require CISCO ID and Official email id for security purposes. We do not sell without these details. We do background verification of the details provided, so request to give us the correct CISCO ID and official email id.

9. The workbooks are in secured pdc format and delivered via email within 24 hours after payment is received.

10. License is provided for only one Device. And we don't give license again if the device crashes or company security policies. Please install license on the device cautiously as the license will not be provided again.

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实验室工作簿政策

1. 我们强烈反对共享工作簿，因此工作簿被映射到笔记本电脑/台式机MAC地址。如果有人试图在注册的MAC之外的其他台式机或笔记本电脑上打开工作簿
地址；帐户将被锁定，我们不会以任何理由解锁。

2. 工作簿没有打印权限；请不要请求启用打印访问。但是，您将可以永久访问您购买的工作簿。

3. 自购买之日起120天内，用户将获得免费更新，并需要续订帐户才能访问最新更新。然而，人们将继续访问他们现有的工作簿。如果您在120天内通过实验室，则没有资格获得进一步的更新。

4. 如果希望续订订阅/帐户，您需要在120天内或帐户到期前续订。120天后，用户可以续订帐户，但续订将被视为新购买。因此，我们鼓励您在购买后120天内续订。

5. 如果一个人在120天内付款，续费费用为999美元，如果一个人没有续费，那么费用将相当于新购买的费用。（续费价格可随时更改，无需通知客户）

6. 每个工作簿都为每个用户使用隐藏单词进行唯一标识。如果一个人与其他人共享工作簿，并且系统检测到共享，则该帐户将被禁止，我们不会接受任何形式的解释。

7. 有关问题/解决方案的任何疑问，您可以通过电子邮件与我们联系：
support@chinesedumps.com 或skype@chinesexams@gmail.com. 对任何查询的响应时间为24小时。

8. 出于安全考虑，我们确实需要CISCO ID和官方电子邮件ID。没有这些细节，我们不会出售。我们会对提供的详细信息进行背景验证，因此请提供正确的CISCO ID和官方电子邮件ID。

9. 工作簿采用安全的pdc格式，并在收到付款后24小时内通过电子邮件发送。

10. 许可证仅提供给一台设备。如果设备违反了公司的安全政策，我们不会再发放许可证。
请谨慎地在设备上安装许可证，因为将不再提供许可证。

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11. We do support devices running Windows OS, Mac OS, Android and Mac iOS only

12. We do not provide Refund in any circumstances once the product is sold.

13. This policy is in effect from 23 November 2016 and in immediate effect for new clients and new renewals. Old clients will continue with the old Policies until the accounts get expired.

14. If there is any update, one will receive the update automatically on their registered email id.

15. Design Module will be given only 3 days before the CCIE exam

16. For any future update you can check our 'updates' page.

17. Labs are always published in phases. For e.g. if there is a new lab we publish it as First, Second, Third ... till Final release.

18. Client who have purchased our worbooks and services and wishes to attempt the lab, need to consult our experts before their CCIE Lab.

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11. 我们仅支持运行Windows操作系统、Mac操作系统、Android和Mac iOS的设备

12. 一旦产品售出，我们在任何情况下都不提供退款。

13. 本政策自2016年11月23日起生效，对新客户和新客户立即生效
续约。旧客户端将继续使用旧策略，直到帐户过期。

14. 如果有任何更新，用户将在其注册的电子邮件id上自动收到更新。

15. 设计模块将在CCIE考试前3天提供

16. 对于任何未来的更新，您可以查看我们的“更新”页面。

17. 实验室总是分阶段发布的。例如，如果有一个新的实验室，我们会将其发布为First、Second、Third。..直到最终发布。

18. 购买了我们的工作簿和服务并希望尝试实验室的客户需要在CCIE实验室之前咨询我们的专家。

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CCIE Deploy, Operate and Optimize Guidelines

Before you begin, please read these guidelines:

Overall module guidelines:

1. The network that you will deploy, operate and optimize in this module will be similar, but not necessarily identical, to the network designed in the previous module. All relevant information that is needed to successfully complete this module can be found in this module itself and overrides any information that was provided in the previous module.

2. Before you start, confirm that all devices in your rack are accessible. During the exam, if any device becomes locked or inaccessible, you must recover it.

3. Your equipment is partially preconfigured. Do not change any of the preconfigured parameters unless you are specifically told to.

4. The partial configuration on the devices may deliberately contain mistakes and errors which may need to be corrected, or workarounds applied, in order to complete specific tasks. Therefore, consider troubleshooting as an integral part of this module.

5. Points are awarded only for fully working configurations. No partial scoring is provided. It is recommended that toward the end of the exam, you go back and test the functionality as per all question requirements.

6. If you need clarification on any of the questions, or if you suspect that there might be an issue with your equipment or exam environment, contact the lab proctor as soon as possible.

7. Item-level feedback can be provided at the question level. Feedback will be processed, but Cisco will not reach out to you to discuss any feedback provided. You will not be compensated for the time you spend while providing the feedback.

8. Access to select Cisco online documentation is available from your desktop. Access to select 3rd party product documentation (such as Python) is available from the Resources window under the "External Documentation category".

9. When you finish the lab exam, make sure that all devices are accessible for the grading proctor by having them in EXEC mode and closing the console windows. A device that is not accessible for grading cannot be graded and this may cause you to lose substantial points.

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CCIE部署、操作和优化指南

在开始之前，请阅读以下指南：

总体模块配置线：

1. 您将在本模块中部署、操作和优化的网络将与上一模块中设计的网络相似，但不一定完全相同。成功完成本模块所需的所有相关信息都可以在本模块中找到，并覆盖上一模块中提供的任何信息。

2. 开始之前，请确认机架中的所有设备都可以访问。在检查过程中，如果任何设备被锁定或无法访问，您必须将其恢复。

3. 您的设备已部分预配置。除非有特别指示，否则不要更改任何预配置的参数。

4. 设备上的部分配置可能故意包含错误和错误，为了完成特定任务，可能需要纠正或应用变通方法。因此，将故障排除视为本模块的一个组成部分。

5. 只有完全工作的配置才能获得积分。不提供部分评分。建议在考试接近尾声时，根据所有问题要求返回并测试功能。

6. 如果您需要澄清任何问题，或者您怀疑您的设备或考试环境可能存在问题，请尽快联系实验室监考员。

7. 可以在问题级别提供项目级反馈。反馈将被处理，但思科不会联系您讨论所提供的任何反馈。您在提供反馈时所花费的时间不会得到补偿。

8. 您可以从桌面访问选定的思科在线文档。可以从“外部文档类别”下的“资源”窗口访问选择的第三方产品文档（如Python）。

9. 完成实验室检查后，通过将所有设备置于EXEC模式并关闭控制台窗口，确保分级程序可以访问所有设备。无法进行评分的设备无法评分，这可能会导致您失去大量分数。

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10. You have 5 hours to complete this module. Upon finishing the exam, ensure that all devices are accessible. Any device that is not accessible for grading purposes may cause you to lose substantial points.

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10. 你有5个小时的时间来完成这个模块。完成检查后，确保所有设备都可以访问。任何无法用于评分目的的设备都可能导致您失去大量分数。

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Track specific guidelines:

1. There are several end hosts present in the lab topology, named hostXY (for example, host11). They are all identical and they can all be used at your full discretion, including accessing the GUI of DNA Center, vManage and ISE through Firefox, performing IP connectivity tests, generating or capturing traffic, and performing coding in Python or C.

2. All hostXY devices are configured as DHCP clients. Should it be necessary to force the host to release and renew its DHCP lease, right-click on the icon of the network manager located between CPU utilization and check applets in the bottom task bar, then unselects "Enable Networking", right-click on it again and select "Enable Networking".

3. The web-based GUI of DNA center, vManage and ISE can only be accessed from the hostXY end hosts, using firefox installed on these end hosts. These servers cannot be accessed directly from the desktop you are just now working with. You must always connect to hostXY as a jump host and access the DNA center, vManage or ISE from there. Always ignore any SSL/TLS certificate warnings in Firefox that may be displayed.

4. Devices in the topology may have more interfaces, addresses and routes configured than what is shown in the diagrams and accompanying tables. Ignore such interfaces addresses and routes entirely, unless a task explicitly requires you to use or modify them.

5. Changing or removing parts of initial running configuration on devices, as opposed to adding new configuration, is allowed only if the task allows or requires it explicitly, or if there is no other way of accomplishing the task.

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跟踪特定指南：

1. 实验室拓扑中存在几个名为hostXY的终端主机（例如host11）。它们都是相同的，可以完全自由使用，包括通过Firefox访问DNA Center、vManage和ISE的GUI，执行IP连接测试，生成或捕获流量，以及用Python或C进行编码。

2. 所有主机XY设备都配置为DHCP客户端。如果需要强制主机释放和续订其DHCP租约，请右键单击位于CPU利用率和检查底部任务栏中的小程序之间的网络管理器图标，然后取消选择“启用”
网络”，再次右键单击它并选择“启用网络”。

3. DNA中心、vManage和ISE的基于web的GUI只能通过安装在这些终端主机上的firefox从主机XYend主机访问。这些服务器不能直接从您刚才使用的桌面访问。您必须始终以跳转主机的身份连接到hostXY，并从那里访问DNA中心、vManage或ISE。始终忽略Firefox中可能显示的任何SSL/TLS证书警告。

4. 拓扑中的设备配置的接口、地址和路由可能比图表和附表中显示的更多。完全忽略此类接口地址和路由，除非任务明确要求您使用或修改它们。

5. 只有在任务明确允许或要求的情况下，或者没有其他完成任务的方法时，才允许更改或删除设备上初始运行配置的部分，而不是添加新配置。

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Diagram 01: Complete Topology

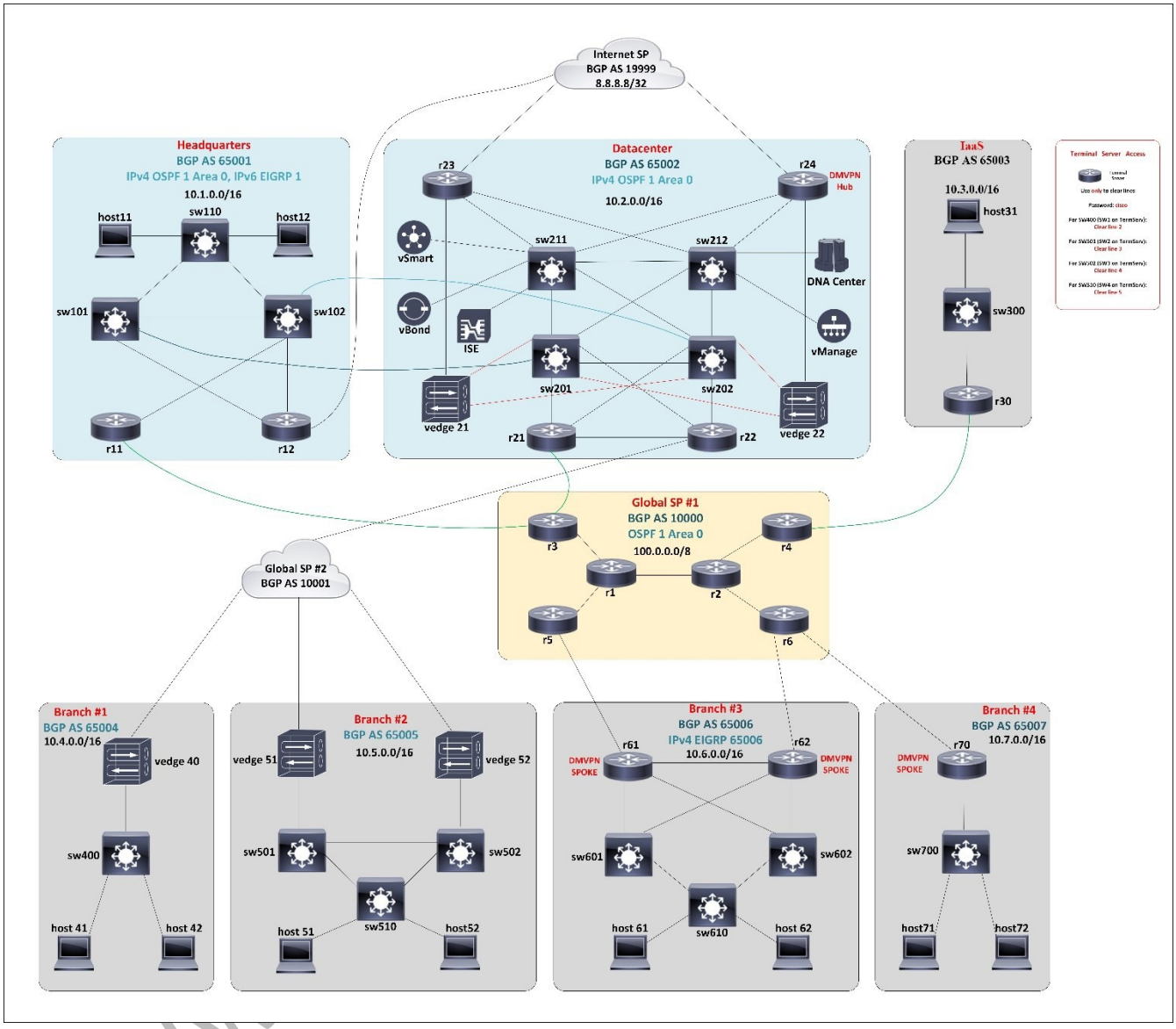


图01：完整拓扑

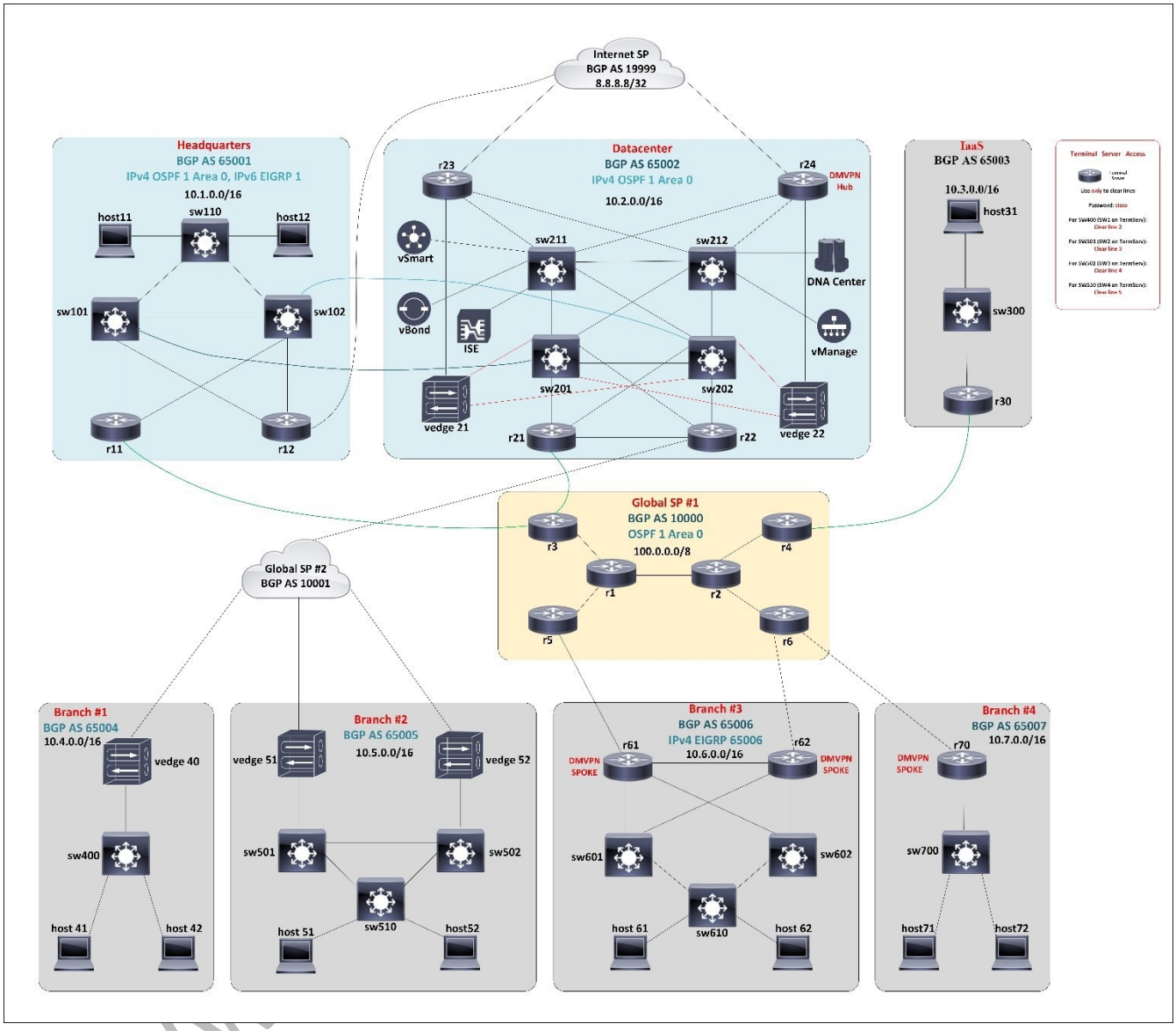

Internet SP
BGP AS 19999
8.8.8.8/32
Headquarters
BGP AS 65001
IPv4 OSPF 1 Area 0, IPv6 EIGRP 1
10.1.0.0/16
host11
sw110
host12
sw101
sw102
r11
r12
Datacenter
BGP AS 65002
IPv4 OSPF 1 Area 0
10.2.0.0/16
r23
r24
DMVPN Hub
vSmart
sw211
sw212
DNA Center
vBond
ISE
vManage
sw201
sw202
vedge 21
vedge 22
r21
r22
IaaS
BGP AS 65003
10.3.0.0/16
host31
sw300
r30
Terminal Server Access
Global SP #1
BGP AS 10000
OSPF 1 Area 0
100.0.0.0/8
r3
r4
r1
r2
r5
r6
Global SP #2
BGP AS 10001
Branch #1
BGP AS 65004
10.4.0.0/16
vedge 40
sw400
host 41
host 42
Branch #2
BGP AS 65005
10.5.0.0/16
vedge 51
vedge 52
sw501
sw502
sw510
host 51
host52
Branch #3
BGP AS 65006
IPv4 EIGRP 65006
10.6.0.0/16
r61
r62
DMVPN SPOKE
DMVPN SPOKE
sw601
sw602
sw610
host 61
host 62
Branch #4
BGP AS 65007
10.7.0.0/16
r70
DMVPN SPOKE
sw700
host71
host72

Diagram 02: Complete Topology with IP addressing

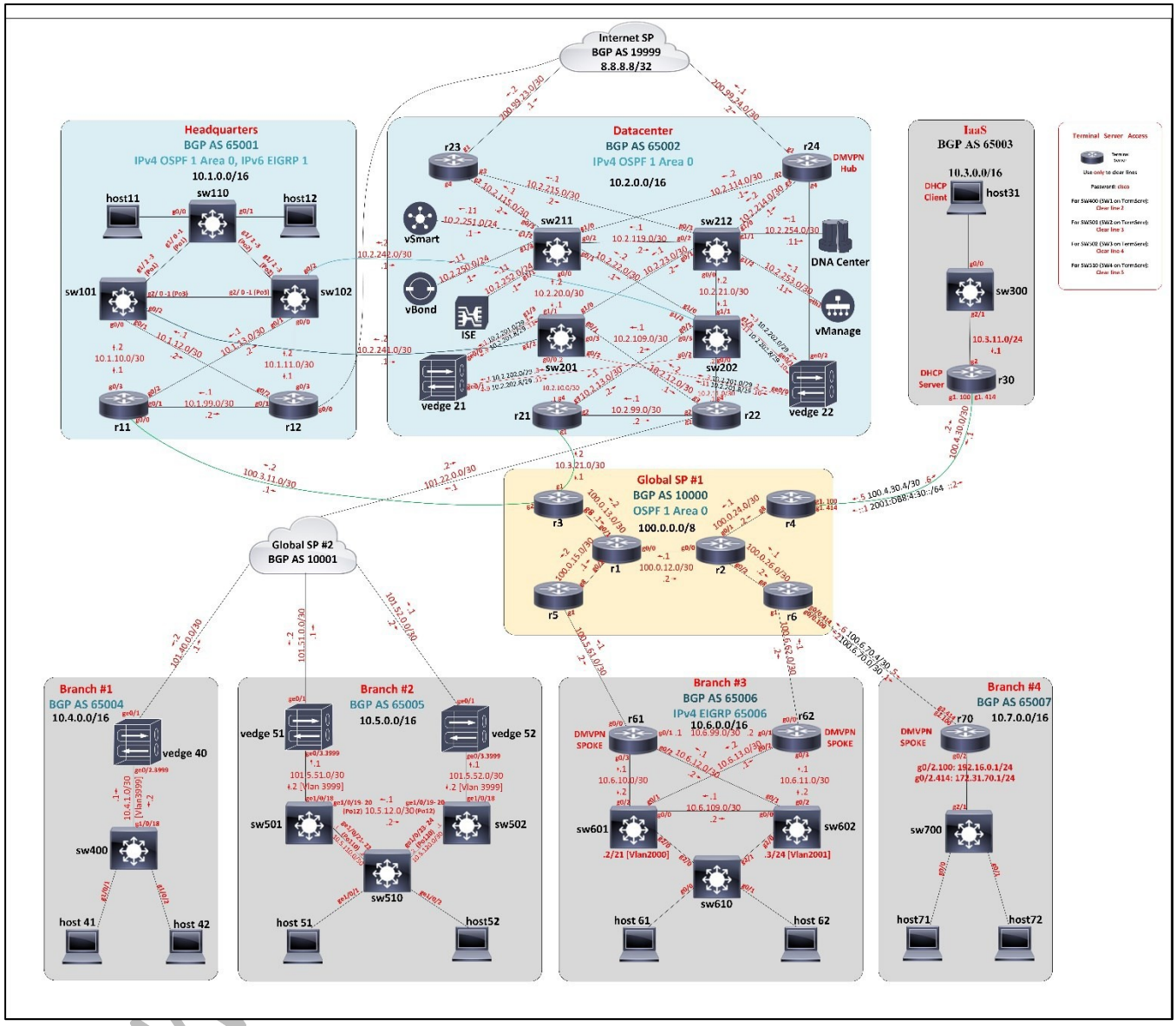
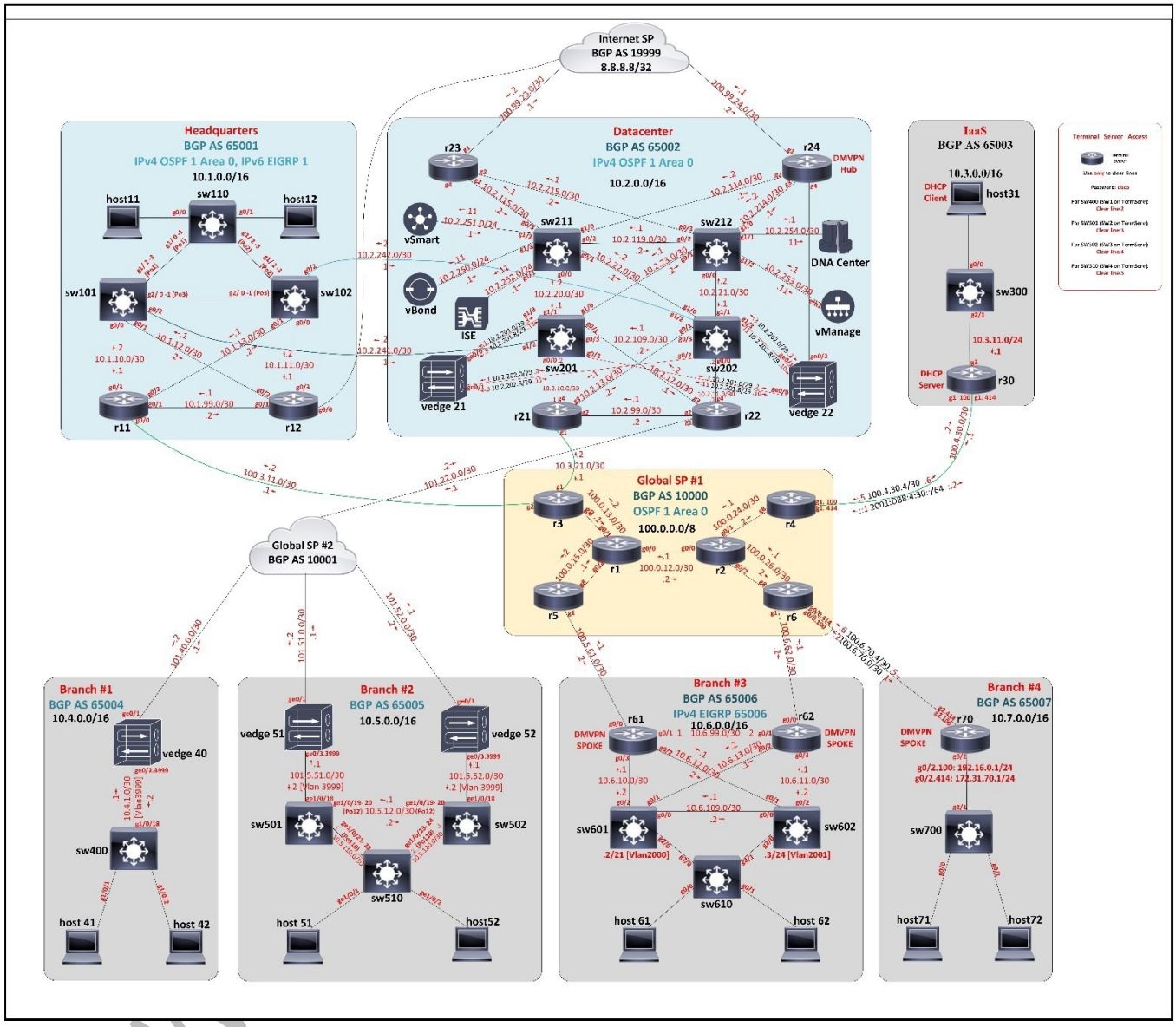


图02：带IP地址的完整拓扑



1.1: Introduction

Welcome back to the FABD2 company!

You will deploy, operate, and optimize our network. The topology you will be working with will be similar, but not necessarily identical to the network that was designed in the previous module and may include technologies and feature sets not touches upon previously.

The best of success!

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1.1：导言

欢迎回到FABD2公司！

您将部署、操作和优化我们的网络。您将使用的拓扑将是
与前一模块中设计的网络类似，但不一定完全相同，可能包括以前没有涉及的技术和功能集。

最好的成功！

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1.2: Layer 2 Technologies in HQ

Complete and correct the EtherChannel configuration between switches sw101, sw102, sw110 according to these requirements:

- At the end of the task, all EtherChannels between switches sw101, sw102, sw110 must be up and operational including all their physical member links.
- Do not create new Port-channel interface; reuse those that already exist on the switches.
- When resolving existing issues, do not change the preconfigured negotiation protocol (if any)
- On EtherChannels that use a negotiation protocol, tune its mode of operation for the shortest link bundling time possible.

Configure spanning tree protocol on switches sw101, sw102, sw110 according to these requirements:

- The STP root for VLAN 2000 must be sw101.
- The STP root for VLAN 2001 must be sw102.
- STP roots must be elected based on bridge priorities.
- On the three switches, have STP perform cost calculations in 32-bit arithmetic.
- On the three switches, use the Rapid STP version and ensure that it can achieve rapid convergence on all interconnections between the switches.
- On Sw110, prevent all current and future access mode interfaces from being affected by the proposal/ Agreement process.

2 Points

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1.2： 总部的第2层技术

根据以下要求完成并纠正交换机sw101、sw102、sw110之间的EtherChannel配置：

- 任务结束时，交换机sw101、sw102、sw110之间的所有以太网通道（包括其所有物理成员链路）必须正常运行。
- 不要创建新的端口通道接口；重用交换机上已经存在的那些。
- 解决现有问题时，不要更改预配置的协商协议（如果有的话）
- 在使用协商协议的EtherChannels上，调整其操作模式以获得尽可能短的链路绑定时间。

根据以下要求在交换机sw101、sw102、sw110上配置生成树协议：

- VLAN 2000的STP根必须是sw101。
- VLAN 2001的STP根必须是sw102。
- STP根必须根据网桥优先级进行选择。
- 在三个交换机上，让STP以32位算术执行成本计算。
- 在三个交换机上，使用Rapid STP版本，并确保它可以在交换机之间的所有互连上实现快速收敛。
- 在Sw110上，防止所有当前和未来的访问模式接口受到提案/协议流程的影响。

2分

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Solution:

Chinese_dumps_sw110:

```
en
sw110#conf t
sw110(config)#spanning-tree mode rapid
sw110(config)#spanning-tree pathcost method long
sw110(config)#spanning-tree portfast edge default

sw110(config)#interface range gi1/2-3
sw110(config-if-range)#channel-group 2 mode active
```

Chinese_dumps_sw101:

```
en
sw101#config t
sw101(config)#spanning-tree mode rapid
sw101(config)#spanning-tree pathcost method long
sw101(config)#spanning-tree vlan 2000 priority 0
sw101(config)#spanning-tree vlan 1-4094 hello-time 1

sw101(config)#interface range gi1/2-3
sw101(config-if-range)#channel-group 1 mode on
```

Chinese_dumps_sw102:

```
en
sw102#conf t
sw102(config)#spanning-tree mode rapid
sw102(config)#spanning-tree pathcost method long
sw102(config)#spanning-tree vlan 2001 priority 0
sw102(config)#spanning-tree vlan 1-4094 hello-time 1

sw102(config)#interface range gi1/2-3
sw102(config-if-range)#channel-group 2 mode active
```

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解决方案：

中文_垃圾_ sw110：

en-sw110#配
置文件
sw110（config）#生成树模式快速sw110

sw110（配置）#接口范围gi1/2-3
sw110（配置if范围）#通道组2模式激活

中文_垃圾_ sw101：

en-sw101#配置
sw101（config）#生成树模式快速sw101
sw101（配置）#生成树vlan 1-4094你好时间1

sw101（配置）#接口范围gi1/2-3 sw101（如
果范围配置）#通道组1模式打开

中文_垃圾_ sw102：

en-sw102#配
置文件
sw102（config）#生成树模式快速sw102
sw102（配置）#生成树vlan 1-4094你好时间1

sw102（配置）#接口范围gi1/2-3
sw102（配置if范围）#通道组2模式激活

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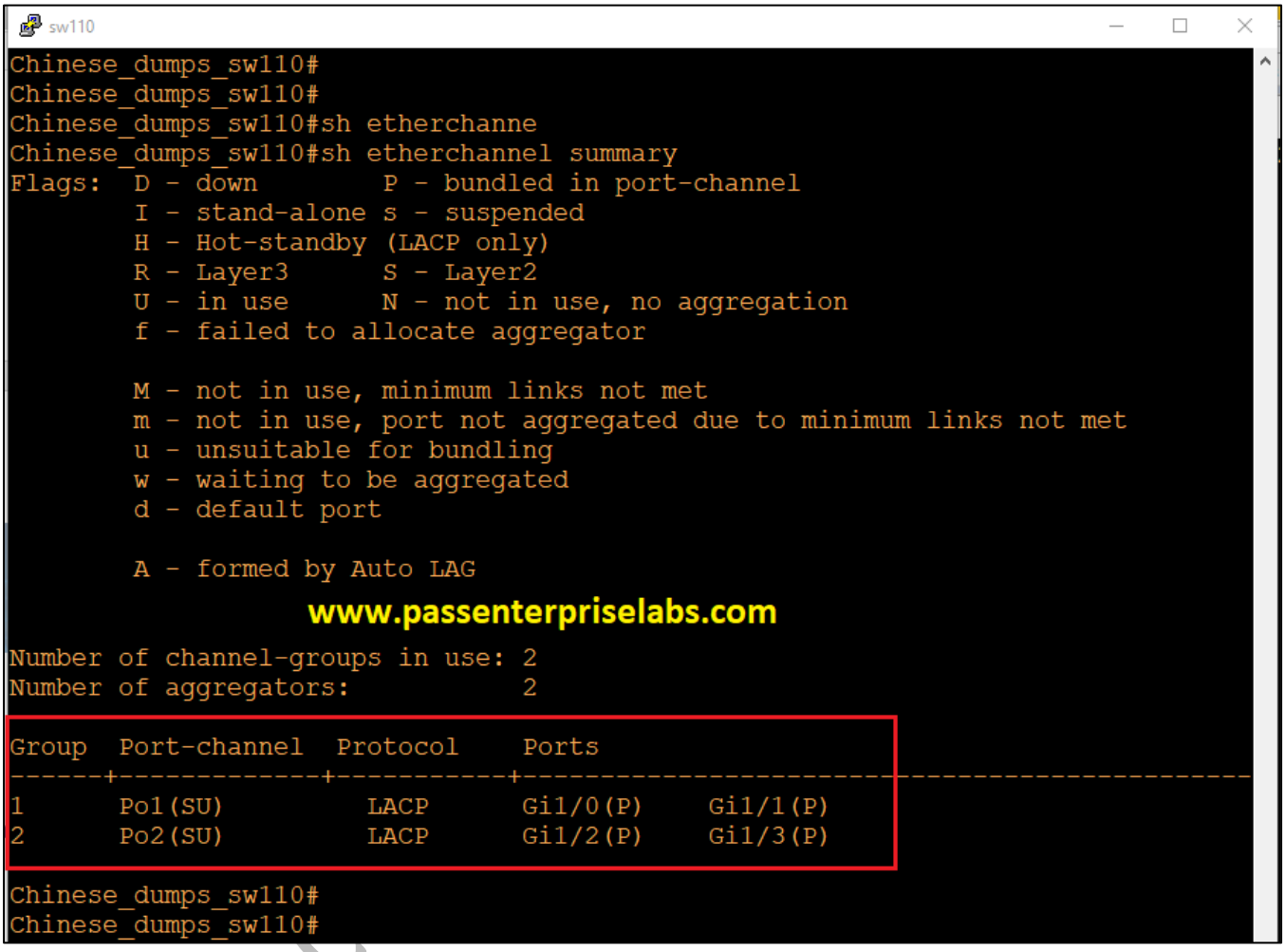
Verification:

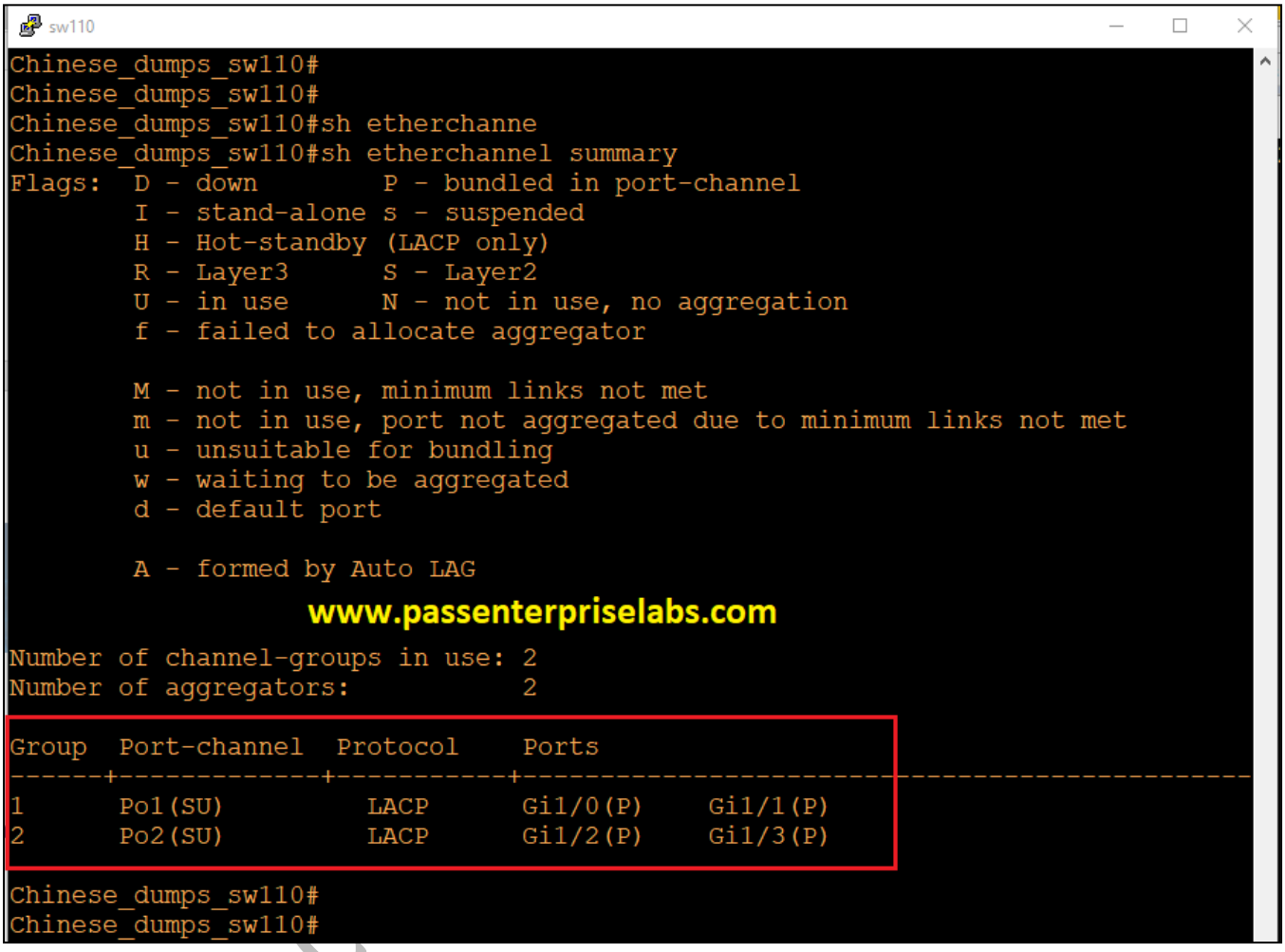
Chinese_dumps_sw110# sh etherchannel summary



www.c

验证：

中文_ dumps_sw110#sh以太坊摘要



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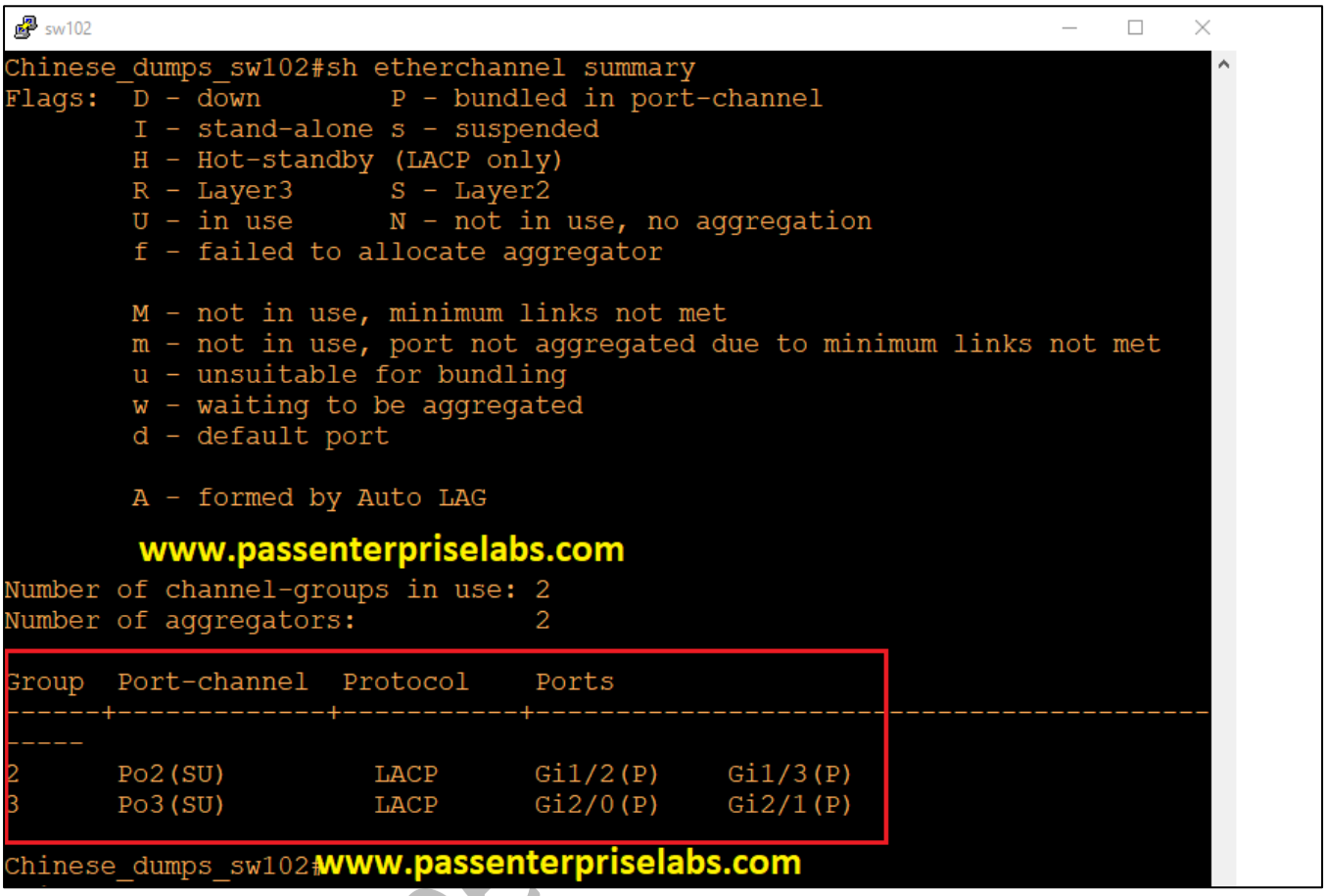
Chinese_dumps_sw102# sh etherchannel summary



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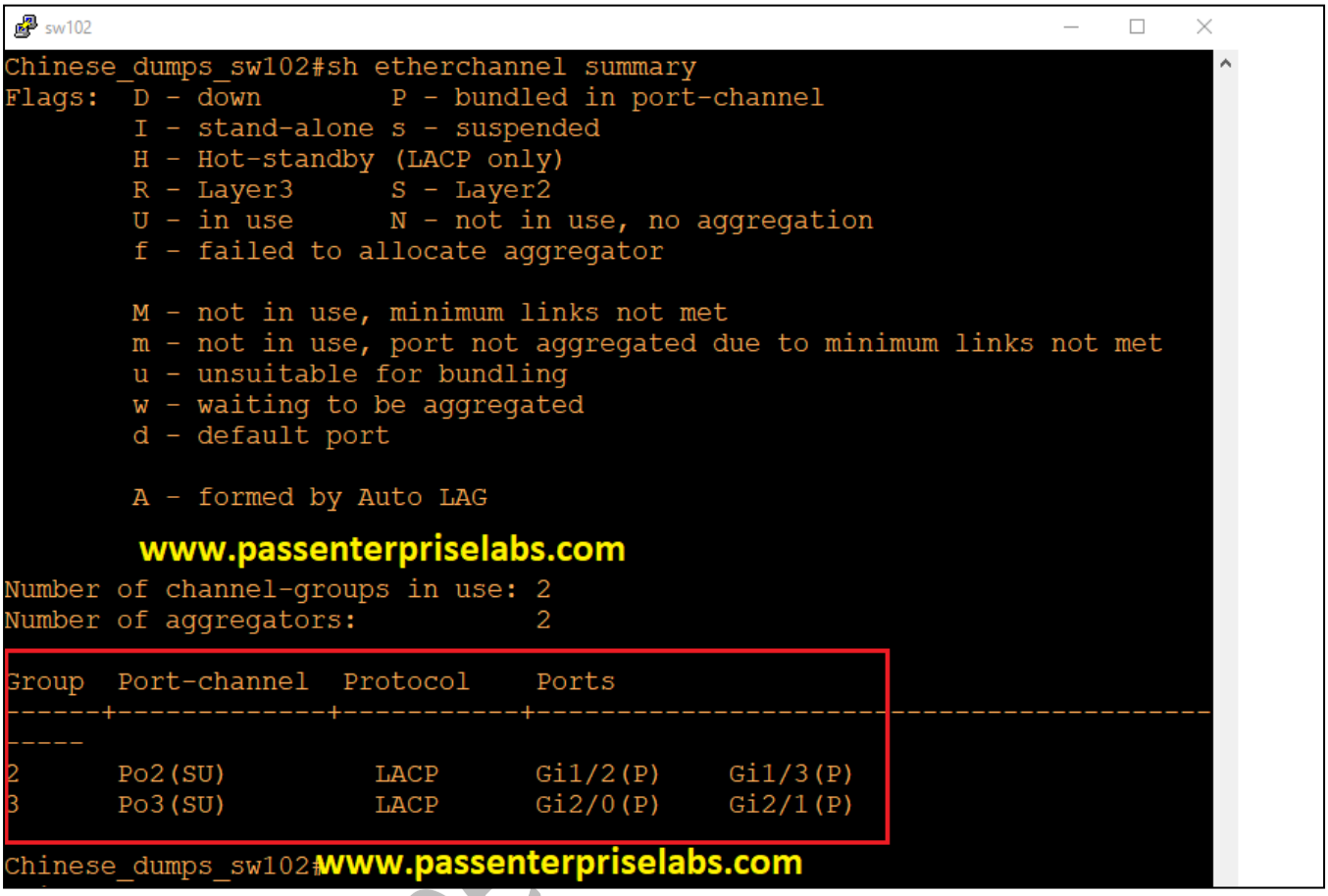
中文_dumps_sw102#sh以太坊频道摘要



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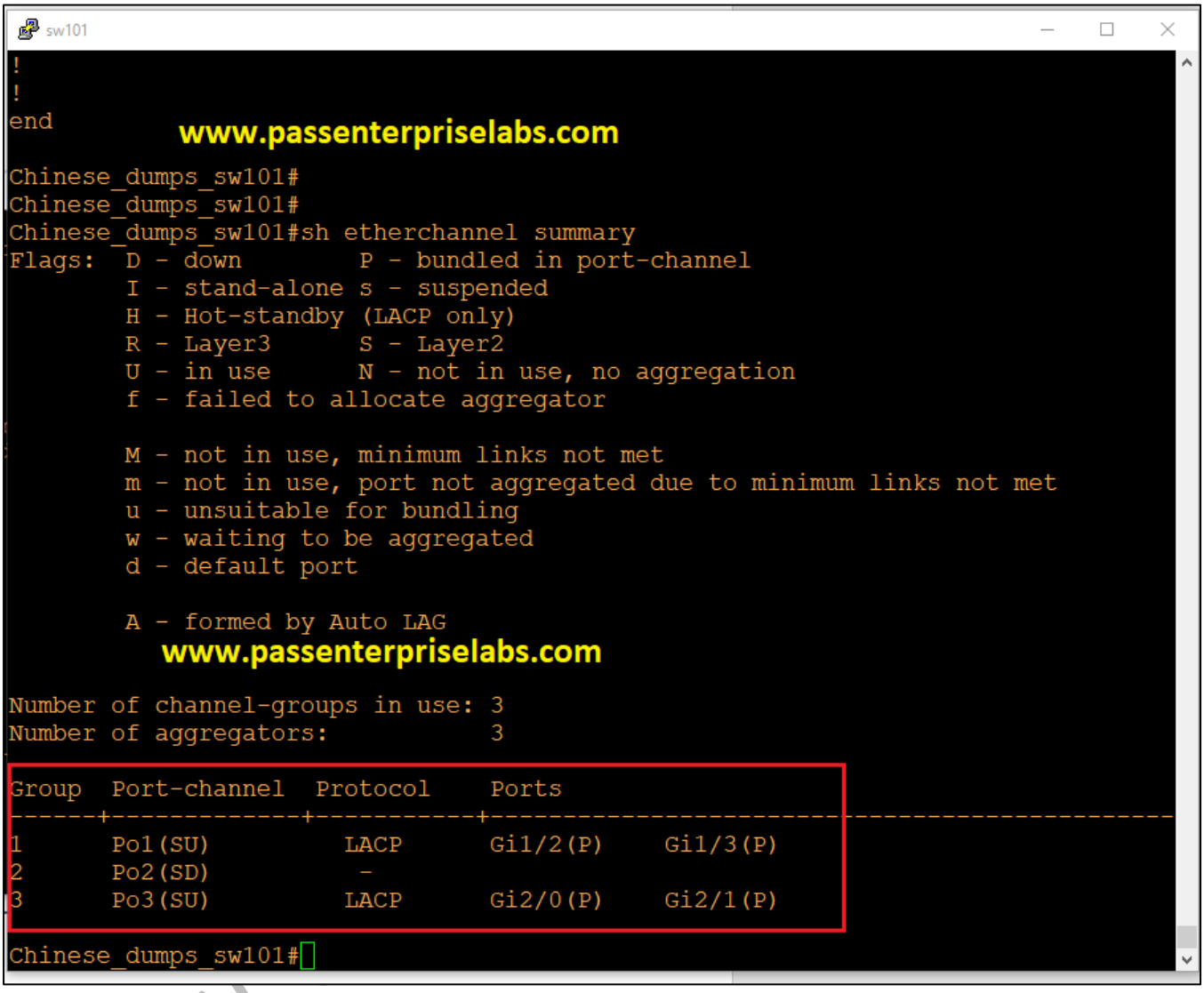
Chinese_dumps_sw101# sh etherchannel summary



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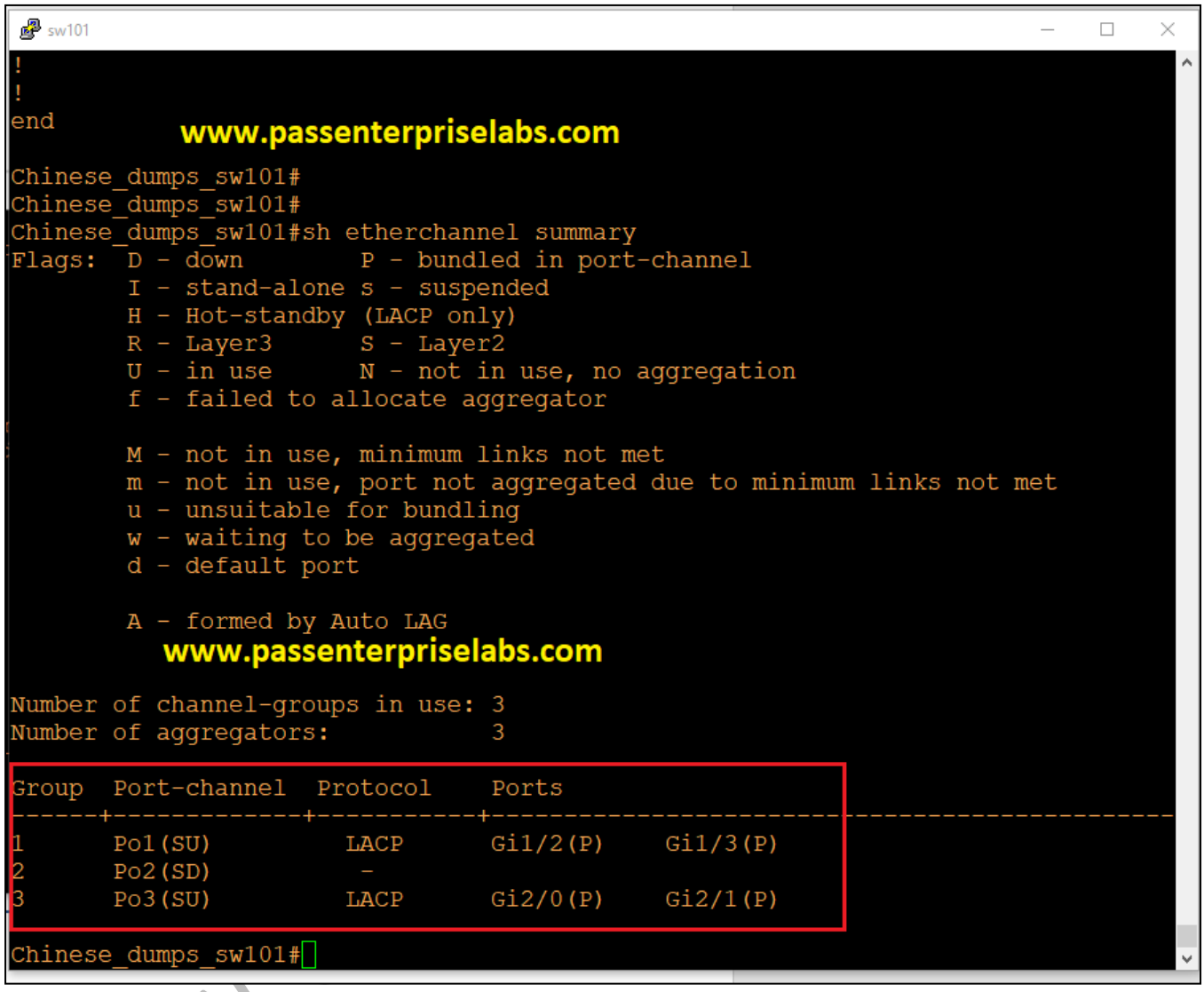
中文_dumps_sw101#sh以太坊频道摘要



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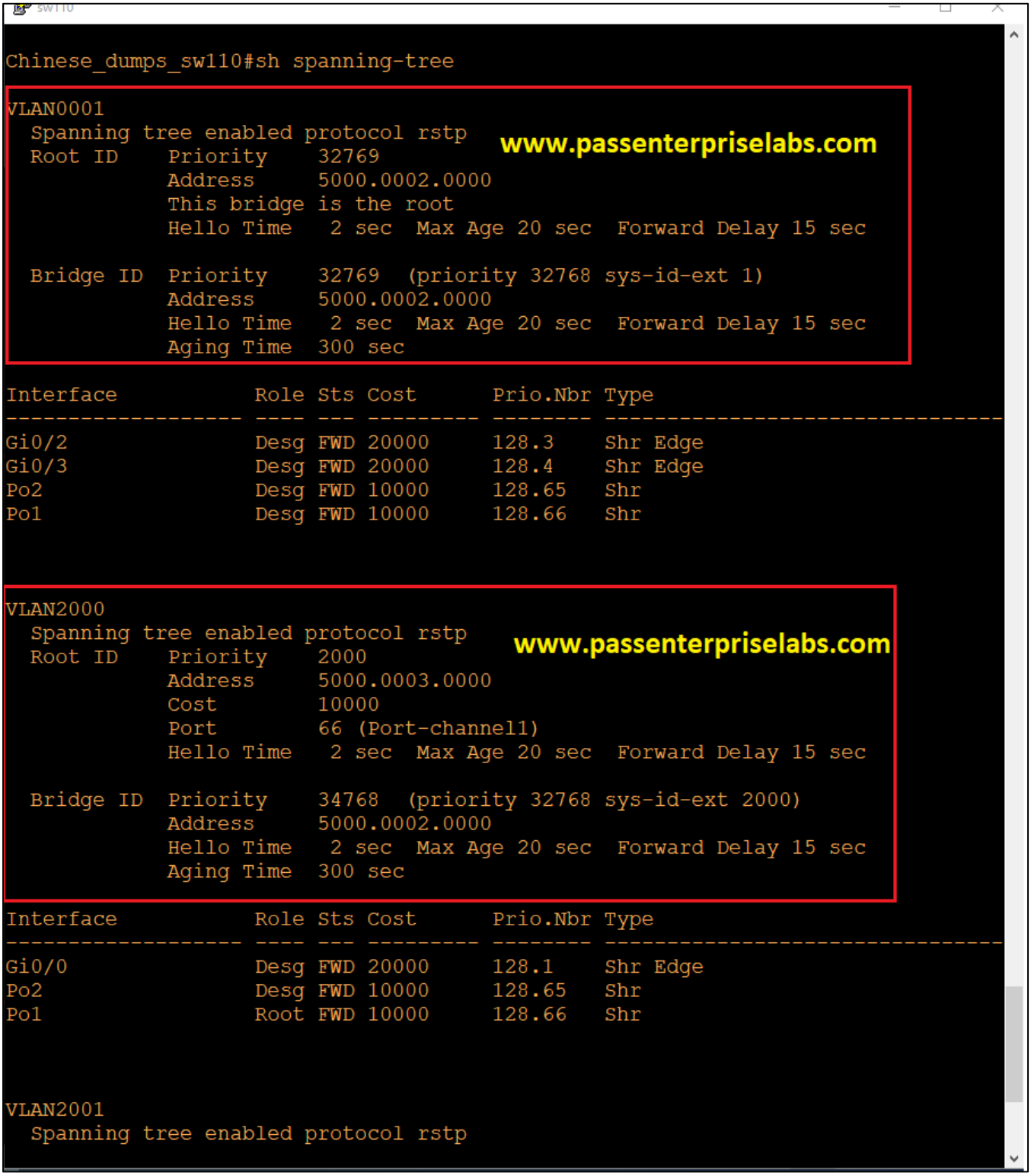
Chinese_dumps_sw110#sh spanning-tree



```
Chinese_dumps_sw110#sh spanning-tree

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID     Priority    32769
              Address     5000.0002.0000
              This bridge is the root
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769  (priority 32768 sys-id-ext 1)
              Address     5000.0002.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/2               Desg FWD 20000     128.3    Shr Edge
Gi0/3               Desg FWD 20000     128.4    Shr Edge
Po2                 Desg FWD 10000     128.65   Shr
Po1                 Desg FWD 10000     128.66   Shr


VLAN2000
  Spanning tree enabled protocol rstp
  Root ID     Priority    2000
              Address     5000.0003.0000
              Cost        10000
              Port        66 (Port-channel1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    34768  (priority 32768 sys-id-ext 2000)
              Address     5000.0002.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Desg FWD 20000     128.1    Shr Edge
Po2                 Desg FWD 10000     128.65   Shr
Po1                 Root FWD 10000     128.66   Shr


VLAN2001
  Spanning tree enabled protocol rstp
```

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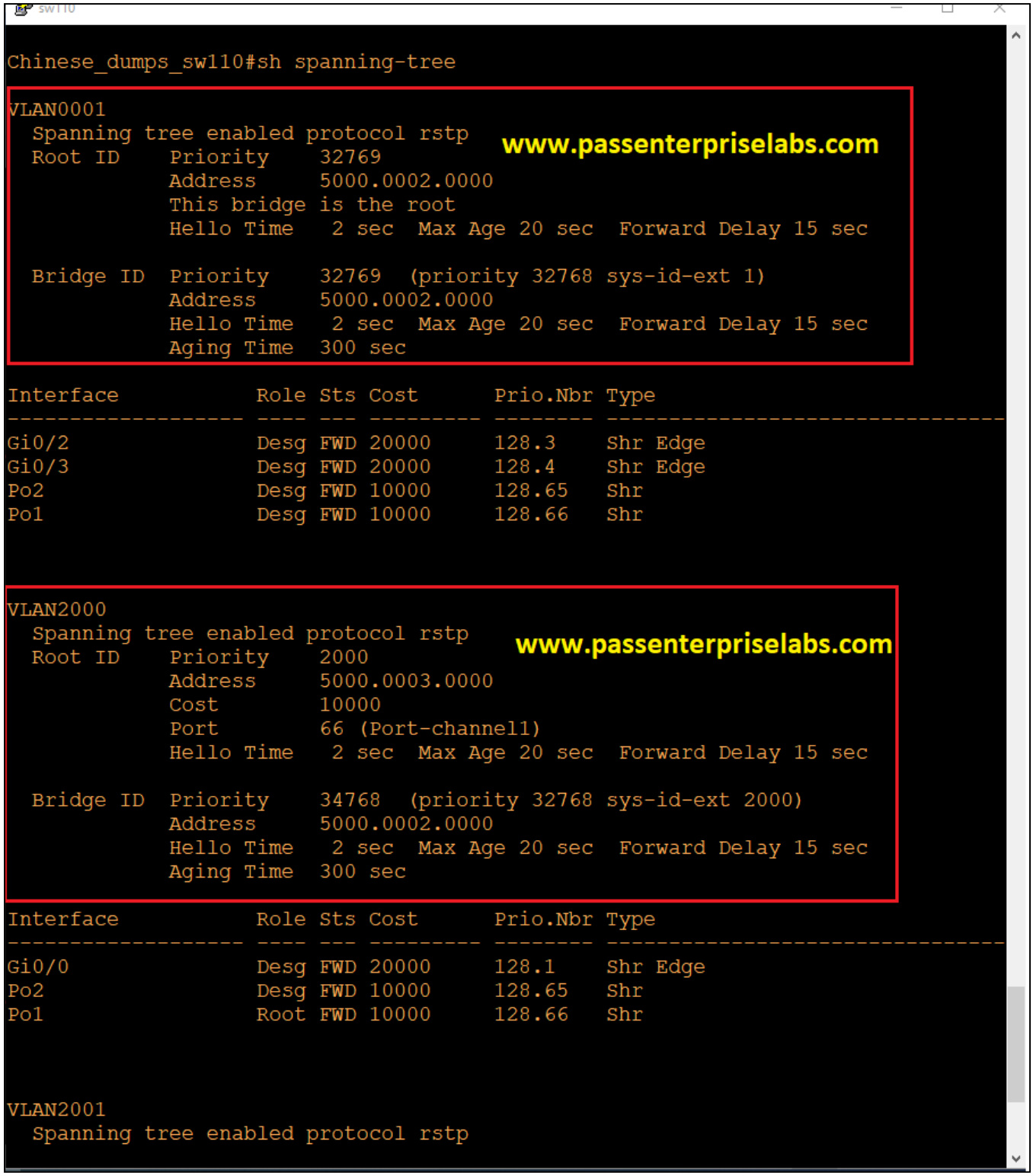
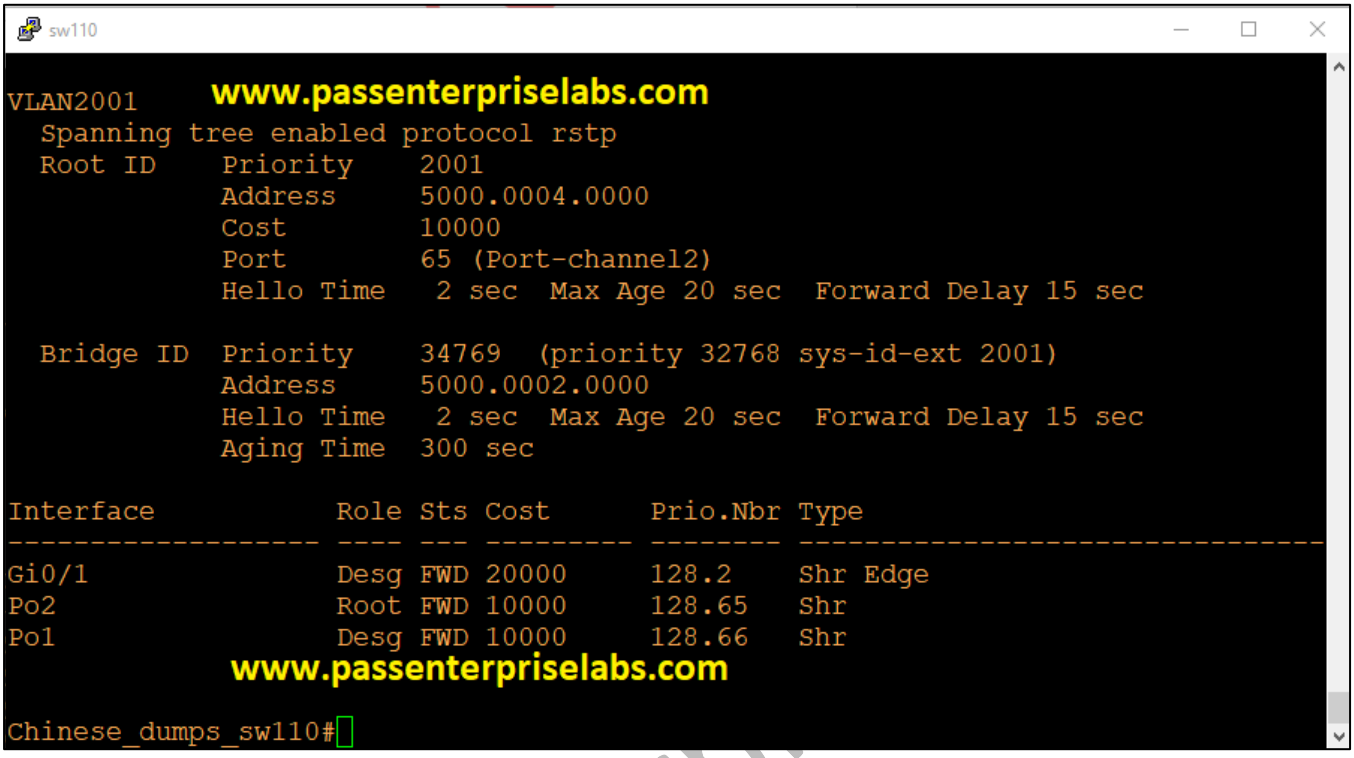
中文_垃圾桶_sw110#sh生成树





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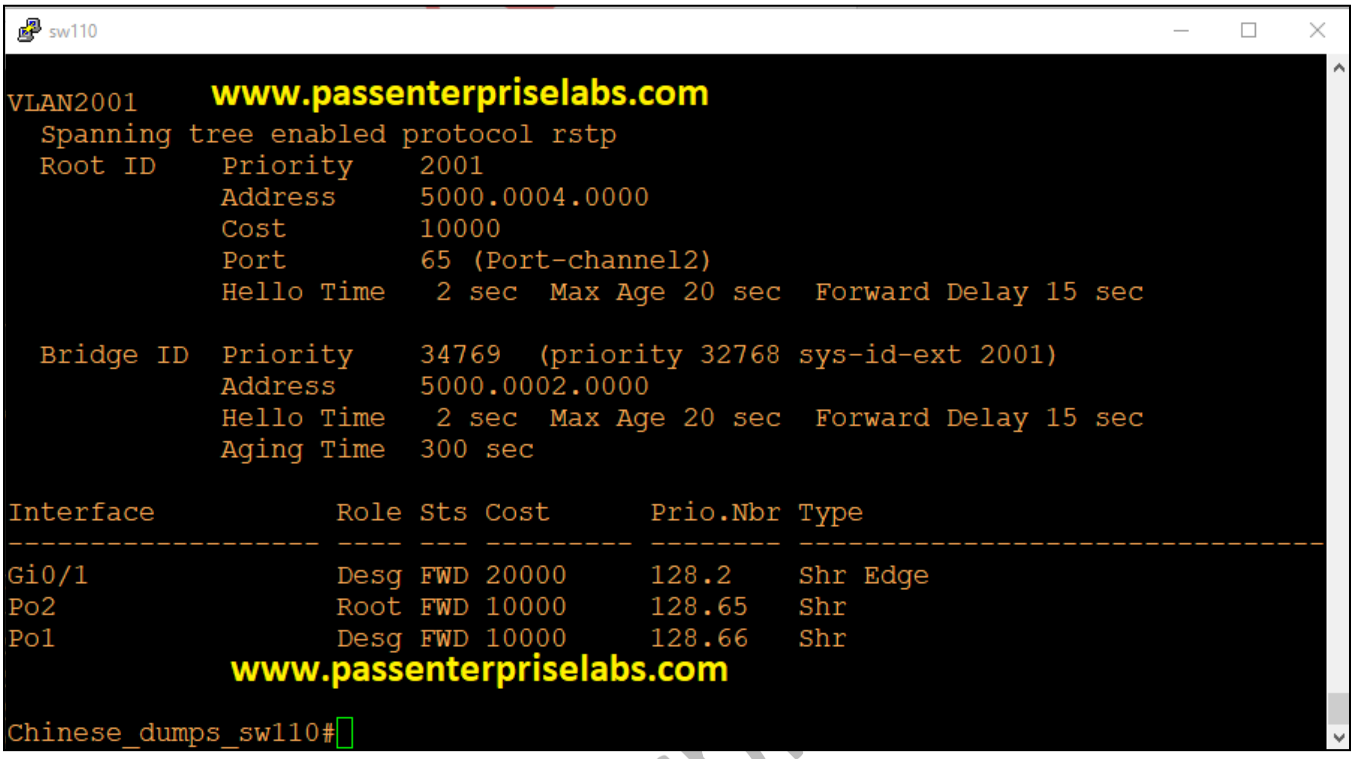

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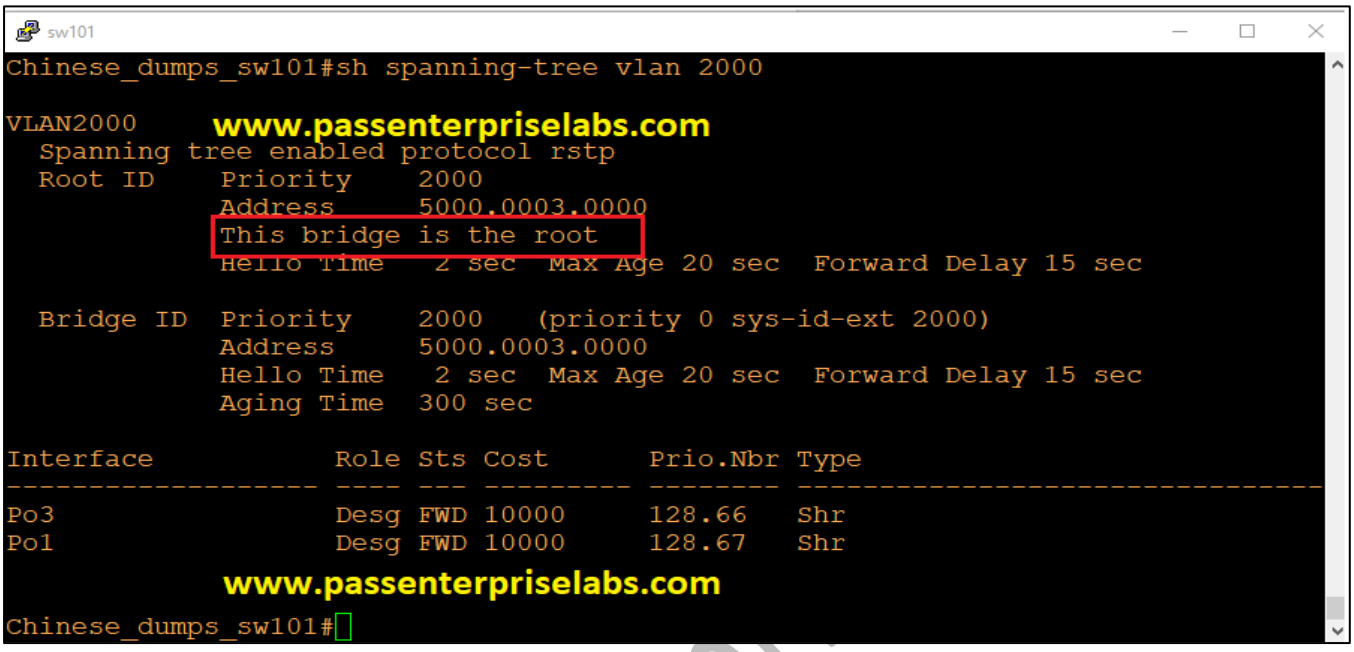
Chinese_dumps_sw101#sh spanning-tree vlan 2000



Chinese_dumps_sw102#sh spanning-tree vlan 2001

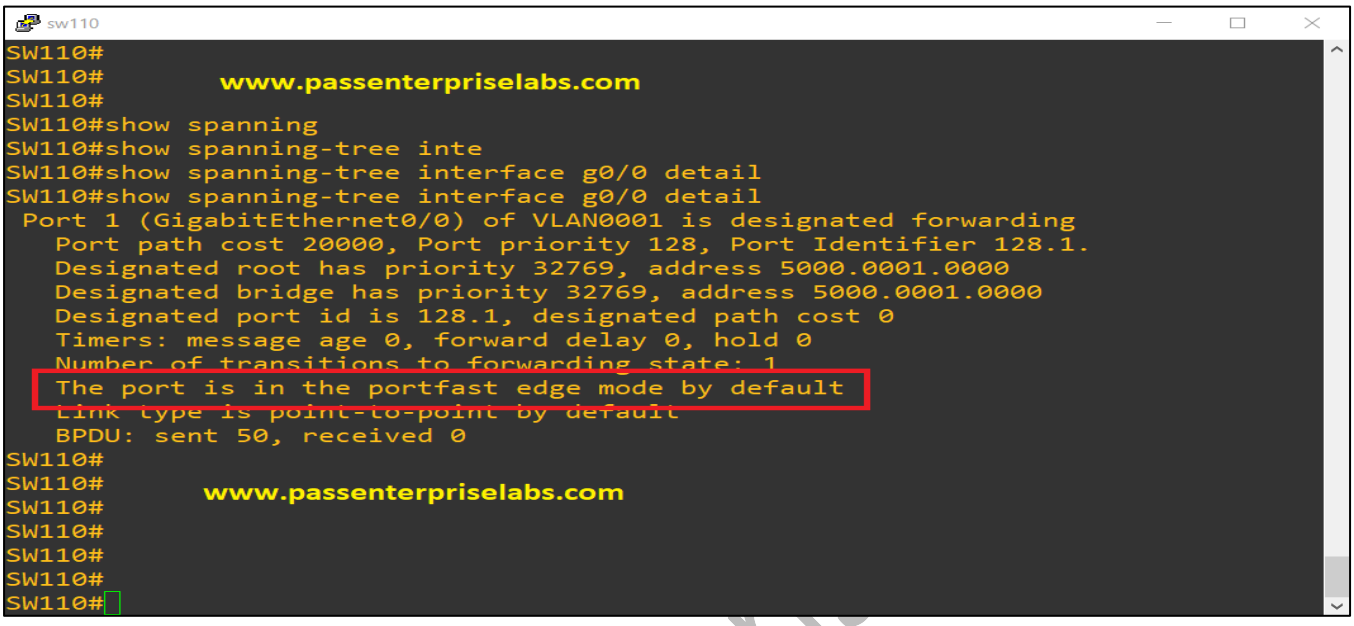
中文_垃圾_ sw101#sh生成树vlan 2000

```
sw101
Chinese_dumps_sw101#sh spanning-tree vlan 2000

VLAN2000     www.passenterpriselabs.com
  Spanning tree enabled protocol rstp
  Root ID     Priority    2000
              Address     5000.0003.0000
              This bridge is the root
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    2000   (priority 0 sys-id-ext 2000)
              Address     5000.0003.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po3                 Desg FWD 10000     128.66   Shr
Po1                 Desg FWD 10000     128.67   Shr
              www.passenterpriselabs.com
Chinese_dumps_sw101#
```

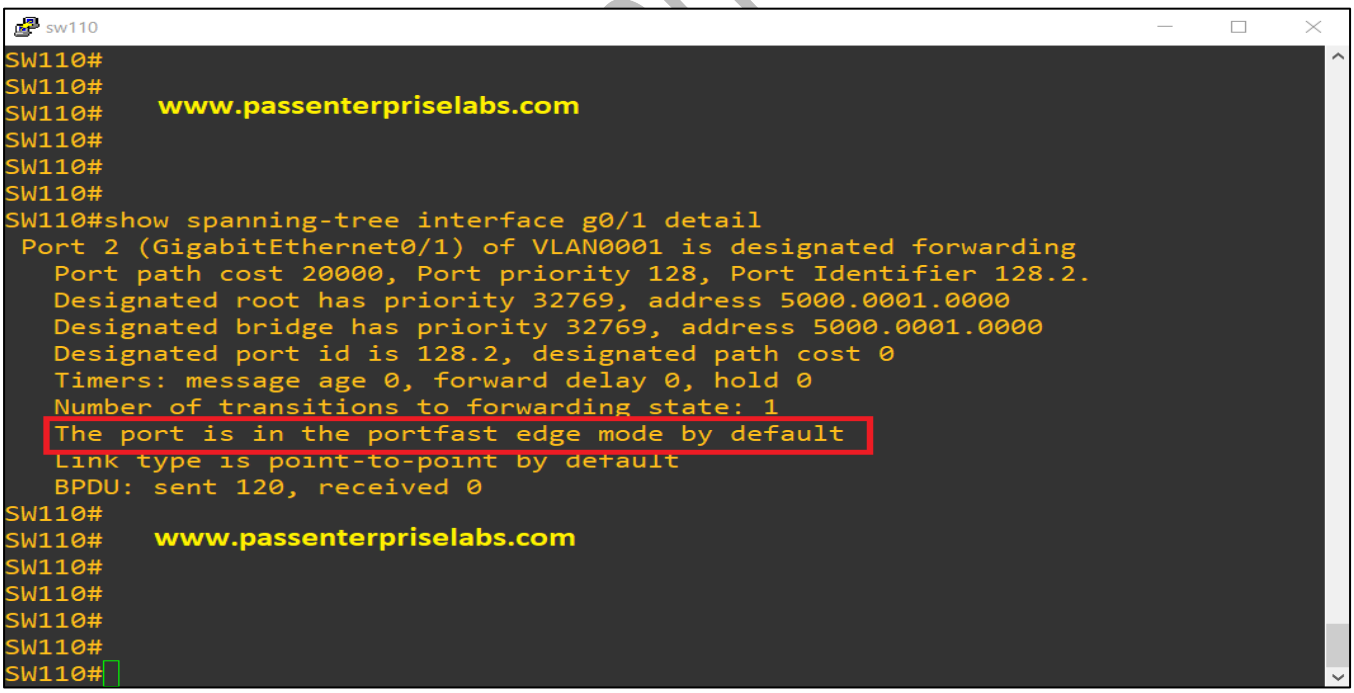
中文_垃圾桶_sw102#sh生成树vlan 2001

```
sw102
Chinese_dumps_sw102#sh spanning-tree vlan 2001
              www.passenterpriselabs.com
VLAN2001
  Spanning tree enabled protocol rstp
  Root ID     Priority    2001
              Address     5000.0004.0000
              This bridge is the root
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    2001   (priority 0 sys-id-ext 2001)
              Address     5000.0004.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po3                 Desg FWD 10000     128.65   Shr
Po2                 Desg FWD 10000     128.66   Shr
              www.passenterpriselabs.com
Chinese_dumps_sw102#
```

Chinese_dumps_sw110#sh spanning-tree interface g0/0 detail

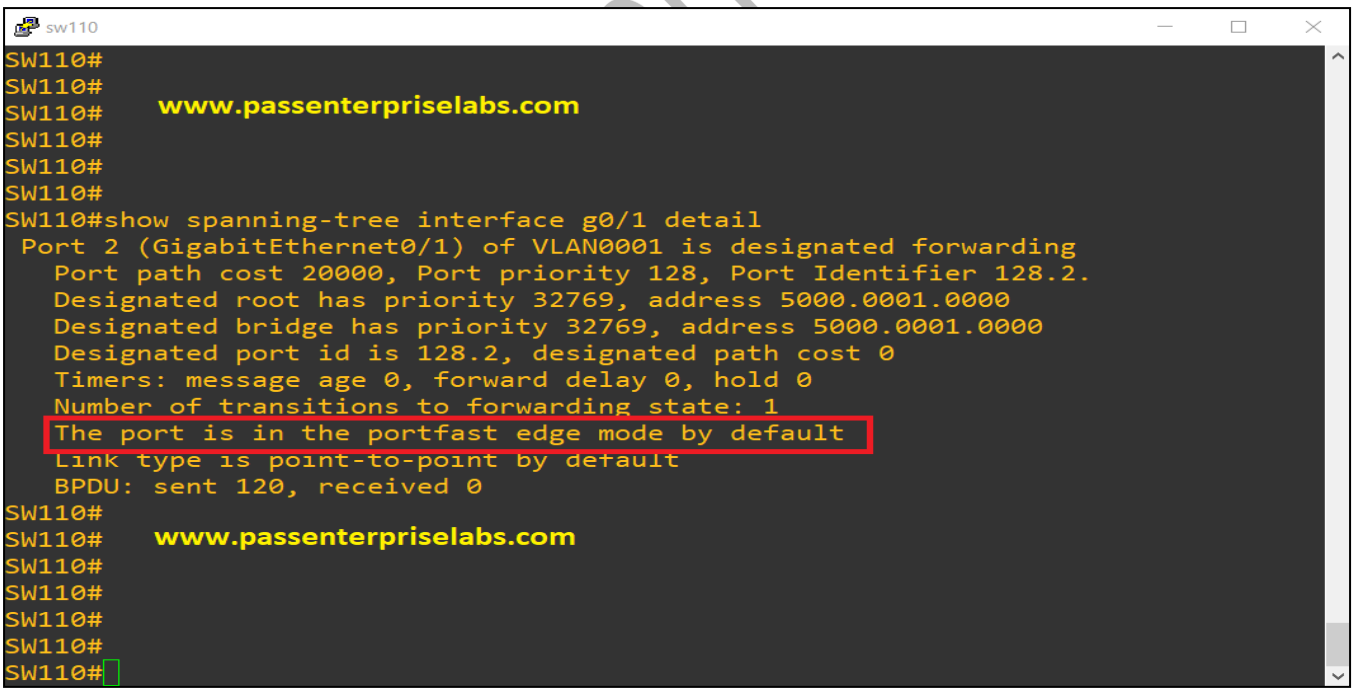


Chinese_dumps_sw110#sh spanning-tree interface g0/1 detail



中文_dumps_sw110#sh生成树接口g0/0详细信息

中文_dumps_sw110#sh生成树接口g0/1详细信息



1.3: First Hop Redundancy Protocol in HQ

For IPv4, implement an FHRP mechanism on sw101 and sw102 for VLANs 2000 and 2001 according to these requirements:

- Use Group number 100 for VLAN 2000 and group number 101 for VLAN 2001.
- Use the first available IPV4 address in the subnet for the address of the virtual router.
- For VLAN 2000, sw101 must be preferred gateway; for VLAN 2001, sw102 must be the preferred gateway. Do not rely on the IPv4 addresses of the switches as role tiebreakers. The role must determine by an explicit configuration solely on the intended preferred gateway.
- Each preferred gateway must monitor the reachability of both routers r11 and r12 using the loopback IPv4 addresses of the routers by an ICMP Echo. The reachability is to be verified every 5 seconds with a timeout of 400 msec. A router must be declared unreachable as soon as it does not respond to three probes in a row. If both r11 and r12 are declared unreachable from a preferred gateway, the other switch must be allowed to assume the gateway role.
- Use the FHRP protocol that allows the virtual IPv4 address to match the IPv4 address of a member router.

2 Points

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1.3：总部第一跳冗余协议

对于IPv4，根据以下要求在sw101和sw102上为VLAN 2000和2001实现FHRP机制：

- VLAN 2000使用组号100，VLAN 2001使用组号101。
- 将子网中第一个可用的IPV4地址用作虚拟路由器的地址。
- 对于VLAN 2000，sw101必须是首选网关；对于VLAN 2001，sw102必须是首选网关。不要依赖交换机的IPv4地址作为角色分层断路器。角色必须仅通过预期首选网关上的显式配置来确定。
- 每个首选网关必须通过ICMP回显使用路由器的环回IPv4地址来监视路由器r11和r12的可达性。每5秒验证一次可达性，超时400毫秒。一旦路由器对连续三个探测没有响应，就必须宣布其不可访问。如果r11和r12都被声明为无法从首选网关访问，则必须允许另一个交换机承担网关角色。
- 使用FHRP协议，该协议允许虚拟IPv4地址与成员路由器的IPv4地址匹配。

2分

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Solution:

Chinese_dumps_sw101:

```
sw101#en
sw101#conf t
sw101(config)#int vlan 2000
sw101(config-if)#vrrp 100 ip 10.1.100.1
sw101(config-if)#vrrp 100 priority 105
sw101(config-if)#ip ospf 1 area 0
sw101(config-if)#vrrp 100 track 100
sw101(config-if)#exit


sw101(config)#int vlan 2001
sw101(config-if)#vrrp 101 ip 10.1.101.1
sw101(config-if)#ip ospf 1 area 0
sw101(config-if)#exit

sw101(config)#ip sla 1
sw101(config-ip-sla)#icmp-echo 10.1.255.11 source-interface vlan 2000
sw101(config-ip-sla-echo)#threshold 400
sw101(config-ip-sla-echo)#timeout 400
sw101(config-ip-sla-echo)#frequency 5
sw101(config-ip-sla-echo)#exit
sw101(config)#ip sla schedule 1 start-time now life forever

sw101(config)#ip sla 2
sw101(config-ip-sla)#icmp-echo 10.1.255.12 source-interface vlan 2000
sw101(config-ip-sla-echo)#threshold 400
sw101(config-ip-sla-echo)#timeout 400
sw101(config-ip-sla-echo)#frequency 5
sw101(config-ip-sla-echo)#exit
sw101(config)#ip sla schedule 2 start-time now life forever

sw101(config)#track 1 ip sla 1
sw101(config-track)#delay down 10
sw101(config)#exit
sw101(config)#track 2 ip sla 2
sw101(config-track)#delay down 10
```

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解决方案：

中文_垃圾_ sw101：

```
sw101#en
sw101#配置
文件
sw101（配置）#int vlan 2000
sw101（配置if）#vrrp 100 ip 10.1.100.1
sw101（配置if）#vrrp 100优先级105
sw101（配置if）#ip ospf 1区域0
sw101（配置if）#vrrp 100轨道
100 sw101（设置if）#退出


sw101（配置）#int vlan 2001
sw101（配置if）#vrrp 101 ip 10.1.101.1
sw101（配置if）#ip-ospf 1
区域0 sw101（设置if）#退出

sw101（配置）#ip sla 1
sw101（配置ip sla）#icmp echo 10.1.255.11源接口vlan 2000
sw101（配置ip sla回声）#阈值400
sw101（配置ip sla echo）#超时400
sw101（配置ip sla回声）#频率5
sw101（设置ip sla回波）#退出
sw101（配置）#ip sla时间表1开始时间现在永远有效

sw101（配置）#ip sla 2
sw101（配置ip sla）#icmp echo 10.1.255.12源接口vlan 2000
sw101（配置ip sla回声）#阈值400
sw101（配置ip sla echo）#超时400
sw101（配置ip sla回声）#频率5
sw101（设置ip sla回波）#退出
sw101（配置）#ip sla计划2开始时间现在永远有效

sw101（配置）#轨道1 ip sla 1
sw101（配置轨道）#延迟10 sw101
（config）#退出sw101（config）
#轨道2 ip sla 2
sw101（配置轨道）#延迟下降10
```

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```
sw101(config)#track 100 list boolean OR
sw101(config-track)#object 1
sw101(config-track)#object 2

sw101(config)#router ospf 1
sw101(config-router)#passive-interface vlan 2000
sw101(config-router)#passive-interface vlan 2001
```

Chinese_dumps_sw102:

```
sw102>en
sw102#conf t
sw102(config)#int vlan 2000
sw102(config-if)#vrrp 100 ip 10.1.100.1
sw102(config-if)#ip ospf 1 area 0
sw102(config-if)#exit

sw102(config)#int vlan 2001
sw102(config-if)#vrrp 101 ip 10.1.101.1
sw102(config-if)#vrrp 101 priority 105
sw102(config-if)#vrrp 101 track 101
sw102(config-if)#ip ospf 1 area 0
sw102(config-if)#exit

sw102(config)#ip sla 1
sw102(config-ip-sla)#icmp-echo 10.1.255.11 source-interface vlan 2001
sw102(config-ip-sla-echo)#threshold 400
sw102(config-ip-sla-echo)#timeout 400
sw102(config-ip-sla-echo)#frequency 5
sw102(config)#exit

sw102(config)#ip sla schedule 1 start-time now life forever
```

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sw101（config）#track 100列表布
尔值或sw101（config-track）#对象
1
sw101（配置轨道）#对象2

sw101（配置路由器）#无源接口vlan2000
sw101（配置路由器）#无源接口vlan 2001

中文_垃圾_ sw102：

sw102>en-sw102#配
置文件
sw102（配置）#int vlan 2000
sw102（配置if）#vrrp 100 ip
10.1.100.1
sw102（配置if）#ip-ospf 1

sw102（配置）#int vlan 2001
sw102（配置if）#vrrp 101 ip
10.1.101.1
sw102（配置if）#vrrp 101优先级
105
sw102（配置if）#vrrp 101轨道101

sw102（配置）#ip sla 1
sw102（配置ip sla）#icmp echo 10.1.255.11源接口vlan 2001
sw102（配置ip sla回声）#阈值400
sw102（配置ip sla echo）#超时400
sw102（配置ip sla回声）#频率5 sw102（设置）#退
出

sw102（配置）#ip sla时间表1开始时间现在永远有效

sw101（配置）#路由器ospf 1

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```
sw102(config)#ip sla 2
sw102(config-ip-sla-echo)#icmp-echo 10.1.255.12 source-interface vlan 2001
sw102(config-ip-sla-echo)#threshold 400
sw102(config-ip-sla-echo)#timeout 400
sw102(config-ip-sla-echo)#frequency 5
sw102(config-ip-sla-echo)#exit

sw102(config)#ip sla schedule 2 start-time now life forever

sw102(config)#track 1 ip sla 1
sw102(config-track)#delay down 10
sw102(config-track)#exit

sw102(config)#track 2 ip sla 2
sw102(config-track)#delay down 10
sw102(config-track)#exit

sw102(config)#track 101 list Boolean OR
sw102(config-track)#object 1
sw102(config-track)#object 2


sw102(config)#router ospf 1
sw102(config-router)#passive-interface vlan 2000
sw102(config-router)#passive-interface vlan 2001
```

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sw102（配置）#ip sla 2
sw102（配置ip sla echo）#icmp echo 10.1.255.12源接口vlan 2001
sw102（配置ip sla回声）#阈值400
sw102（配置ip sla echo）#超时400
sw102（配置ip sla回声）#频率5
sw102（配置ip sla echo）#退出

sw102（配置）#ip sla调度2开始时间现在永远有效

sw102（配置）#轨道1 ip sla 1
sw102（配置轨道）#延迟10
sw102（设置轨道）#退出

sw102（配置）#轨道2 ip sla 2
sw102（配置轨道）#延迟下降10
sw102（配置跟踪）#退出

sw102（config）#track 101列表布
尔OR
sw102（配置轨迹）#对象1

sw102（配置）#路由器ospf 1
sw102（配置路由器）#无源接口vlan2000
sw102（配置路由器）#无源接口vlan 2001

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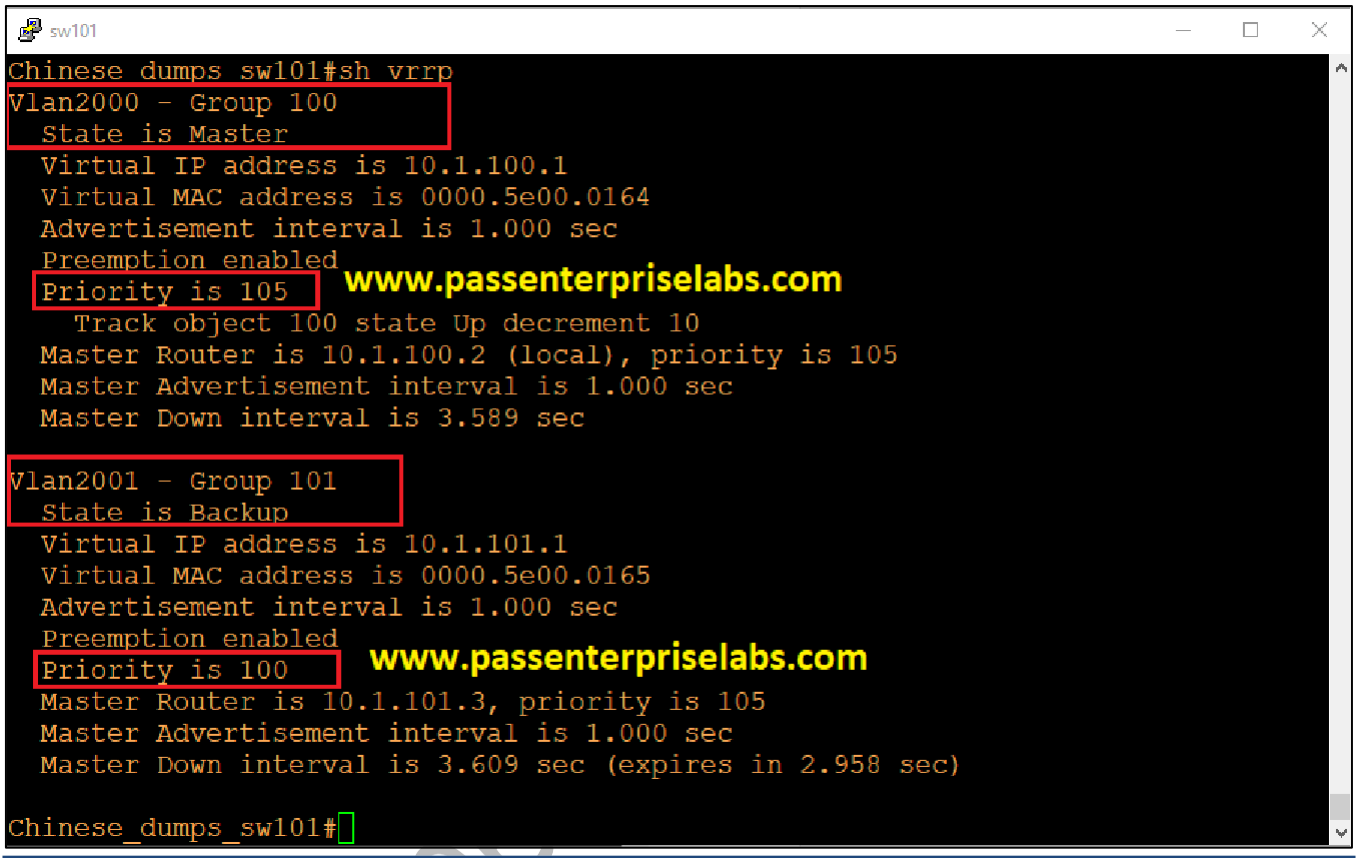
Verification:

Chinese_dumps_sw101#sh vrrp



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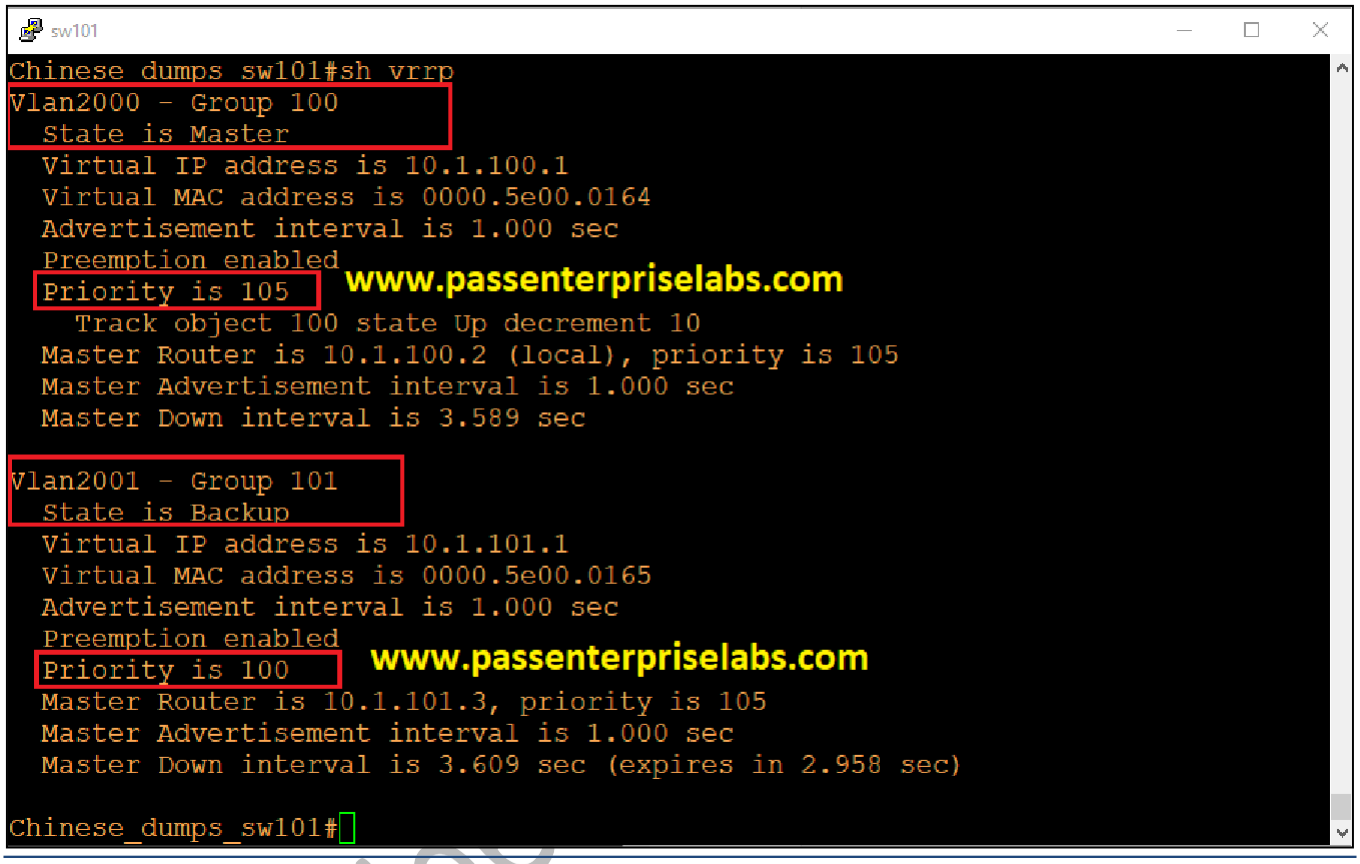
验证：

中文_垃圾桶_sw101#sh vrrp



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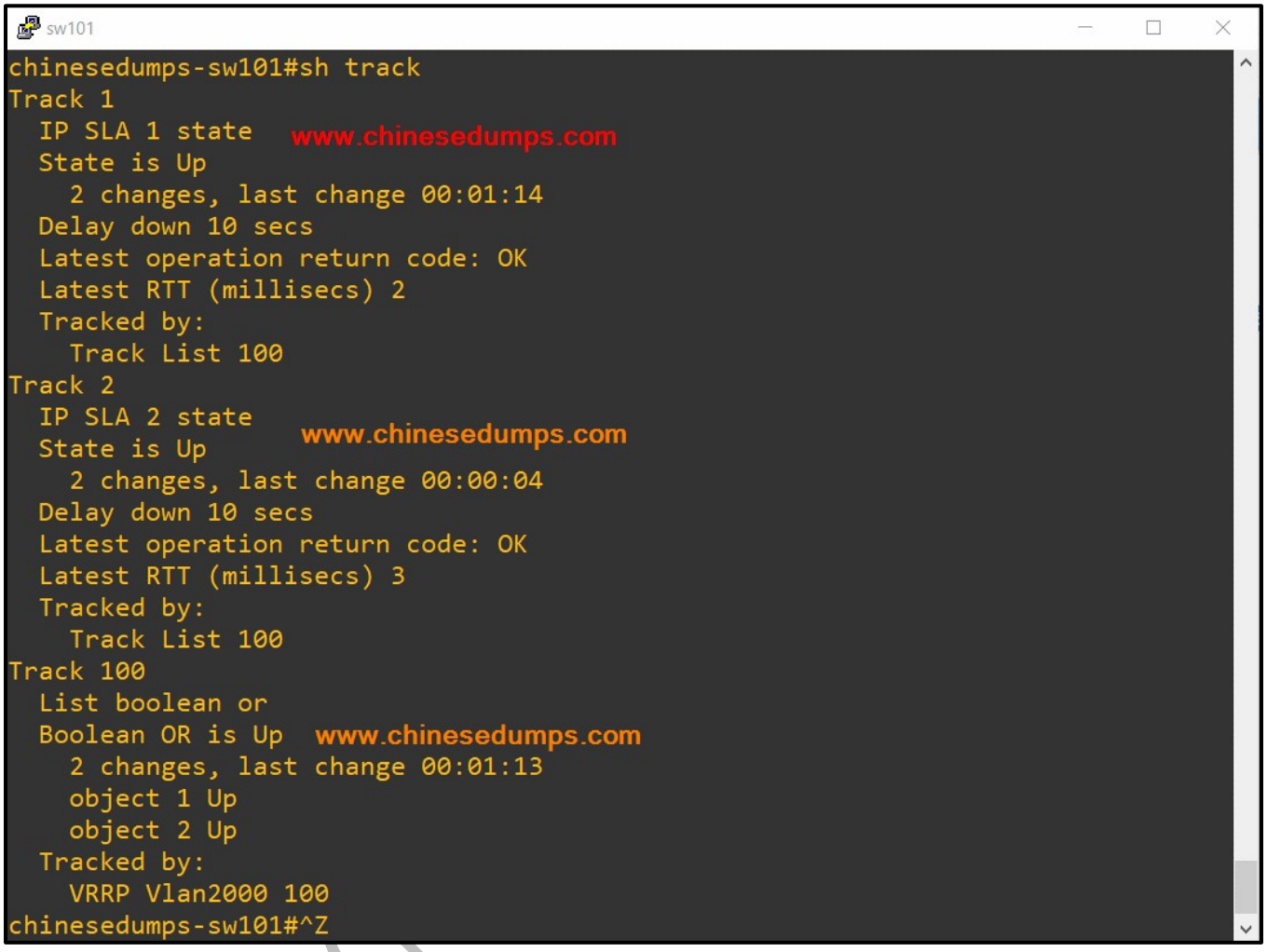
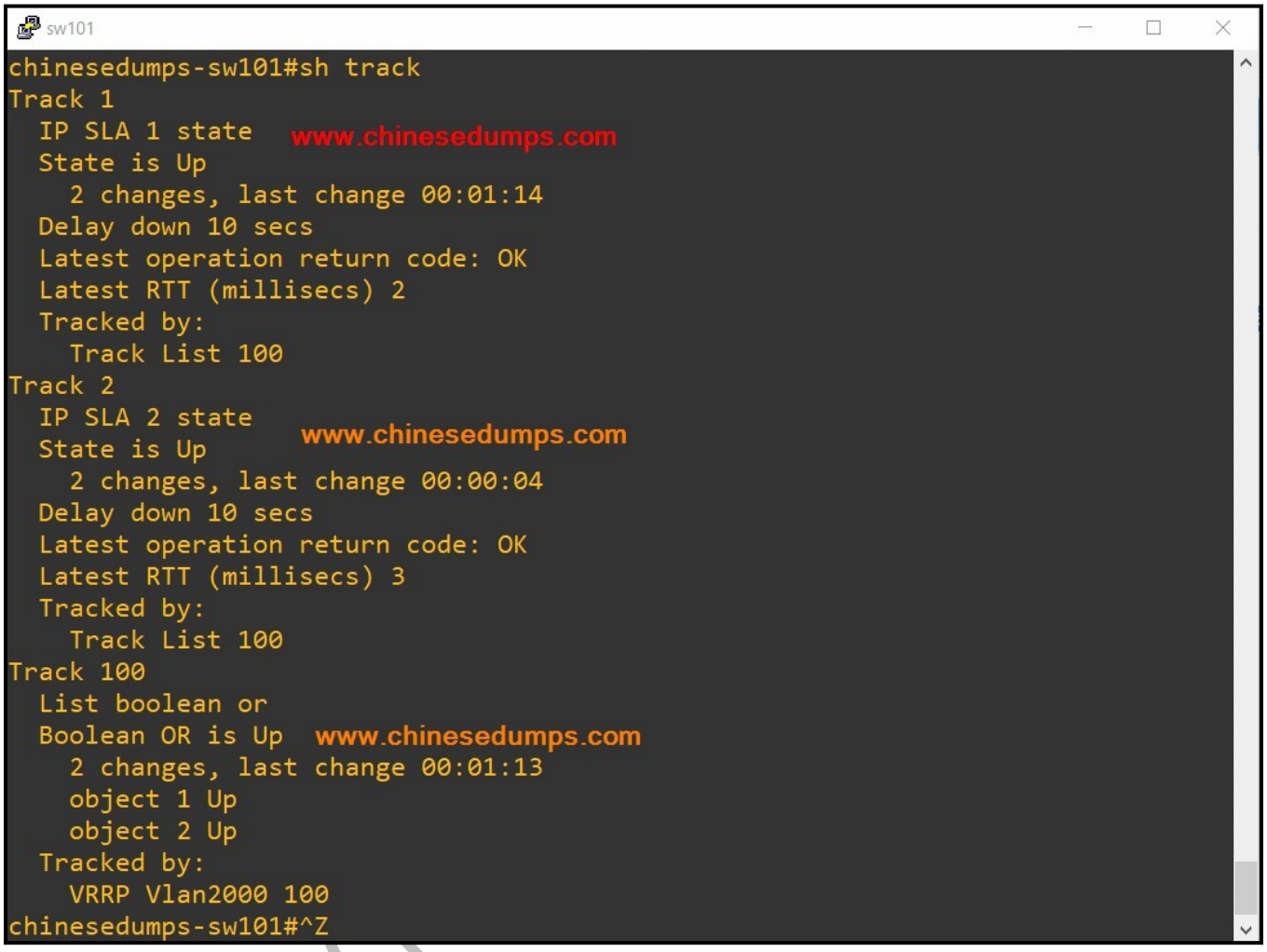
Chinese_dumps_sw101#sh track



www.CC

中文_垃圾桶_ sw101#sh轨道



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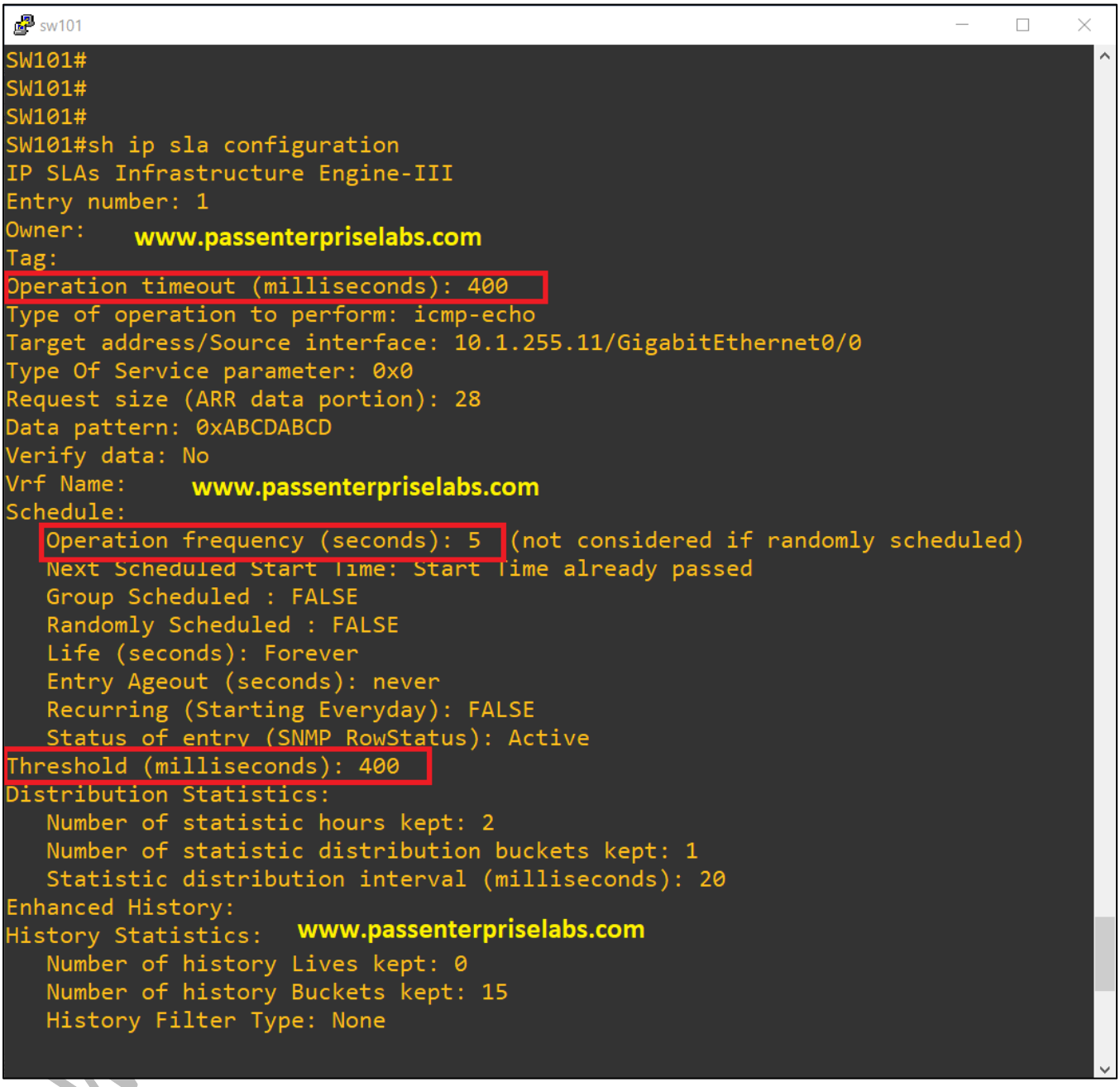
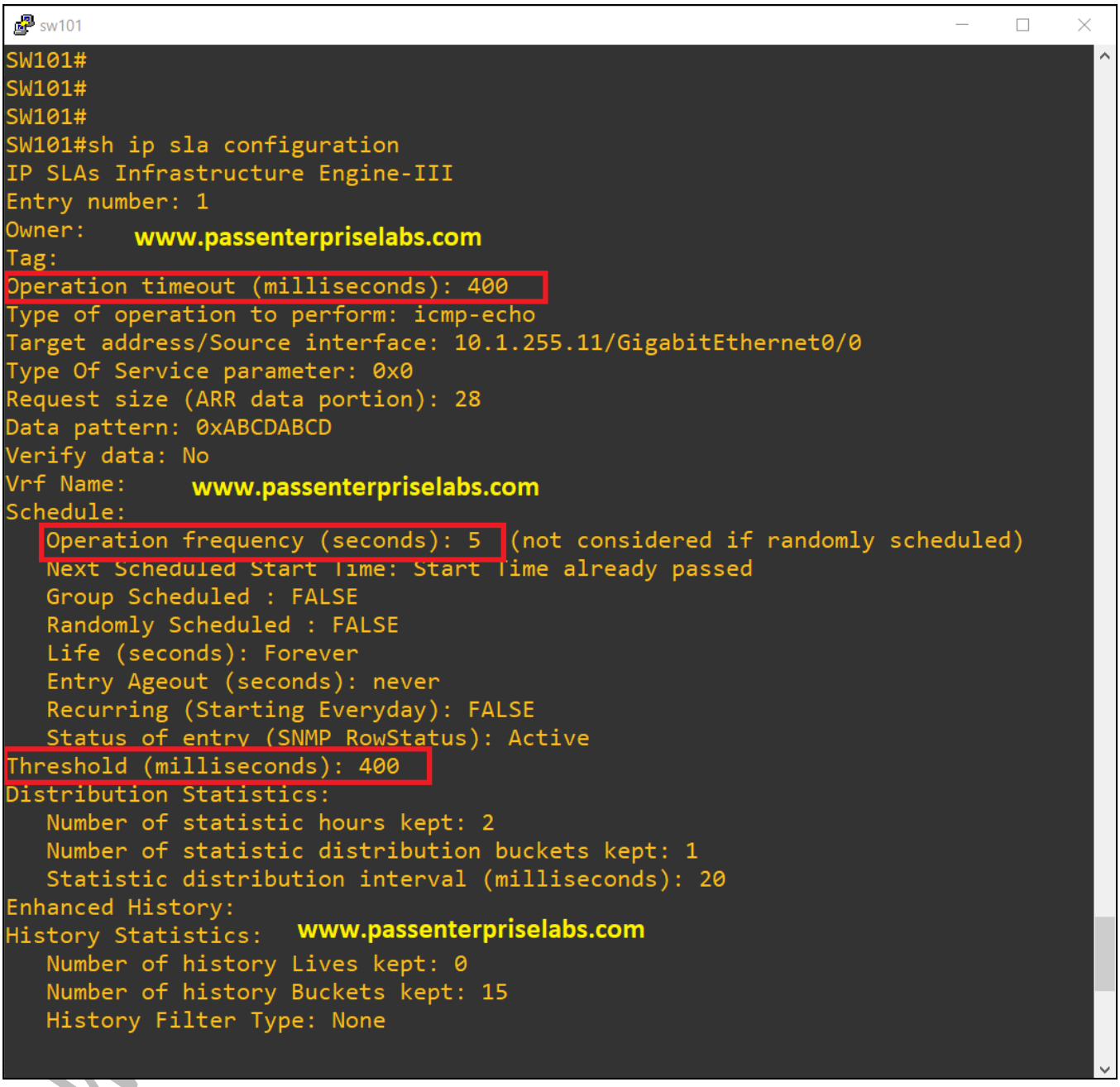
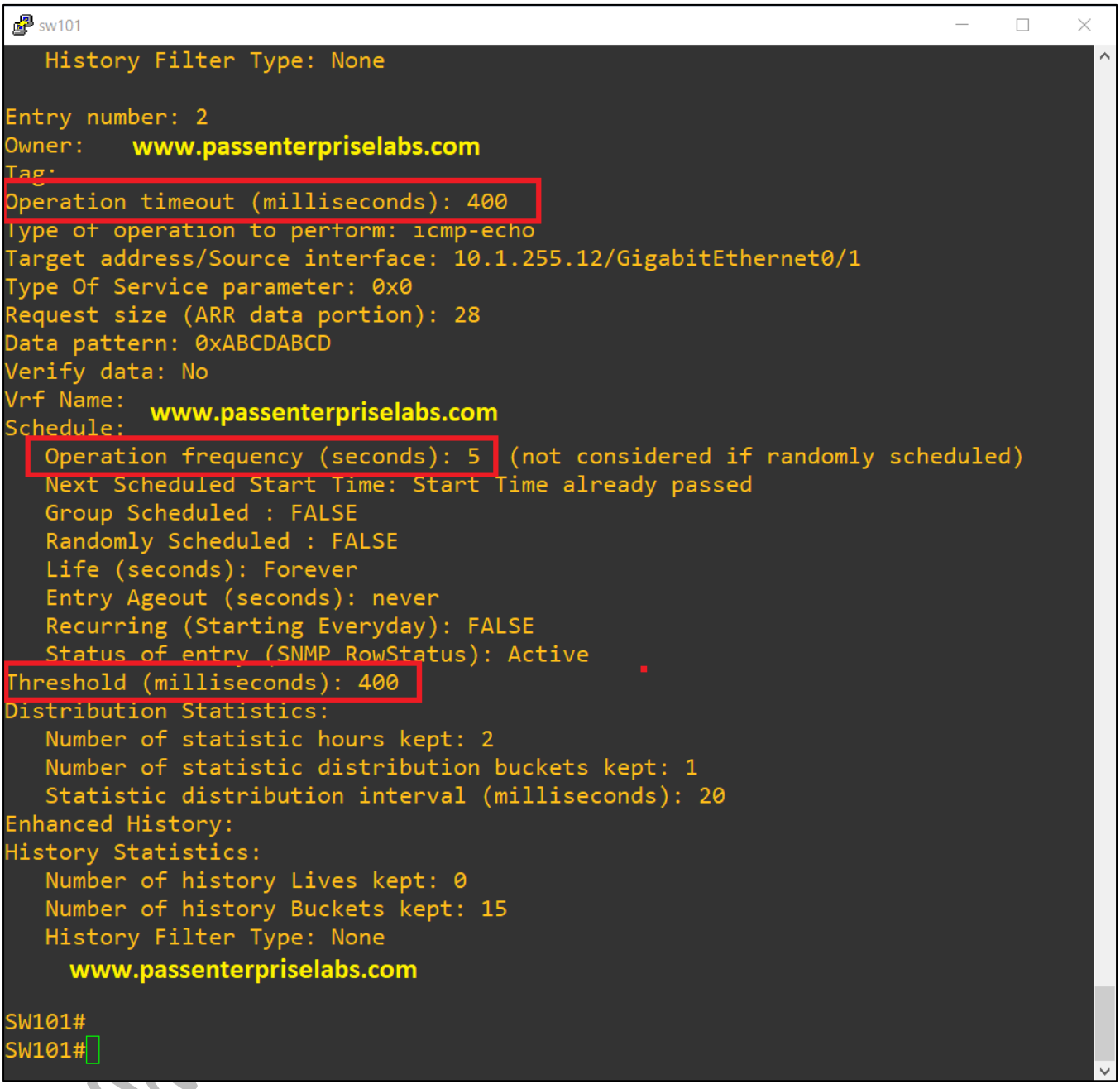
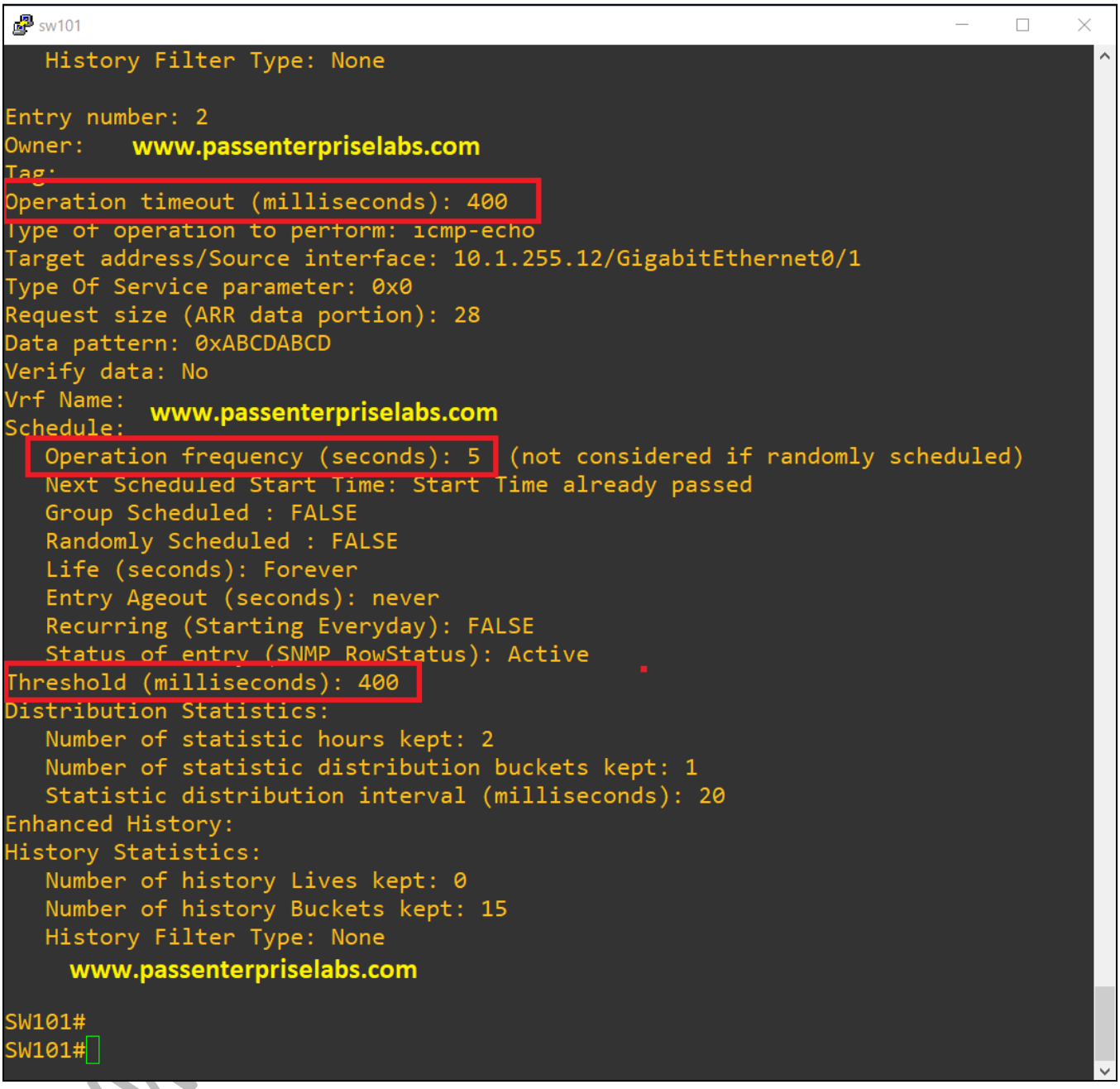
Chinese_dumps_sw101#sh ip sla configuration



中文_dumps_sw101#sh ip sla配置




sw101
History Filter Type: None
Entry number: 2
Owner:
www.passenterpriselabs.com
Tag:
Operation timeout (milliseconds): 400
Type of operation to perform: icmp-echo
Target address/Source interface: 10.1.255.12/GigabitEthernet0/1
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Data pattern: 0xABCDABCD
Verify data: No
Vrf Name:
www.passenterpriselabs.com
Schedule:
Operation frequency (seconds): 5 (not considered if randomly scheduled)
Next Scheduled Start Time: Start Time already passed
Group Scheduled : FALSE
Randomly Scheduled : FALSE
Life (seconds): Forever
Entry Ageout (seconds): never
Recurring (Starting Everyday): FALSE
Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 400
Distribution Statistics:
Number of statistic hours kept: 2
Number of statistic distribution buckets kept: 1
Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
Number of history Lives kept: 0
Number of history Buckets kept: 15
History Filter Type: None
www.passenterpriselabs.com
SW101#
SW101#


sw101
History Filter Type: None
Entry number: 2
Owner:
www.passenterpriselabs.com
Tag:
Operation timeout (milliseconds): 400
Type of operation to perform: icmp-echo
Target address/Source interface: 10.1.255.12/GigabitEthernet0/1
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Data pattern: 0xABCDABCD
Verify data: No
Vrf Name:
www.passenterpriselabs.com
Schedule:
Operation frequency (seconds): 5 (not considered if randomly scheduled)
Next Scheduled Start Time: Start Time already passed
Group Scheduled : FALSE
Randomly Scheduled : FALSE
Life (seconds): Forever
Entry Ageout (seconds): never
Recurring (Starting Everyday): FALSE
Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 400
Distribution Statistics:
Number of statistic hours kept: 2
Number of statistic distribution buckets kept: 1
Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
Number of history Lives kept: 0
Number of history Buckets kept: 15
History Filter Type: None
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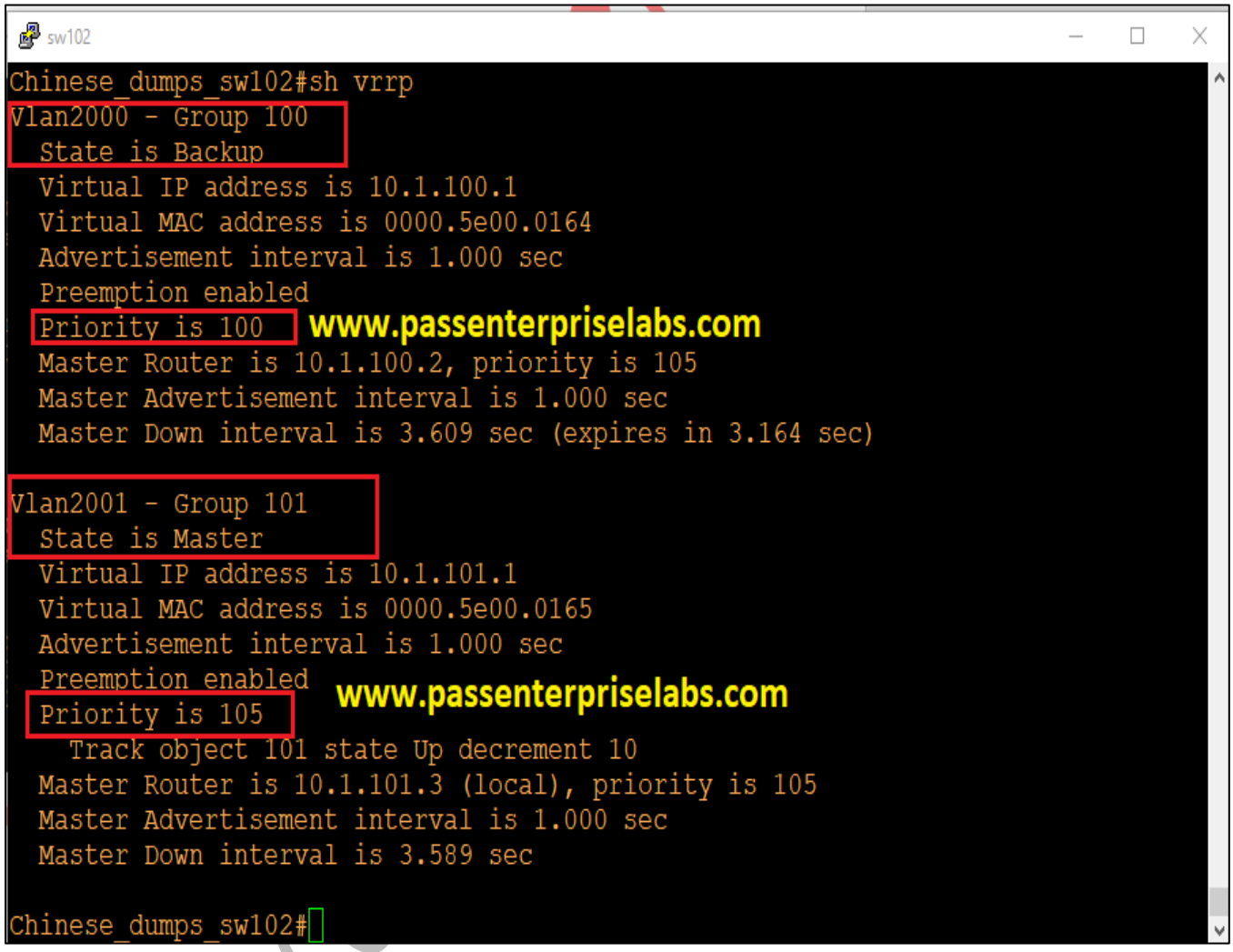
SW101#
SW101#

Chinese_dumps_sw102#sh vrrp



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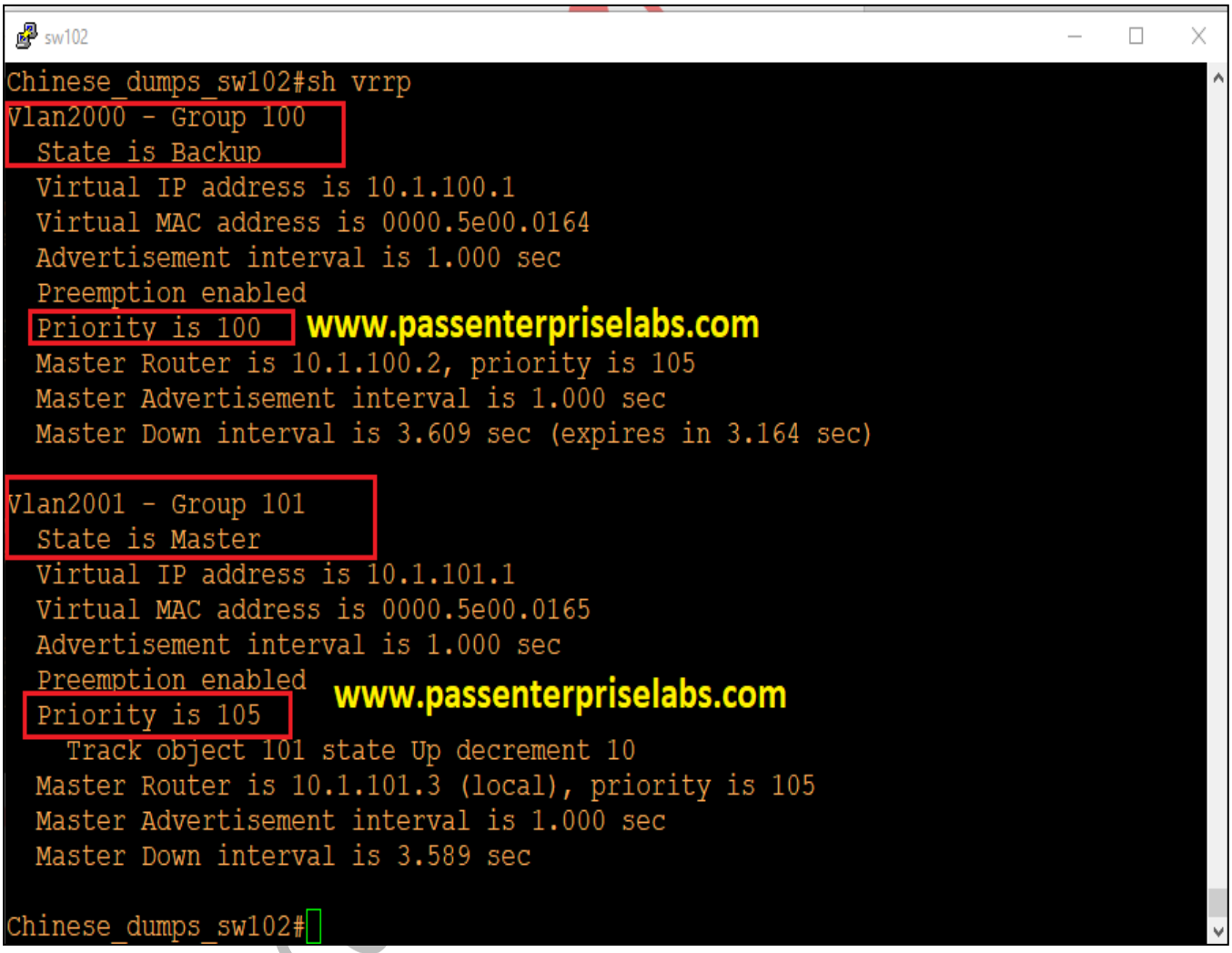
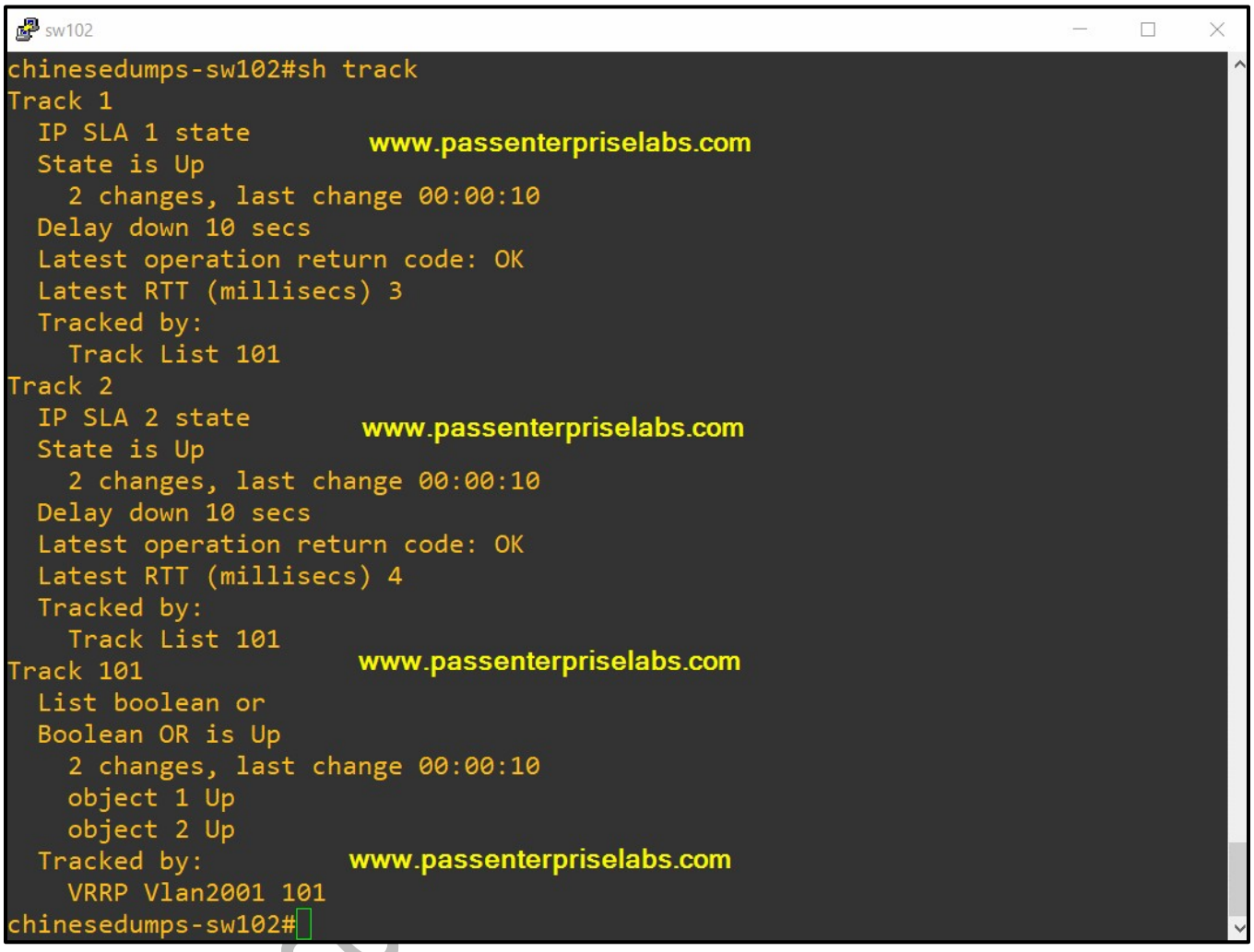
中文_垃圾桶_ sw102#sh vrrp



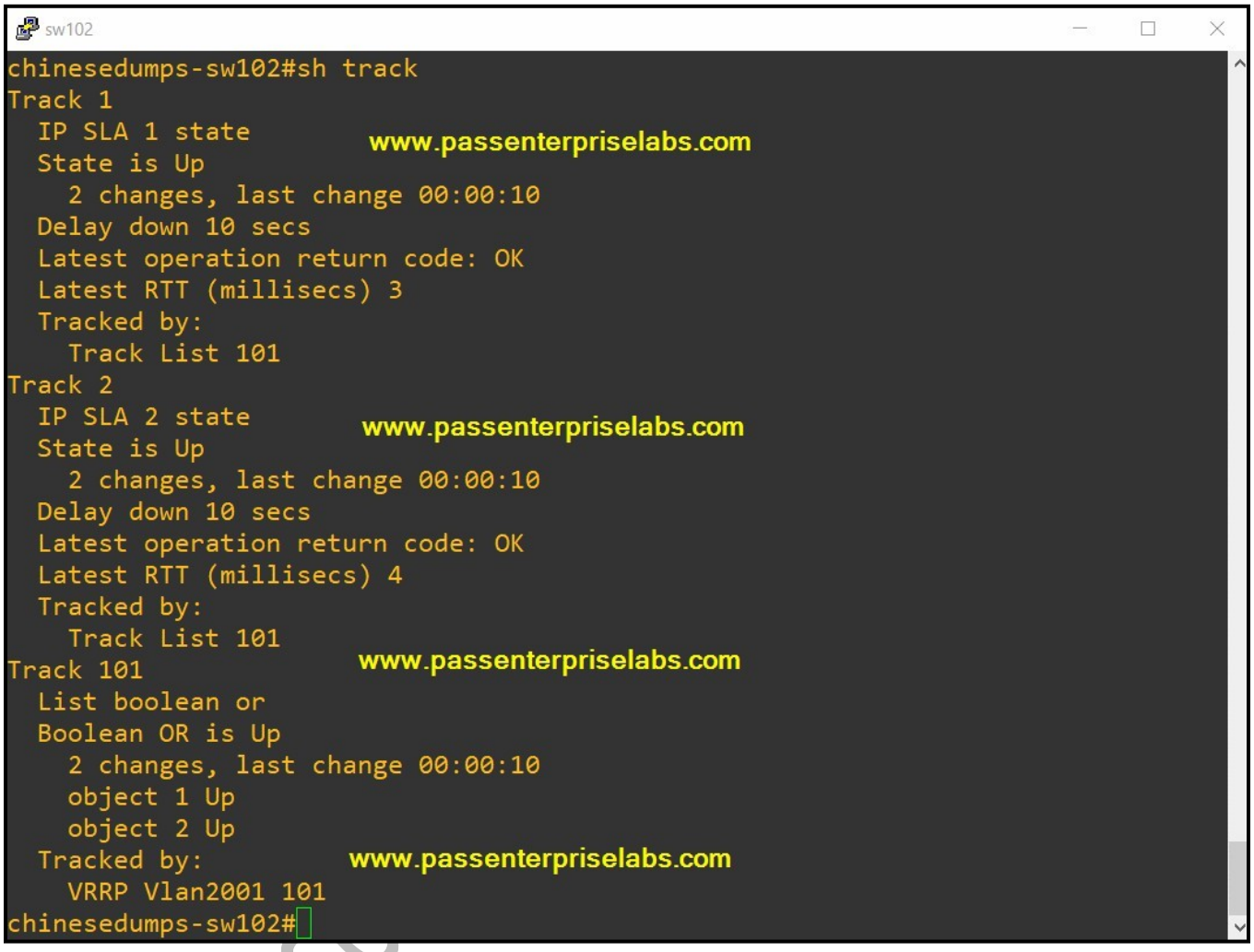
WWW.C

Chinese_dumps_sw102#sh track



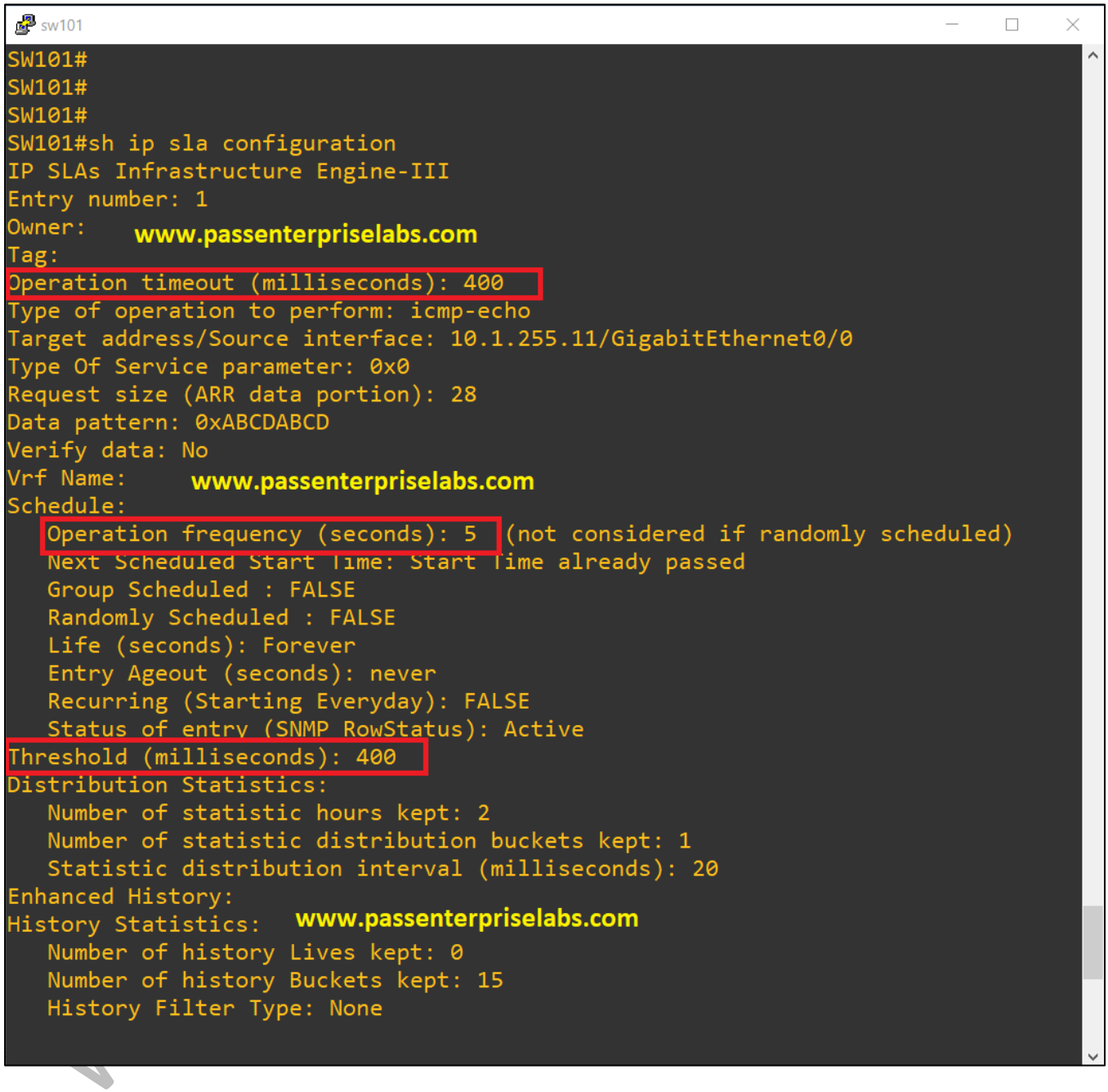
www.co

中文_垃圾_ sw102#sh轨道

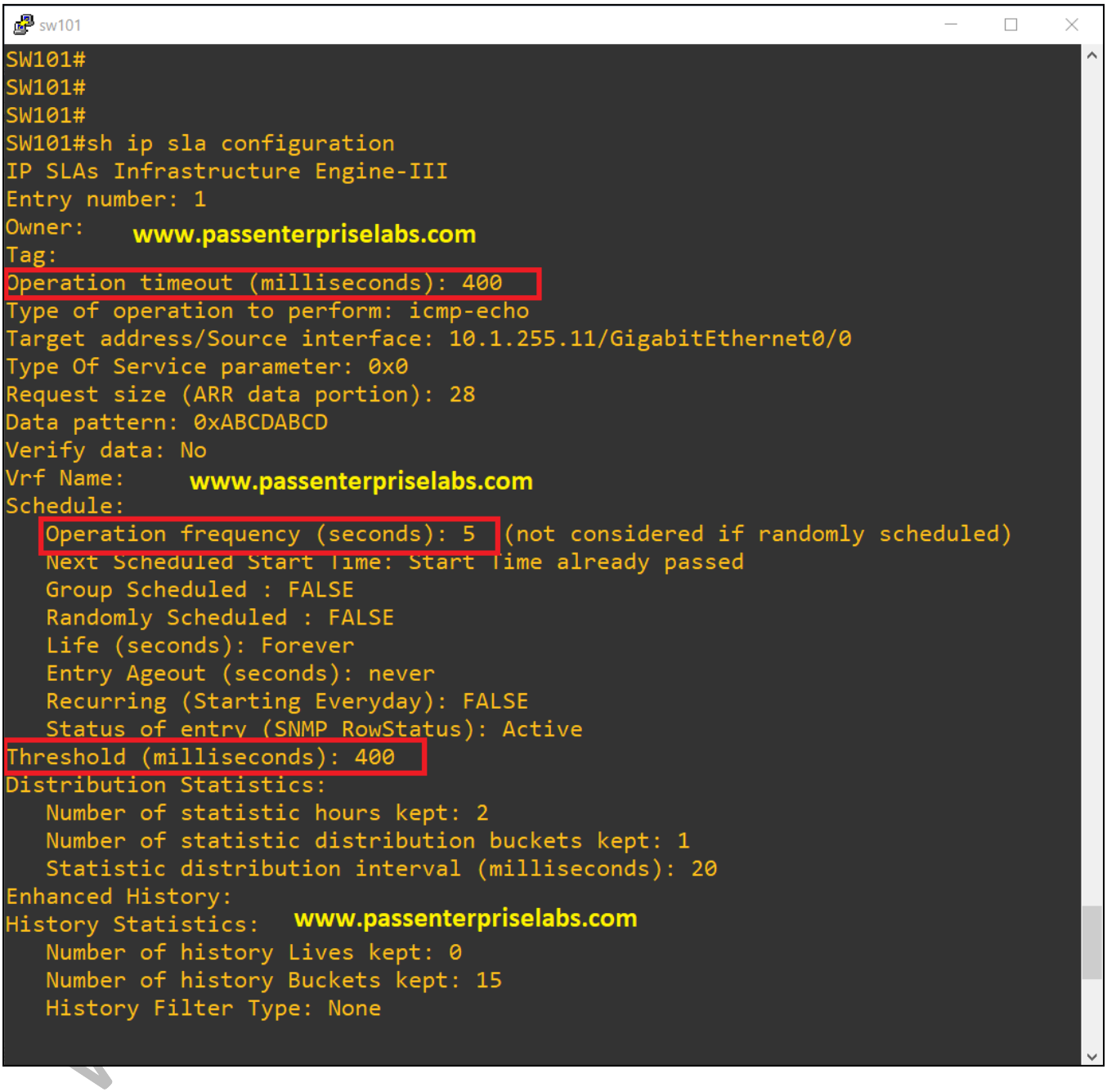


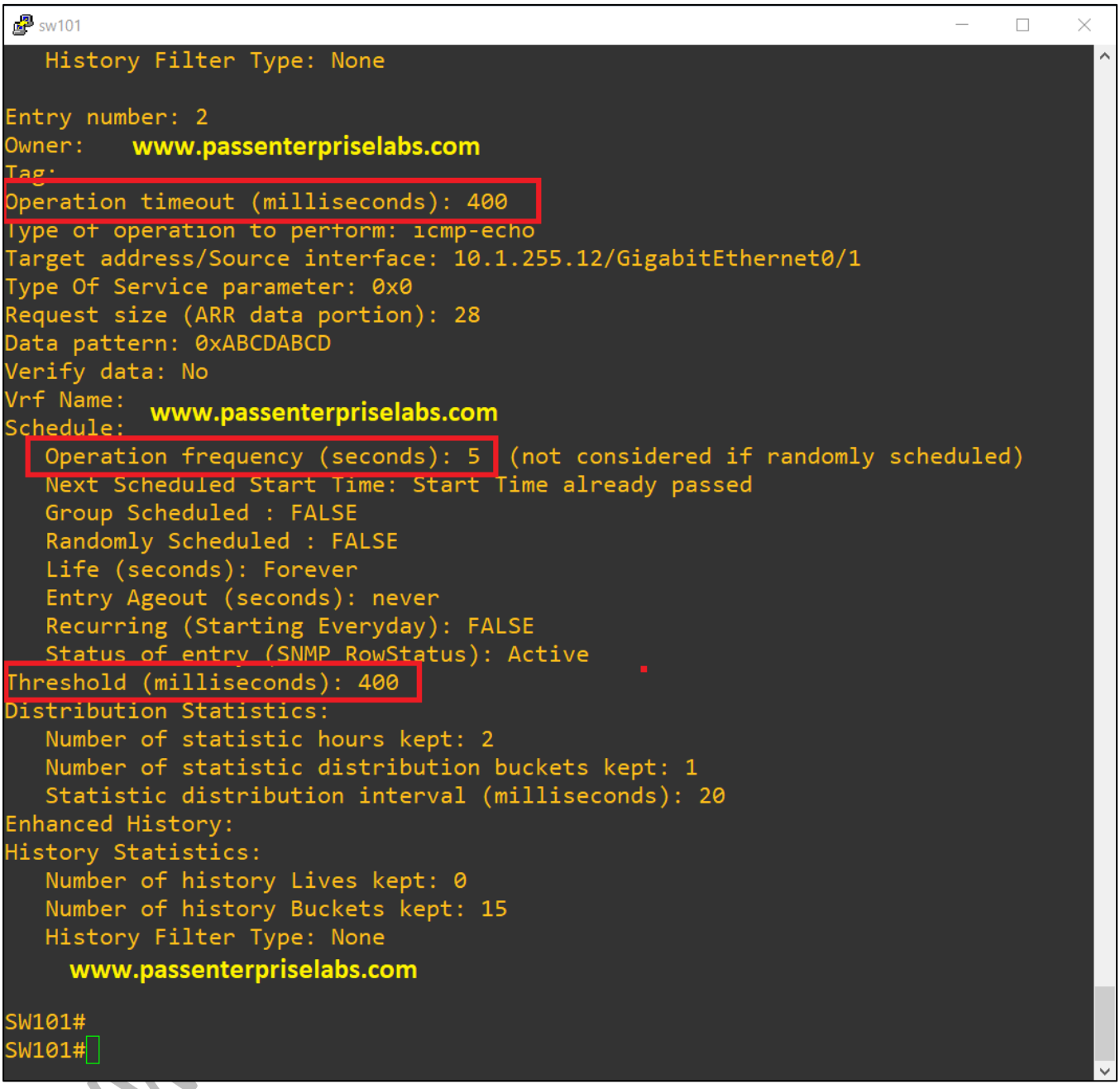
WWW.C

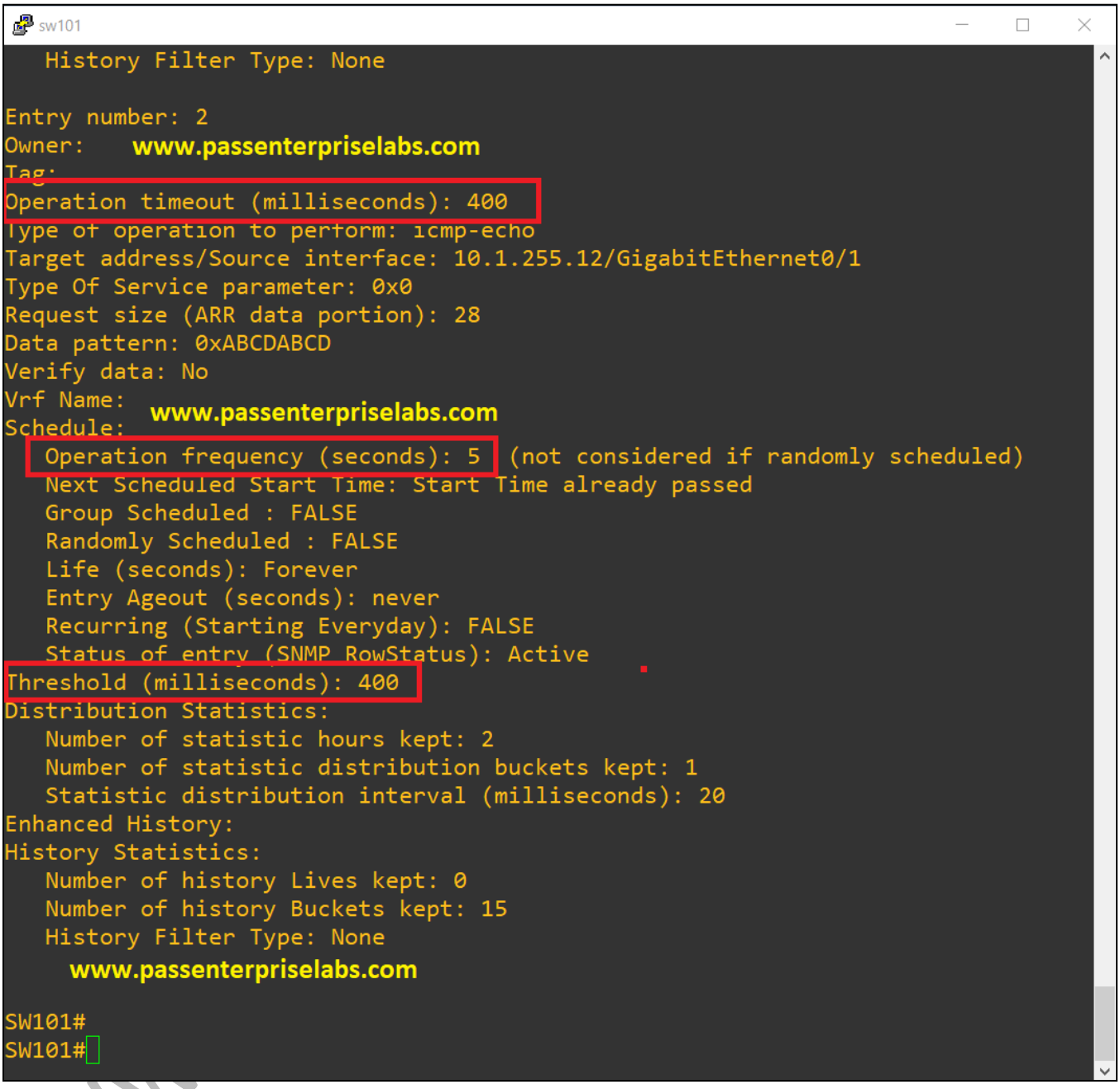
Chinese_dumps_sw102#sh ip sla configuration



中文_dumps_sw102#sh ip sla配置




sw101
History Filter Type: None
Entry number: 2
Owner:
www.passenterpriselabs.com
Tag:
Operation timeout (milliseconds): 400
Type of operation to perform: icmp-echo
Target address/Source interface: 10.1.255.12/GigabitEthernet0/1
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Data pattern: 0xABCDABCD
Verify data: No
Vrf Name:
www.passenterpriselabs.com
Schedule:
Operation frequency (seconds): 5 (not considered if randomly scheduled)
Next Scheduled Start Time: Start Time already passed
Group Scheduled : FALSE
Randomly Scheduled : FALSE
Life (seconds): Forever
Entry Ageout (seconds): never
Recurring (Starting Everyday): FALSE
Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 400
Distribution Statistics:
Number of statistic hours kept: 2
Number of statistic distribution buckets kept: 1
Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
Number of history Lives kept: 0
Number of history Buckets kept: 15
History Filter Type: None
www.passenterpriselabs.com
SW101#
SW101#


sw101
History Filter Type: None
Entry number: 2
Owner:
www.passenterpriselabs.com
Tag:
Operation timeout (milliseconds): 400
Type of operation to perform: icmp-echo
Target address/Source interface: 10.1.255.12/GigabitEthernet0/1
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Data pattern: 0xABCDABCD
Verify data: No
Vrf Name:
www.passenterpriselabs.com
Schedule:
Operation frequency (seconds): 5 (not considered if randomly scheduled)
Next Scheduled Start Time: Start Time already passed
Group Scheduled : FALSE
Randomly Scheduled : FALSE
Life (seconds): Forever
Entry Ageout (seconds): never
Recurring (Starting Everyday): FALSE
Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 400
Distribution Statistics:
Number of statistic hours kept: 2
Number of statistic distribution buckets kept: 1
Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
Number of history Lives kept: 0
Number of history Buckets kept: 15
History Filter Type: None
www.passenterpriselabs.com
SW101#
SW101#

To verify the gateway role:

sw101:

- Make the int g0/0 , g0/1 down (shutdown) and check "sh vrrp" , sw102 should be masterfor both the vlan (2000,2001)

sw102: sh vrrp

Note : If you make the interface of sw101 up , again it should retain its originalrole

要验证网关角色，请执行以下操作：

sw101：

- 关闭int g0/0、g0/1（关机）并检查“sh-vrrp”，sw102应为主控两个vlan（20002001）

sw102:sh-vrrp

注意：如果你启动了sw101的接口，它也应该保留其原始角色

sw102:

- Make the int g0/0 , g0/1 down (shutdown) and check "sh vrrp" , sw101 should be masterfor both the vlan (2000,2001)

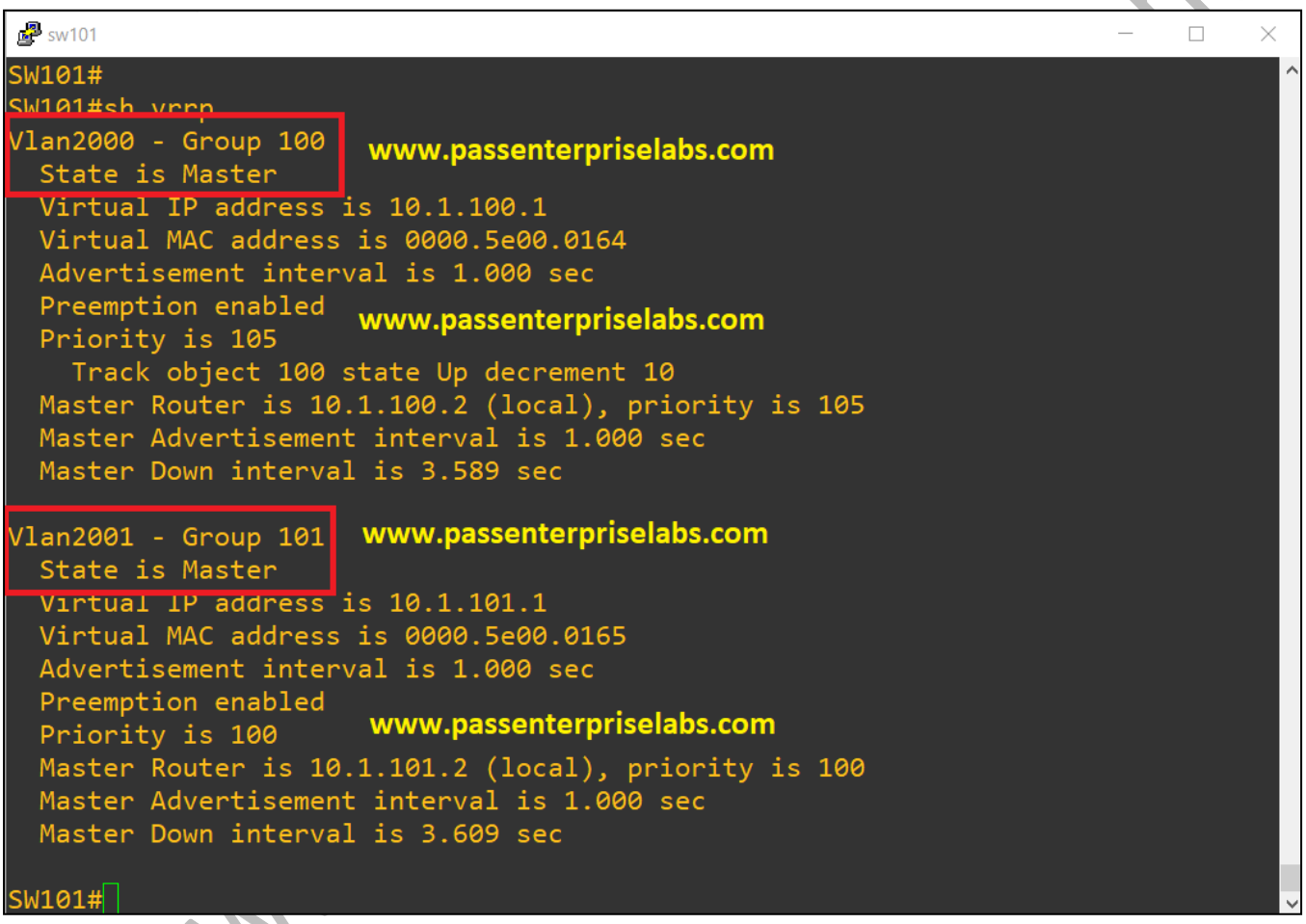
sw101: sh vrrp

Note : If you make the interfaces of sw102 up , again it should retain its original role

sw102：

- 关闭int g0/0、g0/1（关机）并检查“sh-vrrp”，sw101应同时控制vlan（20002001）

sw101:sh-



注意：如果你创建了sw102的接口，它也应该保留其原始角色

1.4: OSPFv2 between HQ and DC

Complete and correct the OSPF configuration on the switches sw101, sw102, sw201 and sw202 according to these requirements:

- Enable OSPFv2 on the redundant interconnections between the DC and HQ sites. Make sure that OSPF establishes adjacencies on these interconnections and exchanges routing information between the DC and HQ sites.
- Protect the authenticity and integrity of the OSPFv2 sessions on the redundant interconnections between DC and HQ with the SHA-384 mechanism. Use key ID 1 and a shared secret of “cci3” (without quotes).
- Improve the detection of unreachable OSPFv2 neighbors on the redundant interconnections between DC and HQ so that OSPF can detect the loss of a neighbor within 300 msec, with the probes being sent every 100 msec. it is not allowed to modify OSPF timers to accomplish this requirement.

3 Points

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1.4：HQ和DC之间的OSPFv2

根据以下要求完成并纠正交换机sw101、sw102、sw201和w202上的OSPF配置：

- 在DC和HQ站点之间的冗余互连上启用OSPFv2。确保OSPF在这些互连上建立邻接关系，并在DC和HQ站点之间交换路由信息。
- 使用SHA-384机制保护DC和HQ之间冗余互连上OSPFv2会话的真实性和完整性。使用密钥ID 1和“cci3”的机密（不带引号）。
- 改进DC和HQ之间冗余互连上无法到达的OSPFv2邻居的检测，以便OSPF可以在300毫秒内检测到邻居的丢失，每100毫秒发送一次探测。不允许修改OSPF定时器来完成此操作要求。

3分

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Solution:

Note: Try making the interface default if the interface showing not connect

Chinese_dumps_sw201:

```
sw201>en
sw201#conf t
sw201(config)#key chain DCHQ
sw201(config-keychainn)#key 1
sw201(config-keychain-key)#key-string cci3
sw201(config-keychain-key)#cryptographic-algorithm hmac-sha-384
sw201(config)#exit
sw201(config)#exit

sw201(config)#int g1/2
sw201(config-if)#ip ospf authentication key-chain DCHQ
sw201(config-if)#ip ospf bfd
sw201(config-if)#bfd interval 100 min_rx 100 multiplier 3
sw201(config-if)#exit

sw201#config t
sw201(config)#router ospf 1
sw201(config-router)#no passive-interface g1/2
```

Chinese_dumps_sw202:

```
sw202>en
sw202#conf t
sw202(config)#key chain DCHQ
sw202(config)#key 1
sw202(config-keychain)#key-string cci3
sw202(config-keychain)#cryptographic-algorithm hmac-sha-384
sw202(config-keychain)#exit
sw202(config)#exit

sw202(config)#int g1/2
sw202(config-if)#ip ospf authentication key-chain DCHQ
sw202(config-if)#ip ospf bfd
sw202(config-if)#bfd interval 100 min_rx 100 multiplier 3
sw202(config)#exit
```

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解决方案：

注意：如果界面显示未连接，请尝试将界面设置为默认值

中文_垃圾_ sw201：

```
sw201>en-
sw201#配置
文件
sw201（config）#密钥链DCHQ sw201（
config-keychainn）#密钥1 sw201（-
config-keychain key）#密钥串cci3
sw201（配置密钥链密钥）#加密算法hmac-sha-384 sw201（
config）#退出
sw201（配置）#退出

sw201（配置）#int g1/2
sw201（config-if）#ip-ospf认证密钥链DCHQ
sw201（config.if）#ip-ospf-bfd
sw201（配置if）#bfd间隔100 min_rx 100乘数3
sw201（设置if）#退出

sw201#配置t sw201（
config）#路由器ospf 1
sw201（配置路由器）#无无源接口g1/2
```

中文_转储_sw202：

```
sw202>en-
sw202#配置
文件
sw202（config）#密钥链DCHQ
sw202（配置）#密钥1
sw202（配置钥匙串）#钥匙串cci3
sw202（配置密钥链）#密码算法hmac-sha-384 sw202（设
置密钥链）#exit
sw202（配置）#退出

sw202（配置）#int g1/2
sw202（config-if）#ip-ospf认证密钥链DCHQ
sw202（config.if）#ip-ospf-bfd
sw202（配置if）#bfd间隔100 min_rx 100乘数3 sw202（设置
）#退出
```

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```
sw202#config t
sw202(config)#router ospf 1
sw202(config)#no passive-interface g1/2
```

Chinese_dumps_sw101:

```
sw101>en
sw101#conf t
sw101(config)#key chain HQDC
sw101(config-keychain)#key 1
sw101(config-keychain-key)#key-string cci3
sw101(config-keychain-key)#cryptographic-algorithm hmac-sha-384
sw101(config-keychain-key)#exit
sw101(config)#exit

sw101(config)#int g0/2
sw101(config-if)#ip ospf authentication key-chain HQDC
sw101(config-if)#ip ospf bfd
sw101(config-if)#bfd interval 100 min_rx 100 multiplier 3
```

Chinese_dumps_sw102 :

```
sw102>en
sw102#conf t
sw102(config)#key chain HQDC
sw102(config-keychain)#key 1
sw102(config-keychain-key)#key-string cci3
sw102(config-keychain-key)#cryptographic-algorithm hmac-sha-384
sw102(config-keychain)#exit
sw102(config)#exit

sw102(config)#int g0/2
sw102(config-if)#ip ospf authentication key-chain HQDC
sw102(config-if)#ip ospf bfd
sw102(config-if)#bfd interval 100 min_rx 100 multiplier 3
```

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sw202#配置t sw202（
config）#路由器ospf 1
sw202（配置）#无无源接口g1/2

中文_垃圾_ sw101：

sw101>en-
sw101#配置
文件
sw101（config）#密钥链HQDC sw101（
config-keychain）#密钥1 sw101（-
config-keychain key）#密钥串cci3
sw101（配置密钥链密钥）#密码算法hmac-sha-384 sw101
sw101（配置）#退出

sw101（配置）#int g0/2
sw101（config-if）#ip-ospf认证密钥链HQDC
sw101（config.if）#ip-ospf-bfd
sw101（配置if）#bfd间隔100 min_rx 100乘数3

中文_垃圾_ sw102：

sw102>en-
sw102#配置
文件
sw102（config）#密钥链HQDC sw102（
config-keychain）#密钥1 sw102（-
config-keychain key）#密钥串cci3
sw102（配置密钥链密钥）#密码算法hmac-sha-384 sw102（设
置密钥链）#退出
sw102（配置）#退出

sw102（配置）#int g0/2
sw102（config-if）#ip-ospf认证密钥链HQDC
sw102（config.if）#ip-ospf-bfd
sw102（配置if）#bfd间隔100 min_rx 100乘数3

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Verification:

Chinese_dumps_sw101#sh key chain

```
sw101
SW101#show key chain
Key-chain HQDC:        www.passenterpriselabs.com
    key 1 -- text "cci3"
        accept lifetime (always valid) - (always valid) [valid now]
        send lifetime (always valid) - (always valid) [valid now]
SW101#
SW101#
SW101#                 www.passenterpriselabs.com
SW101#
SW101#
```

Chinese_dumps_sw101#sh ip ospf interface g0/2

```
sw101
SW101#
SW101#sh ip ospf int g0/2    www.passenterpriselabs.com
GigabitEthernet0/2 is up, line protocol is up (connected)
  Internet Address 10.2.241.2/30, Area 0, Attached via Interface Enable
  Process ID 1, Router ID 10.1.255.101, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Enabled by interface config, including secondary ip addresses
  Transmit Delay is 1 sec, State DR, Priority 1, BFD enabled
  Designated Router (ID) 10.1.255.101, Interface address 10.2.241.2
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:06    www.passenterpriselabs.com
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/4/4, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Sending SA: Key 1, Algorithm HMAC-SHA-384 - key chain HQDC
SW101#
```

验证：

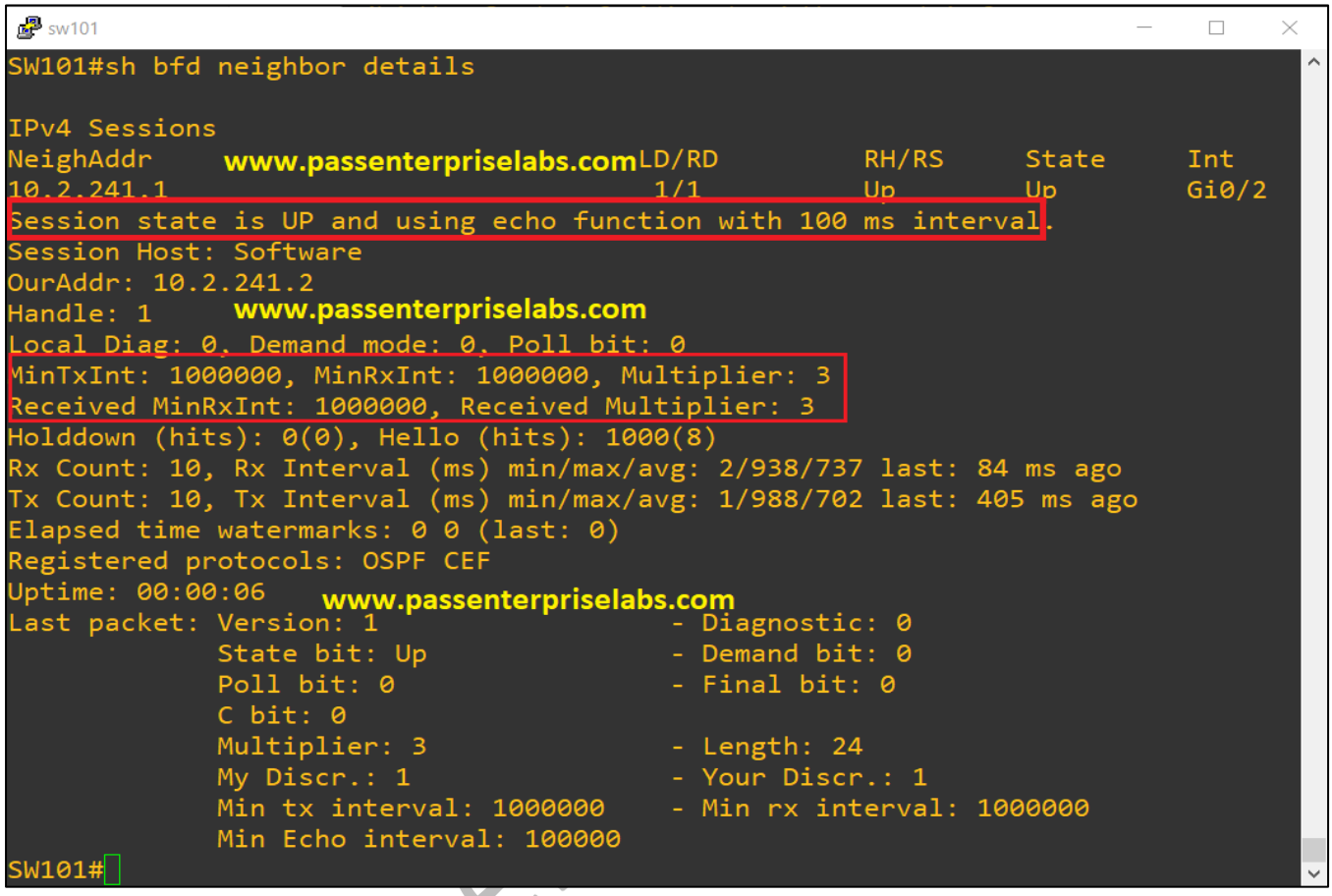
中文_垃圾桶_sw101#sh钥匙扣

```
sw101
SW101#show key chain
Key-chain HQDC:        www.passenterpriselabs.com
    key 1 -- text "cci3"
        accept lifetime (always valid) - (always valid) [valid now]
        send lifetime (always valid) - (always valid) [valid now]
SW101#
SW101#
SW101#                 www.passenterpriselabs.com
SW101#
SW101#
```

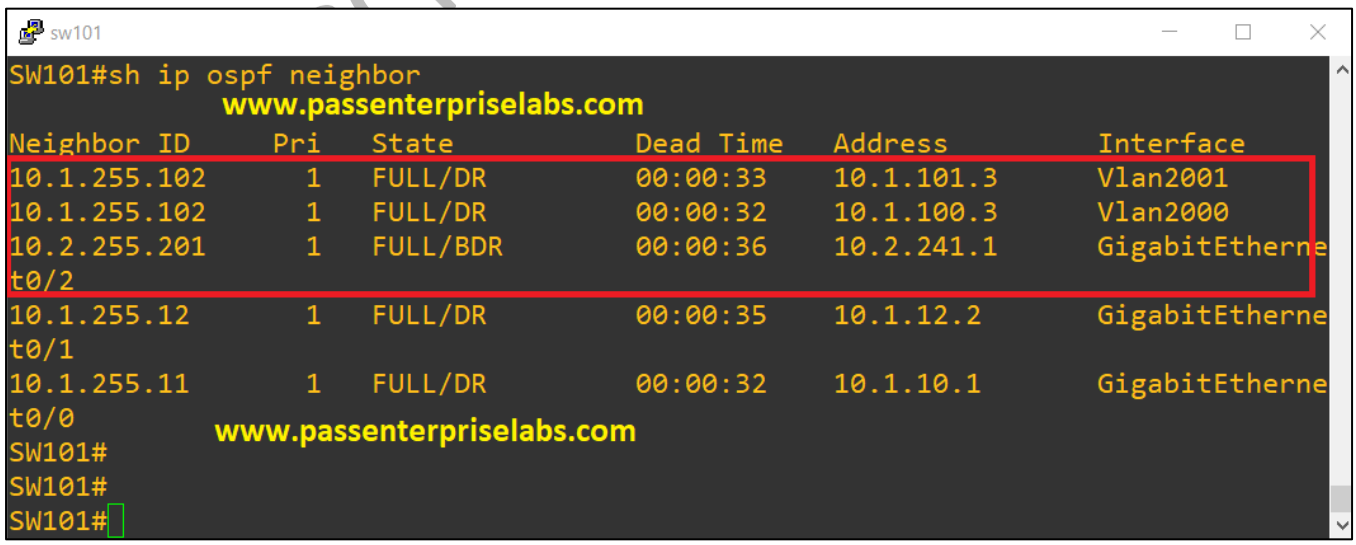
中文_ dumps_sw101#sh ip ospf接口g0/2

```
sw101
SW101#
SW101#sh ip ospf int g0/2    www.passenterpriselabs.com
GigabitEthernet0/2 is up, line protocol is up (connected)
  Internet Address 10.2.241.2/30, Area 0, Attached via Interface Enable
  Process ID 1, Router ID 10.1.255.101, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Enabled by interface config, including secondary ip addresses
  Transmit Delay is 1 sec, State DR, Priority 1, BFD enabled
  Designated Router (ID) 10.1.255.101, Interface address 10.2.241.2
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:06    www.passenterpriselabs.com
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/4/4, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Sending SA: Key 1, Algorithm HMAC-SHA-384 - key chain HQDC
SW101#
```

Chinese_dumps_sw101# sh bfd neighbor details



Chinese_dumps_sw101# sh key ip ospf neighbor



中文_ dumps_sw101#sh bfd邻居详情

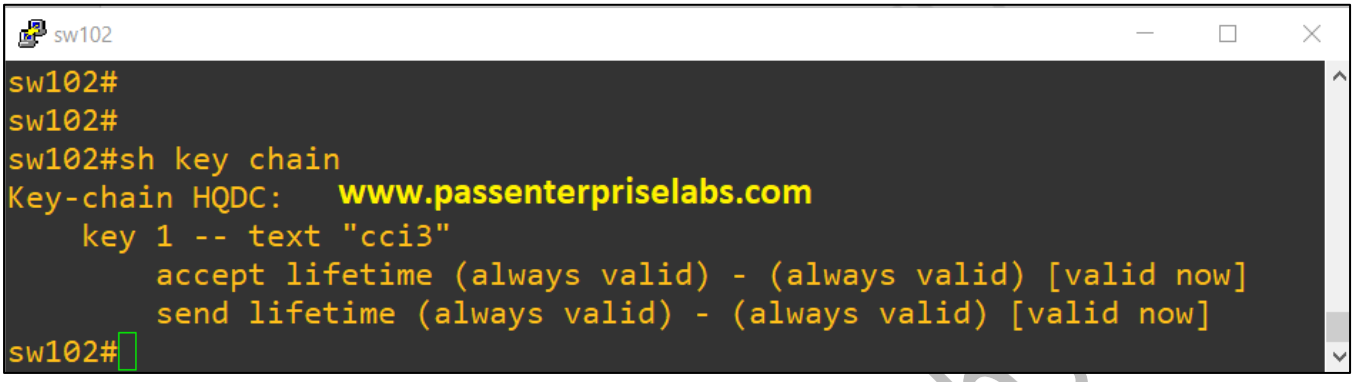
```
sw101
SW101#sh bfd neighbor details

IPv4 Sessions
NeighAddr      www.passenterpriselabs.comLD/RD         RH/RS     State     Int
10.2.241.1                               1/1          Up        Up        Gi0/2
Session state is UP and using echo function with 100 ms interval.
Session Host: Software
OurAddr: 10.2.241.2
Handle: 1      www.passenterpriselabs.com
Local Diag: 0, Demand mode: 0, Poll bit: 0
MinTxInt: 1000000, MinRxInt: 1000000, Multiplier: 3
Received MinRxInt: 1000000, Received Multiplier: 3
Holddown (hits): 0(0), Hello (hits): 1000(8)
Rx Count: 10, Rx Interval (ms) min/max/avg: 2/938/737 last: 84 ms ago
Tx Count: 10, Tx Interval (ms) min/max/avg: 1/988/702 last: 405 ms ago
Elapsed time watermarks: 0 0 (last: 0)
Registered protocols: OSPF CEF
Uptime: 00:00:06    www.passenterpriselabs.com
Last packet: Version: 1                  - Diagnostic: 0
             State bit: Up               - Demand bit: 0
             Poll bit: 0                 - Final bit: 0
             C bit: 0
             Multiplier: 3               - Length: 24
             My Discr.: 1                - Your Discr.: 1
             Min tx interval: 1000000    - Min rx interval: 1000000
             Min Echo interval: 100000
SW101#
```

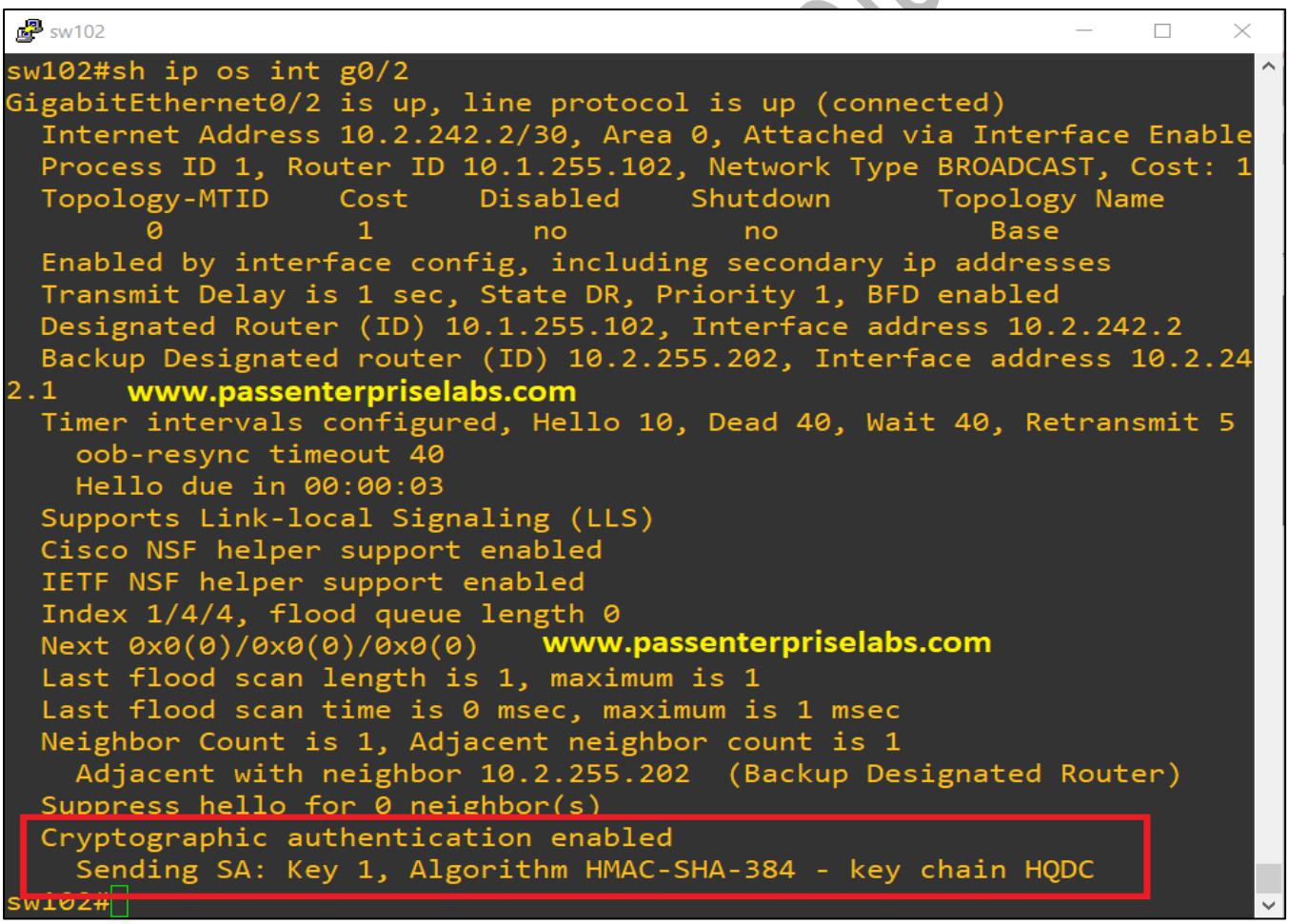
中文_ dumps_sw101#sh密钥ip ospf邻居

```
sw101
SW101#sh ip ospf neighbor
            www.passenterpriselabs.com
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.1.255.102      1   FULL/DR         00:00:33    10.1.101.3      Vlan2001
10.1.255.102      1   FULL/DR         00:00:32    10.1.100.3      Vlan2000
10.2.255.201      1   FULL/BDR        00:00:36    10.2.241.1      GigabitEtherne
t0/2
10.1.255.12       1   FULL/DR         00:00:35    10.1.12.2       GigabitEtherne
t0/1
10.1.255.11       1   FULL/DR         00:00:32    10.1.10.1       GigabitEtherne
t0/0
            www.passenterpriselabs.com
SW101#
SW101#
SW101#
```

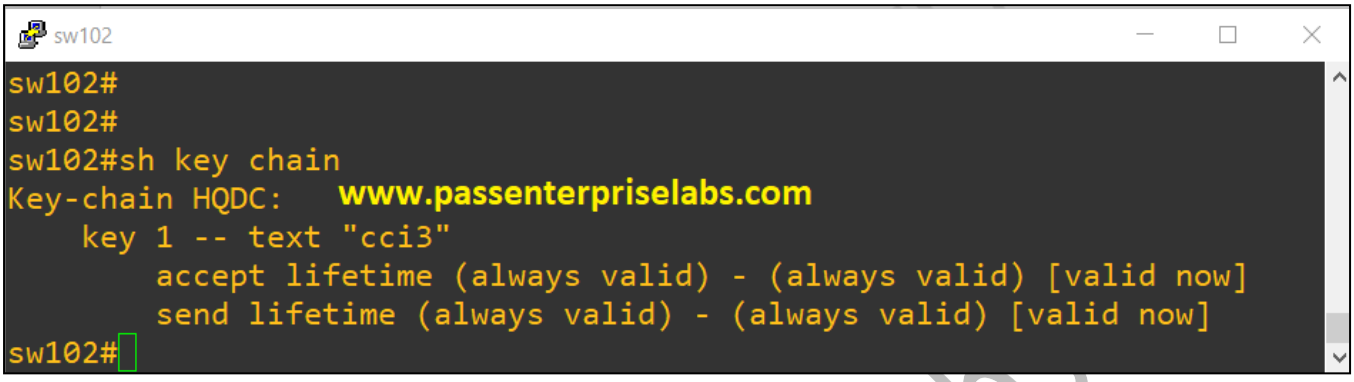
Chinese_dumps_sw102#sh key chain



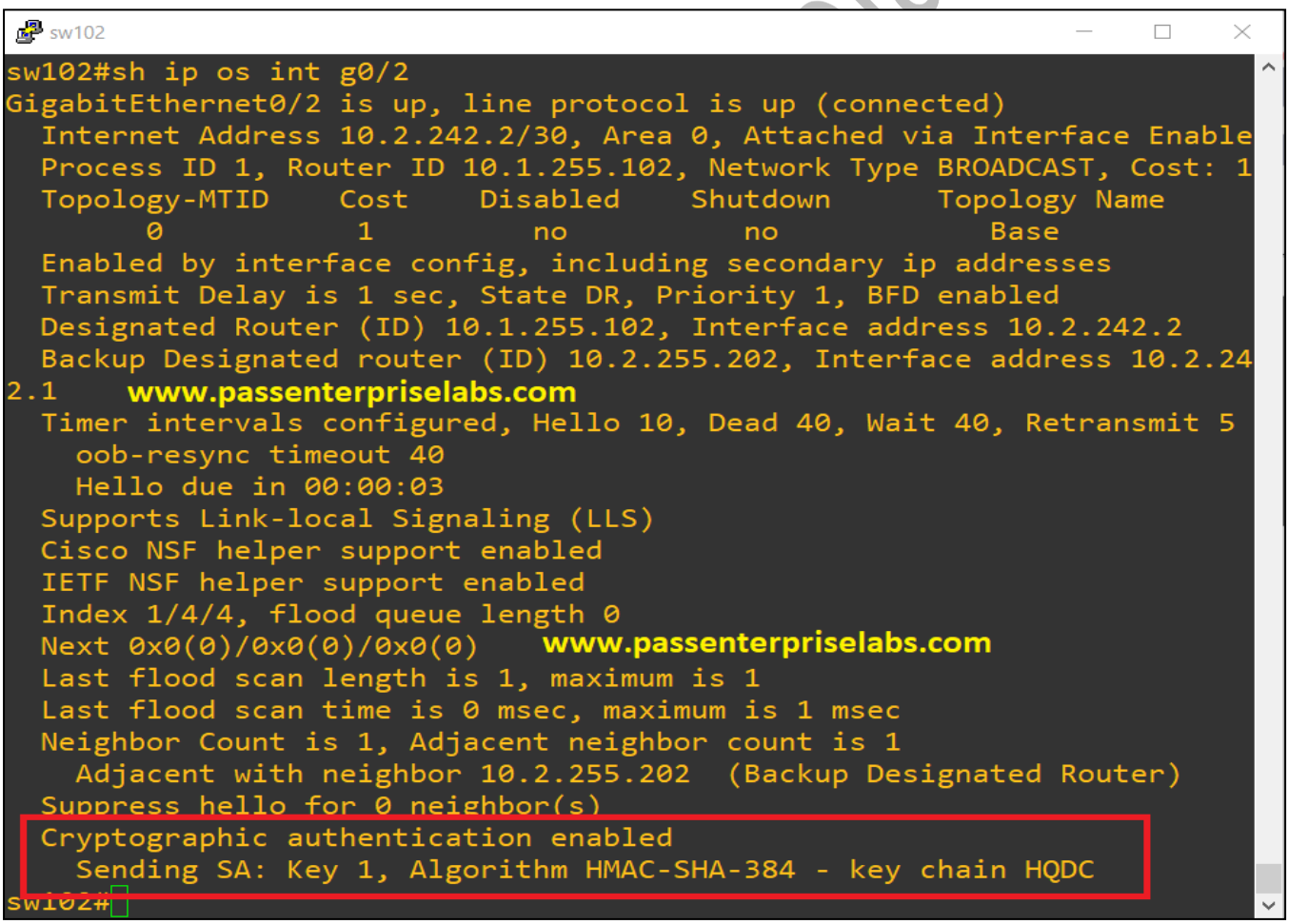
Chinese_dumps_sw102#sh ip os interface g0/2



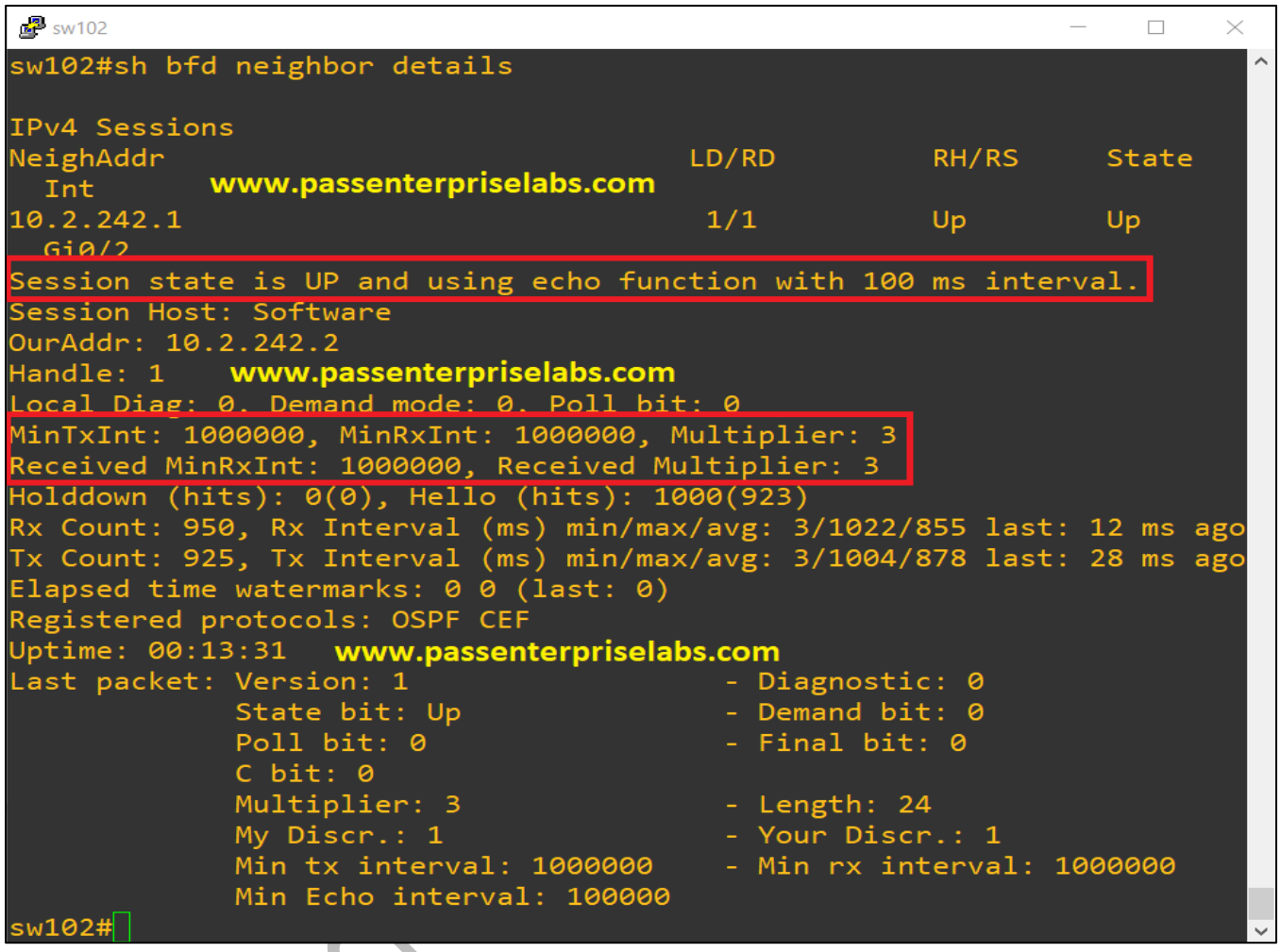
中文_垃圾桶_ sw102#sh钥匙扣



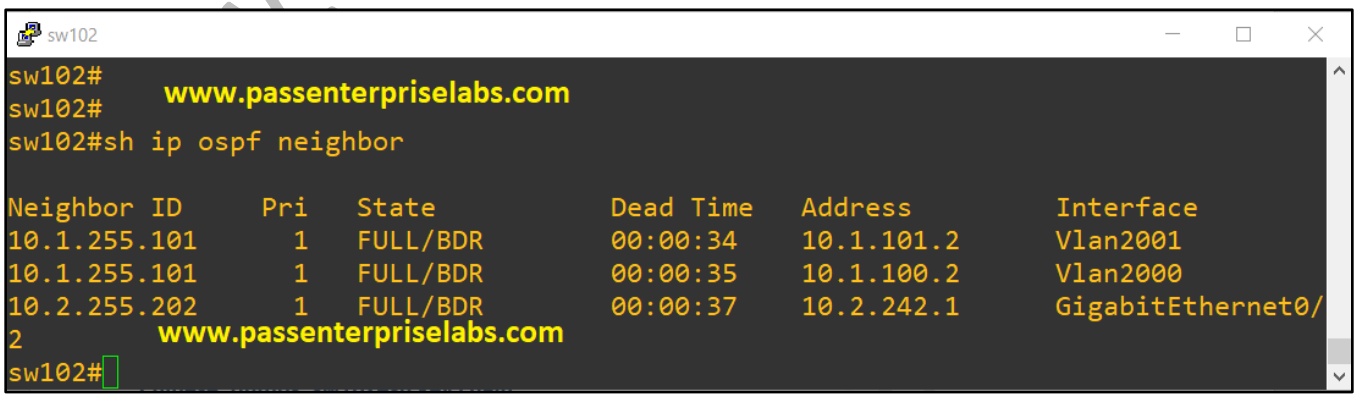
中文_dumps_sw102#sh ip操作系统接口g0/2



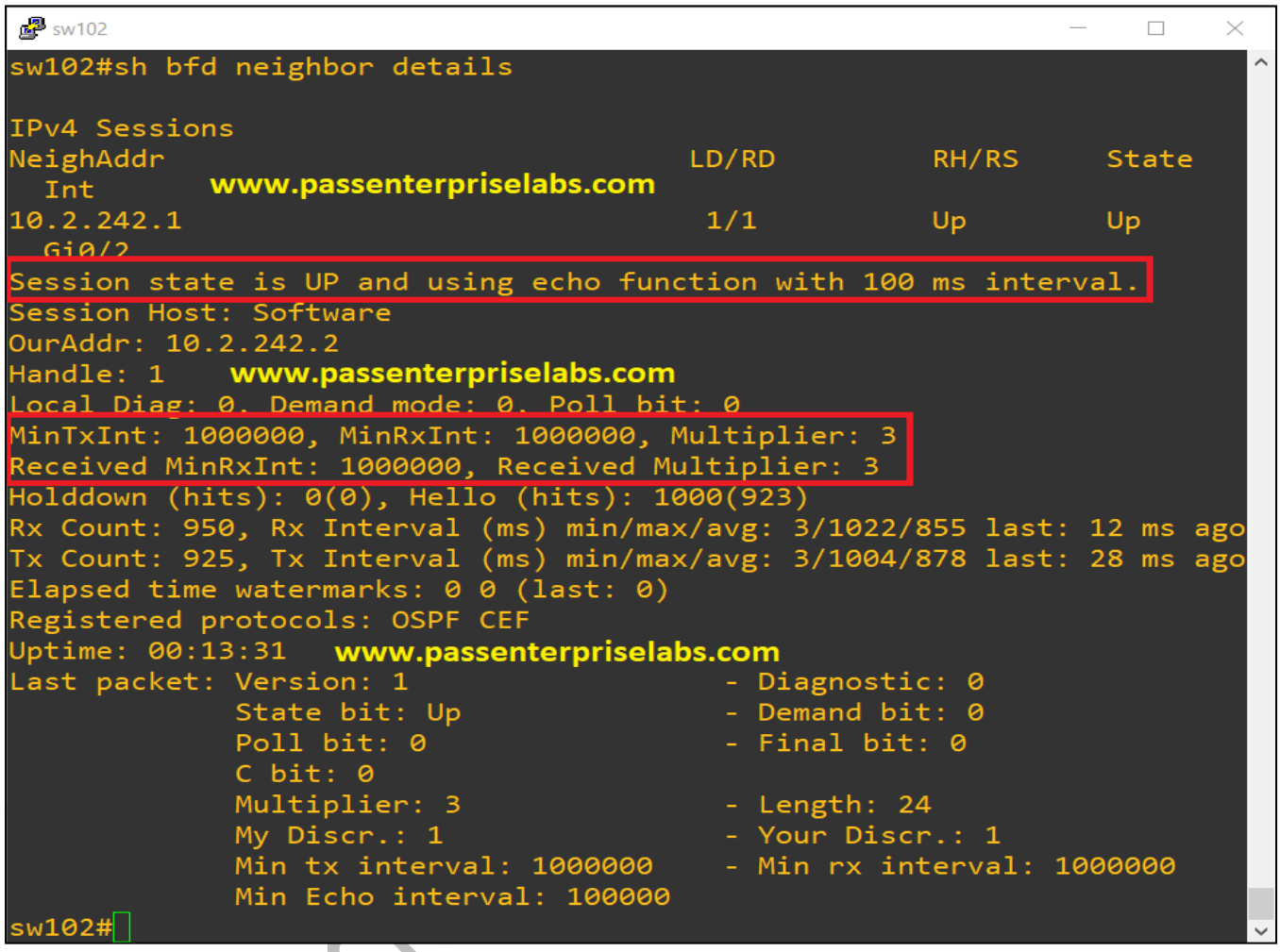
Chinese_dumps_sw102#sh bfd neighbor details



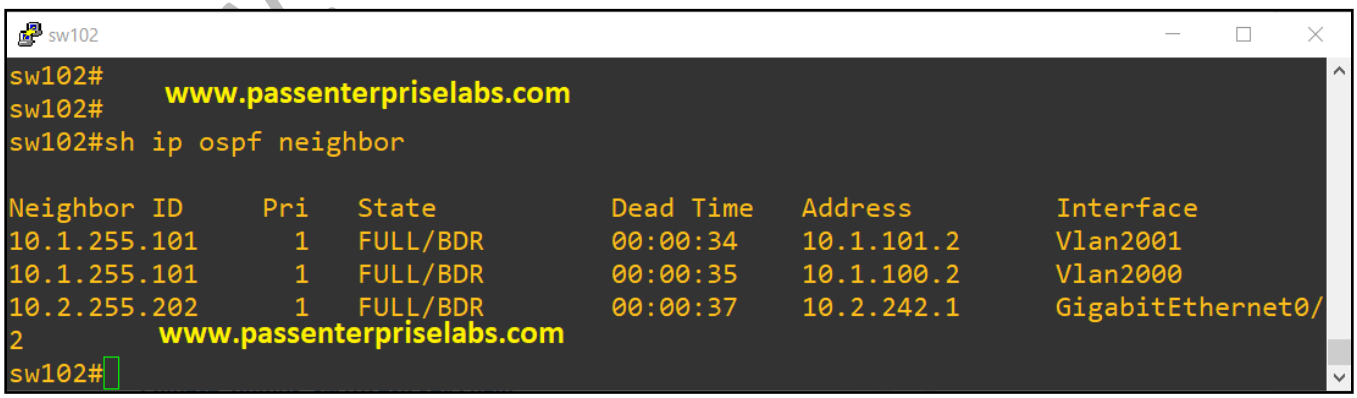
Chinese_dumps_sw102#sh ip ospf neighbor



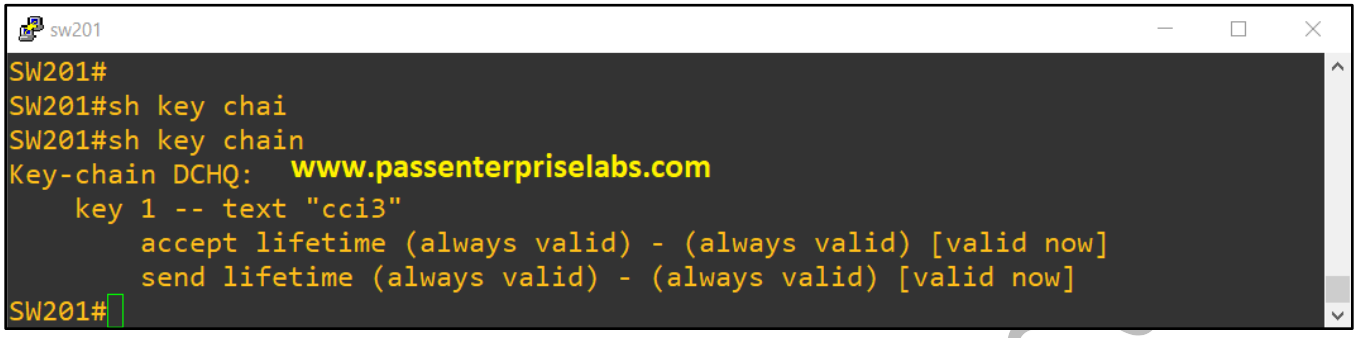
中文_ dumps_sw102#sh bfd邻居详情



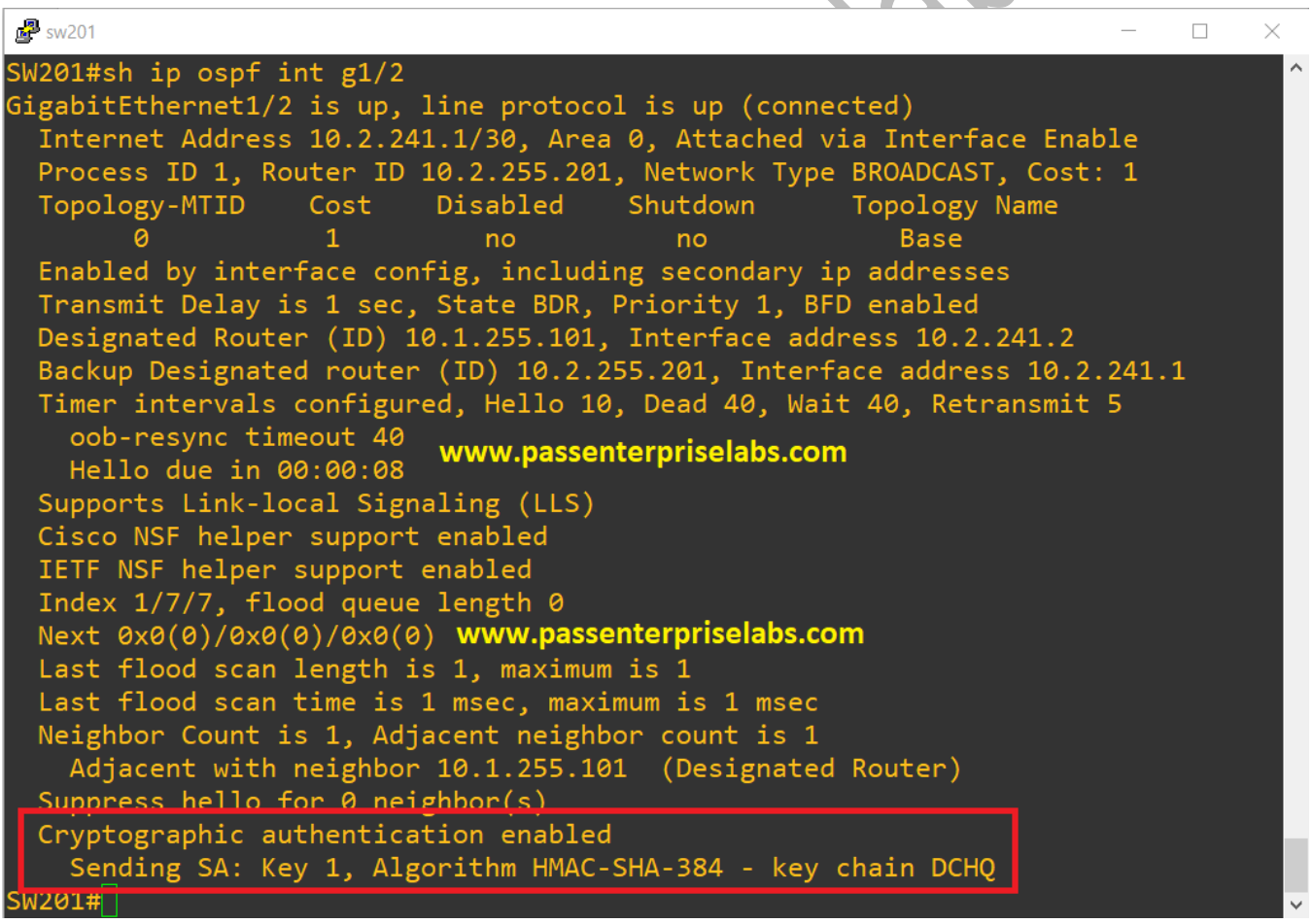
中文_ dumps_sw102#sh ip ospf邻居



Chinese_dumps_sw201#sh key chain

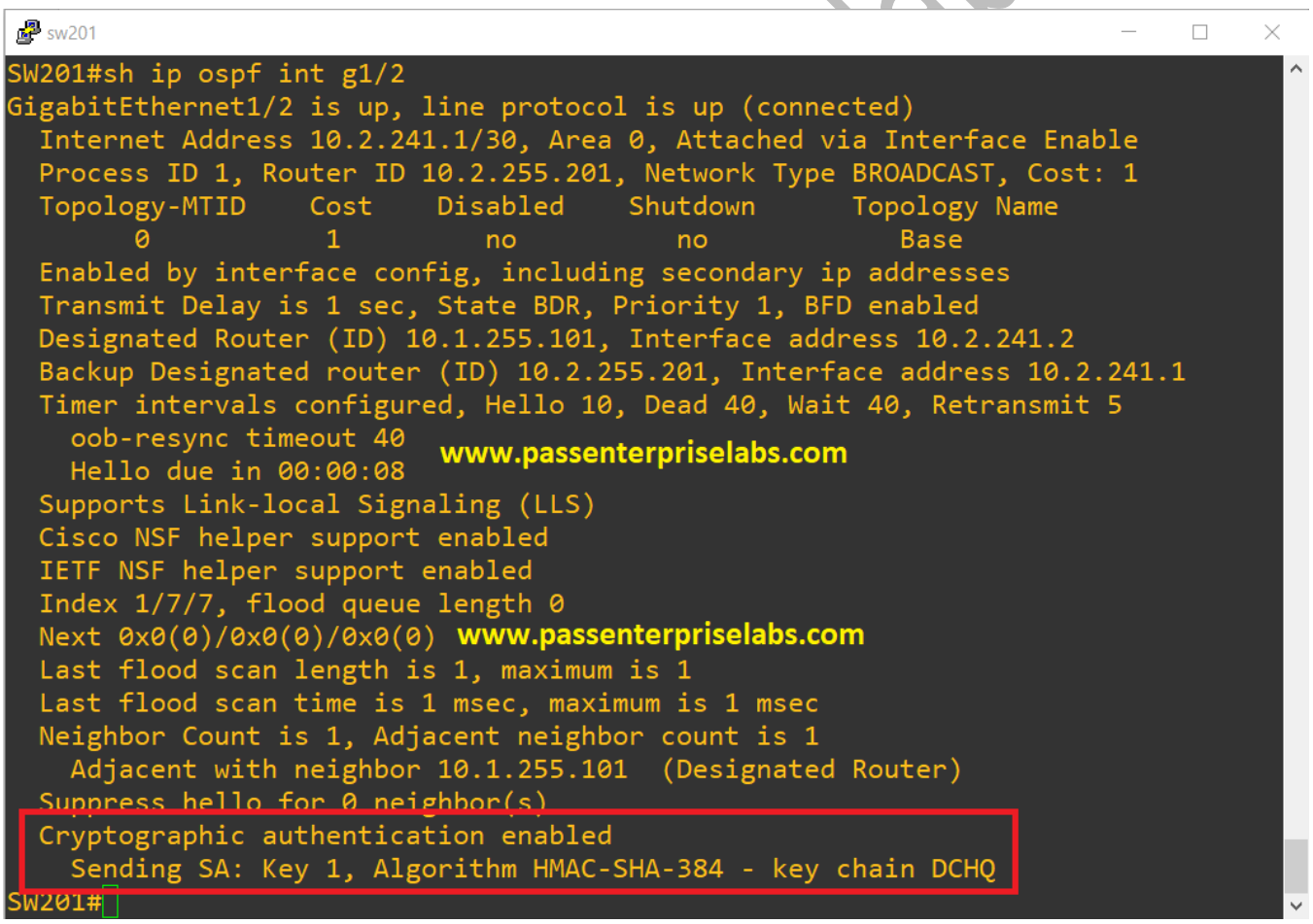


Chinese_dumps_sw201#sh ip ospf int g1/2



中文_垃圾桶_sw201#sh钥匙扣

中文_ dumps_sw201#sh ip ospf int g1/2



Chinese_dumps_sw201#sh bfd neighbor details

```
sw201
SW201#sh bfd neighbor details

IPv4 Sessions
NeighAddr      www.passenterpriselabs.com LD/RD         RH/RS     State     Int
10.2.241.2                                1/1          Up        Up        Gi1/2
Session state is UP and using echo function with 100 ms interval.
Session Host: Software
OurAddr: 10.2.241.1
Handle: 1            www.passenterpriselabs.com
Local Diag: 0, Demand mode: 0, Poll bit: 0
MinTxInt: 1000000, MinRxInt: 1000000, Multiplier: 3
Received MinRxInt: 1000000, Received Multiplier: 3
Holddown (hits): 0(0), Hello (hits): 1000(1529)
Rx Count: 1490, Rx Interval (ms) min/max/avg: 1/1076/901 last: 690 ms ago
Tx Count: 1530, Tx Interval (ms) min/max/avg: 4/1005/878 last: 340 ms ago
Elapsed time watermarks: 0 0 (last: 0)
Registered protocols: OSPF CEF
Uptime: 00:22:23  www.passenterpriselabs.com
Last packet: Version: 1                  - Diagnostic: 0
             State bit: Up               - Demand bit: 0
             Poll bit: 0                 - Final bit: 0
             C bit: 0
             Multiplier: 3               - Length: 24
             My Discr.: 1                - Your Discr.: 1
             Min tx interval: 1000000    - Min rx interval: 1000000
             Min Echo interval: 100000
SW201#
```

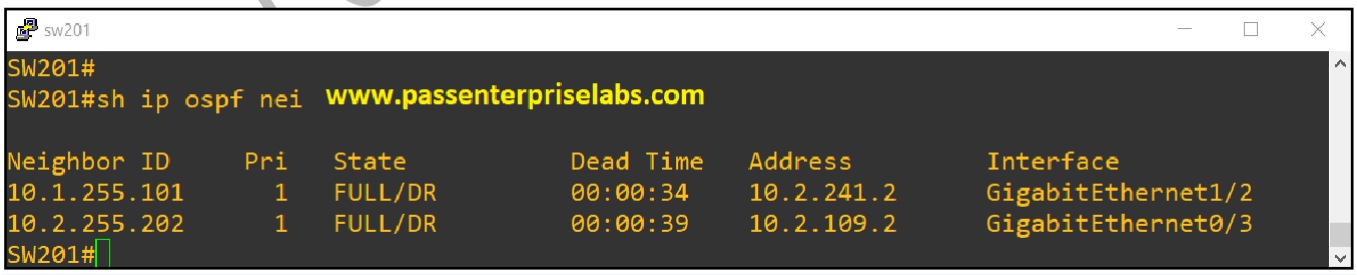
Chinese_dumps_sw201#sh ip ospf neighbor

```
sw201
SW201#
SW201#sh ip ospf nei  www.passenterpriselabs.com

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.1.255.101      1   FULL/DR         00:00:34    10.2.241.2      GigabitEthernet1/2
10.2.255.202      1   FULL/DR         00:00:39    10.2.109.2      GigabitEthernet0/3
SW201#
```

中文_ dumps_sw201#sh bfd邻居详情

中文_ dumps_sw201#sh ip ospf邻居



Chinese_dumps_sw202# sh key chain

```
sw202
SW202#
SW202#sh key chain
Key-chain DCHQ:    www.passenterpriselabs.com
    key 1 -- text "cci3"
        accept lifetime (always valid) - (always valid) [valid now]
        send lifetime (always valid) - (always valid) [valid now]
SW202#
```

Chinese_dumps_sw202# sh ip os int g1/2

```
sw202
SW202#sh ip ospf int g1/2
GigabitEthernet1/2 is up, line protocol is up (connected)
  Internet Address 10.2.242.1/30, Area 0, Attached via Interface Enable
  Process ID 1, Router ID 10.2.255.202, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Enabled by interface config, including secondary ip addresses
  Transmit Delay is 1 sec, State BDR, Priority 1, BFD enabled
  Designated Router (ID) 10.1.255.102, Interface address 10.2.242.2
  Backup Designated router (ID) 10.2.255.202, Interface address 10.2.242.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:00
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/7/7, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 2 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 10.1.255.102  (Designated Router)
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Sending SA: Key 1, Algorithm HMAC-SHA-384 - key chain DCHQ
SW202#
```

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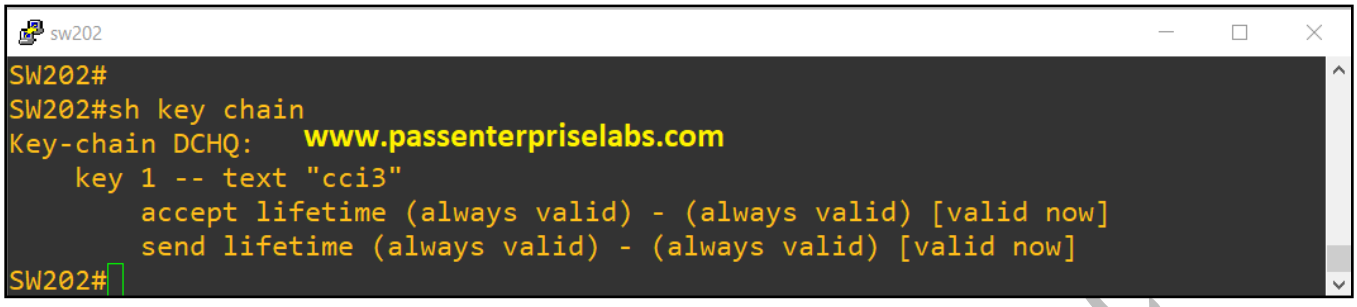
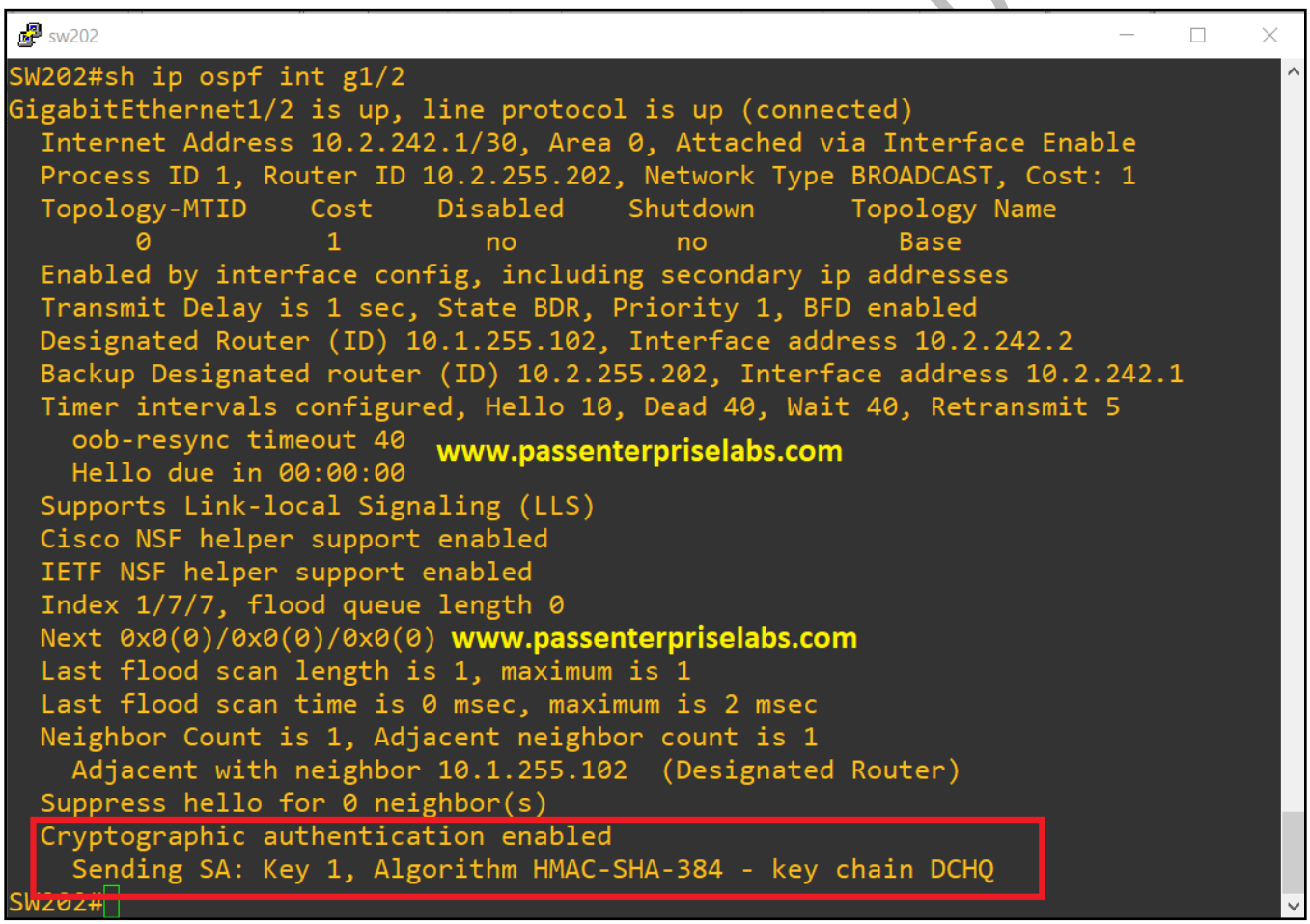
中文_垃圾桶_sw202#sh钥匙扣



中文_ dumps_sw202#sh ip os int g1/2



Chinese_dumps_sw202#sh bfd neighbor details

```
sw202
SW202#sh bfd nei details

IPv4 Sessions
NeighAddr                              LD/RD         RH/RS     State     Int
10.2.242.2                             1/1           Up        Up        Gi1/2
Session state is UP and using echo function with 100 ms interval.
Session Host: Software
OurAddr: 10.2.242.1
Handle: 1
Local Diag: 0, Demand mode: 0, Poll bit: 0
MinTxInt: 1000000, MinRxInt: 1000000, Multiplier: 3
Received MinRxInt: 1000000, Received Multiplier: 3
Holddown (hits): 0(0), Hello (hits): 1000(2165)
Rx Count: 2111, Rx Interval (ms) min/max/avg: 2/1073/898 last: 365 ms ago
Tx Count: 2166, Tx Interval (ms) min/max/avg: 1/1014/877 last: 35 ms ago
Elapsed time watermarks: 0 0 (last: 0)
Registered protocols: OSPF CEF
Uptime: 00:31:36
Last packet: Version: 1                  - Diagnostic: 0
             State bit: Up               - Demand bit: 0
             Poll bit: 0                 - Final bit: 0
             C bit: 0
             Multiplier: 3               - Length: 24
             My Discr.: 1                - Your Discr.: 1
             Min tx interval: 1000000    - Min rx interval: 1000000
             Min Echo interval: 100000
SW202#
```

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Chinese_dumps_sw202: sh ip ospf neighbor

```
sw202
SW202#
SW202#sh ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.1.255.102      1   FULL/DR         00:00:30    10.2.242.2      GigabitEthernet1/2
10.2.255.201      1   FULL/BDR        00:00:36    10.2.109.1      GigabitEthernet0/3
SW202#
```

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中文_ dumps_sw202#sh bfd邻居详情

中文_dumps_sw202:sh ip ospf邻居

1.5: DHCP IPv4 service for HQ

Enable hosts in HQ VLAN 2000 and VLAN 2001 to obtain their IP configuration via DHCP according to these requirements:

- On sw211, create IPv4 DHCP pools named hq_v2000 and hq_v2001 for HQ VLANs 2000 and 2001, respectively. In each subnet, assign addresses from .101 upto .254 inclusively, and the appropriate gateway to clients.
- Enable DHCP snooping on sw110 in VLANs 2000 and 2001 to protect against DHCP-related attacks.
- Place host11 into VLAN 2000.
- Place host12 into VLAN 2001.
- Perform the necessary configuration on switches sw101, sw102, sw110 to enable hosts in VLANs 2000 and 2001 to obtain IPv4 configuration through DHCP. The DHCP server running at sw211 in the DC must be referred to by its loopback IPv4 address 10.2.255.211. Do not disable the Option 82 insertion, and do not enable DHCP snooping on other switches.
- Verify that host11 and host12 have IP connectivity to the Cisco DNA Center, vManage and ISE running in the DC using their internal (In Band Connectivity) addresses.

2 Points

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1.5：HQ的DHCP IPv4服务

使HQ VLAN 2000和VLAN 2001中的主机能够根据以下要求通过DHCP获取其IP配置：

- 在sw211上，分别为hq VLAN 2000和2001创建名为hq_v2000和hq_v2001的IPv4 DHCP池。在每个子网中，为客户端分配从.101到.254的地址以及相应的网关。
- 在VLAN 2000和2001中的sw110上启用DHCP侦听，以防止DHCP相关攻击。
- 将主机11放入VLAN 2000。
- 将主机12放入VLAN 2001。
- 在交换机sw101、sw102、sw110上执行必要的配置，以使VLANs2000和2001中的主机能够通过DHCP获得IPv4配置。DC中sw211上运行的DHCP服务器必须通过其环回IPv4地址10.2.255.211进行引用。不要禁用选项82插入，也不要在其他交换机上启用DHCP侦听。
- 验证主机11和主机12是否使用其内部（带内连接）地址与在DC中运行的Cisco DNA中心、vManage和ISE具有IP连接。

2分

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Solution:

Chinese_dumps_sw110:

```
sw110>en
sw110#conf t
sw110(config)#ip dhcp snooping
sw110(config)#ip dhcp snooping vlan 2000 2001
sw110(config)#int range po1,po2
sw110(config-if)#ip dhcp snooping trust
sw110(config-if)#exit

sw110(config)#int g0/0
sw110(config-if)#switchport mode access
sw110(config-if)#switchport access vlan 2000
sw110(config-if)#exit

sw110(config)#int g0/1
sw110(config-if)#switchport mode access
sw110(config-if)#switchport access vlan 2001
sw110(config-if)#exit
```

Chinese_dumps_sw101:

```
sw101>en
sw101#conf t
sw101(config)#int vlan 2000
sw101(config-if)#ip helper-address 10.2.255.211
sw101(config-if)#ip dhcp relay information trusted

sw101(config)#int vlan 2001
sw101(config-if)#ip helper-address 10.2.255.211
sw101(config-if)#ip dhcp relay information trusted
sw101(config-if)#exit
sw101(config)#exit
```

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解决方案：

中文_垃圾_ sw110：

sw110#配置文件
sw110（配置）#ip-dhcp监听sw110（设置）
#ip-dhcp监听vlan 2000 2001 sw110（设定）
#int范围po1、po2
sw110（配置if）#ip dhcp监听信任
sw110（设置if）#退出

sw110（配置）#int g0/0
sw110（config-if）#交换端口模式访问
sw110（-config-if）#vlan 2000 sw110
（config.if）#退出

sw110（配置）#int g0/1
sw110（config-if）#交换机端口模式访
问sw110（-config-if）#switchport访问
vlan 2001 sw110（config.if）#退出

中文_垃圾_ sw101：

sw101>en-sw101#配
置文件
sw101（配置）#int vlan 2000
sw101（config-if）#ip助手地址
10.2.255.211 sw101（config.if）#ip-dhcp
中继信息可信

sw101（配置）#int vlan 2001
sw101（config-if）#ip助手地址
10.2.255.211 sw101（config.if）#ip-dhcp
中继信息可信sw101（-config-if）#退出
sw110>en

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Chinese_dumps_sw102:

```
sw102>en
sw102#conf t
sw102(config)#int vlan 2000
sw102(config-if)#ip helper-address 10.2.255.211
sw102(config-if)#ip dhcp relay information trusted
sw102(config-if)#exit

sw102(config)#int vlan 2001
sw102(config-if)#ip helper-address 10.2.255.211
sw102(config-if)#ip dhcp relay information trusted
```

Chinesedumps.com-sw211:

```
sw211>en
sw211#conf t
sw211(config)#ip dhcp excluded-address 10.1.100.1 10.1.100.100
sw211(config)#ip dhcp pool hq_v2000
sw211(config-dhcp)#network 10.1.100.0 /24
sw211(config-dhcp)#default-router 10.1.100.1
sw211(config-dhcp)#exit

sw211(config)#ip dhcp excluded-address 10.1.101.1 10.1.101.100
sw211(config)#ip dhcp pool hq_v2001
sw211(config-dhcp)#network 10.1.101.0 /24
sw211(config-dhcp)#default-router 10.1.101.1
sw211(config)#exit
```

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中文_垃圾_ sw102：

sw102>en-
sw102#配置
文件

sw102（配置if）#ip助手地址10.2.255.211
sw102（config-if）#ip-dhcp中继信息可信
sw102（config.if）#退出

sw102（配置）#int vlan 2001
sw102（config-if）#ip助手地址
10.2.255.211 sw102（config.if）#ip-dhcp
中继信息可信

Chinese edumps.com-sw211：

sw211>en-sw211#配
置文件
sw211（配置）#ip-dhcp排除地址10.1.100.1 10.1.100.100 sw211（
设置）#ip-dhcp池hq_v2000
sw211（配置dhcp）#网络10.1.100.0/24
sw211（config dhcp）#默认路由器
10.1.100.1 sw211（config-dhcp）#退出

sw211（配置）#ip-dhcp排除地址10.1.101.1 10.1.101.100 sw211（
设置）#ip-dhcp池hq_v2001
sw211（配置dhcp）#网络10.1.101.0/24
sw211（config dhcp）#默认路由器
10.1.101.1 sw211（配置）#退出

sw102（配置）#int vlan 2000

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Verification:

Chinese_dumps_sw101#sh run int vlan 2000

```
SW101#sh run int vlan 2000
Building configuration...

Current configuration : 251 bytes
!    www.passenterpriselabs.com
interface Vlan2000
 ip dhcp relay information trusted
 ip address 10.1.100.2 255.255.255.0
 ip helper-address 10.2.255.211
 ip ospf 1 area 0
 ipv6 address FE80::101 link-local
 vrrp 100 ip 10.1.100.1
 vrrp 100 priority 105
 vrrp 100 track 100
end  www.passenterpriselabs.com

SW101#
SW101#
```

Chinese_dumps_sw101#sh run int vlan 2001

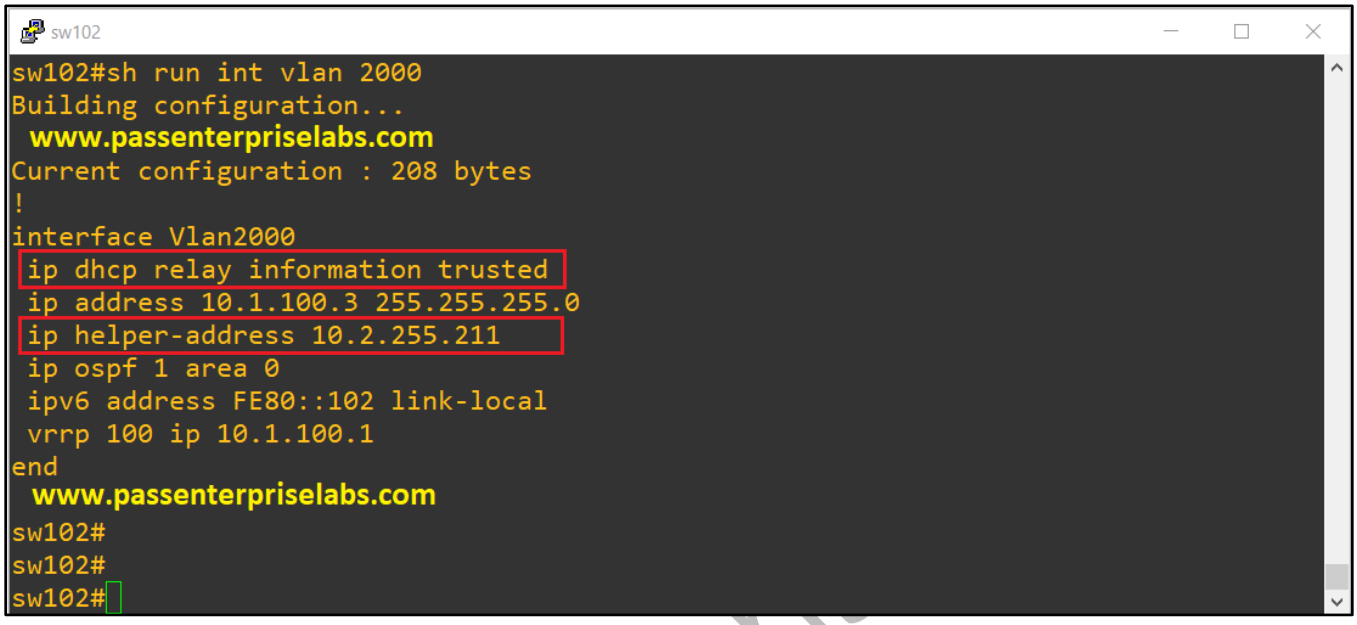
```
SW101#
SW101#sh run int vlan 2001
Building configuration...
  www.passenterpriselabs.com
Current configuration : 208 bytes
!
interface Vlan2001
 ip dhcp relay information trusted
 ip address 10.1.101.2 255.255.255.0
 ip helper-address 10.2.255.211
 ip ospf 1 area 0
 ipv6 address FE80::101 link-local
 vrrp 101 ip 10.1.101.1
end
  www.passenterpriselabs.com
SW101#
SW101#
SW101#
```

验证：

中文. domps_sw101#sh运行于vlan 2000

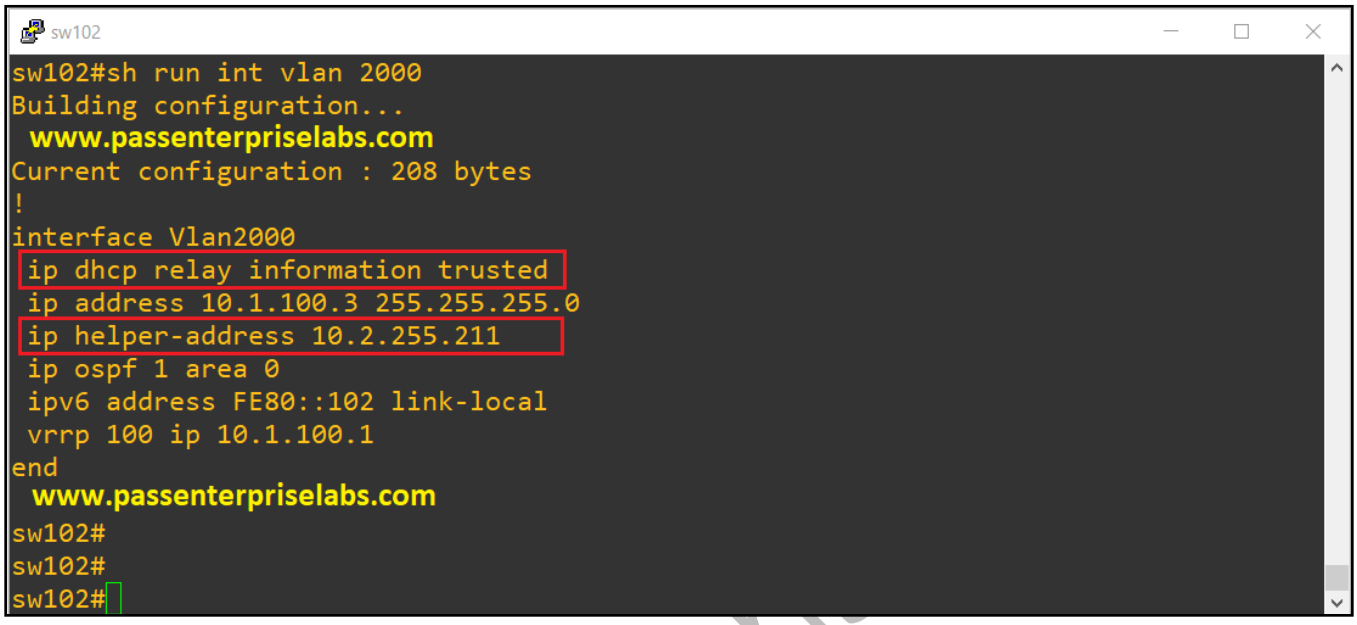
中文_ dumps_sw101#sh运行于2001年

Chinese_dumps_sw102#sh run int vlan 2000

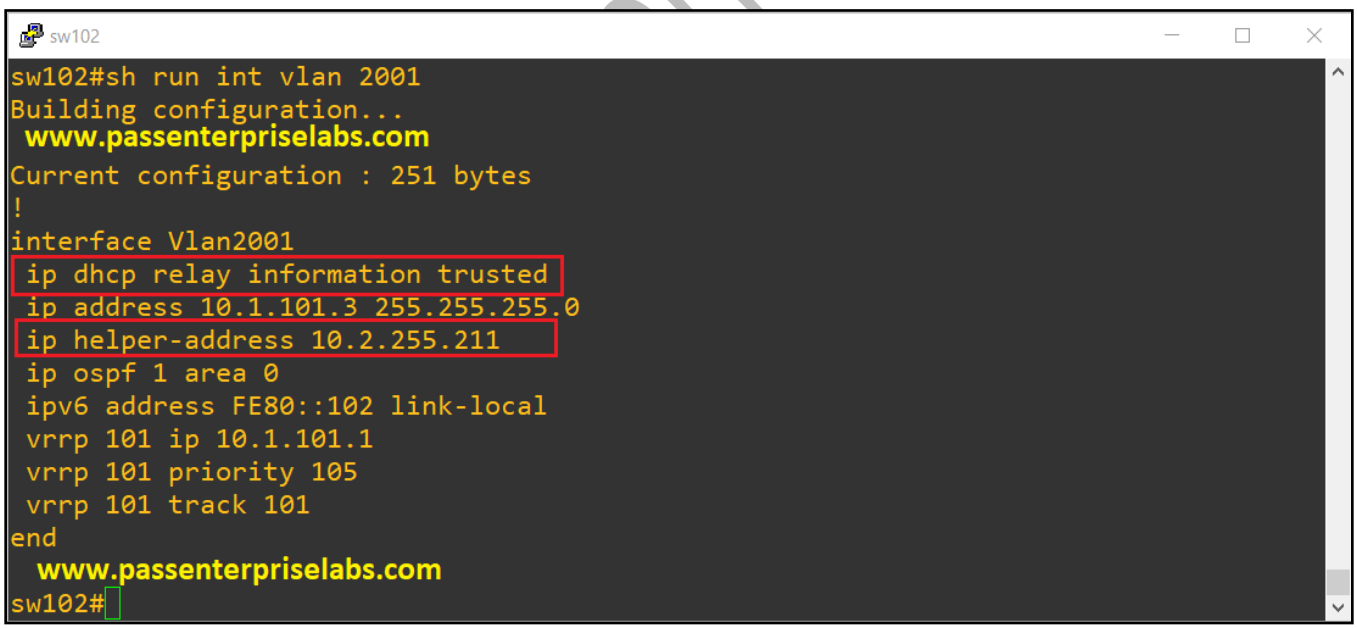


Chinese_dumps_sw102#sh run int vlan 2001

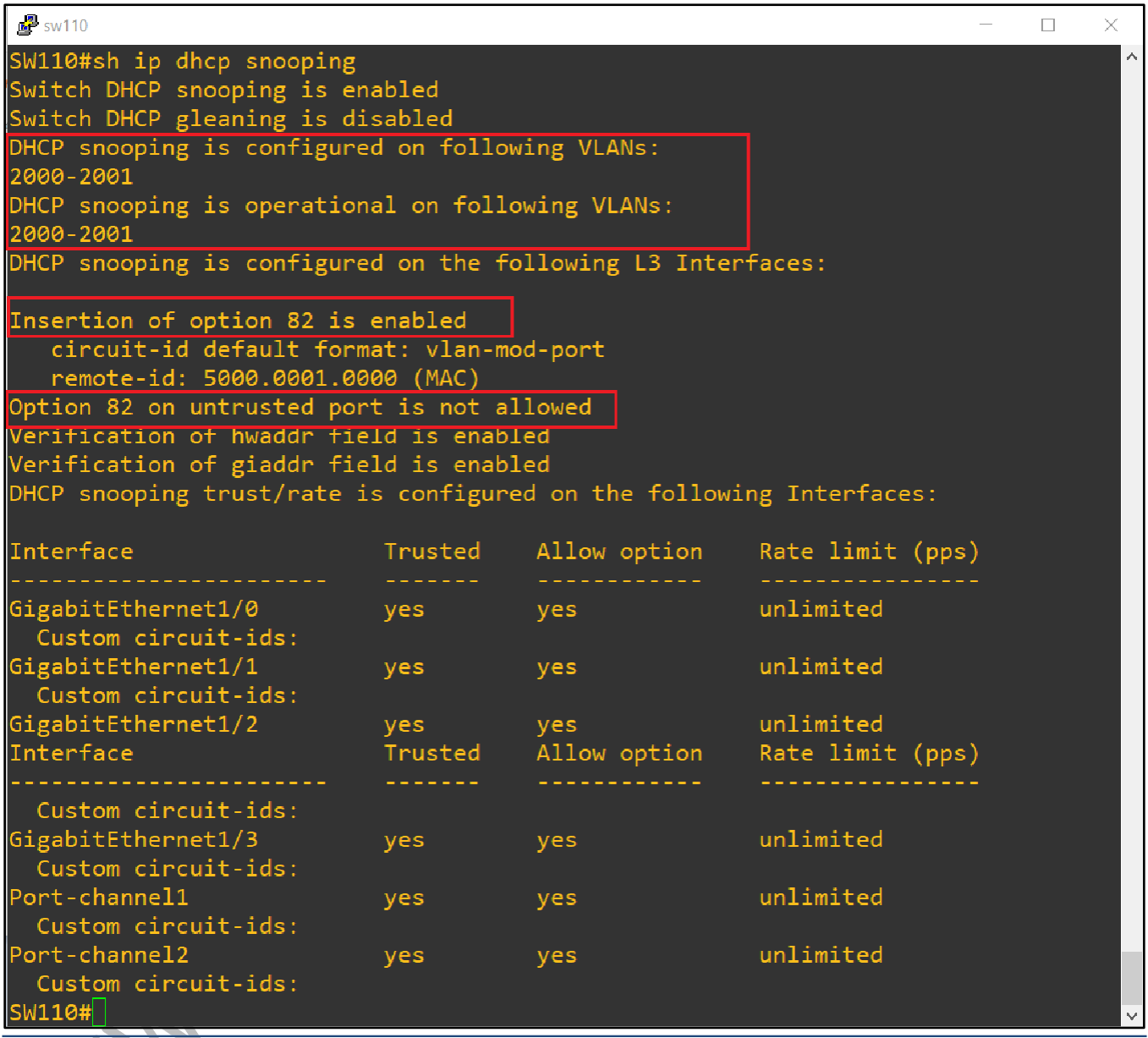
中文.domps_sw102#sh运行于vlan 2000



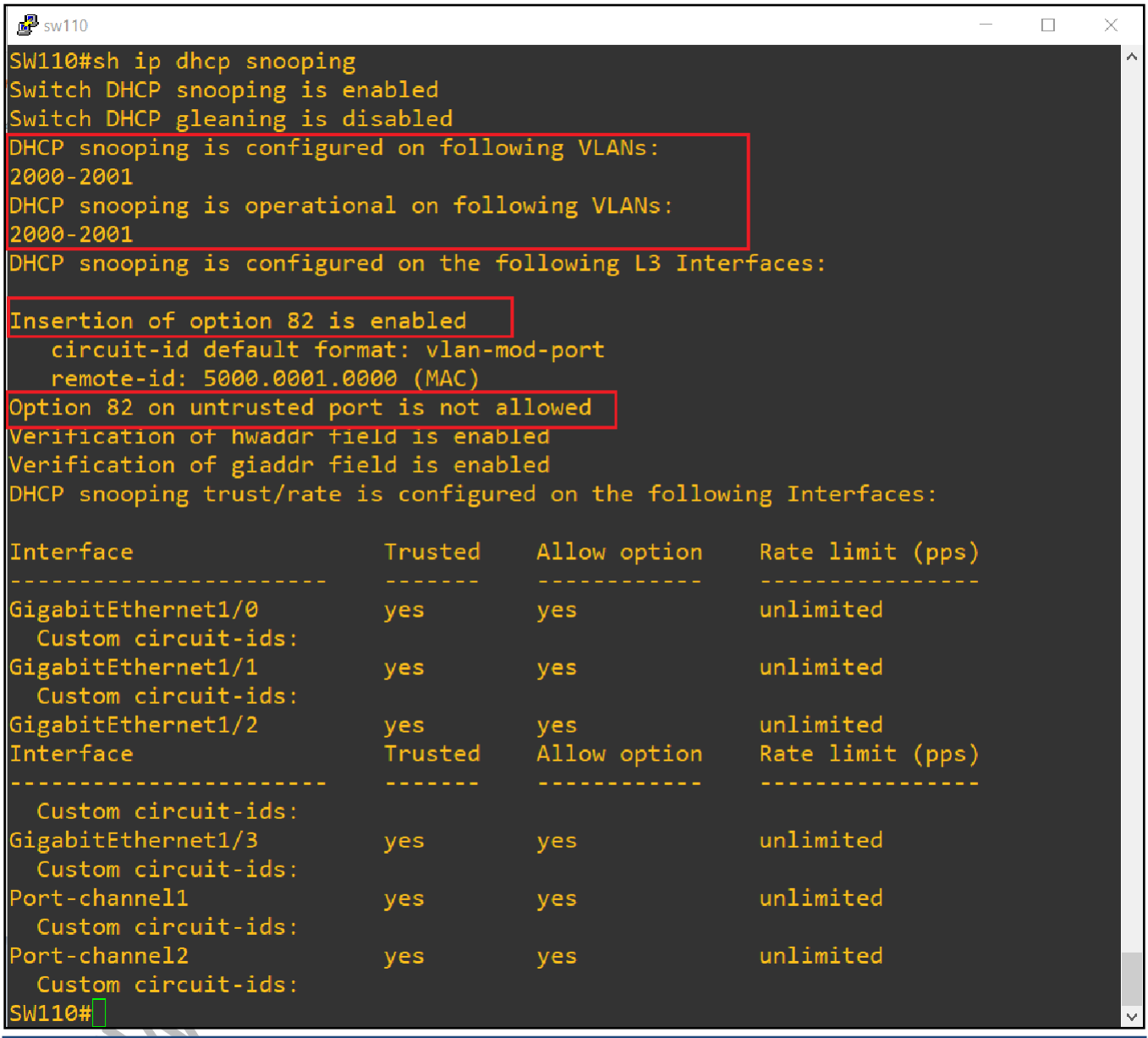
中文_转储_ sw102#sh运行于vlan 2001



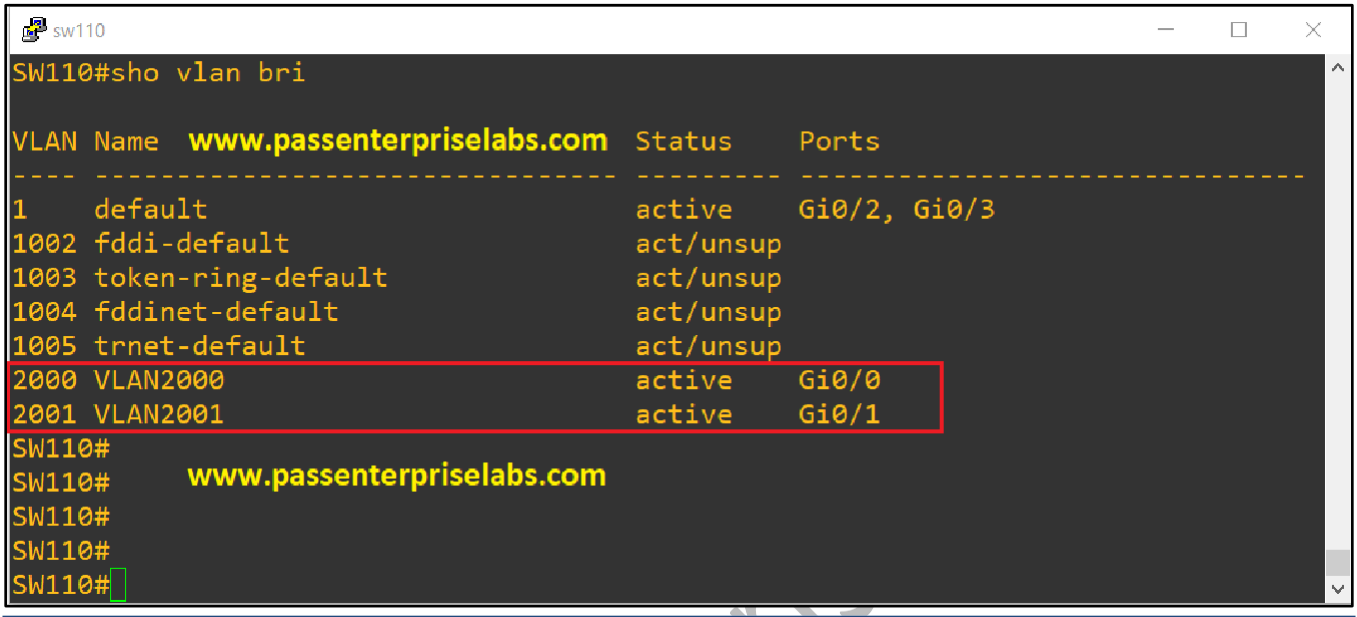
Chinese_dumps_sw110#sh ip dhcp snooping



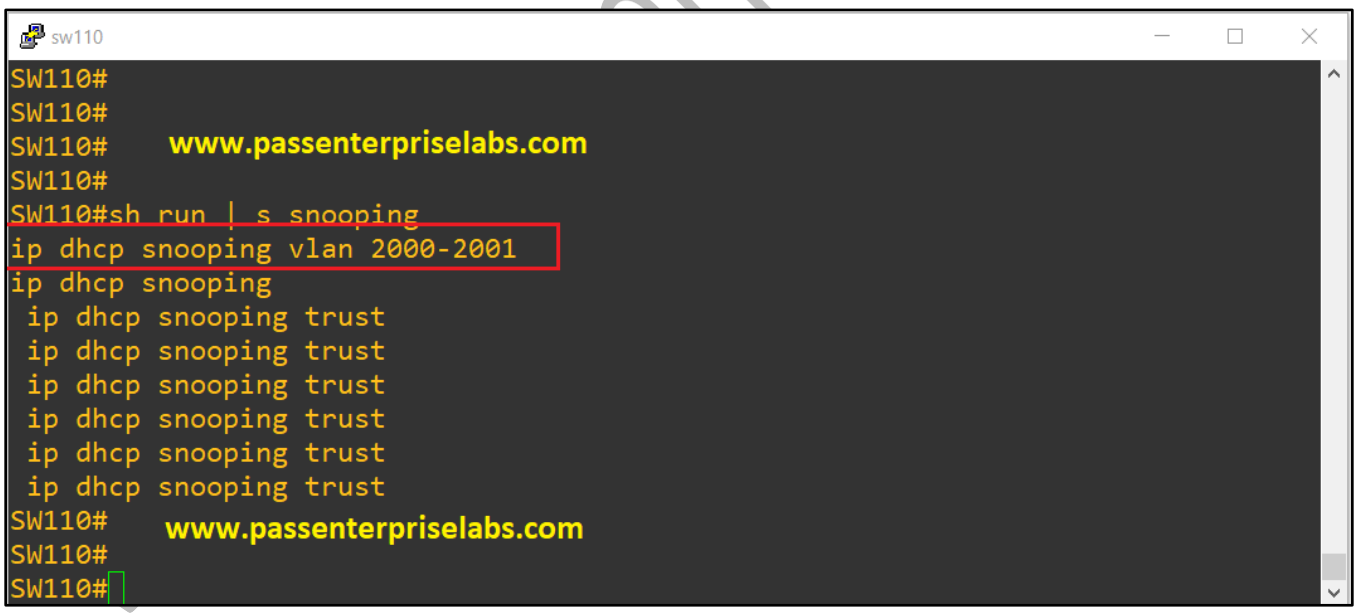
中文_ dumps_sw110#sh ip dhcp监听



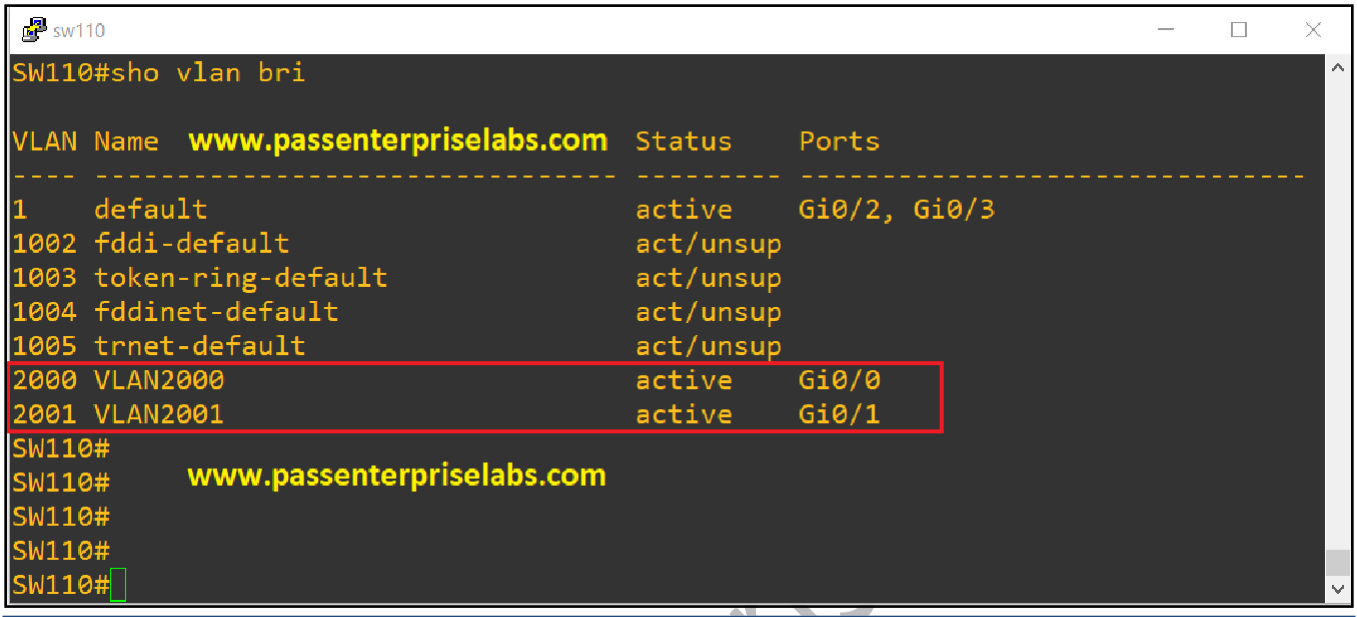
Chinese_dumps_sw110#sh vlan brief



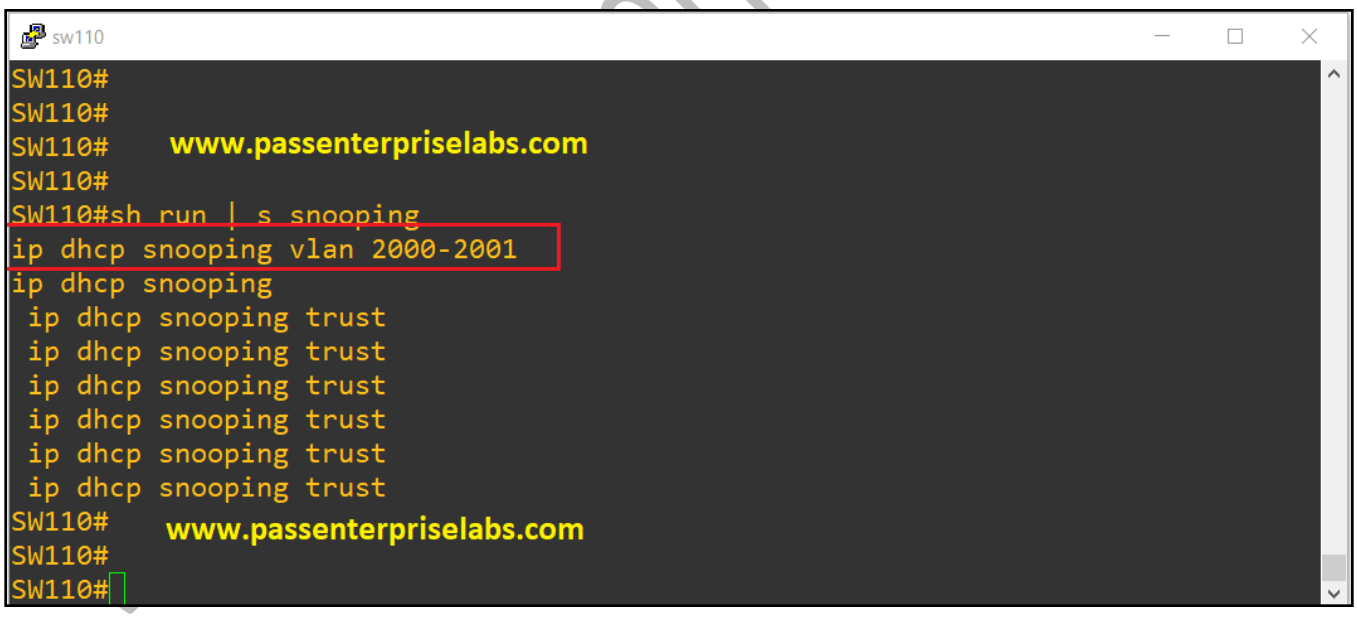
Chinese_dumps_sw110#sh run | s snooping



中文_ dumps_sw110#sh vlan简介



中文_垃圾_ sw110#sh跑的窥探



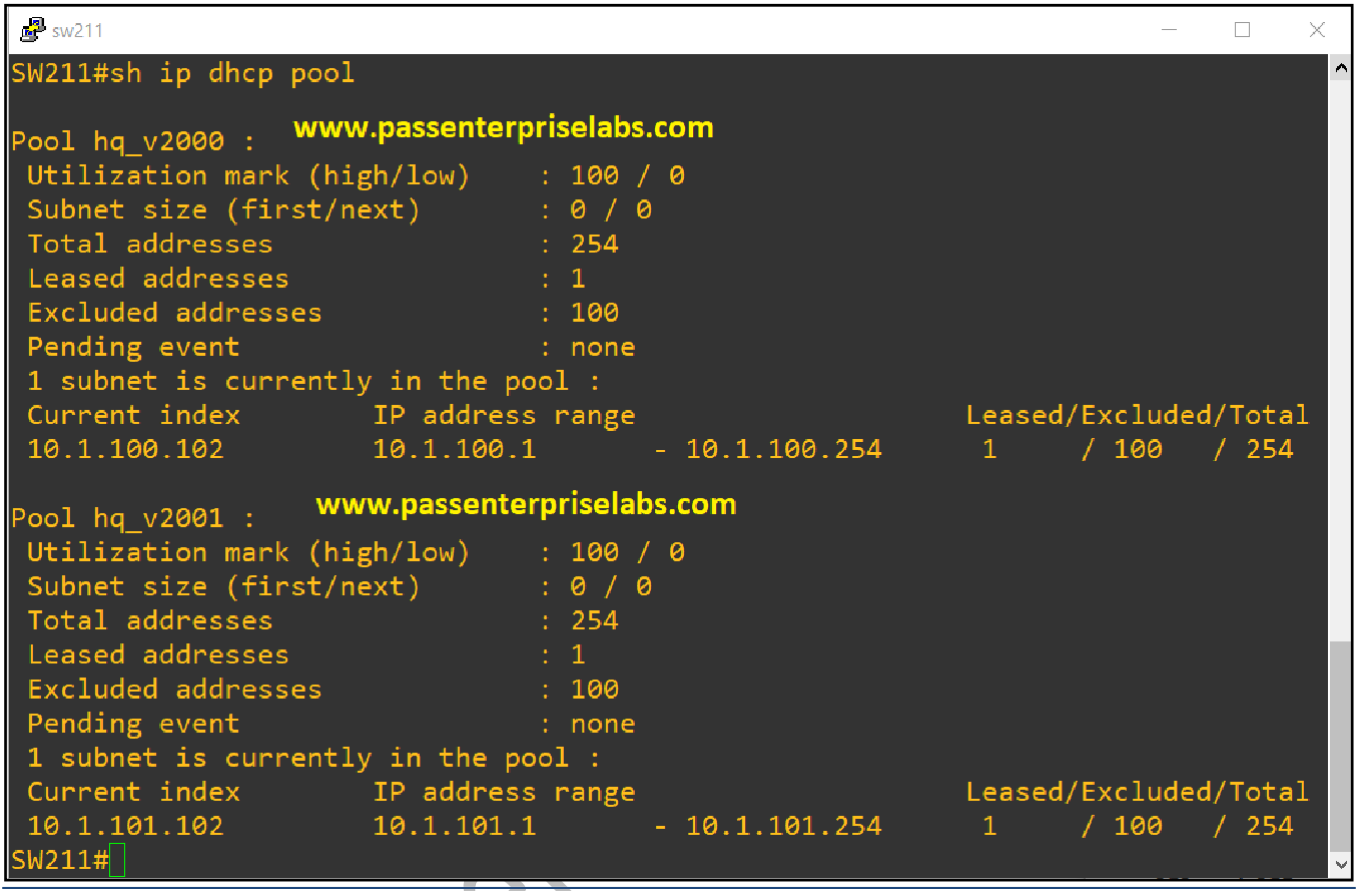
Chinesedumps.com-sw211#sh ip dhcp pool

```
sw211
SW211#sh ip dhcp pool

Pool hq_v2000 :   www.passenterpriselabs.com
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 100
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased/Excluded/Total
 10.1.100.102         10.1.100.1       - 10.1.100.254      1      / 100    / 254

Pool hq_v2001 :   www.passenterpriselabs.com
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 100
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased/Excluded/Total
 10.1.101.102         10.1.101.1       - 10.1.101.254      1      / 100    / 254
SW211#
```

Chinesedumps.com-sw211#sh run | s dhcp

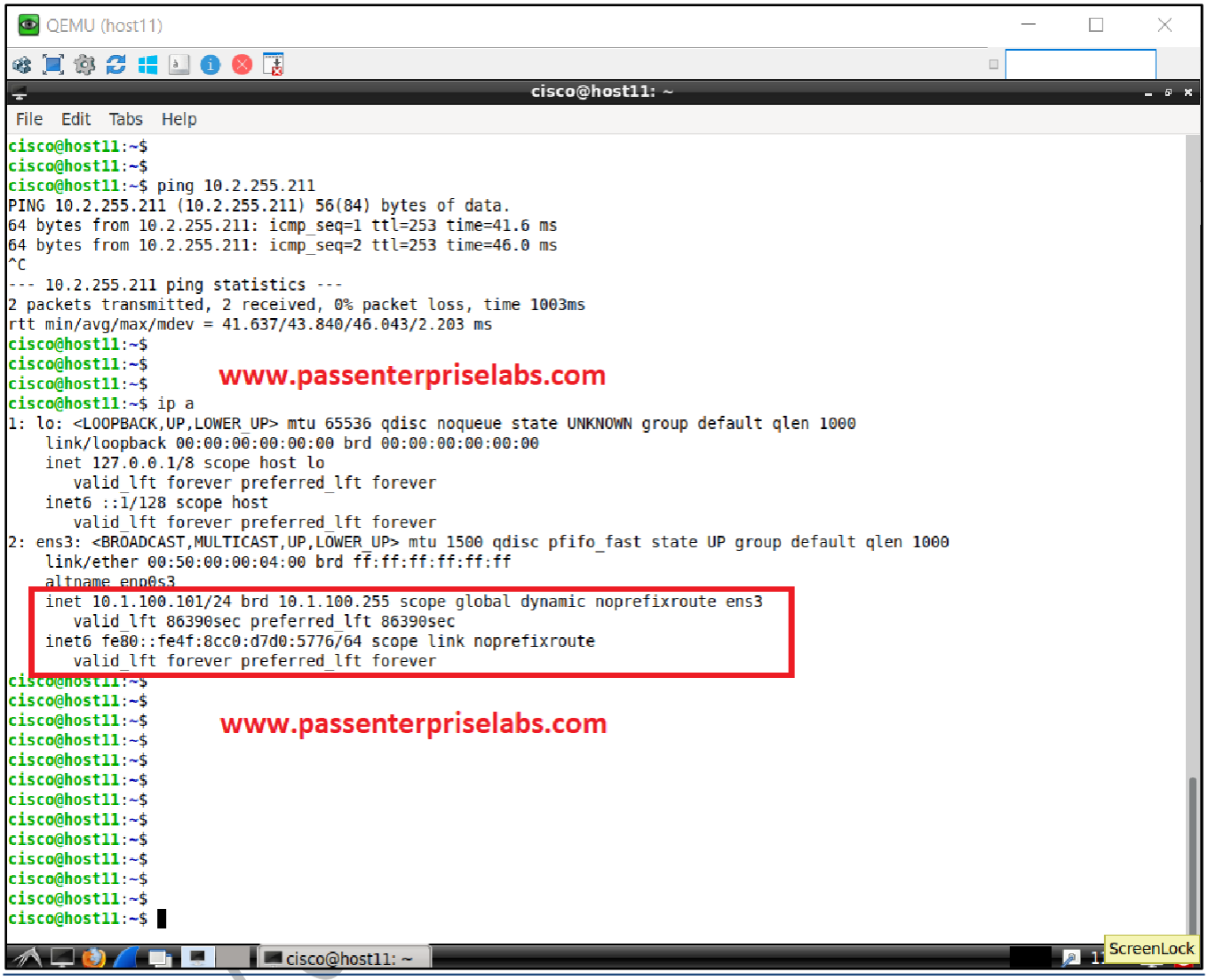
```
sw211
SW211#sh run | s dhcp
ip dhcp excluded-address 10.1.100.1 10.1.100.100
ip dhcp excluded-address 10.1.101.1 10.1.101.100
ip dhcp pool hq_v2000   www.passenterpriselabs.com
 network 10.1.100.0 255.255.255.0
 default-router 10.1.100.1
ip dhcp pool hq_v2001
 network 10.1.101.0 255.255.255.0
 default-router 10.1.101.1
SW211#
SW211#   www.passenterpriselabs.com
SW211#
SW211#
```

Chinese edumps.com-sw211#sh ip dhcp池



Chinese edumps.com-sw211#sh运行dhcp

Host 11:



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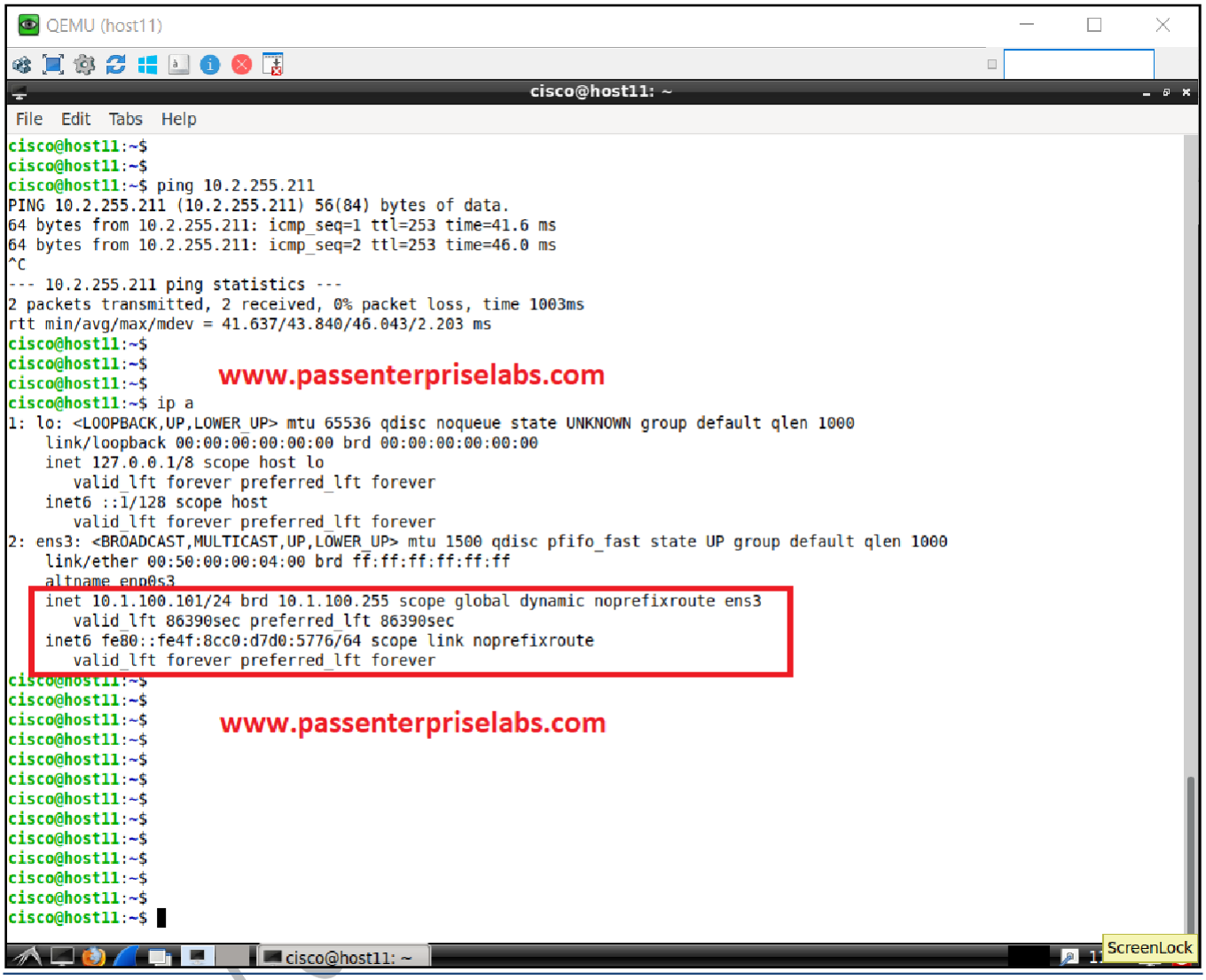
主持人11：



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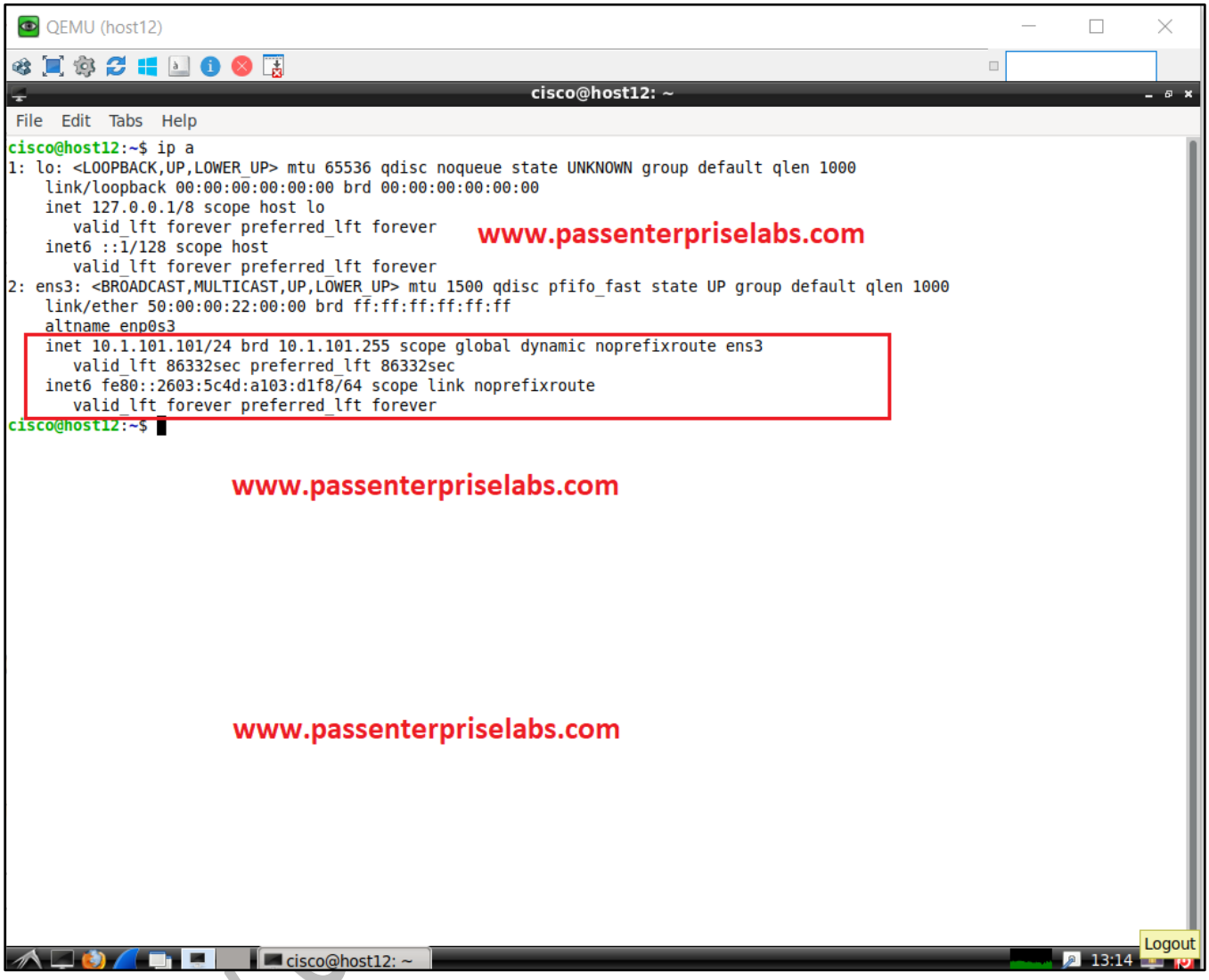
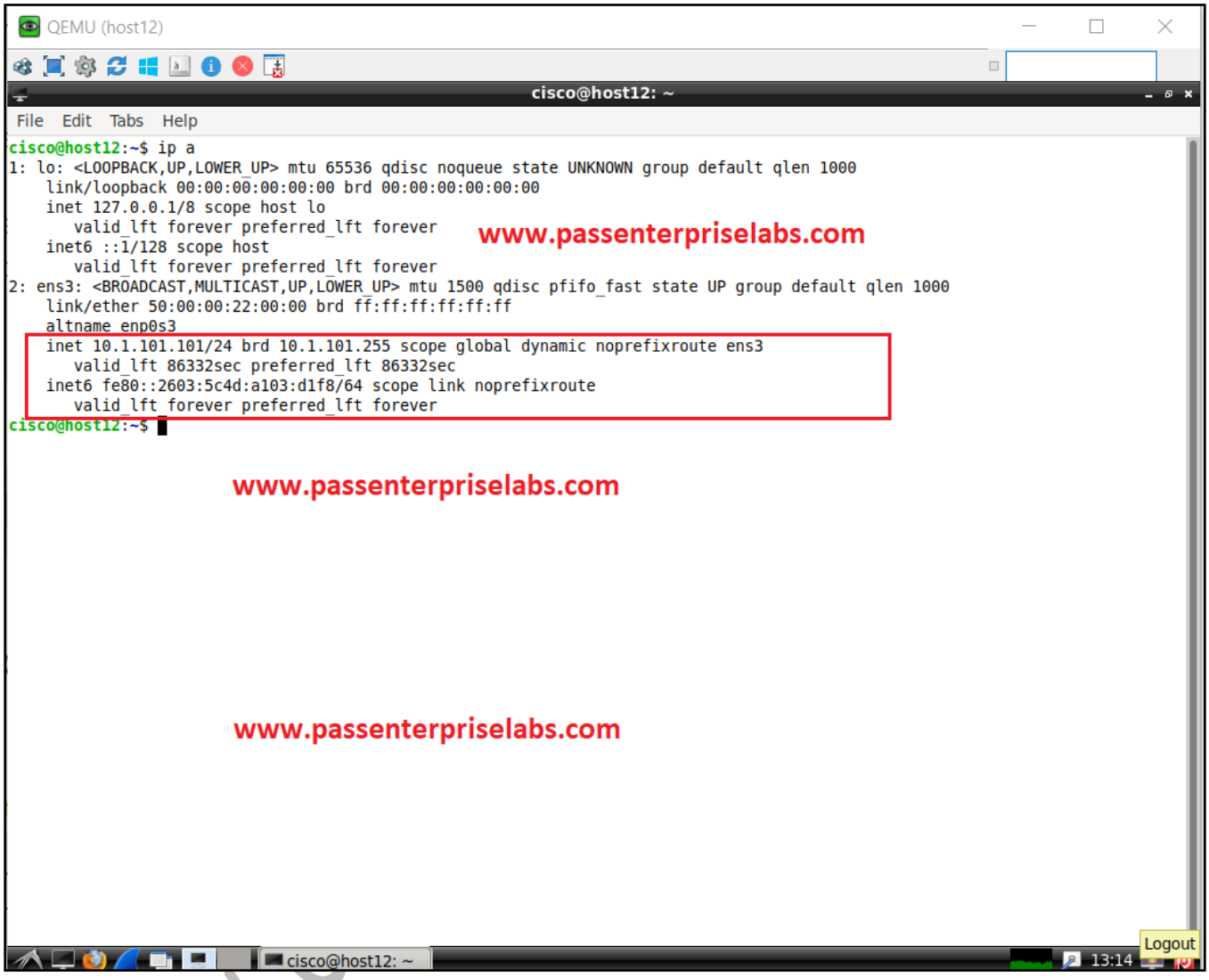
Host 12:


QEMU (host12)
cisco@host12: ~
File Edit Tabs Help
cisco@host12:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 50:00:00:22:00:00 brd ff:ff:ff:ff:ff:ff
altname enp0s3
inet 10.1.101.101/24 brd 10.1.101.255 scope global dynamic noprefixroute ens3
valid_lft 86332sec preferred_lft 86332sec
inet6 fe80::2603:5c4d:a103:d1f8/64 scope link noprefixroute
valid_lft forever preferred_lft forever
cisco@host12:~$
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@host12: ~
13:14
Logout

WWW.

主持人12：


QEMU (host12)
cisco@host12: ~
File Edit Tabs Help
cisco@host12:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 50:00:00:22:00:00 brd ff:ff:ff:ff:ff:ff
altname enp0s3
inet 10.1.101.101/24 brd 10.1.101.255 scope global dynamic noprefixroute ens3
valid_lft 86332sec preferred_lft 86332sec
inet6 fe80::2603:5c4d:a103:d1f8/64 scope link noprefixroute
valid_lft forever preferred_lft forever
cisco@host12:~$
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@host12: ~
13:14
Logout

WWW.

1.6: IPv6 in HQ

Implement IPv6 on sw101 and sw102 for switch virtual interfaces (SVIs) Vlan 2000 and Vlan 2001 according to these requirements:

- Sw101
  Interface Vlan2000: 2001:db8:1:100::1/64
  Interface Vlan2001: 2001:db8:1:101::1/64
- Sw102
  Interface Vlan2000: 2001:db8:1:100::2/64
  Interface Vlan2001: 2001:db8:1:101::2/64
- The configuration must enable hosts in these VLANs to obtain their IPv6 configuration via SLAAC and keep a stable connectivity with other IPv6 networks.
- Use native IPv6 means to provide gateway redundancy, with sw101 being the preferred gateway in VLAN 2000 and sw102 being the preferred gateway in VLAN 2001. The role must be determined by an explicit configuration solely on the intended preferred gateway.
- Hosts must be able to detect the failure of the preferred gateway in as little as 3 seconds.

1 Point

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1.6： 总部IPv6

根据以下要求，在交换机虚拟接口（SVI）Vlan 2000和Vlan 2001的sw101和sw102上实现IPv6：

- 瑞士法郎101
  接口Vlan2000:2001:db8:1:100::1/64
  接口Vlan2001:2001:db8:1:101::1/64
- 瑞士法郎102
  接口Vlan2000:2001:db8:1:100::2/64
  接口Vlan2001:2001:db8:1:101::2/64
- 配置必须使这些VLAN中的主机能够通过SLAAC获得其IPv6配置，并与其他IPv6网络保持稳定的连接。
- 使用本地IPv6手段提供网关冗余，sw101是VLAN 2000中的首选网关，sw102是VLAN 2001中的首选路由器。该角色必须仅通过预期首选网关上的显式配置来确定。
- 主机必须能够在短短3秒内检测到首选网关的故障。

1分

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Solution:

Chinese_dumps_sw101:

```
sw101>en
sw101#conf t
sw101(config)#int vlan 2000
sw101(config-if)#ipv6 enable
sw101(config-if)#ipv6 address 2001:db8:1:100::1/64
sw101(config-if)#ipv6 nd router-preference high
sw101(config-if)#ipv6 nd ra lifetime 3
sw101(config-if)#ipv6 nd ra interval msec 1000

sw101(config)#int vlan 2001
sw101(config-if)#ipv6 enable
sw101(config-if)#ipv6 address 2001:db8:1:101::1/64
sw101(config-if)#ipv6 nd ra lifetime 3
sw101(config-if)#ipv6 nd ra interval msec 1000
```

Chinese_dumps_sw102:

```
sw102>en
sw102(config)#conf t
sw102(config-if)#int vlan 2000
sw102(config-if)#ipv6 enable
sw102(config-if)#ipv6 address 2001:db8:1:100::2/64
sw102(config-if)#ipv6 nd ra lifetime 3
sw102(config-if)#ipv6 nd ra interval msec 1000

sw102(config)#int vlan 2001
sw102(config-if)#ipv6 enable
sw102(config-if)#ipv6 address 2001:db8:1:101::2/64
sw102(config-if)#ipv6 nd router-preference high
sw102(config-if)#ipv6 nd ra lifetime 3
sw102(config-if)#ipv6 nd ra interval msec 1000
```

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解决方案：

中文_垃圾_ sw101：

```
sw101>en-
sw101#配置
文件
sw101（配置）#int vlan
2000 sw101（如果配置）
#ipv6启用
sw101（配置if）#ipv6地址2001:db8:1:100::
1/64 sw101
sw101（配置if）#ipv6和ra间隔毫秒1000

sw101（配置）#int vlan
2001 sw101（如果配置）
#ipv6启用
sw101（配置if）#ipv6地址2001:db8:1:101::
1/64 sw101（设置if）#ipv6-nd-ra生存期3
sw101（配置if）#ipv6和ra间隔毫秒1000
```

中文_垃圾_ sw102：

```
sw102>en-sw102（配置）
#conf-tsw102（如果配置）
#int-vlan 2000 sw102（if
配置）#ipv6启用
sw102（配置if）#ipv6地址2001:db8:1:100:: 2/64
sw102（配置if）#ipv6 ndra生存期3
sw102（设置if）#ipv6 ndra间隔毫秒1000

sw102（配置）#int vlan
2001 sw102（如果配置）
#ipv6启用
sw102（配置if）#ipv6地址2001:db8:1:101::
2/64 sw102
sw102（配置if）#ipv6和ra间隔毫秒1000
```

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Verification:

Chinese_dumps_sw101#sh run int vlan 2000

Chinese_dumps_sw101#sh run int vlan 2001

验证：

中文.domps_sw101#sh运行于vlan 2000

中文_ dumps_sw101#sh运行于2001年

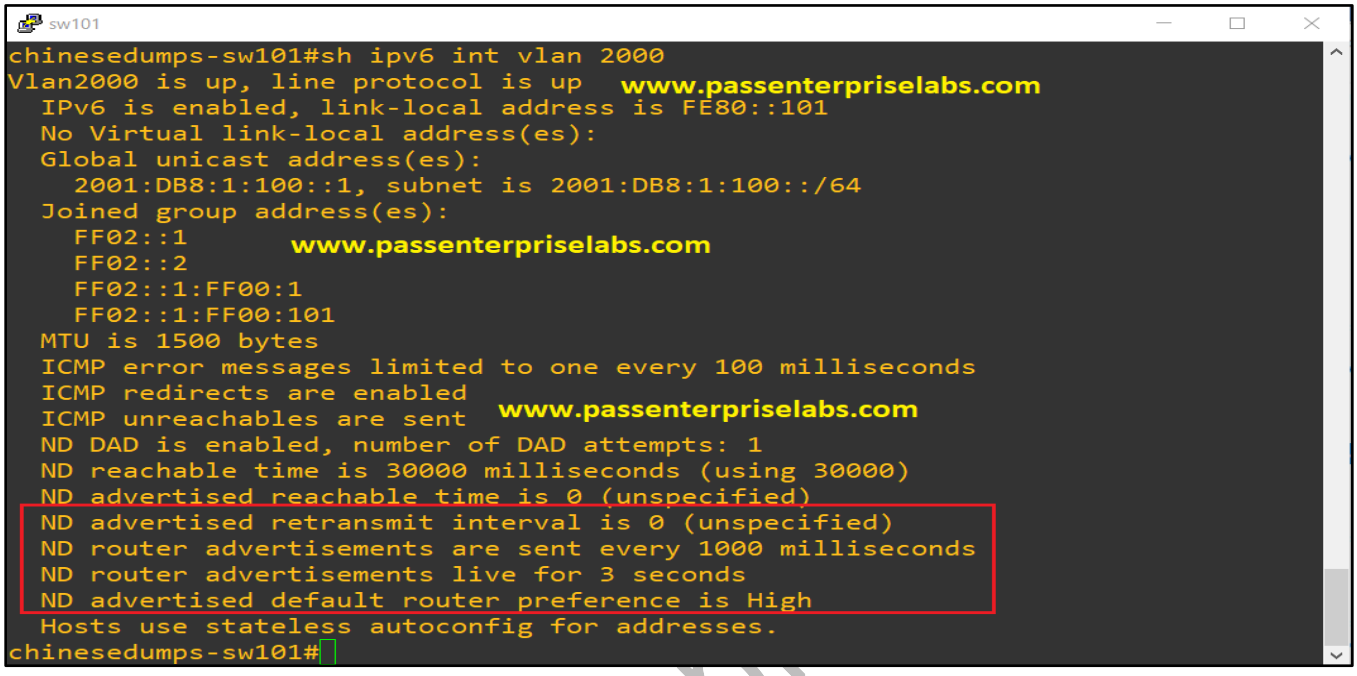
Chinese_dumps_sw102#sh run int vlan 2000

Chinese_dumps_sw102#sh run int vlan 2001

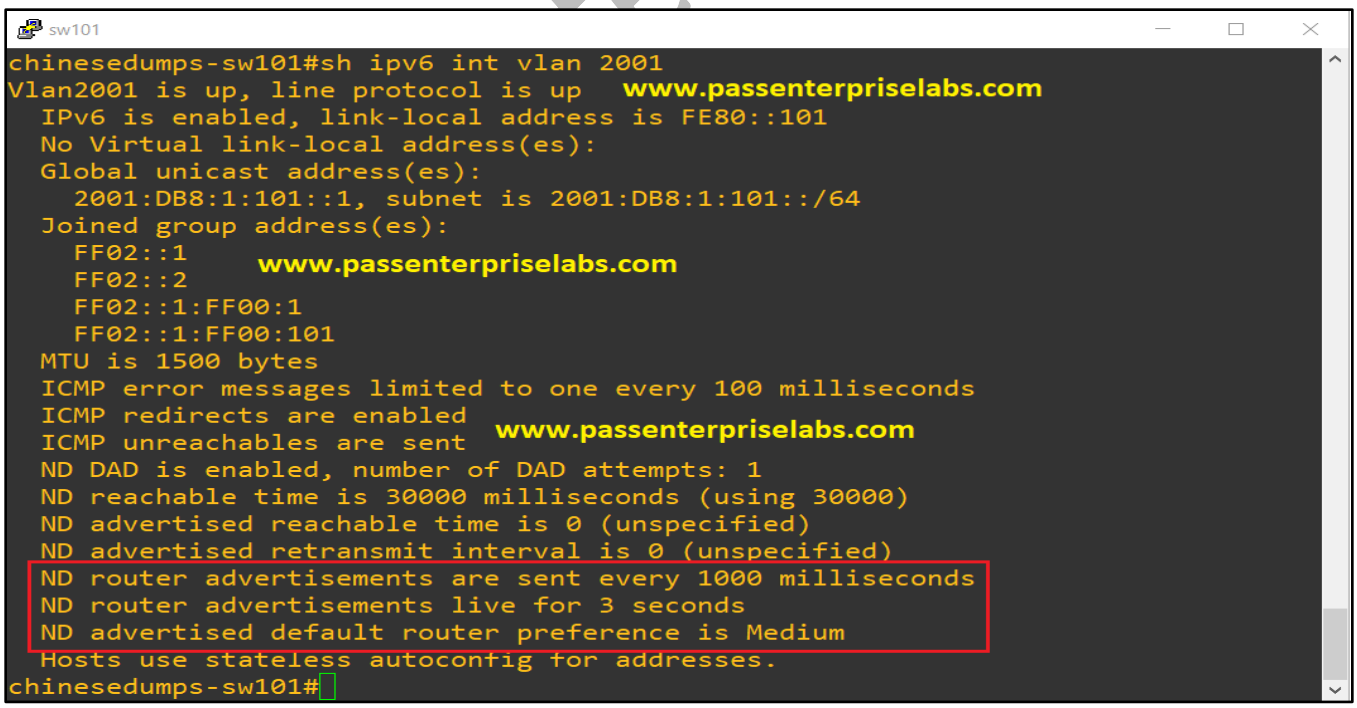
中文.domps_sw102#sh运行于vlan 2000

中文_转储_ sw102#sh运行于vlan 2001

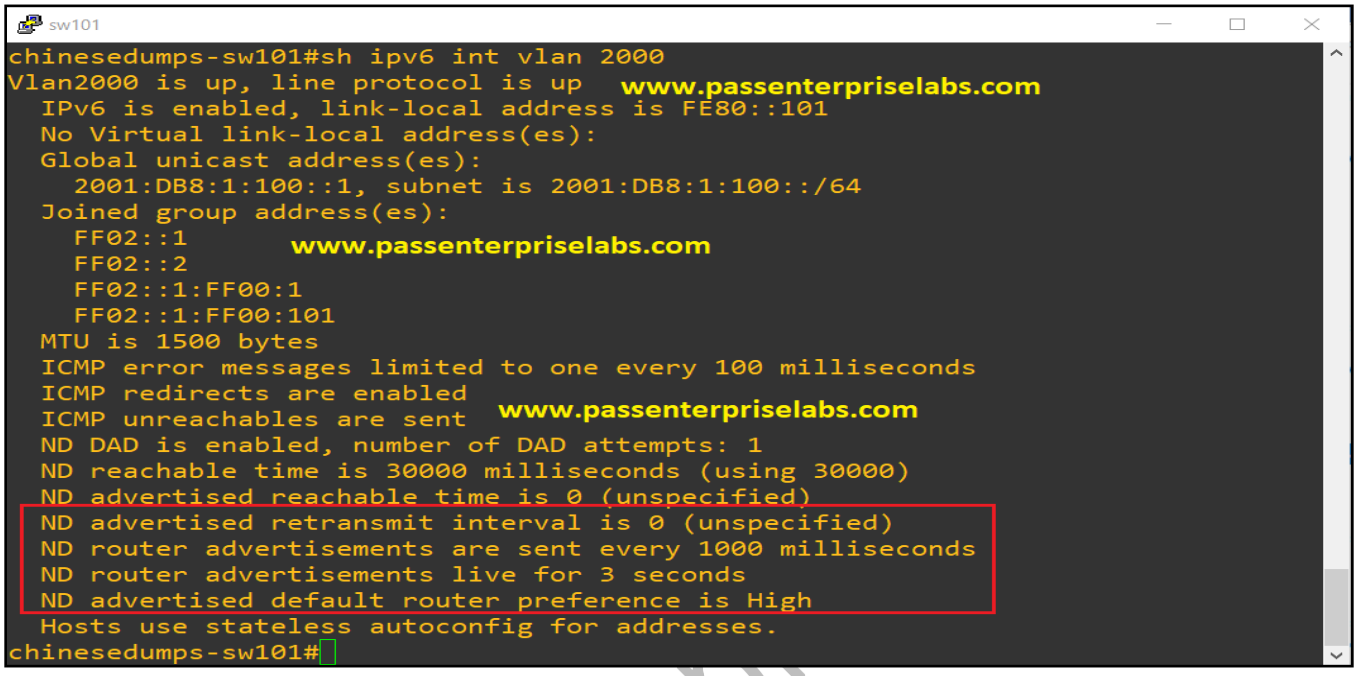
Chinese_dumps_sw101#sh ipv6 int vlan 2000



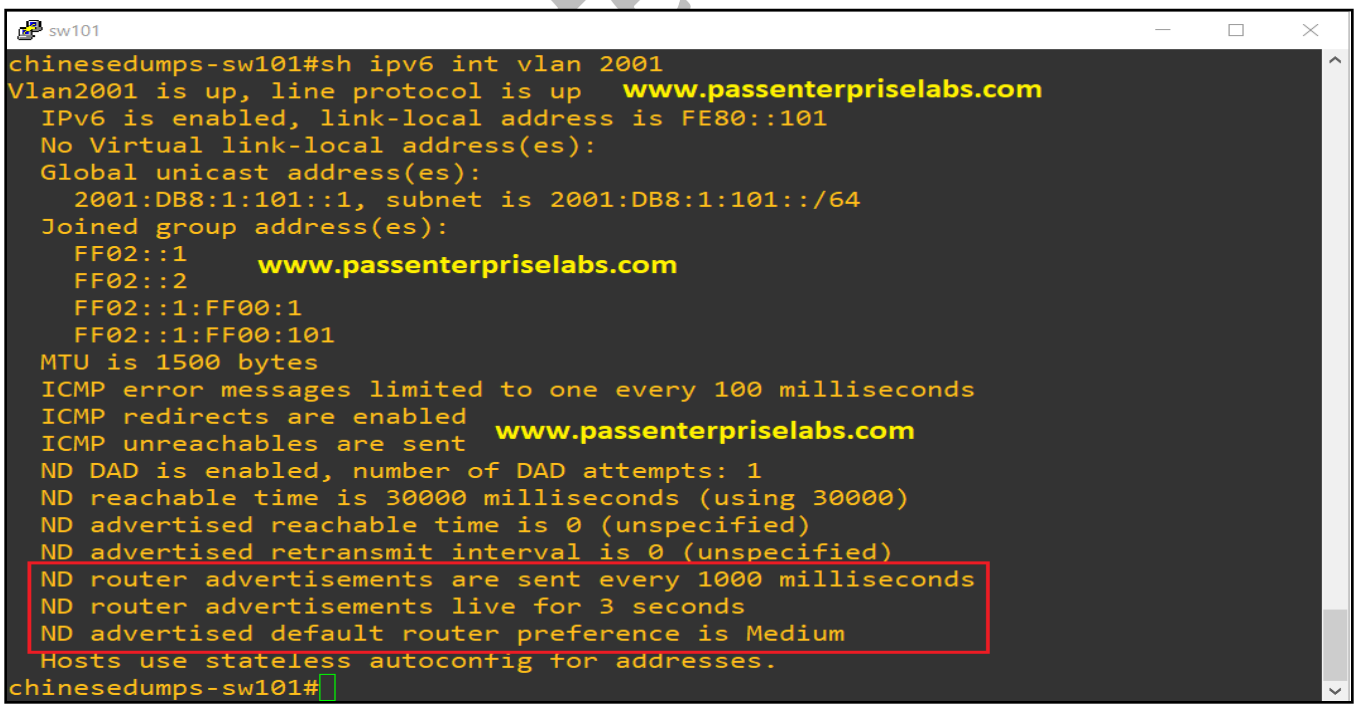
Chinese_dumps_sw101#sh ipv6 int vlan 2001



中文_ dumps_sw101#sh ipv6接口2000



中文_ dumps_sw101#sh ipv6局域网2001



Chinese_dumps_sw102#sh ipv6 int vlan 2000

sw102

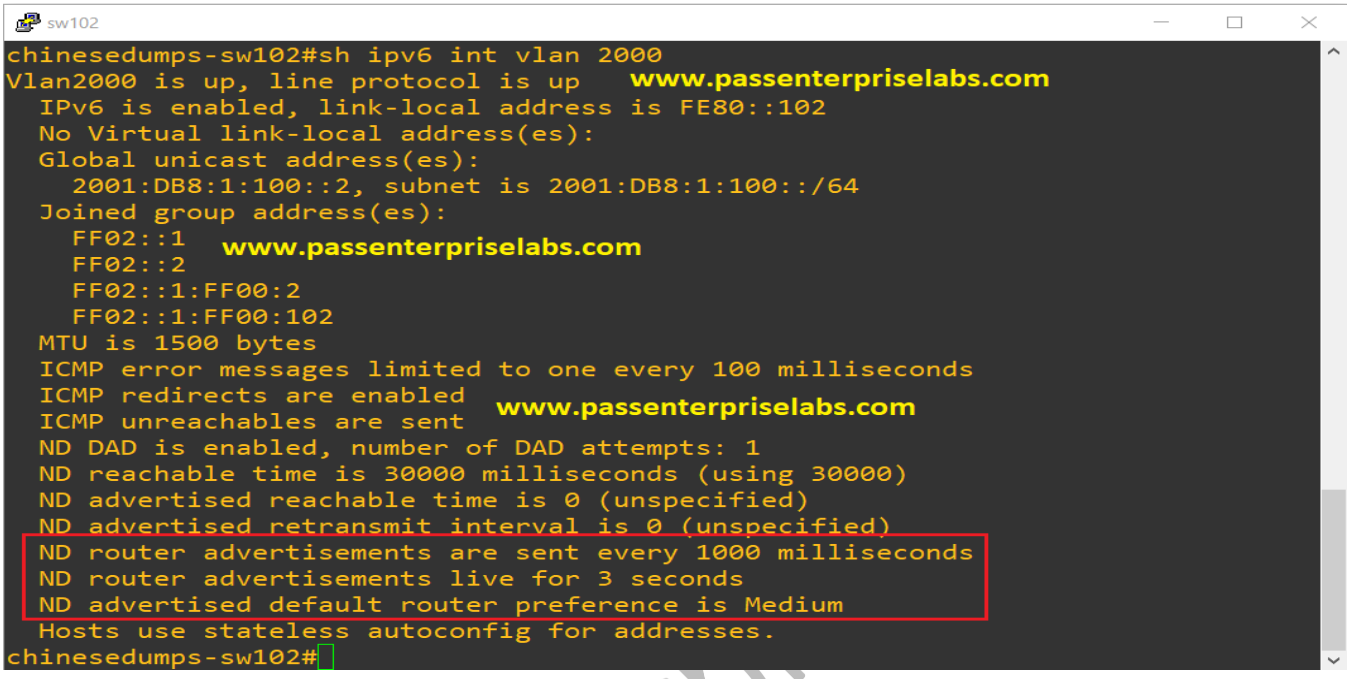
```
chinesedumps-sw102#sh ipv6 int vlan 2000
Vlan2000 is up, line protocol is up    www.passenterpriselabs.com
  IPv6 is enabled, link-local address is FE80::102
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:1:100::2, subnet is 2001:DB8:1:100::/64
  Joined group address(es):
    FF02::1
    FF02::2          www.passenterpriselabs.com
    FF02::1:FF00:2
    FF02::1:FF00:102
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent          www.passenterpriselabs.com
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 1000 milliseconds
  ND router advertisements live for 3 seconds
  ND advertised default router preference is Medium
  Hosts use stateless autoconfig for addresses.
chinesedumps-sw102#
```

Chinese_dumps_sw102#sh ipv6 int vlan 2001

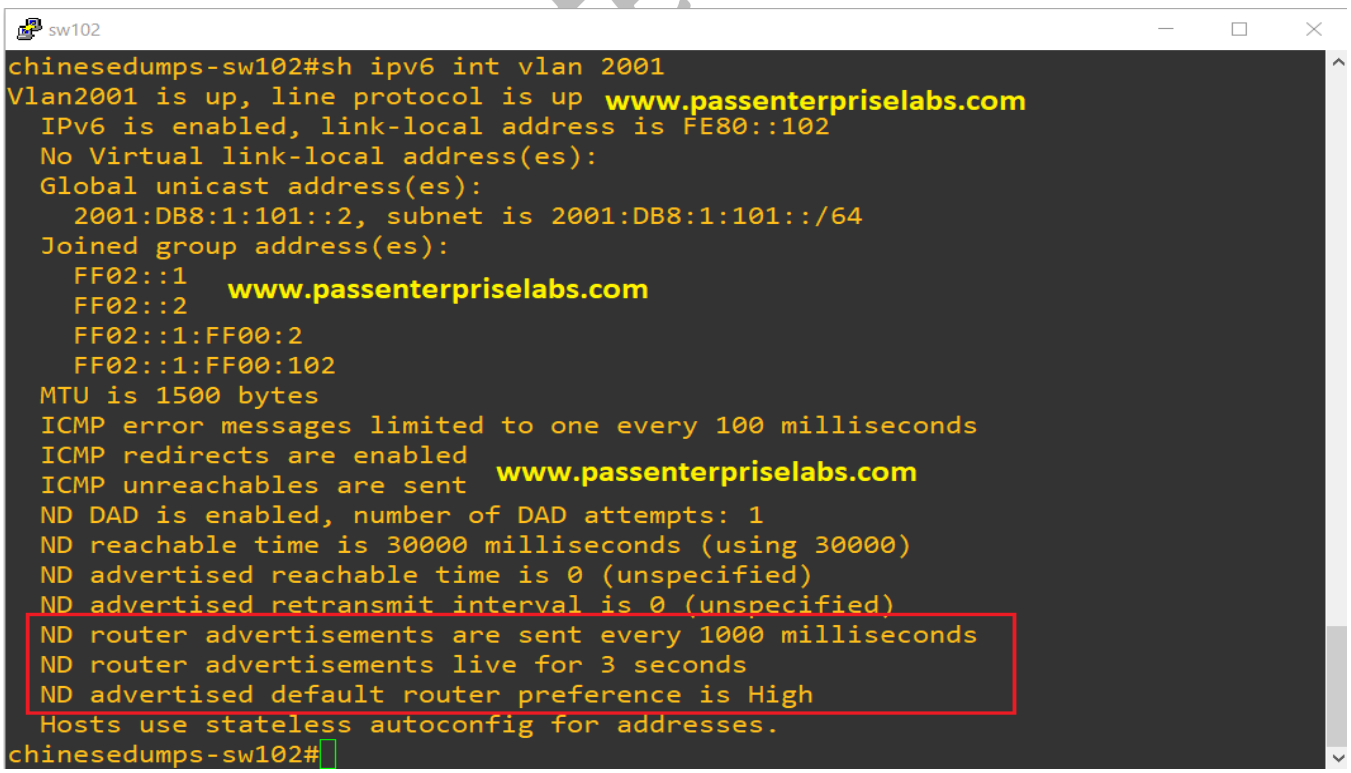
sw102

```
chinesedumps-sw102#sh ipv6 int vlan 2001
Vlan2001 is up, line protocol is up  www.passenterpriselabs.com
  IPv6 is enabled, link-local address is FE80::102
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:1:101::2, subnet is 2001:DB8:1:101::/64
  Joined group address(es):
    FF02::1
    FF02::2          www.passenterpriselabs.com
    FF02::1:FF00:2
    FF02::1:FF00:102
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent          www.passenterpriselabs.com
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 1000 milliseconds
  ND router advertisements live for 3 seconds
  ND advertised default router preference is High
  Hosts use stateless autoconfig for addresses.
chinesedumps-sw102#
```

中文_ dumps_sw102#sh ipv6接口vlan 2000

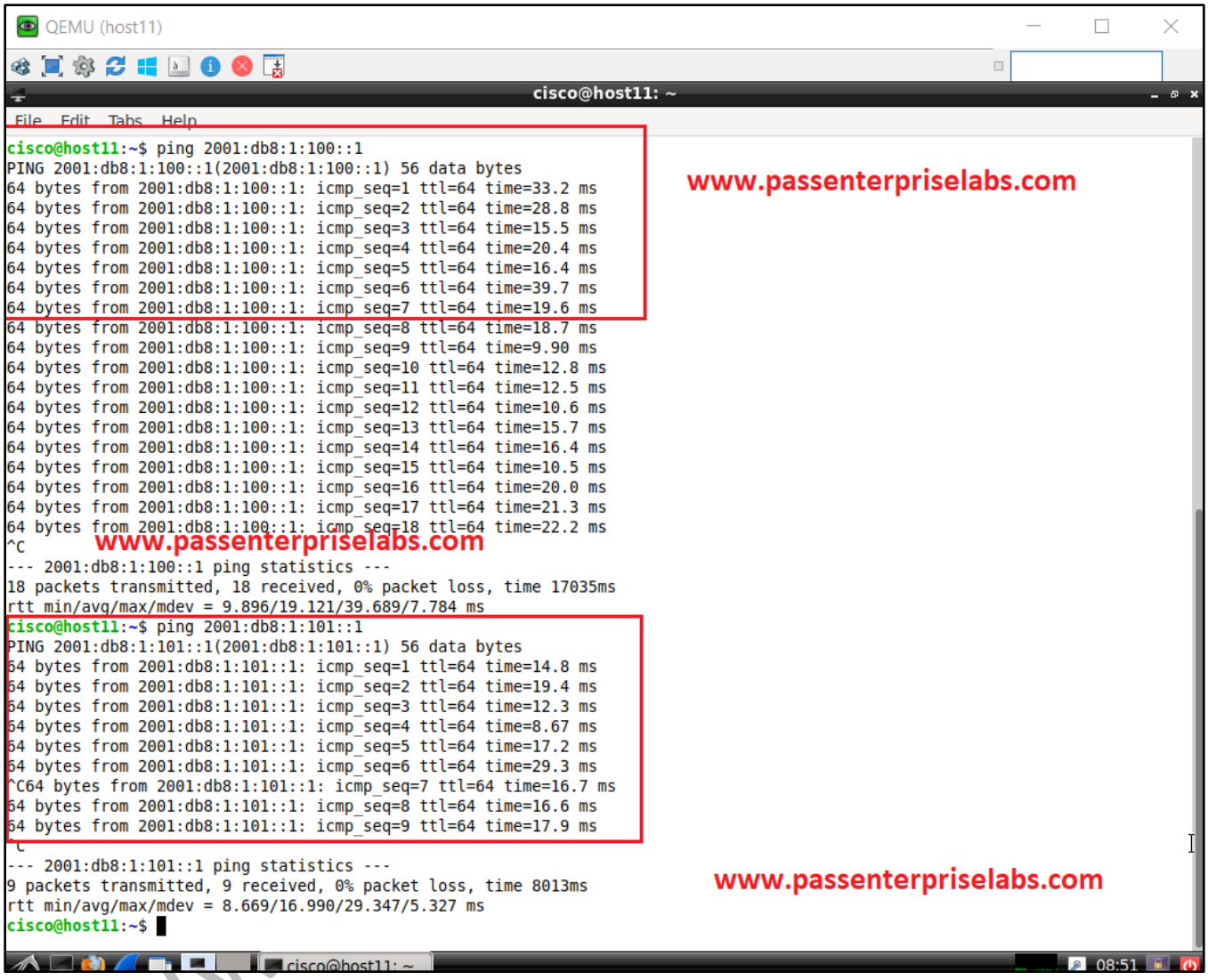


中文_ dumps_sw102#sh ipv6局域网2001



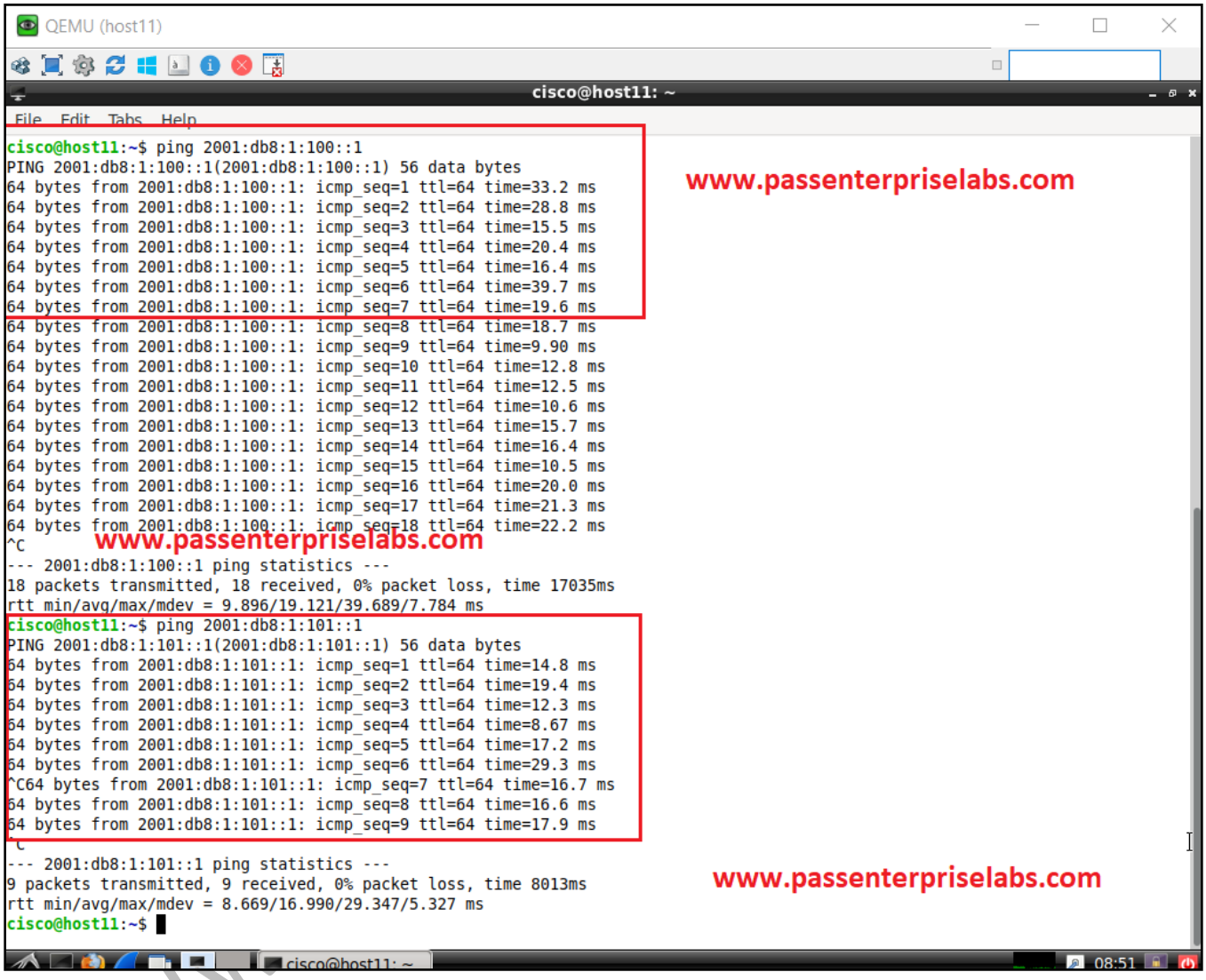
To verify the task:

- Ping from host-11 to both the ipv6 vlan 2000 and 2001(2001:db8:1:100::1 , 2001:db8:1:101::1)

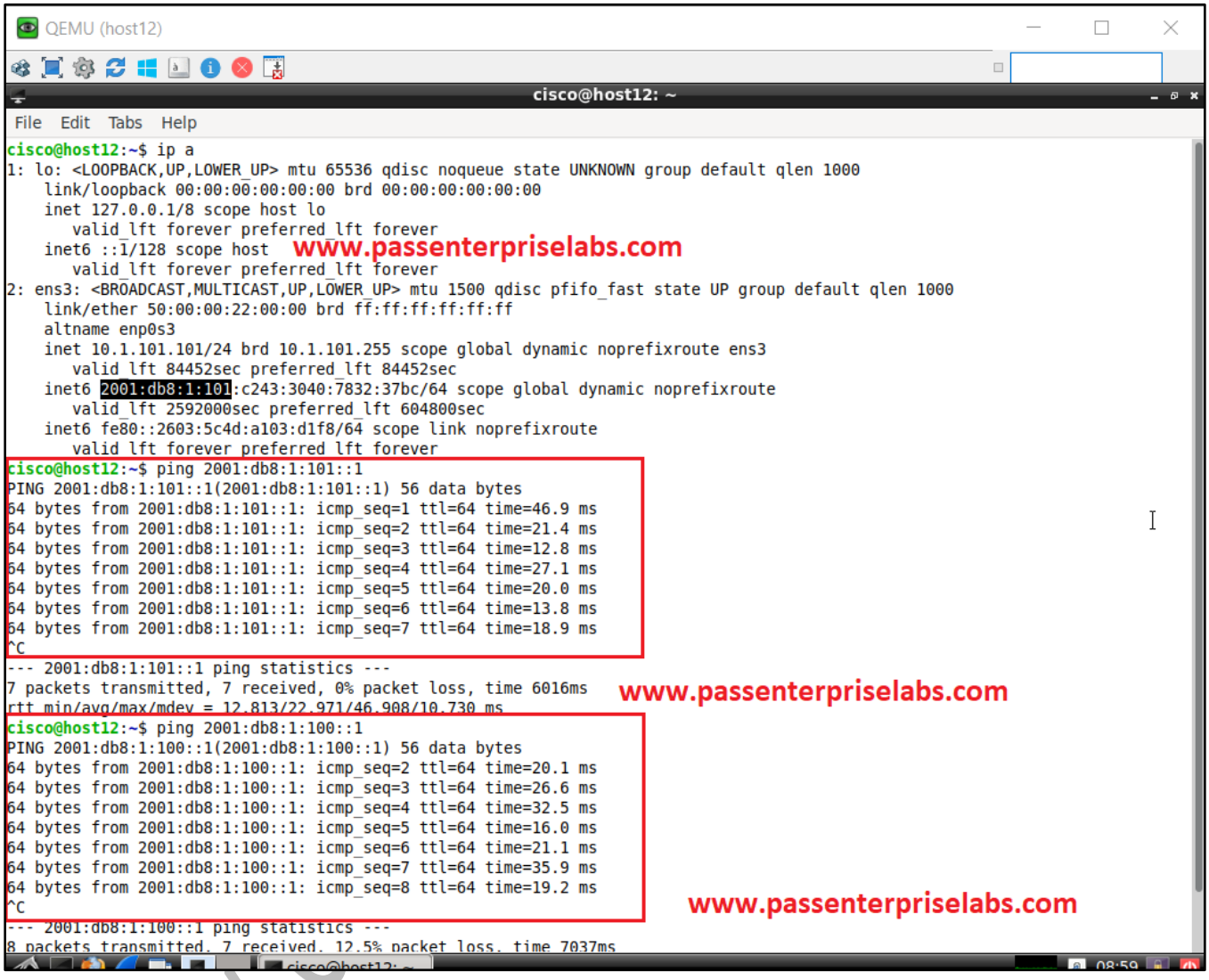


要验证任务，请执行以下操作：

- 从主机-11到ipv6 vlan 2000和2001的Ping（2001:db8:1:100:: 12001:db8:1:101:: 1）



- Ping from host-12 to both the ipv6 vlan 2000 and 2001(2001:db8:1:100::1 , 2001:db8:1:101::1)



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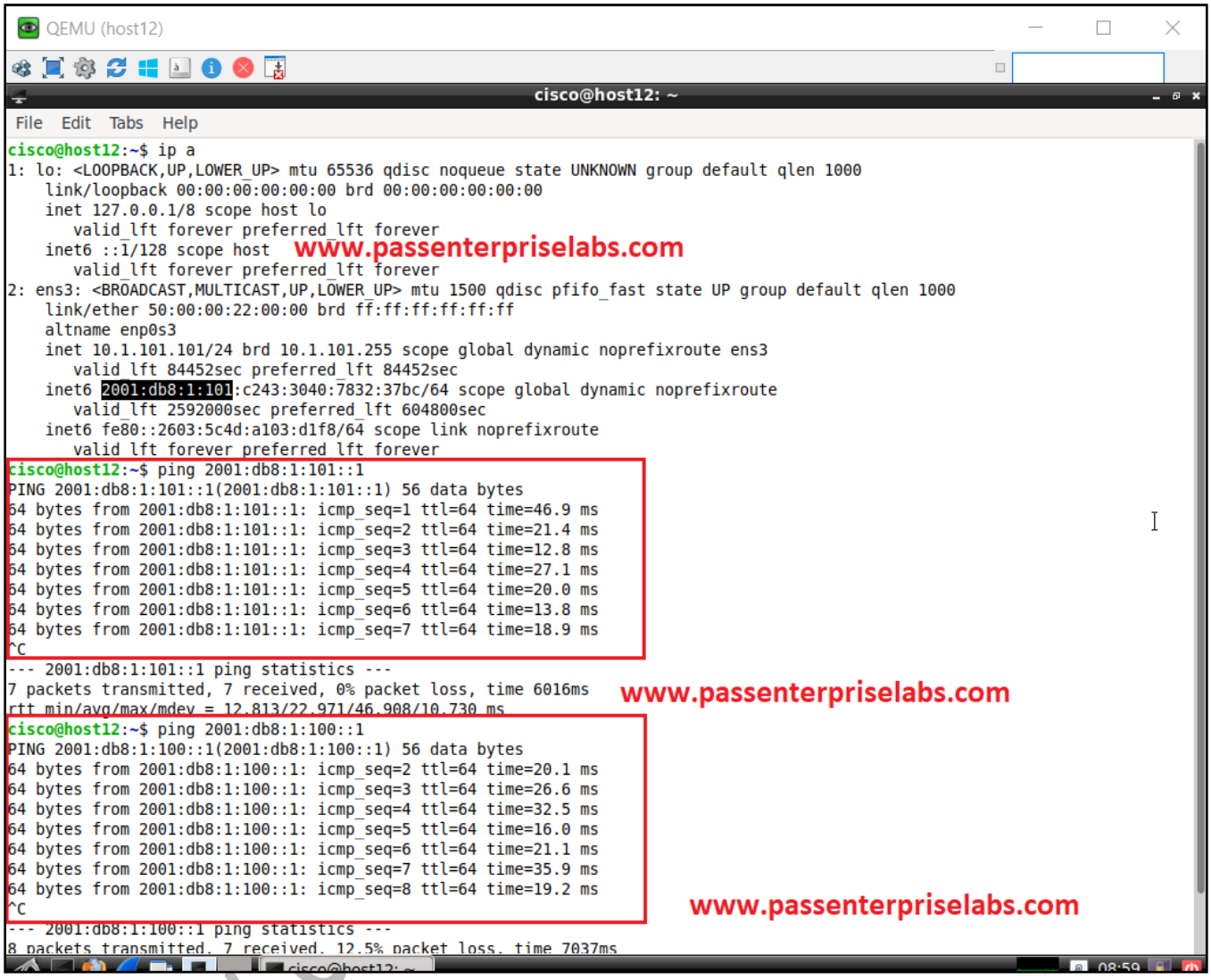
- Ping从主机-12到ipv6 vlan 2000和2001（2001:db8:1:100:: 12001:db8:1:101:: 1）



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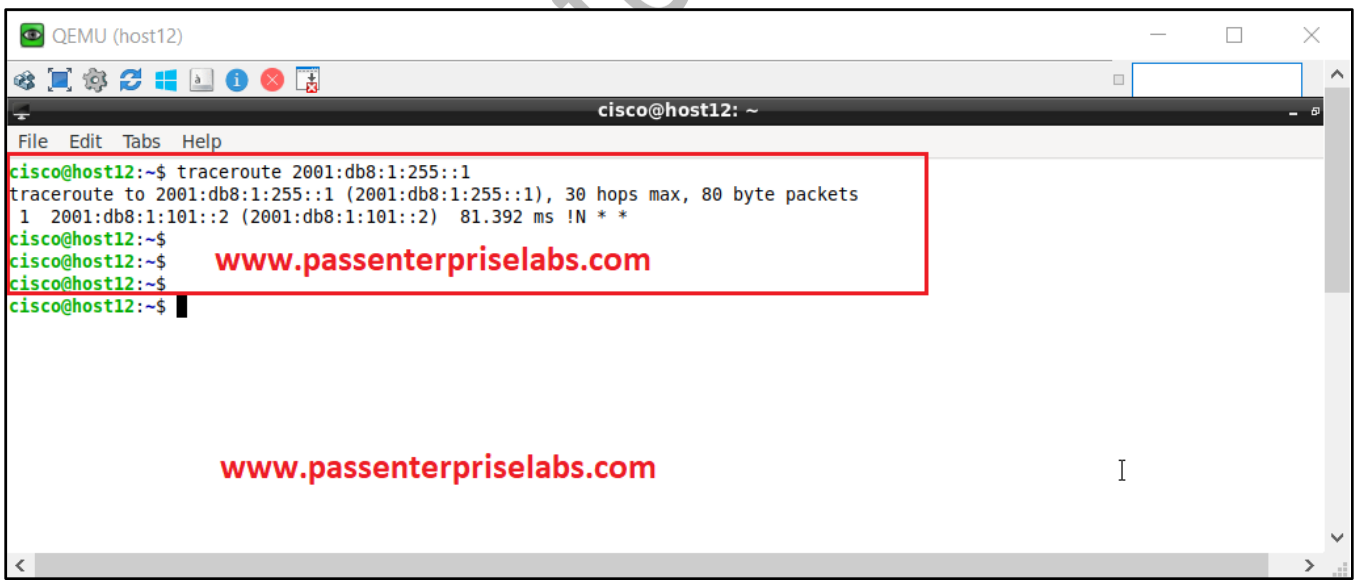
To verify the switching of gateway roles:

- Traceroute from both host-11 & host-12 to ipv6 loopback address of r11 (2001:db8:1:255::11)

In the above output you can see host11 is using sw101 – vlan 2000 to reach r11

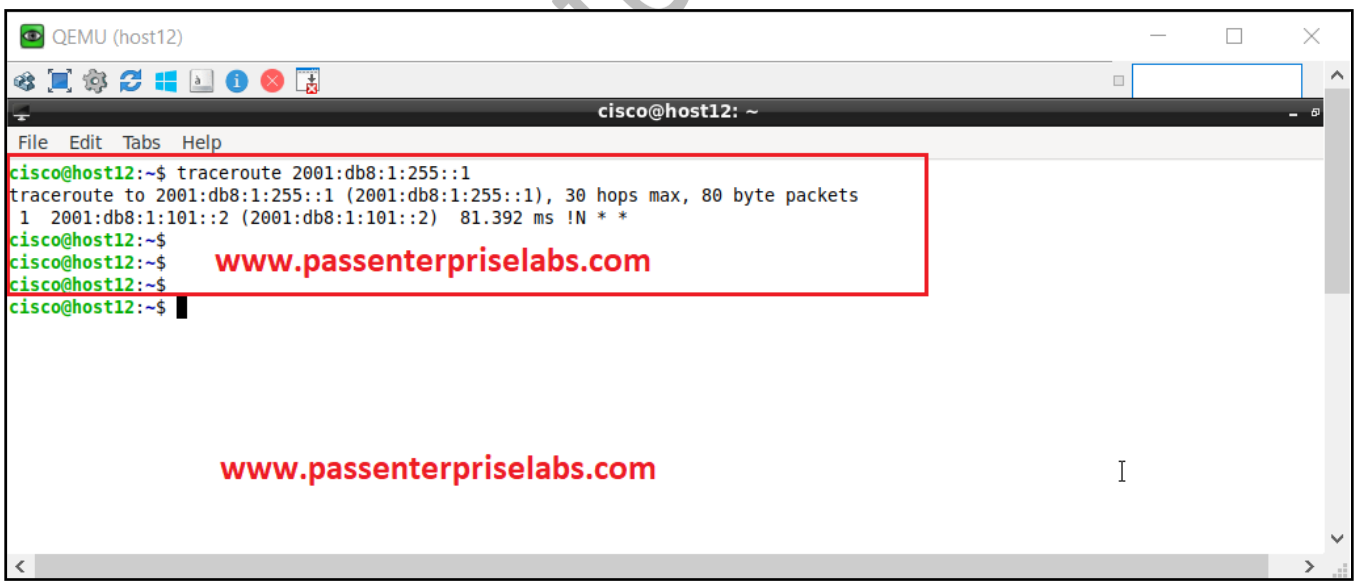


In the above output you can see host12 is using sw102 – vlan 2001 to reach r11

要验证网关角色的切换：

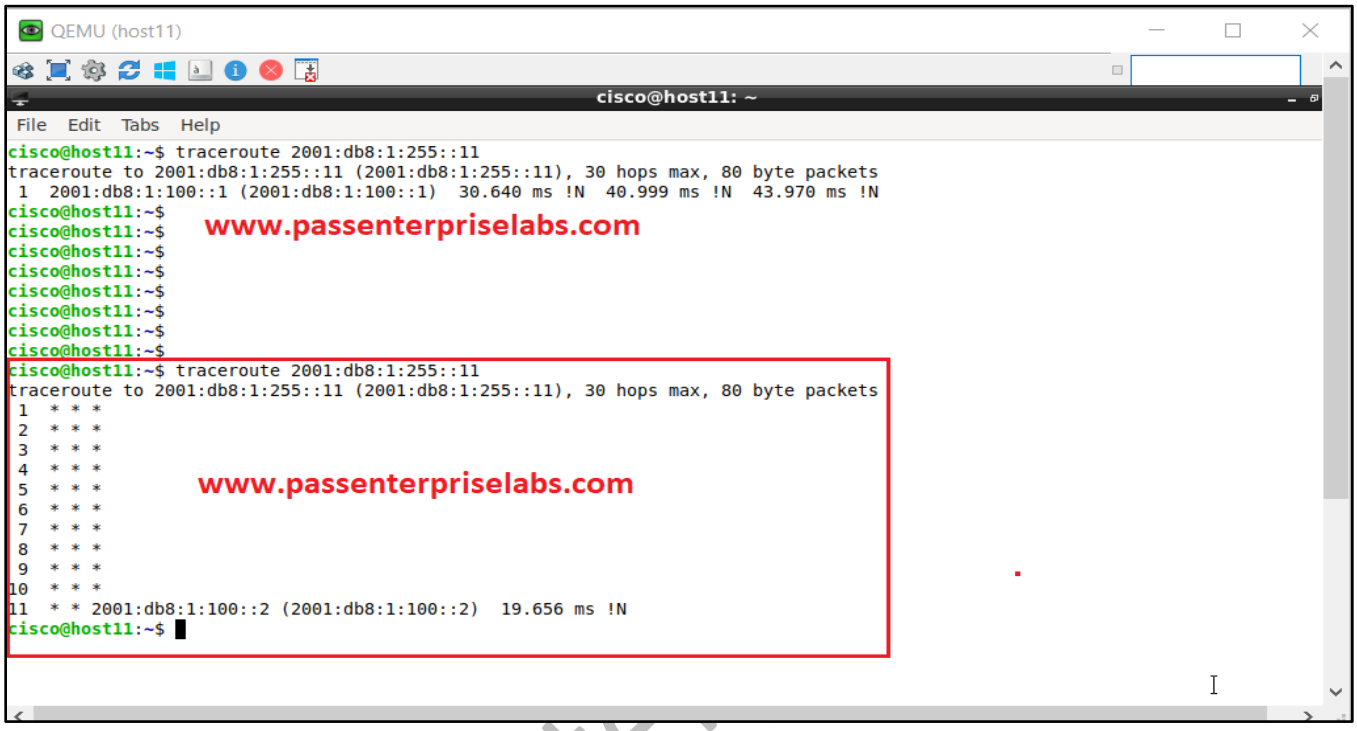
- 从主机-11和主机-12到r11的ipv6环回地址的跟踪路由（2001:db8:1:255::11）

在上面的输出中，您可以看到host11正在使用sw101-vlan 2000来达到r11



在上面的输出中，您可以看到host12正在使用sw102-vlan 2001来访问11

- After we shutdown the vlan 2000 on sw101 , the host11 will switch its gateway to sw102 –vlan 2000



- After we shutdown the vlan 2001 on sw102 , the host12 will switch its gateway to sw101 – vlan 2001

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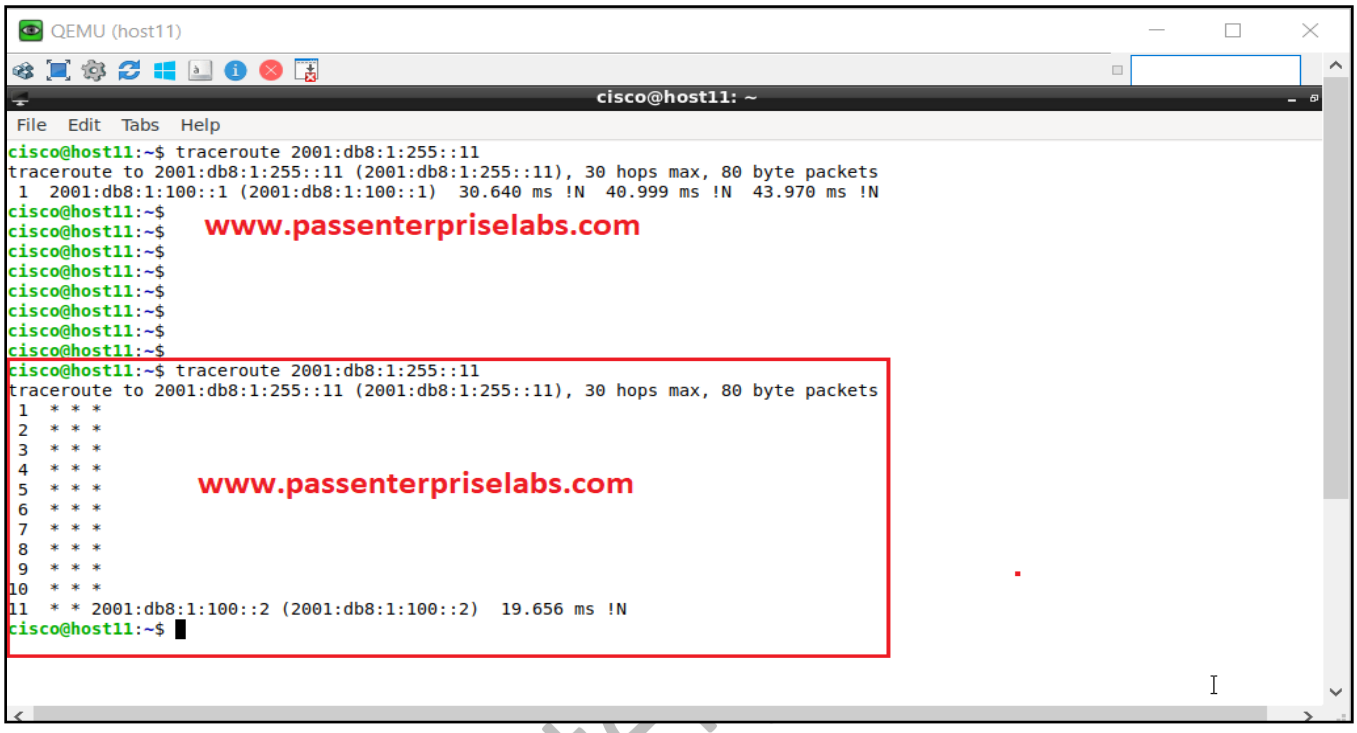
- 在我们关闭sw101上的vlan 2000后，主机11将其网关切换到sw102-vlan 2000



-在我们关闭sw102上的vlan 2001后，主机12将其网关切换到sw101-vlan 2001

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1.7: IPv6 EIGRP in HQ

In HQ, enable EIGRP for IPv6 on r11, r12, sw101 and sw102 according to theserequirements:

- Use process name “ccie”(without the quotes) and AS number 65001.
- Do not configure any additional IPv6 addresses.
- IPv6 EIGRP may form adjacencies only over the physical Layer3 interfaces between r11, r12, sw101 and sw102.
- Prevent IPv6 EIGRP from automatically running on, or advertising attached prefixes from, new IPv6-enabled interfaces in the future unless allowed explicitly.
- Ensure that the attached IPv6 prefixes on SVIs Vlan2000 and Vlan2001 onsw101 and sw102 are advertised in IPv6 EIGRP and learned on r11 and r12.
- No route filtering is allowed to accomplish this entire task.

2 Points

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1.7： 总部IPv6 EIGRP

在总部，根据以下要求在r11、r12、sw101和sw102上启用IPv6的EIGRP：

- 使用流程名称“ccie”（不带引号）和AS编号65001。
- 不要配置任何其他IPv6地址。
- IPv6 EIGRP可能仅在r11、r12、sw101和sw102之间的物理层3接口上形成邻接。
- 除非明确允许，否则防止IPv6 EIGRP将来在启用IPv6的新接口上自动运行，或从中通告附加前缀。
- 确保SVI Vlan2000和Vlan2001上的IPv6前缀在IPv6 EIGRP中通告，并在r11和r12上学习。
- 不允许路由过滤来完成整个任务。

2分

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Solution:

Chinese_dumps_sw101:

```
sw101>en
sw101#conf t
sw101(config)#router eigrp ccie
sw101(config-router)#address-family ipv6 unicast AS 65001
sw101(config-router-af)#af-interface default
sw101(config-router)#shutdown
sw101(config-router)#exit-af-interface
sw101(config-router)#af-interface vlan 2000
sw101(config-router-interface)#no shutdown
sw101(config-router-interface)#passive
sw101(config-router-interface)#af-interface vlan 2001
sw101(config-router-interface)#no shutdown
sw101(config-router-interface)#passive
sw101(config-router-interface)#af-interface g0/0
sw101(config-router-interface)#no shutdown
sw101(config-router-interface)#af-interface g0/1
sw101(config-router-interface)#no shutdown
```

Chinese_dumps_sw102:

```
sw102>en
sw102#conf t
sw102(conifg)#router eigrp ccie
sw102(config-router)#address-family ipv6 unicast AS 65001
sw102(config-router-af)#af-interface default
sw102(conifg-router-af)#shutdown
sw102(conifg-router-af)#af-interface vlan 2000
sw102(conifg-router-af)#no shutdown
sw102(conifg-router-af)#passive
sw102(conifg-router-af)#af-interface vlan 2001
sw102(conifg-router-af)#no shutdown
sw102(conifg-router-af)#passive
sw102(conifg-router-af)#af-interface g0/0
sw102(conifg-router-af)#no shutdown
sw102(conifg-router-af)#af-interface g0/1
sw102(conifg-router-af)#no shutdown
```

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解决方案：

中文_垃圾_ sw101：

```
sw101>en-
sw101#配置
文件
sw101（配置）#路由器eigrp-ccie
sw101（配置路由器）#地址系列ipv6单播AS 65001
sw101（设置路由器af）#af接口默认值
sw101（配置路由器）#关闭sw101（设置路
由器）#退出af接口sw101
sw101（配置路由器接口）#af接口vlan 2001
sw101
sw101（配置路由器接口）#af接口g0/0 sw101
```

中文_垃圾_ sw102：

```
sw102>en-
sw102#配置
文件
sw102（conifg）#路由器eigrp-ccie
sw102（配置路由器）#地址系列ipv6单播AS 65001
sw102（设置路由器af）#af接口默认值
sw102（conifg路由器af）#关闭sw102（
coifg路由器av）#af接口vlan 2000 sw102
sw102（conifg路由器af）/af接口vlan
2001 sw102
sw102（conifg路由器af）#af接口g0/0
sw102（coifg路由器av）#无关机sw102
```

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Chinese_dumps_r11:

```
r11>en
r11#conf t
r11(config)#router eigrp ccie
r11(config-router)#address-family ipv6 unicast AS 65001
r11(config-router-af)#af-interface default
r11(config-router-af)#shutdown
r11(config-router-af)#exit-af-interface

r11(config-router)#af-interface g0/2
r11(config-router-af)#no shutdown
r11(config-router-af)#af-interface g0/1
r11(config-router-af)#no shutdown
r11(config-router-af)#exit-af-interface

r11(config-router)#af-interface g0/3
r11(config-router-af)#no shutdown
```

Chinesedumps.com-r12:

```
r12>en
r12#conf t
r12(config)#router eigrp ccie
r12(config-router)#address-family ipv6 unicast AS 65001
r12(config-router-af)#af-interface default
r12(config-router-af)#shutdown
r12(config-router-af)#exit-af-interface

r12(config-router)#af-interface g0/2
r12(config-router-af)#no shutdown
r12(config-router)#af-interface g0/1
r12(config-router-af)#no shutdown
r12(config-router)#af-interface g0/3
r12(config-router-af)#no shutdown
```

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中文_dumps_r11：

r11>和
r11#conf t
r11（配置）#路由器eigrp-ccie
r11（配置路由器）#地址系列ipv6单播AS 65001
r11（设置路由器af）#af接口默认值
r11（config-router-af）#关闭
r11（config-router-af）#关闭接
口

r11（配置路由器）#af-interface
g0/2 r11（配置路由器）#no
shutdown r11（配置路由器）#af-
interface g0/1 r11（配置路由器
）#no shutdown r11（配置路由器
）#exit-af-interface

r11（配置路由器）#af接口g0/3
r11（设置路由器af）#不关机

中文教育网-r12：

r12>和
r12#conf t
r12（配置）#路由器eigrp-ccie
r12（配置路由器）#地址系列ipv6单播AS 65001
r12（设置路由器af）#af接口默认值
r12（config-router-af）#关闭
r12（config-router-af）#关闭接
口

r12（配置路由器）#af接口g0/2
r12（设置路由器af）#不关机r12

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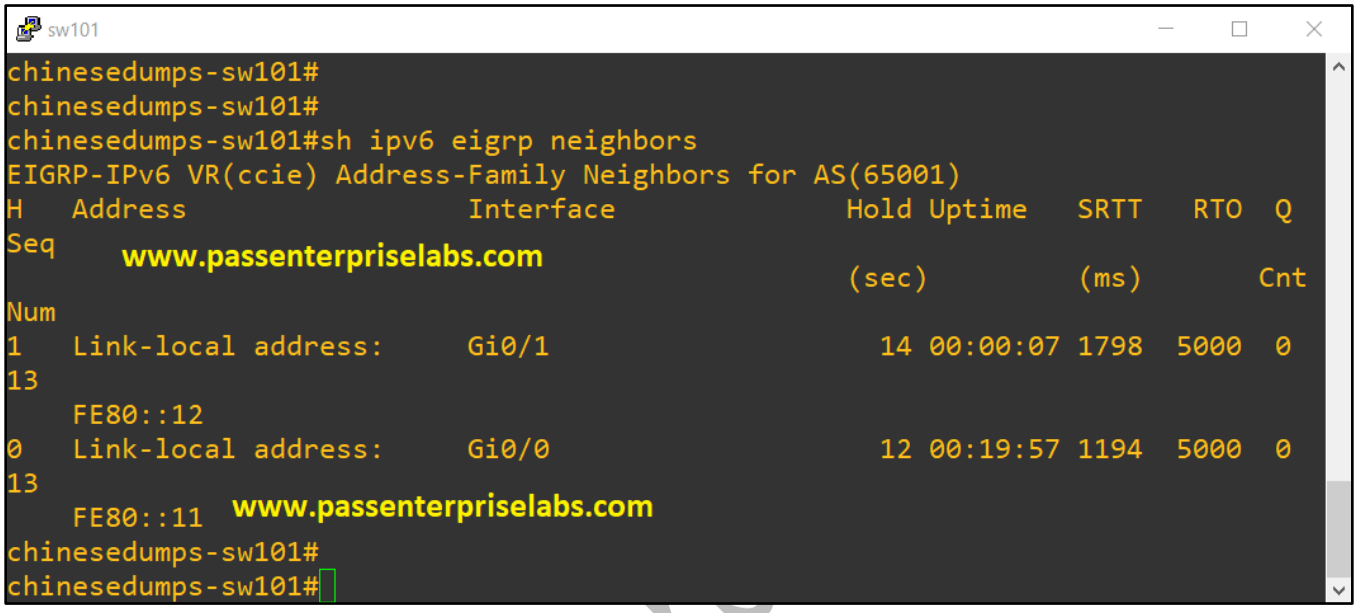
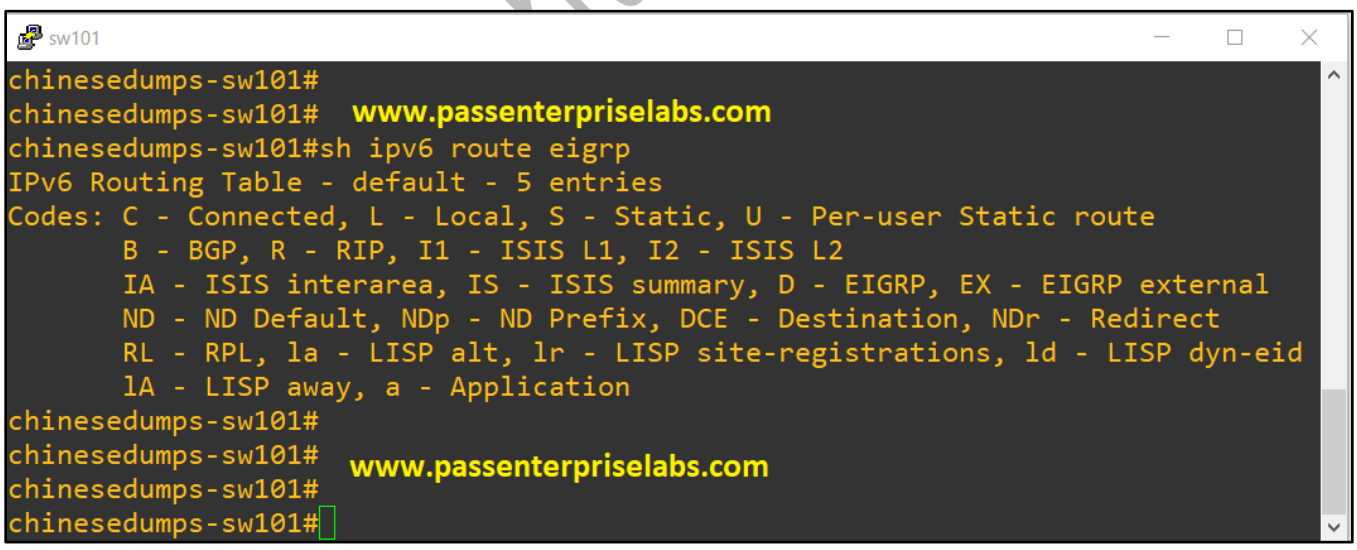
Verification:

Chinese_dumps_sw101:show ipv6 eigrp neighbor



Chinese_dumps_sw101:show ipv6 route eigrp

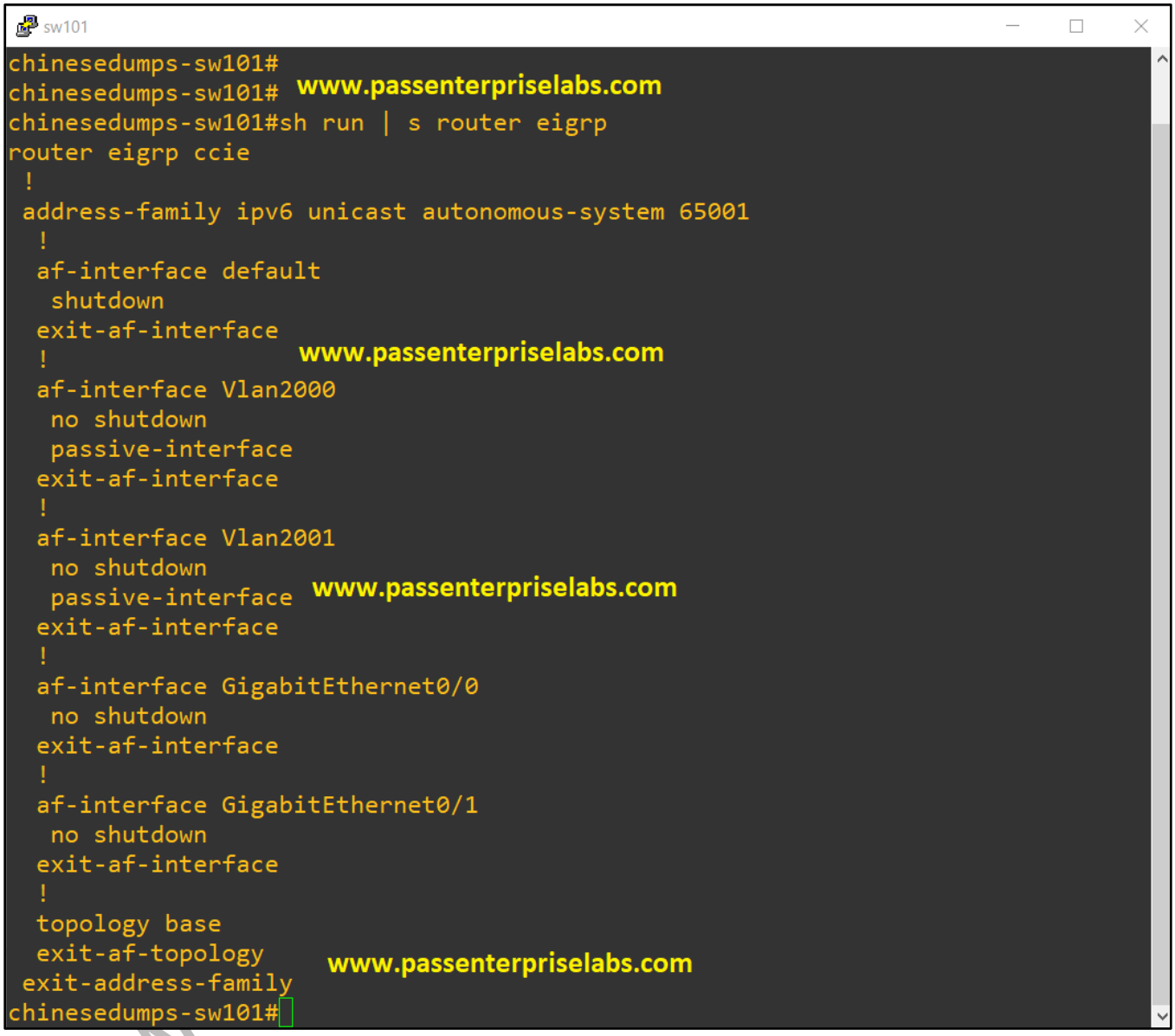


验证：

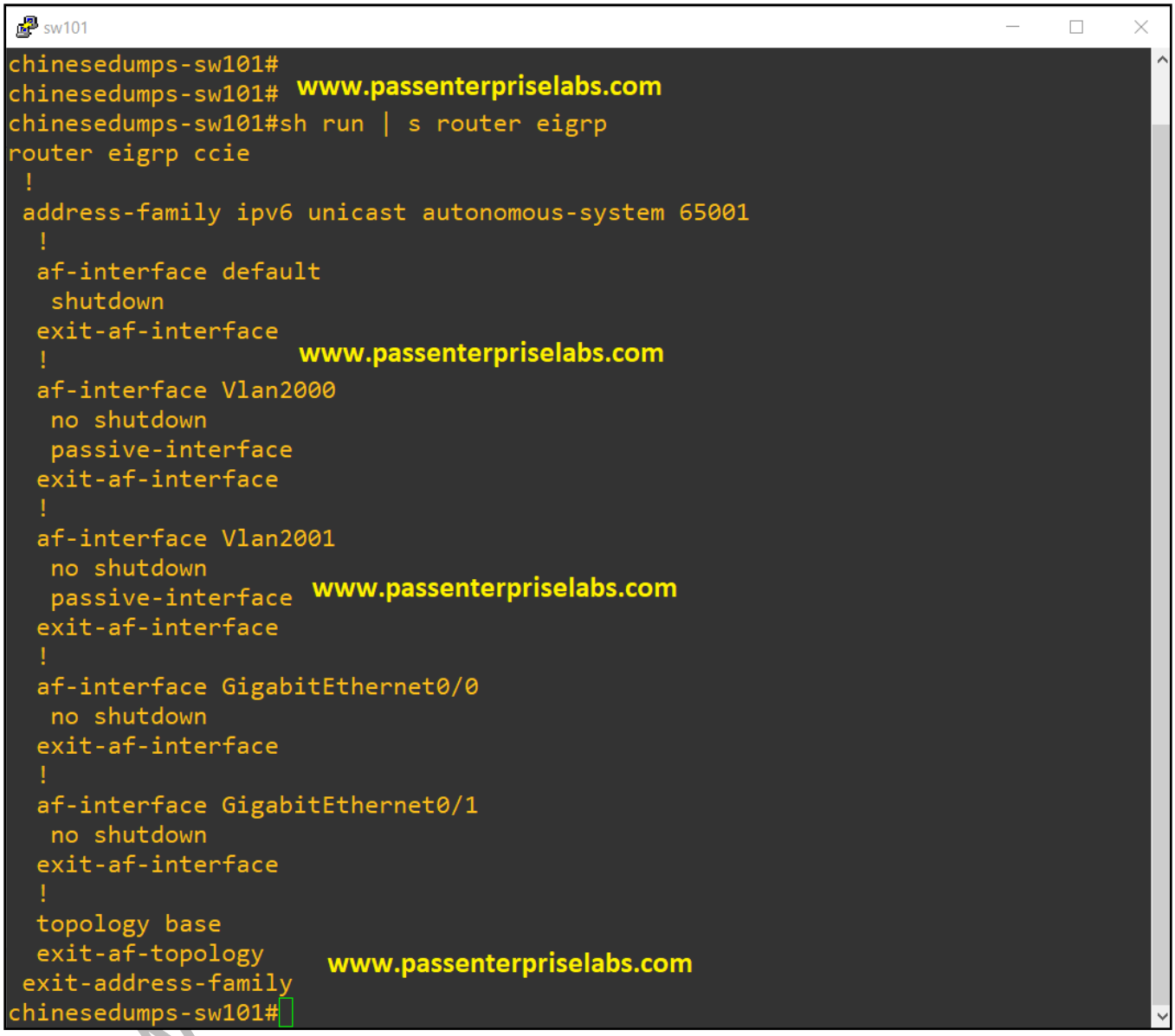
中文_dumps_sw101：显示ipv6 eigrp邻居

中文_dumps_sw101：显示ipv6路由eigrp

Chinese_dumps_sw101:show run | s router eigrp

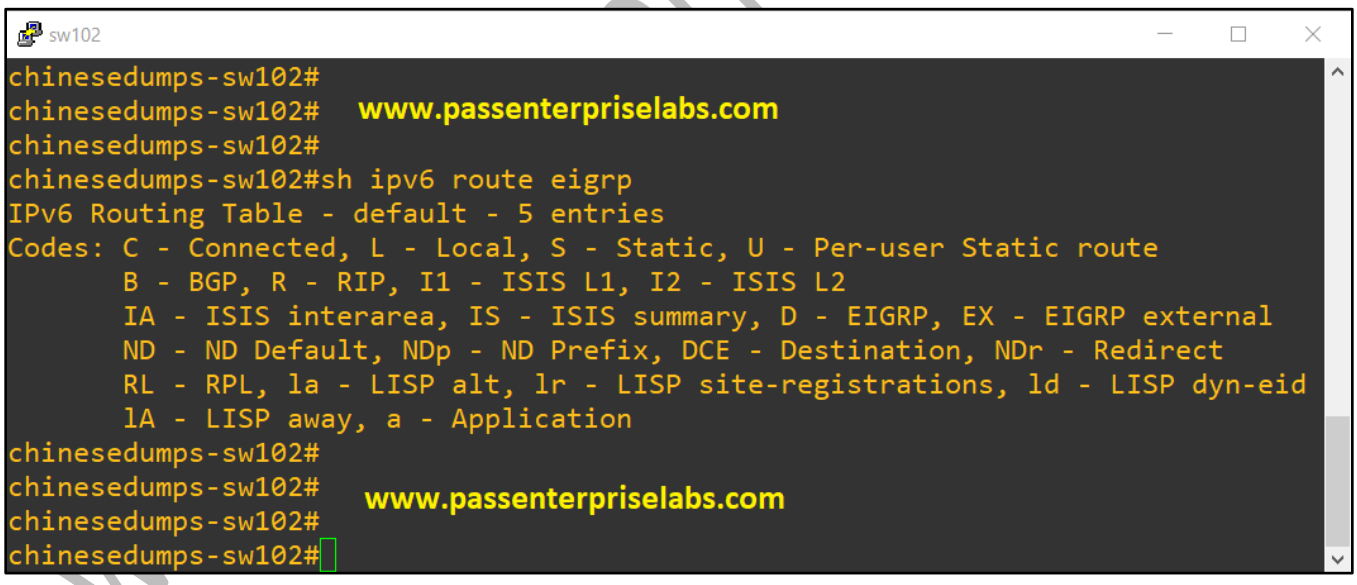


Chinese_dumps_sw101：显示运行的路由器eigrp

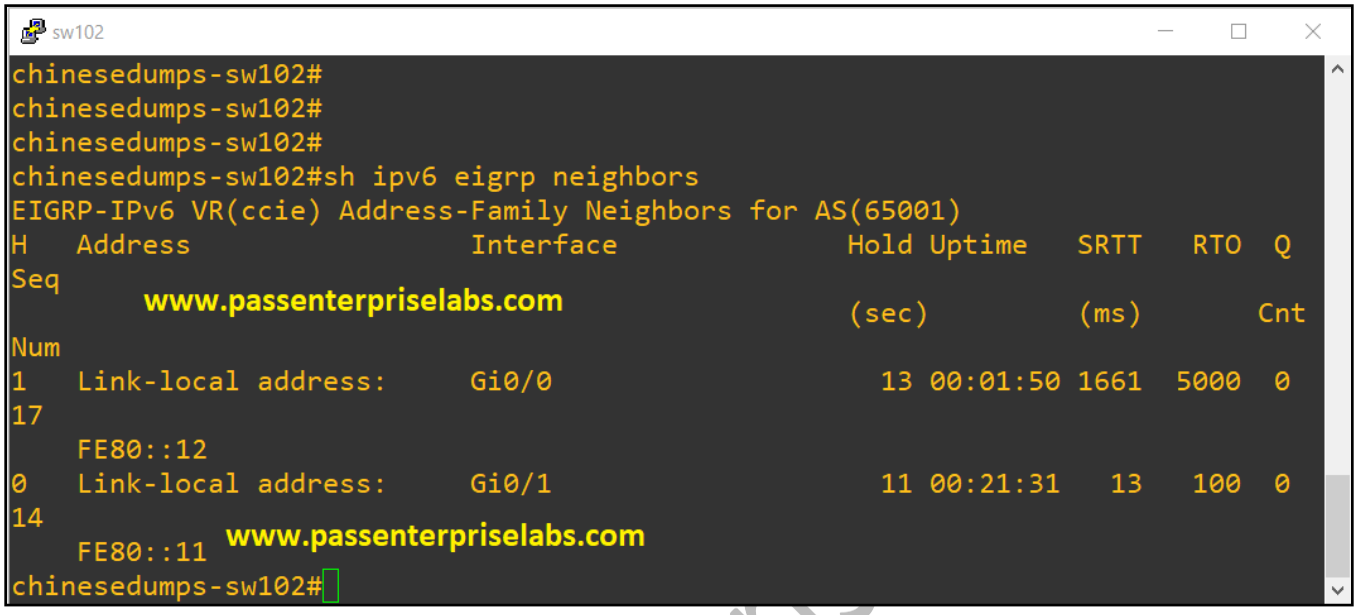


Chinese_dumps_sw102:show ipv6 eigrp neighbor

Chinese_dumps_sw101:show ipv6 route eigrp

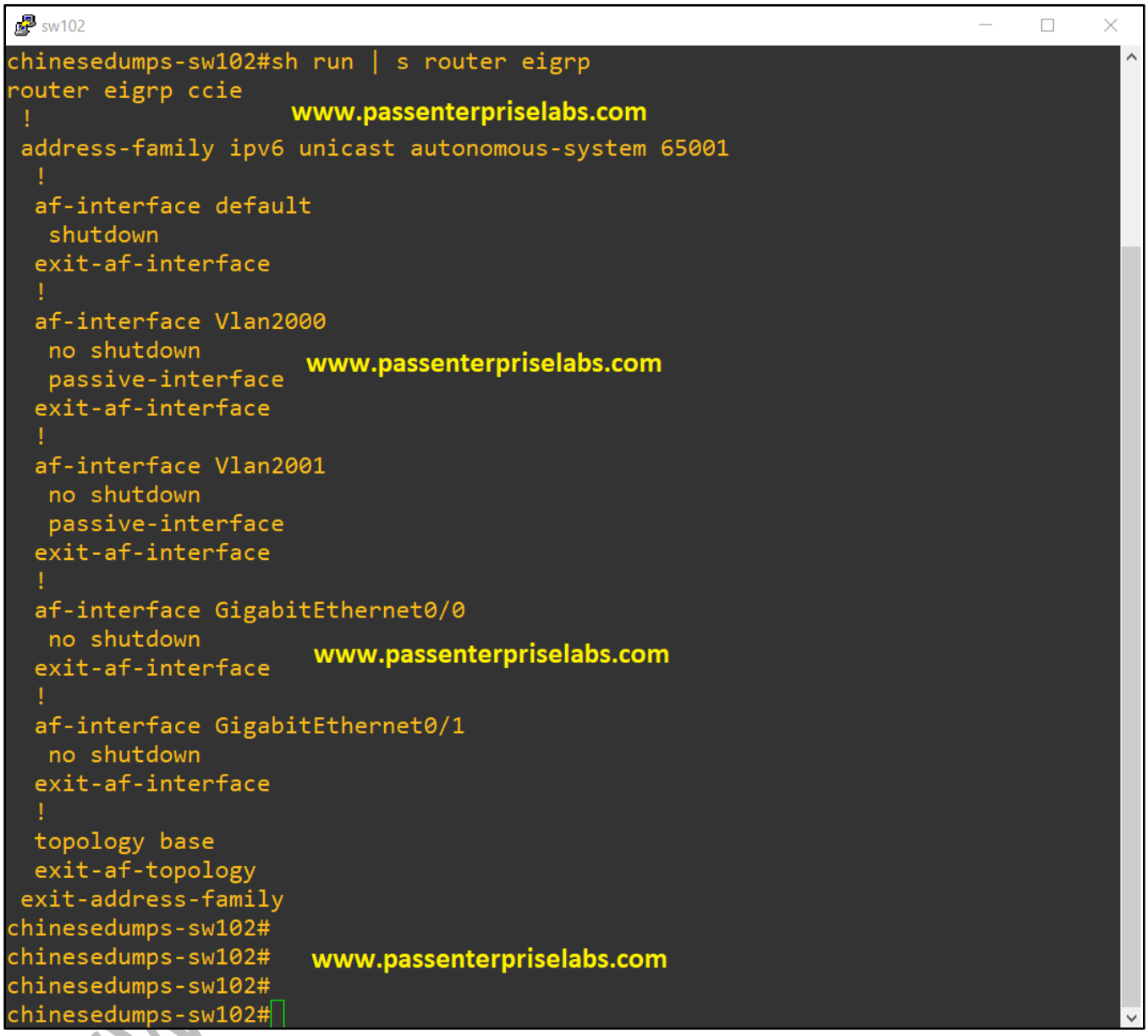


Chinese_dumps_sw102：显示ipv6 eigrp邻居

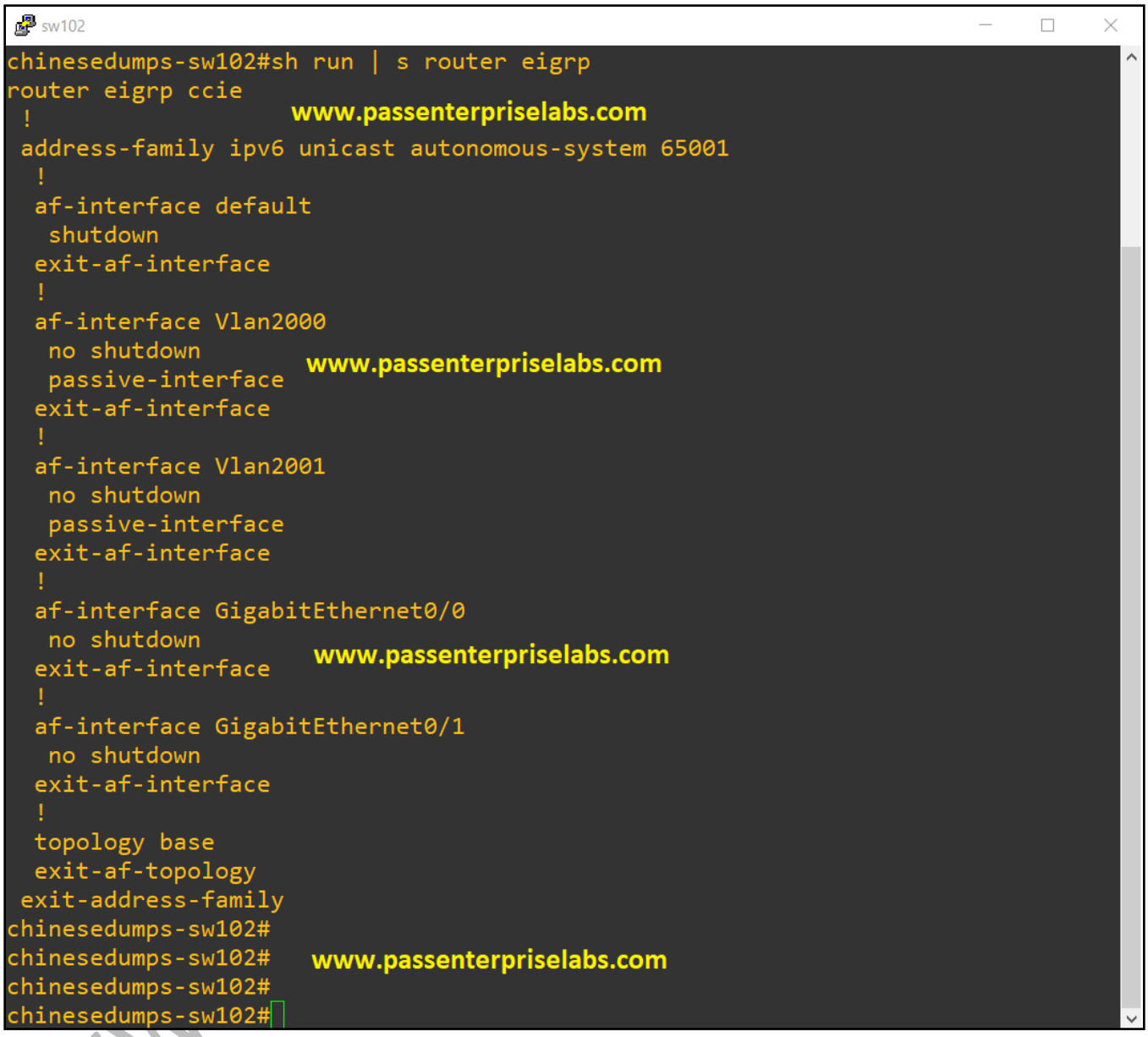


中文_dumps_sw101：显示ipv6路由eigrp

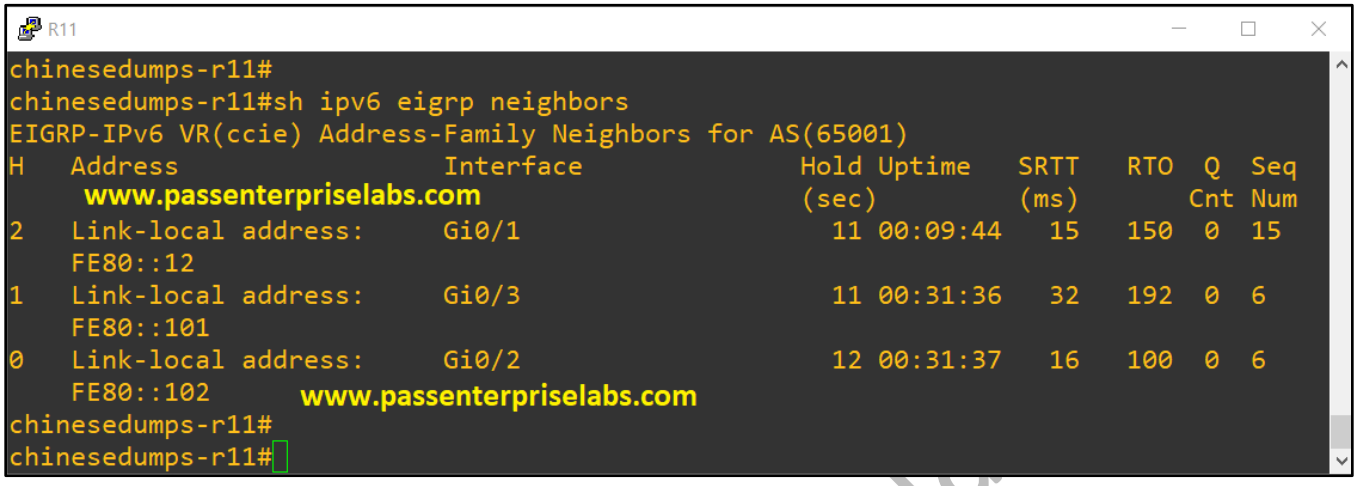
Chinese_dumps_sw101:show run | s router eigrp



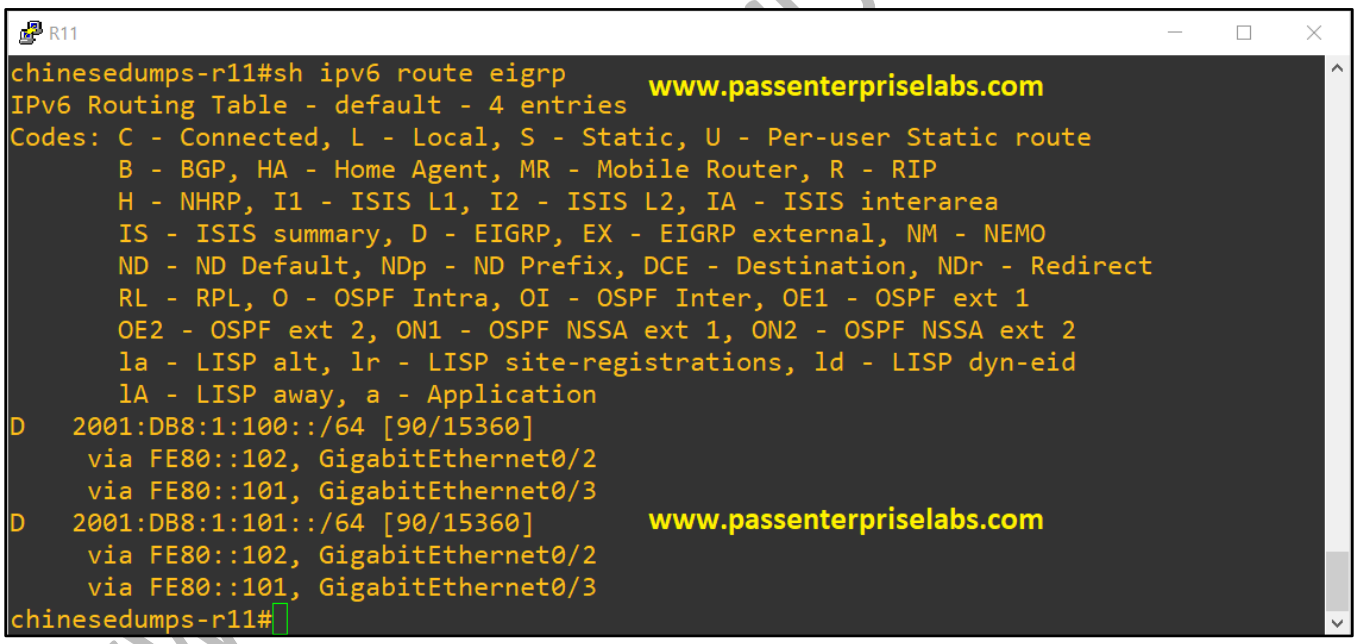
Chinese_dumps_sw101：显示运行的路由器eigrp



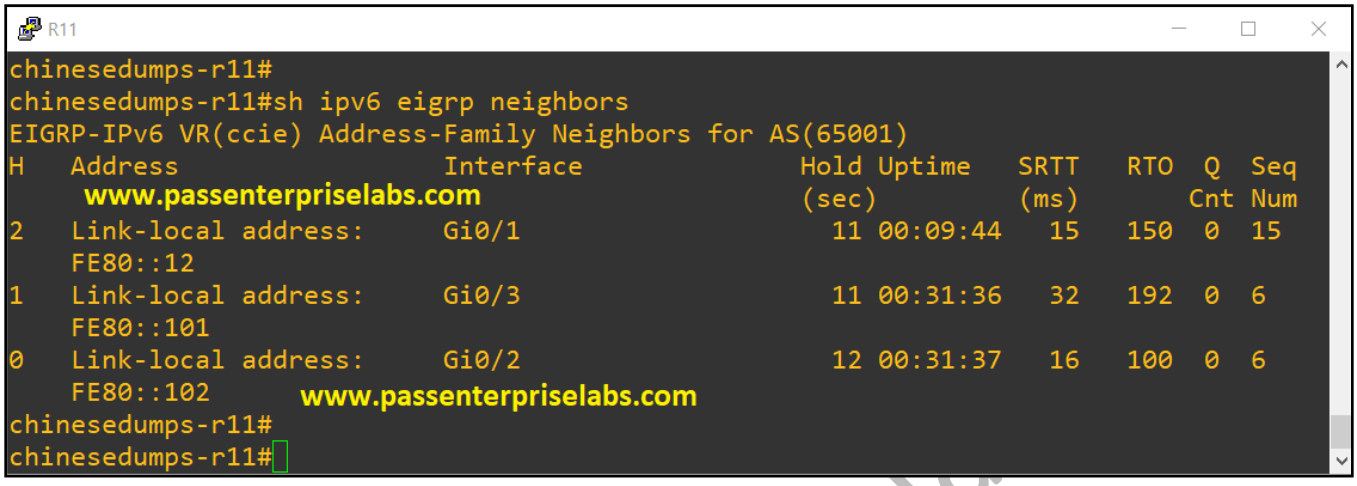
Chinesedumps.com-r11:show ipv6 eigrp neighbor



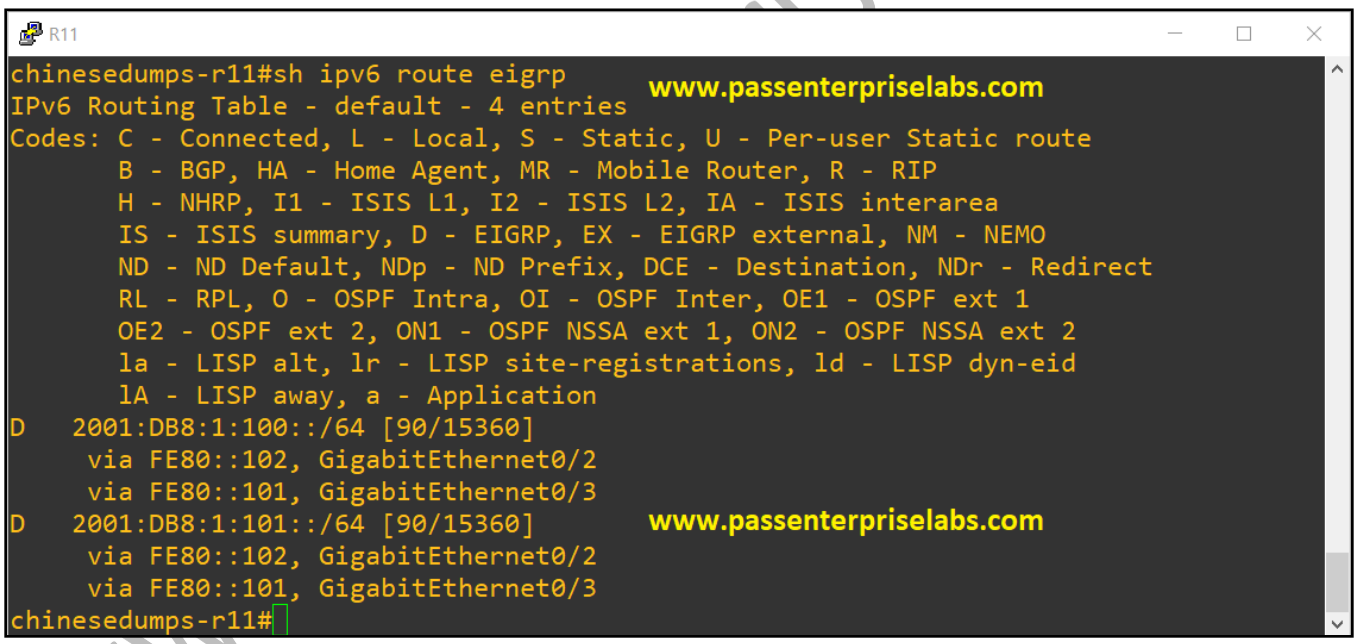
Chinesedumps.com-r11:show ipv6 route eigrp



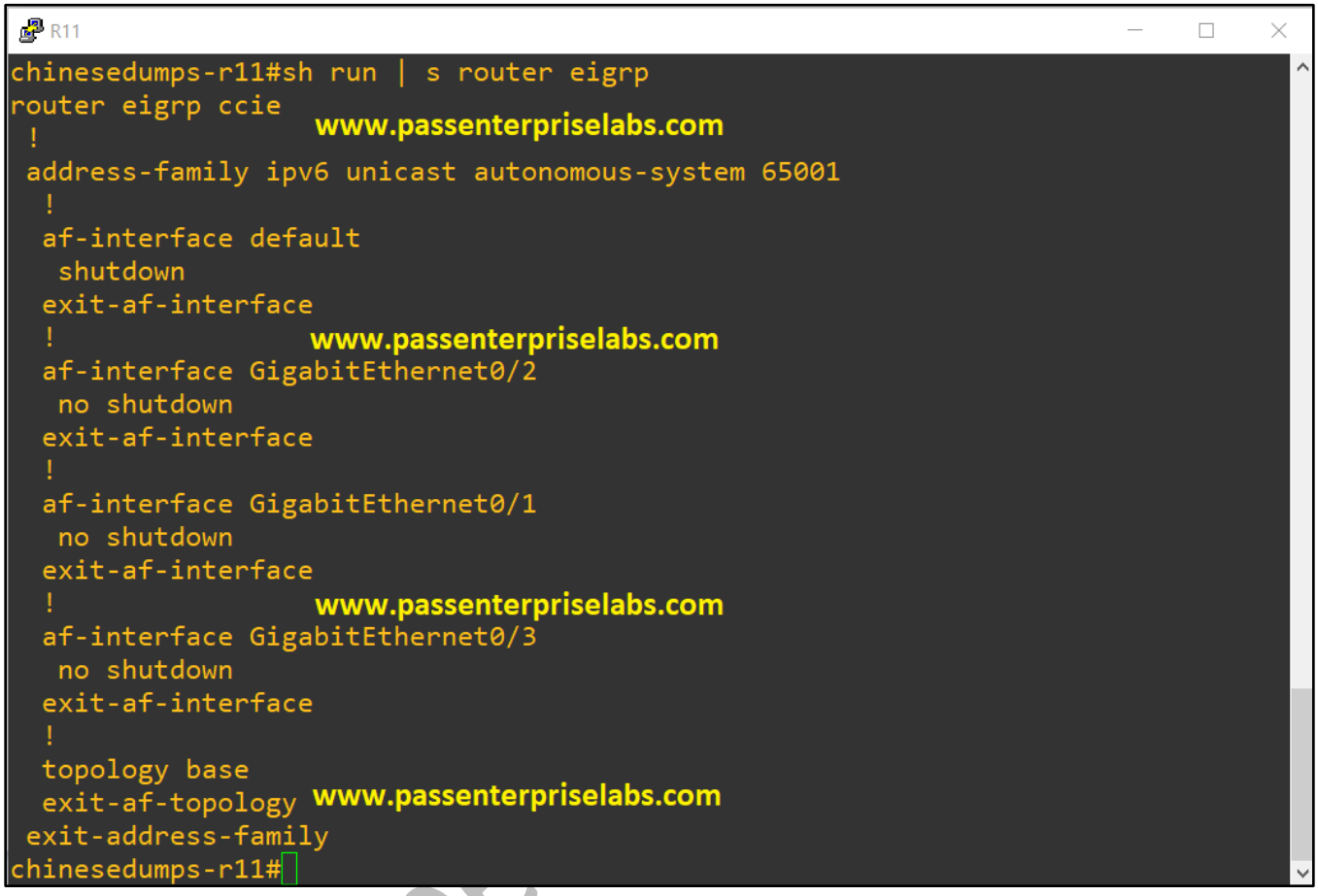
中文教程com-r11：显示ipv6 eigrp邻居



中文教程com-r11：显示ipv6路由eigrp



Chinesedumps.com-r11:show run | s router eigrp



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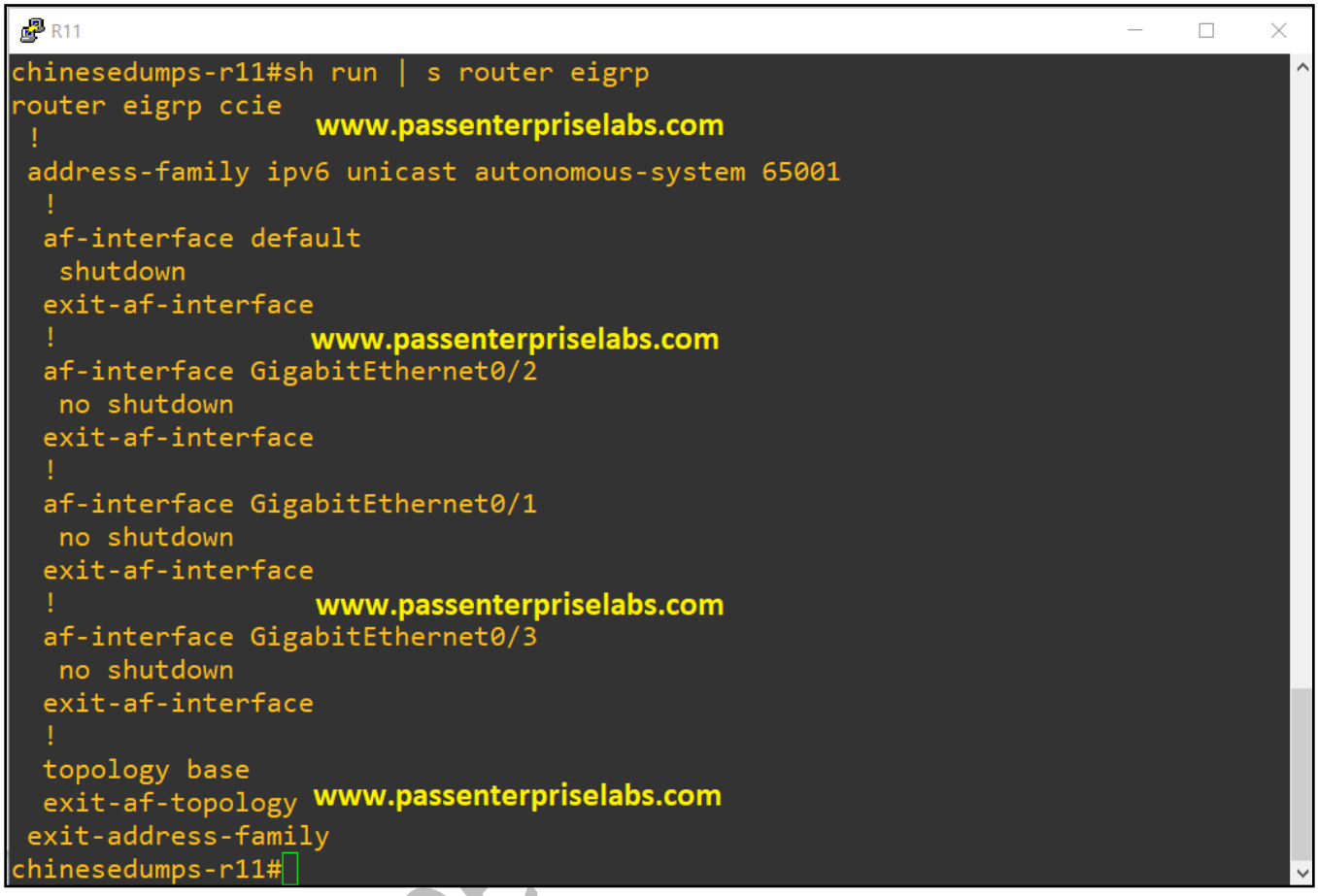
Chinese edumps.com-r11：显示运行的路由器eigrp



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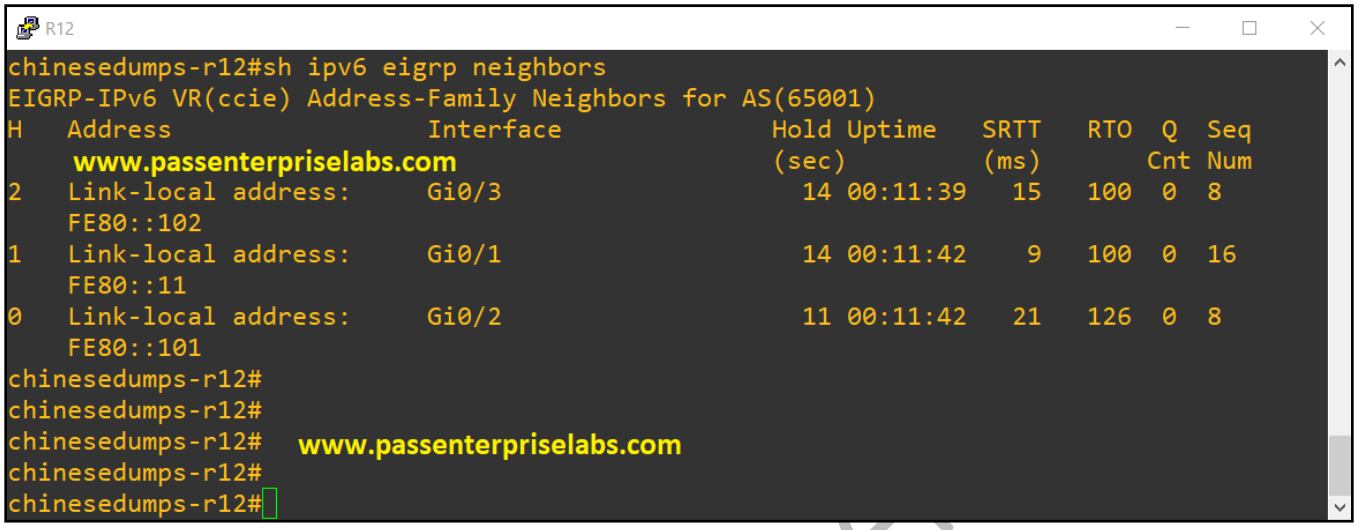
Chinesedumps.com-r12:show ipv6 eigrp neighbor

```
R12
chinesedumps-r12#sh ipv6 eigrp neighbors
EIGRP-IPv6 VR(ccie) Address-Family Neighbors for AS(65001)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
    www.passenterpriselabs.com                     (sec)         (ms)       Cnt Num
2   Link-local address:     Gi0/3                    14 00:11:39   15   100  0  8
    FE80::102
1   Link-local address:     Gi0/1                    14 00:11:42    9   100  0  16
    FE80::11
0   Link-local address:     Gi0/2                    11 00:11:42   21   126  0  8
    FE80::101
chinesedumps-r12#
chinesedumps-r12#
chinesedumps-r12#   www.passenterpriselabs.com
chinesedumps-r12#
chinesedumps-r12#
```

Chinesedumps.com-r11:show ipv6 route eigrp

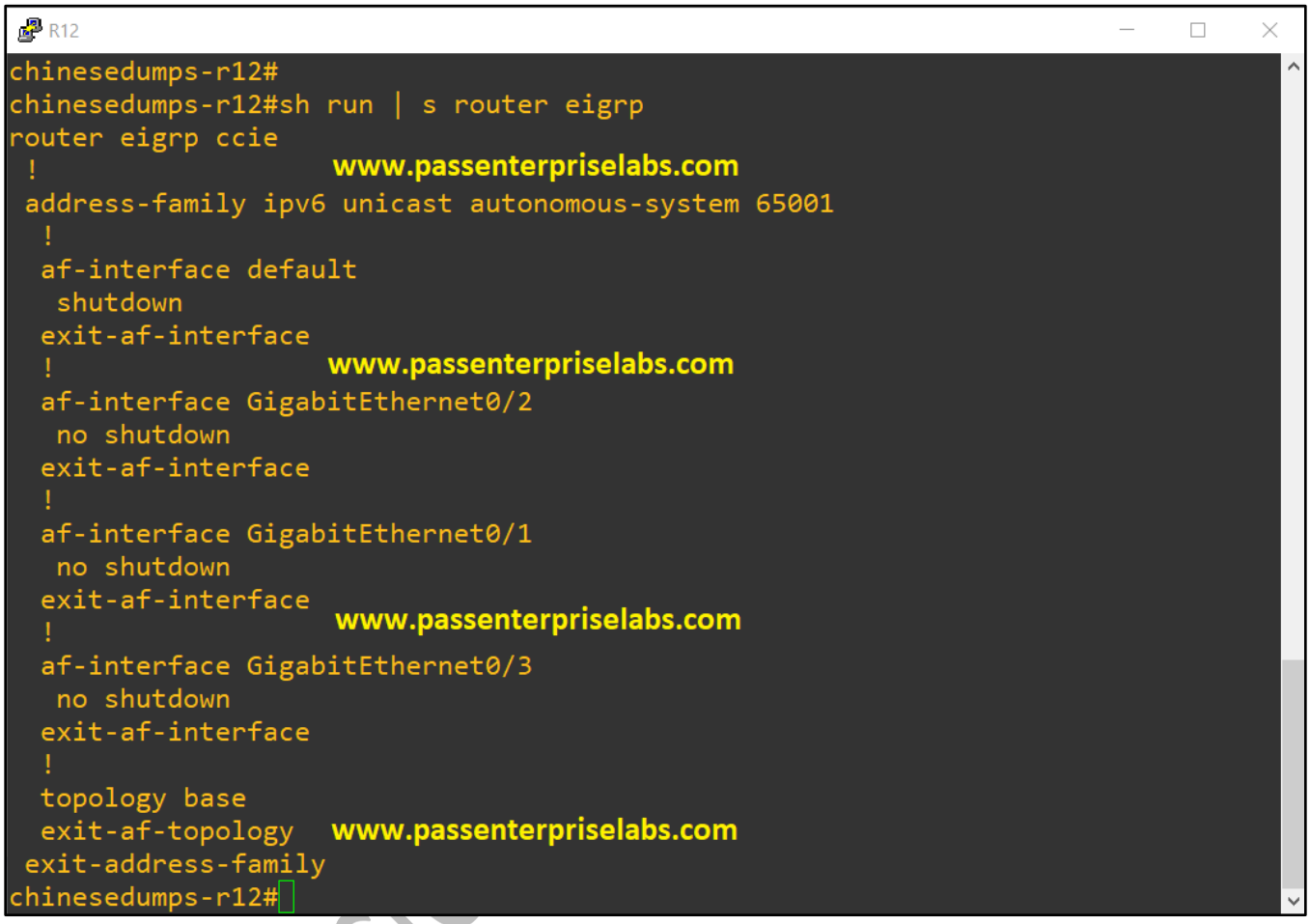
```
R12
chinesedumps-r12#
chinesedumps-r12#sh ipv6 route eigrp     www.passenterpriselabs.com
IPv6 Routing Table - default - 4 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
D   2001:DB8:1:100::/64 [90/15360]
     via FE80::101, GigabitEthernet0/2
     via FE80::102, GigabitEthernet0/3
D   2001:DB8:1:101::/64 [90/15360]       www.passenterpriselabs.com
     via FE80::101, GigabitEthernet0/2
     via FE80::102, GigabitEthernet0/3
chinesedumps-r12#
```

Chinese edumps.com-r12：显示ipv6 eigrp邻居



中文教程com-r11：显示ipv6路由eigrp

Chinesedumps.com-r11:show run | s router eigrp



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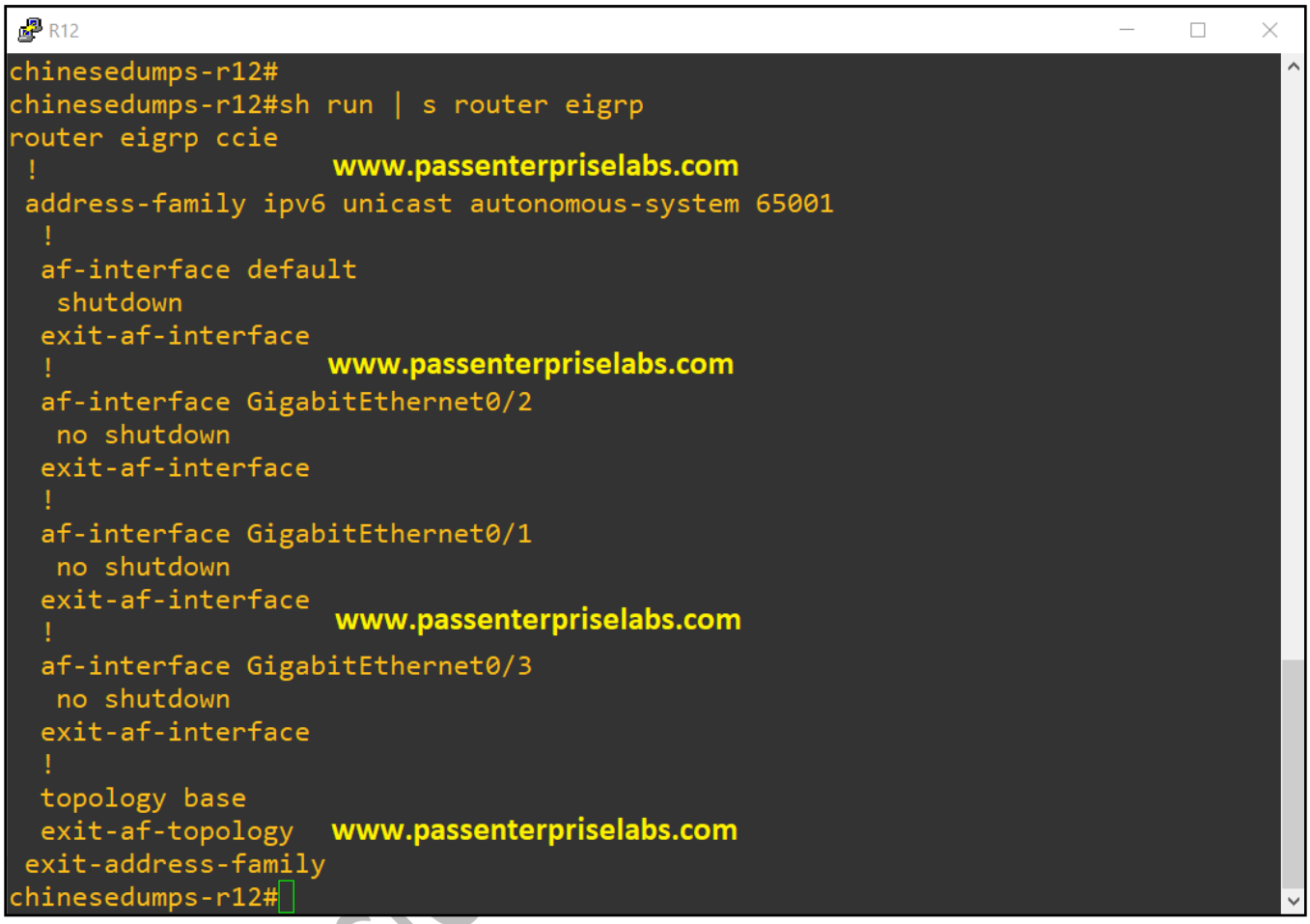
Chinese edumps.com-r11：显示运行的路由器eigrp



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To verify the task:

- As you can see in the above screenshots for sw101 and sw 102 , there are routes advertised in eigrp because only the physical links are advertised in eigrp on r11 and r12
- To get the routes in eigrp on sw101 and sw102 , add the loopback interface on r11 to eigrp (just for testing purpose)

On r11 :

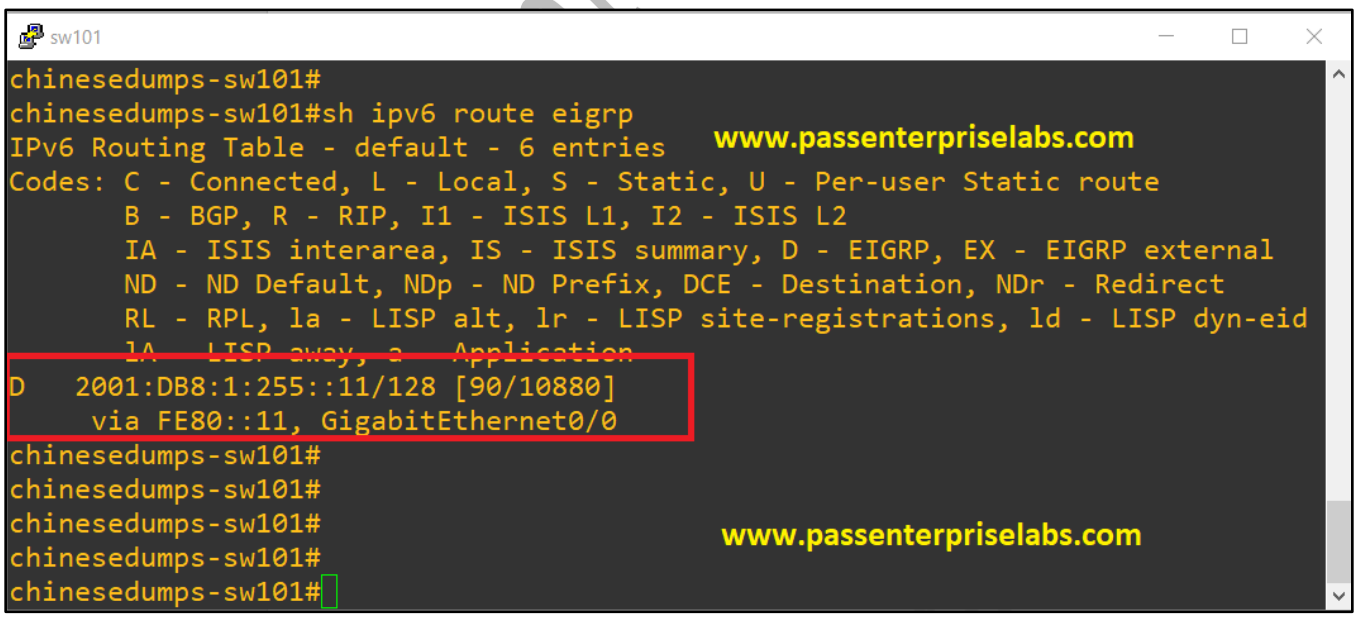
router eigrp ccie

address-family ipv6 unicast autonomous-system 65001

af-interface Loopback0
no shutdown
exit-af-interface

Output:

Sw101:



- In the above output you can see there is a route present in the eigrp routing table

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要验证任务，请执行以下操作：

- 正如您在上面的sw101和sw 102的屏幕截图中看到的，在r11和r12上，eigrp中只通告了物理链路，因此有路由通告
- 要获取sw101和sw102上eigrp中的路由，请将r11上的环回接口添加到eigrp（仅用于测试目的）

在r11：

路由器eigrp-ccie

地址族ipv6单播自治系统65001 af接口Loopback0
无关机退出af
界面

输出：

Sw101：

- 在上面的输出中，您可以看到eigrp路由表中存在一条路由

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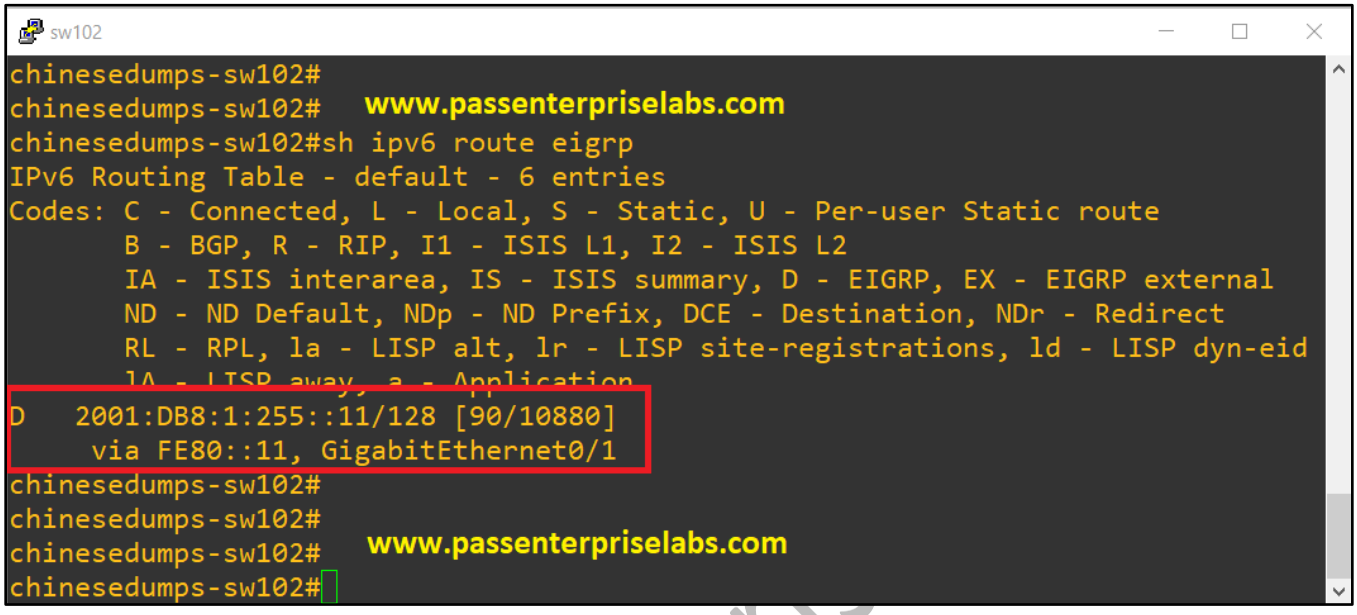
Sw102:



- In the above output you can see there is a route present in the eigrp routing table

NOTE : After testing please remove the loopback 0 on r11 from eigrp

On r11 :

router eigrp ccie

address-family ipv6 unicast autonomous-system 65001

af-interface Loopback0
shutdown
exit-af-interface

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瑞士法郎102：

- 在上面的输出中，您可以看到eigrp路由表中存在一条路由

注意：测试后，请从eigrp中删除r11上的环回0

在r11：

路由器eigrp-ccie

地址族ipv6单播自治系统65001 af接口Loopback0
关闭
退出af界面

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1.8: OSPFv2 in DC

Configure devices in the DC according to these requirements:

- Switches sw201 and sw202 must establish a stable OSPF adjacency in the FULL state with vedge21 and vedge22 on interface Vlan3999. Any configuration changes and corrections necessary to meet this requirement may be performed only on the switches, and any mismatched parameters causing the issue must be changed to exactly match the configuration of the vEdges.
- All OSPF speakers in the DC running Cisco IOS and IOS-XE software must be configured to keep the number of advertised internal routes to an absolute minimum while not impacting the reachability of the services. This included the reachability of ISE, DNA center, vManage, vBond and vSmart on their internal (in Band Connectivity) addresses, as well as any existing and future devices in VLAN 4000 on sw201 and sw202. The configuration of this requirement must be completed exclusively within the "router ospf" and "interface vlan" contexts without causing any impact to existing OSPF adjacencies.
- Router r24 must advertise two prefixes, 10.6.0.0/15 and 10.200.0.0/24, as Type-5 LSAs in OSPFv2 to provide HQ and DC with the reachability to the DMVPN tunnel and branches #3 and #4. The configuration of this requirement must be completed exclusively within the "router ospf" context.
- Any route from the 10.2.0.0/16 range that keeps being advertised in OSPF must continue being advertised as an intra-area route.
- It is not allowed to modify existing areas to accomplish this entire task.

4 Points

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1.8：DC中的OSPFv2

根据以下要求配置DC中的设备：

- 交换机sw201和sw202必须在FULL状态下与接口Vlan3999上的vedge21和vedge22建立稳定的OSPF邻接。满足此要求所需的任何配置更改和更正只能在交换机上执行，并且必须更改导致问题的任何不匹配参数，以完全匹配vEdges的配置。
- 运行Cisco IOS和IOS-XE软件的DC中的所有OSPF扬声器必须配置为将通告的内部路由数量保持在绝对最小值，同时不影响
  服务的可达性。这包括ISE、DNA中心、vManage、vBond和vSmart在其内部（带内连接）地址上的可达性，以及sw201和sw202上VLAN 4000中的任何现有和未来设备。此要求的配置必须仅在“路由器ospf”和“接口vlan”上下文中完成，而不会导致
  对现有OSPF邻接的任何影响。
- 路由器r24必须通告两个前缀10.6.0.0/15和10.200.0.0/24作为OSPFv2中的Type-5 LSA，以向HQ和DC提供对DMVPN隧道和分支#3和#4的可达性。此要求的配置必须仅在“路由器ospf”上下文中完成。
- 在OSPF中持续通告的10.2.0.0/16范围内的任何路由都必须继续作为区域内路由进行通告。
- 不允许修改现有区域来完成整个任务。

4分

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Solution:

All DC DEVICES except Vedge21 , 22 , vsmart , vbond
---
router ospf 1
prefix-suppression

Chinese_Dumps_sw201, sw202

#conf t
(config)#int vlan 3999
(config-if)#ip mtu 1496

(config)#interface vlan 4000
(config-if)#ip ospf prefix-suppression disable

Chinese_Dumps_ r24

r24#conf t
r24(config)#router ospf 1
r24(config-router)#redistribute eigrp 65006 subnets
r24(config-router)#summary-address 10.6.0.0 255.254.0.0 tag 123

Chinese_Dumps_SW211:

sw211#conf t
sw211(config)#router ospf 1
sw211(config-router)#passive-interface gi1/1
sw211(config-router)#passive-interface gi1/2
sw211(config-router)#passive-interface gi1/3

Chinese_Dumps_SW212:

sw212#conf t
sw212(config)#router ospf 1
sw212(config-router)#passive-interface gi1/1
sw212(config-router)#passive-interface gi1/2

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解决方案：

除Vedge21、22、vsmart、vbond外的所有直流设备

路由器ospf 1
前缀抑制

中文_转储_sw201、sw202

#conft
（配置）#int vlan 3999
（配置if）#ip mtu 1496

（config）#接口vlan 4000
（config-if）#ip-ospf前缀抑制禁用

中文_转储_ r24

r24#配置文件r24（
config）#路由器ospf
1
r24（配置路由器）#重新分配eigrp 65006子网

中文_转储_SW211：

sw211#配置sw211#路由器
ospf 1
sw211（配置路由器）#无源接口gi1/1
sw211（配路由器）#有源接口gi1/2
sw211（设路由器）#被动接口gi1/3

中文_转储_SW212：

sw212配置文件sw212路由
器ospf 1
sw212（配置路由器）#无源接口gi1/1

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Verification:

Chinese_Dumps_sw201#show ip ospf neighbors

```
sw201
chinesedumps-sw201#
chinesedumps-sw201#sh ip ospf neighbor
                    www.passenterpriselabs.com          www.passenterpriselabs.com
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.1.255.101      1   FULL/DR         00:00:39    10.2.241.2      GigabitEthernet1/2
1.1.2.1           1   FULL/BDR        00:00:39    10.2.201.1      Vlan4000
1.1.2.2           1   FULL/DROTHER    00:00:38    10.2.201.2      Vlan4000
1.99.2.1          1   FULL/BDR        00:00:38    10.2.201.9      Vlan3999
1.99.2.2          1   FULL/DROTHER    00:00:38    10.2.201.10     Vlan3999
10.2.255.211      1   FULL/BDR        00:00:38    10.2.20.2       GigabitEthernet1/1
10.2.255.212      1   FULL/BDR        00:00:38    10.2.23.2       GigabitEthernet1/0
10.2.255.202      1   FULL/DR         00:00:39    10.2.109.2      GigabitEthernet0/3
10.2.255.22       1   FULL/BDR        00:00:38    10.2.12.1       GigabitEthernet0/1
10.2.255.21       1   FULL/BDR        00:00:39    10.2.10.1       GigabitEthernet0/0
chinesedumps-sw201#
```

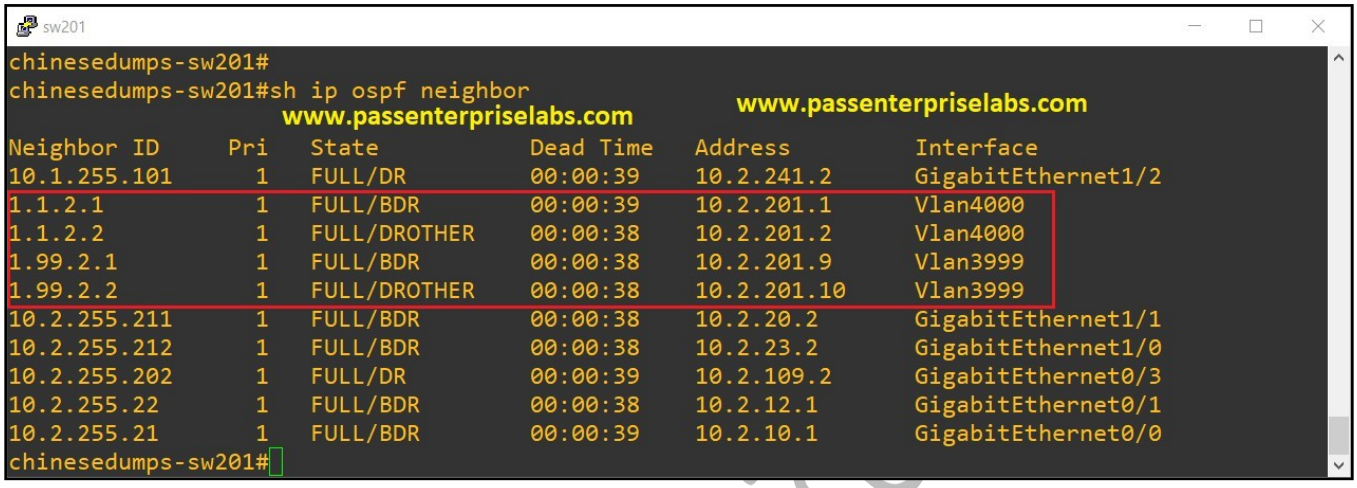
Chinese_Dumps_sw202#show ip ospf neighbors

```
sw202
chinesedumps-sw202#sh ip os nei
                    www.passenterpriselabs.com
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.1.255.102      1   FULL/DR         00:00:39    10.2.242.2      GigabitEthernet1/2
1.1.2.1           1   FULL/BDR        00:00:37    10.2.202.1      Vlan4000
1.1.2.2           1   FULL/DROTHER    00:00:36    10.2.202.2      Vlan4000
1.99.2.1          1   FULL/BDR        00:00:37    10.2.202.9      Vlan3999
1.99.2.2          1   FULL/DROTHER    00:00:36    10.2.202.10     Vlan3999
10.2.255.212      1   FULL/BDR        00:00:35    10.2.21.2       GigabitEthernet1/1
10.2.255.211      1   FULL/BDR        00:00:38    10.2.22.2       GigabitEthernet1/0
10.2.255.201      1   FULL/BDR        00:00:38    10.2.109.1      GigabitEthernet0/3
10.2.255.21       1   FULL/BDR        00:00:34    10.2.13.1       GigabitEthernet0/1
10.2.255.22       1   FULL/BDR        00:00:39    10.2.11.1       GigabitEthernet0/0
chinesedumps-sw202#   www.passenterpriselabs.com
chinesedumps-sw202#
```

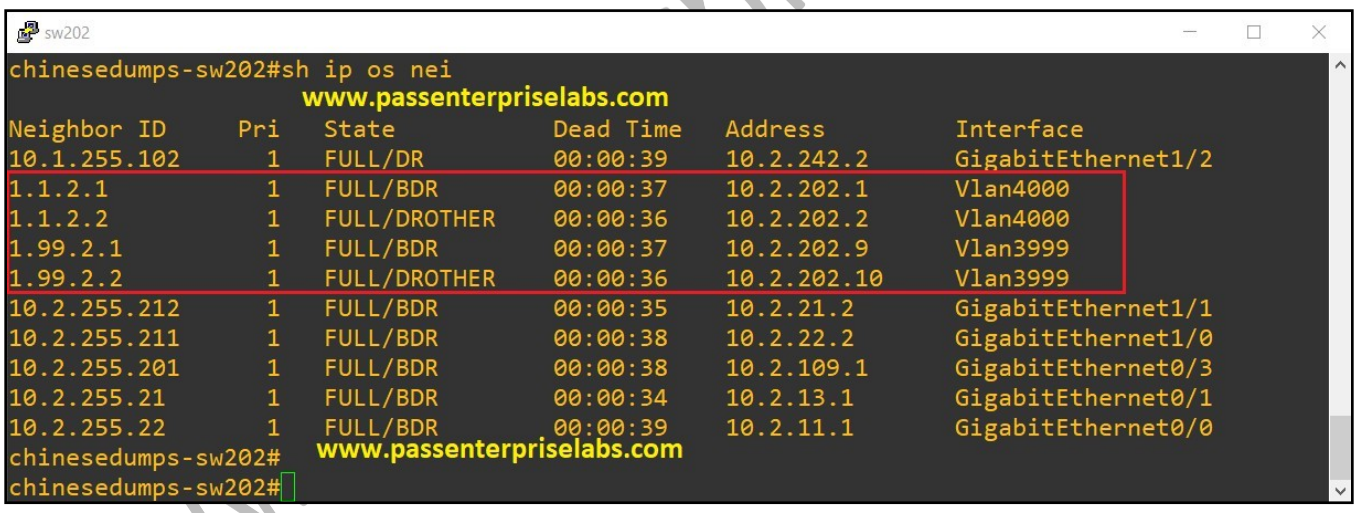
www.passenterpriselabs.com

验证：

中文_Dumps_sw201#显示ip ospf邻居



中文_Dumps_sw202#显示ip ospf邻居



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On all IOS and IOS-XE device in DC:

Chinesedumps.com-swXXX#show ip route ospf | in /30

The routes will be suppressed as we have implemented prefix-suppression

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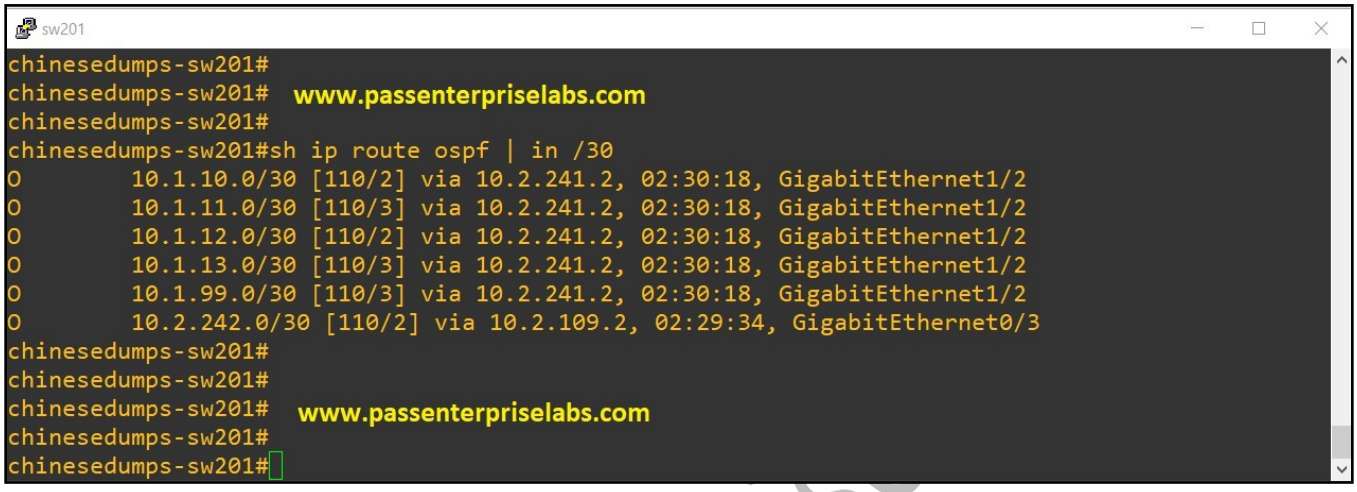
在DC的所有IOS和IOS-XE设备上：

Chinese edumps.com swXXX#显示ip路由ospf|in/30



由于我们已经实施了前缀抑制，路由将被抑制

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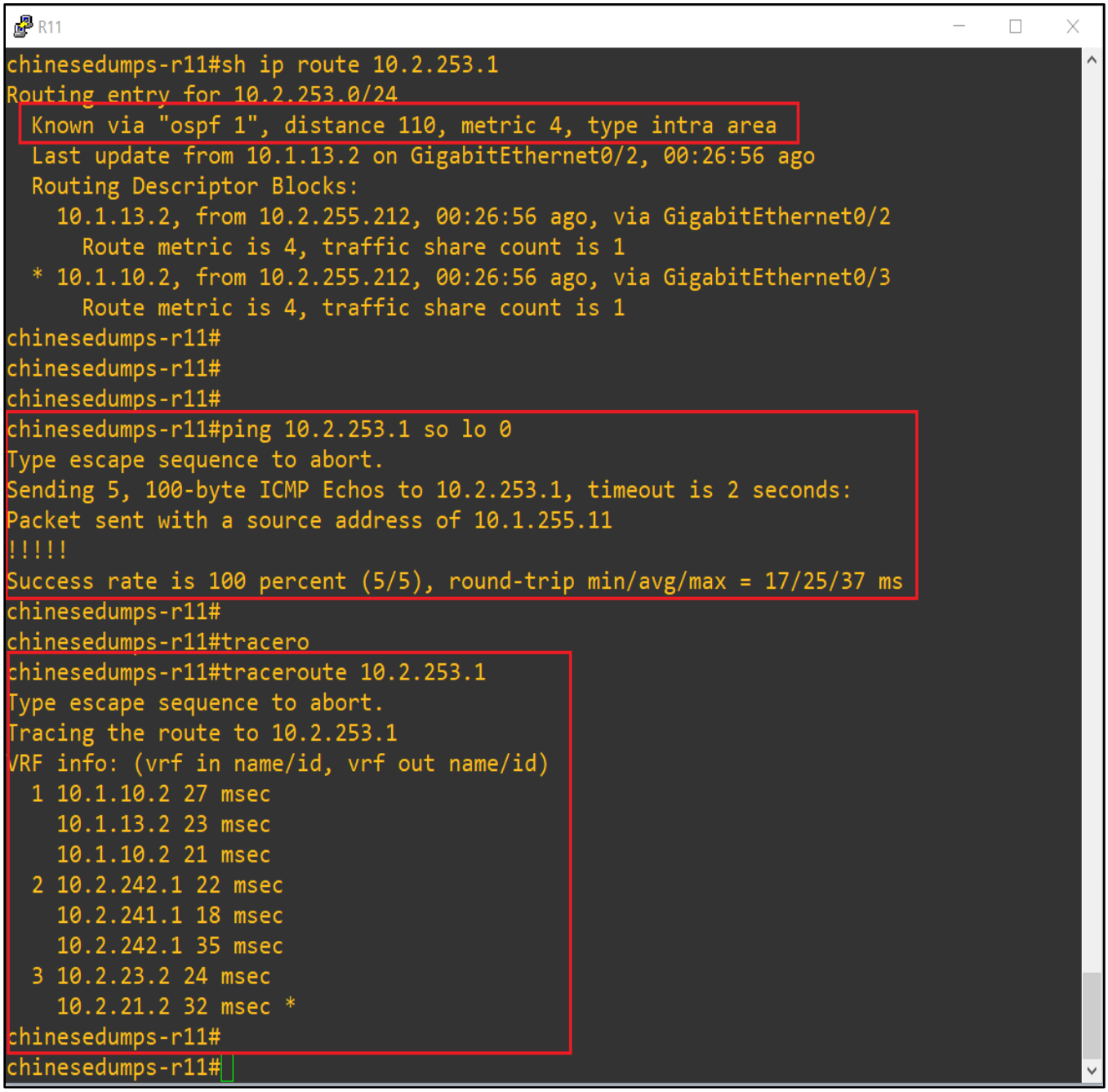
To verify the reachability go in HQ on any device and try to ping

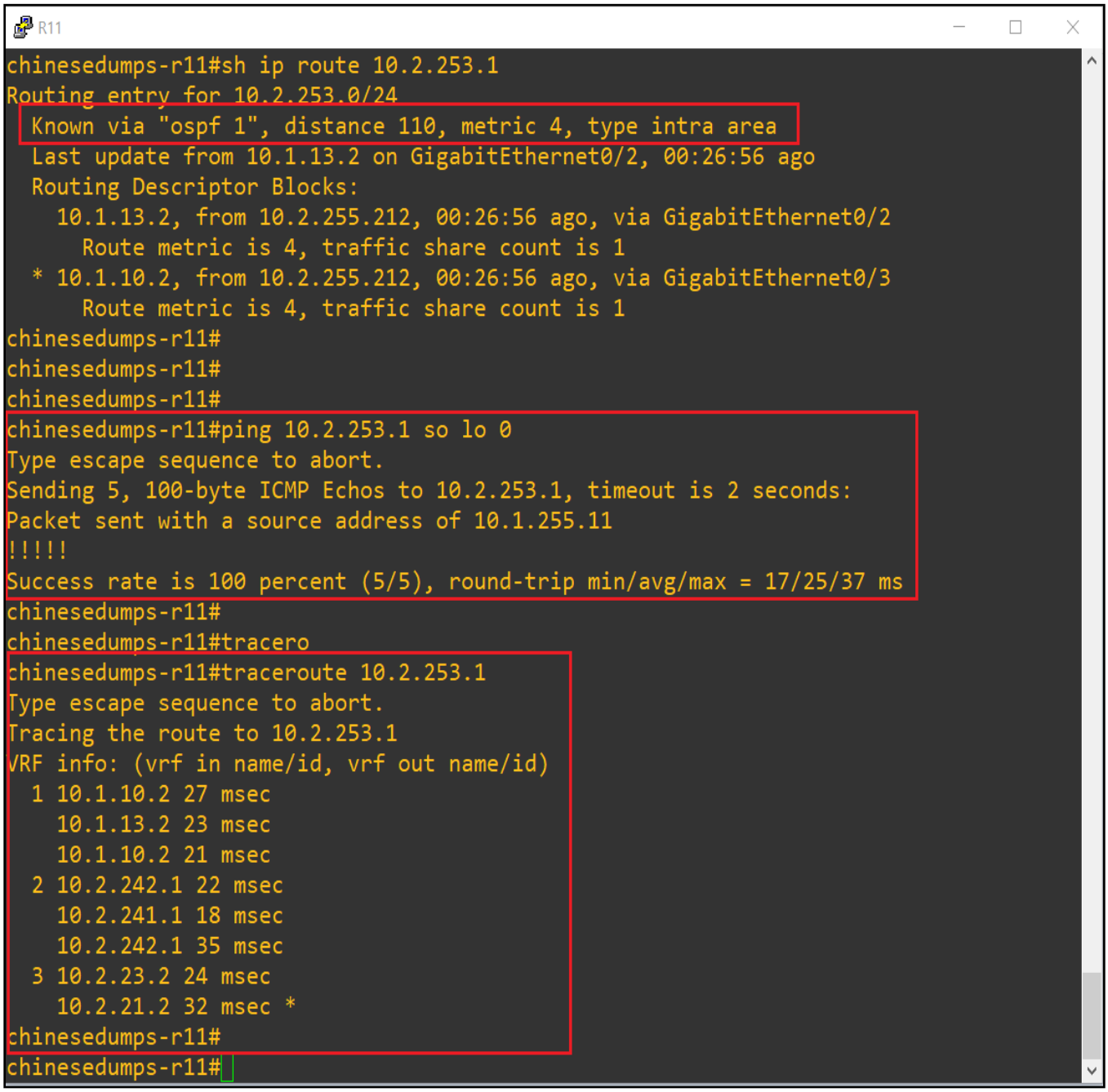
Chinese_Dumps_r11#show ip ospf neighbors



NOTE : The above subnet that is reachable is of vmanage

要验证可达性，请在任何设备上进入HQ，并尝试ping

Chinese_Dumps_r11#show ip ospf邻居



注意：上述可访问的子网是vmanage

1.9: BGP between HQ/DC and service providers

Configure the BGP peerings between HQ/DC and Global SP#1 and Global SP#2 according to these requirements:

- Bring up the BGP peering between HQ r11 and SP#1 r3
- Bring up the BGP peering between DC r21 and SP#1 r3
- Bring up the BGP peering between DC r22 and SP#2
- Ensure that the routes learned over eBGP sessions and further advertised in iBGP will be considered reachable even if the networks on inter-AS links are not advertised in OSPF. The configuration of this requirement must be completed exclusively within the "router bgp" context.
- On r11, r21 and r22 perform mutual redistribution between OSPFv2 and BGP. However, prevent routes that were injected into OSPF from BGP to be reinjected back into BGP. This requirement must be solved on r11, r21 and r22 using only a single route-map on each of the routers and without any reference to ACLs, prefix lists, or route types.
- Prevent HQ and DC from ever communicating through SP#1 r3. All communication between HQ and DC must occur only over the direct sw101/sw201 and sw102/sw202 interconnections. Any other communication must remain unaffected. This requirement must be solved on r21 and r22 by route filtering based on a well-known mandatory attribute without the use of route maps.
- No command may be removed from the configuration on r11 to accomplish this entire task.
- It is allowed to modify existing configuration commands on r21 and r22 to accomplish this entire task.

3 Points

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1.9： 总部/DC和服务提供商之间的BGP

根据以下要求配置HQ/DC与全局SP#1和全局SP#2之间的BGP对等：

- 打开HQ r11和SP#1 r3之间的BGP对等连接
- 打开DC r21和SP#1 r3之间的BGP对等连接
- 启动DC r22和SP#2之间的BGP对等连接
- 确保通过eBGP会话学习并在iBGP中进一步通告的路由将被视为可访问的，即使AS间链路上的网络未在OSPF中通告。此要求的配置必须仅在“routergp”上下文中完成。
- 在r11上，r21和r22执行OSPFv2和BGP之间的相互重新分配。但是，要防止从BGP注入OSPF的路由被重新注入BGP。必须在r11、r21和r22上解决此要求，在每个路由器上只使用一个路由图，而不参考ACL、前缀列表或路由类型。
- 阻止HQ和DC通过SP#1 r3进行通信。HQ和DC之间的所有通信必须仅通过直接的sw101/sw201和sw102/sw202互连进行。任何其他通信都必须不受影响。在r21和r22上，必须通过基于众所周知的强制属性的路由过滤来解决这一要求，而无需使用路由映射。
- 在r11上，不能从配置中删除任何命令来完成整个任务。
- 允许修改r21和r22上的现有配置命令以完成整个任务。

3分

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Solution:

Chinesedumps-r11:

```
r11(config)#router bgp 65001
r11(config-router)#address-family ipv4
r11(config-router)#neighbor 100.3.11.1 remote-as 10000
r11(config-router)#neighbor 100.3.11.1 act
r11(config-router)#distance 200 100.3.11.1 0.0.0.0

r11(config)#router ospf 1
r11(config-router)#redistribute bgp 65001 subnets tag 123

r11(config-router)#route-map DENY deny 10
r11(config-router)#match tag 123
r11(config-router)#route-map DENY permit 20

r11(config)#router bgp 65001
r11(config-router)#address-family ipv4
r11(config-router)#redistribute ospf 1 match internal external 1 external 2 route-map DENY
```

Chinesedumps-r21:

```
r21#conf t
r21(config)#route-map  DENY  deny  10
r21(config-route-map)#match  tag  123
r21(config)#route-map DENY permit 20

r21(config)#ip as-path access-list 100 deny
_65001$ r21(config)#ip as-path access-list 100
permit .*

r21(config)#router bgp 65002
r21(config-router)#neighbor 100.3.21.1 remote-as 10000
r21(config-router)#neighbor 10.2.255.22 remote-as 65002
r21(config-router)#neighbor 10.2.255.22 next-hop-self
r21(config-router)#neighbor 10.2.255.22 update-source lo0
r21(config-router)#redistribute ospf 1 match internal external 1 external 2 route-map DENY
r21(config-router)#neighbor 100.3.21.1 filter-list 100 in

r21 (config)#router ospf 1
```

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解决方案：

中文教程-r11：

r11（配置）#路由器bgp65001r11（
配置路由器）#地址家族ipv4
r11（配置路由器）#邻居100.3.11.1远程为10000
r11（配置路由器）#邻居100.3.11.1行为
r11（配置路由器）#距离200 100.3.11.1 0.0.0.0

r11（配置）#路由器ospf 1
r11（配置路由器）#重新分配bgp 65001子网标签123

r11（配置路由器）#路由映射拒绝10 r11
（设置路由器）#匹配标签123
r11（配置路由器）#路由图DENY permit 20

r11（配置）#路由器bgp65001r11（
配置路由器）#地址家族ipv4
r11（配置路由器）#重新分配ospf 1匹配内部外部1外部2路由图DENY

中文教程-r21：

r21#配置文件
r21（config）#路由映射拒绝10 r21
（配置路由映射）#匹配标签123 r21
（config）#路由地图拒绝许可20

r21（config）#ip作为路径访问列表100拒绝
_65001$r21（配置）#ip作为路径访问列表100允
许的。*

r21（配置）#路由器bgp 65002
r21（配置路由器）#邻居100.3.21.1远程为10000
r21（配置路由器）#邻居10.2.255.22远程为65002
r21（配置路由器）#邻居10.2.255.22下一跳自r21
r21（配置路由器）#重新分配ospf 1匹配内部外部1外部2路由图DENY r21（设置路由
器）#邻居100.3.21.1过滤器列表100 in

r21（配置）#路由器ospf 1

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r21 (config-router)#redistribute bgp 65002 subnets tag 123

r21（配置路由器）#重新分配bgp 65002子网标签123

Chinesedumps-r22:

```
r22#conf t

r22(config)#route-map  DENY  deny  10
r22(config-route-map)#match  tag  123
r22(config)#route-map DENY permit 20

r22(config)#router ospf 1
r22(config-router)#redistribute bgp 65002 subnets tag 123

r22(config)#router bgp 65002
r22(config-router)#neighbor 101.22.0.1 remote-as 10001
r22(config-router)#redistribute ospf 1 match internal external 1 external 2 route-map DENY
```

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中文教程-r22：

r22#配置文件

r22（配置路由图）#匹配标签123
r22（配置）#路线图DENY许可证20

r22（配置）#路由器ospf 1
r22（配置路由器）#重新分配bgp 65002子网标签123

r22（配置）#路由器bgp 65002
r22（配置路由器）#邻居101.22.0.1远程为10001
r22（配置路由器）#重新分配ospf 1匹配内部外部1外部2路由图DENY

r22（config）#路由图拒绝10

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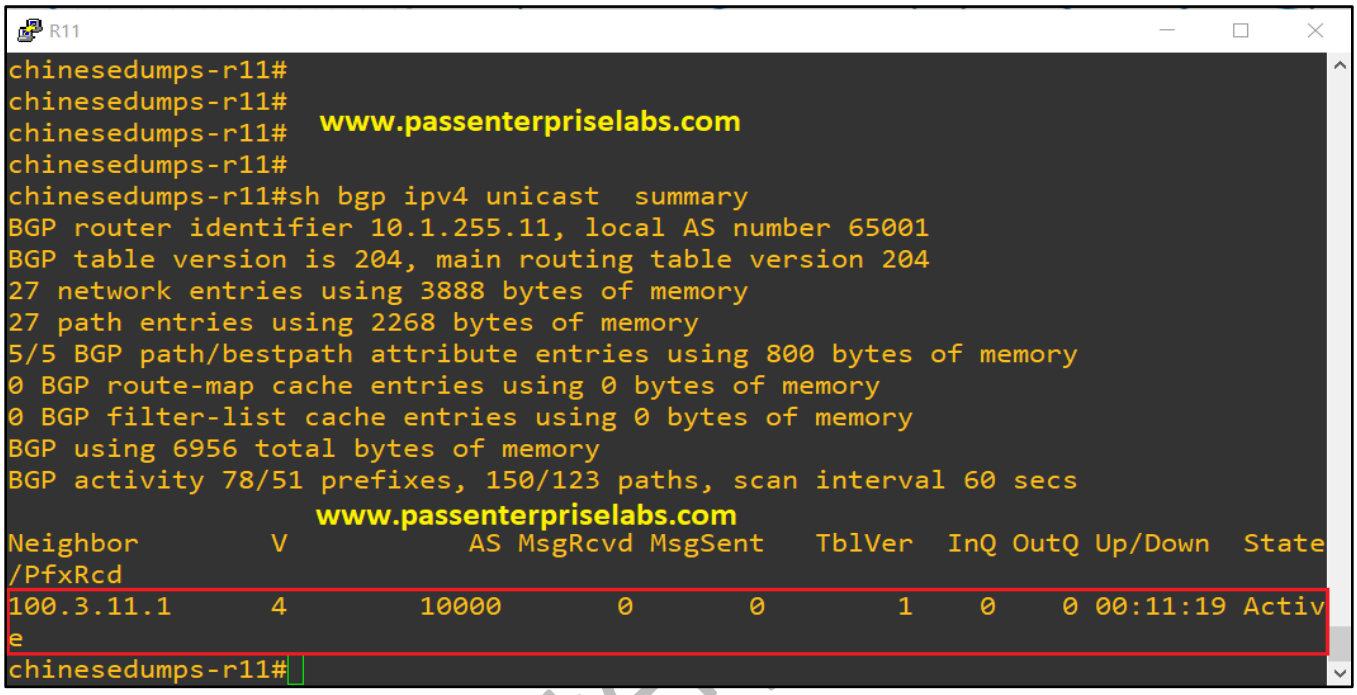
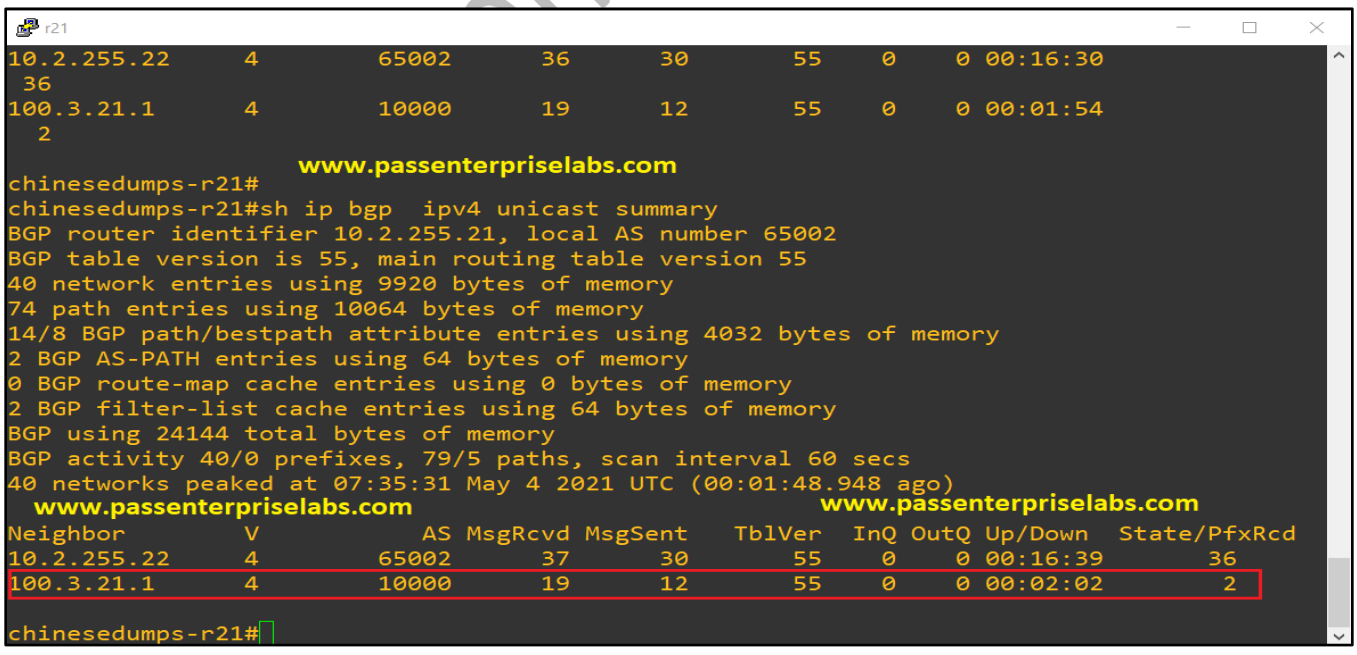
Verification:

Chinese_Dumps_r11#sh bgp ipv4 unicast summary

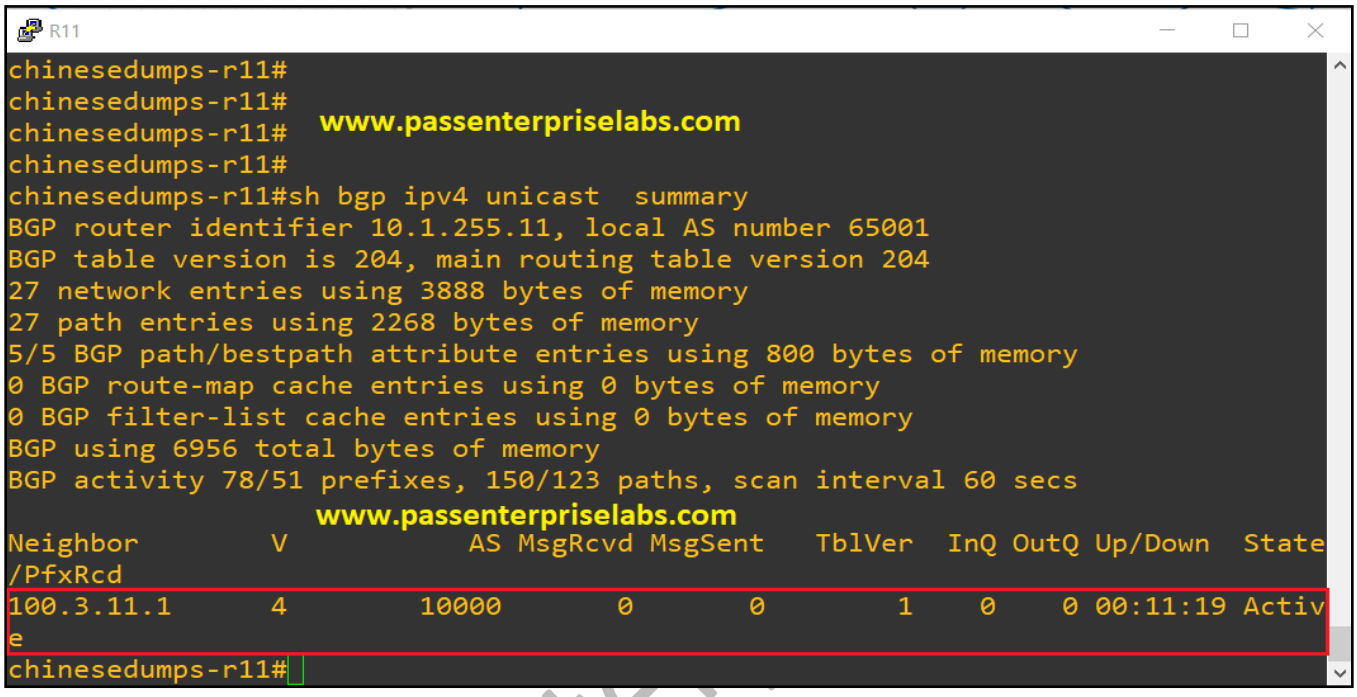


Chinese_Dumps_r21#sh ip bgp ipv4 unicast summary

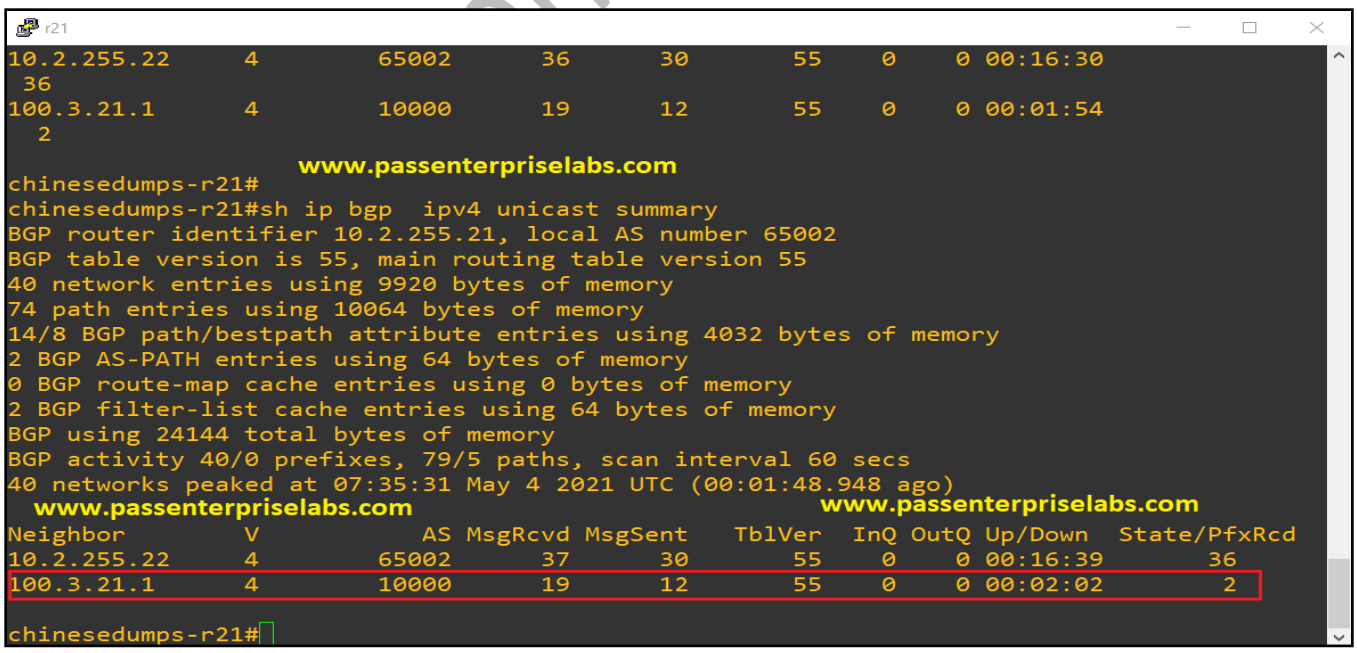


验证：

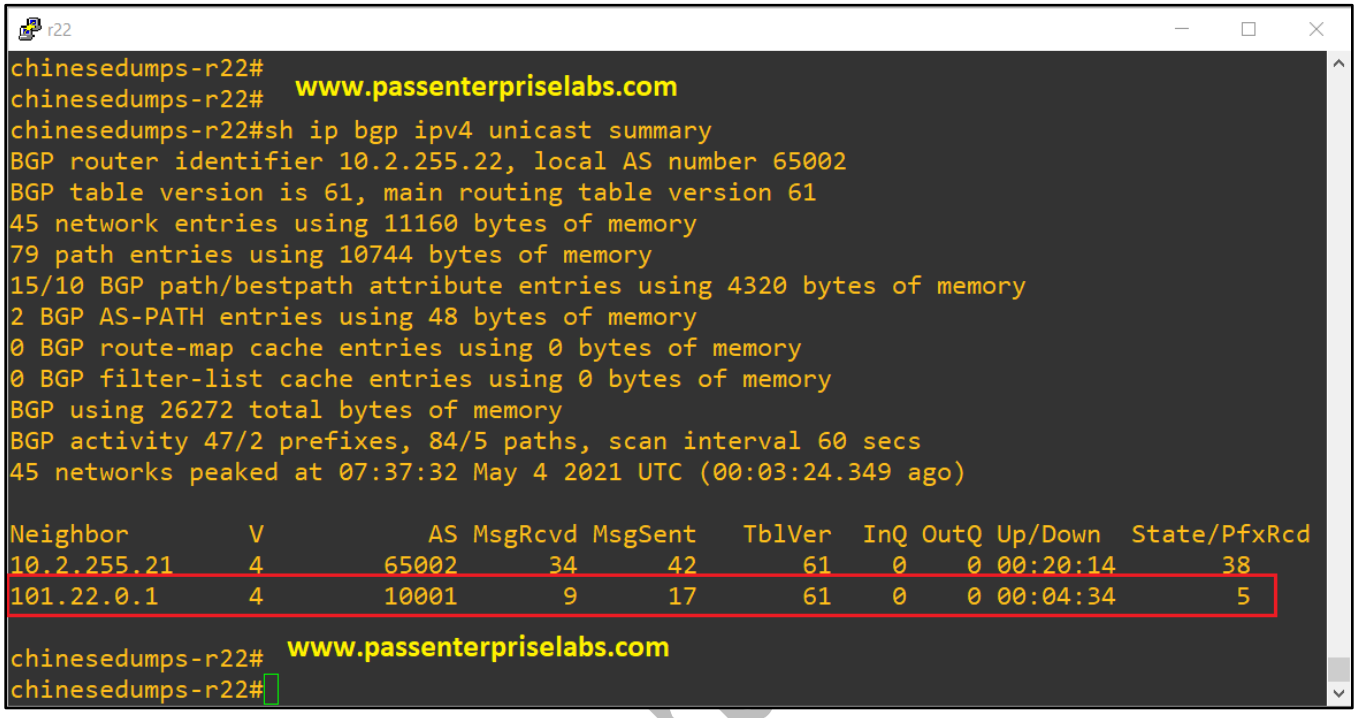
中文_转储_r11#sh bgp ipv4通讯摘要



中文_Dumps_r21#sh ip bgp ipv4单播摘要



Chinese_Dumps_r22# sh ip bgp ipv4 unicast summary



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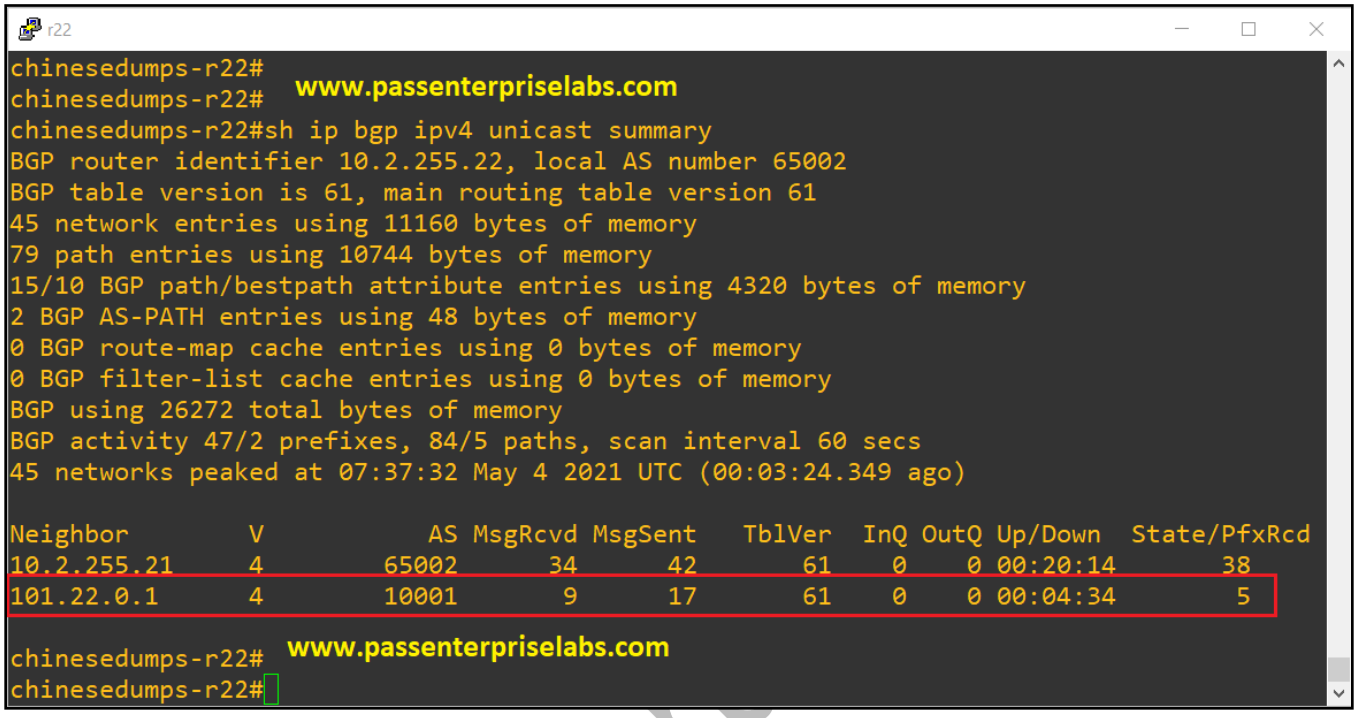
中文_转储_ r22#sh ip bgp ipv4通讯摘要



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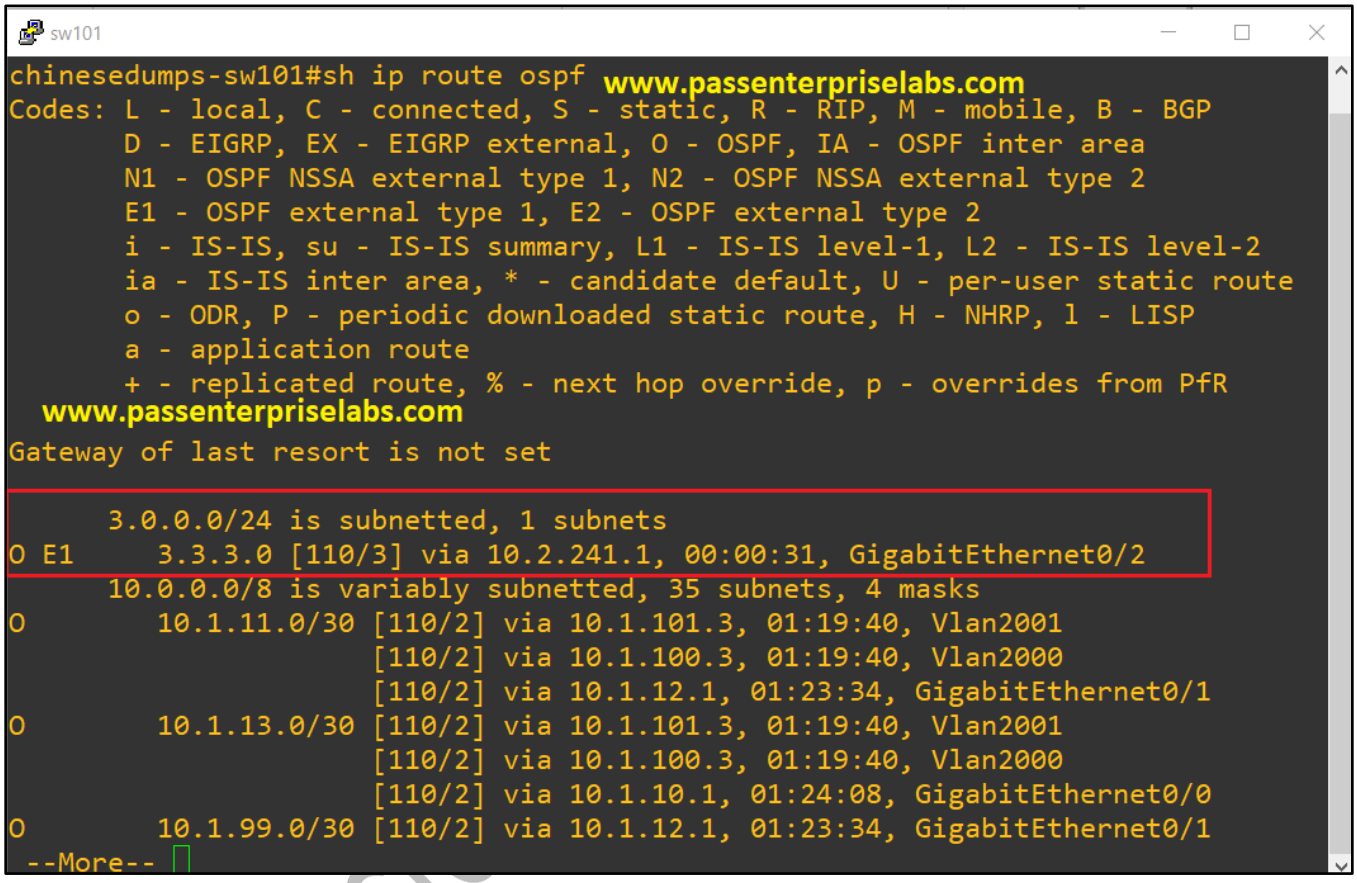
To verify 4th point, we will create a loopback on r3 (int lo1 – 3.3.3.3/24)

- Create int lo1 with ip address 3.3.3.3/24
- Go on sw101 or sw102 or r12 and check the OSPF routes



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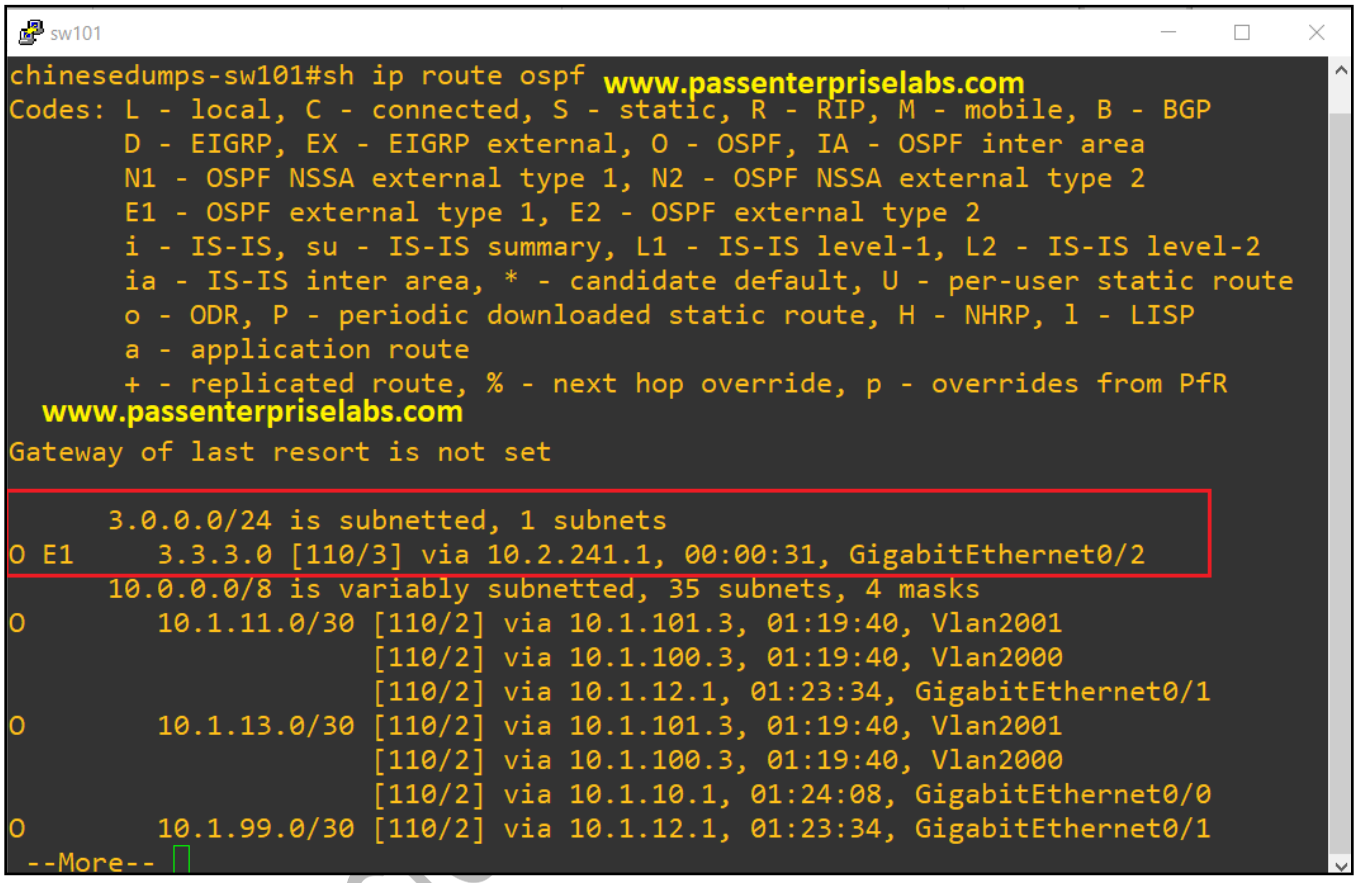
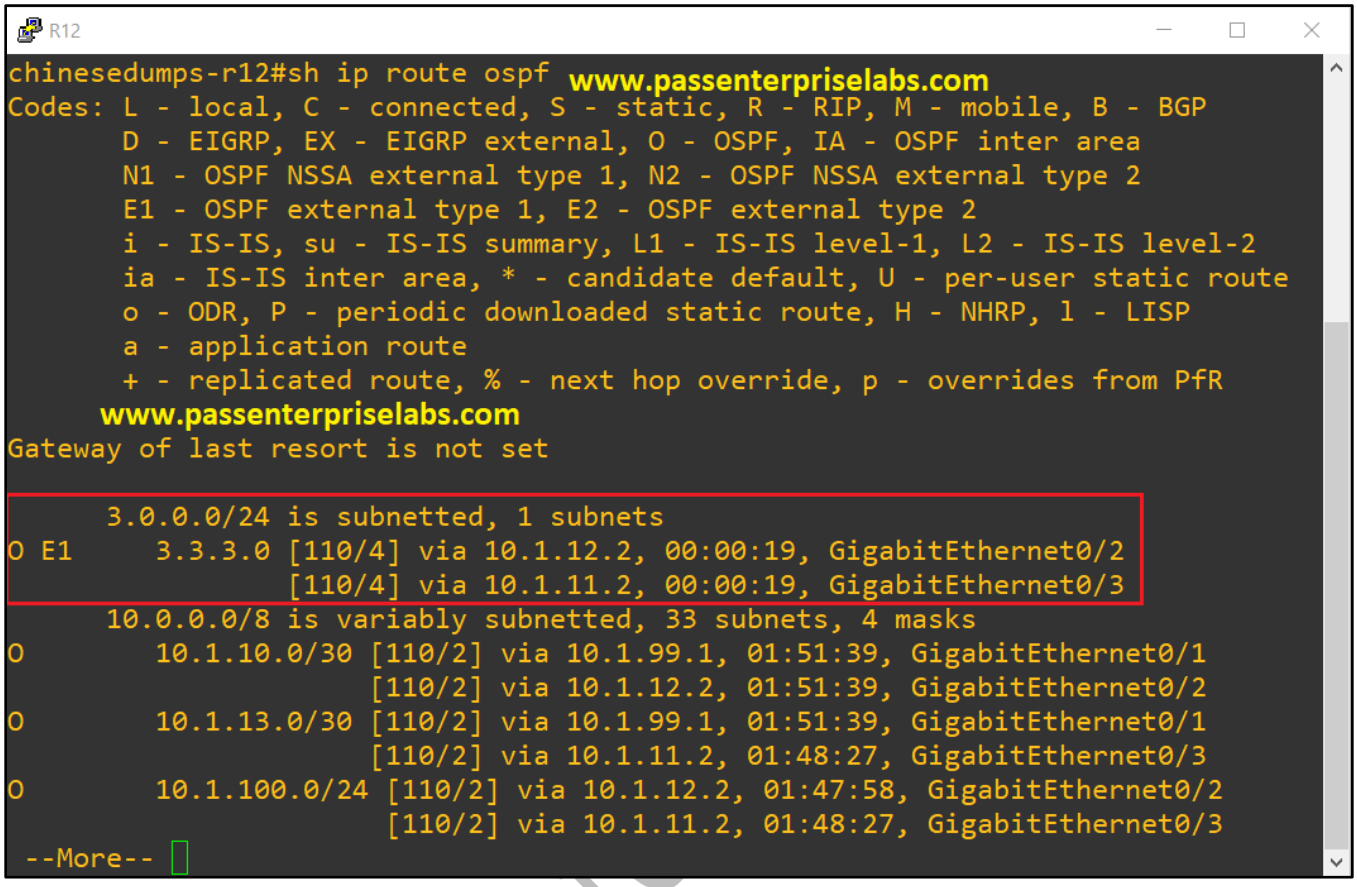
为了验证第四点，我们将在r3上创建一个环回（int lo1-3.3.3.3/24）

- 使用ip地址3.3.3.3/24创建int lo1
- 打开sw101、sw102或r12，检查OSPF路由



WWW.CCIE


R12
chinesedumps-r12#sh ip route ospf
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Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
www.passenterpriselabs.com
Gateway of last resort is not set
3.0.0.0/24 is subnetted, 1 subnets
O E1 3.3.3.0 [110/4] via 10.1.12.2, 00:00:19, GigabitEthernet0/2
[110/4] via 10.1.11.2, 00:00:19, GigabitEthernet0/3
10.0.0.0/8 is variably subnetted, 33 subnets, 4 masks
O 10.1.10.0/30 [110/2] via 10.1.99.1, 01:51:39, GigabitEthernet0/1
[110/2] via 10.1.12.2, 01:51:39, GigabitEthernet0/2
O 10.1.13.0/30 [110/2] via 10.1.99.1, 01:51:39, GigabitEthernet0/1
[110/2] via 10.1.11.2, 01:48:27, GigabitEthernet0/3
O 10.1.100.0/24 [110/2] via 10.1.12.2, 01:47:58, GigabitEthernet0/2
[110/2] via 10.1.11.2, 01:48:27, GigabitEthernet0/3
--More--

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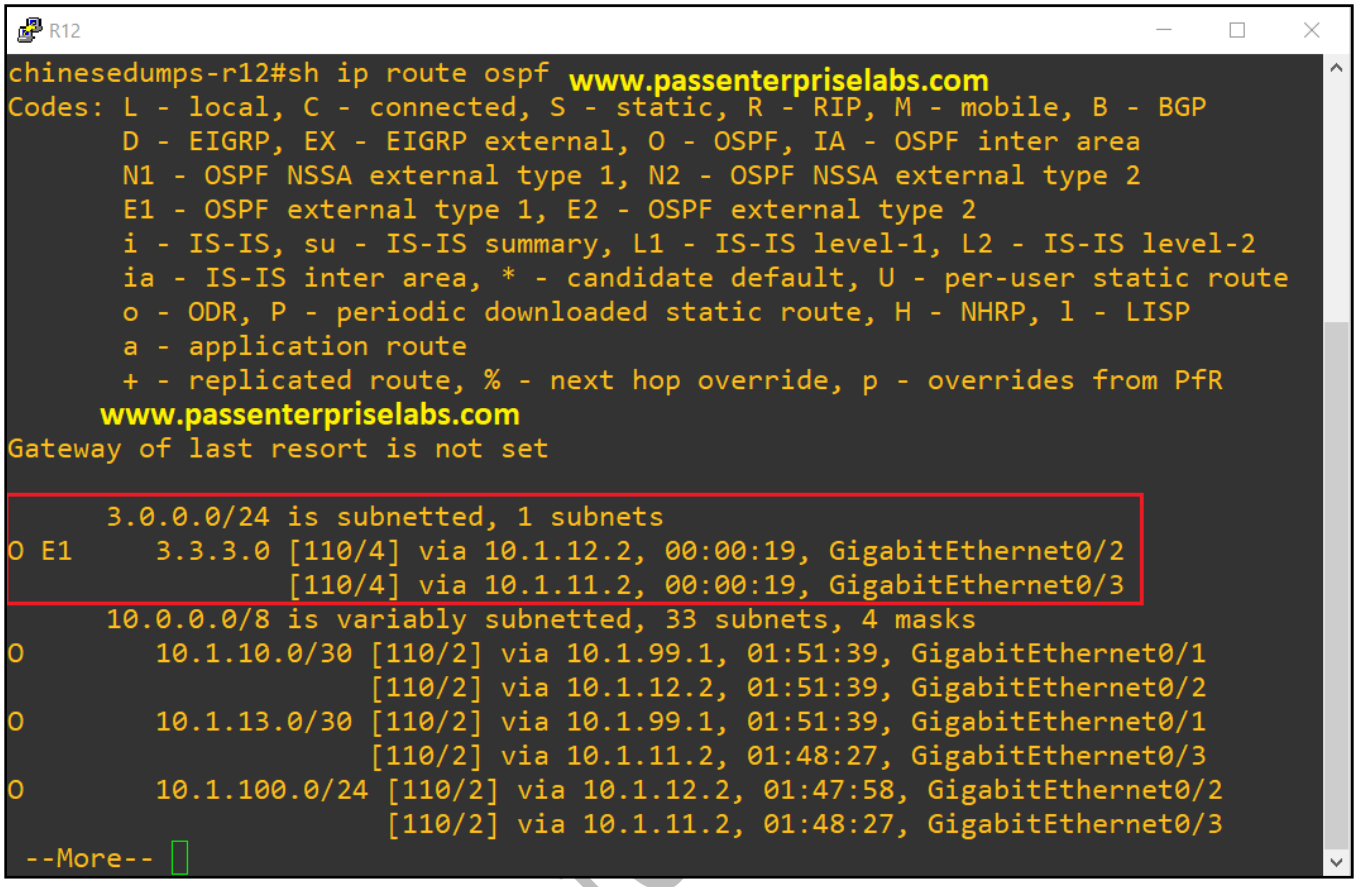
R12
chinesedumps-r12#sh ip route ospf
www.passenterpriselabs.com
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
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Gateway of last resort is not set
3.0.0.0/24 is subnetted, 1 subnets
O E1 3.3.3.0 [110/4] via 10.1.12.2, 00:00:19, GigabitEthernet0/2
[110/4] via 10.1.11.2, 00:00:19, GigabitEthernet0/3
10.0.0.0/8 is variably subnetted, 33 subnets, 4 masks
O 10.1.10.0/30 [110/2] via 10.1.99.1, 01:51:39, GigabitEthernet0/1
[110/2] via 10.1.12.2, 01:51:39, GigabitEthernet0/2
O 10.1.13.0/30 [110/2] via 10.1.99.1, 01:51:39, GigabitEthernet0/1
[110/2] via 10.1.11.2, 01:48:27, GigabitEthernet0/3
O 10.1.100.0/24 [110/2] via 10.1.12.2, 01:47:58, GigabitEthernet0/2
[110/2] via 10.1.11.2, 01:48:27, GigabitEthernet0/3
--More--

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To verify 5th point, we will use the previously created loopback on r3 (int lo1 – 3.3.3.3/24)

- int lo1 with ip address 3.3.3.3/24
- Remove next-hop-self command from r21 for neighbor 10.2.255.22
- For verification go on r22 and check bgp routes

- From above output you can see the route is just showing as an ibgp route and also it isnot present in the routing table of bgp

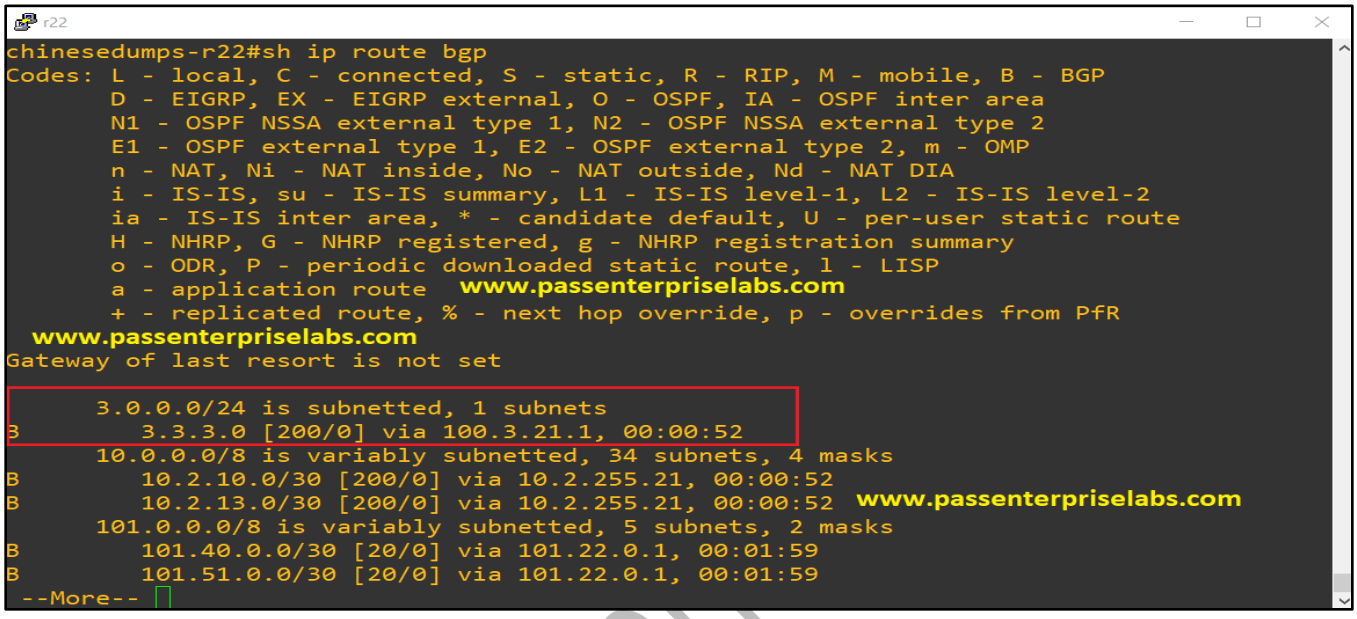
为了验证第五点，我们将使用之前在r3上创建的环回（int lo1-3.3.3.3/24）

- ip地址为3.3.3.3/24的int lo1
- 从r21中删除邻居10.2.255.22的下一跳self命令
- 要进行验证，请转到r22并检查bgp路由

- 从上面的输出中，您可以看到路由只是显示为ibgp路由，而且它也不存在于bgp的路由表中

To get the output as per the requirement

- Add next-hop-self command on r21 for neighbor 10.2.255.22
- Verify with the same commands as above

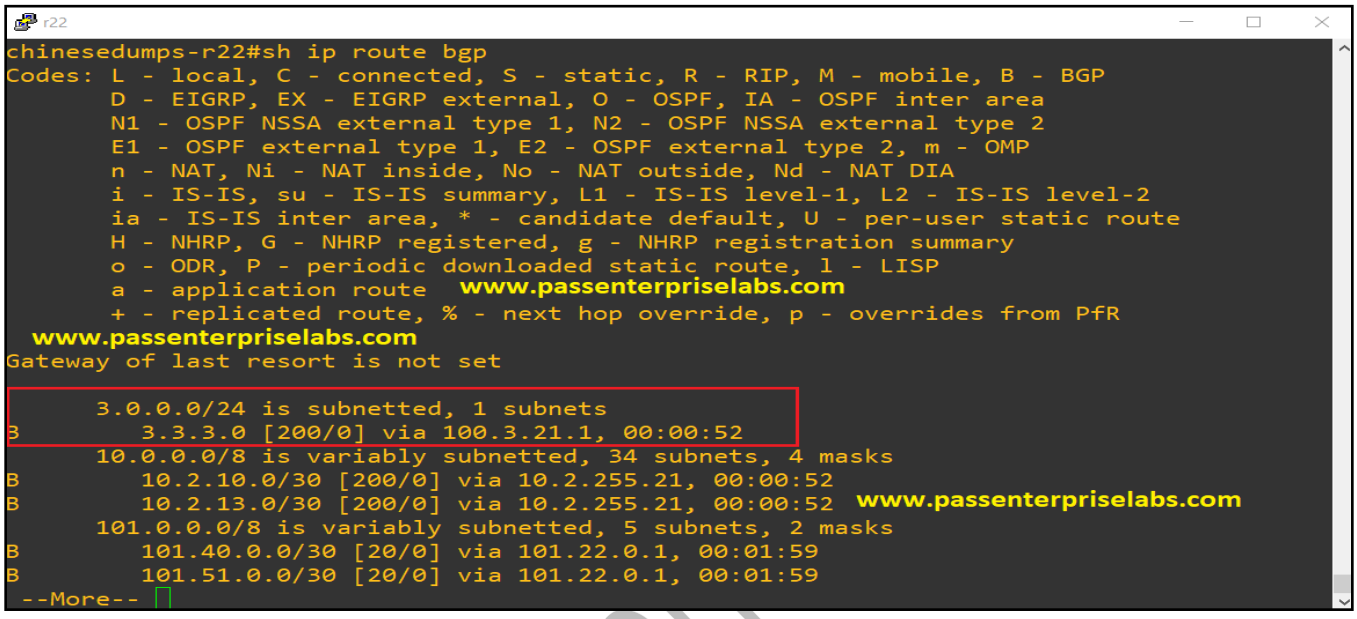


From the above output you can see that the 3.3.3.0 loopback is showing as best route and also it is installed in the routing table.

Note: After performing the verification, please remove the loopback fromr3

按照要求获得输出

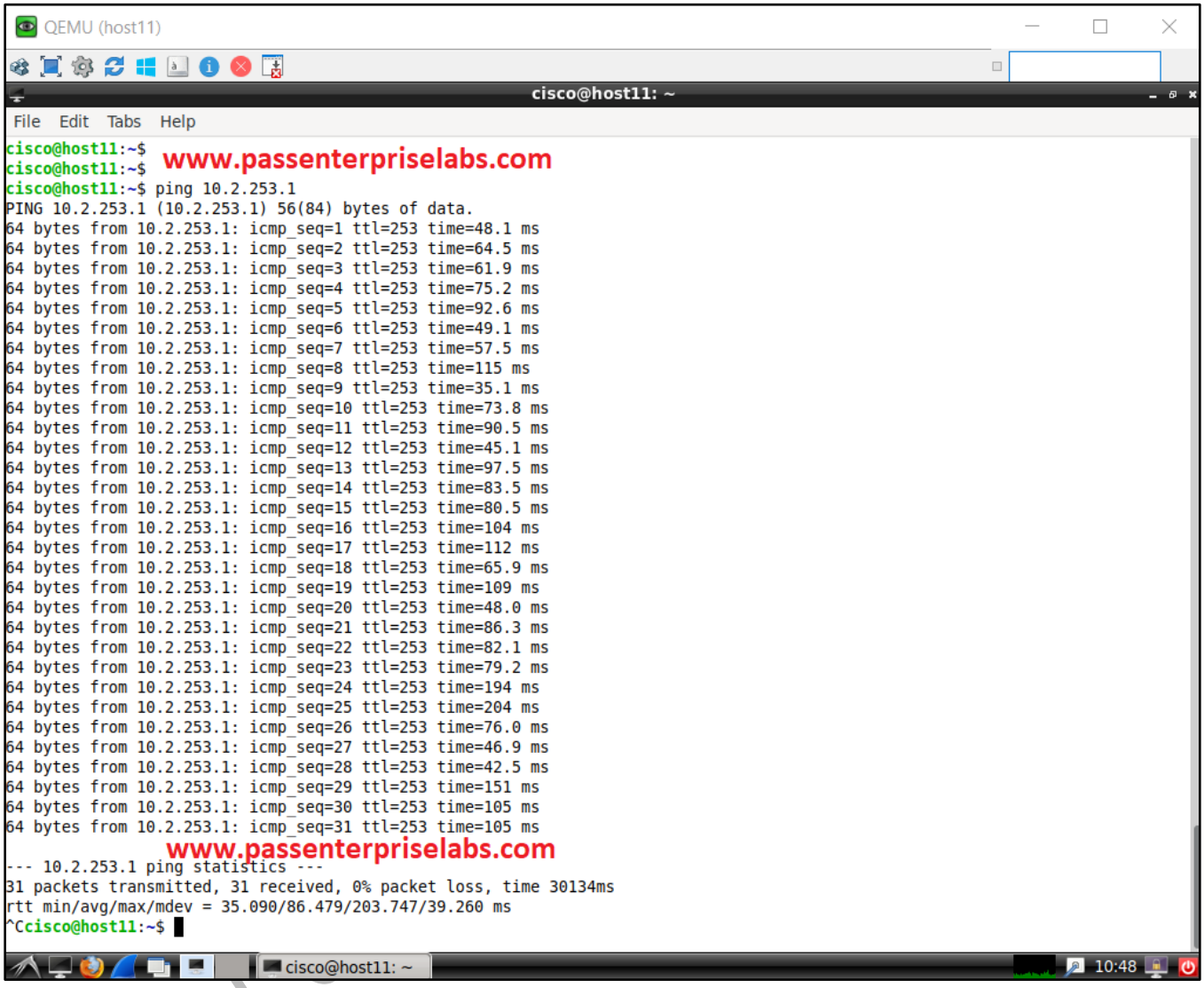
- 在r21上为邻居10.2.255.22添加下一跳self命令
- 使用与上述相同的命令进行验证



从上面的输出中，您可以看到3.3.3.0环回显示为最佳路由，并且它也安装在路由表中。

注意：执行验证后，请从r3中删除环回

6th point :



From above output you can see the HQ devices are preferring links between sw101 – sw201 & sw102 – sw202 to reach any subnet in DC

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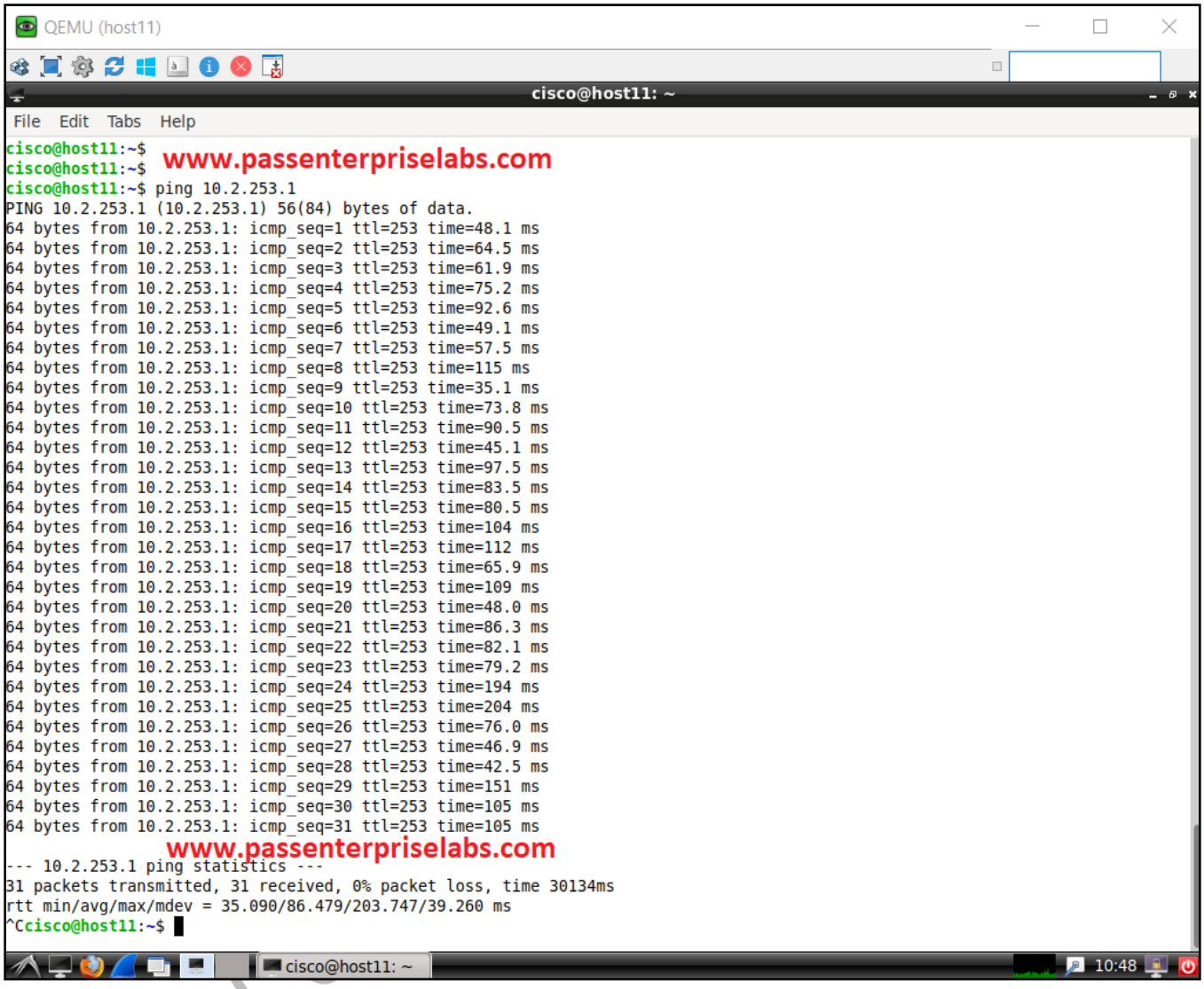
第六点：



从上面的输出中，您可以看到HQ设备更喜欢sw101-sw201和sw102-sw202之间的链接，以到达DC中的任何子网

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To verify 6th point ,

- Shutdown the link between sw101 – sw201 & sw102 – sw202 and check if you are having reachability to DC subnets via SP-1

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From above output you can see the HQ device host 11 cannot reach as the direct link between the switches sw101 – sw201 & sw102 – sw202

为了验证第六点，

- 关闭sw101-sw201和sw102-sw202之间的链接，并检查您是否可以通过SP-1访问DC子网

从上面的输出中，您可以看到HQ设备主机11无法作为交换机sw101-sw201和sw102-sw202之间的直接链路连接

1.10: Bringing up VPNv4/VPNv6 in SP#1

Configure routers r3, r4, r5 and r6 in SP#1 according to these requirements:

- Configure r3 through r6 for mutual VPNv4 and VPNv6 route exchange without the use of a route reflector. Use Lo0 IPv4 addresses for peering's.
- Configure r3 through r6 to assign (allocate/bind) as few unique MPLS labels to all existing and future VPNv4 and VPNv6 routes as possible.
- On Routers r3 through r6, prevent any existing and future customer from discovering details about the inner topology of SP#1. It is not allowed to use ACLs to accomplish this requirement.

3 Point

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1.10： 在SP#1中启动VPNv4/VPNv6

根据以下要求配置SP#1中的路由器r3、r4、r5和r6：

- 配置r3到r6，以便在不使用
  路线反射器。使用Lo0 IPv4地址进行对等连接。
- 配置r3到r6，为所有现有和未来的VPNv4和VPNv6路由分配（分配/绑定）尽可能少的唯一MPLS标签。
- 在路由器r3至r6上，防止任何现有和未来的客户发现详细信息
  关于SP#1的内部拓扑结构。不允许使用ACL来满足此要求。

3分

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Solution:

Chinesedumps.com-r3:

```
r3>en
r3#conf t
r3(config)#router bgp 10000
r3(config-router)#neighbor 100.255.254.4 remote-as 10000
r3(config-router)#neighbor 100.255.254.4 update-source lo0
r3(config-router)#neighbor 100.255.254.5 remote-as 10000
r3(config-router)#neighbor 100.255.254.5 update-source lo0
r3(config-router)#neighbor 100.255.254.6 remote-as 10000
r3(config-router)#neighbor 100.255.254.6 update-source lo0
r3(config-router)#address-family vpnv4
r3(config-router)#neighbor 100.255.254.4 activate
r3(config-router)#neighbor 100.255.254.5 activate
r3(config-router)#neighbor 100.255.254.6 activate
r3(config-router)#address-family vpnv6
r3(config-router)#neighbor 100.255.254.4 activate
r3(config-router)#neighbor 100.255.254.5 activate
r3(config-router)#neighbor 100.255.254.6 activate
```

Chinesedumps.com-r4:

```
r4>en
r4#conf t
r4(config)#router bgp 10000
r4(config-router)#neighbor 100.255.254.3 remote-as 10000
r4(config-router)#neighbor 100.255.254.3 update-source lo0
r4(config-router)#neighbor 100.255.254.5 remote-as 10000
r4(config-router)#neighbor 100.255.254.5 update-source lo0
r4(config-router)#neighbor 100.255.254.6 remote-as 10000
r4(config-router)#neighbor 100.255.254.6 update-source lo0
r4(config-router)#address-family vpnv4
r4(config-router)#neighbor 100.255.254.3 activate
r4(config-router)#neighbor 100.255.254.5 activate
r4(config-router)#neighbor 100.255.254.6 activate
r4(config-router)#address-family vpnv6
r4(config-router)#neighbor 100.255.254.3 activate
r4(config-router)#neighbor 100.255.254.5 activate
r4(config-router)#neighbor 100.255.254.6 activate
```

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解决方案：

中文教育网-r3：

r3>和
r3#conf t
r3（配置）#路由器bgp 10000
r3（配置路由器）#邻居100.255.254.4远程为10000 r3
r3（配置路由器）#邻居100.255.254.4激活
r3（配置路由器）#邻居100.255.254.5已激活
r3（配置路由器）#邻居100.255.254.6激活r3
（设置路由器）#地址系列vpnv6
r3（配置路由器）#邻居100.255.254.4激活
r3（配置路由器）#邻居100.255.254.5已激活
r3（配置路由器）#邻居100.255.254.6激活

Chinese edumps.com-r4：

r4>和
r4#conf t
r4（配置）#路由器bgp 10000
r4（配置路由器）#邻居100.255.254.3远程为10000 r4
（配置路由）#邻居100.0255.254.3更新源lo0 r4
r4（配置路由器）#邻居100.255.254.3激活
r4（配置路由器）#邻居100.255.255.5激活
r4（配置路由器）#邻居100.255.254.6激活r4
（配置路由）#地址族vpnv6
r4（配置路由器）#邻居100.255.254.3激活
r4（配置路由器）#邻居100.255.255.5激活
r4（配置路由器）#邻居100.255.254.6激活

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Chinesedumps.com-r5:

```
r5>en
r5#conf t
r5(config)#router bgp 10000
r5(config-router)#neighbor 100.255.254.3 remote-as 10000
r5(config-router)#neighbor 100.255.254.3 update-source lo0
r5(config-router)#neighbor 100.255.254.4 remote-as 10000
r5(config-router)#neighbor 100.255.254.4 update-source lo0
r5(config-router)#neighbor 100.255.254.6 remote-as 10000
r5(config-router)#neighbor 100.255.254.6 update-source lo0
r5(config-router)#address-family vpnv4
r5(config-router)#neighbor 100.255.254.3 activate
r5(config-router)#neighbor 100.255.254.4 activate
r5(config-router)#neighbor 100.255.254.6 activate
r5(config-router)#address-family vpnv6
r5(config-router)#neighbor 100.255.254.3 activate
r5(config-router)#neighbor 100.255.254.4 activate
r5(config-router)#neighbor 100.255.254.6 activate
```

Chinesedumps.com-r6:

```
r6>en
r6#conf t
r6(config)#router bgp 10000
r6(config-router)#neighbor 100.255.254.3 remote-as 10000
r6(config-router)#neighbor 100.255.254.3 update-source lo0
r6(config-router)#neighbor 100.255.254.4 remote-as 10000
r6(config-router)#neighbor 100.255.254.4 update-source lo0
r6(config-router)#neighbor 100.255.254.5 remote-as 10000
r6(config-router)#neighbor 100.255.254.5 update-source lo0
r6(config-router)#address-family vpnv4
r6(config-router)#neighbor 100.255.254.3 activate
r6(config-router)#neighbor 100.255.254.4 activate
r6(config-router)#neighbor 100.255.254.5 activate
r6(config-router)#address-family vpnv6
r6(config-router)#neighbor 100.255.254.3 activate
r6(config-router)#neighbor 100.255.254.4 activate
r6(config-router)#neighbor 100.255.254.5 activate
```

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中文教育网-r5：

r5>和
r5#conf t
r5（配置）#路由器bgp 10000
r5（配置路由器）#邻居100.255.254.3远程为10000 r5
r5（配置路由器）#邻居100.255.254.3激活
r5（配置路由器）#邻居100.255.254.4激活
r5（配置路由器）#邻居100.255.254.6激活r5
（设置路由器）#地址族vpnv6
r5（配置路由器）#邻居100.255.254.3激活
r5（配置路由器）#邻居100.255.254.4激活
r5（配置路由器）#邻居100.255.254.6激活

中文教育网-r6：

r6>和
r6#conf t
r6（配置）#路由器bgp 10000
r6（配置路由器）#邻居100.255.254.3远程为10000 r6
r6（配置路由器）#邻居100.255.254.3激活
r6（配置路由器）#邻居100.255.254.4激活
r6（配置路由器）#邻居100.255.254.5激活r6
（设置路由器）#地址族vpnv6
r6（配置路由器）#邻居100.255.254.3激活
r6（配置路由器）#邻居100.255.254.4激活
r6（配置路由器）#邻居100.255.254.5已激活

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On all routers r3, r4, r5, r6

```
#conf t
#mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
#mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
#no mpls ip propagate-ttl forwarded
```

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在所有路由器r3、r4、r5、r6上

```
#conft

#mpls标签模式所有vrf协议bgp-vpnv6每个vrf
#没有转发mpls ip传播ttl
```

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```
#mpls标签模式所有vrf协议bgp-vpnv4每个vrf
```

Verification:

Chinesedumps.com-r3#sh bgp vpnv4 unicast allsummary

```
chinesedumps-r3#sh bgp vpnv4 unicast all  summary
BGP router identifier 100.255.254.3, local AS number 10000
BGP table version is 166, main routing table version 166
50 network entries using 12800 bytes of memory
65 path entries using 8840 bytes of memory
23/12 BGP path/bestpath attribute entries using 6992 bytes of memory
4 BGP AS-PATH entries using 112 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 28768 total bytes of memory
BGP activity 67/17 prefixes, 156/91 paths, scan interval 60 secs
50 networks peaked at 11:05:01 May 4 2021 UTC (00:15:08.660 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.3.11.2      4        65001     251     276      166    0    0 03:45:18       15
100.3.21.2      4        65002     325     282      166    0    0 03:44:52       36
100.255.254.4   4        10000      10      20      166    0    0 00:04:55        0
100.255.254.5   4        10000      24      32      166    0    0 00:16:01        3
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.255.254.6   4        10000      22      29      166    0    0 00:12:52        3

chinesedumps-r3#
```

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Chinesedumps.com-r3#sh bgp vpnv6 unicast all summary

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```
chinesedumps-r3#sh bgp vpnv6 unicast all  summary
BGP router identifier 100.255.254.3, local AS number 10000
BGP table version is 1, main routing table version 1
```

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```
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.255.254.4   4        10000      11      21        1    0    0 00:06:19        0
100.255.254.5   4        10000      26      33        1    0    0 00:17:25        0
100.255.254.6   4        10000      23      31        1    0    0 00:14:12        0
```

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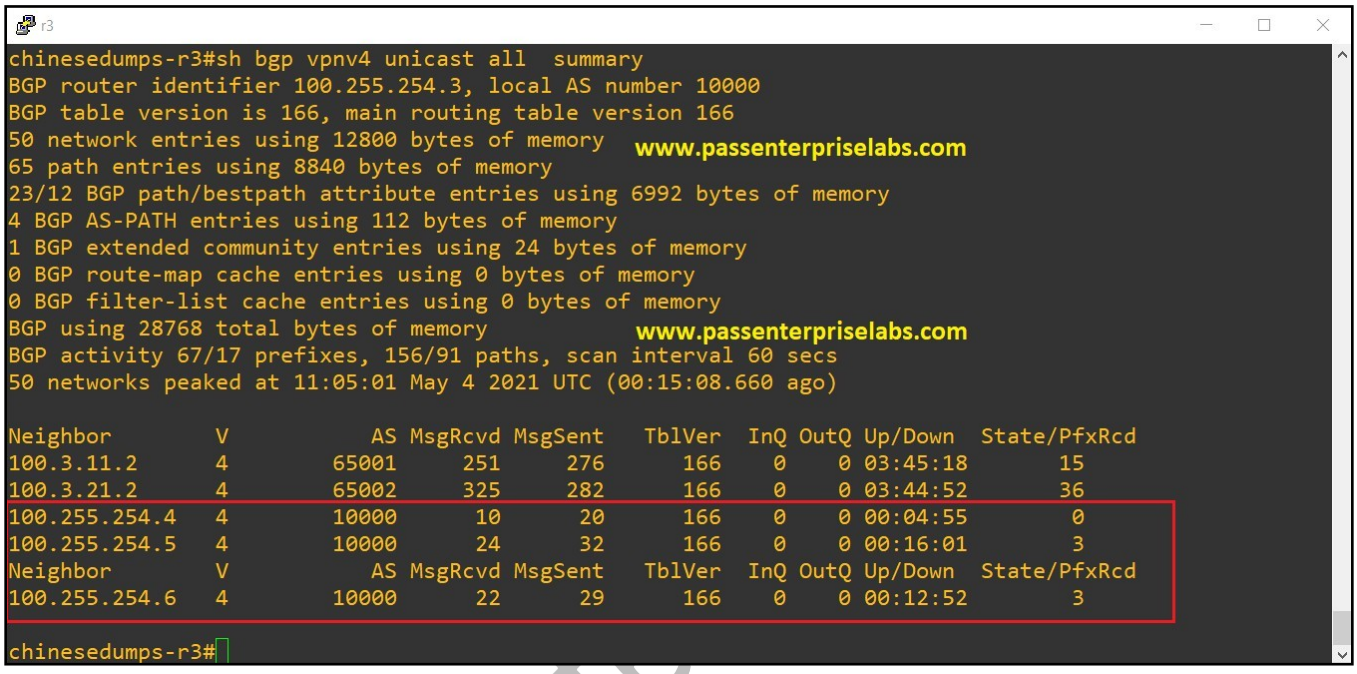
```
chinesedumps-r3#
chinesedumps-r3#
chinesedumps-r3#
chinesedumps-r3#
```

验证：

Chinese edumps.com-r3#sh bgp vpnv4单播全部摘要



```
chinesedumps-r3#sh bgp vpnv4 unicast all  summary
BGP router identifier 100.255.254.3, local AS number 10000
BGP table version is 166, main routing table version 166
50 network entries using 12800 bytes of memory
65 path entries using 8840 bytes of memory
23/12 BGP path/bestpath attribute entries using 6992 bytes of memory
4 BGP AS-PATH entries using 112 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 28768 total bytes of memory
BGP activity 67/17 prefixes, 156/91 paths, scan interval 60 secs
50 networks peaked at 11:05:01 May 4 2021 UTC (00:15:08.660 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.3.11.2      4        65001     251     276      166    0    0 03:45:18       15
100.3.21.2      4        65002     325     282      166    0    0 03:44:52       36
100.255.254.4   4        10000      10      20      166    0    0 00:04:55        0
100.255.254.5   4        10000      24      32      166    0    0 00:16:01        3
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.255.254.6   4        10000      22      29      166    0    0 00:12:52        3

chinesedumps-r3#
```

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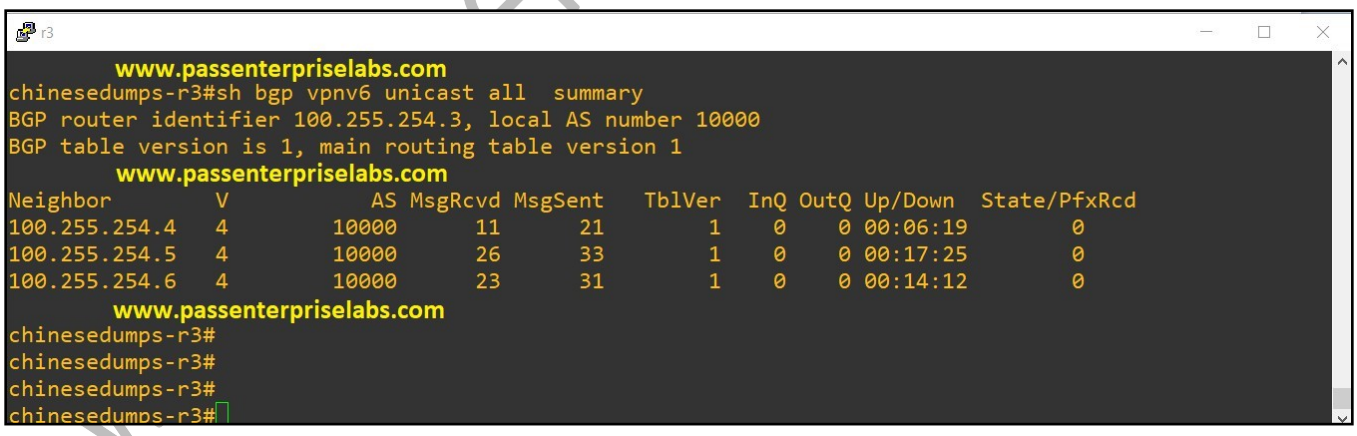
Chinese edumps.com-r3#sh bgp vpnv6单播全部摘要



```
        www.passenterpriselabs.com
chinesedumps-r3#sh bgp vpnv6 unicast all  summary
BGP router identifier 100.255.254.3, local AS number 10000
BGP table version is 1, main routing table version 1
        www.passenterpriselabs.com
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.255.254.4   4        10000      11      21        1    0    0 00:06:19        0
100.255.254.5   4        10000      26      33        1    0    0 00:17:25        0
100.255.254.6   4        10000      23      31        1    0    0 00:14:12        0
        www.passenterpriselabs.com
chinesedumps-r3#
chinesedumps-r3#
chinesedumps-r3#
```

Chinesedumps.com-r4#sh bgp vpnv4 unicast allsummary

```
r4
chinesedumps-r4# www.passenterpriselabs.com
chinesedumps-r4#sh bgp vpnv4 uni all summ
BGP router identifier 100.255.254.4, local AS number 10000
BGP table version is 3, main routing table version 3
1 network entries using 256 bytes of memory
1 path entries using 136 bytes of memory
1/1 BGP path/bestpath attribute entries using 304 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 720 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs
1 networks peaked at 11:15:08 May 4 2021 UTC (00:00:12.603 ago)
                www.passenterpriselabs.com
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.4.30.2      4        65003       0       0        1    0    0 never    Idle
100.255.254.3   4        10000      15       5        3    0    0 00:00:06        0
100.255.254.5   4        10000       7       5        3    0    0 00:00:03        0
100.255.254.6   4        10000       8       5        3    0    0 00:00:12        1

chinesedumps-r4#
chinesedumps-r4# www.passenterpriselabs.com
chinesedumps-r4#
chinesedumps-r4#
```

Chinesedumps.com-r4#sh bgp vpnv6 unicast all summary

```
r4
chinesedumps-r4#
chinesedumps-r4# www.passenterpriselabs.com
chinesedumps-r4#sh bgp vpnv6 uni all summ
BGP router identifier 100.255.254.4, local AS number 10000
BGP table version is 1, main routing table version 1
                www.passenterpriselabs.com
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
2001:DB8:4:30::2
                4        65003       0       0        1    0    0 never    Idle
100.255.254.3   4        10000      19       9        1    0    0 00:03:49        0
100.255.254.5   4        10000      11       8        1    0    0 00:03:46        0
100.255.254.6   4        10000      12       9        1    0    0 00:03:55        0

                www.passenterpriselabs.com
chinesedumps-r4#
chinesedumps-r4#
```

Chinese edumps.com-r4#sh bgp vpnv4单播全部摘要

```
chinesedumps-r4# www.passenterpriselabs.com
chinesedumps-r4#sh bgp vpnv4 uni all summ
BGP router identifier 100.255.254.4, local AS number 10000
BGP table version is 3, main routing table version 3
1 network entries using 256 bytes of memory
1 path entries using 136 bytes of memory
1/1 BGP path/bestpath attribute entries using 304 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 720 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs
1 networks peaked at 11:15:08 May 4 2021 UTC (00:00:12.603 ago)
                www.passenterpriselabs.com
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.4.30.2      4        65003       0       0        1    0    0 never    Idle
100.255.254.3   4        10000      15       5        3    0    0 00:00:06        0
100.255.254.5   4        10000       7       5        3    0    0 00:00:03        0
100.255.254.6   4        10000       8       5        3    0    0 00:00:12        1

chinesedumps-r4#
chinesedumps-r4# www.passenterpriselabs.com
chinesedumps-r4#
chinesedumps-r4#
```

Chinese edumps.com-r4#sh bgp vpnv6单播全部摘要

```
chinesedumps-r4#
chinesedumps-r4# www.passenterpriselabs.com
chinesedumps-r4#sh bgp vpnv6 uni all summ
BGP router identifier 100.255.254.4, local AS number 10000
BGP table version is 1, main routing table version 1
                www.passenterpriselabs.com
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
2001:DB8:4:30::2
                4        65003       0       0        1    0    0 never    Idle
100.255.254.3   4        10000      19       9        1    0    0 00:03:49        0
100.255.254.5   4        10000      11       8        1    0    0 00:03:46        0
100.255.254.6   4        10000      12       9        1    0    0 00:03:55        0

                www.passenterpriselabs.com
chinesedumps-r4#
chinesedumps-r4#
```

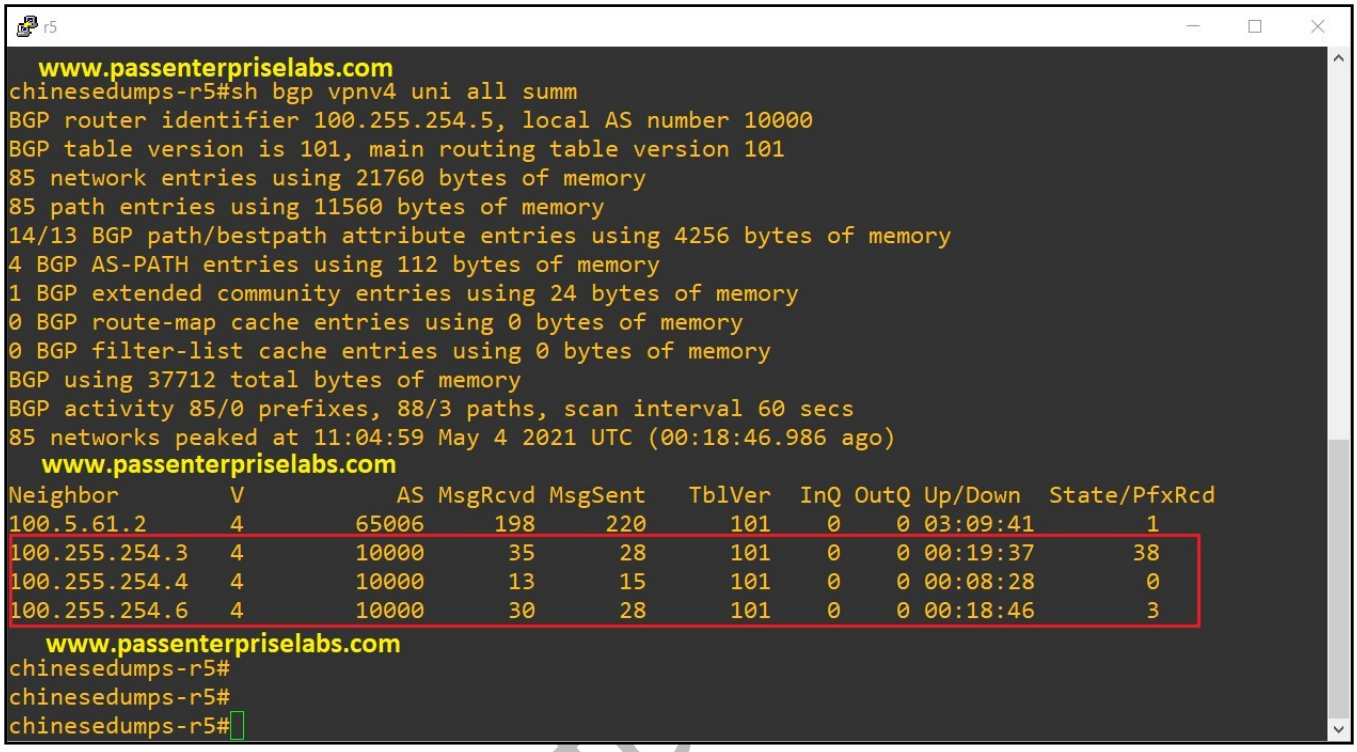
Chinesedumps.com-r5#sh bgp vpnv4 unicast allsummary

```
r5
www.passenterpriselabs.com
chinesedumps-r5#sh bgp vpnv4 uni all summ
BGP router identifier 100.255.254.5, local AS number 10000
BGP table version is 101, main routing table version 101
85 network entries using 21760 bytes of memory
85 path entries using 11560 bytes of memory
14/13 BGP path/bestpath attribute entries using 4256 bytes of memory
4 BGP AS-PATH entries using 112 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 37712 total bytes of memory
BGP activity 85/0 prefixes, 88/3 paths, scan interval 60 secs
85 networks peaked at 11:04:59 May 4 2021 UTC (00:18:46.986 ago)
www.passenterpriselabs.com
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.5.61.2      4        65006     198     220      101    0    0 03:09:41        1
100.255.254.3   4        10000      35      28      101    0    0 00:19:37       38
100.255.254.4   4        10000      13      15      101    0    0 00:08:28        0
100.255.254.6   4        10000      30      28      101    0    0 00:18:46        3
www.passenterpriselabs.com
chinesedumps-r5#
chinesedumps-r5#
chinesedumps-r5#
```

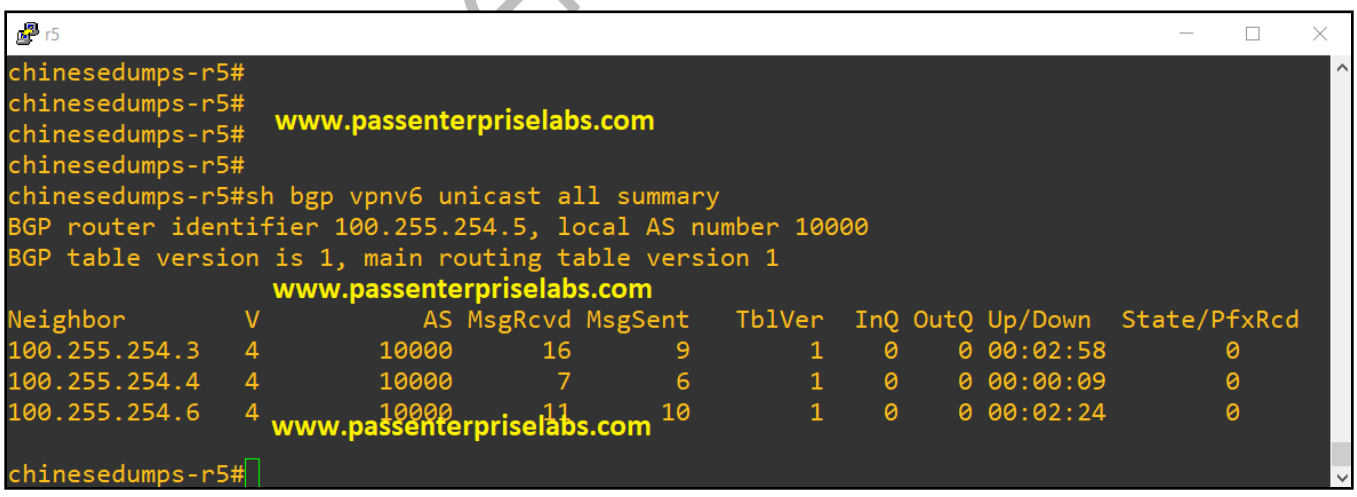
Chinesedumps.com-r5#sh bgp vpnv6 unicast all summary

```
r5
chinesedumps-r5#
chinesedumps-r5#
chinesedumps-r5#
chinesedumps-r5#
www.passenterpriselabs.com
chinesedumps-r5#sh bgp vpnv6 unicast all summary
BGP router identifier 100.255.254.5, local AS number 10000
BGP table version is 1, main routing table version 1
www.passenterpriselabs.com
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.255.254.3   4        10000      16       9        1    0    0 00:02:58        0
100.255.254.4   4        10000       7       6        1    0    0 00:00:09        0
100.255.254.6   4        10000      11      10        1    0    0 00:02:24        0
www.passenterpriselabs.com
chinesedumps-r5#
```

Chinese edumps.com-r5#sh bgp vpnv4单播全部摘要



Chinese edumps.com-r5#sh bgp vpnv6单播全部摘要



Chinesedumps.com-r6#sh bgp vpnv4 unicast allsummary

```
r6
chinesedumps-r6#
chinesedumps-r6#  www.passenterpriselabs.com
chinesedumps-r6#sh bgp vpnv4 unicast all summary
BGP router identifier 100.255.254.6, local AS number 10000
BGP table version is 86, main routing table version 86
76 network entries using 19456 bytes of memory
76 path entries using 10336 bytes of memory
11/11 BGP path/bestpath attribute entries using 3344 bytes of memory
3 BGP AS-PATH entries using 88 bytes of memory
2 BGP extended community entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 33272 total bytes of memory
BGP activity 76/0 prefixes, 80/4 paths, scan interval 60 secs
76 networks peaked at 09:11:52 May 5 2021 UTC (00:02:28.375 ago)
           www.passenterpriselabs.com
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.6.62.2      4        65006       0       0        1    0    0 never    Idle
100.6.70.2      4        65007       0       0        1    0    0 never    Active
100.255.254.3   4        10000      19      11       86    0    0 00:04:59        34
100.255.254.4   4        10000       9      10       86    0    0 00:02:28         1
100.255.254.5   4        10000      12      14       86    0    0 00:04:42         2
                  www.passenterpriselabs.com
chinesedumps-r6#
```

Chinesedumps.com-r6#sh bgp vpnv6 unicast all summary

```
r6
   www.passenterpriselabs.com
chinesedumps-r6#sh bgp vpnv6 unicast all summary
BGP router identifier 100.255.254.6, local AS number 10000
BGP table version is 1, main routing table version 1

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.255.254.3   4        10000       0       0        1    0    0 never    (NoNeg)
100.255.254.4   4        10000      11      12        1    0    0 00:03:48        0
100.255.254.5   4        10000      14      15        1    0    0 00:06:02        0
   www.passenterpriselabs.com
chinesedumps-r6#
```

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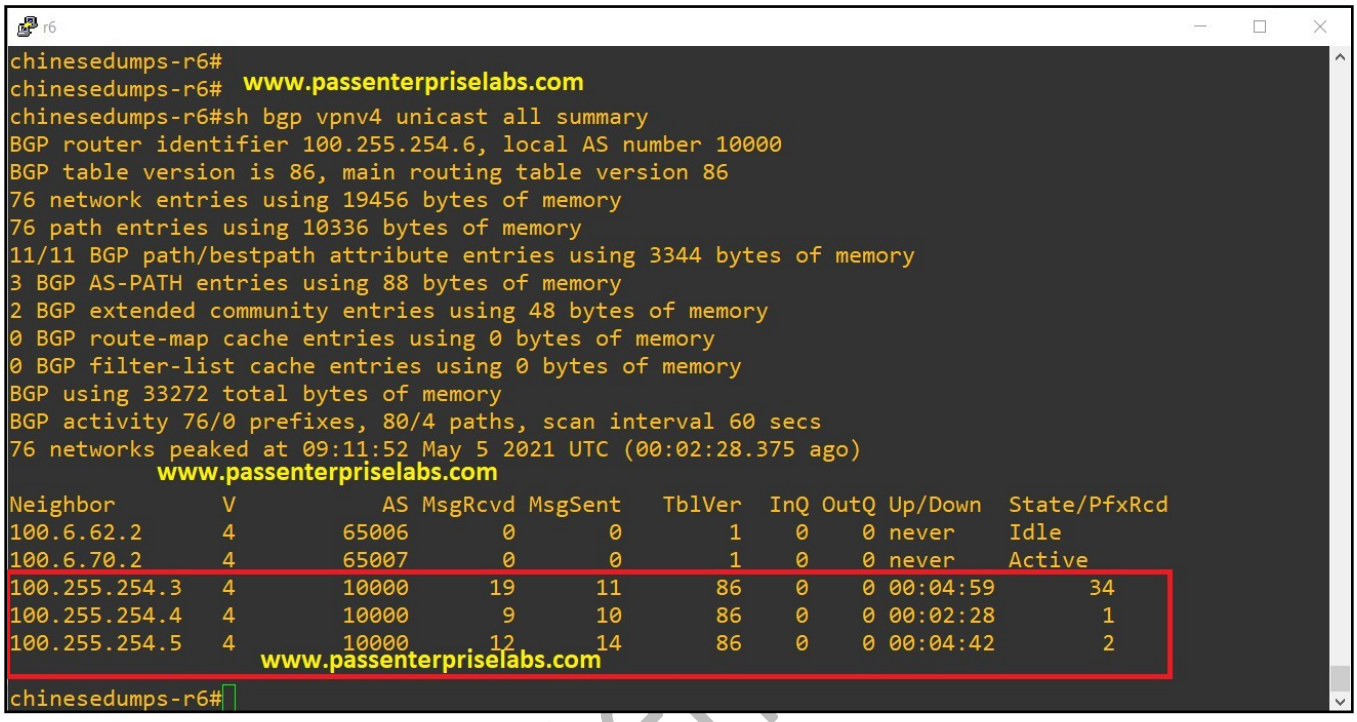
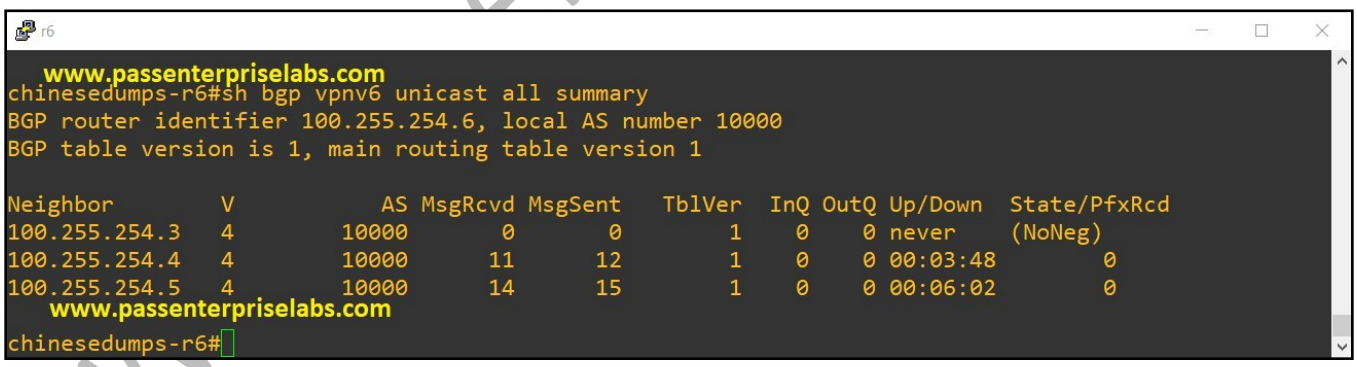
Chinese edumps.com-r6#sh bgp vpnv4单播全部摘要



Chinese edumps.com-r6#sh bgp vpnv6单播全部摘要



1.11: Fixing Broken DMVPN between Dc and Branches #3 and #4

Correct the configuration issues resulting in broken DMVPN tunnel connectivity between DC, Branch3 and Branch4 according to these requirements:

- The DMVPN must operate in IPSec-protected phase 3 mode.
- Using the FVRF approach, safeguard the DMVPN operation against any potential recursive routing issues involving the tunnel.
- Do not create any new VRFs.
- Do not change the tunnel source commands on Tunnel interfaces.
- On Spokes, do not add new BGP neighbors; reuse those that are currently up while changing their VRF membership as needed.
- It is not allowed to modify configuration on DC r24 to complete this entire task.

3 Point

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1.11： 修复Dc和分支之间损坏的DMVPN#3和#4

根据以下要求更正导致DC、Branch3和Branch4之间DMVPN隧道连接中断的配置问题：

- DMVPN必须在IPSec保护的第3阶段模式下运行。
- 使用FVRF方法，保护DMVPN操作免受涉及隧道的任何潜在递归路由问题的影响。
- 不要创建任何新的VRF。
- 不要更改隧道接口上的隧道源命令。
- 在辐条上，不要添加新的BGP邻居；在根据需要更改其VRF成员资格的同时重用当前正在运行的那些。
- 不允许修改DC r24上的配置以完成整个任务。

3分

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Solution:

Chinesedumps.com-r61:

```
r61>en
r61#conf t
r61(config)#int lo 0
r61(config-if)#vrf forwarding WAN
r61(config-if)#ip address 10.6.255.61 255.255.255.255

r61(config)#int g0/0
r61(config-if)#vrf forwarding WAN
r61(config-if)#ip address 100.5.61.2 255.255.255.252

r61(config)#router bgp 65006
r61(config-router)#address-family ipv4 vrf WAN
r61(config-router)#network 10.6.255.61 mask 255.255.255.255
r61(config-router)#neighbor 100.5.61.1 remote-as 10000
r61(config-router)#neighbor 100.5.61.1 act

r61(config)#interface tunnel 0
r61(config-if)#no  ip nhrp map 10.2.255.24 10.200.0.1
r61(config-if)#ip nhrp map 10.200.0.1 10.2.255.24
r61(config-if)#no ip nhrp redirect
r61(config-if)#ip nhrp shortcut
r61(config-if)#tunnel vrf WAN
r61(config-if)#ip mtu 1440

r61(config)#crypto isakmp policy 10
r61(config)#no hash md5
r61(config)#no crypto isakmp key cisco address 0.0.0.0
r61(config)#crypto keyring KR vrf WAN
r61(config)#pre-shared-key address 0.0.0.0 0.0.0.0 key cisco
r61(config)#crypto ipsec profile prof
```

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解决方案：

中文教育网-r61：

r61>和
r61#conf t
r61（配置）#int lo 0
r61（配置if）#vrf转发广域网
r61（配置if）#ip地址10.6.255.61 255.255.255

r61（配置）#int g0/0
r61（配置if）#vrf转发广域网
r61（配置if）#ip地址100.5.61.2 255.255.255.252

r61（配置）#路由器bgp 65006
r61（配置路由器）#地址家族ipv4 vrf广域网
r61（配置路由器）#网络10.6.255.61掩码255.255.255
r61（配置路由器）#邻居100.5.61.1远程为10000
r61（配置路由器）#邻居100.5.61.1行为

r61（config）#接口隧道0
r61（配置if）#无ip nhrp映射10.2.255.24 10.200.0.1
r61（配置if）#ip nhrp映射10.200.0.1
10.2.255.24 r61（设置if）#无ip nhrp重定
向
r61（配置if）#ip-nhrp快
捷方式r61（设置if）#隧道
vrf WAN r61（设定if）#ip
mtu 1440

r61（配置）#加密isakmp策略10 r61（设置）
#无哈希md5
r61（配置）#无加密isakmp密钥思科地址0.0.0.0 r61（设置）
#加密密钥环KR vrf WAN
r61（config）#预共享密钥地址0.0.0.0 0.0.0.0密钥cisco
r61（配置）#加密ipsec配置文件配置文件

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```
r61(config)#router eigrp ccie
r61(config-router)#address-family ipv4 unicast autonomous-system 65006
r61(config-af-interface)#af-interface default
r61(config-af-interface)#passive-interface
r61(config-router)#exit-af-interface
r61(config-af-interface)#af-interface Tunnel0
r61(config-af-interface)#no passive-interface
r61(config-af-interface)#exit-af-interface
```

Chinesedumps.com-r62:

```
r62>en
r62#conf t
r62(config)#int lo 0
r62(config-if)#vrf forwarding WAN
r62(config-if)#ip address 10.6.255.62 255.255.255.255

r62(config)#int g0/0
r62(config-if)#vrf forwarding WAN
r62(config-if)#ip address 100.6.62.2 255.255.255.252
r62(config-if)#router bgp 65006
r62(config-if)#address-family ipv4 vrf WAN
r62(config-if)#network 10.6.255.62 mask 255.255.255.255
r62(config-if)#neighbor 100.6.62.1 remote-as 10000

r62(config)#int tunnel 0
r62(config-if)#no  ip nhrp map 10.2.255.24 10.200.0.1
r62(config-if)#ip nhrp map 10.200.0.1 10.2.255.24
r62(config-if)#no ip nhrp redirect
r62(config-if)#ip nhrp network-id 1010
r62(config-if)#tunnel vrf WAN
r62(config-if)#ip mtu 1440
r62(config-if)#ip nhrp shortcut


r62(config)#crypto isakmp policy 10
r62(config)#no hash md5
r62(config)#no crypto isakmp key cisco address 0.0.0.0
r62(config)#crypto keyring KR vrf WAN
r62(config)#pre-shared-key address 0.0.0.0 0.0.0.0 key cisco
r62(config)#crypto ipsec profile prof
```

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r61（配置）#路由器eigrp-ccie
r61（配置路由器）#地址家族ipv4单播自治系统65006 r61（设置af
接口）#af接口默认值
r61（配置af接口）#无源接口r61（设置
路由器）#退出af接口r61

中文教育网-r62：

r62>和
r62#conf t
r62（配置）#int lo 0
r62（配置if）#vrf转发广域网
r62（配置if）#ip地址10.6.255.62 255.255.255

r62（配置）#int g0/0
r62（配置if）#vrf转发广域网
r62（配置if）#ip地址100.6.62.2 255.255.252
r62（配置if）#路由器bgp 65006
r62（配置if）#地址家族ipv4 vrf广域网
r62（配置if）#网络10.6.255.62掩码255.255.255
r62（配置if）#邻居100.6.62.1远程为10000

r62（config）#int隧道0
r62（配置if）#无ip nhrp映射10.2.255.24 10.200.0.1
r62（配置if）#ip nhrp映射10.200.0.1
10.2.255.24 r62（设置if）#无ip nhrp重定
向
r62（配置if）#ip nhrp网络id
1010 r62（设置if）#隧道vrf WAN
r62（设定if）#ip-mtu 1440
r62（配置if）#ip nhrp快捷方式

r62（配置）#加密isakmp策略10 r62（设置）
#无哈希md5
r62（配置）#无加密isakmp密钥思科地址0.0.0.0 r62（设置）
#加密密钥环KR vrf WAN
r62（config）#预共享密钥地址0.0.0.0 0.0.0.0密钥cisco
r62（配置）#加密ipsec配置文件配置文件

www.cciepenterpriselabs.com

r61(config)#router eigrp ccie
r61(config-router)#address-family ipv4 unicast autonomous-system 65006
r61(config-af-interface)#af-interface default
r61(config-af-interface)#passive-interface
r61(config-router)#exit-af-interface
r61(config-af-interface)#af-interface Tunnel0
r61(config-af-interface)#no passive-interface
r61(config-af-interface)#exit-af-interface

Chinesedumps.com-r70:

r70>en
r70#conf t
r70(config)#int tunnel 0
r70(config-if)#tunnel vrf WAN
r70(config-if)#ip mtu 1440
r70(config-if)#no ip nhrp redirect
r70(config-if)#ip nhrp shortcut

r70(config)#crypto isakmp policy 10
r70(config)#no hash md5
r70(config)#crypto ipsec profile prof

r70(config)#router eigrp ccie
r70(config-router)#address-family ipv4 unicast autonomous-system 65006
r70(config-router)#af-interface Tunnel0
r70(config-af-interface)#no passive-interface
r70(config-router)#exit-af-interface

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r61（配置）#路由器eigrp-ccie
r61（配置路由器）#地址家族ipv4单播自治系统65006 r61（设置af
接口）#af接口默认值
r61（配置af接口）#无源接口r61（设
置路由器）#退出af接口
r61（配置af接口）#af接口Tunnel0
r61（配置af接口）#无被动接口r61（设
置af接口）#exit af接口

中文教育网-r70：

r70>和
r70#conf t
r70（配置）#int隧道0 r70（
如果配置）#隧道vrf WAN r70

r70（配置）#加密isakmp策略10 r70（
设置）#无哈希md5 r70（设定）#加密
ipsec配置文件配置

r70（配置）#路由器eigrp-ccie
r70（配置路由器）#地址家族ipv4单播自治系统65006 r70（设置路
由器）#af接口Tunnel0
r70（配置af接口）#无无源接口r70（设
置路由器）#退出af接口

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Solution Steps:

Match the crypto configs on r24 and then configure the spokes (r70 , r61 ,r62)

Faults On Spoke :

1. Hashing needs to be removed on spokes as there is no hash configured on r24 in pre-configuration

```
crypto isakmp policy 10
 no hash md5
```

2.

a. On r24 legacy key is configured
b. on spokes we will need to complete the key ring configuration for isakmp

```
crypto keyring KR vrf WAN
 pre-shared-key address 0.0.0.0 0.0.0.0 key cisco
```

3. The DMVPN must operate in IPSec-protected phase 3 mode.

a. On hub (r24)" **ip nhrp redirect** " : - This command says that we are running phase 3 mode
b. On Spokes (r61,r62,r70) " **ip nhrp shortcut** "

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解决方案步骤：

匹配r24上的加密配置，然后配置轮辐（r70、r61、r62）轮辐故障：

1. 需要删除轮辐上的哈希，因为在预配置中r24上没有配置哈希
   加密isakmp策略10无
   哈希md5

2.
a. 在r24上配置了旧密钥
b. 在辐条上，我们需要完成isakmp的钥匙圈配置
   加密钥匙圈KR vrf WAN
   预共享密钥地址0.0.0.0 0.0.0.0密钥cisco
3. DMVPN必须在IPSec保护的第3阶段模式下运行。
a. 在hub（r24）上“ip-nhrp-redirect”：-此命令表示我们正在运行第3阶段模式
b. 关于辐条（r61、r62、r70）“ip-nhrp快捷方式”

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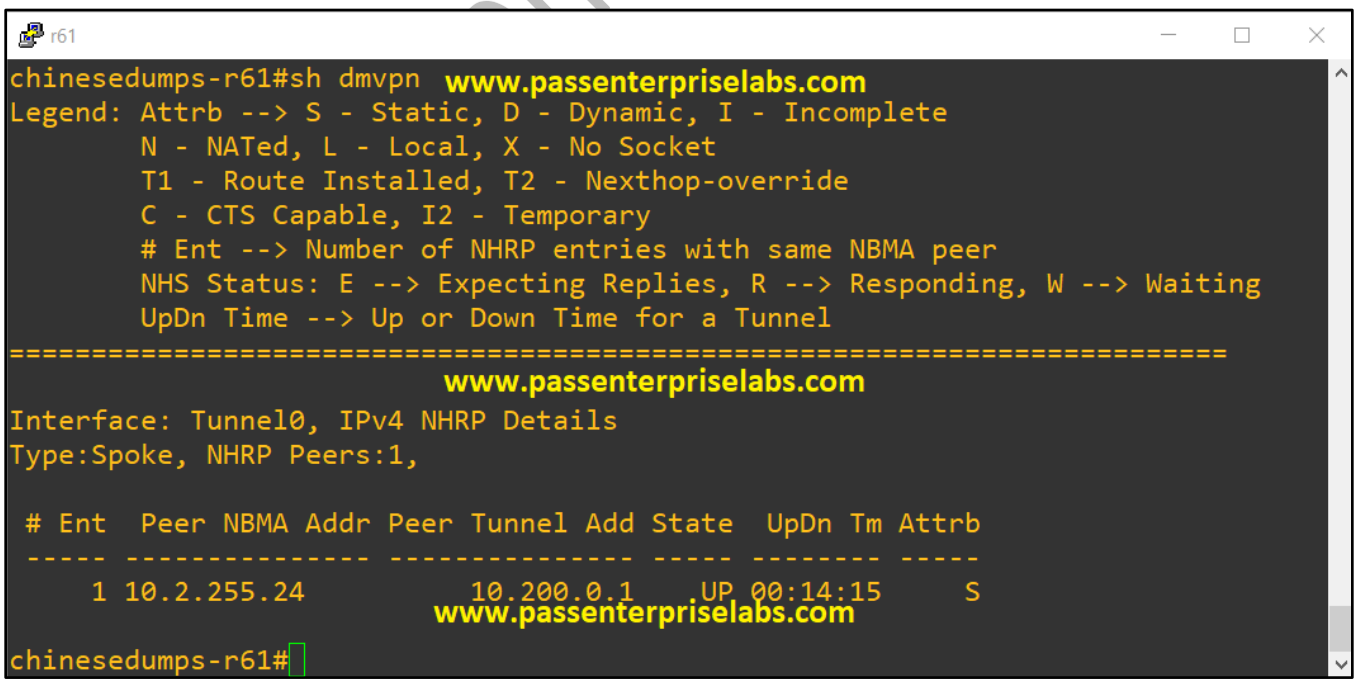
Verification:

Chinesedumps.com-r61#sh crypto session

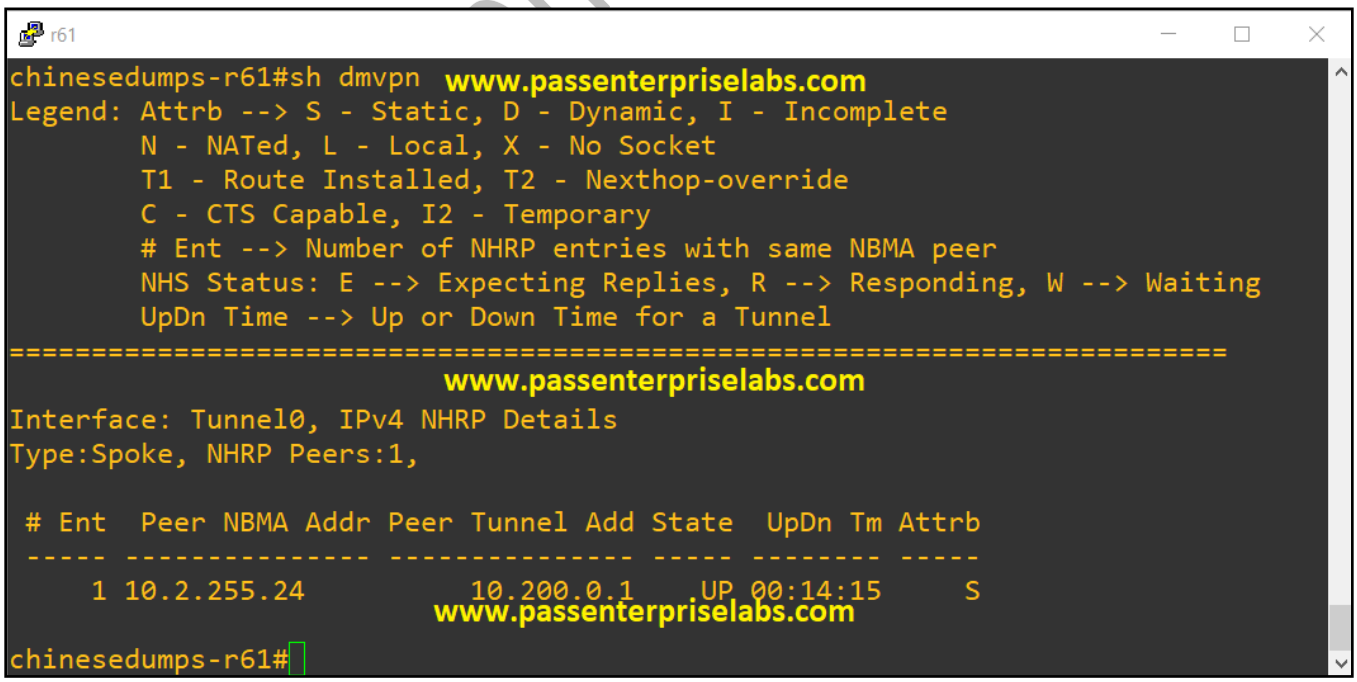
Chinesedumps.com-r61#sh dmvpn



验证：

Chineseedumps.com-r61#sh加密会话

Chinese edumps.com-r61#sh dmvpn



Chinesedumps.com-r61#sh ip eigrp neighbor

```
r61
EIGRP-IPv4 VR(ccie) Address-Family Neighbors for AS(65006)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q
Seq
                                                   (sec)         (ms)       Cnt
Num
0   10.200.0.1              Tu0                      14 01:55:28  309  1854  0
9
chinesedumps-r61#sh ip eigrp neighbors
EIGRP-IPv4 VR(ccie) Address-Family Neighbors for AS(65006)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
0   10.200.0.1              Tu0                      13 01:55:33  309  1854  0  9
chinesedumps-r61#
```

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Chinesedumps.com-r61#sh ip nhrp detail

```
r61
chinesedumps-r61#
chinesedumps-r61#
chinesedumps-r61#
chinesedumps-r61#
chinesedumps-r61#sh ip nhrp detail
10.200.0.1/32 via 10.200.0.1
   Tunnel0 created 00:30:01, never expire
   Type: static, Flags: used
   NBMA address: 10.2.255.24
   Preference: 255
chinesedumps-r61#
```

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Chinesedumps.com-r61#ping 10.200.0.70

```
r61
chinesedumps-r61#ping 10.200.0.70
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.200.0.70, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 119/209/373 ms
chinesedumps-r61#
chinesedumps-r61#
chinesedumps-r61#traceroute  10.200.0.70
Type escape sequence to abort.
Tracing the route to 10.200.0.70
VRF info: (vrf in name/id, vrf out name/id)
  1 10.200.0.70 99 msec *  36 msec
chinesedumps-r61#
chinesedumps-r61#
chinesedumps-r61#
```

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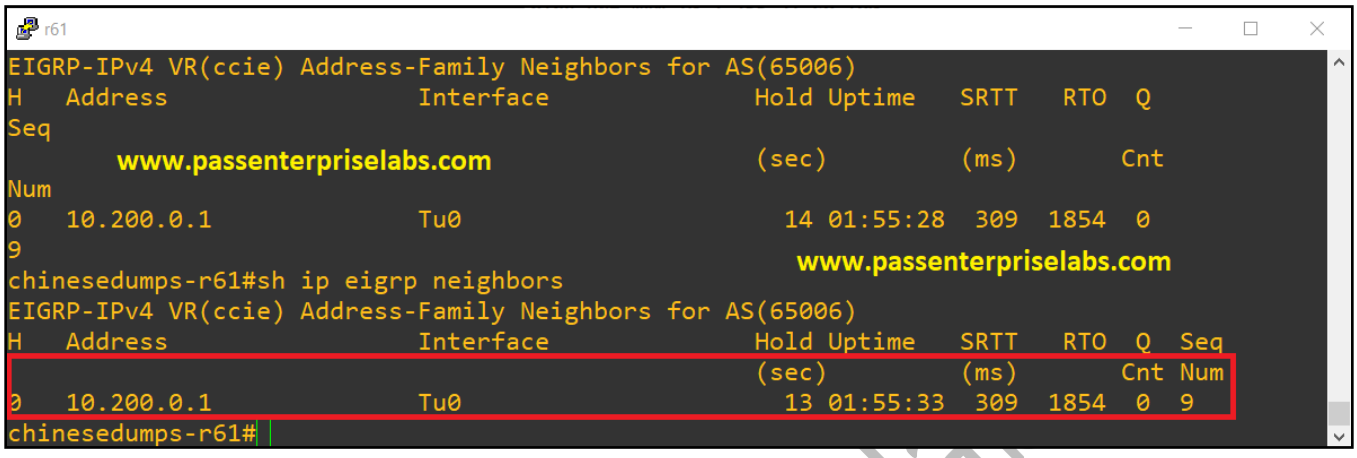
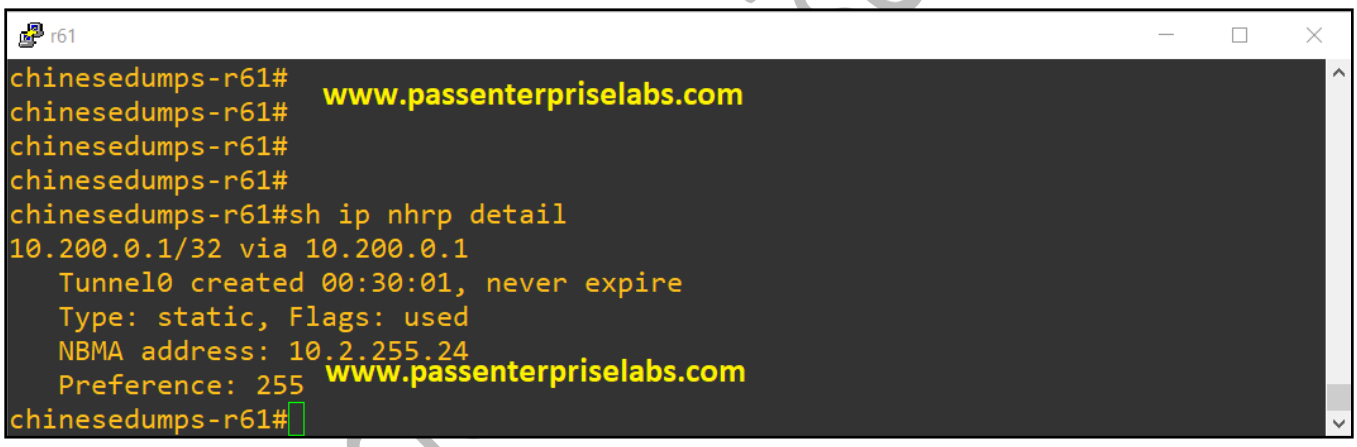
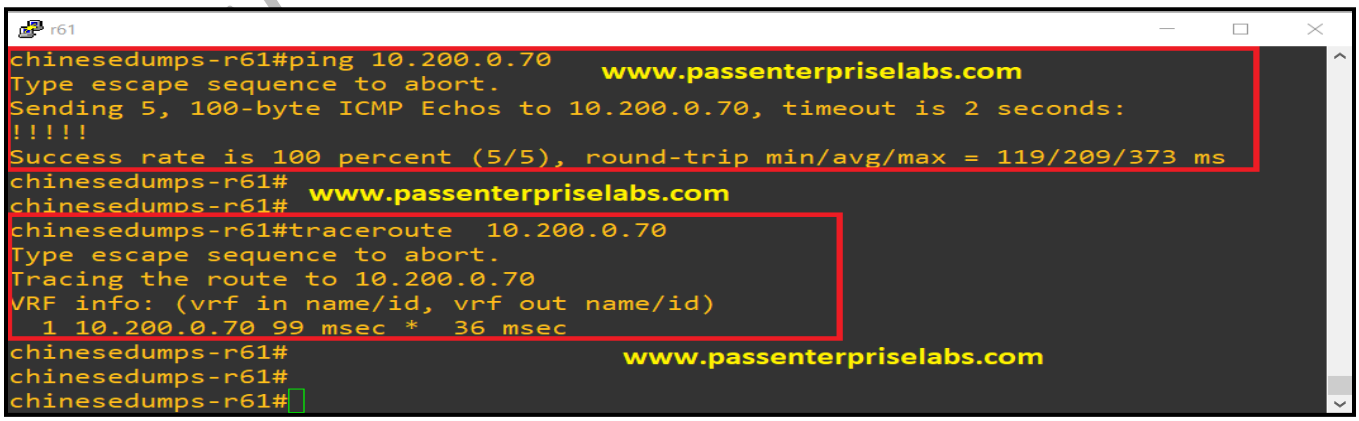
Chinese edumps.com-r61#sh ip eigrp邻居



Chinese edumps.com-r61#sh ip nhrp详细信息



中文教程com-r61#ping 10.200.0.70



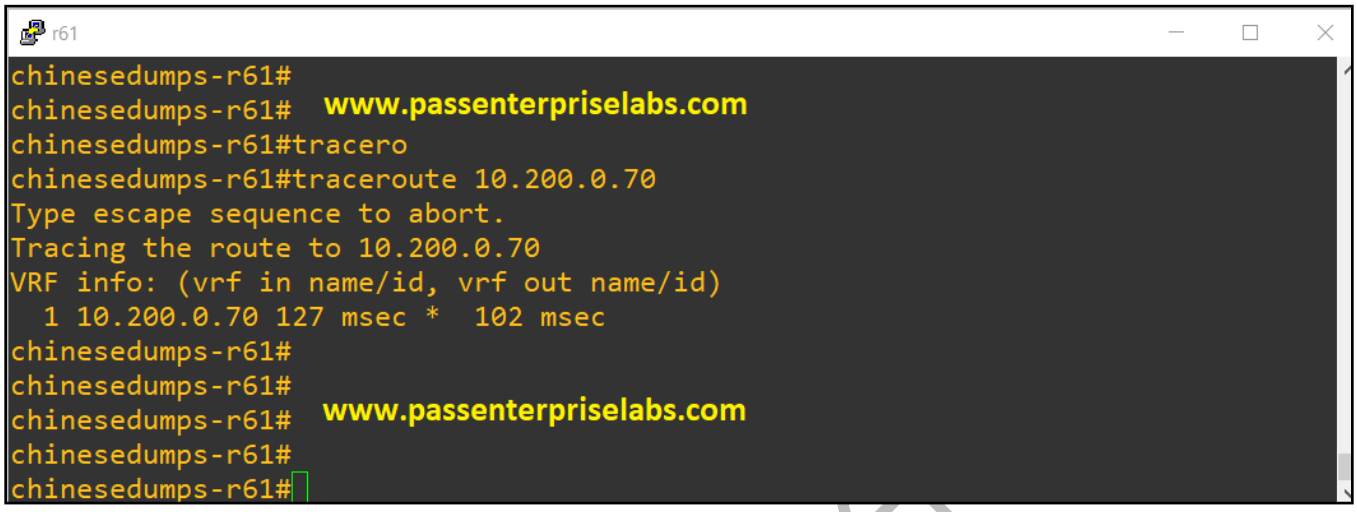
Chinesedumps.com-r61#traceroute 10.200.0.70

```
r61
chinesedumps-r61#
chinesedumps-r61#   www.passenterpriselabs.com
chinesedumps-r61#tracero
chinesedumps-r61#traceroute 10.200.0.70
Type escape sequence to abort.
Tracing the route to 10.200.0.70
VRF info: (vrf in name/id, vrf out name/id)
  1 10.200.0.70 127 msec *  102 msec
chinesedumps-r61#
chinesedumps-r61#
chinesedumps-r61#   www.passenterpriselabs.com
chinesedumps-r61#
chinesedumps-r61#
```

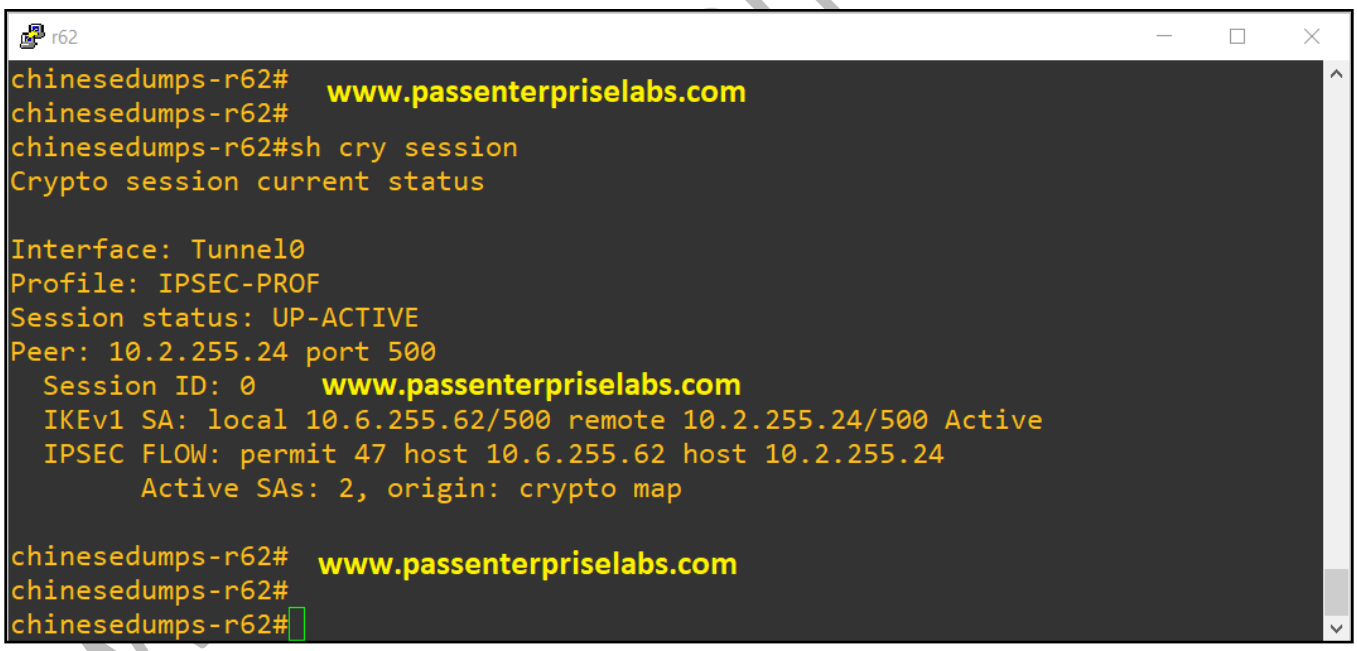
Chinesedumps.com-r62#sh crypto session

```
r62
chinesedumps-r62#   www.passenterpriselabs.com
chinesedumps-r62#
chinesedumps-r62#sh cry session
Crypto session current status

Interface: Tunnel0
Profile: IPSEC-PROF
Session status: UP-ACTIVE
Peer: 10.2.255.24 port 500
  Session ID: 0    www.passenterpriselabs.com
  IKEv1 SA: local 10.6.255.62/500 remote 10.2.255.24/500 Active
  IPSEC FLOW: permit 47 host 10.6.255.62 host 10.2.255.24
        Active SAs: 2, origin: crypto map

chinesedumps-r62#   www.passenterpriselabs.com
chinesedumps-r62#
chinesedumps-r62#
```

Chinese edumps.com-r61#追踪路线10.200.0.70

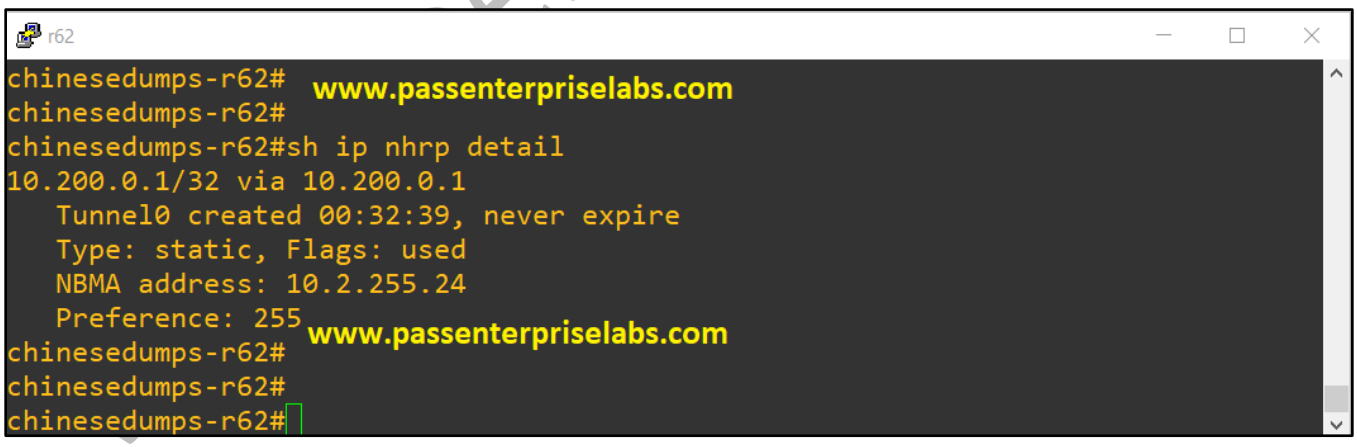


Chineseedumps.com-r62#sh加密会话



Chinesedumps.com-r62#sh dmvpn

Chinesedumps.com-r62#sh ip nhrp detail



Chinese edumps.com-r62#sh dmvpn

```
chinesedumps-r62#sh dmvpn  www.passenterpriselabs.com
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================
                            www.passenterpriselabs.com
Interface: Tunnel0, IPv4 NHRP Details
Type:Spoke, NHRP Peers:1,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
     1 10.2.255.24           10.200.0.1    UP 00:02:13     S
                            www.passenterpriselabs.com
chinesedumps-r62#
```

Chinese edumps.com-r62#sh ip nhrp详细信息

```
chinesedumps-r62#  www.passenterpriselabs.com
chinesedumps-r62#
chinesedumps-r62#sh ip nhrp detail
10.200.0.1/32 via 10.200.0.1
   Tunnel0 created 00:32:39, never expire
   Type: static, Flags: used
   NBMA address: 10.2.255.24
   Preference: 255
                            www.passenterpriselabs.com
chinesedumps-r62#
chinesedumps-r62#
chinesedumps-r62#
```

Chinesedumps.com-r70#sh crypto session

```
r70
chinesedumps-r70#  www.passenterpriselabs.com
chinesedumps-r70#sh cry session
Crypto session current status

Interface: Tunnel0
Profile: IPSEC-PROF
Session status: UP-ACTIVE
Peer: 10.2.255.24 port 500
  Session ID: 0    www.passenterpriselabs.com
  IKEv1 SA: local 10.7.255.70/500 remote 10.2.255.24/500 Active
  IPSEC FLOW: permit 47 host 10.7.255.70 host 10.2.255.24
        Active SAs: 6, origin: crypto map

chinesedumps-r70#
chinesedumps-r70#  www.passenterpriselabs.com
chinesedumps-r70#
chinesedumps-r70#
```

Chinesedumps.com-r70#sh dmvpn

```
r70
chinesedumps-r70#sh dmvpn    www.passenterpriselabs.com
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================
                              www.passenterpriselabs.com
Interface: Tunnel0, IPv4 NHRP Details
Type:Spoke, NHRP Peers:1,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
     1 10.2.255.24          10.200.0.1    UP 00:02:11     S
                              www.passenterpriselabs.com
chinesedumps-r70#
```

Chineseedumps.com-r70#sh加密会话

```
r70
chinesedumps-r70# www.passenterpriselabs.com
chinesedumps-r70#sh cry session
Crypto session current status

Interface: Tunnel0
Profile: IPSEC-PROF
Session status: UP-ACTIVE
Peer: 10.2.255.24 port 500
  Session ID: 0   www.passenterpriselabs.com
  IKEv1 SA: local 10.7.255.70/500 remote 10.2.255.24/500 Active
  IPSEC FLOW: permit 47 host 10.7.255.70 host 10.2.255.24
        Active SAs: 6, origin: crypto map

chinesedumps-r70#
chinesedumps-r70# www.passenterpriselabs.com
chinesedumps-r70#
chinesedumps-r70#
```

Chinese edumps.com-r70#sh dmvpn

```
r70
chinesedumps-r70#sh dmvpn    www.passenterpriselabs.com
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================
                            www.passenterpriselabs.com
Interface: Tunnel0, IPv4 NHRP Details
Type:Spoke, NHRP Peers:1,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
     1 10.2.255.24          10.200.0.1    UP 00:02:11     S
                            www.passenterpriselabs.com
chinesedumps-r70#
```

Chinesedumps.com-r70#sh ip nhrp detail

```
r70
chinesedumps-r70#
chinesedumps-r70#  www.passenterpriselabs.com
chinesedumps-r70#
chinesedumps-r70#sh ip nhrp detail
10.200.0.1/32 via 10.200.0.1
   Tunnel0 created 00:31:23, never expire
   Type: static, Flags: used
   NBMA address: 10.2.255.24
   Preference: 255 www.passenterpriselabs.com
chinesedumps-r70#
chinesedumps-r70#
```

Chinesedumps.com-r24#sh ip eigrp neighbor

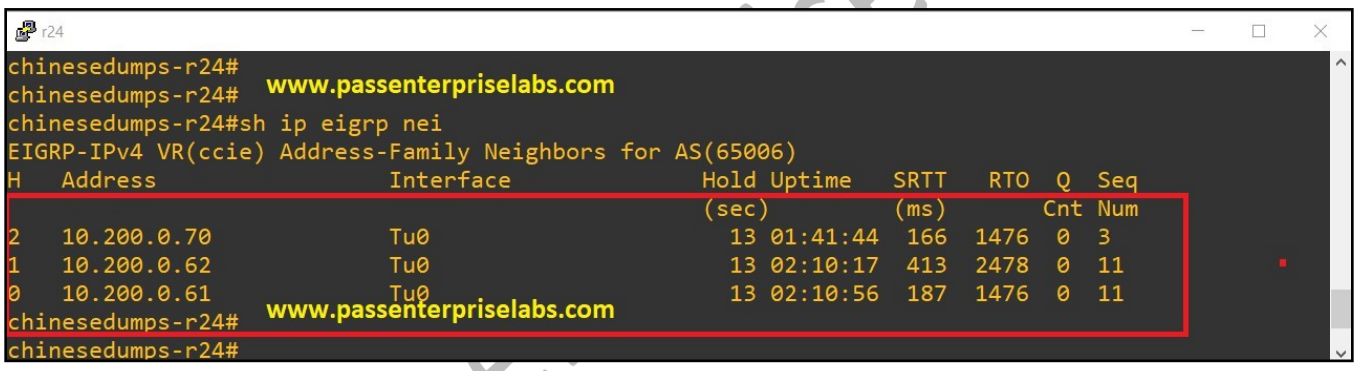
```
r24
chinesedumps-r24#
chinesedumps-r24#  www.passenterpriselabs.com
chinesedumps-r24#sh ip eigrp nei
EIGRP-IPv4 VR(ccie) Address-Family Neighbors for AS(65006)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
2   10.200.0.70             Tu0                      13 01:41:44  166  1476  0  3
1   10.200.0.62             Tu0                      13 02:10:17  413  2478  0  11
0   10.200.0.61             Tu0                      13 02:10:56  187  1476  0  11
chinesedumps-r24#  www.passenterpriselabs.com
chinesedumps-r24#
```

Chinesedumps.com-r24#sh ip route eigrp

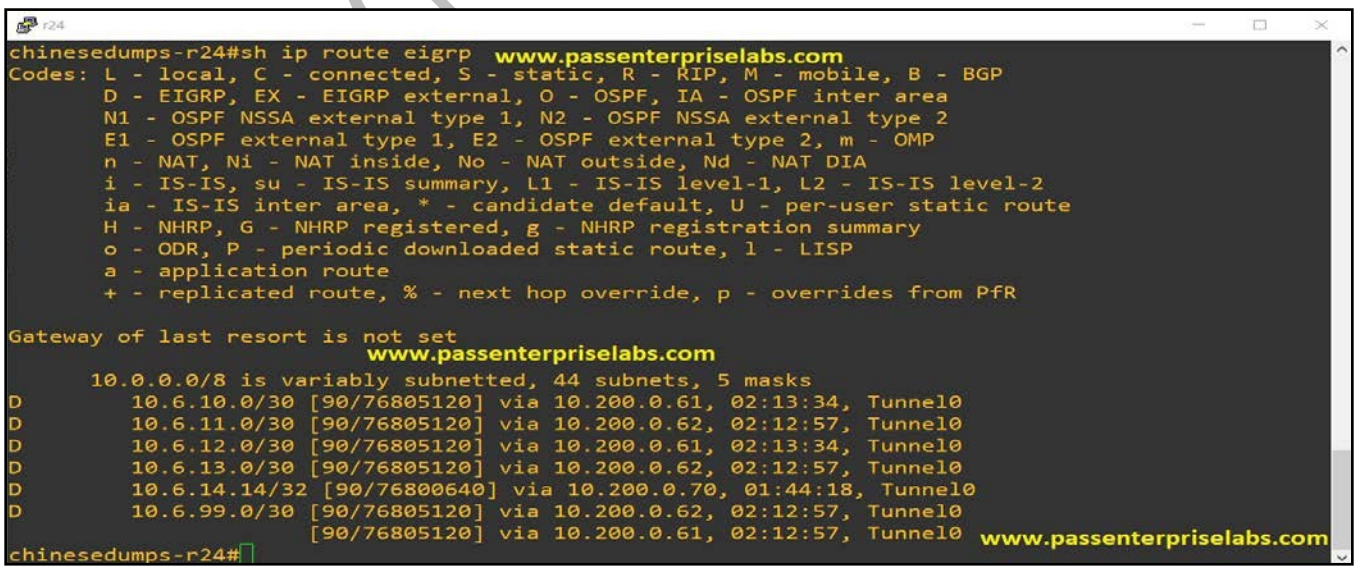
```
r24
chinesedumps-r24#sh ip route eigrp  www.passenterpriselabs.com
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set
                        www.passenterpriselabs.com
      10.0.0.0/8 is variably subnetted, 44 subnets, 5 masks
D        10.6.10.0/30 [90/76805120] via 10.200.0.61, 02:13:34, Tunnel0
D        10.6.11.0/30 [90/76805120] via 10.200.0.62, 02:12:57, Tunnel0
D        10.6.12.0/30 [90/76805120] via 10.200.0.61, 02:13:34, Tunnel0
D        10.6.13.0/30 [90/76805120] via 10.200.0.62, 02:12:57, Tunnel0
D        10.6.14.14/32 [90/76800640] via 10.200.0.70, 01:44:18, Tunnel0
D        10.6.99.0/30 [90/76805120] via 10.200.0.62, 02:12:57, Tunnel0
                      [90/76805120] via 10.200.0.61, 02:12:57, Tunnel0
chinesedumps-r24#                                  www.passenterpriselabs.com
```

Chinese edumps.com-r70#sh ip nhrp详细信息

Chinese edumps.com-r24#sh ip eigrp邻居



Chinese edumps.com-r24#sh ip路由igrp



Chinesedumps.com-r24#sh ip nhrp detail

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Chinese edumps.com-r24#sh ip nhrp详细信息

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1.12: Tuning EIGRP on DMVPN and DMVPN-enabled Sites

Optimize the DMVPN operation according to these requirements:

- Ensure that Branches #3 and #4 receive only a default route over EIGRP in DMVPN.
- The default route origination must be done on DC r24 without the use of any static routes, redistribution, or route filtering.
- It is not allowed to modify the configuration of r61 and r62 in Branch #3 to accomplish this task;
- It is allowed to add commands to the configuration of r70 in branch #4 to accomplish this task; none of the existing configuration on r70 may be removed to accomplish this task.

Configure sw601 and sw602 at Branch#3 according to these requirements:

- Routers r61 and r62 must not send EIGRP queries to sw601 and sw602.
- Switches sw601 and sw602 must allow advertising any current or future directly connected network to r61 and r62 after the network is added to EIGRP.
- Switches Sw601 and sw602 must continue to propagate the default route received from r61 and r62 to each other. To select the default route, use a prefix list with a “permit”- type entry only.
- Switches sw601 and sw602 must not propagate the default route back to r61 and r62.
- If the prefix list that allows the propagation of selected EIGRP-learned networks between sw601 and sw602 is modified in the future, the same set of networks must be disallowed from being advertised back to r61 and r62 automatically, without any additional commands

3 Points

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1.12： 在DMVPN和启用DMVPN的站点上调整EIGRP

根据以下要求优化DMVPN操作：

- 确保分支#3和#4仅通过DMVPN中的EIGRP接收默认路由。
- 默认路由发起必须在DC r24上完成，而不使用任何统计冲突、重新分发或路由过滤。
- 不允许修改分支#3中r61和r62的配置来完成此任务；
- 允许在分支#4的r70配置中添加命令以完成此任务；r70上的现有配置都不能被删除以完成此任务。

根据这些要求在3号分支配置sw601和sw602：

- 路由器r61和r62不得向sw601和sw602发送EIGRP查询。
- 交换机sw601和sw602必须允许在将网络添加到EIGRP后，向r61和r62通告任何当前或未来直接连接的网络。
- 交换机Sw601和sw602必须继续将从r61和r62接收到的默认路由相互传播。要选择默认路由，请使用仅包含“permit”类型条目的前缀列表。
- 交换机sw601和sw602不得将默认路由传播回r61和r62。
- 如果将来修改了允许在sw601和sw602之间传播选定EIGRP学习网络的前缀列表，则必须禁止在没有任何额外命令的情况下自动将同一组网络通告回r61和r62

3分

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Solution:

Chinesedumps-r24:

```
r24>en
r24#conf t
r24(config)#router eigrp ccie
r24(config-router)#address-family ipv4 as 65006
r24(config-router)#network 10.200.0.0 0.0.0.255
r24(config-router)#af-interface tunnel 0
r24(config-router)#summary-address 0.0.0.0/0
r24(config-router)#topology base
```

Chinesedumps-r70:

```
r70#conf t
r70(config)#router eigrp ccie
r70(config-router)#address-family ipv4 as 65006
r70(config-router)#af-interface tunnel 0
r70(config-router)#no passive
```

Chinesedumps-sw601# & sw602:

```
sw>en
sw#conf t
sw(config)#ip prefix-list ALLOW-DEF seq 5 permit 0.0.0.0/0
sw(config)#route-map ALLOW-DEF permit 10
sw (config)#match ip address prefix-list ALLOW-DEF

sw (config)##route-map BLOCK-DEF deny 10
sw (config)##match ip address prefix-list ALLOW-DEF
sw (config)##route-map BLOCK-DEF permit 20
```

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解决方案：

中文教程-r24：

r24#配置文件
r24（配置）#路由器eigrp-ccie
r24（配置路由器）#地址家族ipv4为65006
r24（设置路由器）#网络10.200.0.0
0.0.0.255
r24（配置路由器）#af接口隧道0 r24（配
路由器）#摘要地址0.0.0.0/0 r24（设路

中文教材-r70：

r70#配置r70#路由器
eigrp-ccie
r70（配置路由器）#地址家族ipv4为65006
r70（设置路由器）#af接口隧道0 r70（设
定路由器）#无无源

中文教程-sw601#&sw602：

sw>en-sw
配置文件
sw（config）#ip前缀列表ALLOW-DEF seq 5 permit
0.0.0.0/0 sw（config）#路由映射ALLOW-DFE permit
10
sw（config）#匹配ip地址前缀列表ALLOW-DEF

sw（config）##路线图BLOCK-DEF拒绝10
sw（config）##匹配ip地址前缀列表ALLOW-DEF sw

r24>en

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```
sw (config)#router eigrp ccie
sw (config)#address-family ipv4 unicast autonomous-system 65006
sw (config-router)##eigrp stub connected leak-map  ALLOW-DEF
sw (config-router)#af-interface vlan 2000
sw (config-router)#passive
sw (config)#af-interface vlan 2001
sw (config-if)#passive
sw (config-if)#topology base
sw (config-if)#distribute-list route-map BLOCK-DEF out GigabitEthernet0/1
sw (config-if)#distribute-list route-map BLOCK-DEF out GigabitEthernet0/2
```

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sw（config）#路由器eigrp-ccie
sw（config）#地址家族ipv4单播自治系统65006 sw（配置路由器）##eigrp存根连接泄漏图ALLOW-DEF sw（设置路由器）#af接口vlan 2000

sw（config）#af接口vlan2001
sw（配置if）#被动
sw（config-if）#拓扑基础
sw（如果配置）#分发列表路由图BLOCK-DEF outGigabitEthernet0/1
sw（if配置）#分配列表路由图LOCK-DEF out千兆以太网0/2

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sw（配置路由器）#被动

Verification:

Chinesedumps-r61:

```
r61
chinesedumps-r61#
chinesedumps-r61#   www.passenterpriselabs.com
chinesedumps-r61#
chinesedumps-r61#sh ip route eig
chinesedumps-r61#sh ip route eigrp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
                   www.passenterpriselabs.com
Gateway of last resort is 10.200.0.1 to network 0.0.0.0

D*    0.0.0.0/0 [90/76800640] via 10.200.0.1, 00:06:17, Tunnel0
chinesedumps-r61#
chinesedumps-r61#
```

Chinesedumps-r62:

```
r62
chinesedumps-r62#
chinesedumps-r62#   www.passenterpriselabs.com
chinesedumps-r62#sh ip ro ei
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
                   www.passenterpriselabs.com
Gateway of last resort is 10.200.0.1 to network 0.0.0.0

D*    0.0.0.0/0 [90/76800640] via 10.200.0.1, 00:08:08, Tunnel0
chinesedumps-r62#
chinesedumps-r62#
```

验证：

中文教程-r61：

```
r61
chinesedumps-r61#
chinesedumps-r61#   www.passenterpriselabs.com
chinesedumps-r61#
chinesedumps-r61#sh ip route eig
chinesedumps-r61#sh ip route eigrp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
                   www.passenterpriselabs.com
Gateway of last resort is 10.200.0.1 to network 0.0.0.0

D*    0.0.0.0/0 [90/76800640] via 10.200.0.1, 00:06:17, Tunnel0
chinesedumps-r61#
chinesedumps-r61#
```

中文版本r62：

```
r62
chinesedumps-r62#
chinesedumps-r62#   www.passenterpriselabs.com
chinesedumps-r62#sh ip ro ei
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
                   www.passenterpriselabs.com
Gateway of last resort is 10.200.0.1 to network 0.0.0.0

D*    0.0.0.0/0 [90/76800640] via 10.200.0.1, 00:08:08, Tunnel0
chinesedumps-r62#
chinesedumps-r62#
```

Chinesedumps-r70:

```
r70
chinesedumps-r70#
chinesedumps-r70#
chinesedumps-r70#sh ip ro ei
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is 10.200.0.1 to network 0.0.0.0

D*    0.0.0.0/0 [90/76800640] via 10.200.0.1, 00:09:32, Tunnel0
chinesedumps-r70#
chinesedumps-r70#
```

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While checking the default route information is present on sw601 & sw602

Chinesedumps-sw601:sh ip route

```
sw601
chinesedumps-sw601#sh ip route
*May  6 05:24:01.914: %SYS-5-CONFIG_I: Configured from console by pel on console
chinesedumps-sw601#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 14 subnets, 3 masks
C        10.6.10.0/30 is directly connected, GigabitEthernet0/2
L        10.6.10.2/32 is directly connected, GigabitEthernet0/2
D        10.6.11.0/30 [90/15360] via 10.6.109.2, 00:00:05, GigabitEthernet0/0
D        10.6.12.0/30 [90/15360] via 10.6.109.2, 00:00:05, GigabitEthernet0/0
C        10.6.13.0/30 is directly connected, GigabitEthernet0/1
L        10.6.13.2/32 is directly connected, GigabitEthernet0/1
C        10.6.100.0/24 is directly connected, Vlan2000
L        10.6.100.2/32 is directly connected, Vlan2000
C        10.6.101.0/24 is directly connected, Vlan2001
L        10.6.101.2/32 is directly connected, Vlan2001
C        10.6.109.0/30 is directly connected, GigabitEthernet0/0
L        10.6.109.1/32 is directly connected, GigabitEthernet0/0
```

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中文教材-r70：

```
r70
chinesedumps-r70#
                    www.passenterpriselabs.com
chinesedumps-r70#
chinesedumps-r70#sh ip ro ei
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
                    www.passenterpriselabs.com
Gateway of last resort is 10.200.0.1 to network 0.0.0.0

D*    0.0.0.0/0 [90/76800640] via 10.200.0.1, 00:09:32, Tunnel0
chinesedumps-r70#
chinesedumps-r70#
```

在检查sw601和sw602上是否存在默认路由信息时

中文教程-sw601:sh-ip路由

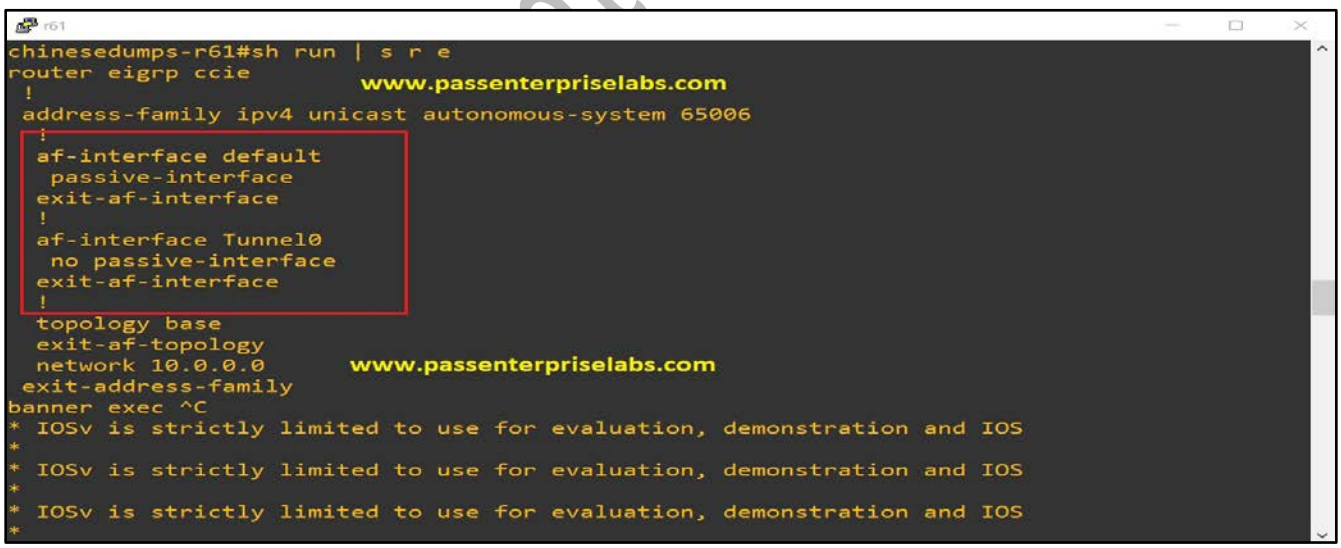
```
sw601
chinesedumps-sw601#sh ip route
*May  6 05:24:01.914: %SYS-5-CONFIG_I: Configured from console by pel on console
chinesedumps-sw601#sh ip route                    www.passenterpriselabs.com
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
                    www.passenterpriselabs.com
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 14 subnets, 3 masks
C        10.6.10.0/30 is directly connected, GigabitEthernet0/2
L        10.6.10.2/32 is directly connected, GigabitEthernet0/2
D        10.6.11.0/30 [90/15360] via 10.6.109.2, 00:00:05, GigabitEthernet0/0
D        10.6.12.0/30 [90/15360] via 10.6.109.2, 00:00:05, GigabitEthernet0/0
C        10.6.13.0/30 is directly connected, GigabitEthernet0/1
L        10.6.13.2/32 is directly connected, GigabitEthernet0/1
C        10.6.100.0/24 is directly connected, Vlan2000
L        10.6.100.2/32 is directly connected, Vlan2000
C        10.6.101.0/24 is directly connected, Vlan2001   www.passenterpriselabs.com
L        10.6.101.2/32 is directly connected, Vlan2001
C        10.6.109.0/30 is directly connected, GigabitEthernet0/0
L        10.6.109.1/32 is directly connected, GigabitEthernet0/0
```

Found that r61 was not sending default route on sw601 (same for sw602)

Issue :

- While working performed a check on r61 (sh ip eigrp neighbor)
- Found that there was no neighborship between r61 and sw601 , sw602

- From the above output you can see that sw 601 is only neighbor with sw 602
- To confirm the neighborship issue on r61 : sh run | s router eigrp

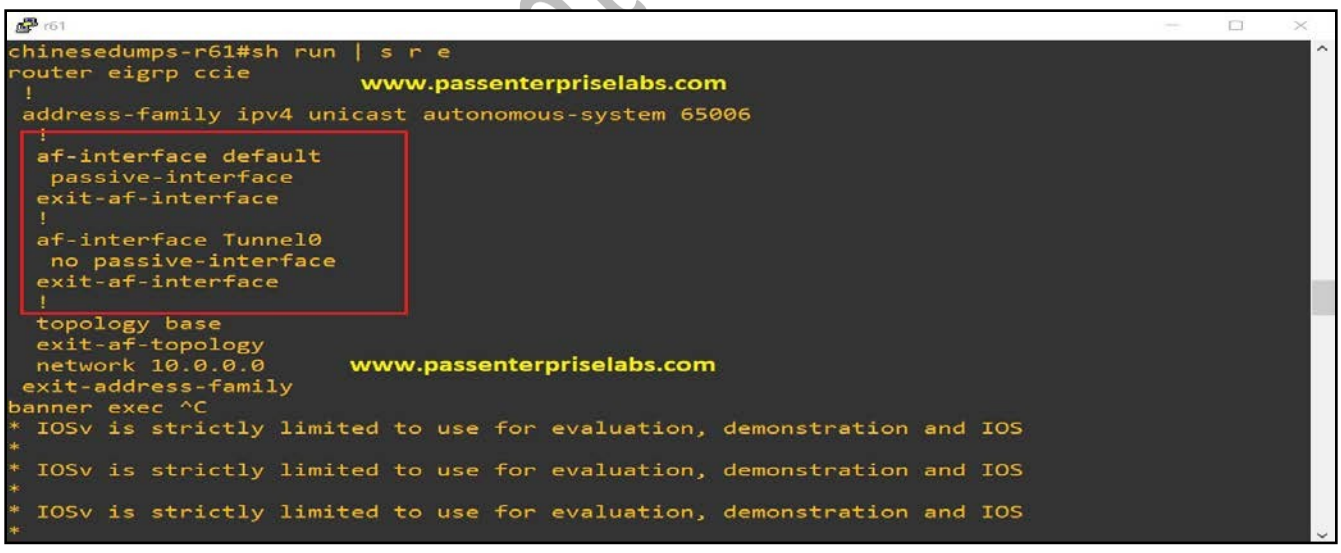


- From above output you can see that in eigrp af-interface default is made passive and only tunnel interface is removed from passive state

发现r61没有在sw601上发送默认路由（与sw602相同）问题：

- 工作时对r61（sh-ip-eigrp邻居）进行了检查
- 发现r61和sw601、sw602之间没有邻居关系

- 从上面的输出中可以看出，sw601是sw602的唯一邻居
- 确认r61:sh-run|s路由器eigrp上的邻居问题



- 从上面的输出中可以看出，在eigrp中，af接口默认设置为被动，只有隧道接口从被动状态中删除

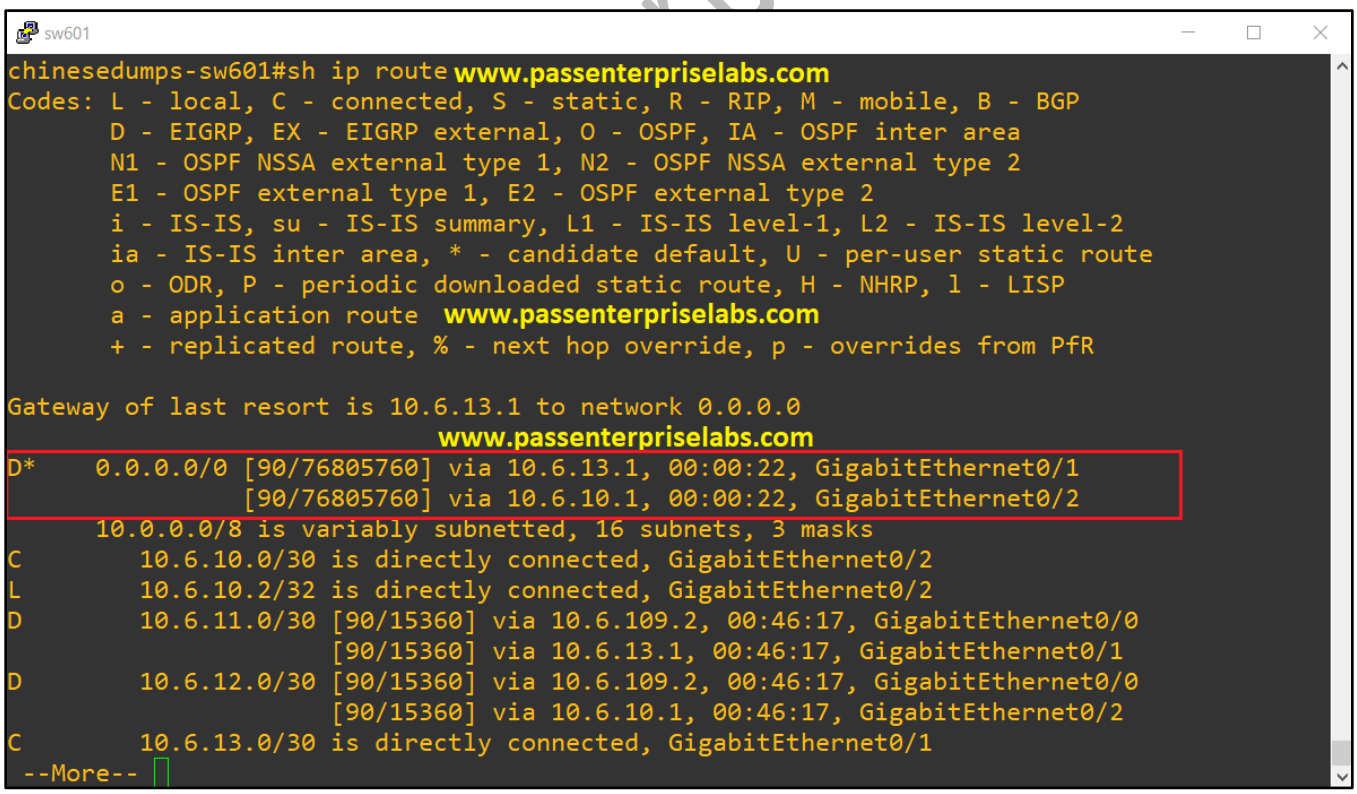
NOTE : Whenever we make an interface passive in eigrp the router stops sending eigrp messages including Hello messages and the devices need Hello messages for neighborship

- **To resolve the neighborship issue :**

  On r61 and r62:

```
router eigrp ccie
  address-family ipv4 unicast autonomous-system 65006
    af-interface GigabitEthernet0/2
      no passive-interface
    exit-af-interface

    af-interface GigabitEthernet0/3
      no passive-interface
    exit-af-interface
```

- After removing the connected interface on r61 and r62 to the downstream switches from passive-interface we are getting the default route information on sw601 and sw602



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注意：每当我们在eigrp中将接口设置为被动时，路由器就会停止发送包括Hello消息在内的eigrp消息，设备需要Hello消息进行邻居关系

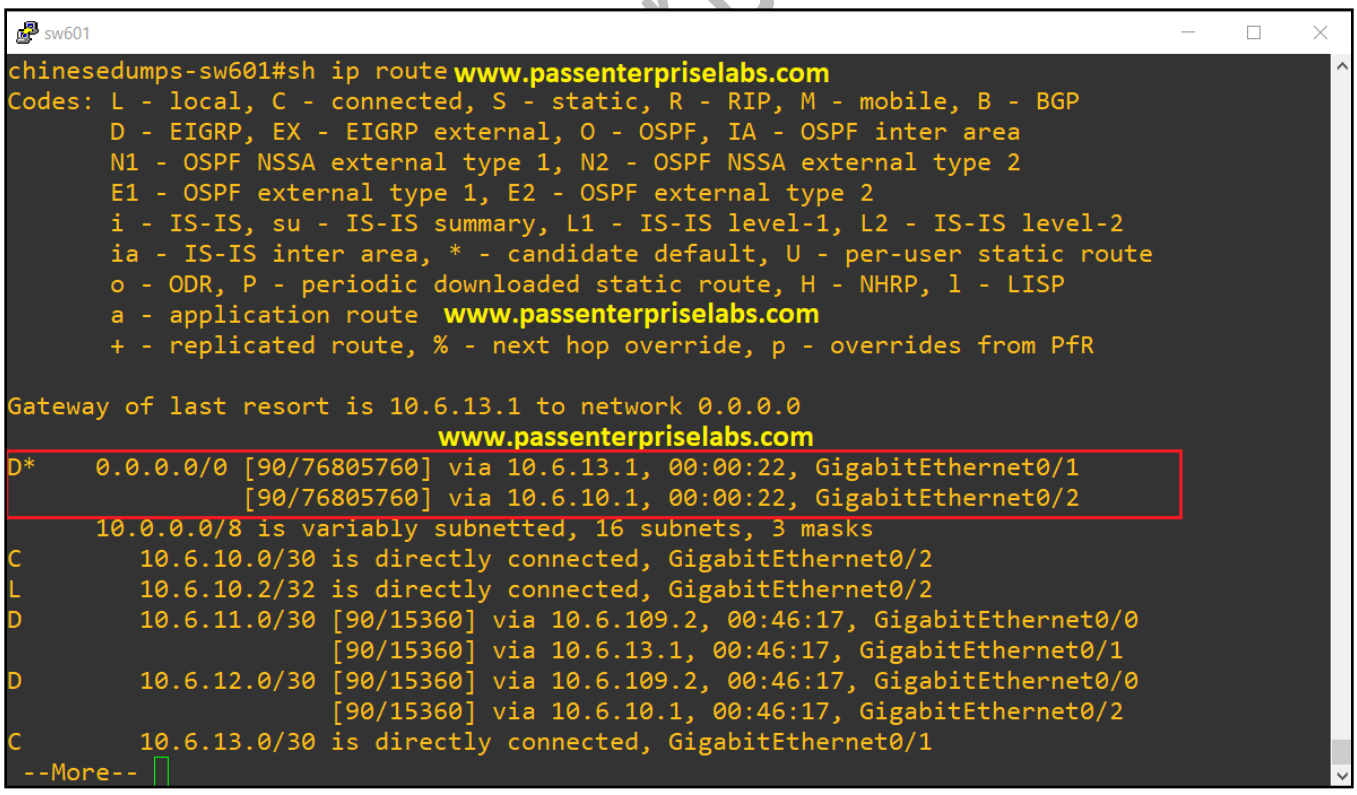
- 解决睦邻问题：

在r61和r62上：

路由器eigrp-ccie
地址家族ipv4单播自治系统65006 af接口千兆以太网0/2
无无源接口退出af接口

af接口千兆以太网0/3无无源接口
退出af界面

- 从被动接口删除r61和r62上连接到下游交换机的接口后，我们将获得sw601和sw602上的默认路由信息



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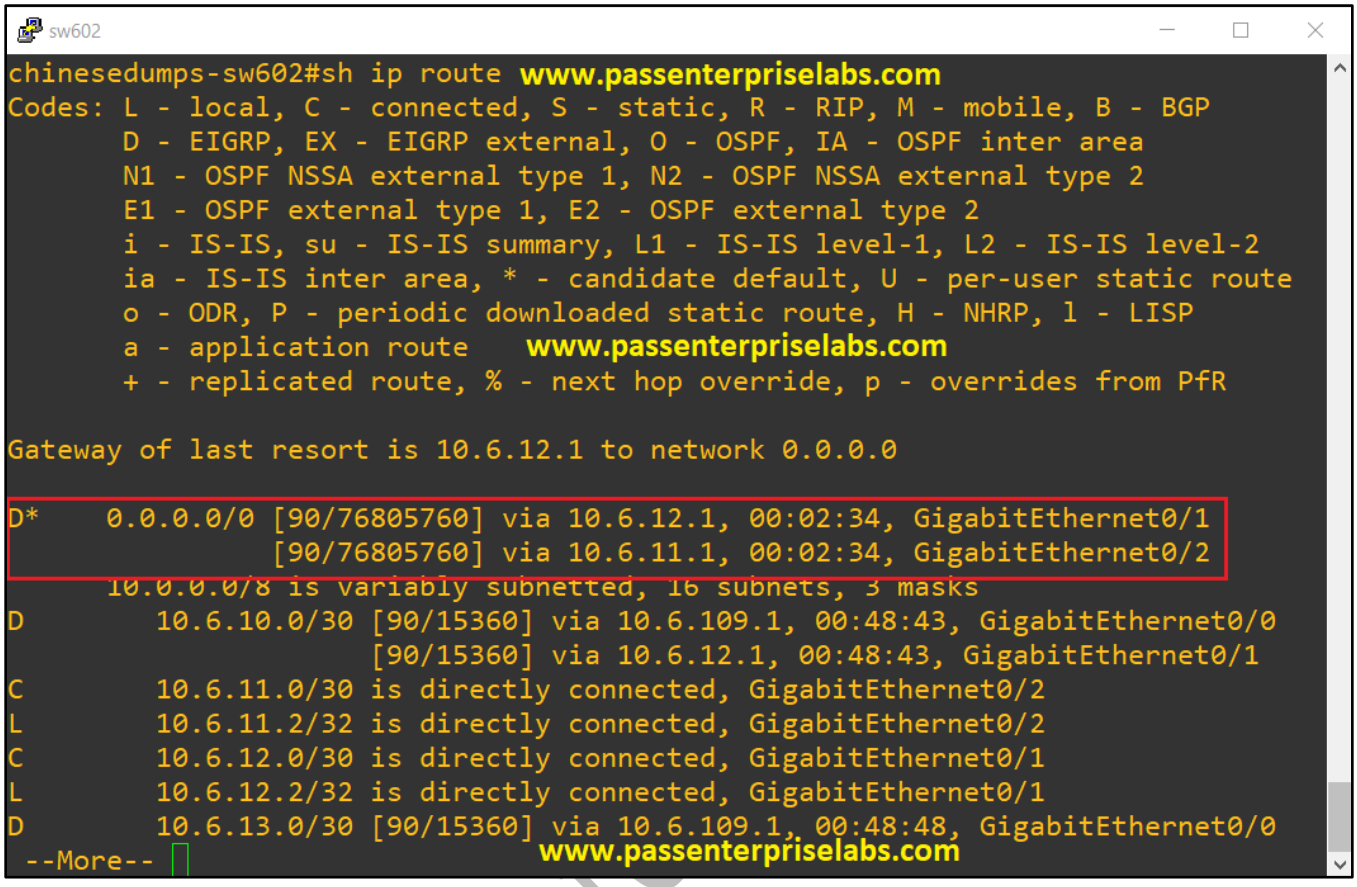
sw602
chinesedumps-sw602#sh ip route www.passenterpriselabs.com
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
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+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is 10.6.12.1 to network 0.0.0.0
D* 0.0.0.0/0 [90/76805760] via 10.6.12.1, 00:02:34, GigabitEthernet0/1
[90/76805760] via 10.6.11.1, 00:02:34, GigabitEthernet0/2
10.0.0.0/8 is variably subnetted, 16 subnets, 3 masks
D 10.6.10.0/30 [90/15360] via 10.6.109.1, 00:48:43, GigabitEthernet0/0
[90/15360] via 10.6.12.1, 00:48:43, GigabitEthernet0/1
C 10.6.11.0/30 is directly connected, GigabitEthernet0/2
L 10.6.11.2/32 is directly connected, GigabitEthernet0/2
C 10.6.12.0/30 is directly connected, GigabitEthernet0/1
L 10.6.12.2/32 is directly connected, GigabitEthernet0/1
D 10.6.13.0/30 [90/15360] via 10.6.109.1, 00:48:48, GigabitEthernet0/0
--More--
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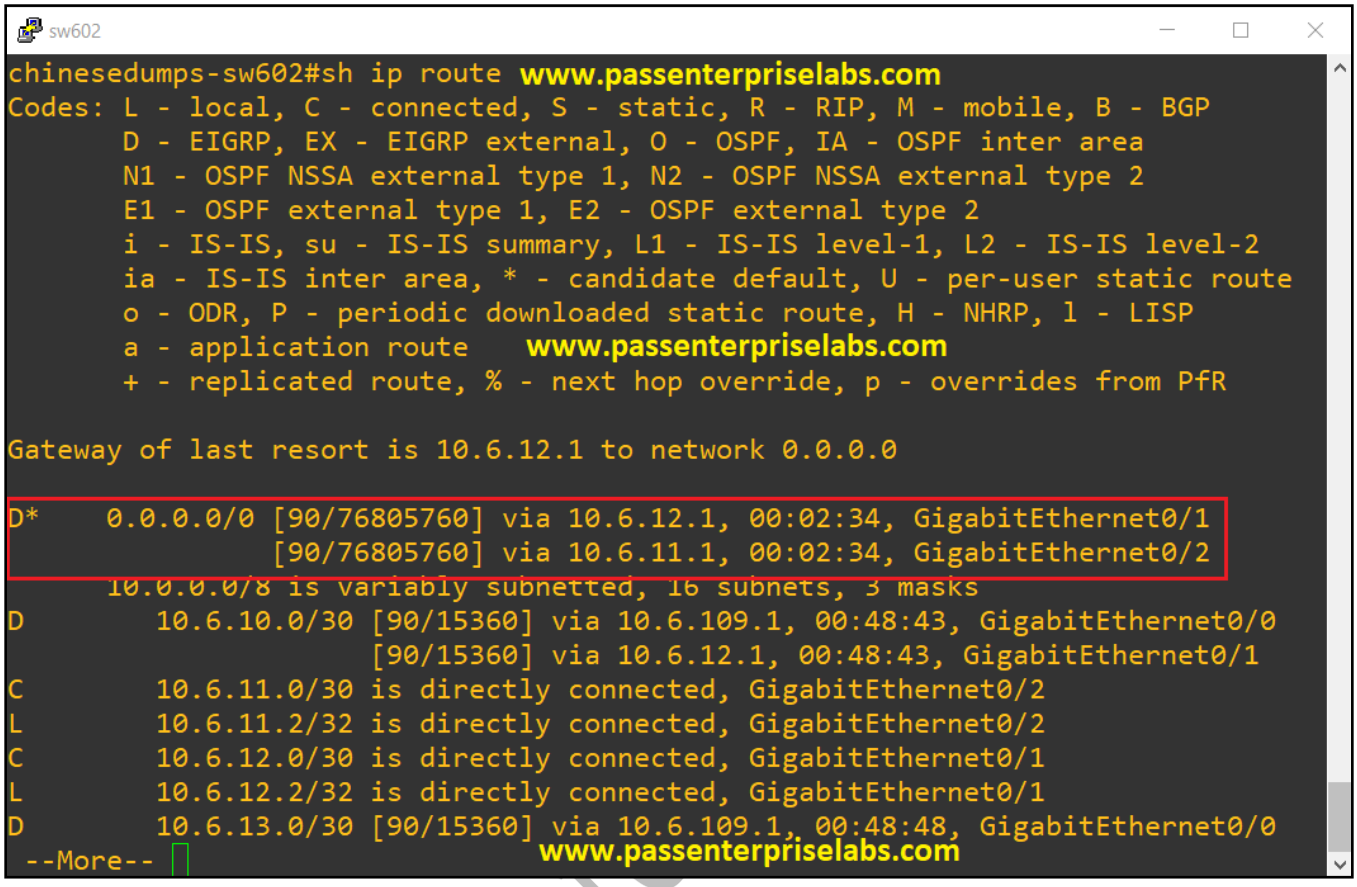
sw602
chinesedumps-sw602#sh ip route www.passenterpriselabs.com
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route www.passenterpriselabs.com
+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is 10.6.12.1 to network 0.0.0.0
D* 0.0.0.0/0 [90/76805760] via 10.6.12.1, 00:02:34, GigabitEthernet0/1
[90/76805760] via 10.6.11.1, 00:02:34, GigabitEthernet0/2
10.0.0.0/8 is variably subnetted, 16 subnets, 3 masks
D 10.6.10.0/30 [90/15360] via 10.6.109.1, 00:48:43, GigabitEthernet0/0
[90/15360] via 10.6.12.1, 00:48:43, GigabitEthernet0/1
C 10.6.11.0/30 is directly connected, GigabitEthernet0/2
L 10.6.11.2/32 is directly connected, GigabitEthernet0/2
C 10.6.12.0/30 is directly connected, GigabitEthernet0/1
L 10.6.12.2/32 is directly connected, GigabitEthernet0/1
D 10.6.13.0/30 [90/15360] via 10.6.109.1, 00:48:48, GigabitEthernet0/0
--More-- www.passenterpriselabs.com

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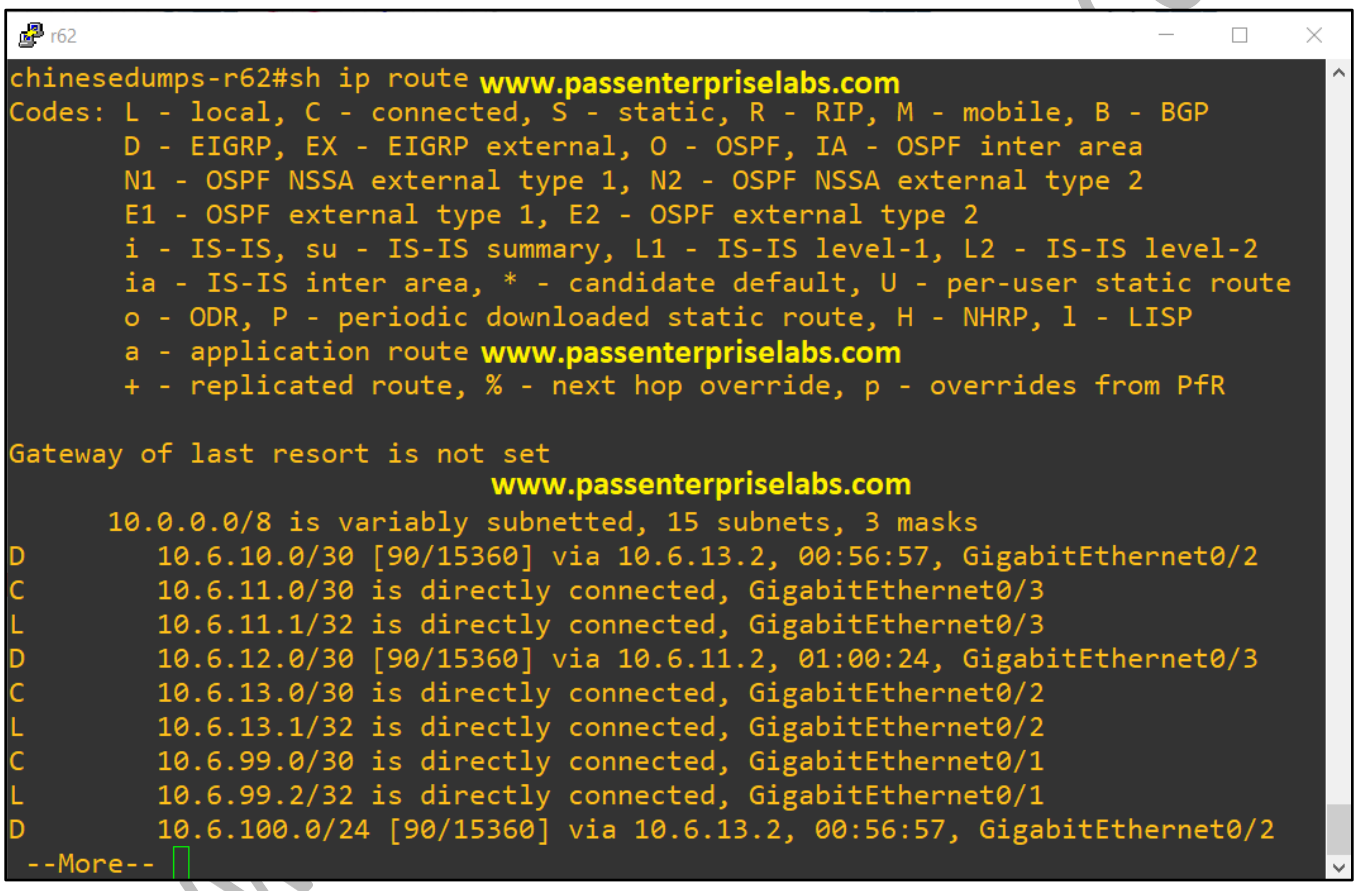
- **To verify "Switches sw601 and sw602 must not propagate the default route back to r61 and r62."**
- Shutdown the interface on r62 that is connected to r6 in SP 1

On r62:

Int g0/0
Shutdown



- After shutting down the interface on r62 we can see from the above output that there is no default route in the routing table

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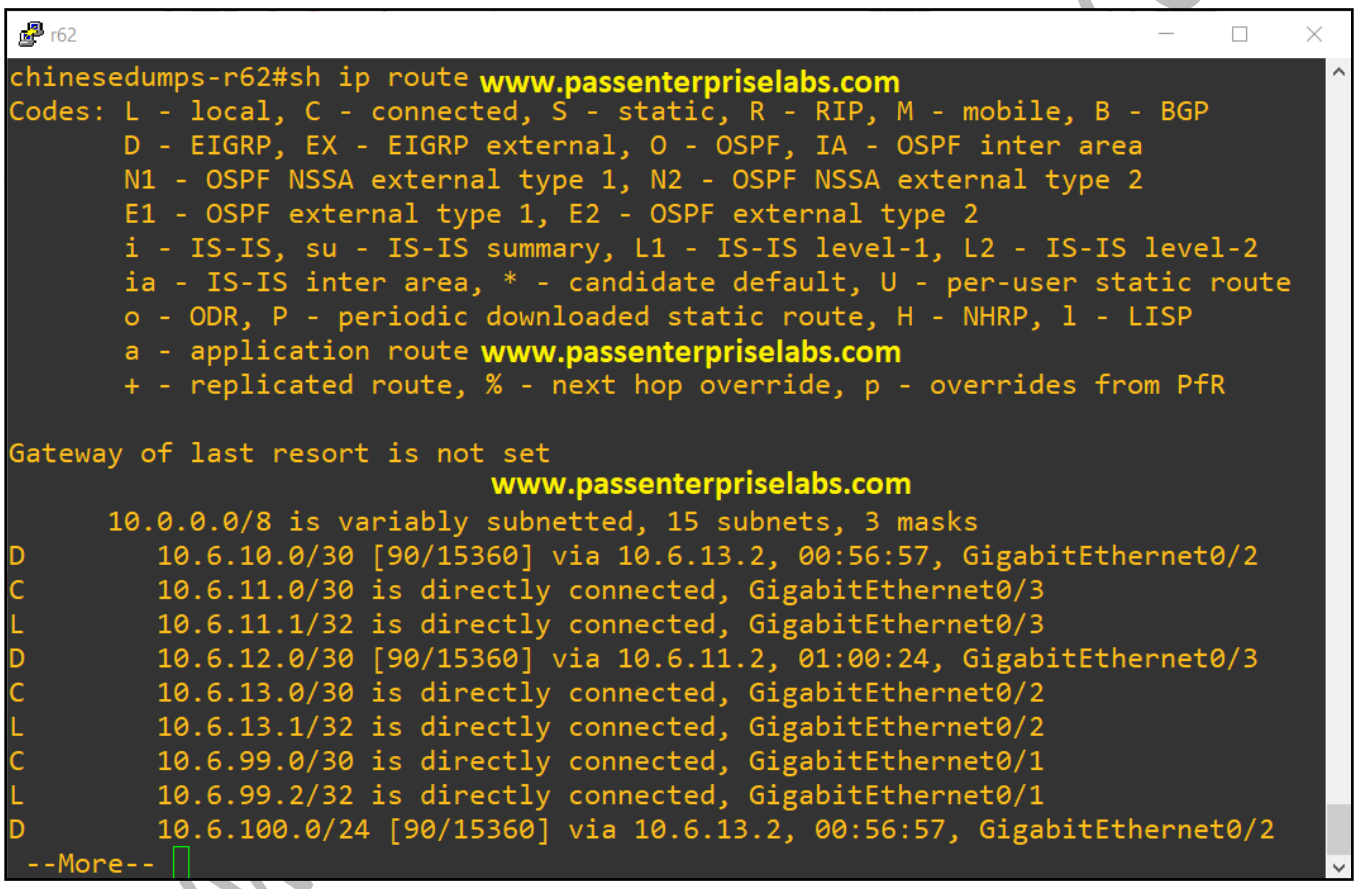
- **验证“交换机sw601和sw602不得将默认路由传播回r61和r62。”**

- 关闭连接到SP 1中r6的r62上的接口

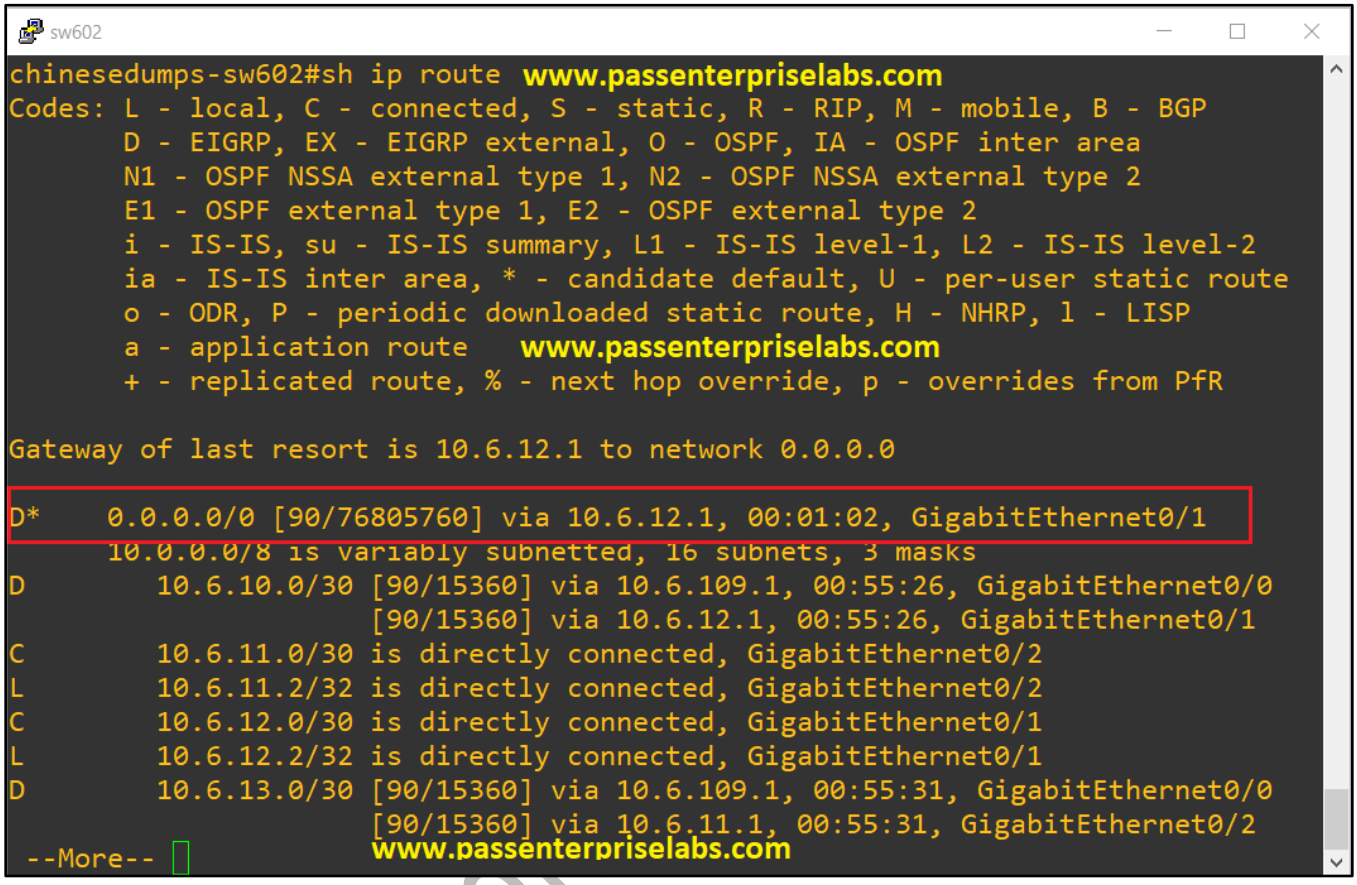
在r62上：

Int g0/0
关机



- 关闭r62上的接口后，我们可以从上面的输出中看到没有路由表中的默认路由

- If we check on sw601 or sw602 they should have the default route present in the routingtable



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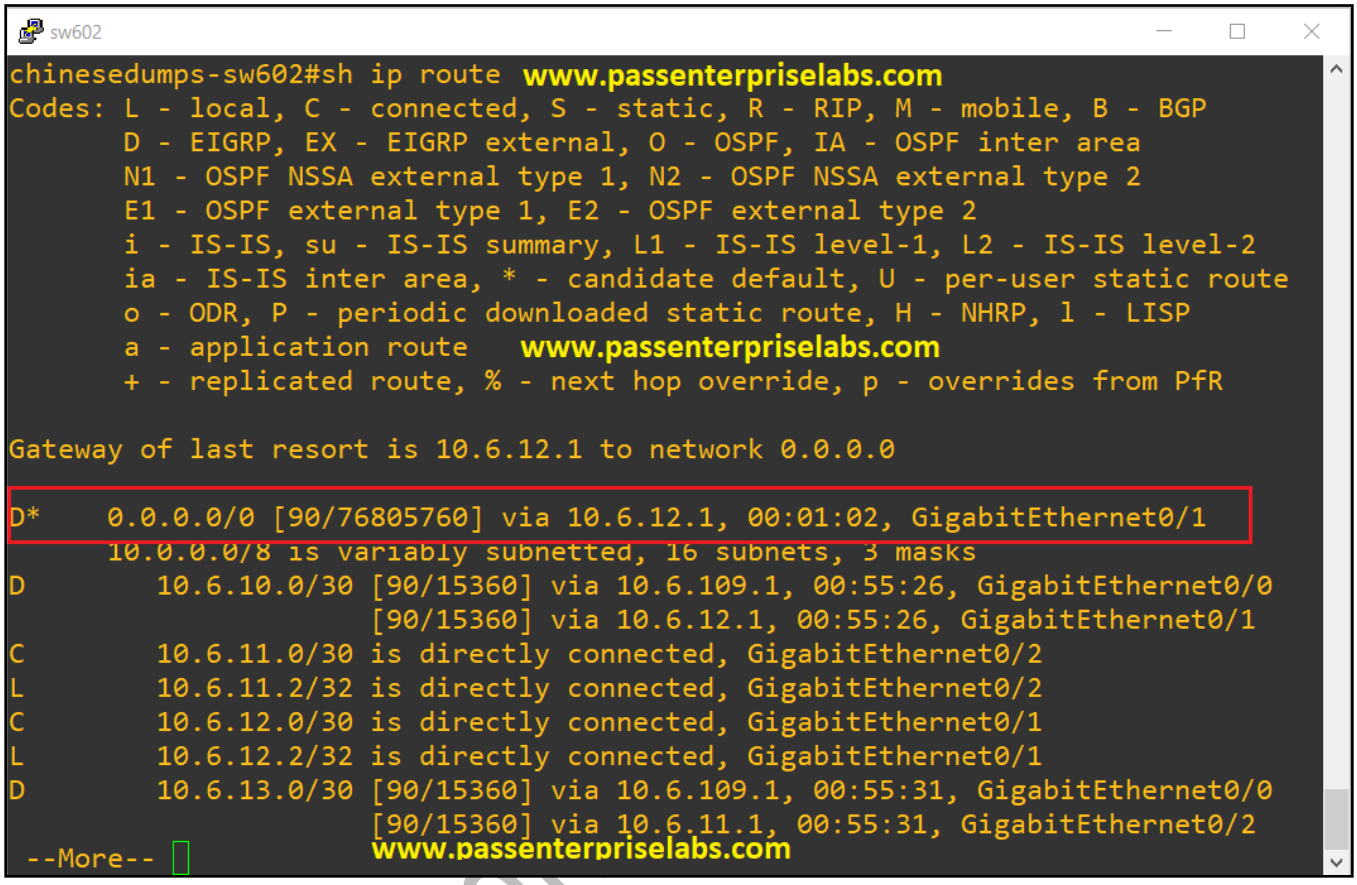
- 如果我们检查sw601或sw602，它们应该在路由表中有默认路由



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1.13: IPv4 Networks on Legacy Branches

On sw211 in DC, complete the DHCP server configuration according to these requirements:

- Create IPv4 DHCP pools named br3_v2000 and br3_v2001 for Branch #3 VLANs 2000 (10.6.100.0/24) and 2001 (10.6.101.0/24) respectively.
- Create IPv4 DHCP pool named br4_v1 for the subnet 10.7.1.0/24 on branch #4.
- In each subnet, assign addresses from .101 up to .254 inclusively, and the appropriate gateway to clients.

On Branch #3, Complete and correct the configuration on switches sw601, sw602 and sw610 to allow HSRP and DHCP relay operation in VLANs 2000 and 2001 according to these requirements:

- HSRP must implicitly use the vMAC address range of 0000.0c9f.f000 through 0000.0c9f.ffff
- The group member must be 100 for VLAN 2000 and 101 for VLAN 2001
- Sw601 must be the Active gateway for VLAN 2000 with a priority of 110; the Active role ownership must be deterministic
- Sw602 must be the Active gateway for VLAN 2001 with a priority of 110; the Active role ownership must be deterministic
- Each active switch must track its uplink interfaces gi0/1 and gi0/2 if either of these interfaces goes down; the active switch must allow the other switch to become Active. However, it is not allowed for the tracking to modify the HSRP priority to accomplish this requirement.
- Both sw601 and sw602 must be configured as DHCP relay agents in both VLANs 2000 and 2001, pointing toward the DHCP server 10.2.255.211 at sw211. However, at any time, only the Active router in the particular VLAN should relay the DHCP messages.
- Place host61 and host62 into VLANs 2000 and 2001, respectively, and make sure they are assigned their correct IPv4 configuration.

It is not permitted to use any kind of scripting to complete this task.

On Branch #4, complete the configuration of the router r70 according to these requirements;

- Assign IP address 10.7.1.1/24 to gi0/2
- Enable DHCP relay on this interface and point it to the DHCP server 10.2.255.211 at sw211
- It is allowed to add one additional missing command to the r70 configuration to allow DHCP clients connected to gi0/2 obtain their IPv4 configuration.
- Make sure that host71 and host72 are assigned their correct IPv4 configuration.

2 Points

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1.13：遗留分支上的IPv4网络

在DC的sw211上，按照以下要求完成DHCP服务器配置：

- 分别为分支#3 VLAN 2000（10.6.100.0/24）和2001（10.6.101.0/24）创建名为br3_v2000和br3_v2001的IPv4 DHCP池。
- 为分支#4上的子网10.7.1.0/24创建名为br4_v1的IPv4 DHCP池。
- 在每个子网中，为客户端分配从.101到.254的地址以及相应的网关。

在3号分支上，完成并纠正交换机sw601、sw602和sw610的配置，以允许根据这些要求在VLAN 2000和2001中进行HSRP和DHCP中继操作：

- HSRP必须隐式使用0000.0c9f.ff000到0000.0c9-f.fff的vMAC地址范围
- VLAN 2000的组成员必须为100，VLAN 2001的组成员为101
- Sw601必须是VLAN 2000的活动网关，优先级为110；Activerole所有权必须是确定的
- Sw602必须是VLAN 2001的活动网关，优先级为110；Activerole所有权必须是确定的
- 如果上行链路接口gi0/1和gi0/2中的任何一个发生故障，则每个活动交换机必须跟踪其上行链路接口gi0/2；活动开关必须允许另一个开关变为活动。但是，不允许跟踪修改HSRP优先级来满足这一要求。
- sw601和sw602都必须在VLAN 2000和2001中配置为DHCP中继代理，指向sw211处的DHCP服务器10.2.255.211。但是，在任何时候，只有特定VLAN中的活动路由器应该中继DHCP消息。
- 将主机61和主机62分别放入VLAN 2000和2001中，并确保为它们分配了正确的IPv4配置。

不允许使用任何类型的脚本来完成此任务。

在4号分支上，根据这些要求完成路由器r70的配置；

- 将IP地址10.7.1.1/24分配给gi0/2
- 在此接口上启用DHCP中继，并将其指向sw211上的DHCP服务器10.2.255.211
- 允许在r70配置中添加一个额外的缺失命令，以允许连接到gi0/2的DHCP客户端获取其IPv4配置。
- 确保为主机71和主机72分配了正确的IPv4配置。

2分

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Solution:

Chinesedumps.com-sw211:

```
sw211>en
sw211#conf t
sw211(config-dhcp)#ip dhcp excluded-address 10.6.100.1 10.6.100.100
sw211(config)#ip dhcp pool br3_v2000
sw211(config-dhcp)#network 10.6.100.0 /24
sw211(config-dhcp)#default-router 10.6.100.1

sw211(config-dhcp)#ip dhcp excluded-address 10.6.101.1 10.6.101.100
sw211(config)#ip dhcp pool br3_v2001
sw211(config-dhcp)#network 10.6.101.0 /24
sw211(config-dhcp)#default-router 10.6.101.1

sw211(config-dhcp)#ip dhcp excluded-address 10.7.1.1 10.7.1.100
sw211(config)#ip dhcp pool br4_v1
sw211(config-dhcp)#network 10.7.1.0 /24
sw211(config-dhcp)#default-router 10.7.1.1
```

Chinesedumps.com-sw601:

```
sw601>en
sw601#conf t
sw601(config)#int vlan 2000
sw601(config-if)#standby version 2
sw601(config-if)#standby 100 ip 10.6.100.1
sw601(config-if)#standby 100 preempt
sw601(config-if)#standby 100 priority 110
sw601(config-if)#standby 100 track 1 shut
sw601(config-if)#standby 100 track 2 shut
sw601(config-if)#standby 100 name VLAN2K
sw601(config-if)#ip helper-address 10.2.255.211 redundancy VLAN2K

sw601(config)#int vlan 2001
sw601(config-if)#standby 101 ip 10.6.101.1
sw601(config-if)#standby version 2
sw601(config-if)#standby 101 name VLAN2K1
sw601(config-if)#ip helper-address 10.2.255.211 redundancy VLAN2K1
```

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解决方案：

Chinese edumps.com-sw211：

```
sw211>en-
sw211#配置
文件
sw211（配置dhcp）#ip dhcp排除地址10.6.100.1 10.6.100.100
sw211（设置）#ip dhcp池br3_v2000
sw211（配置dhcp）#网络10.6.100.0/24
sw211（配置dhcp）#默认路由器10.6.100.1

sw211（配置dhcp）#ip dhcp排除地址10.6.101.1 10.6.101.100
sw211（设置）#ip dhcp池br3_v2001
sw211（配置dhcp）#网络10.6.101.0/24
sw211（配置dhcp）#默认路由器10.6.101.1

sw211（配置dhcp）#ip dhcp排除地址10.7.1.1 10.7.1.100
sw211（设置）#ip dhcp池br4_v1
sw211（配置dhcp）#网络10.7.1.0/24
sw211（配置dhcp）#默认路由器10.7.1.1
```

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```
sw601>en
sw601配置文
件
sw601（配置）#int vlan 2000
sw601（配置if）#备用版本2
sw601（配置if）#备用100 ip 10.6.100.1
sw601（配置if）#待机100抢占
sw601（配置if）#待机100优先级110
sw601（配置if）#待机100轨道1关闭
sw601（配置if）#备用100轨道2关闭
sw601（设置if）#备份100名称VLAN2K
sw601（配置if）#ip助手地址10.2.255.211冗余VLAN2K

sw601（配置）#int vlan 2001
sw601（配置if）#备用101 ip 10.6.101.1
sw601（config if）#备用版本2 sw601（
config-if）#备用101名称VLAN2K1
sw601（配置if）#ip助手地址10.2.255.211冗余VLAN2K1
```

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```
sw601(config-if)#track 1 interface g0/1 line-protocol
sw601(config-if)#track 2 interface g0/0 line-protocol
```

Chinesedumps.com-sw602:

```
sw602>en
sw602#conf t
sw602 (config)#int vlan  2001
sw602 (config-if)#standby version 2
sw602 (config-if)#no standby 0 ip 10.6.101.1
sw602 (config-if)#standby 101 ip 10.6.101.1
sw602 (config-if)#standby 101 priority 110
sw602 (config-if)#standby 101 preempt
sw602 (config-if)#standby 101 track 1 shut
sw602 (config-if)#standby 101 track 2 shut
sw602 (config-if)#standby 101 name VLAN2K1
sw602 (config-if)#ip helper-address 10.2.255.211 redundancy VLAN2K1

sw602 (config)#int vlan  2000
sw602 (config-if)#standby version 2
sw602 (config-if)#standby 100 ip 10.6.100.1
sw602 (config-if)#standby 100 name VLAN2K
sw602 (config-if)#ip helper-address 10.2.255.211 redundancy VLAN2K

sw602 (config-if)#track 1 interface g0/1 line-protocol
sw602 (config-if)#track 2 interface g0/0 line-protocol
```

Chinesedumps.com-sw610:

```
sw610>en
sw610#conf t
sw610(config-if)#int range g2/0-1
sw610(config-if)#switch trunk allowed vlan add 2001

sw610(config)#int g0/0
sw610(config-if)#switchport mode access
sw610(config-if)#switchport access vlan 2000

sw610(config)#int g0/1
sw610(config-if)#switchport mode access
sw610(config-if)#switchport access vlan 2001
```

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sw601（config-if）#轨道1接口g0/1线路协议
sw601（config.if）#轨道2接口g0/0线路协议

Chinese edumps.com-sw602：

sw602>en
sw602配置文
件
sw602（配置）#int vlan 2001
sw602（config if）#备用版本2
sw602（配置if）#无备用0 ip 10.6.101.1
sw602（配置if）#备用101 ip 10.6.101.1
sw602（如果配置）#备用101优先级110
sw602
sw602（配置if）#备用101轨道2关闭sw602
（设置if）#备份101名称VLAN2K1
sw602（配置if）#ip助手地址10.2.255.211冗余VLAN2K1

sw602（配置）#int vlan 2000
sw602（如果配置）#备用版本2
sw602（配置if）#备用100 ip 10.6.100.1
sw602（设置if）#备份100名称VLAN2K
sw602（配置if）#ip助手地址10.2.255.211冗余VLAN2K

sw602（config-if）#轨道1接口g0/1线路协议sw602
（config.if）#轨道2接口g0/0线路协议

中文教育网-sw610：

sw610>en-
sw610#配置
文件
sw610（配置if）#int范围g2/0-1
sw610（配置if）#交换机中继允许vlan添加2001

sw610（配置）#int g0/0
sw610（config-if）#交换端口模式访问
sw610（-config-if）#vlan2000交换端口
访问

sw610（配置）#int g0/1
sw610（config-if）#交换机端口模式访

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Chinesedumps.com-r70:

```
r70>en
r70#conf t
r70(config)#interface GigabitEthernet0/2
r70(config-if)#ip address 10.7.1.1 255.255.255.0
r70(config-if)#ip helper-address 10.2.255.211

r70(config)#router eigrp ccie
r70(config-router)#address-family ipv4 unicast autonomous-system 65006
r70(config-router)#network 10.7.1.0 0.0.0.255
```

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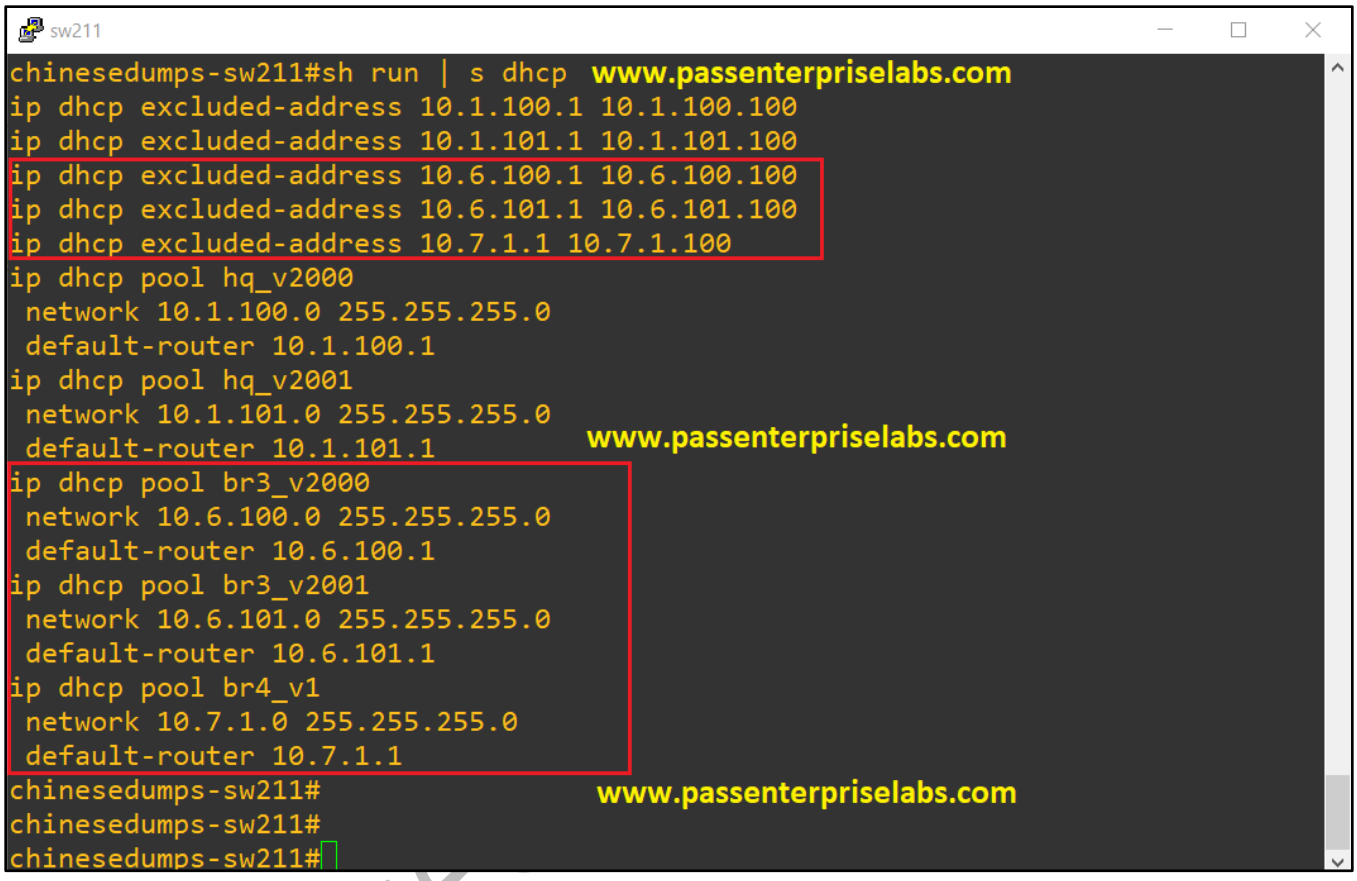
问sw610（config.if）#交换机接口vlan
2001

Verification:

Chinesedumps.com-r211:



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中文教育网-r70：

r70>和
r70#conf t

r70（配置if）#ip地址10.7.1.1 255.255.255.0
r70（配置if）#ip助手地址10.2.255.211

r70（配置）#路由器eigrp-ccie
r70（配置路由器）#地址家族ipv4单播自治系统65006 r70（设置路由器）#网络10.7.1.0 0.0.0.255

r70（配置）#接口千兆以太网0/2

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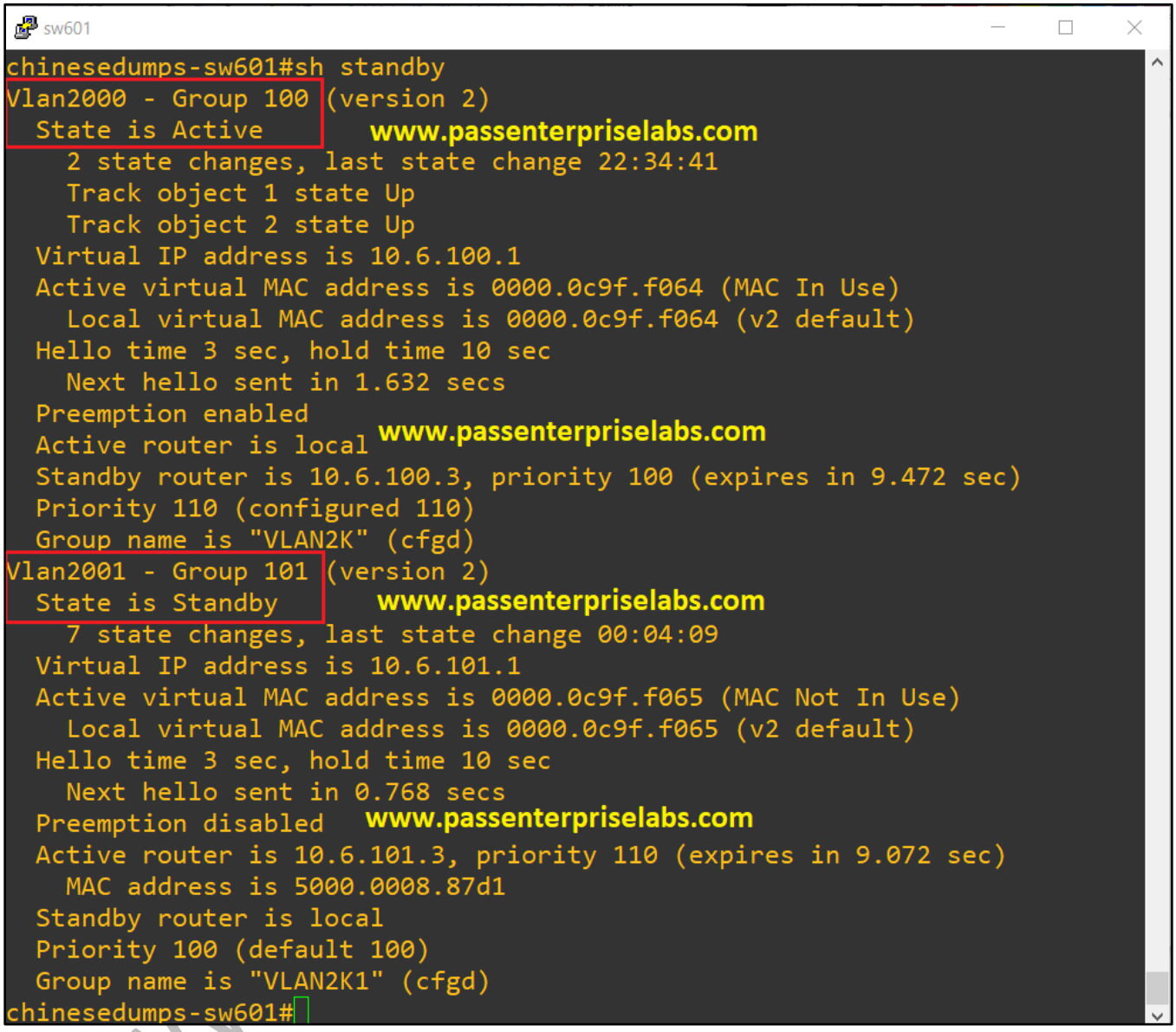
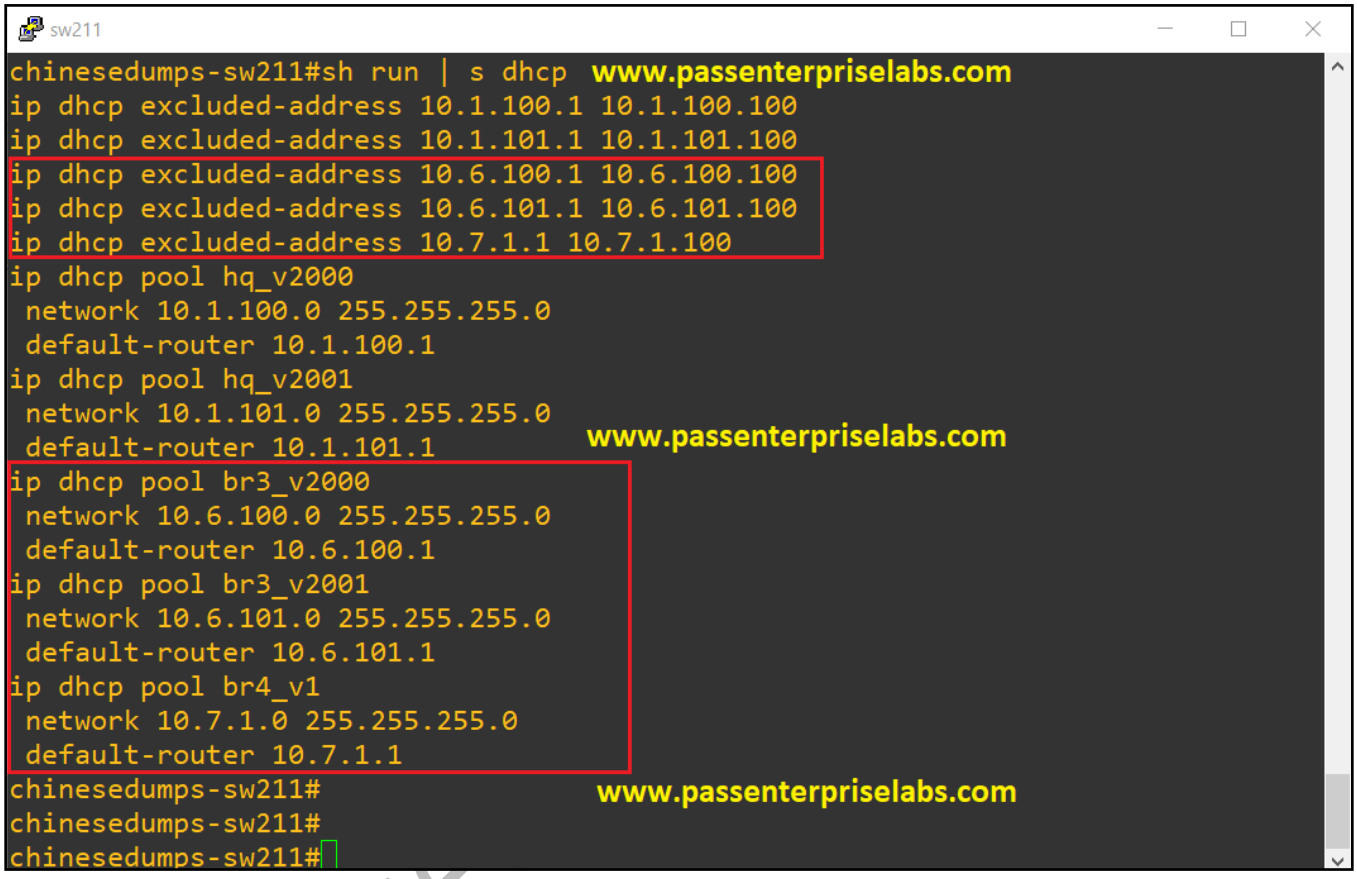
Chinesedumps.com-sw601:



www

验证：

中文教育网-r211：



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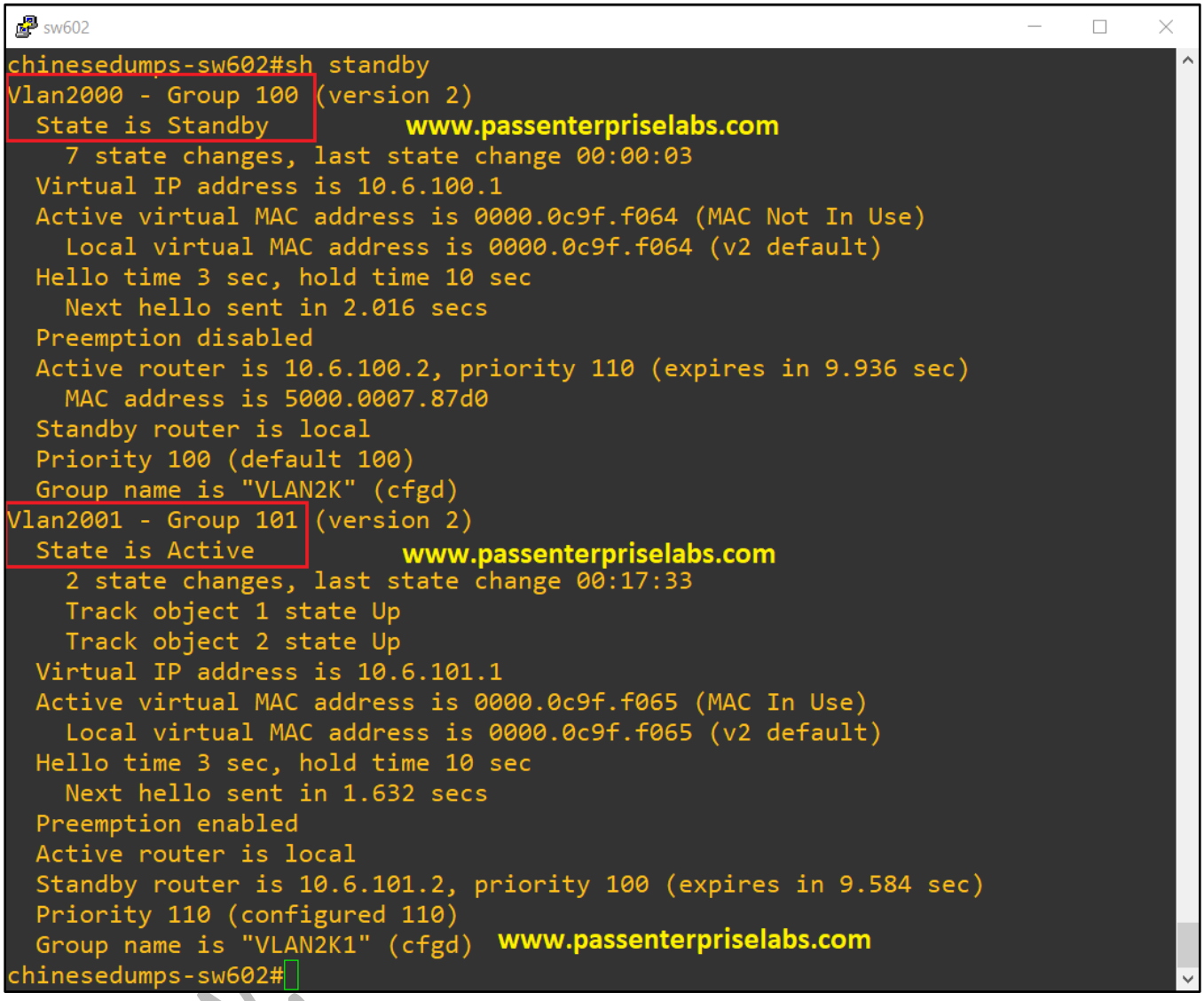
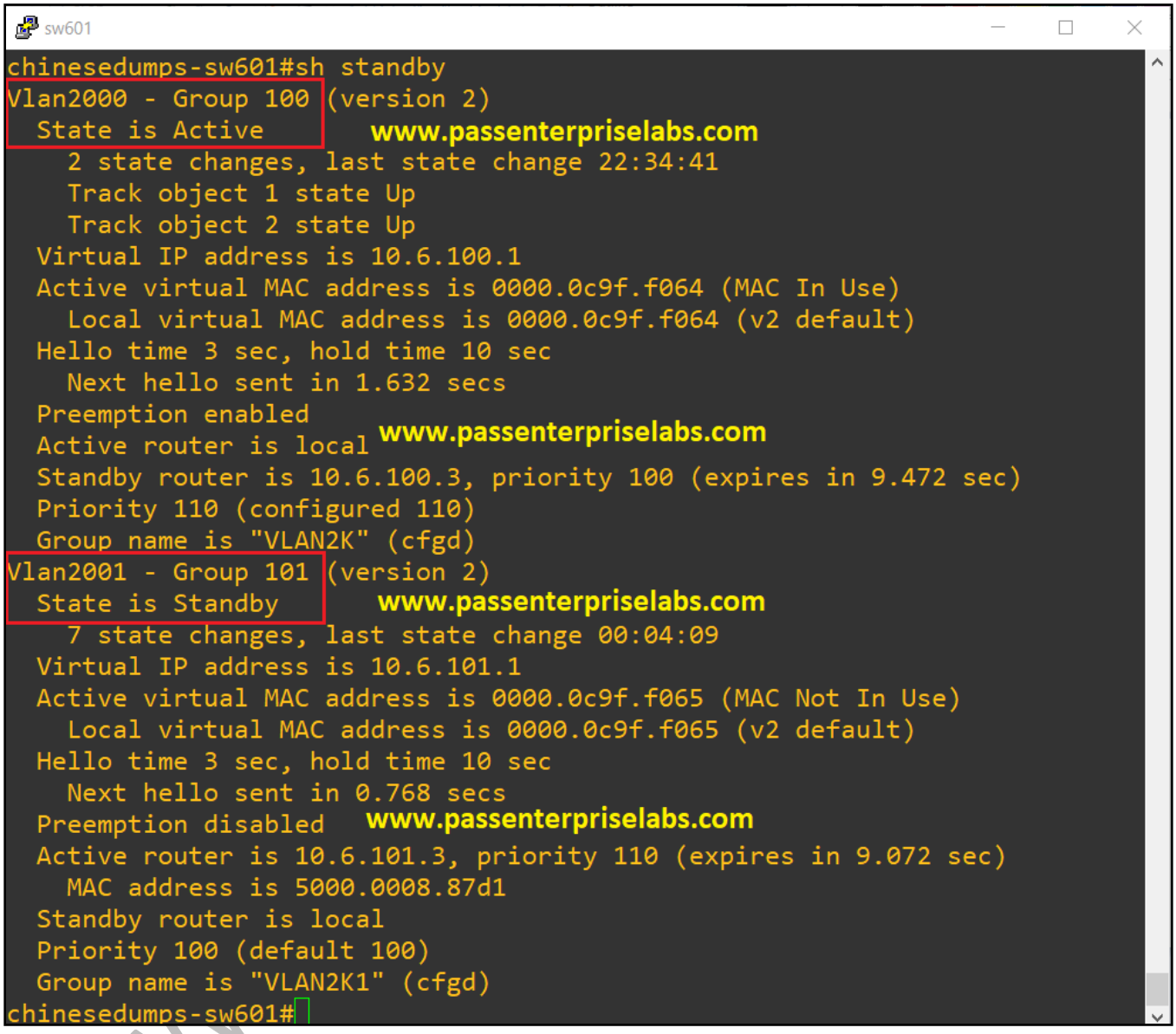
Chinesedumps.com-sw602:



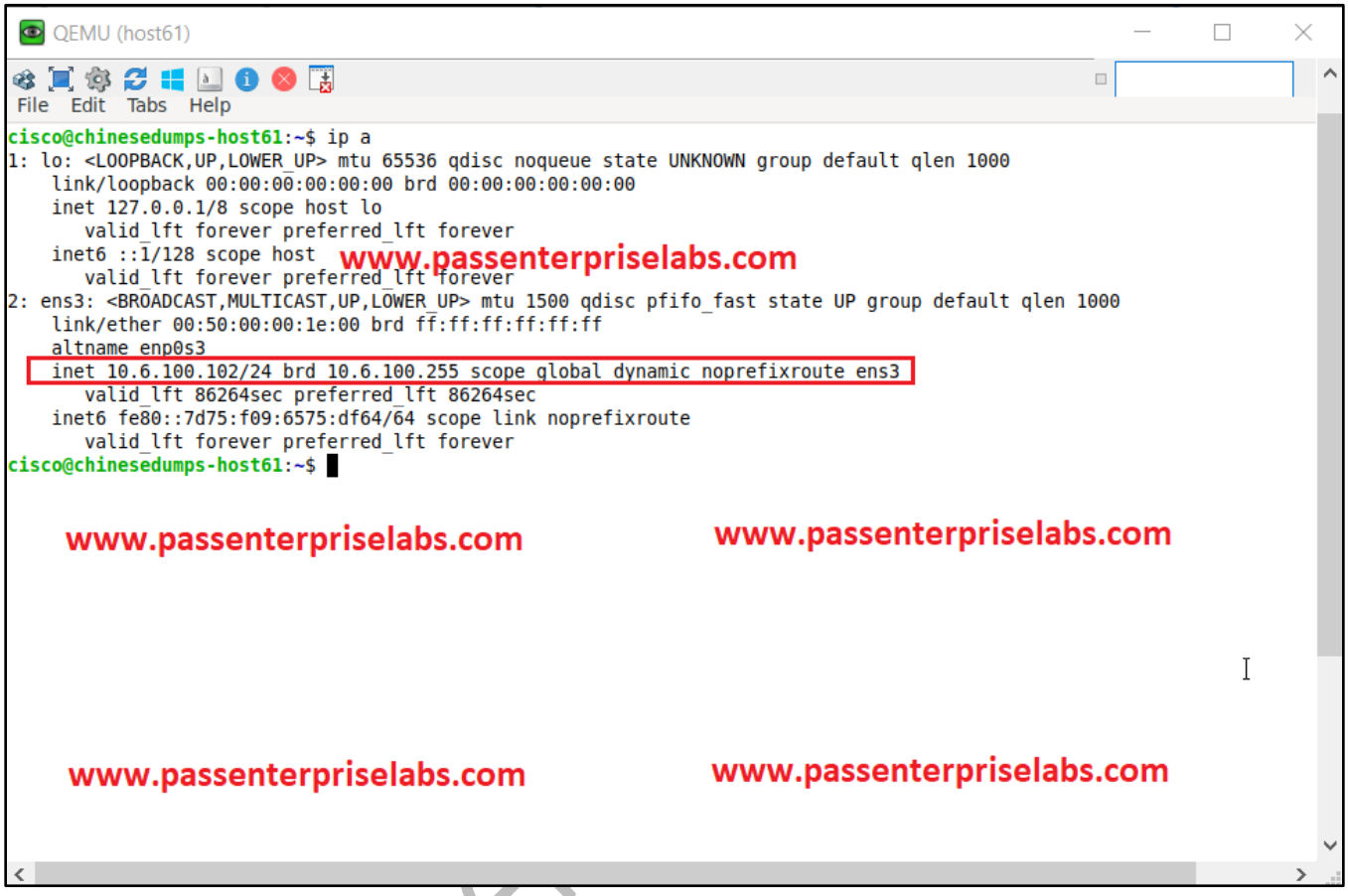
www.

中文教育网-sw601：



WWW

Chinesedumps-host61:



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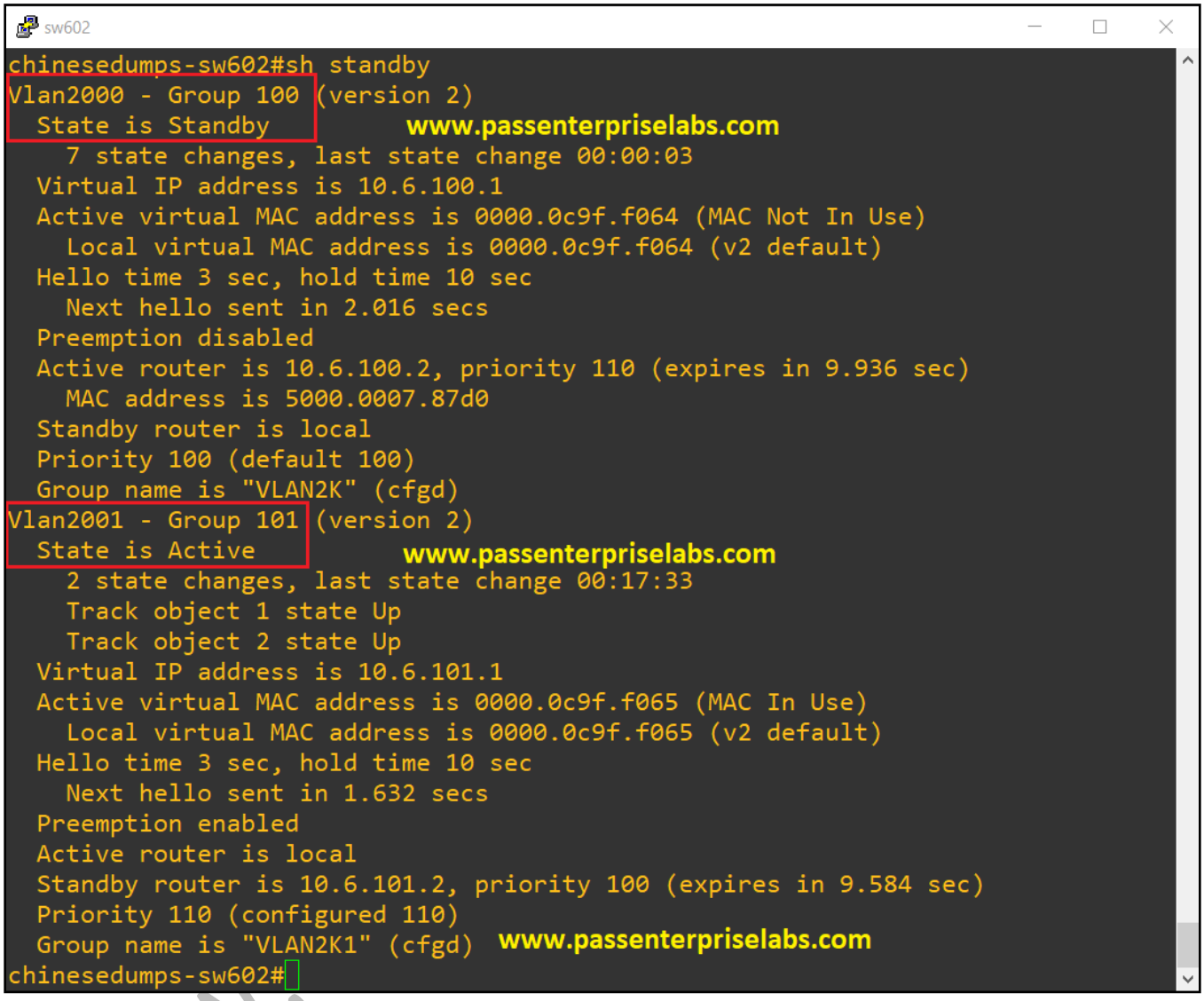
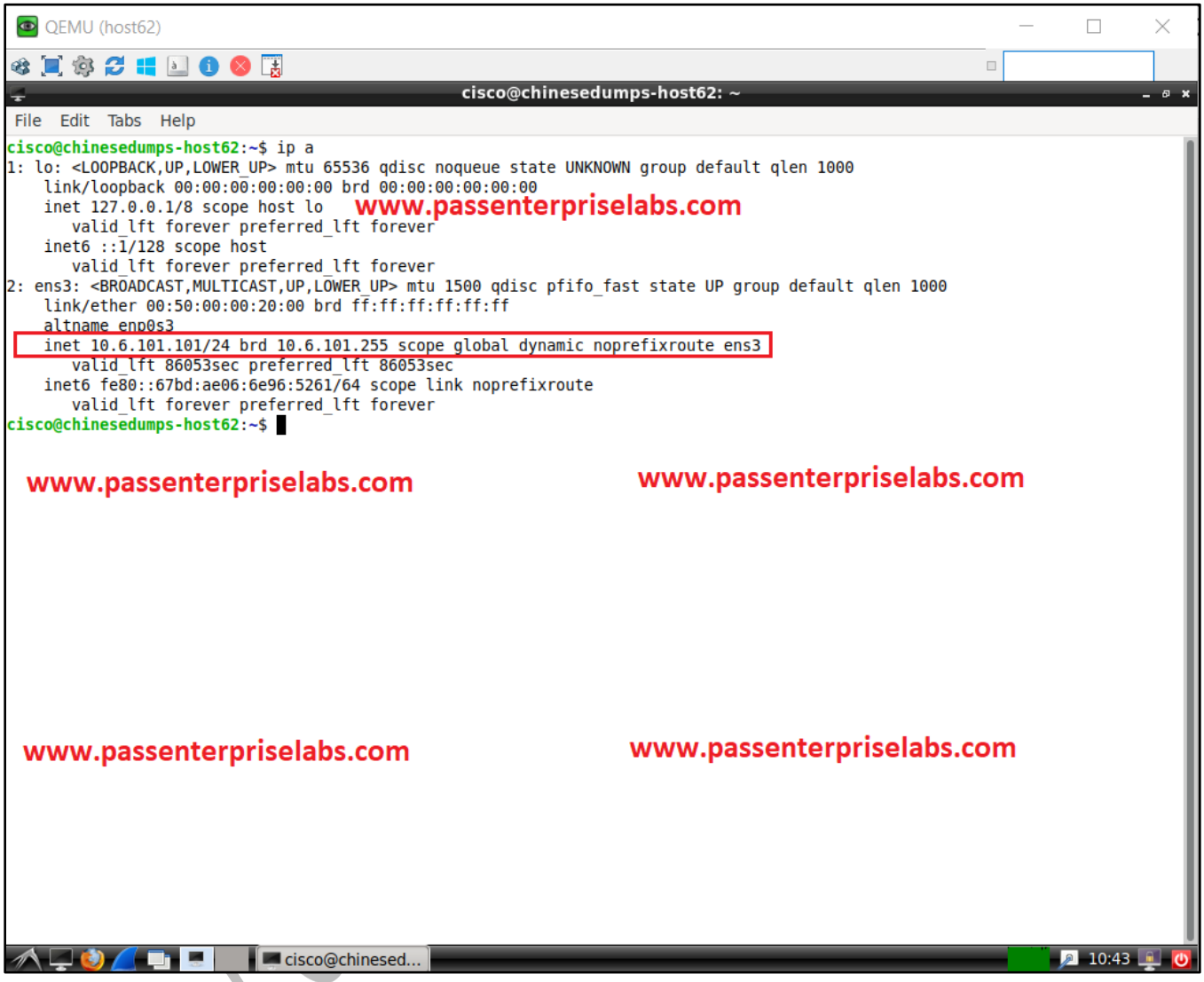
Chinese edumps.com-sw602:



www.

Chinesedumps-host62:



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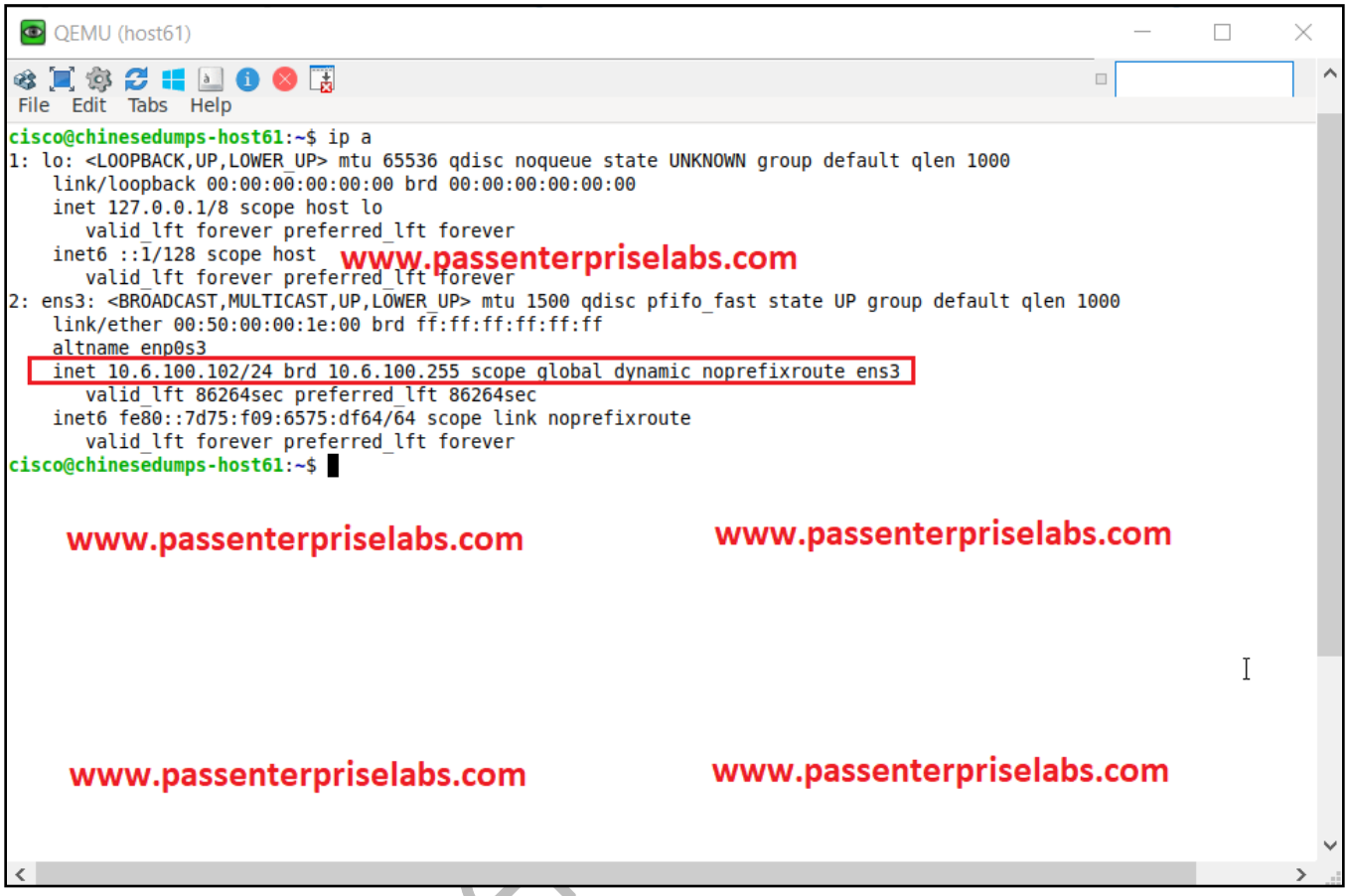
中文主机61：



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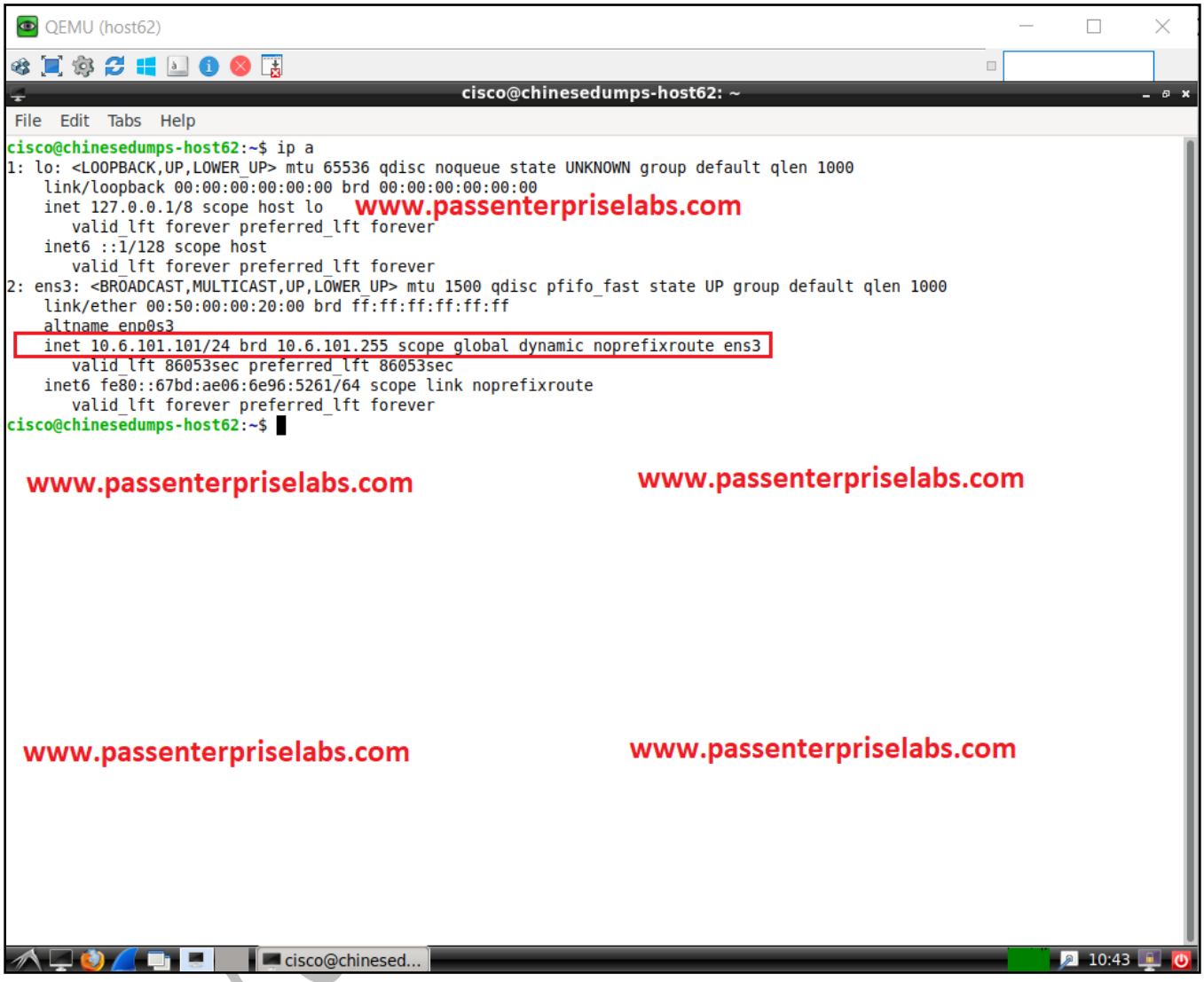
- To verify “Each active switch must track its uplink interfaces gi0/1 and gi0/2/ if either ofthese interfaces goes down; the active switch must allow the other switch to become Active. However,, it is not allowed for the tracking to modify the HSRP priority to accomplish this requirement. “

- Go on sw601 and shutdown the interface g0/2 or g0/1 , and then check on sw602 that it should become the gateway now for vlan 2000

中文主机62：


QEMU (host62)
cisco@chinesedumps-host62: ~
File Edit Tabs Help
cisco@chinesedumps-host62:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 00:50:00:00:20:00 brd ff:ff:ff:ff:ff:ff
altname enp0s3
inet 10.6.101.101/24 brd 10.6.101.255 scope global dynamic noprefixroute ens3
valid_lft 86053sec preferred_lft 86053sec
inet6 fe80::67bd:ae06:6e96:5261/64 scope link noprefixroute
valid_lft forever preferred_lft forever
cisco@chinesedumps-host62:~$
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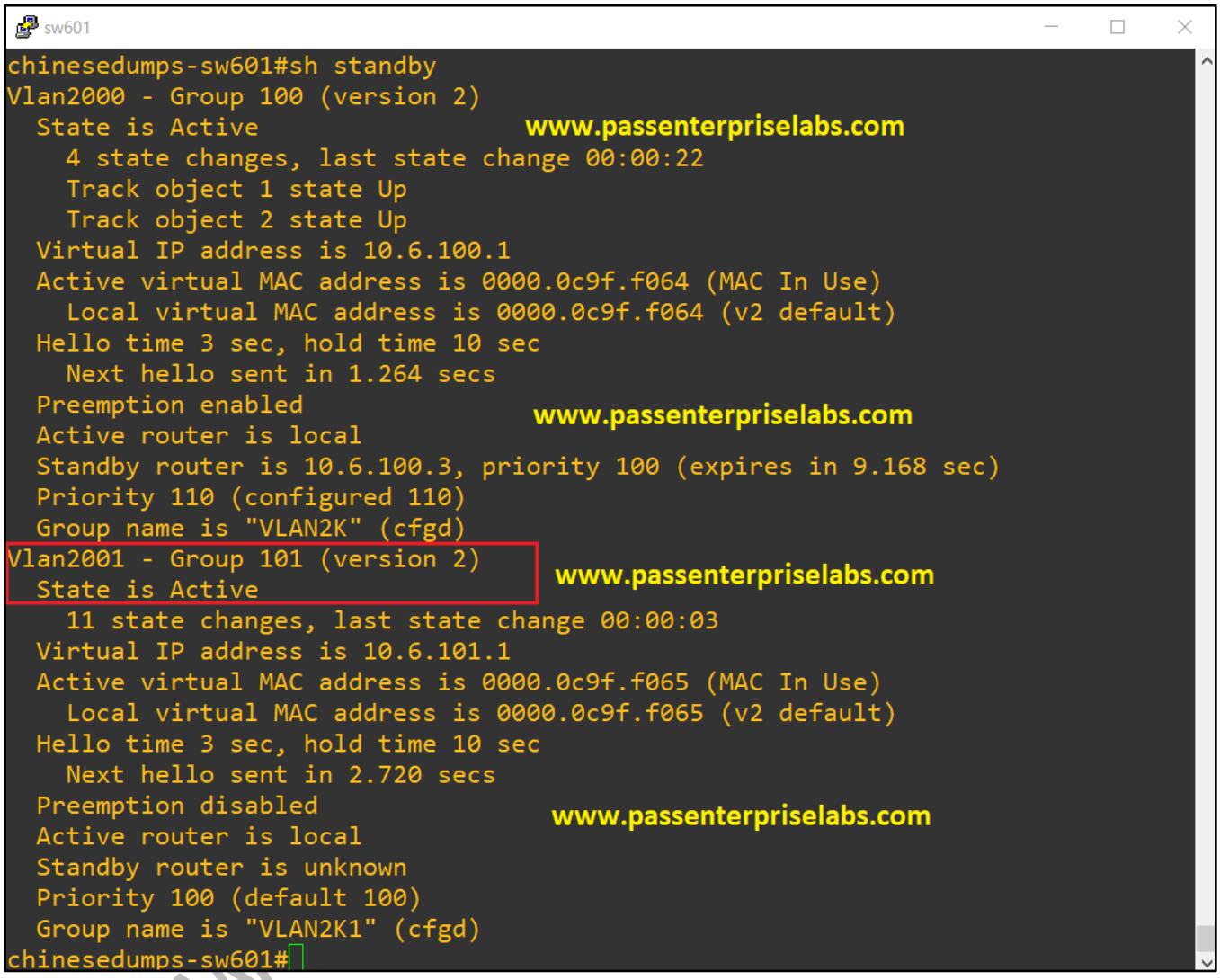
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@chinesed...
10:43

- Go on sw602 and shutdown the interface g0/2 or g0/1, and then check on sw601 that it should become the gateway now for vlan 2001

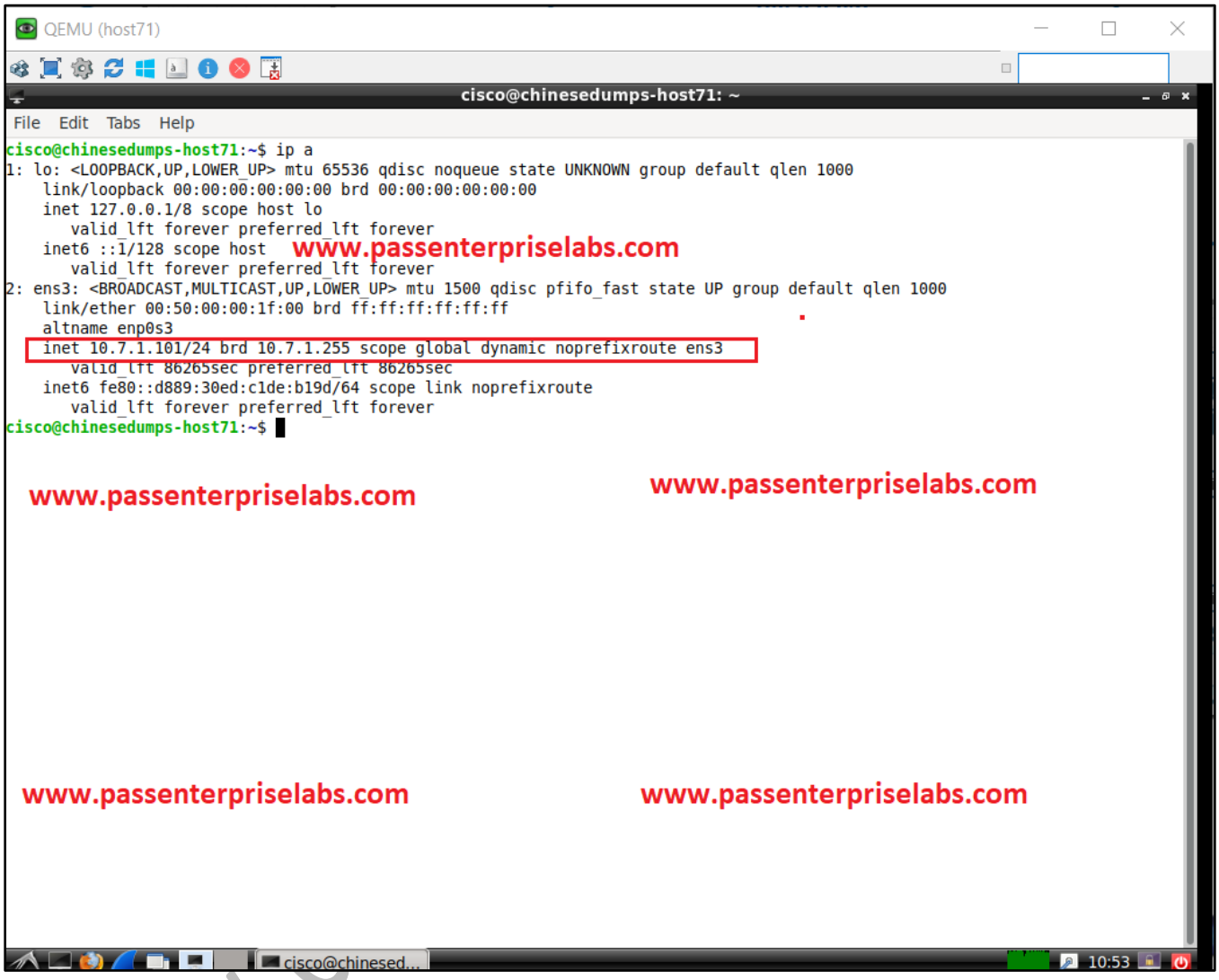


WWW

- 验证“如果其中一个接口发生故障，每个活动交换机必须跟踪其上行链路接口gi0/1和gi0/2/；活动交换机必须允许另一个交换机变为活动状态。但是，不允许跟踪修改HSRP优先级来满足这一要求。”

  - 打开sw601并关闭接口g0/2或g0/1，然后在sw602上检查它现在应该成为vlan 2000的网关

Chinesedumps-host71:



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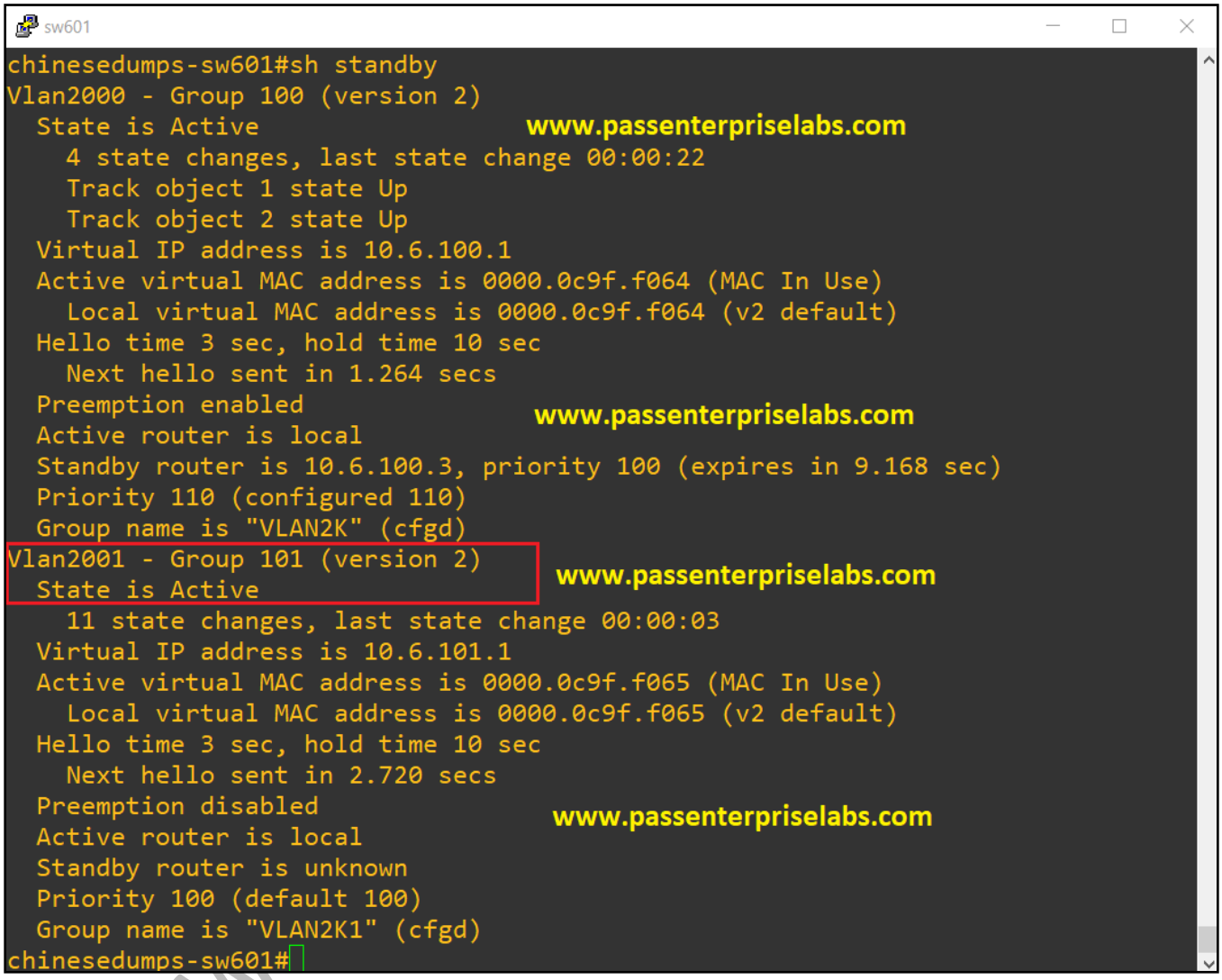
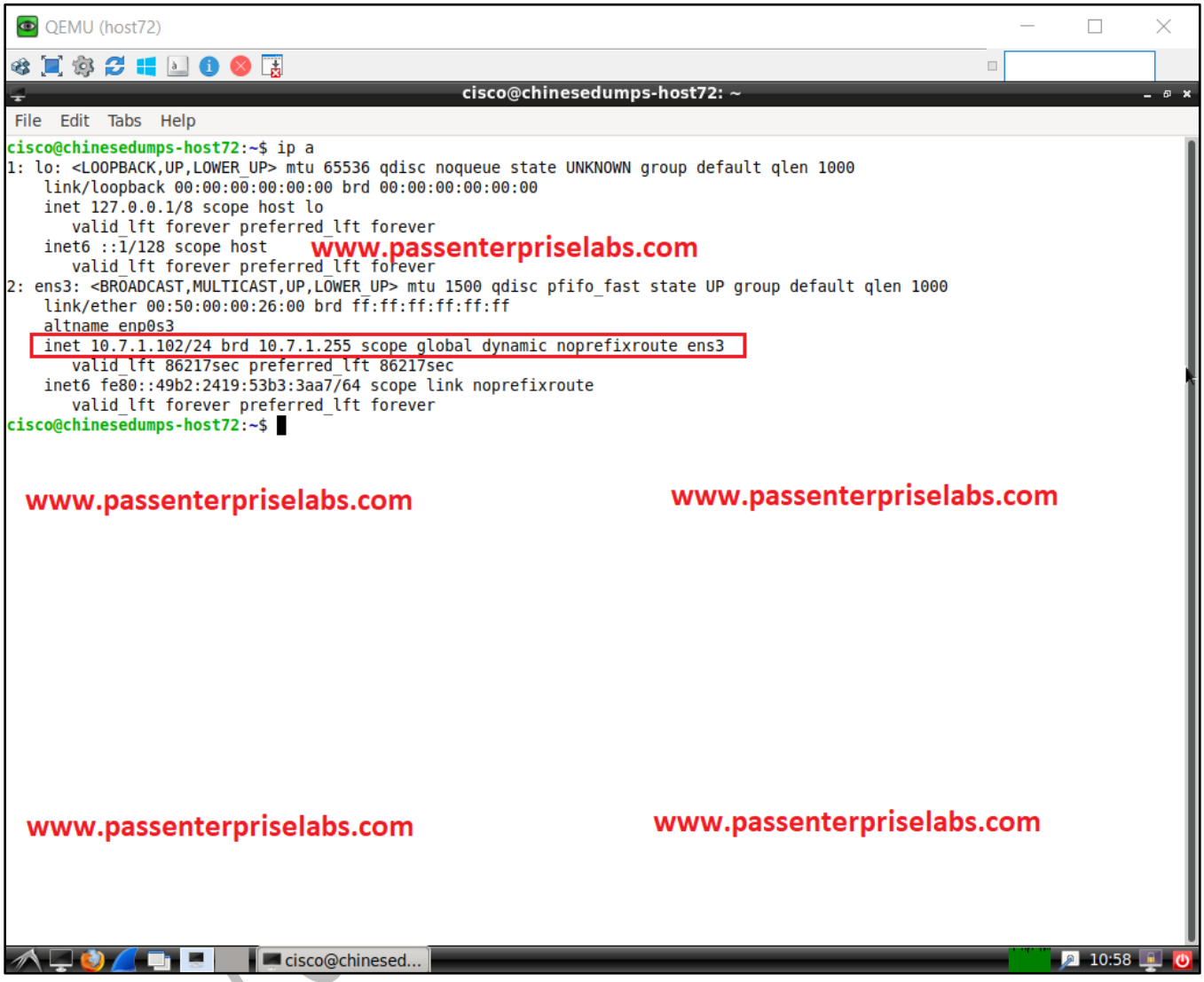
- 打开sw602并关闭接口g0/2或g0/1，然后在sw601上检查它现在应该成为vlan 2001的网关



WWW

Chinesedumps-host72:



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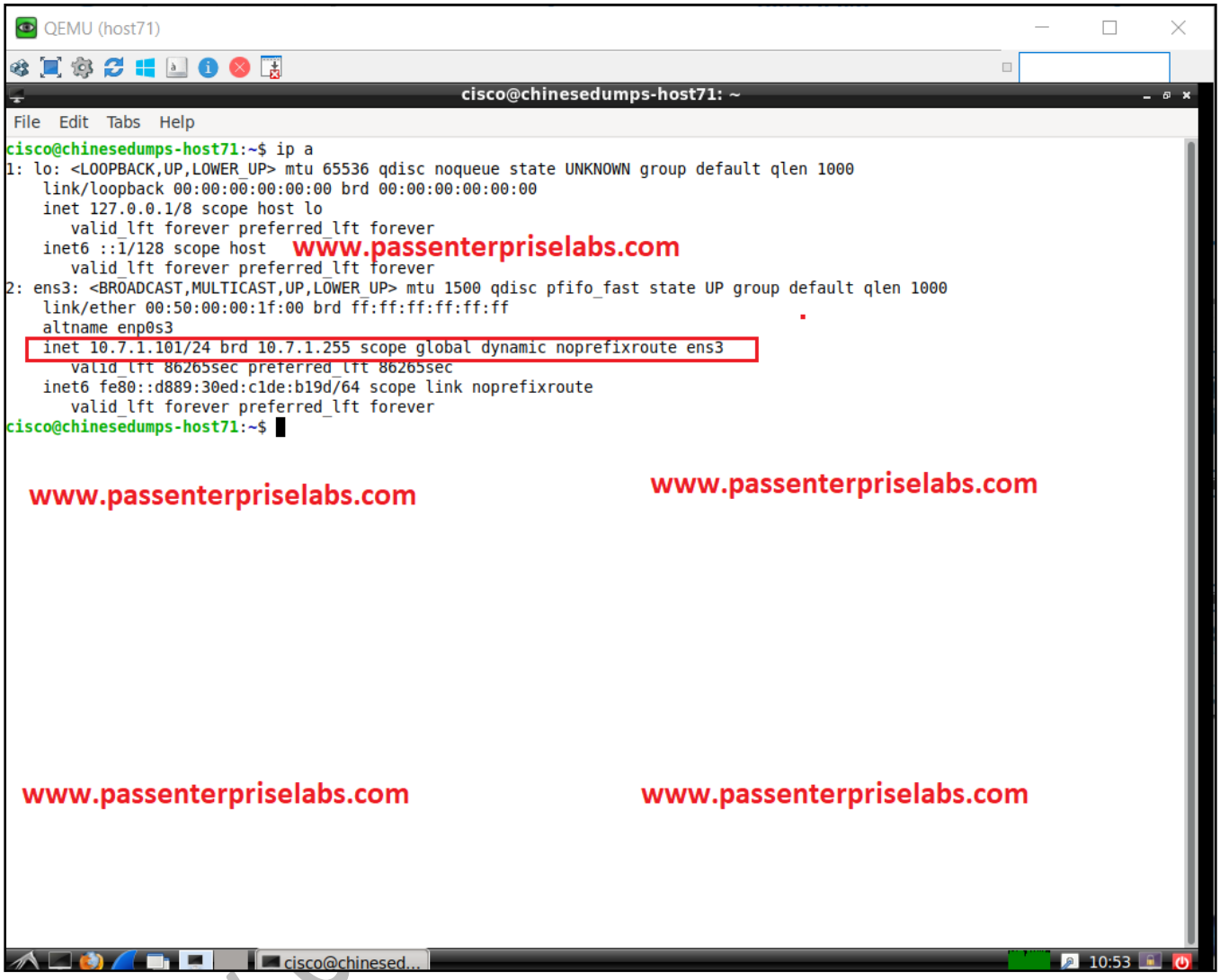
中文主机71：


QEMU (host71)
cisco@chinesedumps-host71: ~
File Edit Tabs Help
cisco@chinesedumps-host71:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 00:50:00:00:1f:00 brd ff:ff:ff:ff:ff:ff
altname enp0s3
inet 10.7.1.101/24 brd 10.7.1.255 scope global dynamic noprefixroute ens3
valid_lft 86265sec preferred_lft 86265sec
inet6 fe80::d889:30ed:c1de:b19d/64 scope link noprefixroute
valid_lft forever preferred_lft forever
cisco@chinesedumps-host71:~$
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www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@chinesed...
10:53

WWW.

1.14: Multicast in FABD2

FABD2 is preparing to enable PIM Sparse mode multicast routing in its network. As a part of validating the runbooks, FABD2 requires a sanity check to prevent inappropriate use of multicast-related configuration commands on different router types:

- First Hop Routers – routers where multicast sources are connected
- Last Hop Routers- routers where multicast receivers (subscribers) are connected
- Intermediary Hop Routers- routers on the path between First Hop and Last Hop routers

In the Table below, for each configuration command, select all router type where the use ofthe command is appropriate. (Select all that apply)

Command	Router Type		
	First Hop Router	Intermediary Hop Router	Last Hop Router
ip pim register-source	☐	☐	☐
ip igmp version	☐	☐	☐
ip pim spt-threshold	☐	☐	☐
ip pim rp-address	☐	☐	☐
ip pim sparse-mode	☐	☐	☐

2 points

Answer:

Command	Router Type		
	First Hop Router	Intermediary Hop Router	Last Hop Router
ip pim register-source	☐	☐	☐
ip igmp version	☐	☐	☐
ip pim spt-threshold	☐	☐	☐
ip pim rp-address	☐	☐	☐
ip pim sparse-mode	☐	☐	☐

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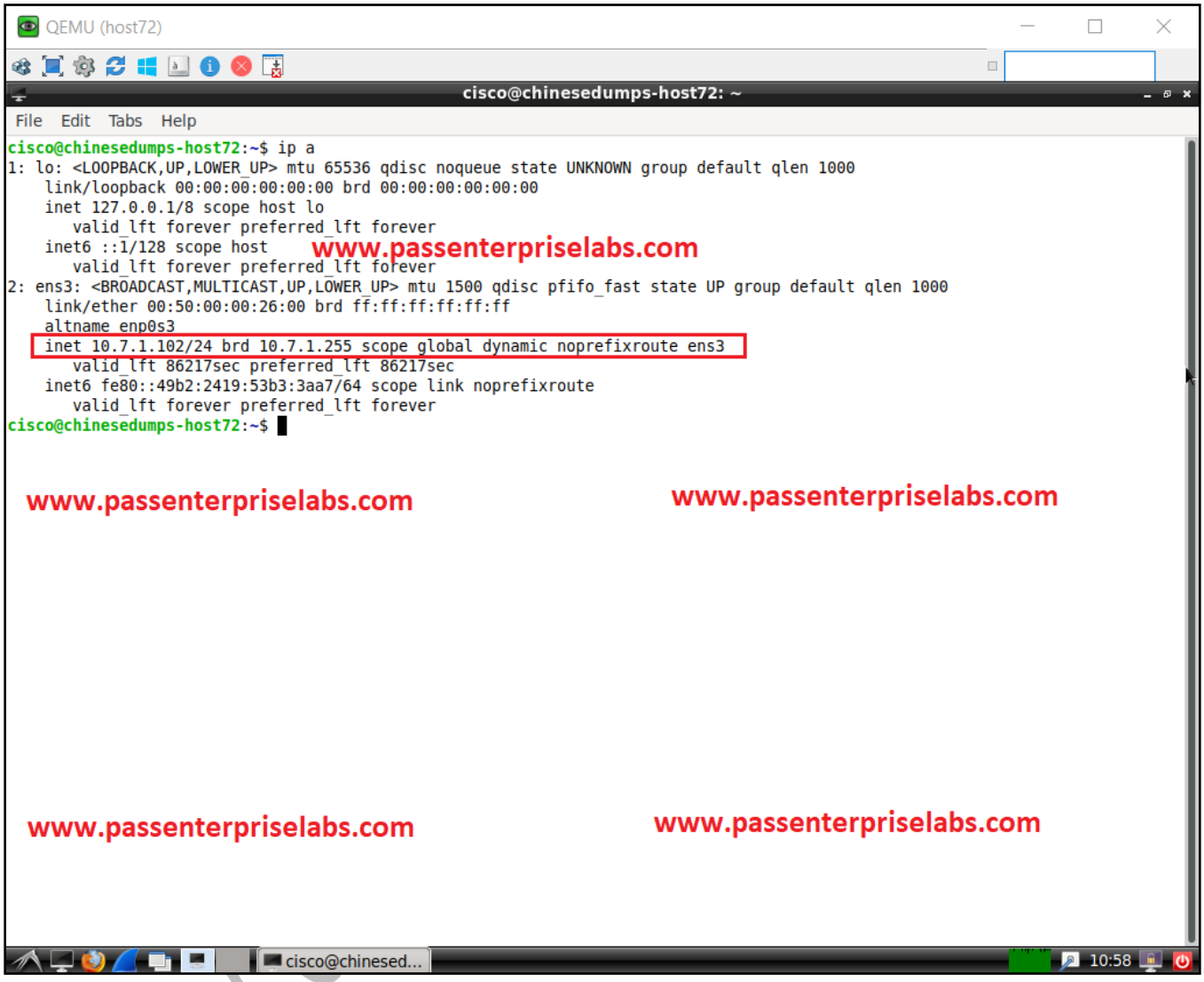
中文主机72：


QEMU (host72)
cisco@chinesedumps-host72: ~
File Edit Tabs Help
cisco@chinesedumps-host72:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 00:50:00:00:26:00 brd ff:ff:ff:ff:ff:ff
altname enp0s3
inet 10.7.1.102/24 brd 10.7.1.255 scope global dynamic noprefixroute ens3
valid_lft 86217sec preferred_lft 86217sec
inet6 fe80::49b2:2419:53b3:3aa7/64 scope link noprefixroute
valid_lft forever preferred_lft forever
cisco@chinesedumps-host72:~$
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www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@chinesed...
10:58

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1.15: Extending Connectivity to IaaS Site

Extend the IPv6 connectivity from HQ through the SP into the giosk VRF on the IaaS site according to these requirements:

Set up global IPv6 addressing on the link between r11 and r3

- On r11, assign 2001:2710:311::2/64 to g0/0
- On r3, assign 2001:2710:311::1/64 to g1
- Enable the existing IPv4 BGP session between r11 and r3 to also advertise IPv6 prefixes. Do not configure a standalone IPv6 BGP session between these two routers.
- Perform bidirectional route redistribution between the IPv6 EIGRP and BGP processes on r11
- Ensure that all current and future IPv6 prefixes advertised between r11 and r3 will be installed into the RIB of these routers with the next hop address set to the proper global unicast address on their interconnection. Any policy that accomplishes this requirement must be applied in the inbound direction.
- The giosk VRF on r4 that extends the IPv6 connectivity from r4 to r30 on the IaaS site is a separate VRF independent of fabd2 VRF. Any route leaking from fabd2 VRF into giosk VRF must be done on per-site basis and only for those FABD2 sites that need connectivity in the IaaS site.
- By configuring r3 and r4 only, ensure that the HQ FABD2 site will have mutual visibility with the IaaS site while preventing
    - Any other FABD2 site from possibly learning about the routes on the IaaS site
    - The IaaS site from possibly learning about the routes on any other FABD2 site
- Use the minimum amount of commands necessary to accomplish this requirement. Do not remove any existing configuration. If necessary, you are allowed to use an additional route target with the value of 10000:3681.
- Verify that host11 and host12 can ping 2001:db8:14::1 located at the IaaS site. It is permitted to modify one existing configuration command on one of the SP routers to meet this requirement

3 Points

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1.14： FABD2中的多播

FABD2正准备在其网络中启用PIM稀疏模式多播路由。作为验证运行手册的一部分，FABD2需要进行健全性检查，以防止在不同的路由类型上不当使用与多播相关的配置命令：

- 第一跳路由器——连接多播源的路由器
- 最后一跳路由器-连接多播接收器（订户）的路由器
- 中间跳路由器-在第一跳和最后跳路由器之间的路径上的路由器

在下表中，对于每个配置命令，选择适合使用该命令的所有路由器类型。（选择所有适用项）

命令	路由器类型		
	第一跳路由器	中间跳 路由器	最后一跳路由器
ip-pim寄存器源	□	□	□
ip igmp版本	□	□	□
ip pim spt阈值	□	□	□
ip pim-rp地址	□	□	□
ip-pim稀疏模式	□	□	□

2分

答案：

命令	路由器类型		
	第一跳路由器	中间跳 路由器	最后一跳路由器
ip-pim寄存器源	□	□	□
ip igmp版本	□	□	□
ip pim spt阈值	□	□	□
ip pim-rp地址	□	□	□
ip-pim稀疏模式	□	□	□

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Solution:

Chinese_Dumps_r11:

```
r11(config-if)#int gi0/0
r11(config-if)#ipv6 address 2001:2710:311::2/64

r11(config-if)#route-map NH-IPV6 per 10
r11(config-if)#set ipv6 next-hop 2001:2710:311::1

r11(config)#router bgp 65001
r11(config-router)#address-family ipv6
r11(config-router)#address-family ipv6 unicast
r11(config-router)#neighbor 100.3.11.1 remote-as 10000
r11(config-router)#neighbor 100.3.11.1 route-map NH-IPV6 in
r11(config-router)#redistribute eigrp 65001

r11(config)#router eigrp ccie
r11(config-router)#address-family ipv6 unicast autonomous-system 65001
 r11(config-router)#topology base
 r11(config-router)#redistribute bgp 65001 metric 100 50 255 1 1500
```

Chinese_Dumps_r4:

```
r4(config)#vrf definition giosk
r4(config)#route-target import 10000:3681

r4(config)#int lo0
r4(config-if)#ip address 100.255.254.4 255.255.255.255
```

Chinese_Dumps_r3:

```
r3(config)#vrf definition fabd2
r3(config)#route-target export 10000:3681
r3(config)#route-target import 10000:414
r3(config-vrf)#address-family ipv6
r3(config-af)#exit-address-family
```

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1.15： 将连接扩展到laaS站点

根据以下要求，将IPv6连接从总部通过SP扩展到laaS站点上的giosk VRF：

在r11和r13之间的链路上设置全局IPv6寻址

- 在r11上，将2001:2710:311::2/64分配给g0/0
- 在r3上，将2001:2710:311::1/64分配给g1
- 启用r11和r3之间的现有IPv4 BGP会话，以同时通告IPv6前缀。请勿在这两个路由器之间配置独立的IPv6 BGP会话。
- 在r11上执行IPv6 EIGRP和BGP进程之间的双向路由重新分配
- 确保在r11和r3之间通告的所有当前和未来的IPv6前缀都将安装到这些路由器的RIB中，下一跳地址设置为其互连上的正确全局单播地址。任何满足此要求的策略都必须应用于入站方向。
- r4上的giosk VRF将laaS站点上的IPv6连接从r4扩展到r30，它是一个独立于fabd2 VRF的单独VRF。从fabd2 VRF泄漏到giosk VRF的任何路线都必须按每个站点进行，并且仅适用于需要在laaSite中连接的fabd2站点。
- 通过仅配置r3和r4，确保总部FABD2站点与laaS站点具有相互可见性，同时防止
    - 任何其他FABD2站点可能无法了解laaS站点上的路线
    - laaS网站可能无法了解任何其他FABD2网站上的路线
- 使用完成此要求所需的最小命令量。不要删除任何现有配置。如有必要，您可以使用值为10000:3681的额外路线目标。
- 验证主机11和主机12是否可以ping位于laaS站点的2001:db8:14::1。允许修改其中一个SP路由器上的一个现有配置命令以满足此要求

3分

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```
r3(config)#int g1
r3(config-if)#ipv6 address 2001:2710:311::1/64
r3(config-if)#route-map NH-IPV6 per 10
r3(config-if)#set ipv6 next-hop 2001:2710:311::2

r3(config-if)#router bgp 10000
r3(config-router)#address-family ipv6 vrf fabd2
r3(config-router)#neighbor 100.3.11.2 remote-as 65001
r3(config-router)#neighbor 100.3.11.2 route-map NH-IPV6 in
```

To test R30 connectivity as per question, I created a loopback interface on R30

```
interface Loopback414
 no ip address
 ipv6 address 2001:DB8:14::1/128
 ipv6 enable
```

NOTE : On r4 there is loopback in pre-configs with /31 mask , we need to correct the loopback mask to /32

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解决方案：

中文_Dumps_r11：

r11（配置if）#int gi0/0
r11（配置if）#ipv6地址2001:2710:311:: 2/64

r11（config-if）#每10个路由映射NH-IPV6
r11（config.if）#设置IPV6下一跳
2001:2710:311:: 1

r11（配置）#路由器bgp65001r11（
配置路由器）#地址族ipv6
r11（配置路由器）#地址家族ipv6单播r11（设置路
由器）#邻居100.3.11.1远程为10000
r11（配置路由器）#邻居100.3.11.1 r11中的路由图NH-
IPV6#重分发eigrp 65001

r11（配置）#路由器eigrp-ccie
r11（配置路由器）#地址族ipv6单播自治系统65001 r11（设置路由
器）#拓扑库
r11（配置路由器）#重新分配bgp 65001指标100 50 255 1 1500

中文_Dumps_r4:

r4（配置）#vrf定义giosk r4（配置"#
路由目标导入10000:3681

r4（配置）#int lo0
r4（配置if）#ip地址100.255.255.4 255.255.255.255

中文_Dumps_r3：

r3（config）#vrf定义fabd2 r3（配置
）#路由目标导出10000:3681
r3（config）#路由目标导入10000:414
r3（config-vrf）#地址族ipv6 r3（
config.af）#退出地址族

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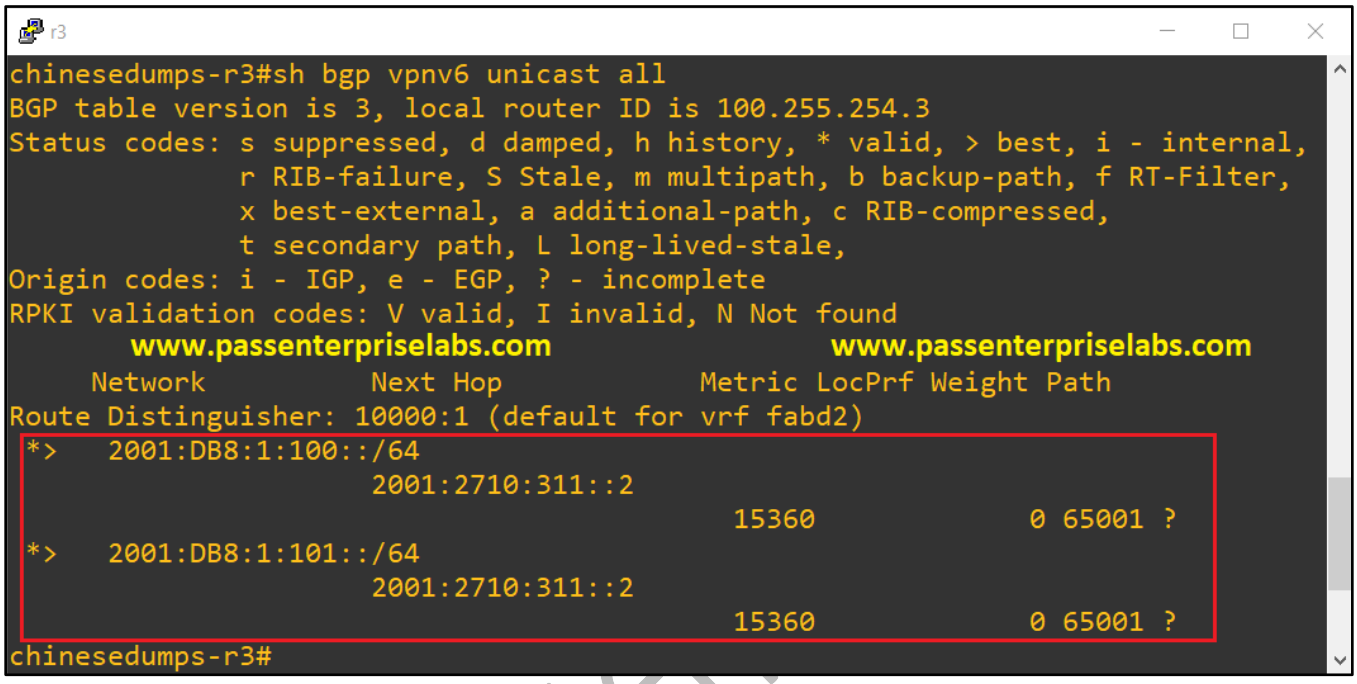
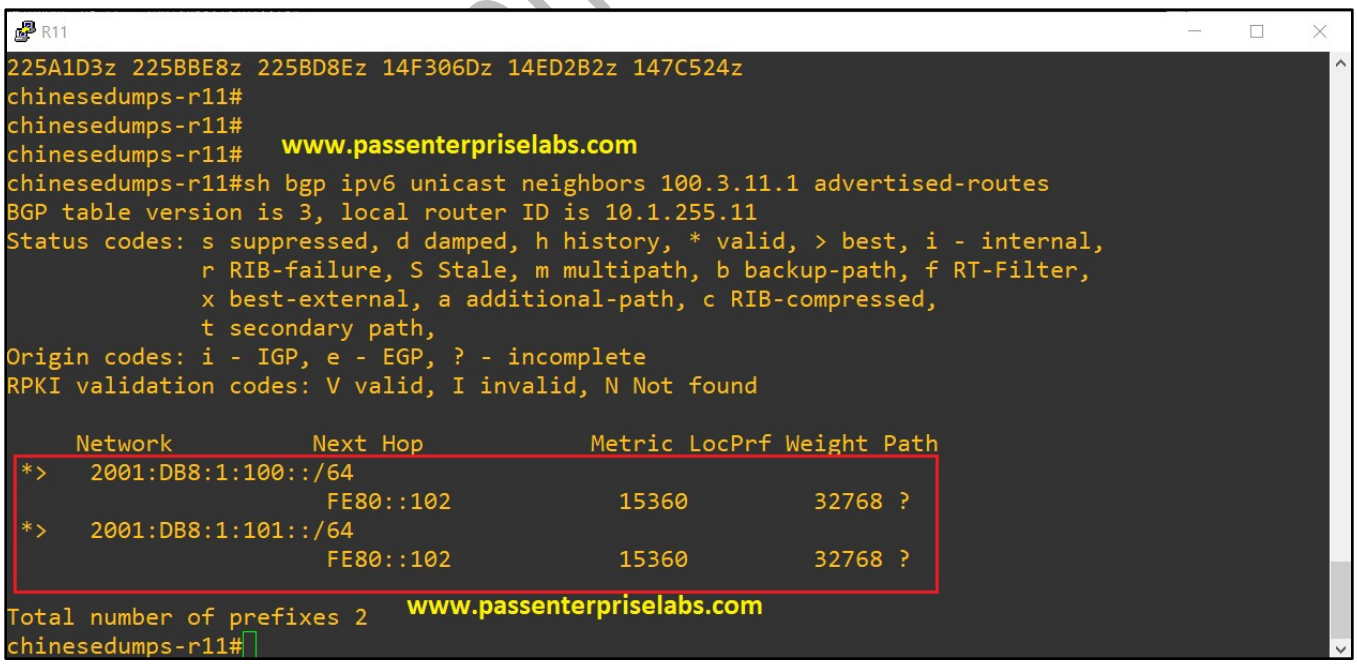
Verification:

Chinesedumps-r3: sh bgp vpnv6 unicast all



Chinesedumps-r11: sh bgp ipv6 unicast neighbors 100.3.11.1 advertised-routes



r3（配置）#int g1
r3（config-if）#ipv6地址2001:2710:311:
：1/64 r3（config.if）#每10个路由映射
NH-ipv6 r3（-config-if）#set ipv6下一跳
2001:2710:11:：2

r3（配置if）#路由器bgp 10000
r3（配置路由器）#地址系列ipv6 vrf fabd2 r3（
设置路由器）#邻居100.3.11.2远程为65001
r3（配置路由器）#邻居100.3.11.2中的NH-IPV6路由图

为了按照问题测试R30的连接性，我在R30上创建了一个环回接口

接口Loopback414没有
ip地址
ipv6地址2001:DB8:14:：1/128启
用ipv6

注意：在r4上，使用/31掩码的预配置中存在环回，我们需要更正环回掩码至/32

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Chinesedumps-r3: sh ip route vrf fabd2 100.4.30.5

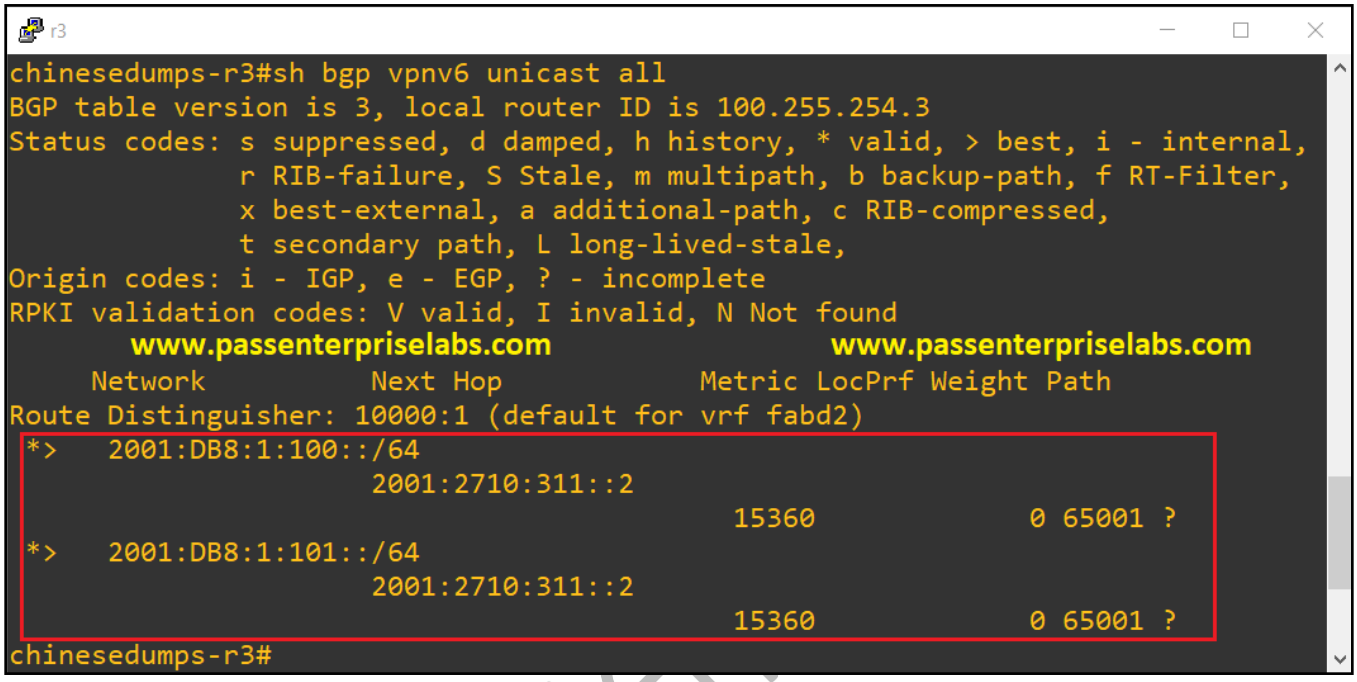
```
r3
chinesedumps-r3#
chinesedumps-r3#   www.passenterpriselabs.com
chinesedumps-r3#sh ip route vrf fabd2 100.4.30.5

Routing Table: fabd2
Routing entry for 100.4.30.4/30
  Known via "bgp 10000", distance 200, metric 0, type internal
  Last update from 100.255.254.4 00:36:47 ago
  Routing Descriptor Blocks:
  * 100.255.254.4 (default), from 100.255.254.4, 00:36:47 ago
      Route metric is 0, traffic share count is 1
      AS Hops 0
      MPLS label: 16
      MPLS Flags: MPLS Required
chinesedumps-r3#   www.passenterpriselabs.com
chinesedumps-r3#
```

- From above output we can see that the giosk network (100.4.30.5) is present on r3 in fabd2
- And in the below output on r5 we can see that the giosk network (100.4.30.5) is not present in fabd2

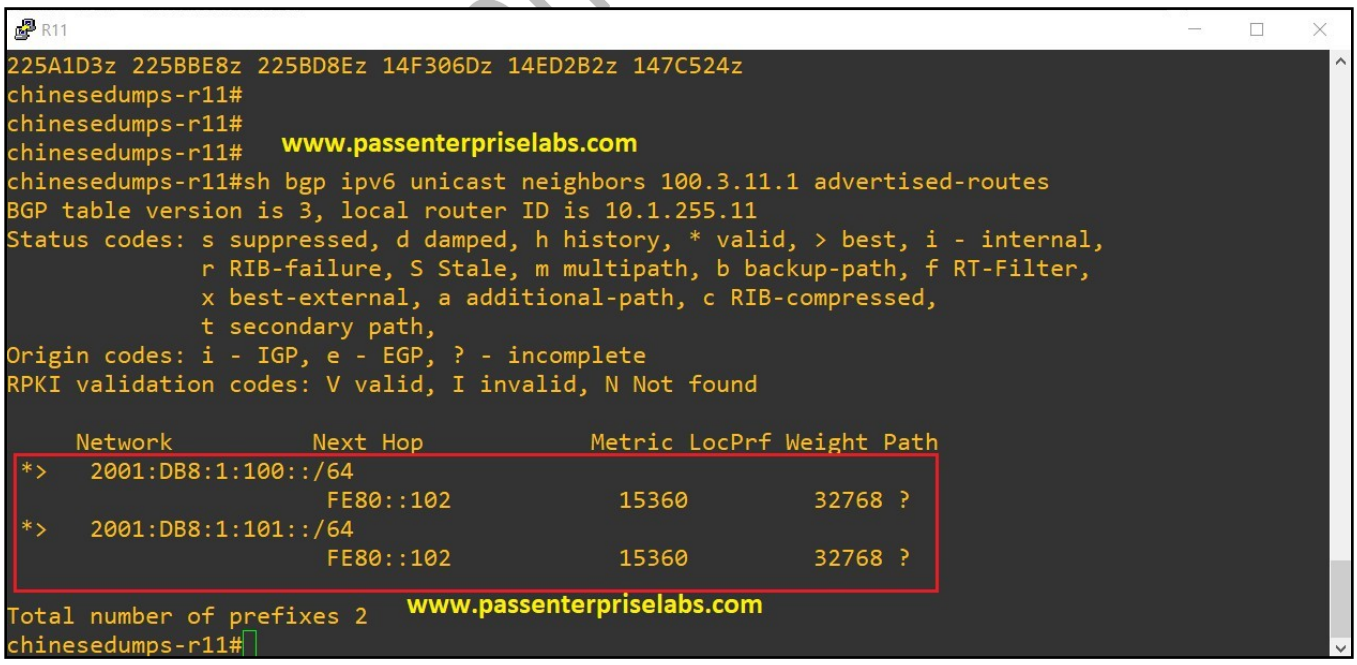
```
r5
chinesedumps-r5#
chinesedumps-r5#   www.passenterpriselabs.com
chinesedumps-r5#
chinesedumps-r5#
chinesedumps-r5# sh ip route vrf fabd 100.4.30.5
% IP routing table vrf fabd does not exist
chinesedumps-r5#
chinesedumps-r5#   www.passenterpriselabs.com
chinesedumps-r5#
chinesedumps-r5#
```

验证：

中文教程-r3:sh bgp vpnv6单机版

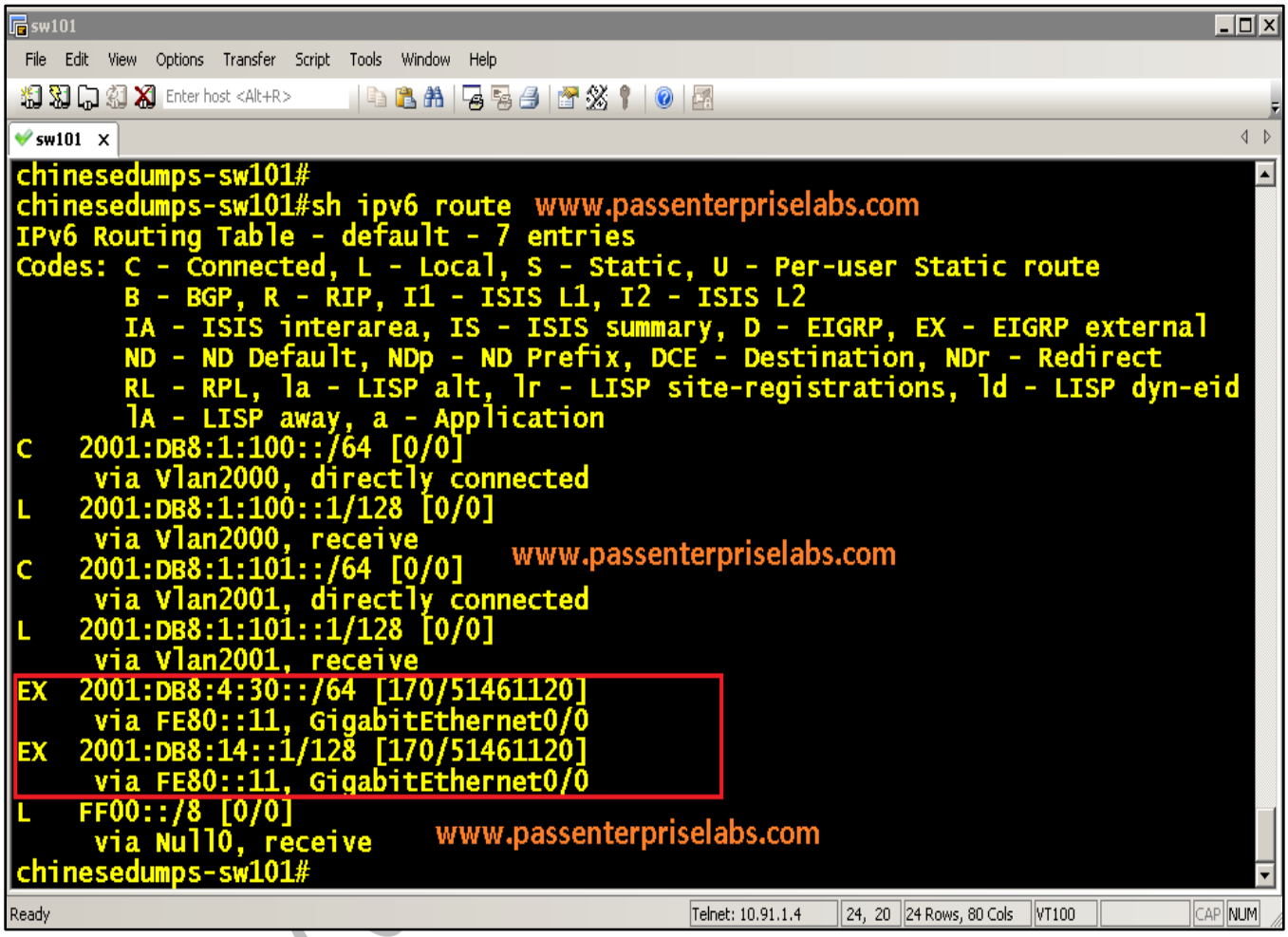


中文教程-r11:sh bgp ipv6单播邻居100.3.11.1通告路由



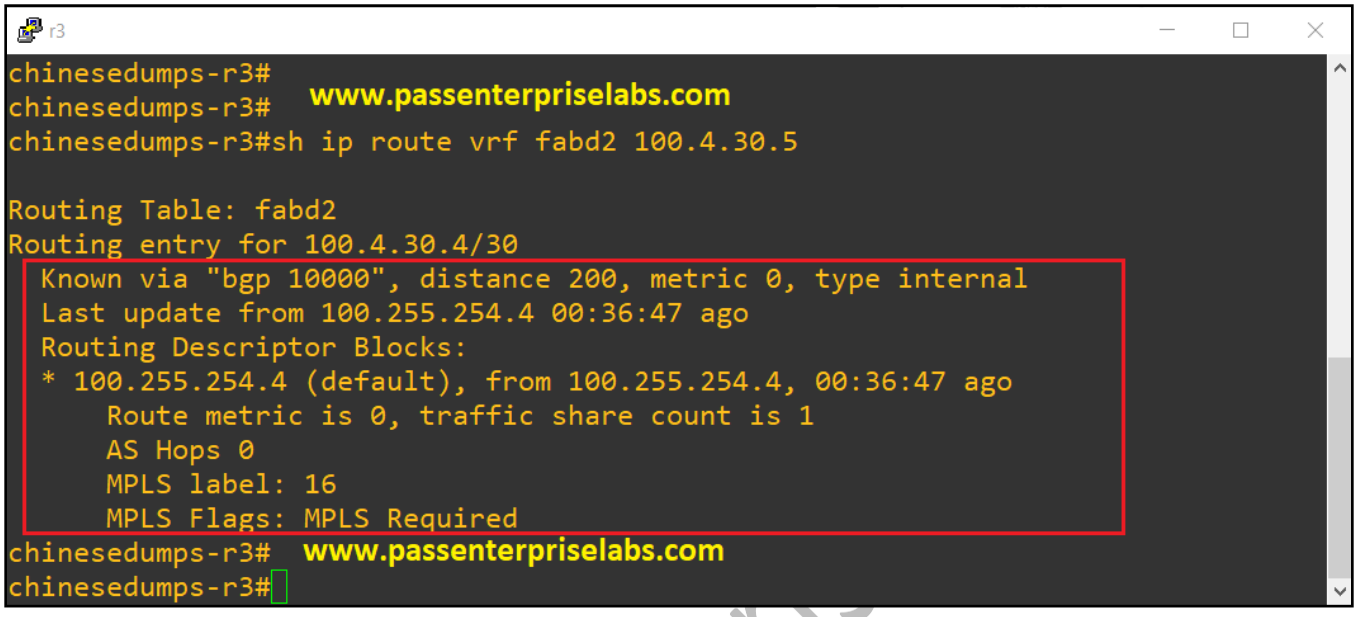
- Verify that host11 and host12 can ping 2001:db8:14::1 located at the IaaS site.

Chinesedumps-sw101: sh ipv6 route

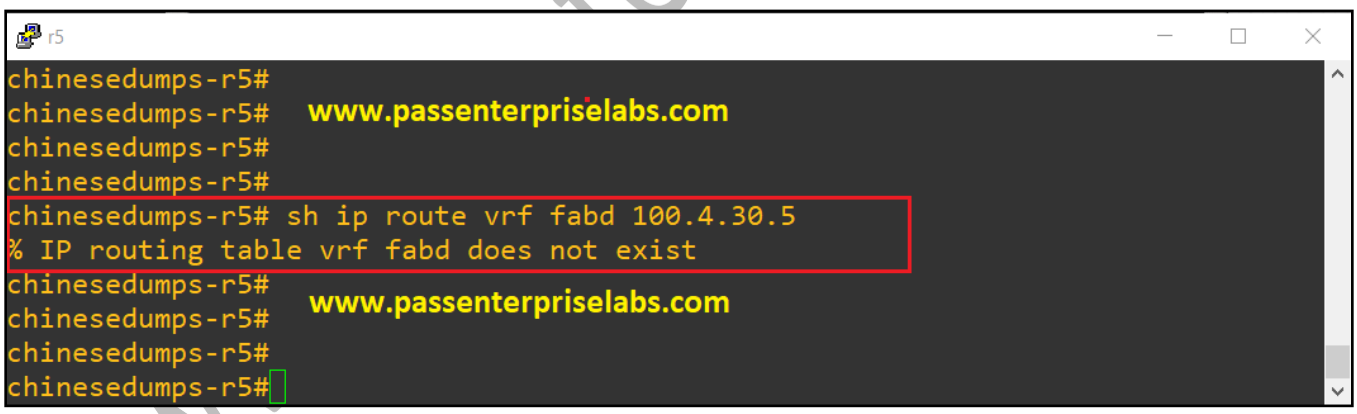


www.Ch

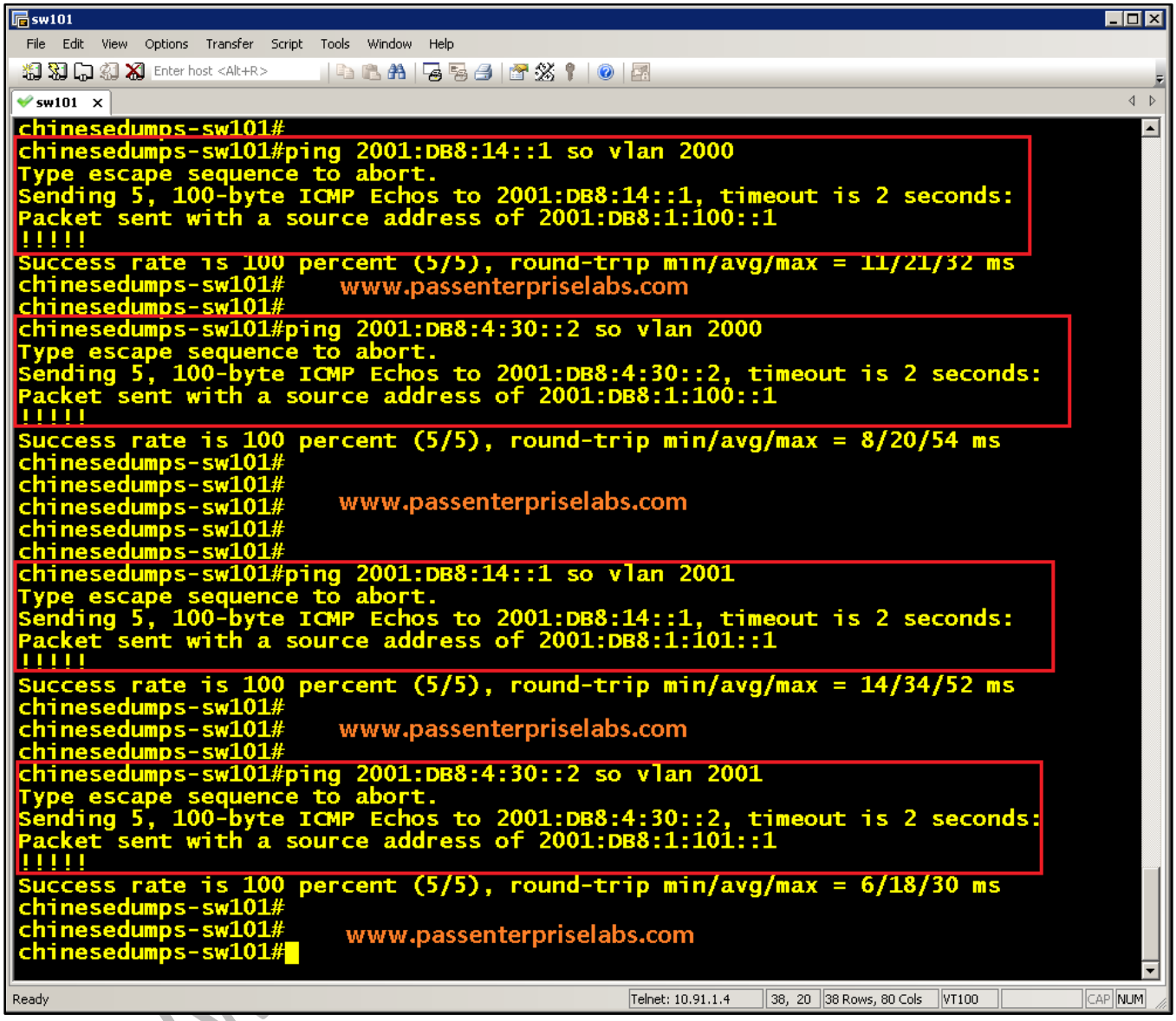
中文教程-r3:sh-ip路由vrf fabd2 100.4.30.5



- 从上面的输出中我们可以看到，giosk网络（100.4.30.5）存在于fabd2中的r3上
- 在r5的以下输出中，我们可以看到giosk网络（100.4.30.5）在fabd2中不存在



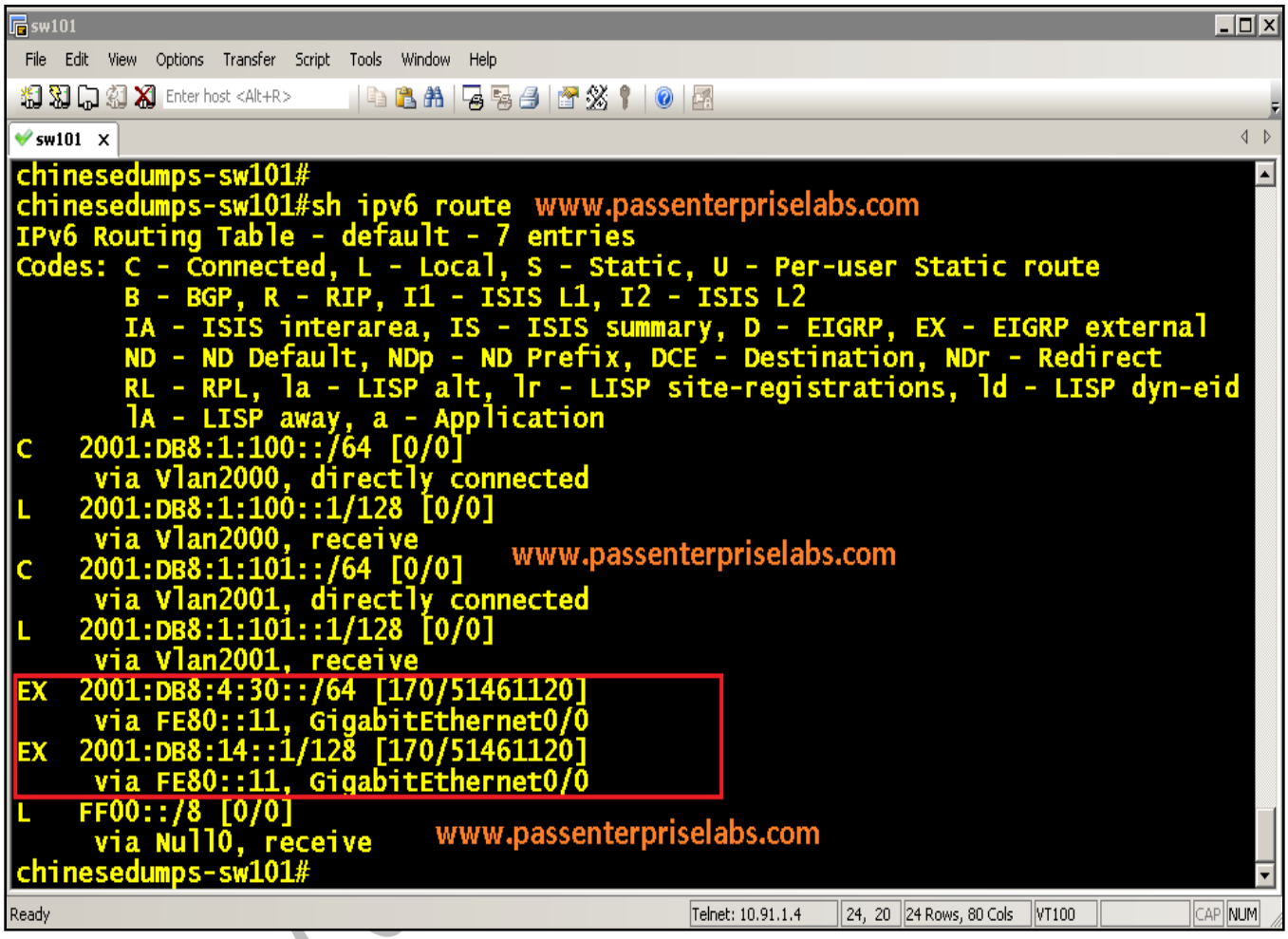
Chinesedumps-sw101:



www

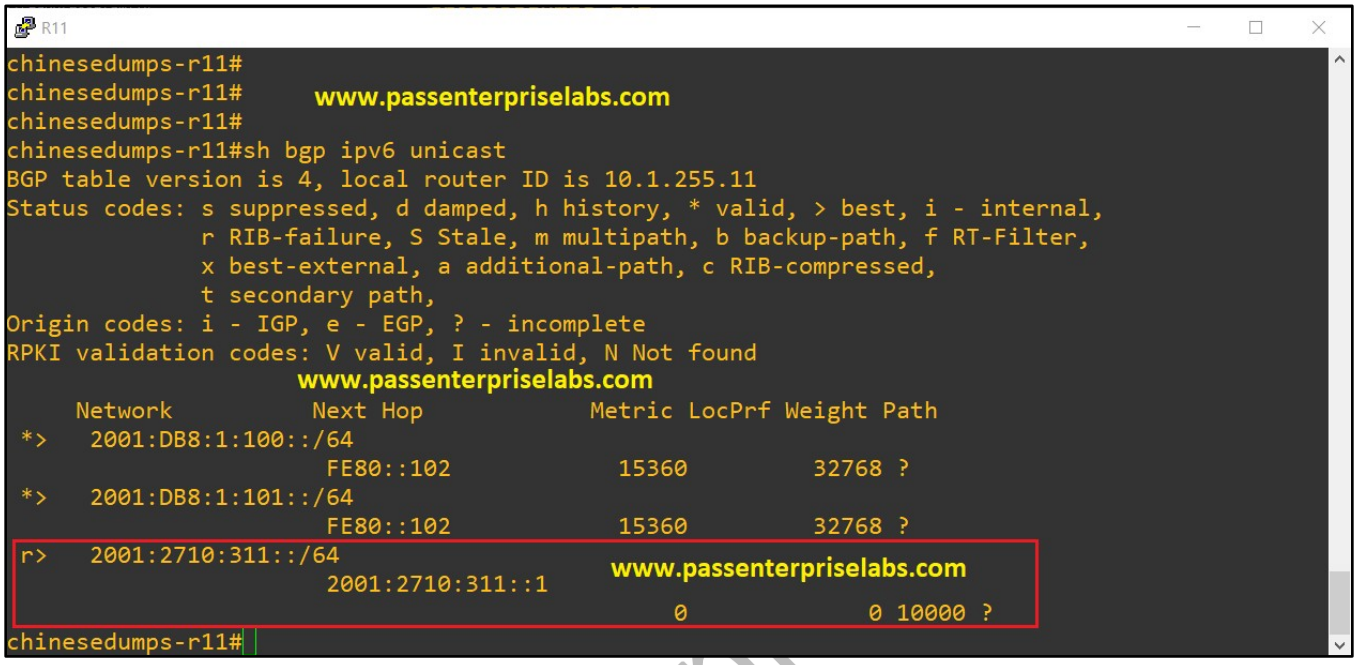
- 验证主机11和主机12是否可以ping位于IaaS站点的2001:db8:14:：1。

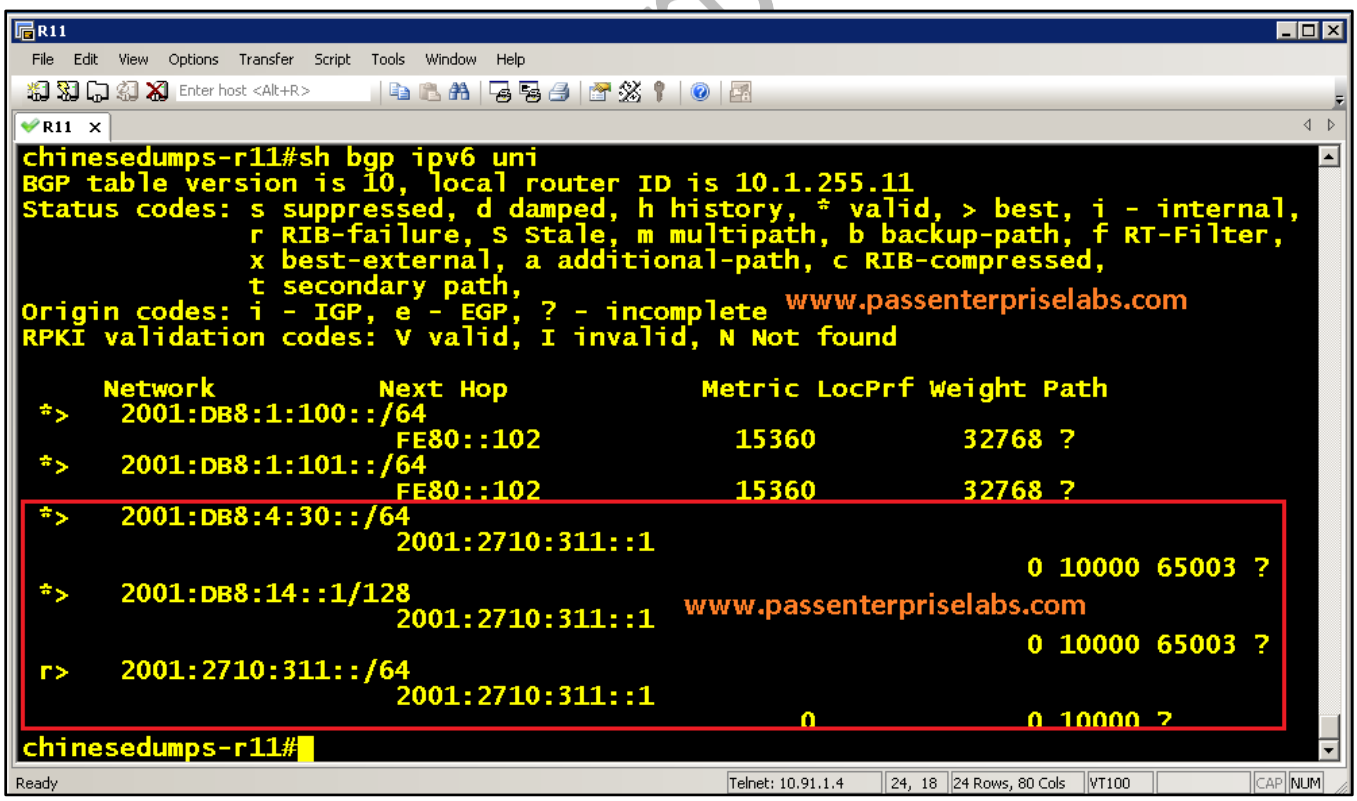
中文教程-sw101:sh ipv6路由



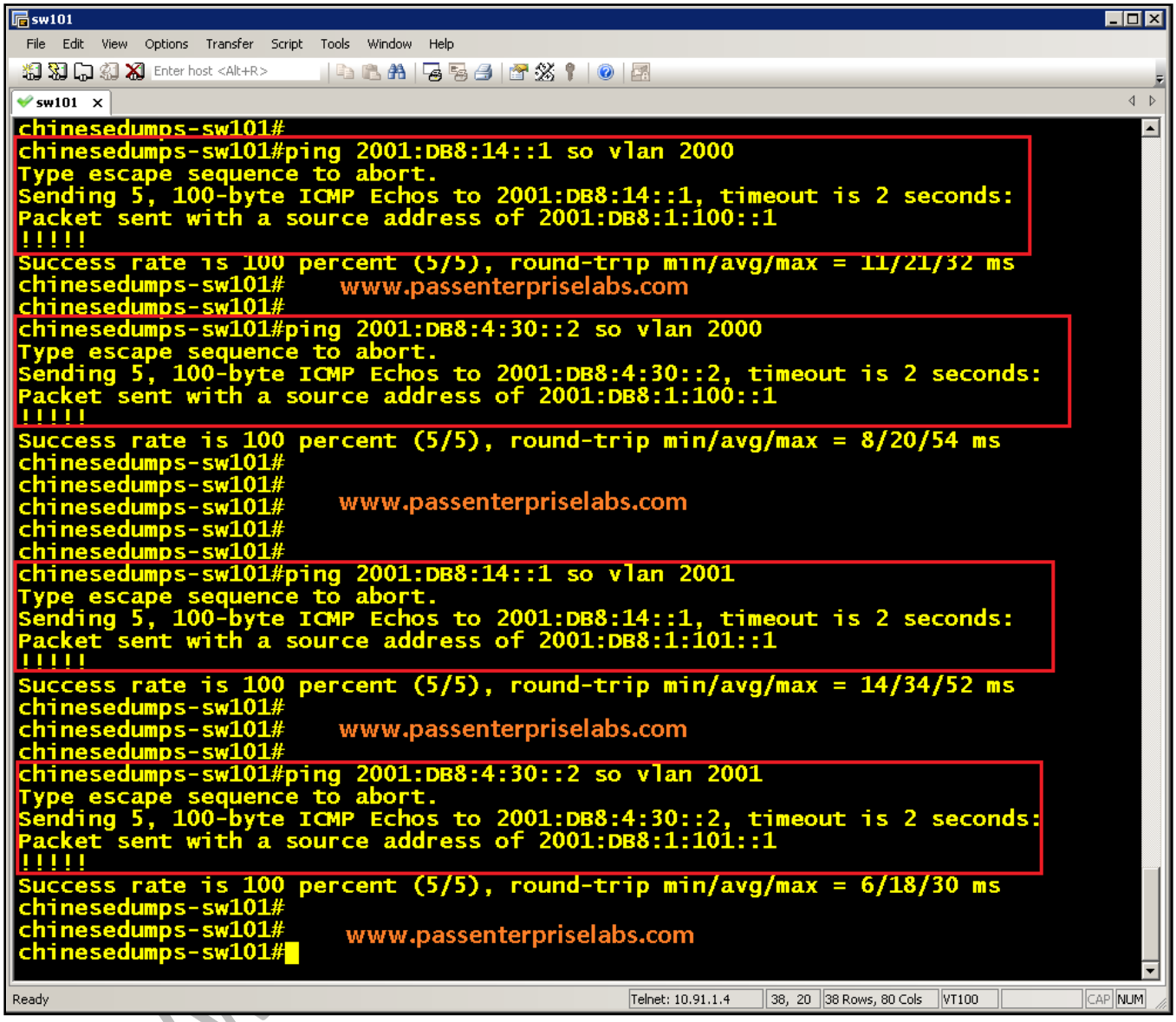
www.c

Chinesedumps-r11: sh bgp ipv6 unicast



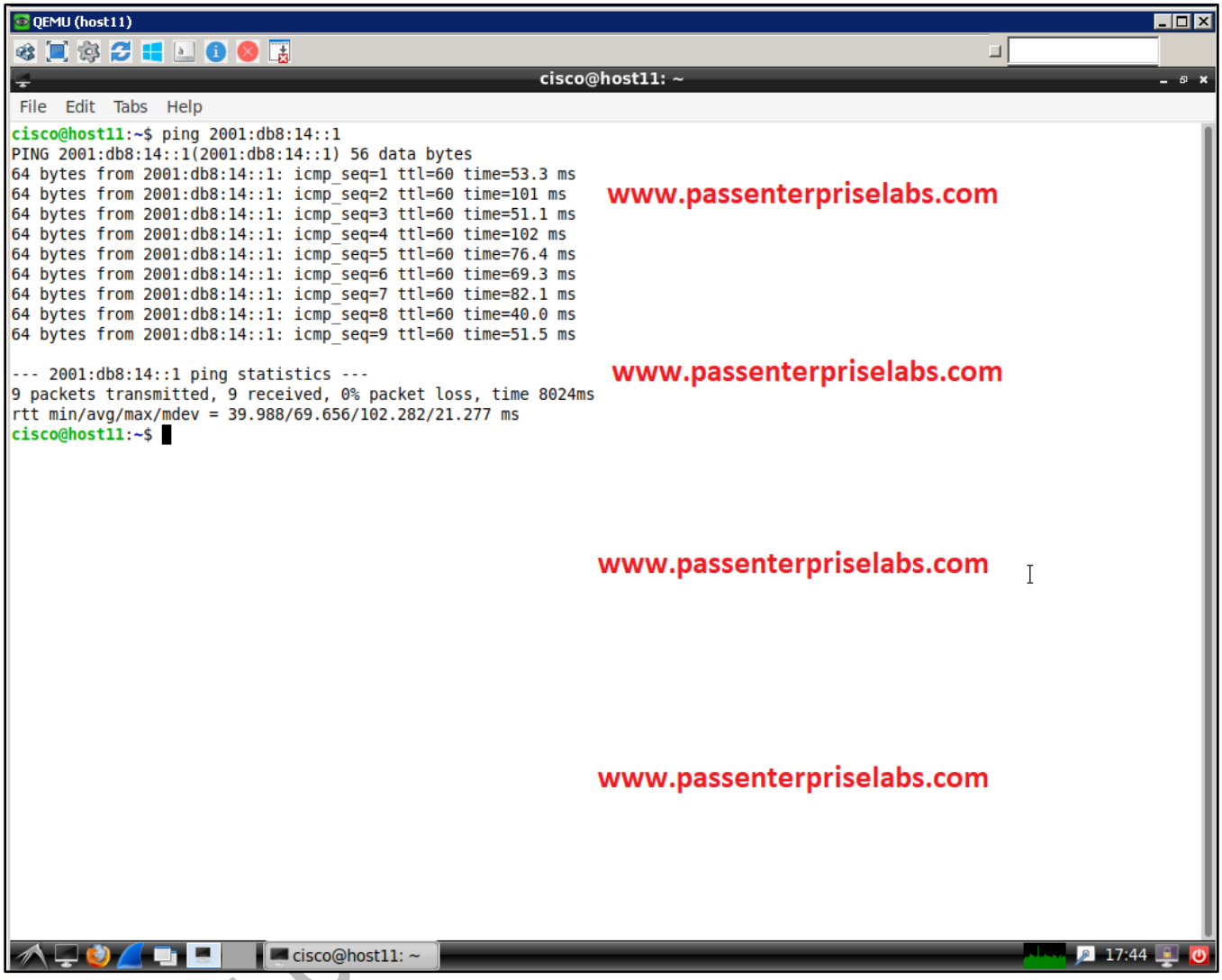


中文教程sw101：



WWW

Chinesedumps-host11:



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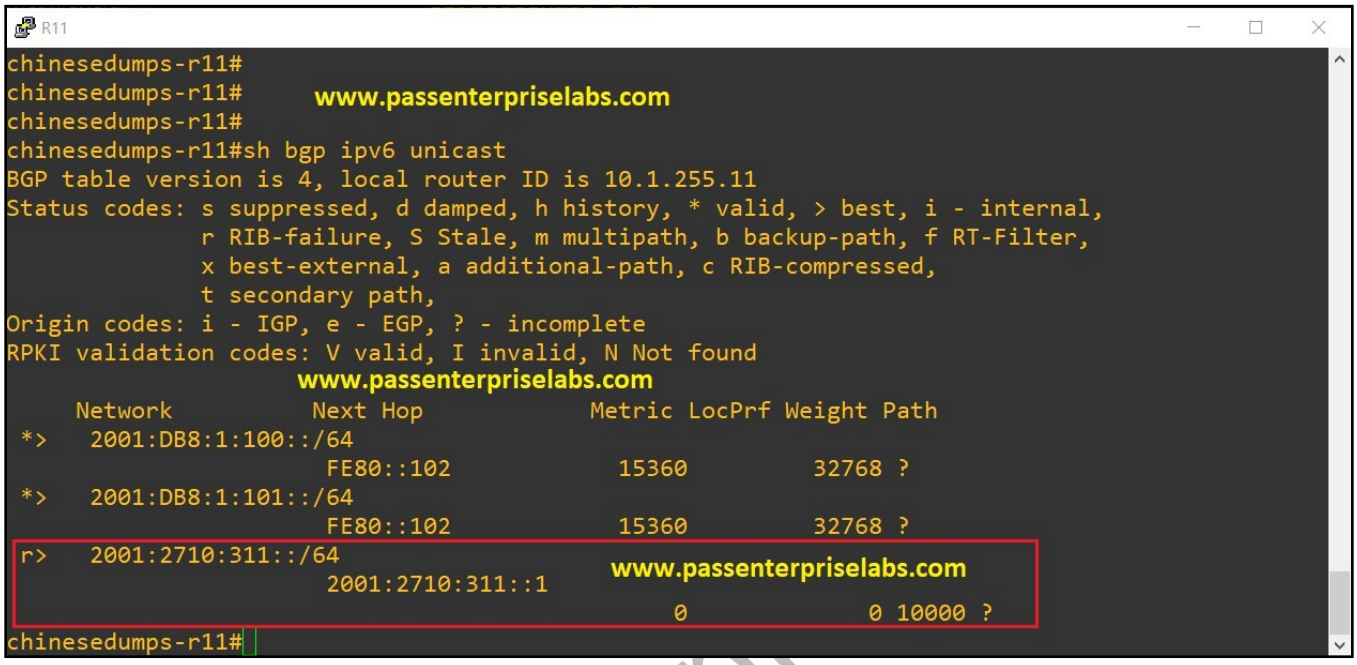
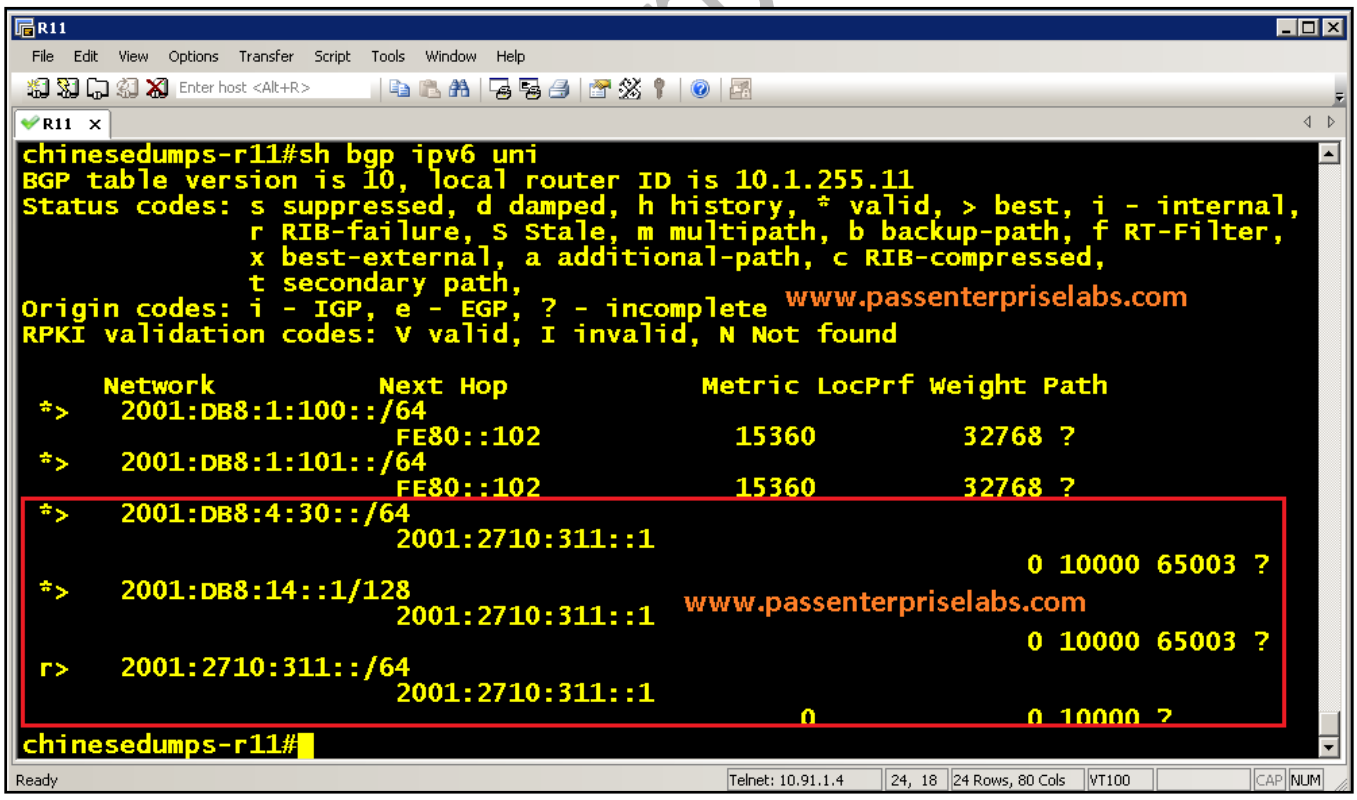
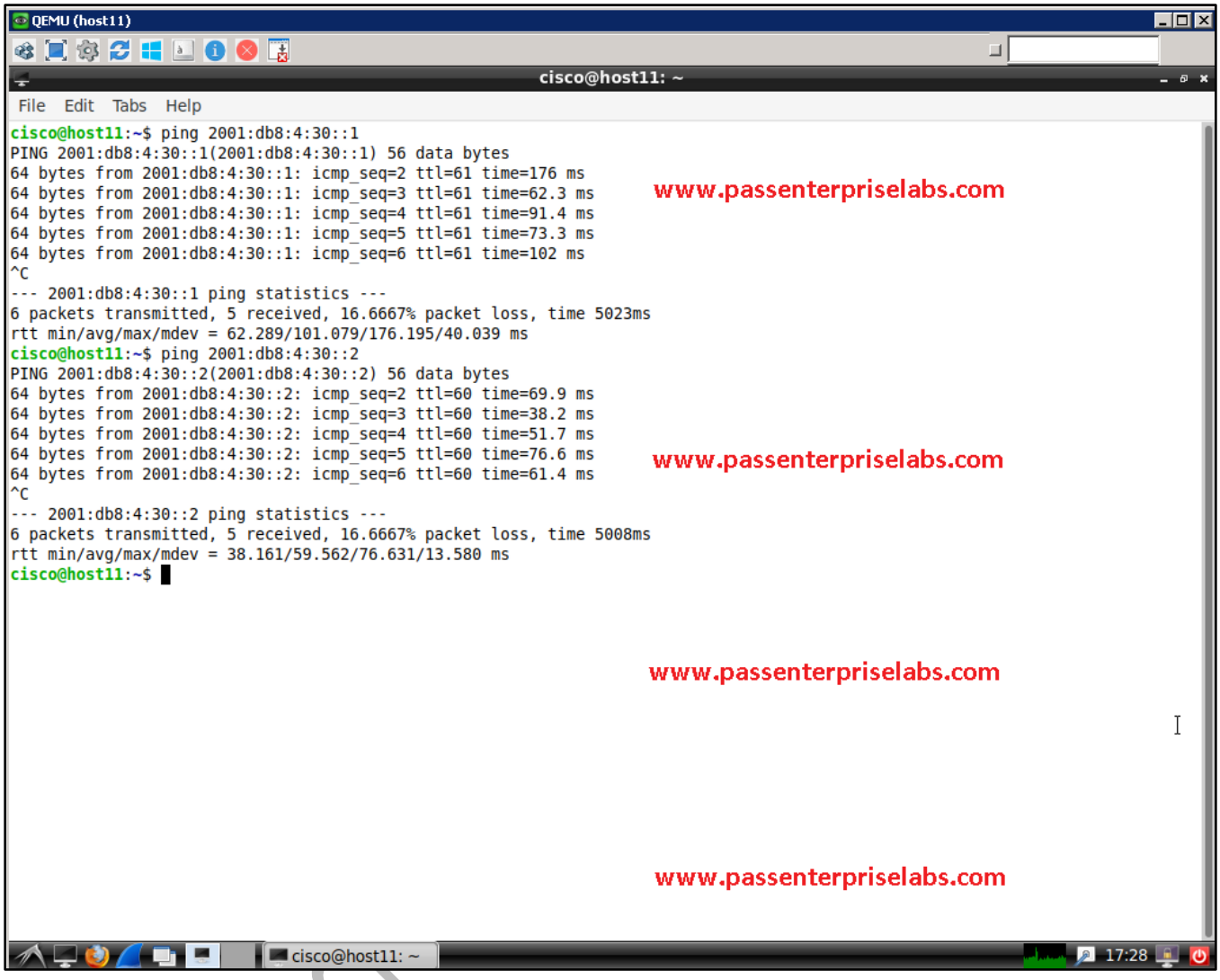
中文教程-r11:sh bgp ipv6单播






QEMU (host11)
cisco@host11: ~
File Edit Tabs Help
cisco@host11:~$ ping 2001:db8:4:30::1
PING 2001:db8:4:30::1(2001:db8:4:30::1) 56 data bytes
64 bytes from 2001:db8:4:30::1: icmp_seq=2 ttl=61 time=176 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=3 ttl=61 time=62.3 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=4 ttl=61 time=91.4 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=5 ttl=61 time=73.3 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=6 ttl=61 time=102 ms
^C
--- 2001:db8:4:30::1 ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5023ms
rtt min/avg/max/mdev = 62.289/101.079/176.195/40.039 ms
cisco@host11:~$ ping 2001:db8:4:30::2
PING 2001:db8:4:30::2(2001:db8:4:30::2) 56 data bytes
64 bytes from 2001:db8:4:30::2: icmp_seq=2 ttl=60 time=69.9 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=3 ttl=60 time=38.2 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=4 ttl=60 time=51.7 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=5 ttl=60 time=76.6 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=6 ttl=60 time=61.4 ms
^C
--- 2001:db8:4:30::2 ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5008ms
rtt min/avg/max/mdev = 38.161/59.562/76.631/13.580 ms
cisco@host11:~$
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@host11: ~
17:28

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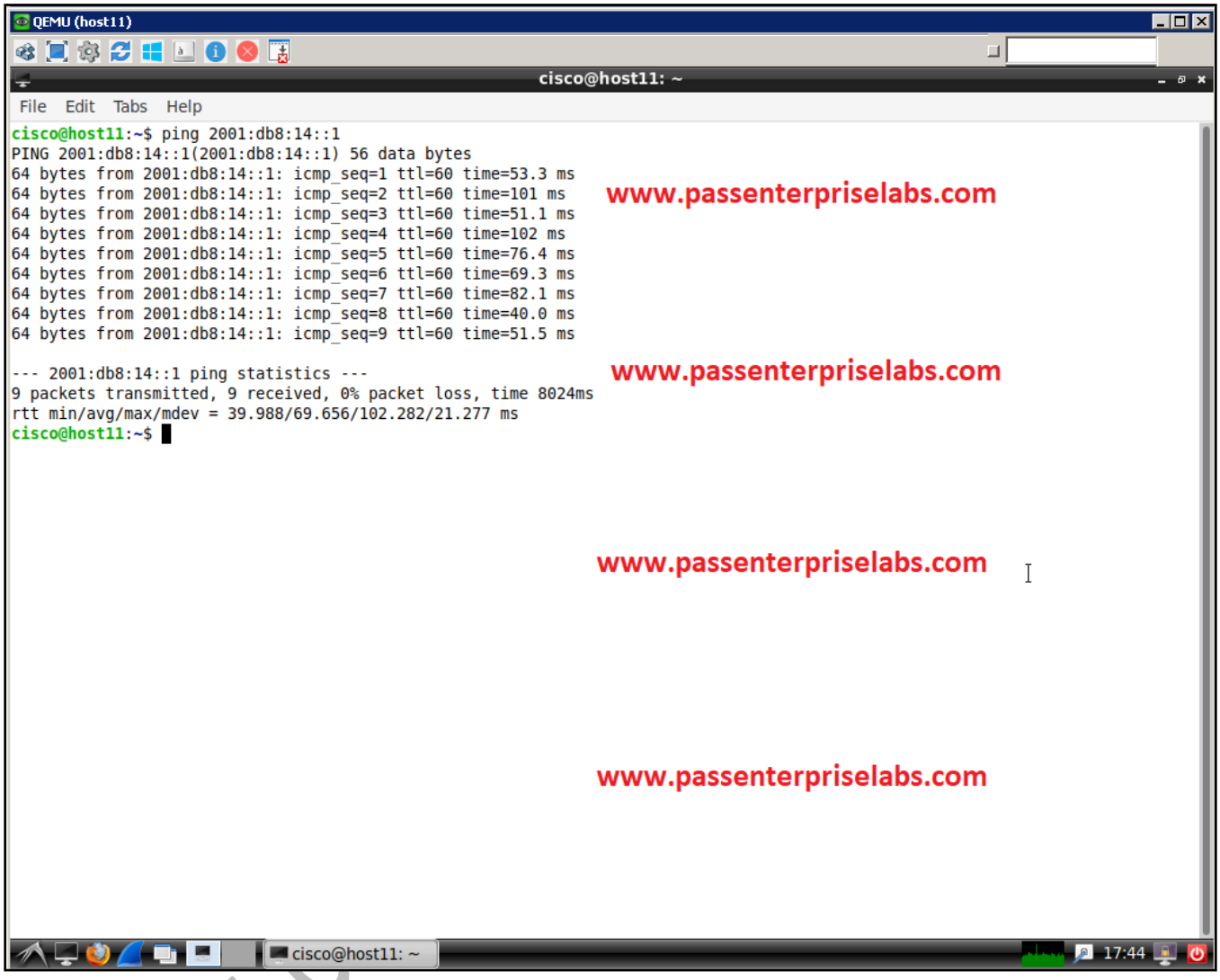
中文主机11：


QEMU (host11)
cisco@host11: ~
File Edit Tabs Help
cisco@host11:~$ ping 2001:db8:14::1
PING 2001:db8:14::1(2001:db8:14::1) 56 data bytes
64 bytes from 2001:db8:14::1: icmp_seq=1 ttl=60 time=53.3 ms
64 bytes from 2001:db8:14::1: icmp_seq=2 ttl=60 time=101 ms
64 bytes from 2001:db8:14::1: icmp_seq=3 ttl=60 time=51.1 ms
64 bytes from 2001:db8:14::1: icmp_seq=4 ttl=60 time=102 ms
64 bytes from 2001:db8:14::1: icmp_seq=5 ttl=60 time=76.4 ms
64 bytes from 2001:db8:14::1: icmp_seq=6 ttl=60 time=69.3 ms
64 bytes from 2001:db8:14::1: icmp_seq=7 ttl=60 time=82.1 ms
64 bytes from 2001:db8:14::1: icmp_seq=8 ttl=60 time=40.0 ms
64 bytes from 2001:db8:14::1: icmp_seq=9 ttl=60 time=51.5 ms
--- 2001:db8:14::1 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8024ms
rtt min/avg/max/mdev = 39.988/69.656/102.282/21.277 ms
cisco@host11:~$
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@host11: ~
17:44

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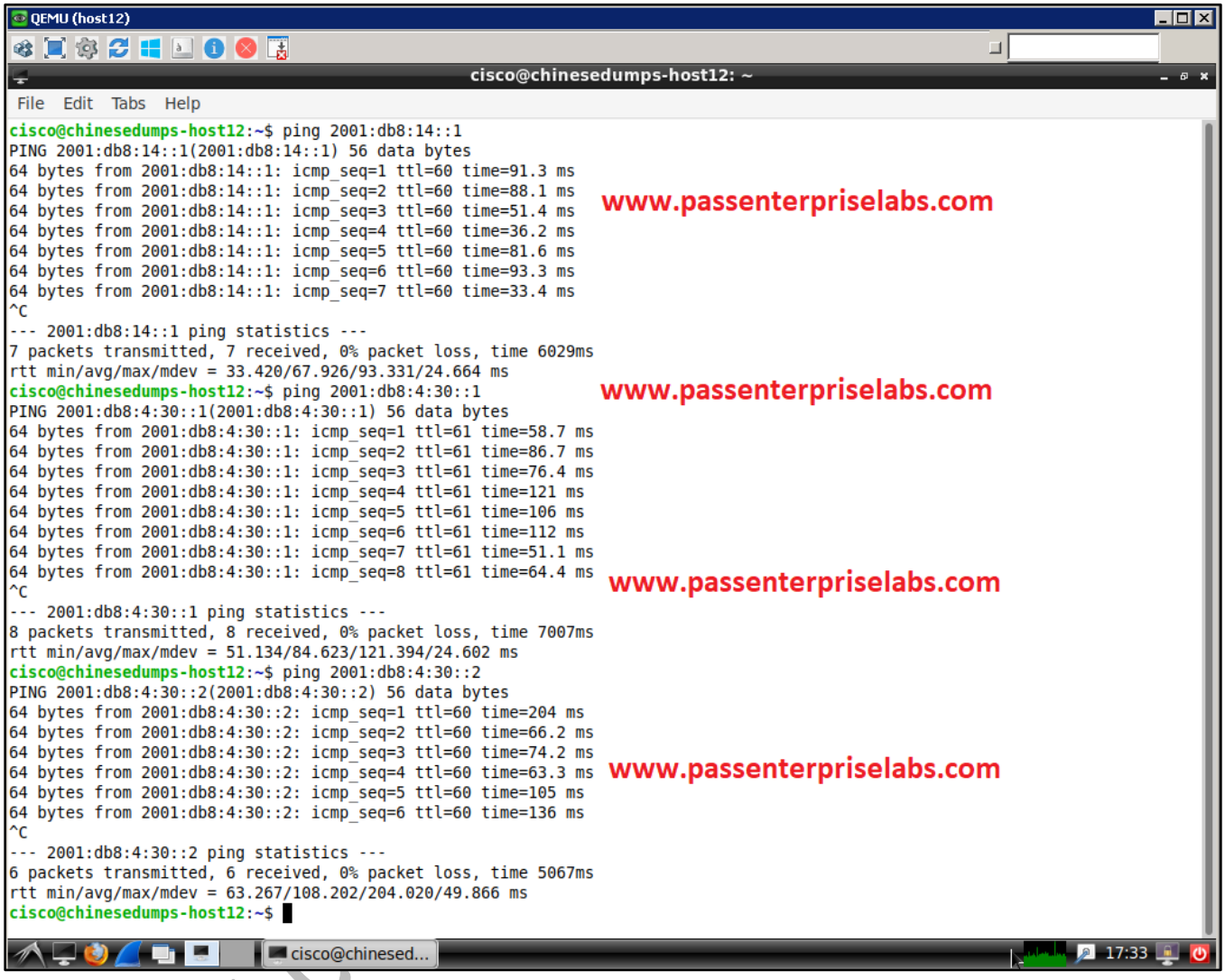
Chinesedumps-host12:



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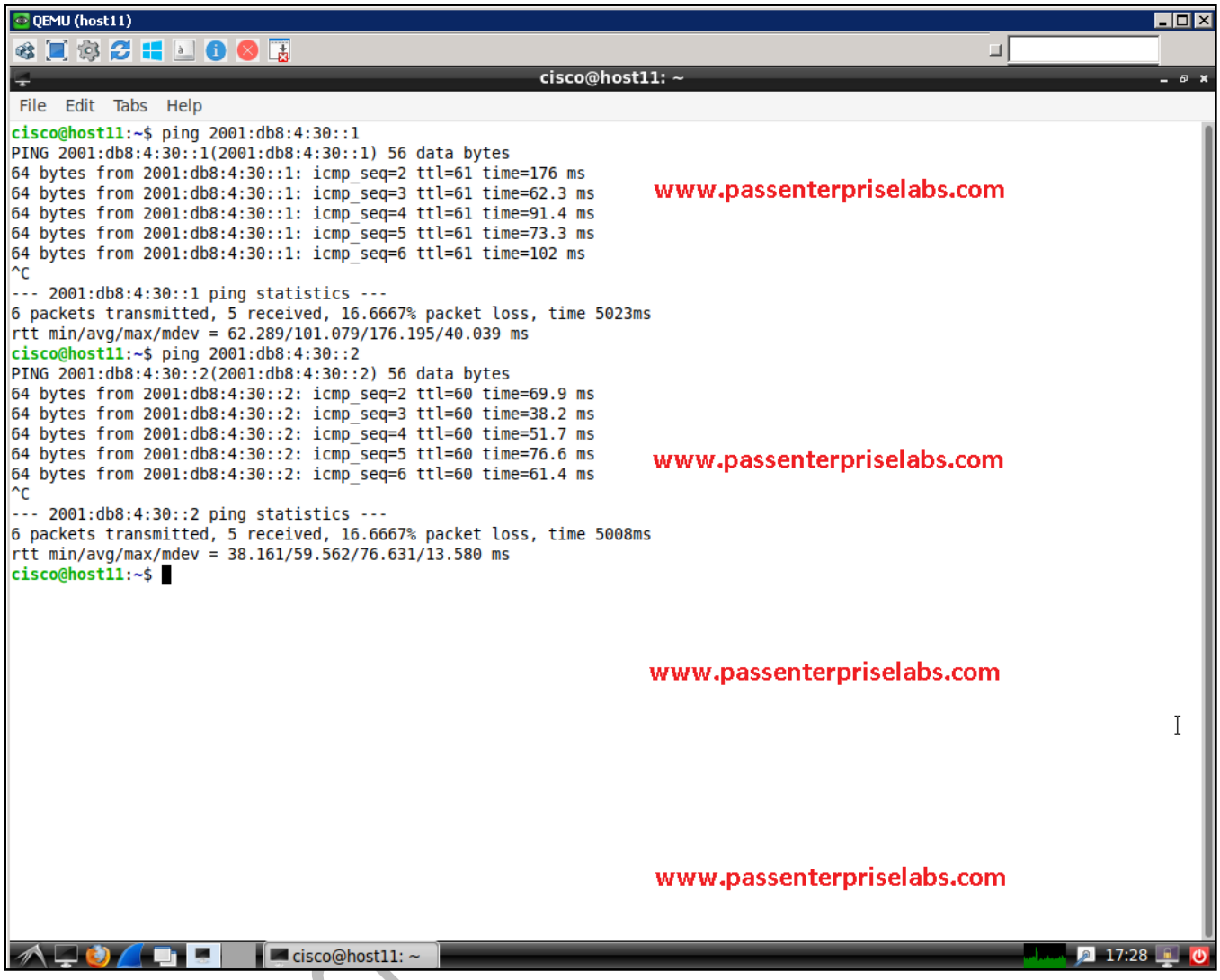
QEMU (host11)
cisco@host11: ~
File Edit Tabs Help
cisco@host11:~$ ping 2001:db8:4:30::1
PING 2001:db8:4:30::1(2001:db8:4:30::1) 56 data bytes
64 bytes from 2001:db8:4:30::1: icmp_seq=2 ttl=61 time=176 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=3 ttl=61 time=62.3 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=4 ttl=61 time=91.4 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=5 ttl=61 time=73.3 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=6 ttl=61 time=102 ms
^C
--- 2001:db8:4:30::1 ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5023ms
rtt min/avg/max/mdev = 62.289/101.079/176.195/40.039 ms
cisco@host11:~$ ping 2001:db8:4:30::2
PING 2001:db8:4:30::2(2001:db8:4:30::2) 56 data bytes
64 bytes from 2001:db8:4:30::2: icmp_seq=2 ttl=60 time=69.9 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=3 ttl=60 time=38.2 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=4 ttl=60 time=51.7 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=5 ttl=60 time=76.6 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=6 ttl=60 time=61.4 ms
^C
--- 2001:db8:4:30::2 ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5008ms
rtt min/avg/max/mdev = 38.161/59.562/76.631/13.580 ms
cisco@host11:~$
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www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@host11: ~
17:28

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1.16: Enabling Internet Access for FABD2

Enable highly available internet access for the FABD2 company network according to these requirements:

- On routers r12, r23 and r24, bring up IPv4 BGP peerings with the ISP, make sure that a default route is received over these peerings
- On router r12 and r23, inject default route into OSPF if it present in the routing table from a different routing source than the OSPFv2 process 1. On each router, this requirement must be completed using the minimum possible number of commands
- On route r24, inject default route into OSPF if any only if it is learned from ISP over BGP. To accomplish this requirement, it is allowed to use a route-map that referenced both a prefix-list and tag. This requirement must be completed using the minimum possible number of commands
- Router r12 may be used as an internet exit for the FABD2 company network only if neither r23 nor r24 are advertising the default route in OSPF. This requirement must be accomplished exclusively in "router ospf" mode on router r12 without changing the default parameters on routers r23 and r24
- On routers r12, r23 and r24, configure PAT and translate the entire FABD2 internal network 10.0.0.0/8 to the router address on the link toward the ISP. Create a standard ACL named NAT for this purpose. Do not use NAT pools
- Ensure that the internet connectivity of the FABD2 company network makes use of the highly availability provided by r12, r23 and r24.

1 Point

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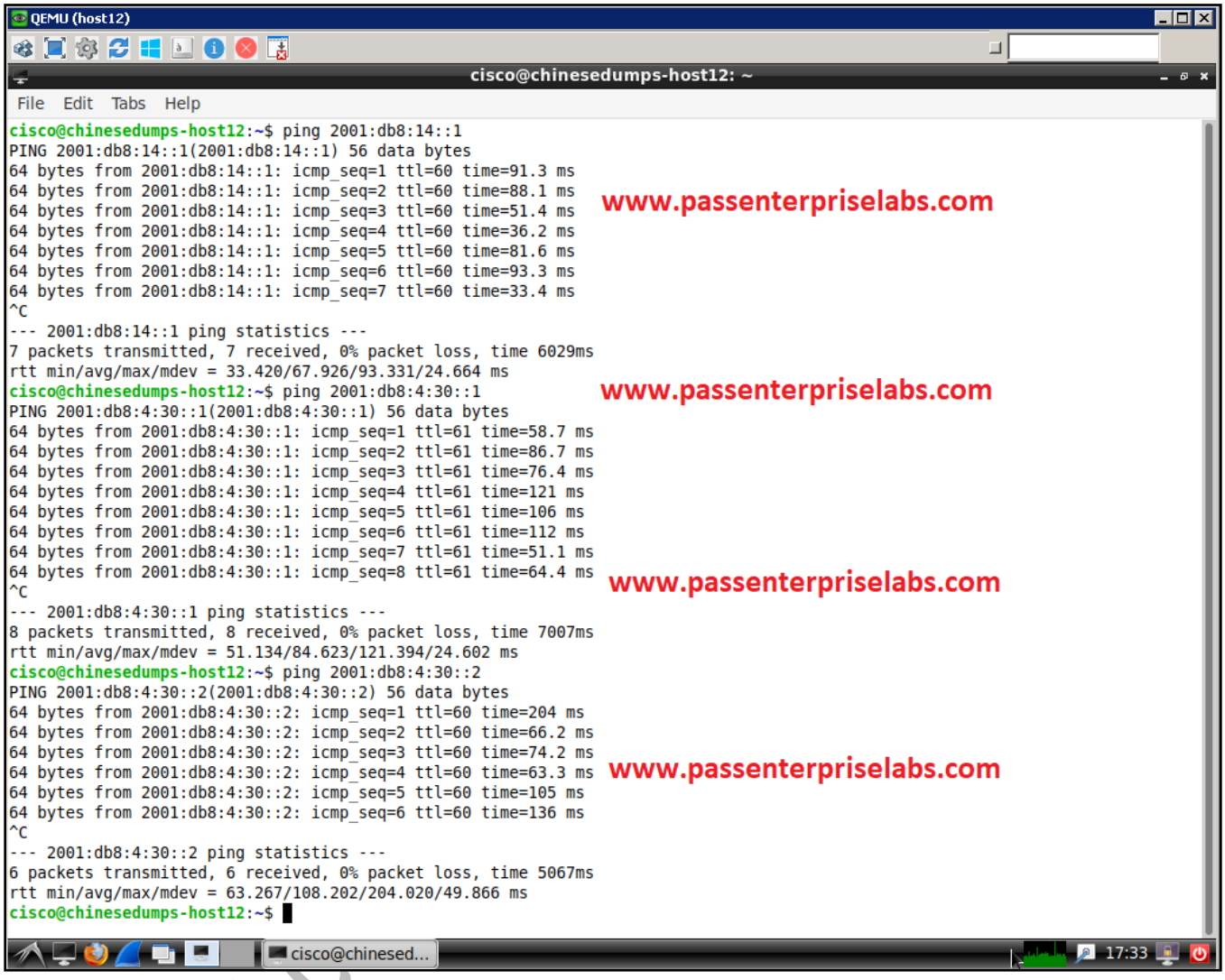
中文主机12：


QEMU (host12)
cisco@chinesedumps-host12: ~
File Edit Tabs Help
cisco@chinesedumps-host12:~$ ping 2001:db8:14::1
PING 2001:db8:14::1(2001:db8:14::1) 56 data bytes
64 bytes from 2001:db8:14::1: icmp_seq=1 ttl=60 time=91.3 ms
64 bytes from 2001:db8:14::1: icmp_seq=2 ttl=60 time=88.1 ms
64 bytes from 2001:db8:14::1: icmp_seq=3 ttl=60 time=51.4 ms
64 bytes from 2001:db8:14::1: icmp_seq=4 ttl=60 time=36.2 ms
64 bytes from 2001:db8:14::1: icmp_seq=5 ttl=60 time=81.6 ms
64 bytes from 2001:db8:14::1: icmp_seq=6 ttl=60 time=93.3 ms
64 bytes from 2001:db8:14::1: icmp_seq=7 ttl=60 time=33.4 ms
^C
--- 2001:db8:14::1 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6029ms
rtt min/avg/max/mdev = 33.420/67.926/93.331/24.664 ms
cisco@chinesedumps-host12:~$ ping 2001:db8:4:30::1
PING 2001:db8:4:30::1(2001:db8:4:30::1) 56 data bytes
64 bytes from 2001:db8:4:30::1: icmp_seq=1 ttl=61 time=58.7 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=2 ttl=61 time=86.7 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=3 ttl=61 time=76.4 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=4 ttl=61 time=121 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=5 ttl=61 time=106 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=6 ttl=61 time=112 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=7 ttl=61 time=51.1 ms
64 bytes from 2001:db8:4:30::1: icmp_seq=8 ttl=61 time=64.4 ms
^C
--- 2001:db8:4:30::1 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7007ms
rtt min/avg/max/mdev = 51.134/84.623/121.394/24.602 ms
cisco@chinesedumps-host12:~$ ping 2001:db8:4:30::2
PING 2001:db8:4:30::2(2001:db8:4:30::2) 56 data bytes
64 bytes from 2001:db8:4:30::2: icmp_seq=1 ttl=60 time=204 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=2 ttl=60 time=66.2 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=3 ttl=60 time=74.2 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=4 ttl=60 time=63.3 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=5 ttl=60 time=105 ms
64 bytes from 2001:db8:4:30::2: icmp_seq=6 ttl=60 time=136 ms
^C
--- 2001:db8:4:30::2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5067ms
rtt min/avg/max/mdev = 63.267/108.202/204.020/49.866 ms
cisco@chinesedumps-host12:~$
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
www.passenterpriselabs.com
cisco@chinesed...
17:33

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Solution:

Chinesedumps.com-r12:

```
r12(config)#router bgp 65001
r12(config-router)#address-family ipv4 unicast
r12(config-router)#neighbor 200.99.12.1 remote-as 19999
r12(config-router)#neighbor 200.99.12.1 activate


r12(config)#router ospf 1
r12(config-router)#default-information originate

r12(config)#int g0/0
r12(config-if)#ip nat outside
r12(config)#int range g0/1-3
r12(config-if)#ip nat inside
r12(config)#int lo0
r12(config-if)#ip nat inside
r12(config-if)#ip access-list standard NAT
r12(config-std-acl)#permit 10.0.0.0 0.255.255.255
r12(config)#ip nat inside source list NAT interface g0/0 overload
```

Chinesedumps.com-r23:

```
r23(config)#router bgp 65002
r23(config-router)#neighbor 200.99.23.1 remote-as 19999
r23(config)#router ospf 1
r23(config-router)#default-information originate metric-type 1

r23(config)#int g1
r23(config-if)#ip nat outside
r23(config)#int range g2-4
r23(config-if)#ip nat inside
r23(config)#int lo0
r23(config-if)#ip nat inside
r23(config-if)#ip access-list standard NAT
r23(config-if)#permit 10.0.0.0 0.255.255.255
r23(config-if)#ip nat inside source list NAT interface g1 overload
```

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1.16： 为FABD2启用互联网接入

根据以下要求，为FABD2公司网络启用高可用互联网接入：

- 在路由器r12、r23和r24上，与ISP建立IPv4 BGP对等，确保通过这些对等接收到故障路由
- 在路由器r12和r23上，如果路由表中存在来自与OSPFv2进程1不同的路由源的默认路由，则将其注入OSPF。在每个路由器上，必须使用尽可能少的命令来完成此要求
- 在路由r24上，仅当通过BGP从ISP获知默认路由时，才将默认路由注入OSPF（如果有的话）。为了满足这一要求，允许使用同时引用aprefix列表和标签的路线图。必须使用尽可能少的命令数量来完成此要求
- 只有当r23和r24都没有在OSPF中通告默认路由时，路由器r12才能用作FABD2公司网络的互联网出口。此要求必须仅在路由器r12的“路由器ospf”模式下完成，而不更改路由器r23和r24的默认参数
- 在路由器r12、r23和r24上，配置PAT并将整个FABD2内部网络10.0.0.0/8转换为指向ISP的链路上的路由器地址。为此目的创建一个名为NAT的标准ACL。不使用NAT池
- 确保FABD2公司网络的互联网连接利用了r12、r23和r24提供的高可用性。

1分

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Chinesedumps.com-r24:

```
r24(config)#router bgp 65002
r24(config-router)#neighbor 200.99.24.1 remote-as 19999

r24(config)#ip prefix-list DEFAULT permit 0.0.0.0/0

r24(config)#route-map ISP permit 10
r24(config-route-map)#match ip address prefix-list DEFAULT
r24(config-route-map)#match tag 19999

r24(config)#router ospf 1
r24(config-router)#default-information originate route-map ISP metric-type 1

r24(config)#interface g1
r24(config-if)#ip nat outside
r24(config-if)#int range g2-4,tunnel 0
r24(config-if)#ip nat inside
r24(config-if)#int lo0
r24(config-if)#ip nat inside
r24(config-if)#ip access-list standard NAT
r24(config-if)#permit 10.0.0.0 0.255.255.255
r24(config-if)#ip nat inside source list NAT interface g1 overload

r24(config)#router eigrp ccie
r24(config-router)#address-family ipv4 unicast autonomous-system 65006
r24(config-router-af)#topology base
r24(config-router-af)#summary-metric 0.0.0.0/0 distance 254
```

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解决方案：

中文教育网-r12：

r12（配置）#路由器bgp 65001
r12（配置路由器）#地址家族ipv4单播
r12（配置路由器）#邻居200.99.12.1远程为19999
r12（配置路由器）#邻居200.99.12.1激活

r12（配置）#路由器ospf 1
r12（配置路由器）#默认信息来源

r12（config）#int g0/0
r12（config-if）#r12外
部的ip nat（config）
#int范围g0/1-3 r12（-
config-if）#r12内部的
ip nat
r12（config if）#ip访问列表标准NAT r12（
配置std acl）#许可10.0.0.0 0.255.255.255
r12（config）#ip-nat内源列表nat接口g0/0过载

中文教育网-r23：

r23（配置）#路由器bgp 65002
r23（配置路由器）#邻居200.99.23.1远程为19999
r23（配置）#路由器ospf 1
r23（配置路由器）#默认信息来源度量类型1

r23（config）#int g1
r23（config-if）#r23外
部的ip nat（config）
#int范围g2-4 r23（-
config-if）#r23内部的
ip nat
r23（config-if）#ip访问列表标准NAT
r23（config.if）#permit 10.0.0.0
0.255.255.255
r23（config-if）#源列表nat接口gloverload内的ip-nat

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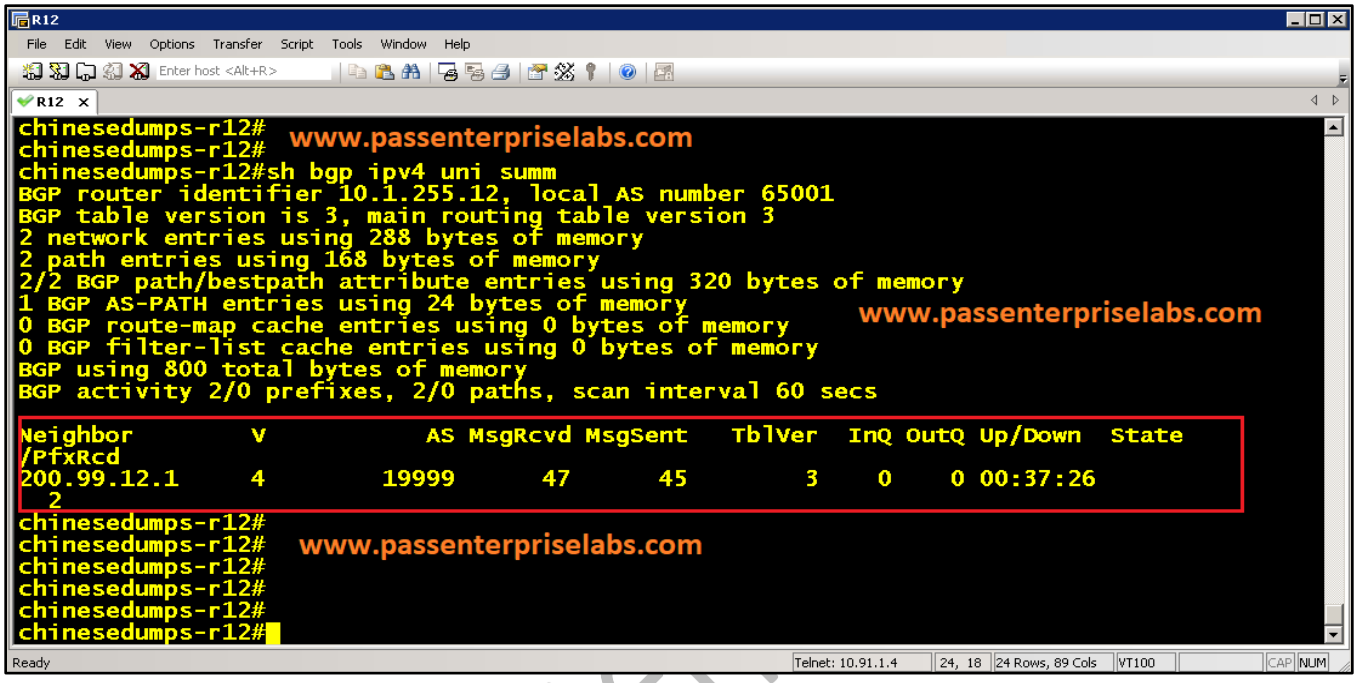
Verification:

Chinesedumps-r12: sh bgp ipv4 unicast summary



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中文教育网-r24：

r24（配置）#路由器bgp 65002
r24（配置路由器）#邻居200.99.24.1远程为19999

r24（config）#ip前缀列表默认许可

0.0.0.0/0 r24（config）#路由映射ISP许可

10
r24（配置路由映射）#匹配ip地址前缀列表DEFAULT
r24（设置路由映射）#match标签19999

r24（配置）#路由器ospf 1
r24（配置路由器）#默认信息来源路由映射ISP度量类型1

r24（config）#接口g1
r24（config-if）#外部
ip nat
r24（config-if）#int范围g2-4，
隧道0 r24（config.if）#r24内的
ip-nat（config-ife）#int lo0
r24（配置if）#内部ip nat
r24（config-if）#ip访问列表标准NAT
r24（config.if）#permit 10.0.0.0
0.255.255.255
r24（config-if）#源列表nat接口gloverload内的ip-nat

r24（配置）#路由器eigrp-ccie
r24（配置路由器）#地址家族ipv4单播自治系统65006 r24（设置路
由器af）#拓扑基础
r24（配置路由器af）#汇总度量0.0.0.0/0距离254

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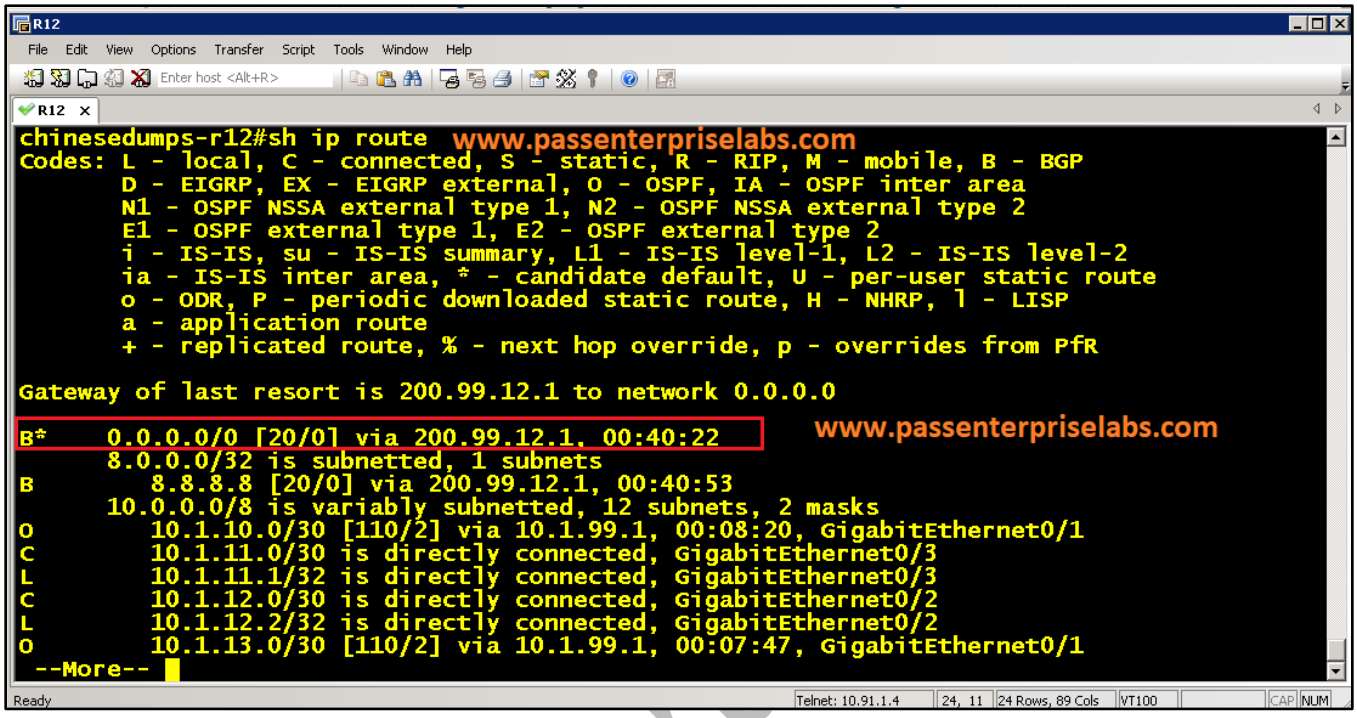
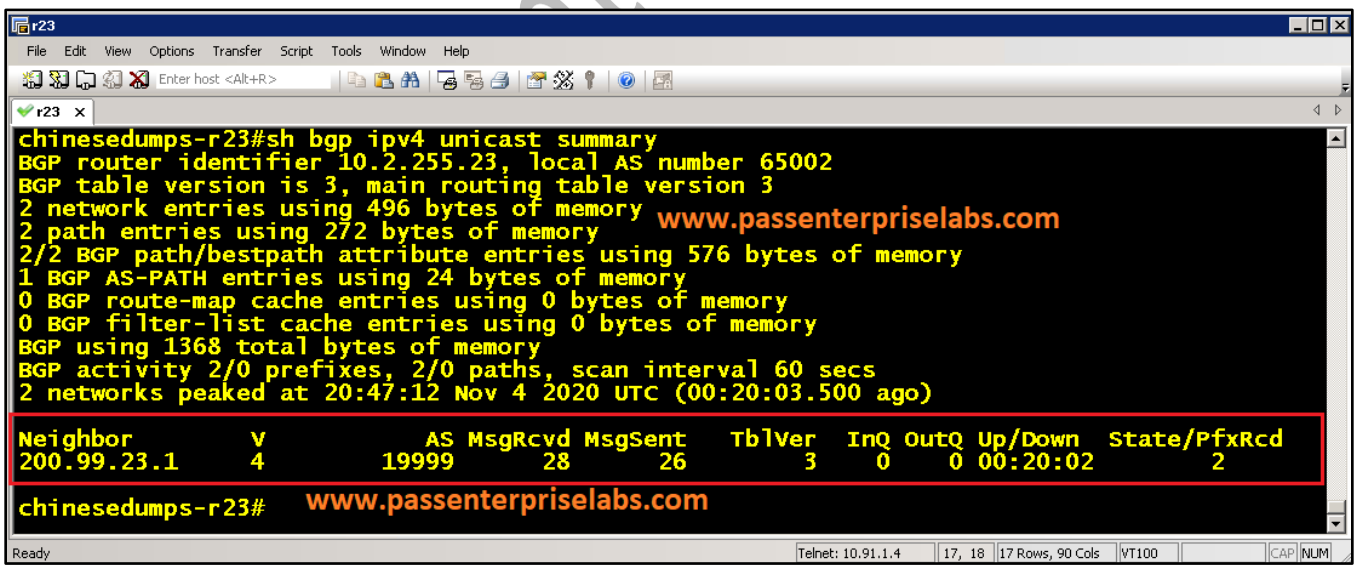
Chinesedumps.com-r24:sh ip route

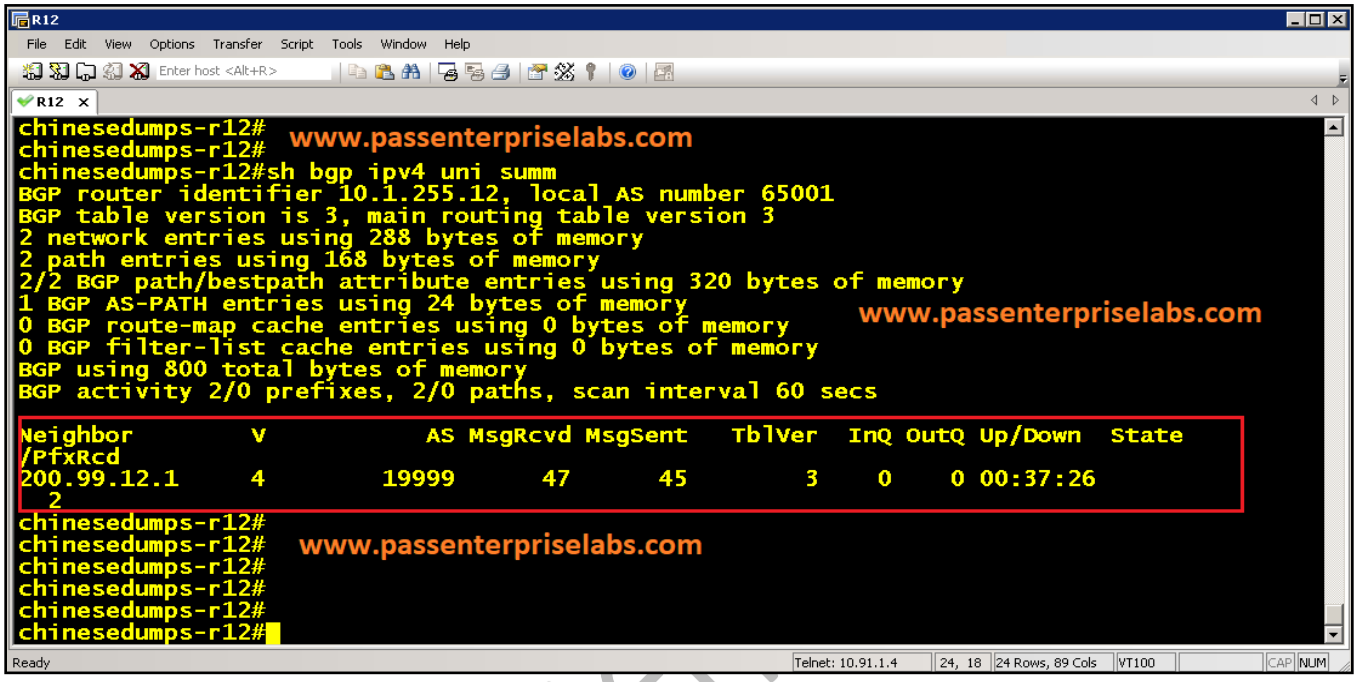


Chinesedumps.com-r23:sh bgp ipv4 unicast summary



验证：

中文教程-r12:sh bgp ipv4通讯摘要



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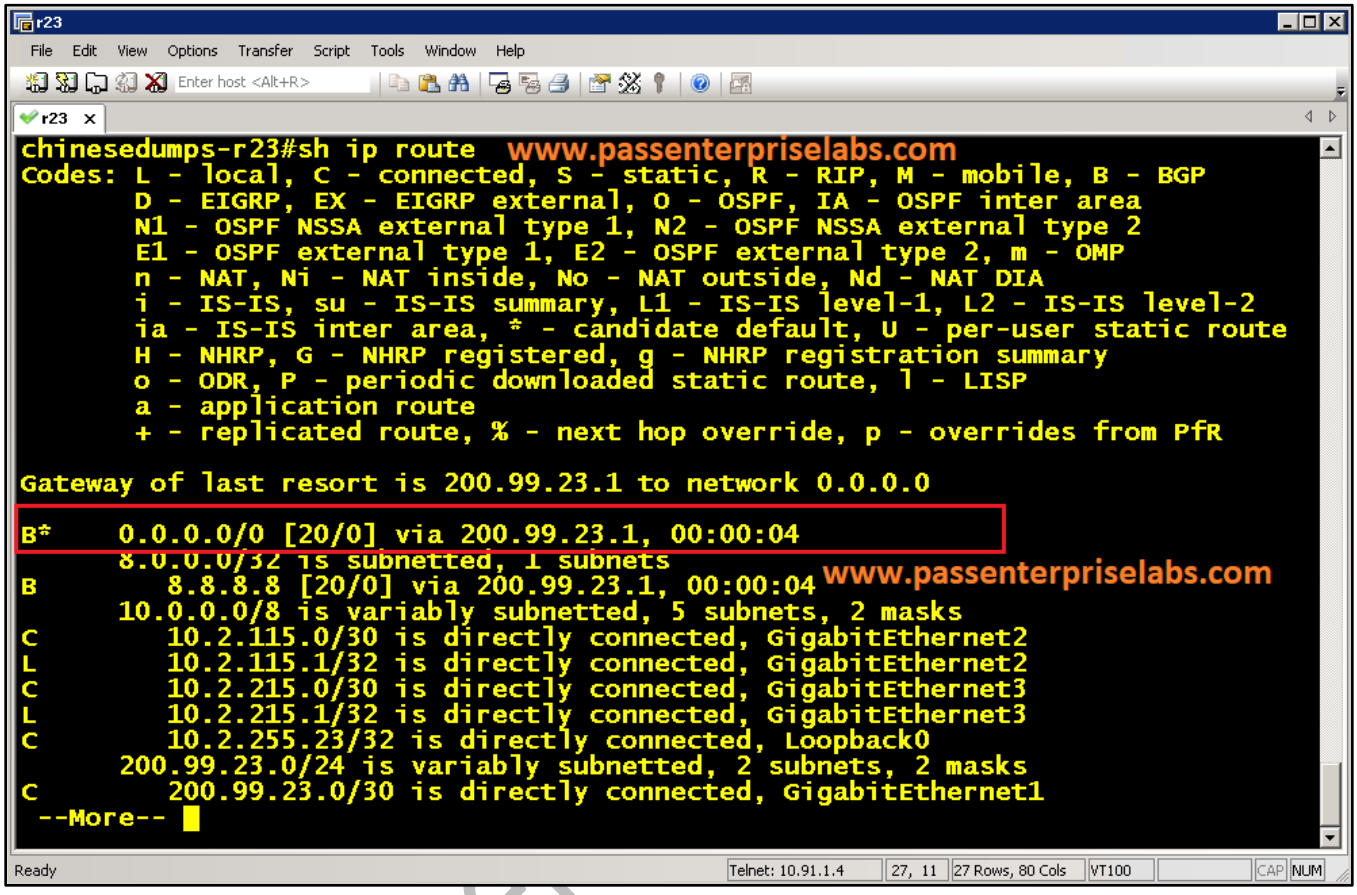
Chinesedumps.com-r23:sh ip route



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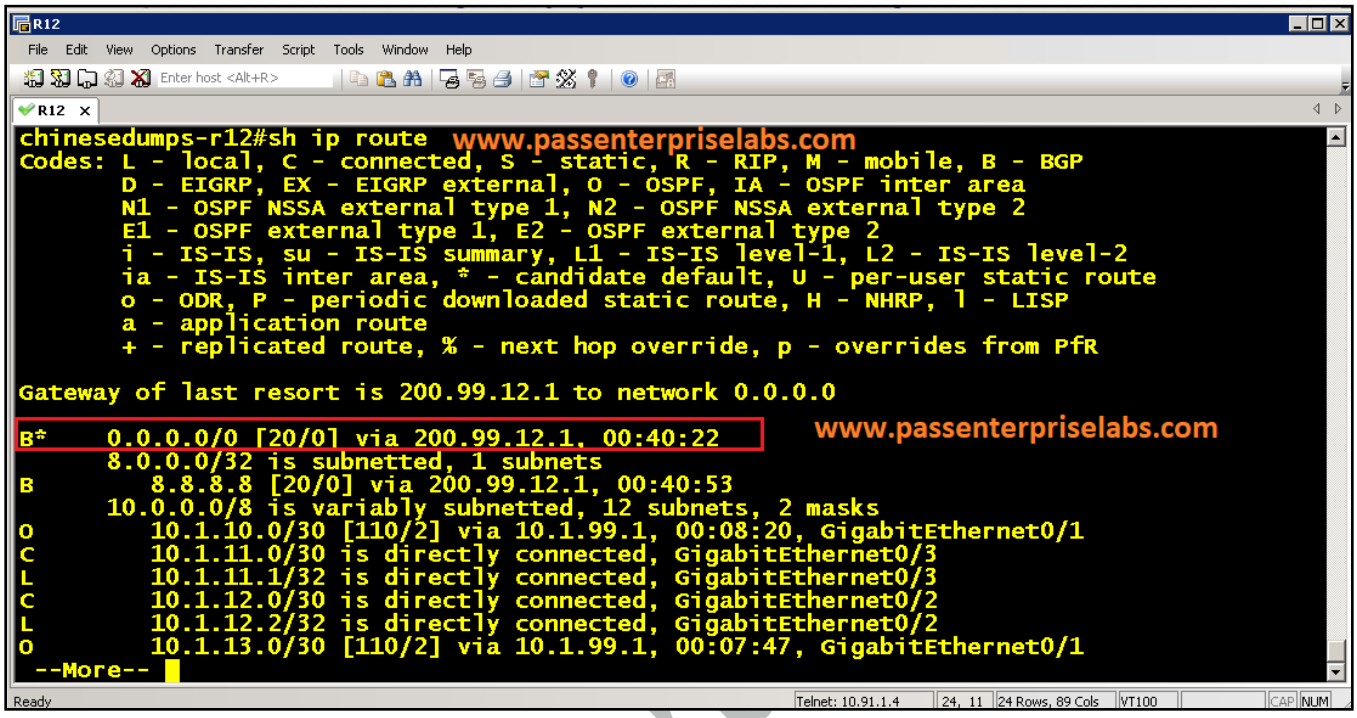
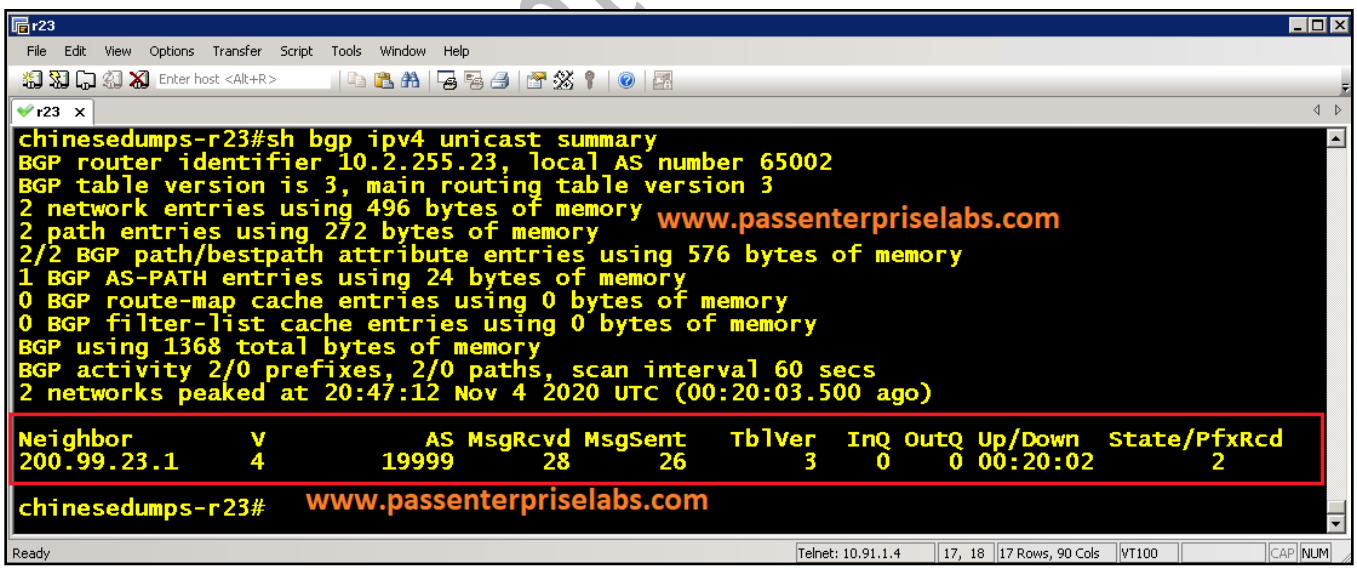
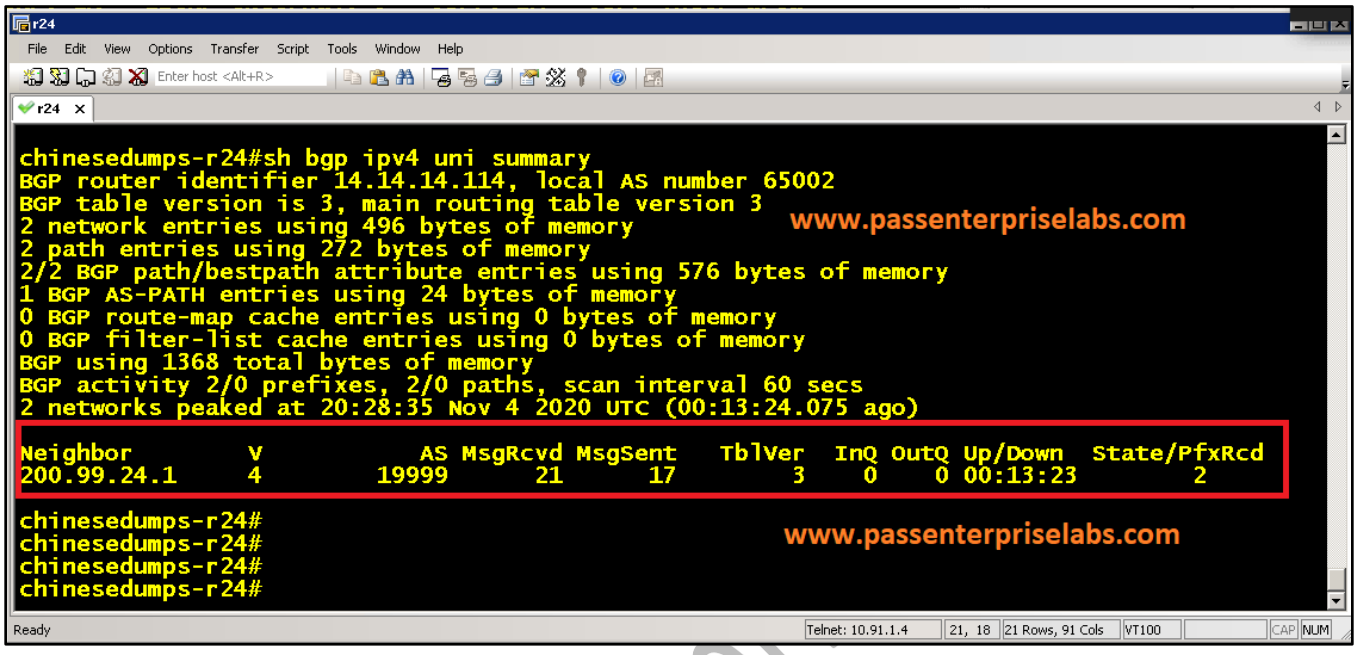
Chinese edumps.com-r24:sh-ip路由



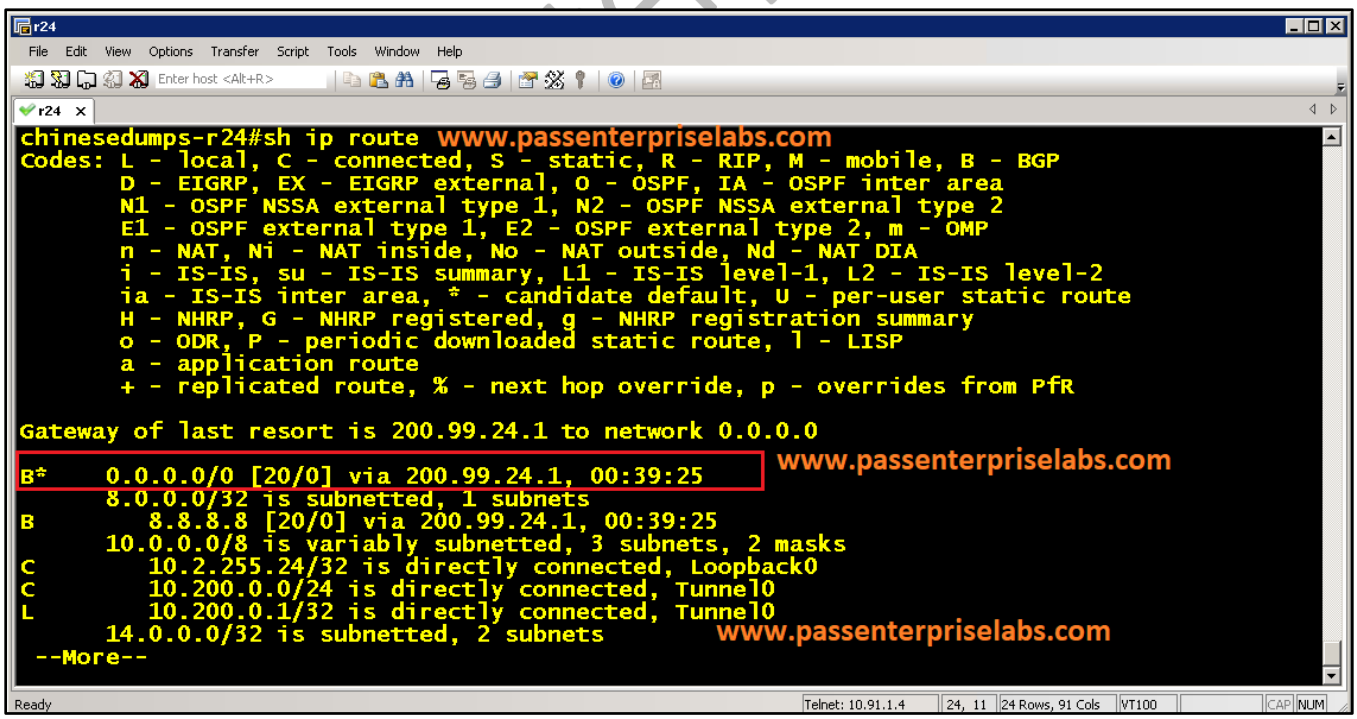
Chinese edumps.com-r23:sh bgp ipv4单播摘要



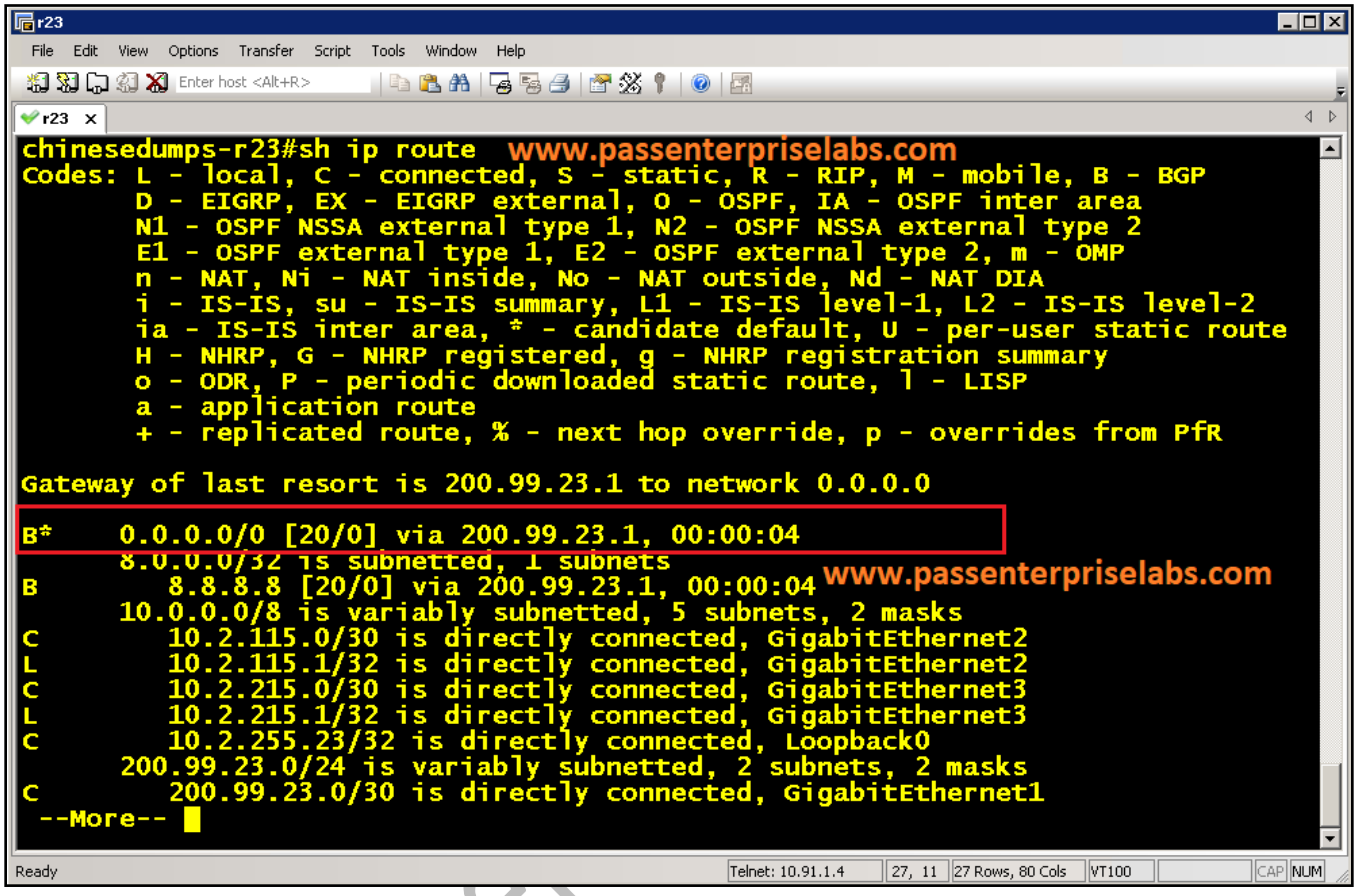
Chinesedumps.com-r24: sh bgp ipv4 unicastsummary



Chinesedumps.com-r24: sh ip route



Chinese edumps.com-r23:sh-ip路由



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2.1: Correcting the IP addresses of Managed switches in DNA center

After Cisco DNA center first achieves IP connectivity with the managed switches in Branches #1 and #2, it will place them into maintenance mode due to their serial number being different from the one DNA center remember. In addition, their management IP addresses in DNA Center will be automatically changed by appending them with the ".dummy.com" string. As a result, after an initial contact, DNA Center will lose connectivity with the switches unless their management IP addresses are corrected in the DNA center settings.

Correct the IP addresses of managed switches in the DNA center according to the following requirements:

- Use any host, such as host11, to access the DNA Center GUI website at https://203.0.113.11 URL.
- Execute the Provision – Devices – Inventory – Global – Actions – Inventory - Resync Device action in DNA Center on all switches before proceeding further.
- DNA Center API reference and sandbox is available at https://203.0.113.11/dna/apitester URL.
- The /network/device/update-maintenance-device-ip-address API call description and sandbox are available in the Inventory section of the API reference.
- Use the /network-device/update-maintenance-device-ip address API call to correct the IP addresses of the switches in Branches #1 and #2 by removing the appended text.

Note: These IP addresses cannot be changed from DNA Center GUI directly because they will become automatically invalidated again. This is a built-in DNA Center behavior.

3 points

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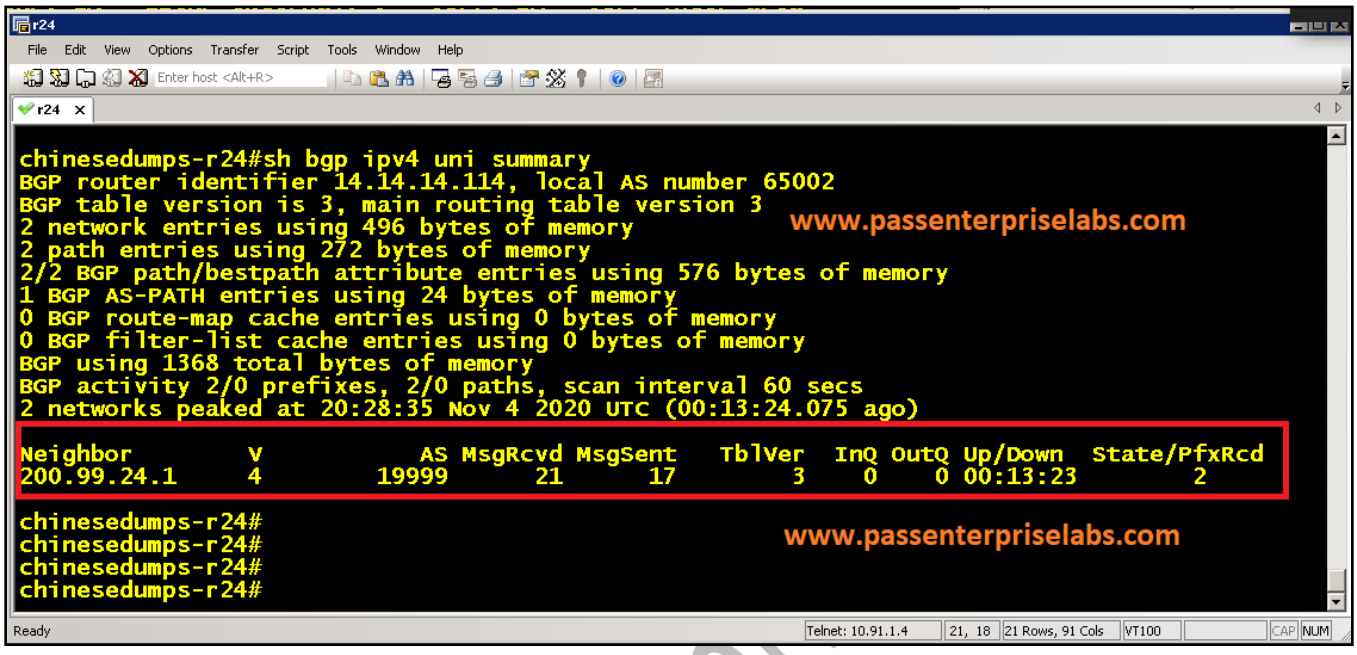
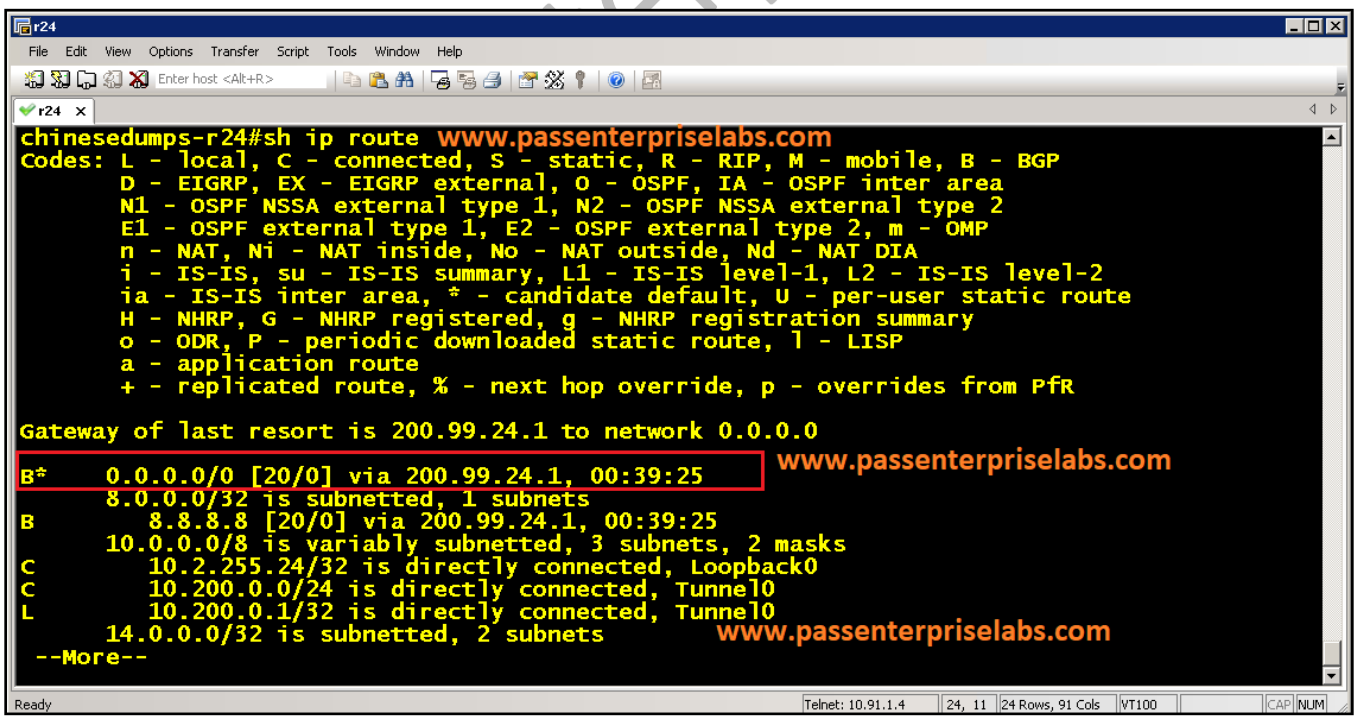
Chinese edumps.com-r24:sh bgp ipv4通讯摘要



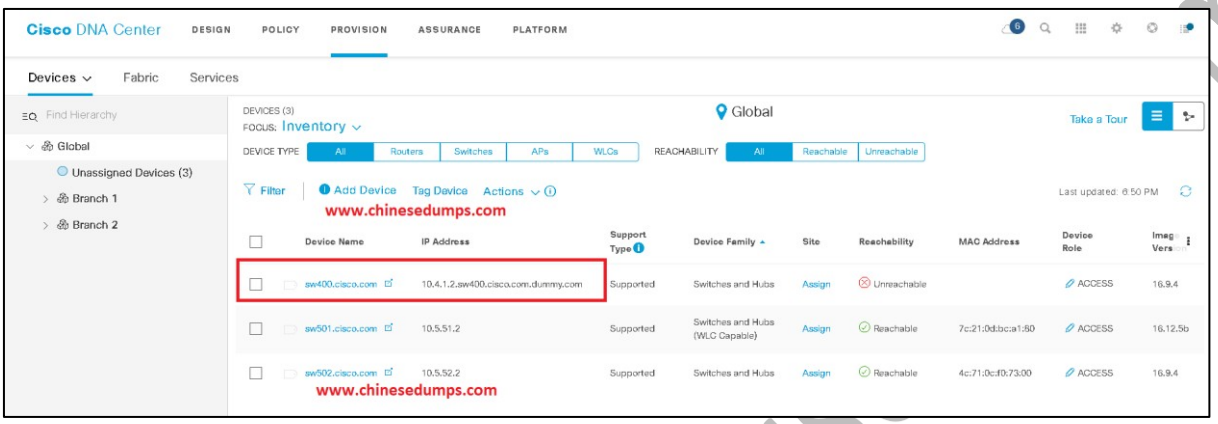
Chineseedumps.com-r24:sh-ip路由



Solution:

Step 1:

Access DNA Center and then go to PROVISION

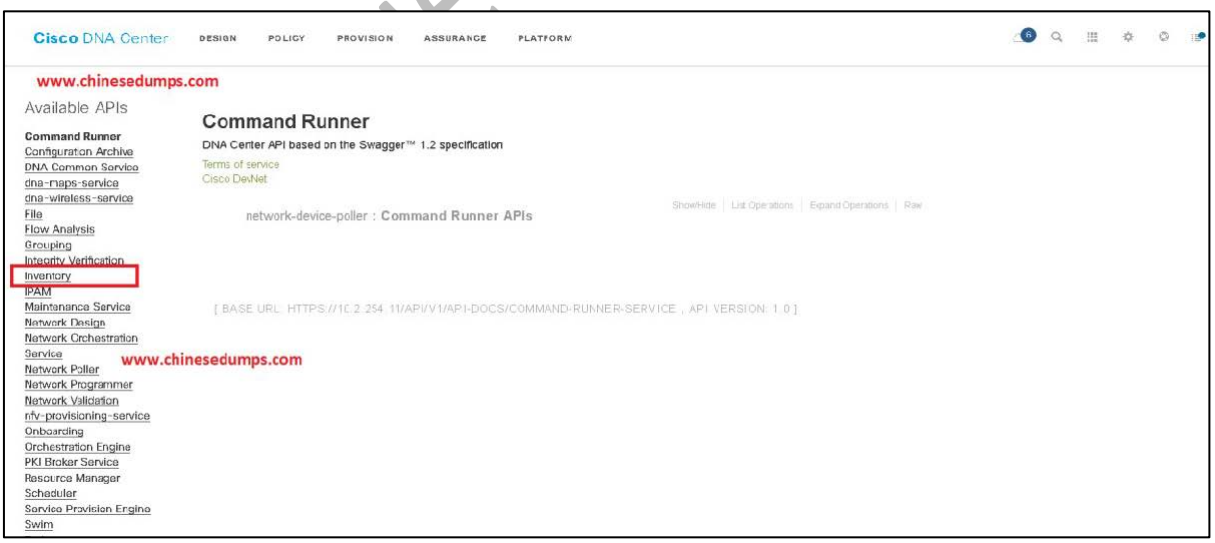


Above Output will be displayed for all the 3 switches (sw400 , sw501 , sw502)

Step 2:

To complete the task you have to go to below URL

https://10.2.254.11/dna/apitester



Click on **Inventory** option

2.1： 更正DNA中心托管交换机的IP地址

在思科DNA中心首次与分支#1和#2中的受管交换机实现IP连接后，由于它们的序列号与oneDNA中心记住的序列号不同，它将把它们置于维护模式。此外，他们在DNA中心的管理IP地址将通过附加".dumpy.com"字符串自动更改。因此，在初始联系后，DNA中心将失去与交换机的连接，除非在DNA中心设置中更正了他们的管理IP位址。

根据以下要求更正DNA中心中受管交换机的IP地址：

- 使用任何主机（如host11）访问DNA中心GUI网站https://203.0.113.11 URL。
- 在进一步操作之前，在DNA中心的所有交换机上执行配置-设备-库存-全局-操作-库存-重新同步设备操作。
- DNA中心API参考和沙箱可在https://203.0.113.11/dna/apitester URL。
- /network/device/update-matenance-device-ip-address API调用描述和沙盒可在API参考资料的库存部分中找到。
- 使用/network-device/update-matenance-device-ip address API调用，通过删除附加文本来更正分支#1和#2中交换机的ip地址。

注意：这些IP地址不能直接从DNA中心GUI更改，因为它们将再次自动失效。这是一种内置的DNA中心行为。

3分

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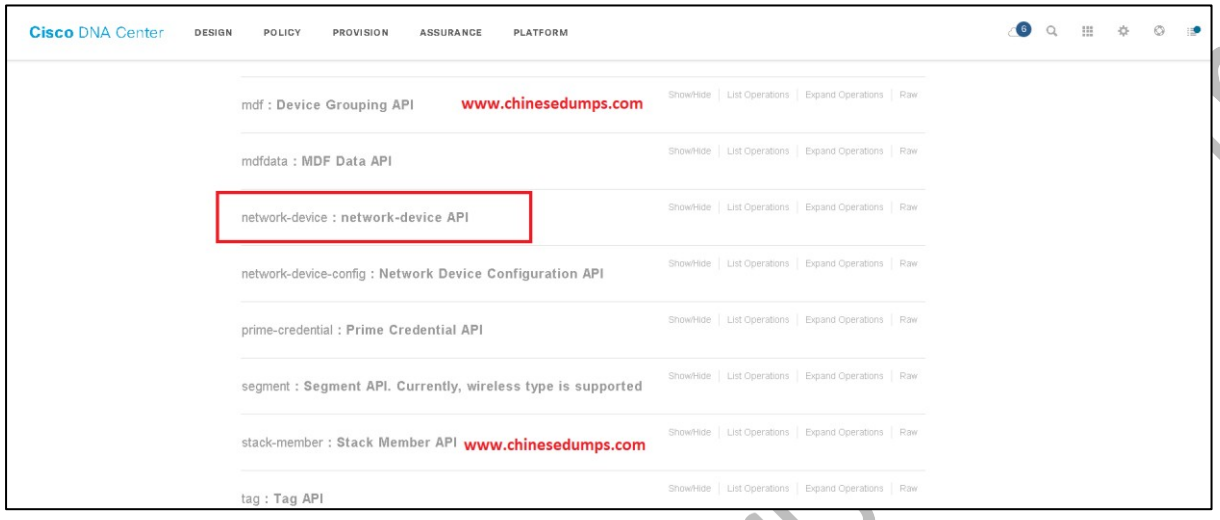
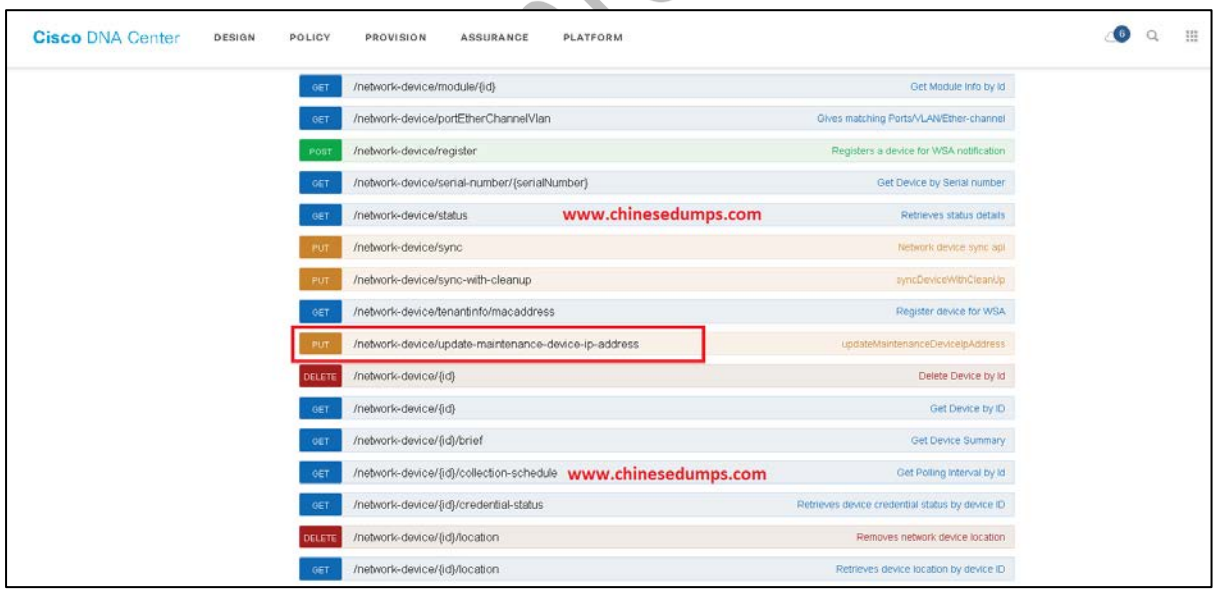
Step 3:

Scroll down and Click on **network-device**



Step 4:

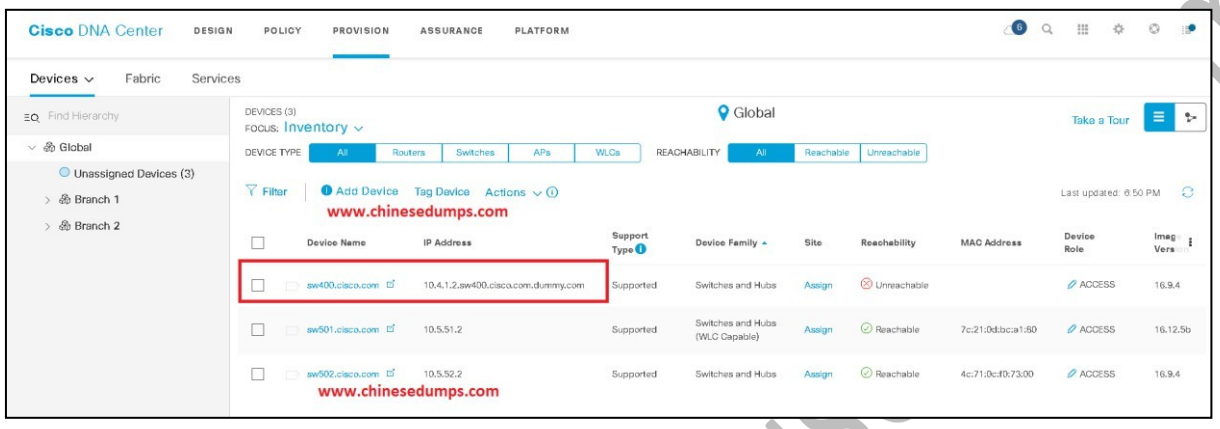
- Scroll down and click on **/network/device-update-maintainance-device-ip-address**



解决方案：

第一步：

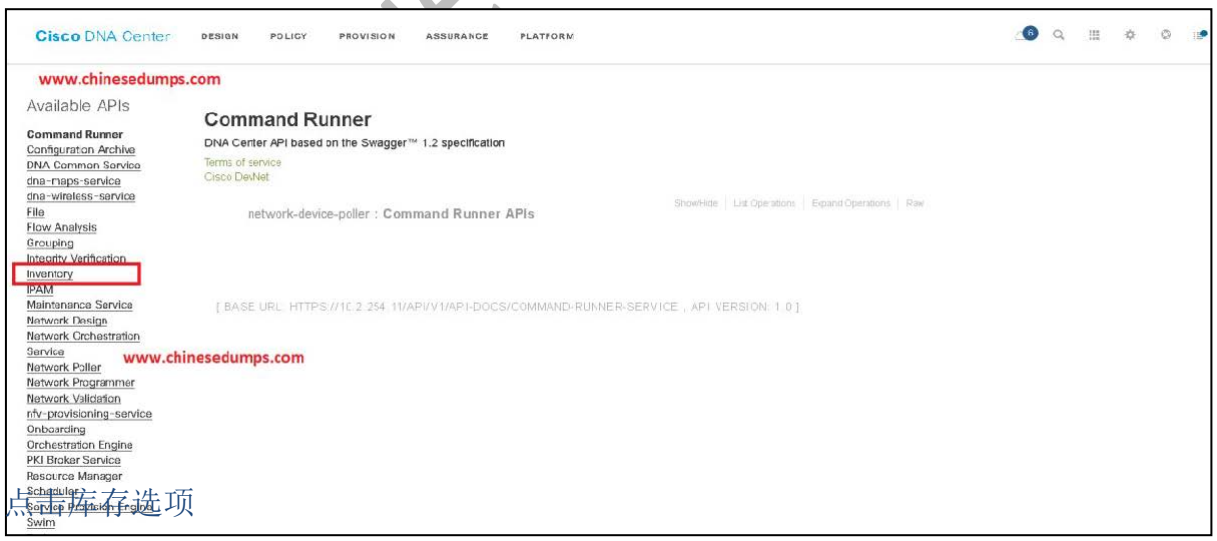
访问DNA中心，然后转到提供



以上输出将显示所有3个开关（sw400、sw501、sw502）的步骤2：

要完成任务，您必须转到以下URL

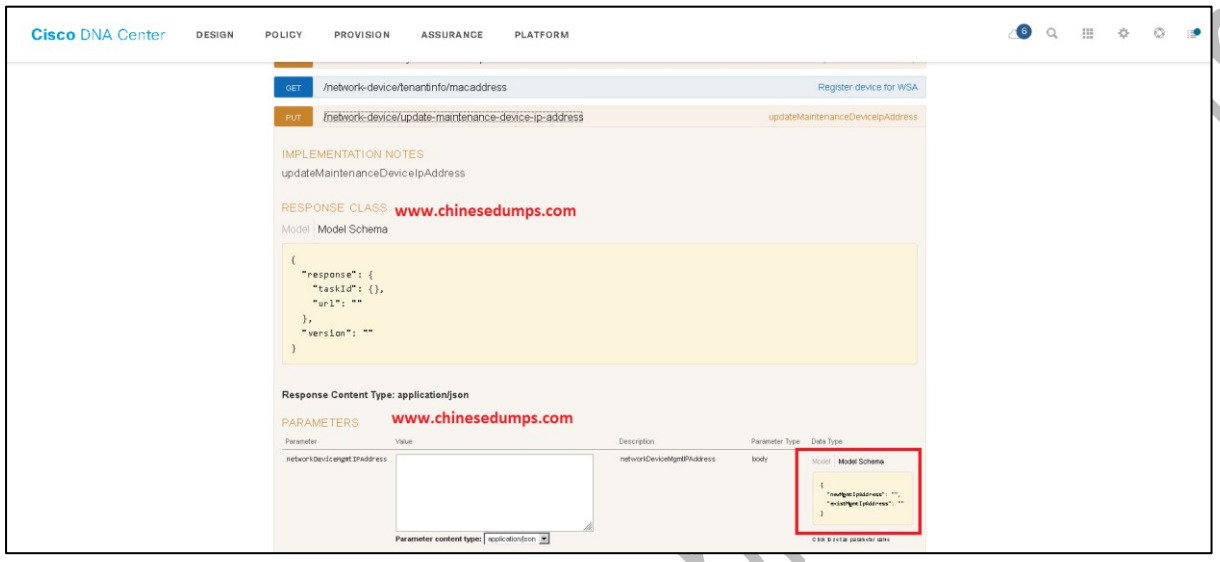
https://10.2.254.11/dna/apitester



点击库存选项

Step 5:

- Scroll down and at the right hand side you will see **Model Schema**
- Just click inside the box where the code is written



Step 6:

- The code will be copied in the right side blank box
- Then change the **NewMgmtIpAddress – "10.4.255.11" ,**
  existMgmtIpAddress – "10.4.255.11.dummy.com"

```
{
 "newMgmtIpAddress": "10.4.1.2",
 "existMgmtIpAddress": "10.4.255.11dummy.com"
}
```

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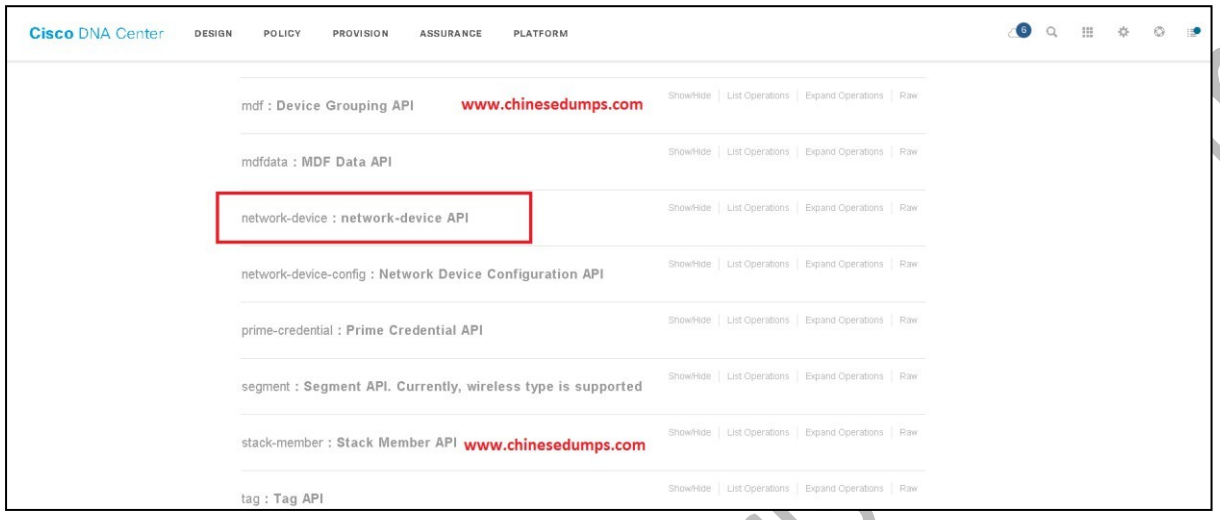
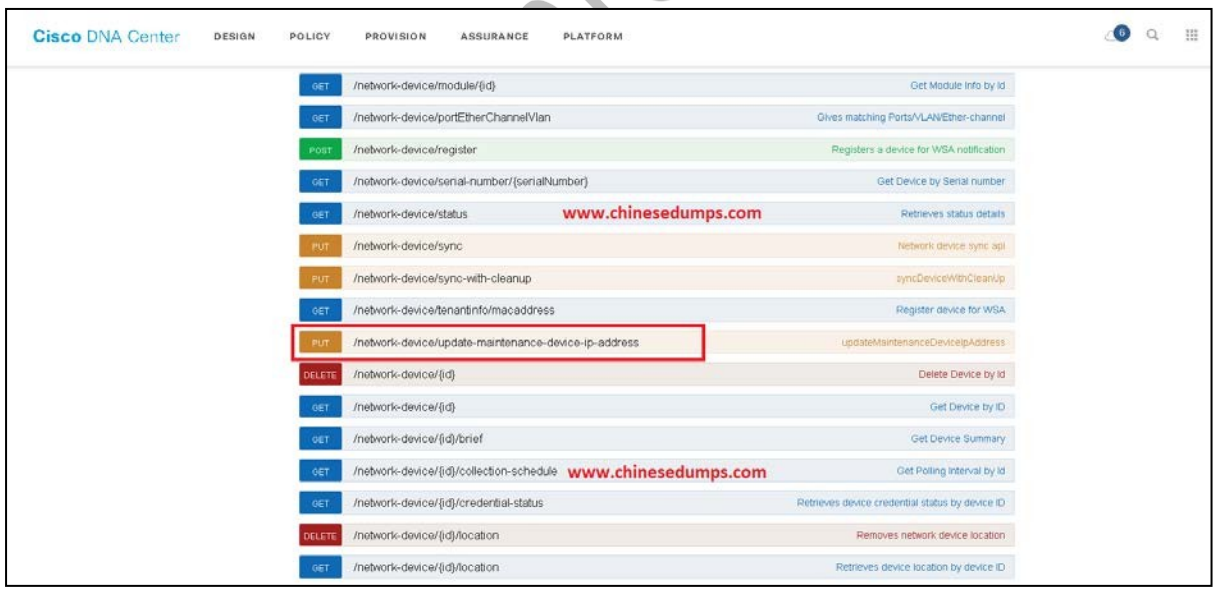
步骤3：

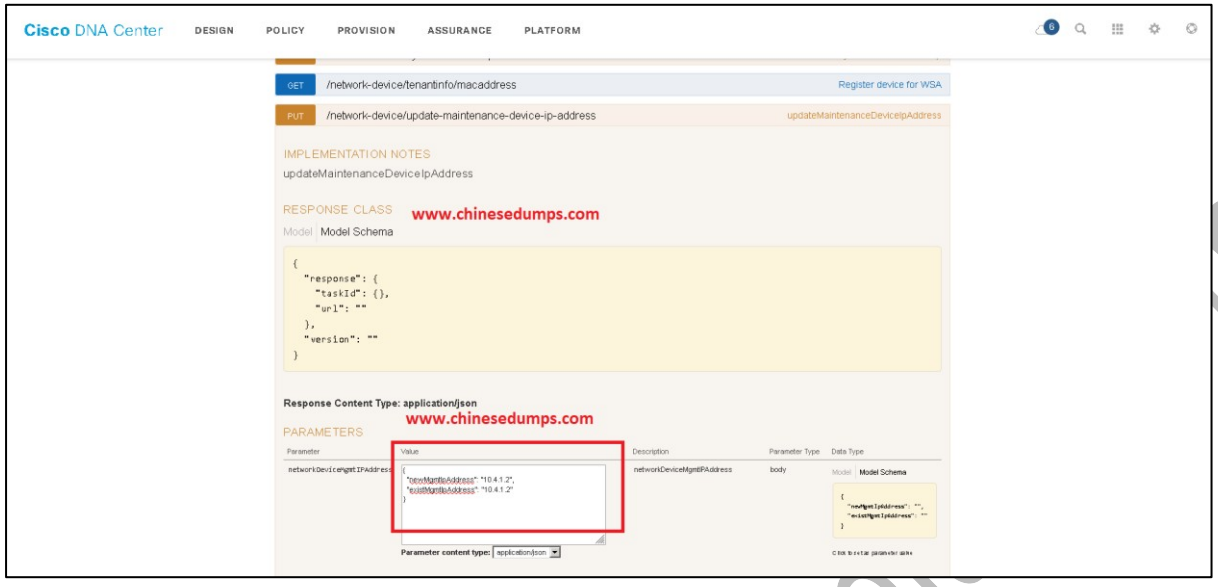
向下滚动并单击网络设备



步骤4：

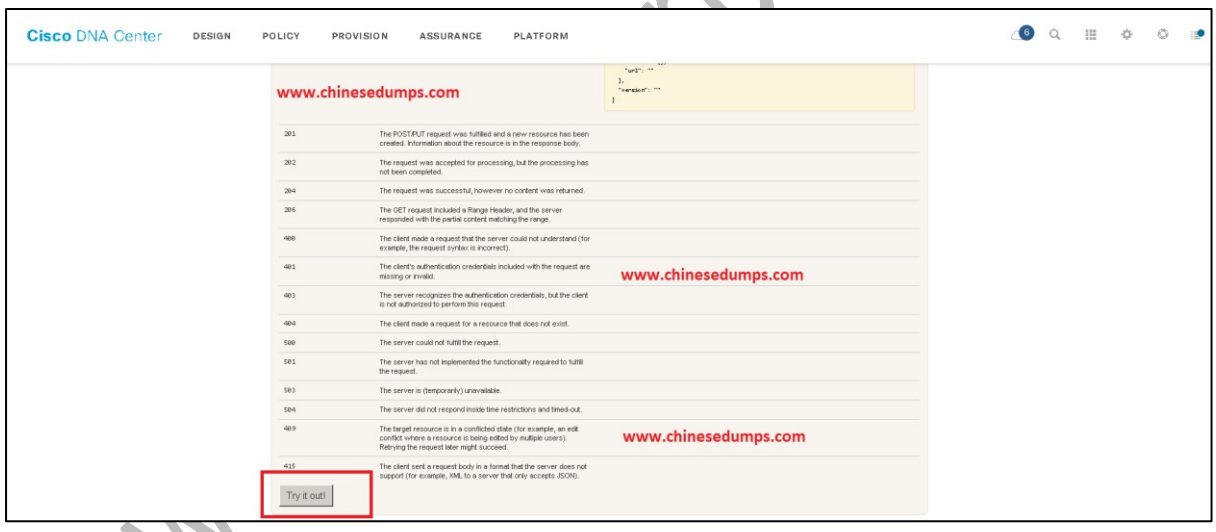
- 向下滚动并单击/网络/设备更新维护设备ip地址





Step 7:

- Scroll down and click **Try it out** button



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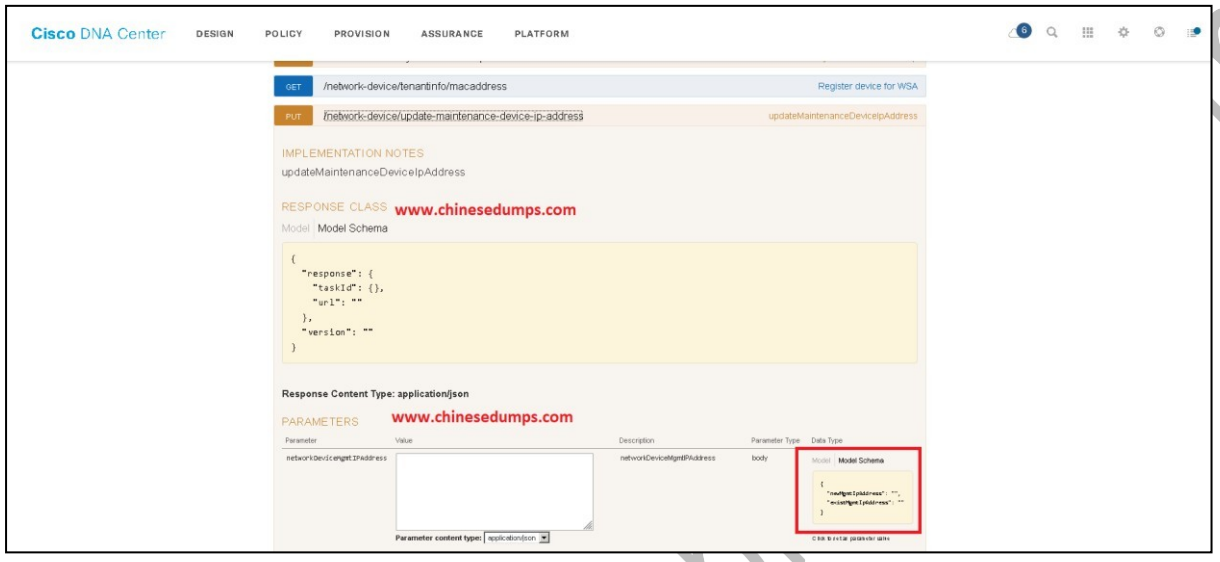
步骤5：

- 向下滚动，您将在右侧看到模型模式
- 只需在编写代码的框内单击



步骤6：

- 代码将被复制到右侧空白框中
- 然后更改NewMgmtIpAddress-“10.4.255.11”,
  existMgmtIpAddress-“10.4.255.11.pdummy.com”

```
{
 “newMgmtIpAddress”: “10.4.1.2”,
 “existMgmtIpAddress:”10.4.255.11dummy.com“
}
```

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2.2: Completing VN Configuration in DNA center

Using the DNA Center GUI, perform configuration tasks according to these requirements:

- Add new virtual Network named IoT for the internet-of-things network on the Branches #1 &#2
- Create new address pools for the IoT VN named Branch1-For IoT and Branch2-For IoT on the global level, and branch1-IoT and Branch2-IoT on the Branch level.
- For Branch #1 IoT VN, allocate the subnet 10.4.198.0/24 and the gateway IP address 10.4.198.1.
- For Branch #2 IoT VN, allocate the subnet 10.5.198.0/24, and the gateway IP address 10.5.198.1.
- Associate the Branch1-IoT and Branch2-IoT pools with the IoT VN on the respective branches.
- Complete the configuration of the address pools for the Guest VN in the DNA Center so that Branch #1 and Branch #2 can accommodate guest connections. If a new address pool needs to be created and an address range allocated to it, follow the established addressing plan.
- Correct the addressing information currently defined for the Branch2-For Employees and Branch2-Employees address pool.
- For all address pools, use the DHCP server 10.2.255.211 to allocate addresses to clients.

On sw211, complete the DHCP server configuration according to these requirements:

- Create four new DHCP pools for the IoT and Employees VNs on respective branches
    - Pool named br1_iot for Branch #1 IoT VN
    - Pool named br1_emp for Branch #1 Employees VN
    - Pool named br2_iot for Branch #2 IoT VN
    - Pool named br2_emp for Branch #2 Employees VN
- In each subset, assign addresses from .101 up to .254 inclusively, and the appropriate gateway to clients.

3 Points

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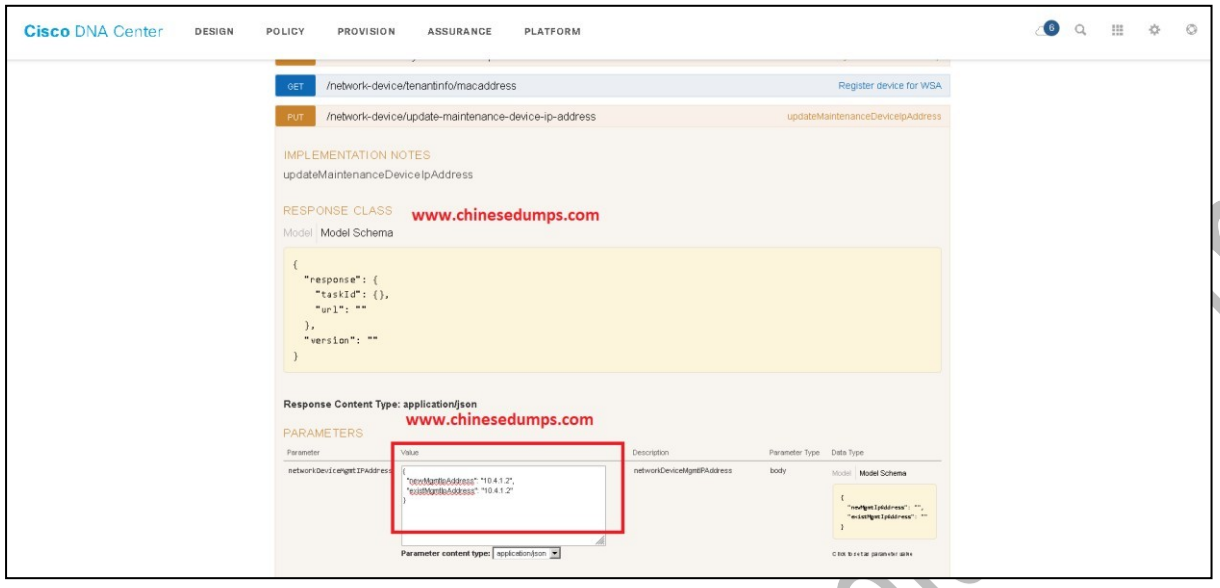
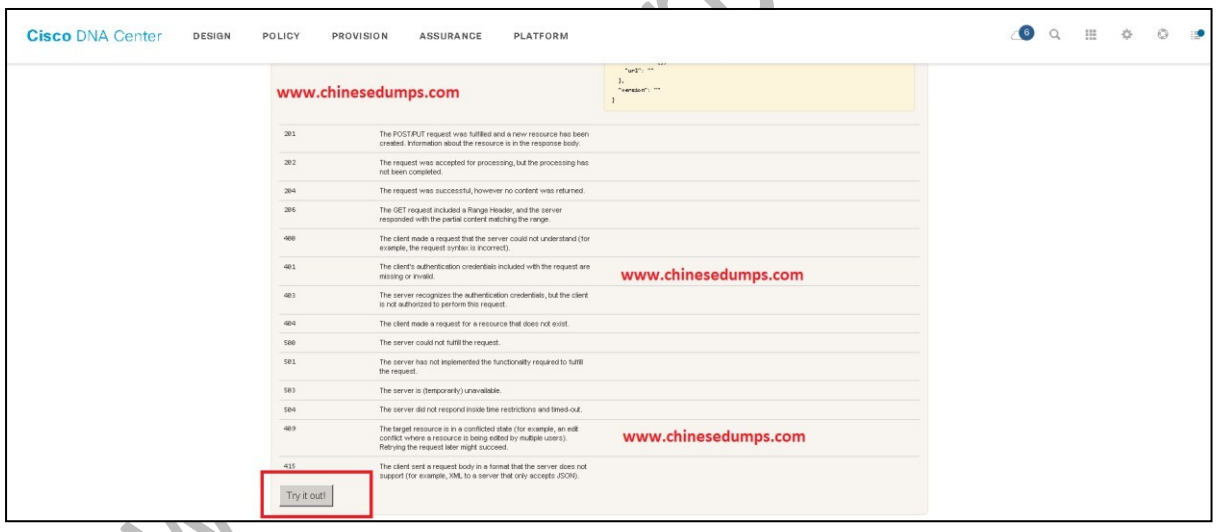


步骤7：

- 向下滚动并单击“试用”按钮



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Solution:

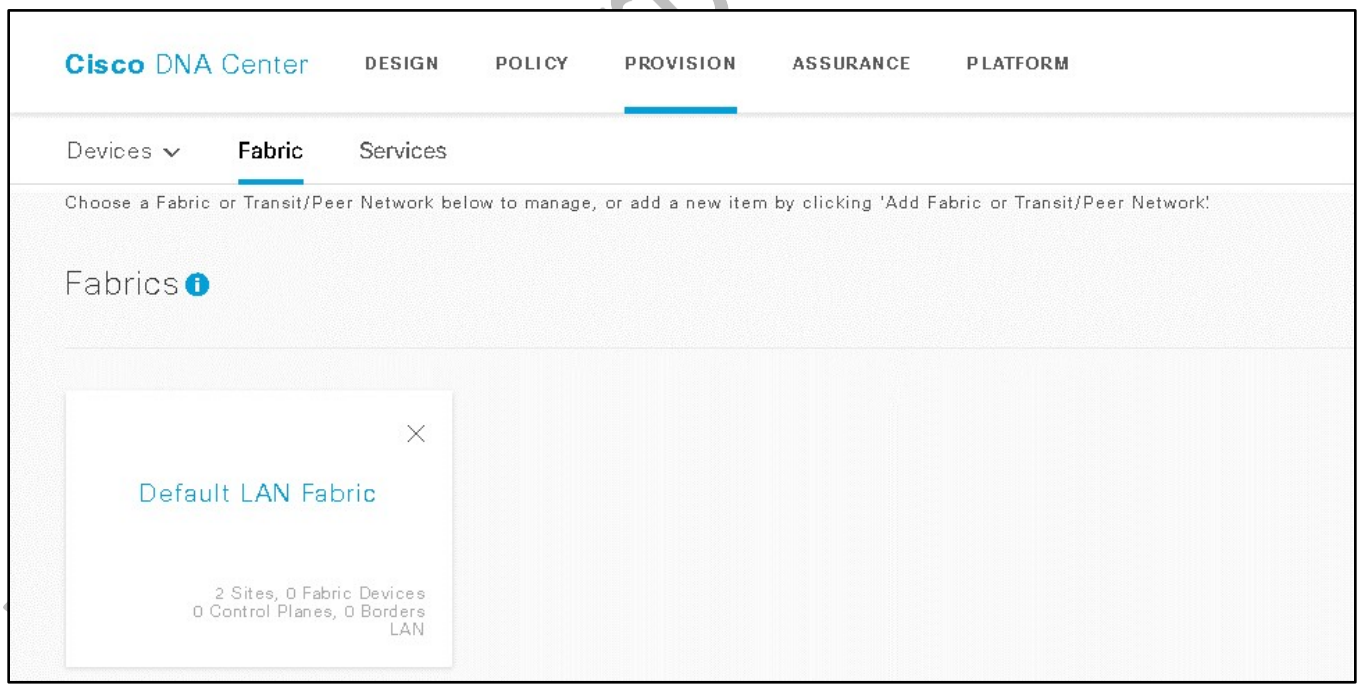
Step 1:

Configure DHCP server under network setting 10.2.255.211. Click Save.

Note: The pools that will be pre-configured will not have DHCP server attached with them , so we need to add DHCP server before moving forward to create other pools

How to add DHCP to preconfigured IP Pools :

a. Go under Provision > Fabric > Default LAN_FABRIC



2.2： 完成DNA中心的VN配置

使用DNA中心GUI，根据以下要求执行配置任务：

- 在分支机构#1和#2上为物联网网络添加名为IoT的新虚拟网络
- 在全局级别为名为Branch1 for IoT和Branch2 for IoT的IoT VN创建新的地址池，在分支级别为Branch1 IoT和Branch2 IoT。
- 对于分支#1 IoT VN，分配子网10.4.198.0/24和网关IP地址10.4.198.1。
- 对于分支#2 IoT VN，分配子网10.5.198.0/24和网关IP地址10.5.198.1。
- 将Branch1 IoT和Branch2 IoT池与各自分支上的IoT VN相关联。
- 完成DNA中心访客VN的地址池配置，以便分支#1和分支#2可以容纳访客连接。如果需要创建新的地址池并为其分配地址范围，请按照既定的地址计划进行操作。
- 更正当前为Branch2 for Employees和Branch2 Employees地址池定义的地址信息。
- 对于所有地址池，使用DHCP服务器10.2.255.211为客户端分配地址。

在sw211上，按照以下要求完成DHCP服务器配置：

- 为各自分支上的IoT和Employees VN创建四个新的DHCP池
  - 名为br1_iot的池用于分支#1 IoT VN
  - 为第一分公司员工VN命名为br1_emp的池
  - 名为br2_iot的池用于分支#2 IoT VN
  - 为2号分公司员工VN命名为br2_emp的池
- 在每个子集中，为客户端分配从.101到.254的地址，以及相应的网关。

3分

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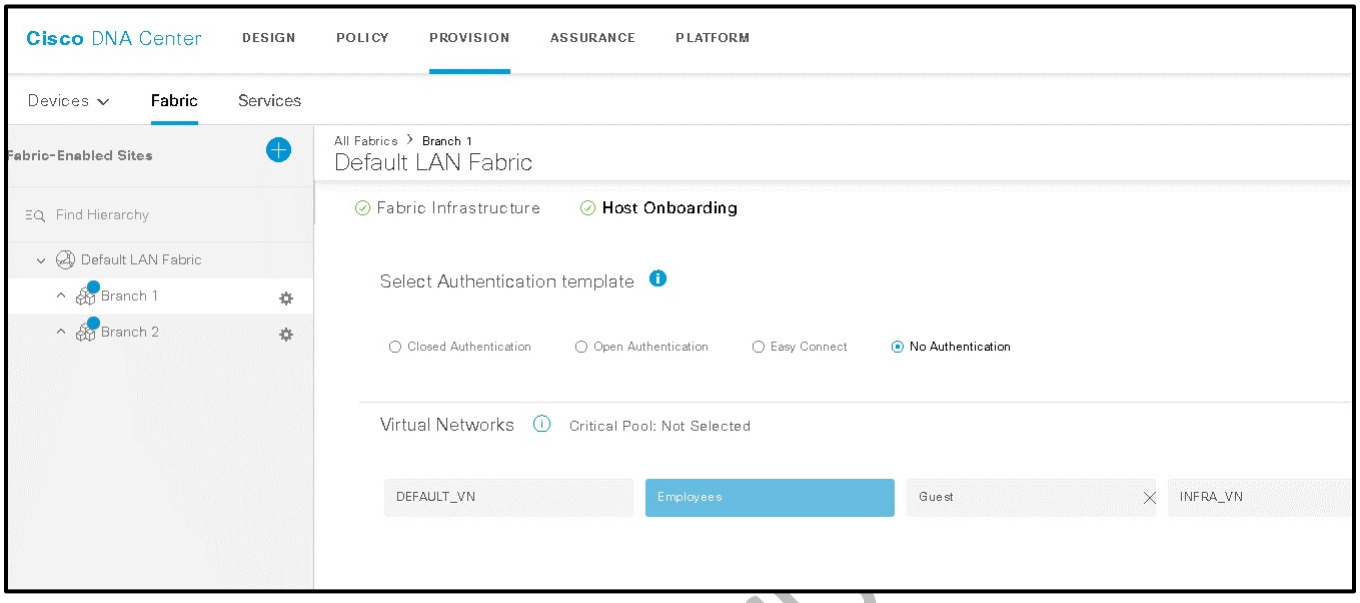
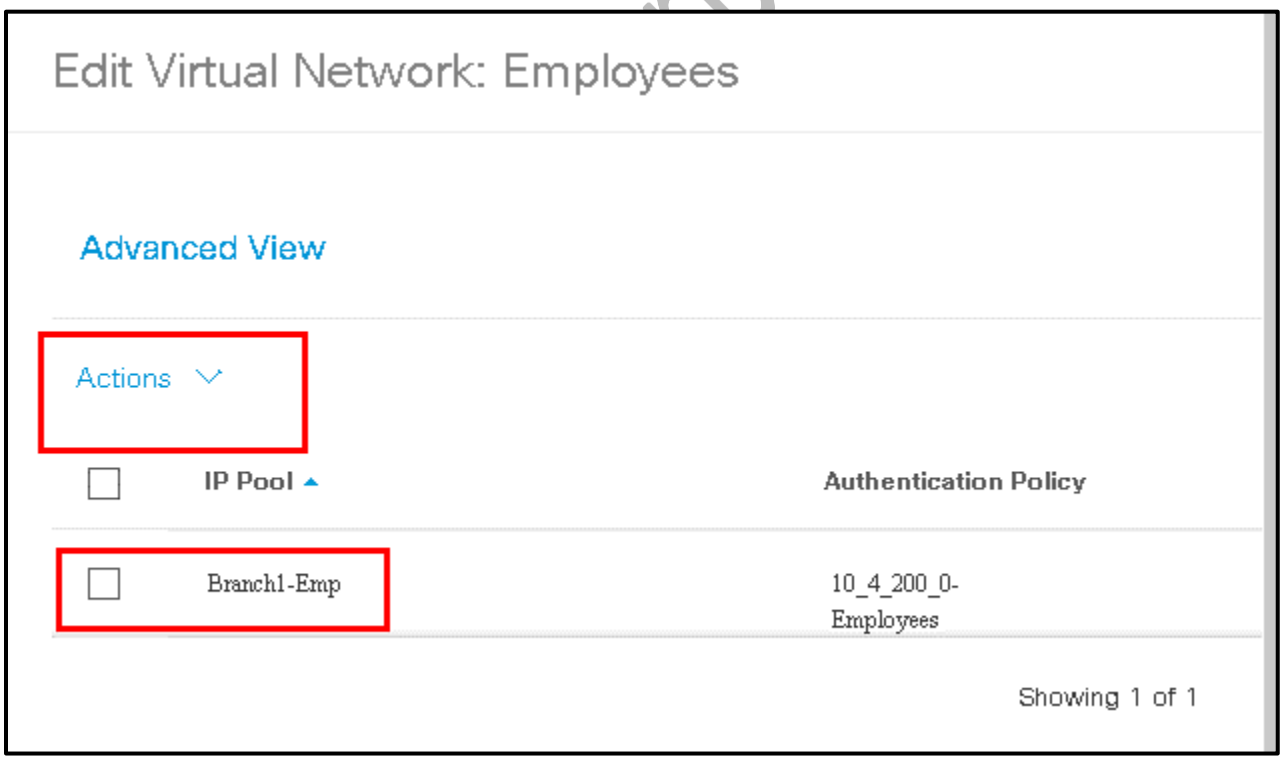
b. Select Branch1 and go under Host Onboarding and Select Employees VN



c. Then got Actions and delete the associated to Employee VN



解决方案：

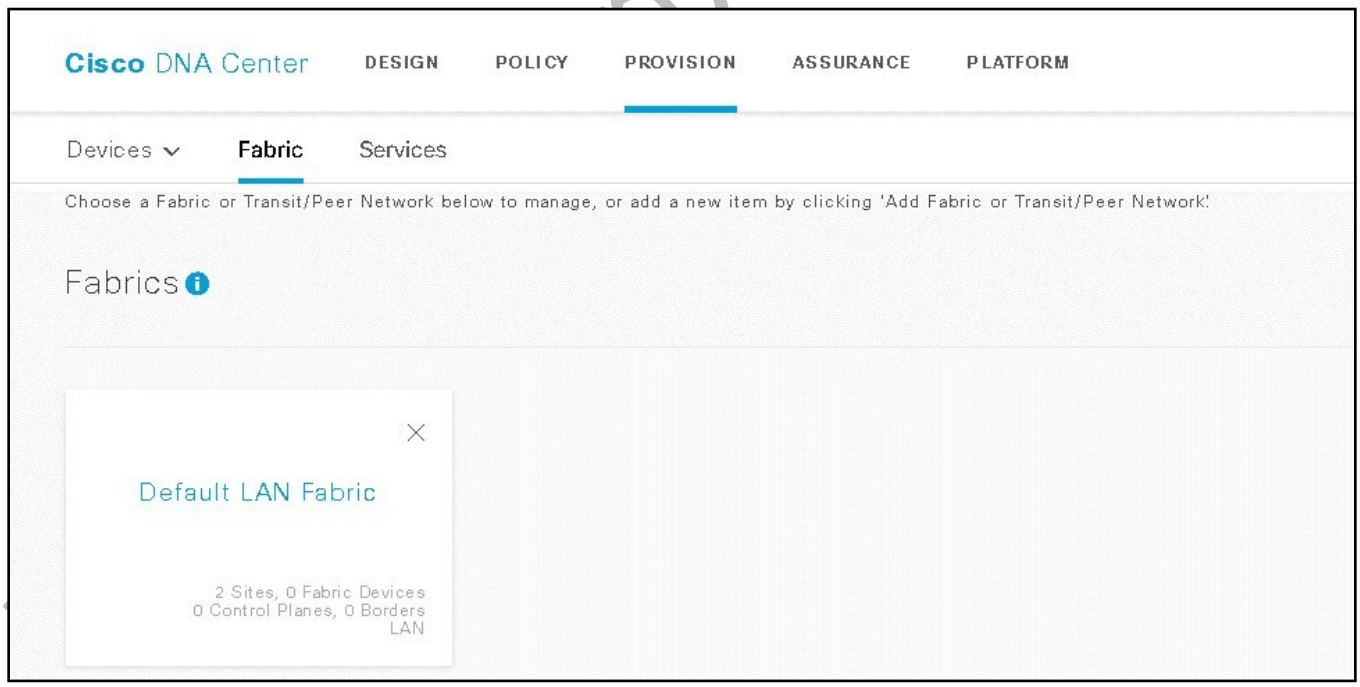
步骤1：

在网络设置10.2.255.211下配置DHCP服务器。单击“保存”。

注意：将要预配置的池将不会连接DHCP服务器，因此我们需要在继续创建其他池之前添加DHCP服务器

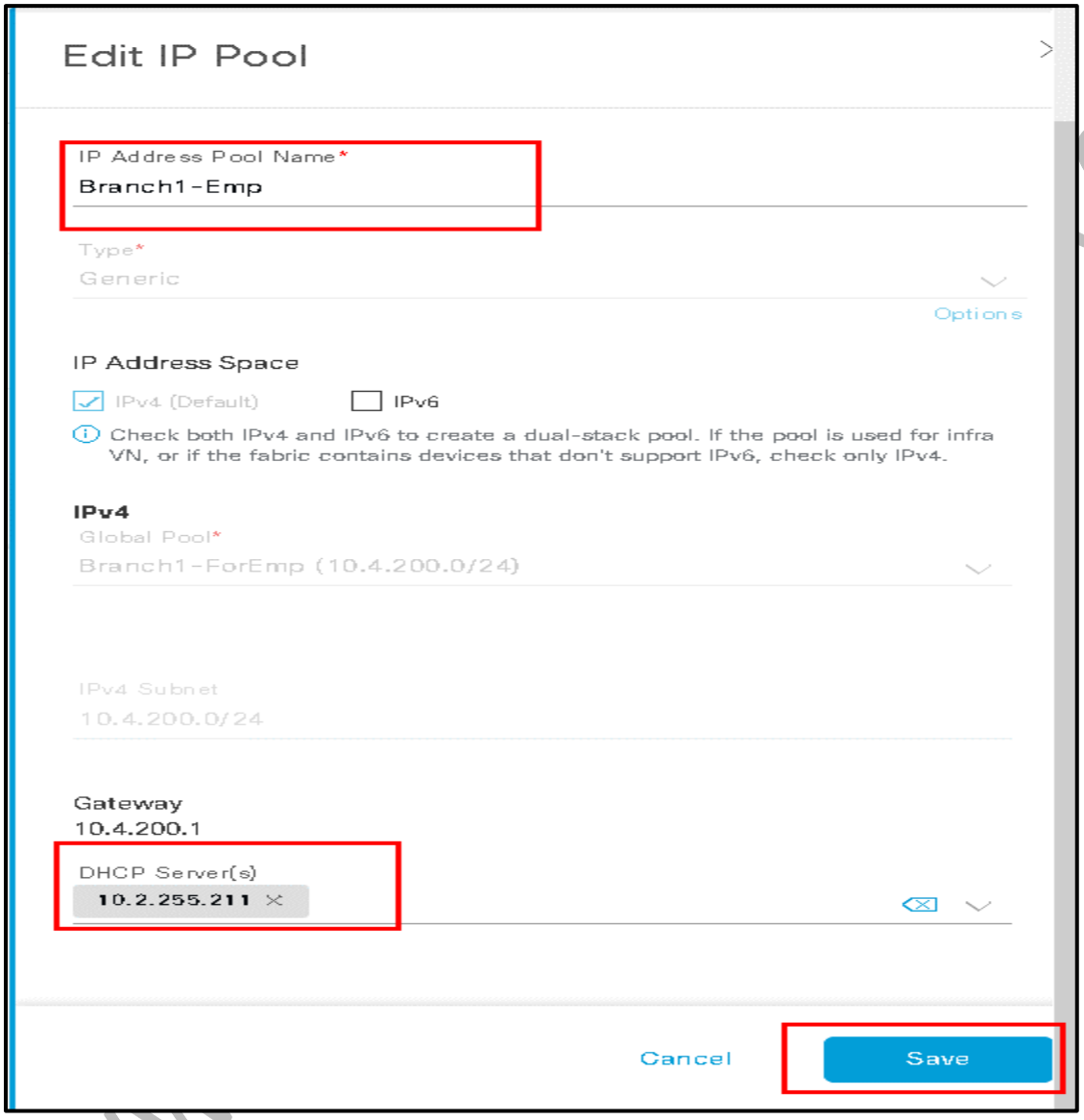
如何将DHCP添加到预配置的IP池：

a. 进入配置>结构>默认LAN_Fabric



d. After deleting go under Design > Network Settings > IP pools

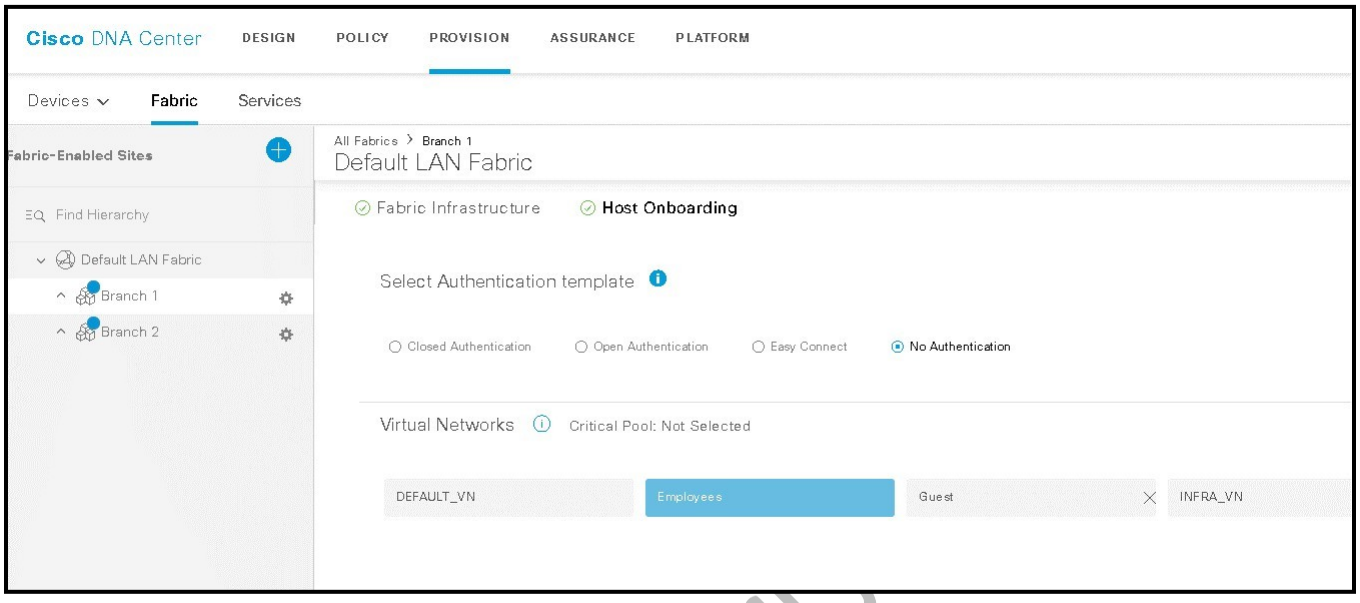
e. Select Branch1_Emp pool and add DHCP server and reserve it to Fabric again



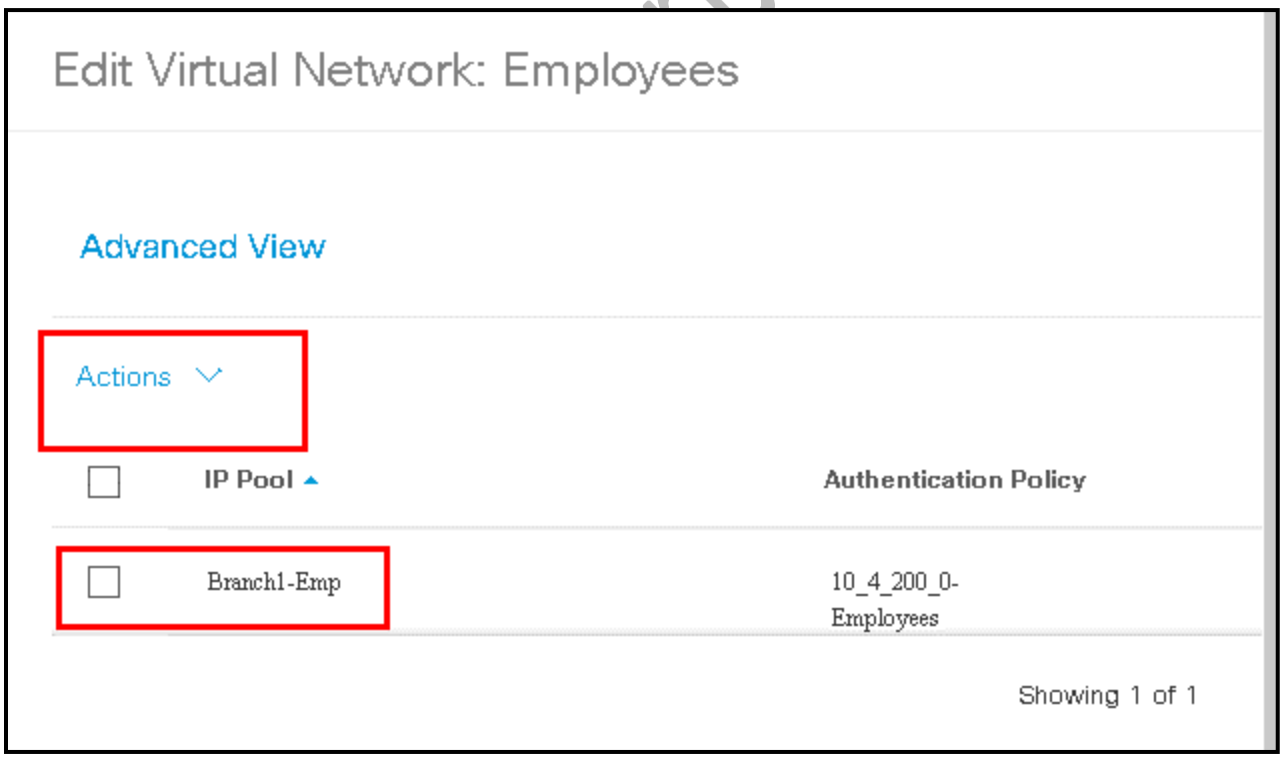
f. Edit the Branch1_Emp pool and add DHCP as shown and then save it

g. Just again add the pool to fabric

b. 选择分支机构1，进入“员工入职”和“选择员工VN”下



c. 然后获取操作并删除与员工VN关联的



Step 2:

Under IP Address pool on global level

Click Add

Fill in info Branch1-ForIoT

IP 10.4.198.0/24

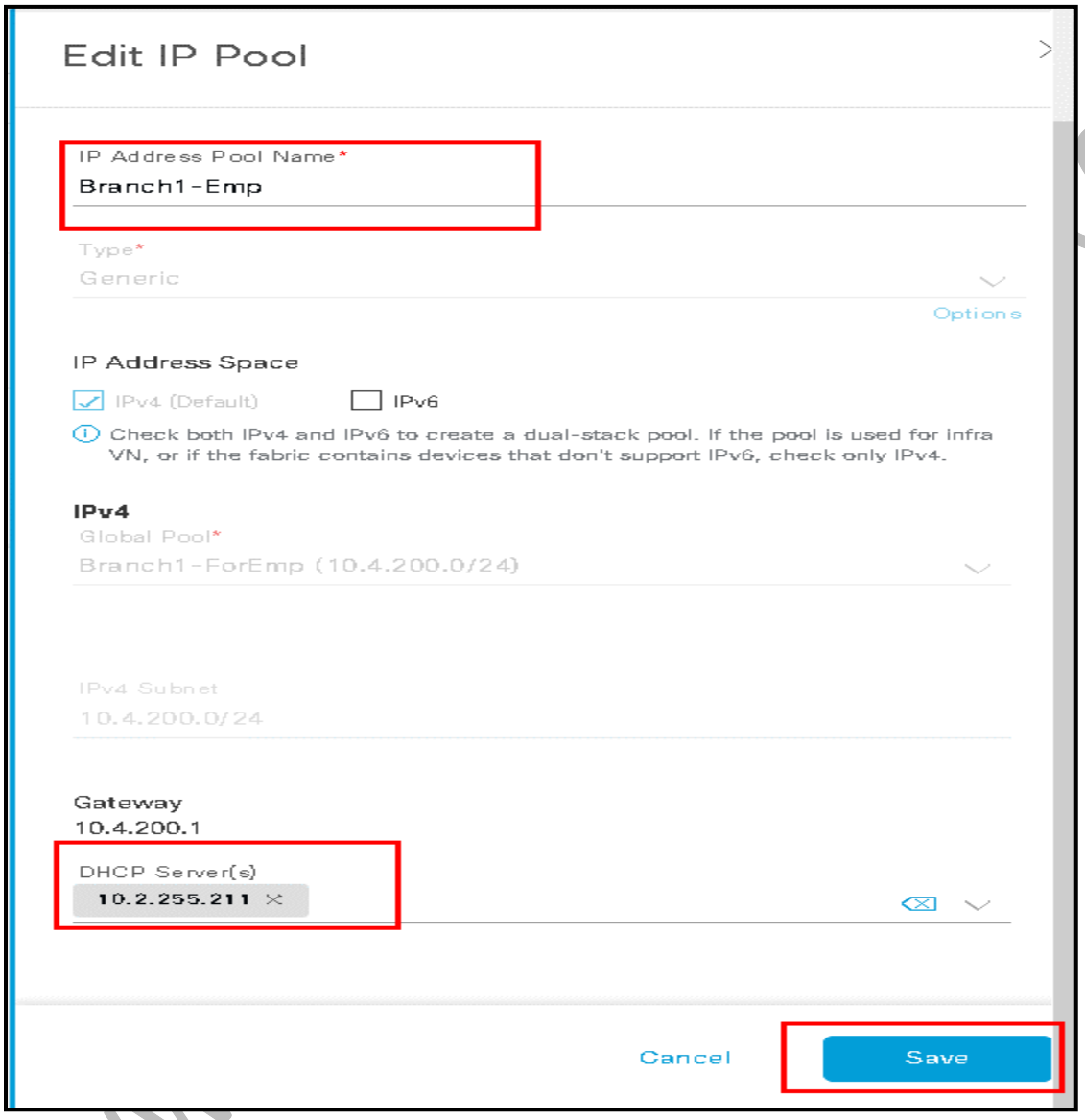
Select the DHCP pool created in step 1. Repeat for Branch 2.

Step 3:

Go under each branch under Global and click Reserve

d. 删除后，转到“设计”>“网络设置”>“IP池”下

e. 选择Branch1_Emp池并添加DHCP服务器，然后将其再次保留到结构



f. 编辑Branch1_Emp池，如图所示添加DHCP，然后保存

g. 只需再次将游泳池添加到织物中

Name the IP pool Branch1-IoT

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Select the Global pool created in step 2 for Branch1.

Select prefix length /24

Select the DHCP server 10.2.255.211

Click reserve.

Repeat for Branch 2

(Verify that the employee IP pool have the DHCP server selected and have the correct gateway and is reserved under Branch 1 and 2)

(If a new Guest IP pool is needed to be created. Configure them in the same way as IoT but different IP range)

Step 4:

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第二步：

在全局级别的IP地址池下单击“添加

”

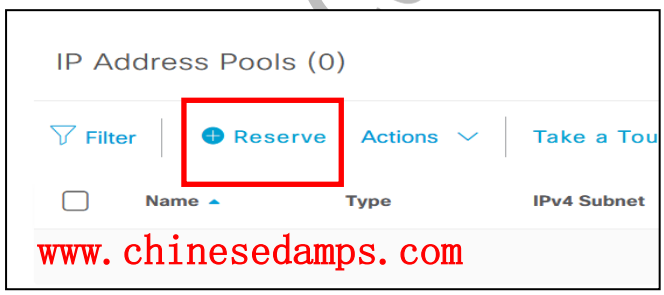
填写信息Branch1 for IoT

IP 1998年4月10日/24日

选择在步骤1中创建的DHCP池。对分支2重复上述步骤。

步骤3：

转到Global下的每个分支下，然后单击Reserve



Under Policy>Virtual Network. Click Create Virtual Network.

Enter the name IoT

Step 5:

Under Provisioning>Fabric select Branch 1.

Select Host Onboarding > Virual Networks and click Add Virtual Network

Select the VN IoT and click update. The click on the Grey VN Iot

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将IP池命名为Branch1 IoT

为Branch1选择在步骤2中创建的全局池。选择前

缀长度/24

选择DHCP服务器10.2.255.211单

击保留。

对分支2重复上述步骤

（验证员工IP池是否已选择DHCP服务器，是否具有正确的网关，是否在分支1和2下保留）

（如果需要创建新的访客IP池。以与物联网相同的方式配置它们，但IP范围不同）

步骤4：

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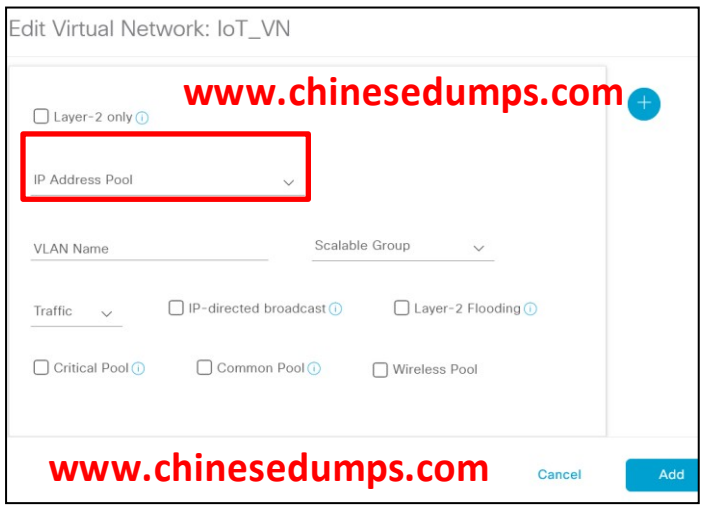

Select the IP Address pool for Branch1-IoT

Repeat for Branch 2

(If needed add the created IP pool to the Guest VN)

Step 6:

DCHP server on sw211

ip dchp exclude10.4.198.1 10.1.100.100
ip dchp exclude 10.5.198.1 10.1.101.100

ip dhcp pool br1_iot
network 10.4.198.0 /24
default-router 10.4.198.1
exit

ip dhcp pool br2_iot
network 10.5.198.0 /24
default-router 10.5.198.1

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在“策略”>“虚拟网络”下。单击“创建虚拟网络”。输入名称

IoT

步骤5：

在配置>结构下，选择分支1。

选择主机板载>虚拟网络，然后单击添加虚拟网络

选择VN IoT并单击更新。点击灰色VN-Iot

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2.3: Mapping SDA VNs to SD-WAN VPNs

Using vManage GUI, perform configuration tasks according to these requirements:

- Use any host, such as host11, to access the vManage GUI website at https://203.0.113.21 URL.
- Create three new SD-WAN VPNs to carry the SDA VN traffic
  - VPN ID 198 for IoT VN
  - VPN ID 199 for Guest VN
  - VPN ID 200 for Employees VN
- On Branch #1 and Branch #2 vEdges, for each of these VPNs:
  - Create a new sub-interface on the interface toward the SDA border switch. Align the VLAN ID and IP address on the sub interface with the configuration generated by DNA Center on the border switches for the appropriate VN.
  - Peer the vEdge and the SDA border switch using iBGP. Ensure full reachability between all locations of the same VPN.

4 Points

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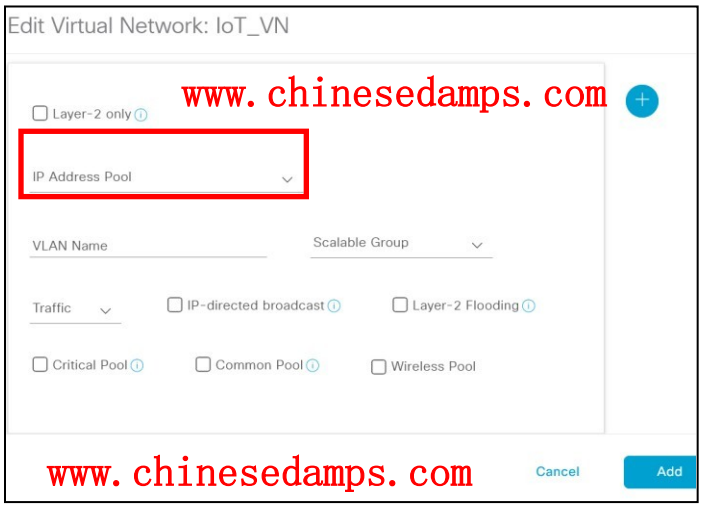

为Branch1 IoT选择IP地址池

对分支2重复上述步骤

（如果需要，将创建的IP池添加到Guest VN）

步骤6：

sw211上的DCHP服务器

ip dchp除外10.4.198.1
10.1.100.100 ip dchp不包括
10.5.198.1 10.1.11.100

ip dhcp池br1_iot网络
10.4.198.0/24
默认路由器10.4.198.1退
出

ip dhcp池br2_iot网络
10.5.198.0/24
默认路由器10.5.198.1

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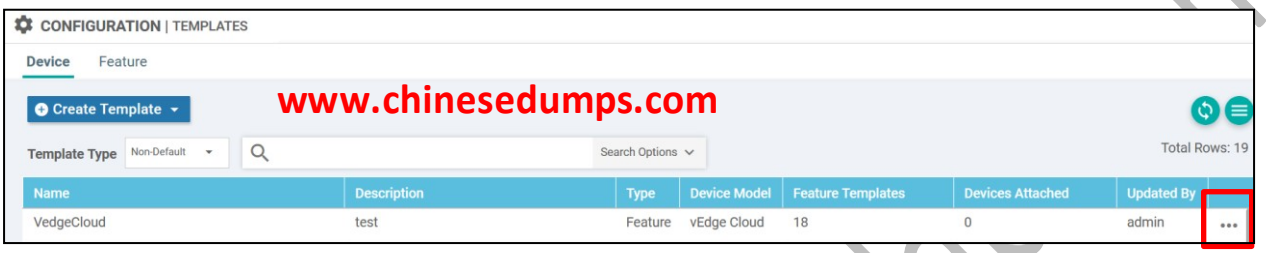
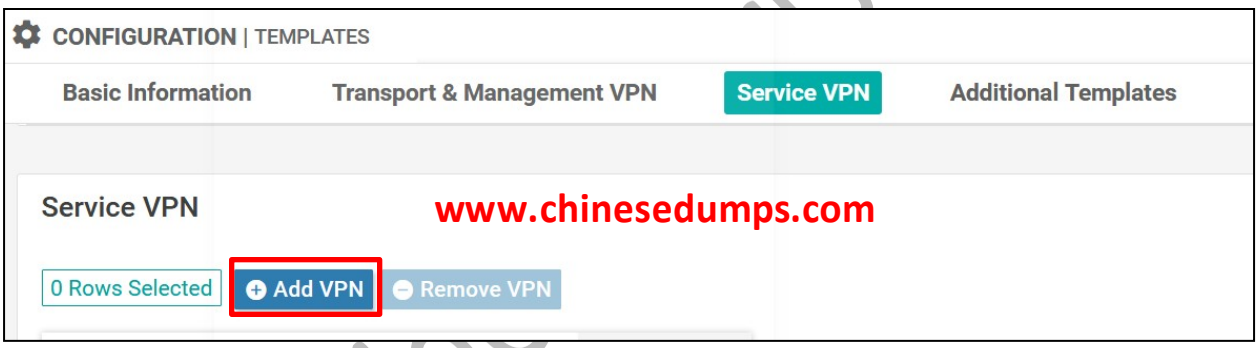
Solution:

Step 1:

Select the device template where the Branch Vedges are attached. Click the Three dots <...> and Edit Device Template.



Go under Service VPN and click Add VPN



On the bottom of the page click Create VPN Template

2.3：将SDA VN映射到SD-WAN VPN

使用vManage GUI，根据以下要求执行配置任务：

- 使用任何主机（如host11）访问vManage GUI网站https://203.0.113.21 URL。
- 创建三个新的SD-WAN VPN来承载SDA VN流量
  - IoT VN的VPN ID 198
  - 访客VN的VPN ID 199
  - 员工VPN ID 200 VN
- 在分支#1和分支#2 vEdges上，对于这些VPN中的每一个：
  - 在朝向SDA边界交换机的接口上创建一个新的子接口。将子接口上的VLAN ID和IP地址与边界交换机上DNA中心为相应VN生成的配置对齐。
  - 使用iBGP对等vEdge和SDA边界交换机。确保两者之间完全可达同一VPN的所有位置。

4分

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Give the template a name and add the VPN number 198

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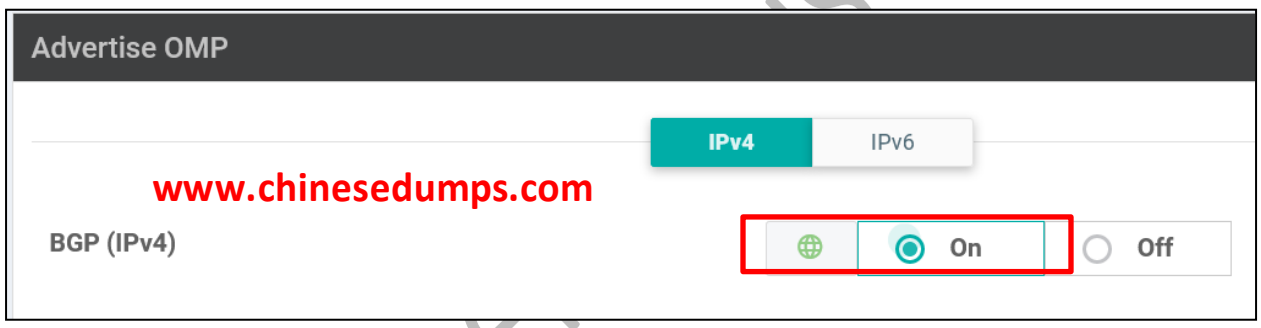
Under Advertise OMP Click on for BGP



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Click Save and repeat for VPN 199 and 200

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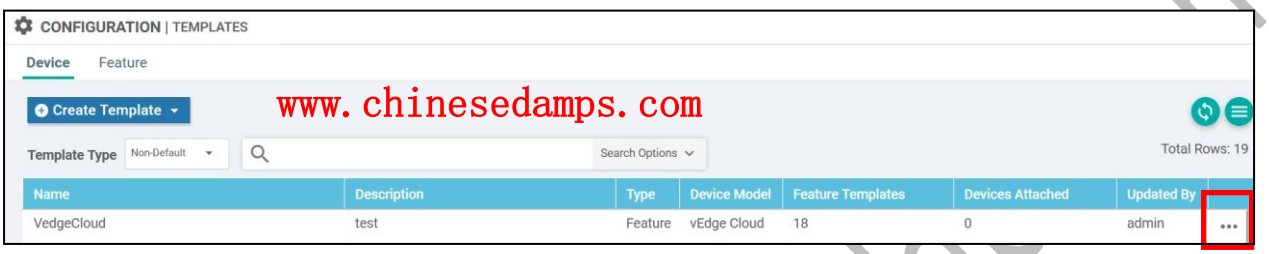
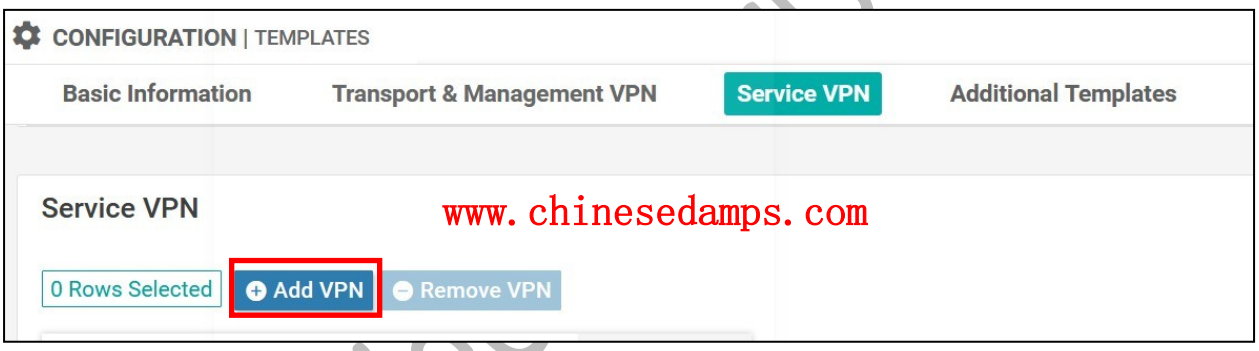
解决方案：

第一步：

选择连接分支视频的设备模板。单击三个点<...>并编辑设备模板。



转到"服务VPN"下，然后单击"添加



在页面底部单击创建VPN模板

Step 2:

Select all three VPNs and click Next

Pick BGP and VPN Interface Ethernet

Press the BGP drop down bar and press create template.

Give the template a name.

Leave BGP AS number to Device specific

Under Unicast Address Family click New Redistribute and under Protocol choose OMP and click Add.

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为模板命名并添加VPN号码198

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在广告OMP下点击BGP

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单击“保存”，然后对VPN 199和200重复

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Under Neighbor Click New Neighbor.

Change Address to device specific and click Add. Then Click Save

Step 3:

a. Create the VPN interface Template
b. Click Create new VPN Interface Ethernet Template
c. Give the template a name and description.
d. Change Shutdown to be Global No
e. Change Interface Name to be Device specific

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第二步：

选择所有三个VPN，然后单击下一步

选择BGP和VPN接口以太网

按BGP下拉栏，然后按创建模板。

为模板命名。

将BGP AS号码留给特定设备

在“单播地址族”下，单击“新建重新分发”，在“协议”下选择“OMP”，然后单击“添加”。

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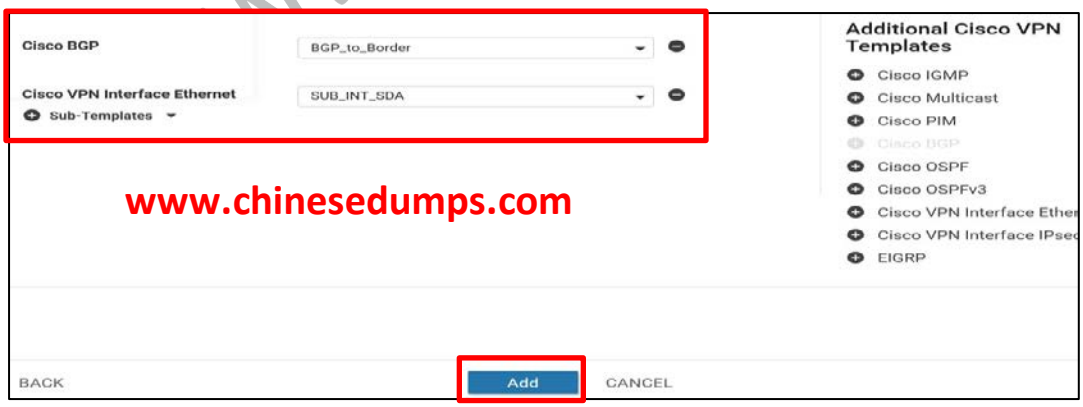
Change IPv4 Address to be device specific.

Under Advanced change the MTU to be Global 1496 and click Save.

Add both template that was created and click Add and the press Update\



在“邻居”下，单击“新建邻居”。

将地址更改为特定于设备的地址，然后单击添加。然后单击保存

步骤3：

a. 创建VPN接口模板
b. 单击创建新的VPN接口以太网模板
c. 为模板命名和描述。
d. 将关机更改为全局否
e. 将接口名称更改为特定于设备的名称

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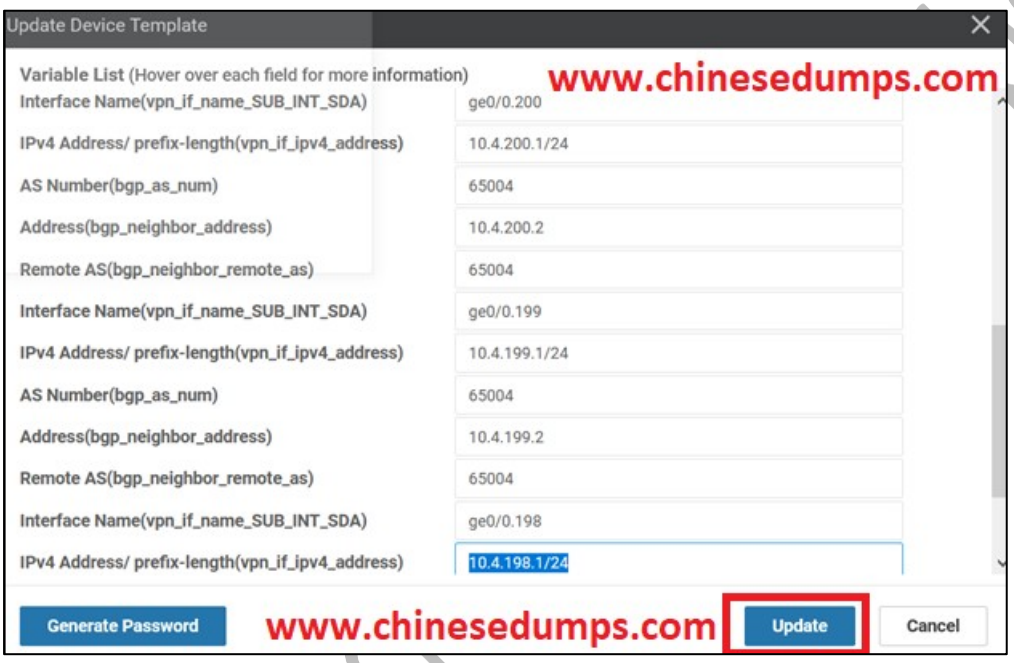
Click the three dots on the right to add the deivces values <…> For branch1 Router vEdge 40 Set BGP AS number to 65004 and Vedge 51 and 52 AS 65005

Place cursor on the three dots <…> to verify the interfaces.



Repeat for Vedge 51 and 52.

Then click Update and Next and push the config to the devices.

Step 4:

*Configure the SDA Border switches Sw 400, Sw 501 and Sw 502.

*DNA center should have generated VRFs for Employee VN, Guest VN and IoT VN.

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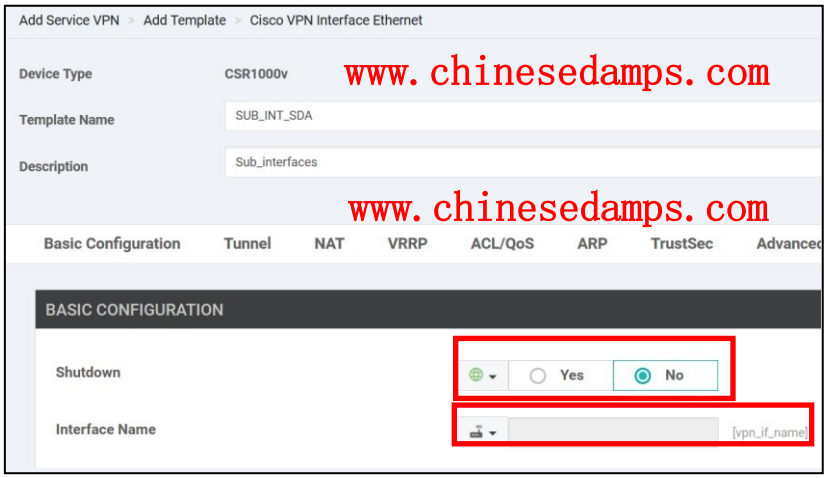
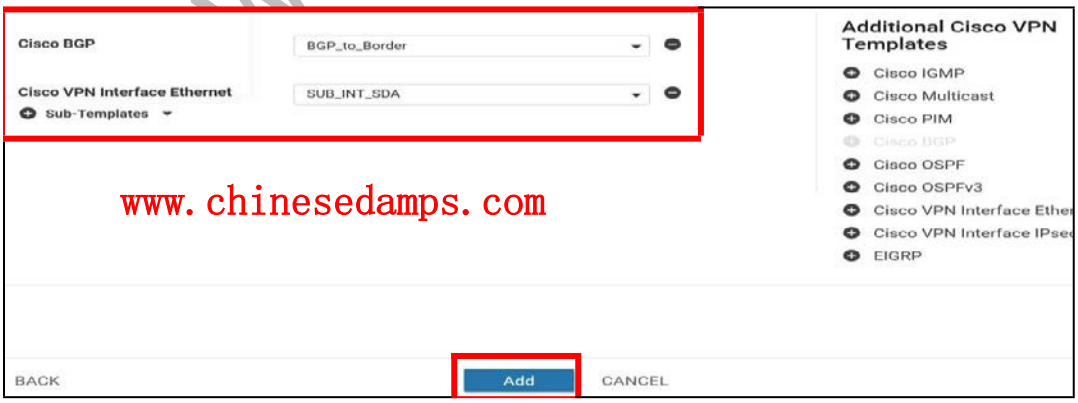


将IPv4地址更改为特定于设备的地址。

在“高级”下，将MTU更改为“全局1496”，然后单击“保存”。

添加已创建的两个模板，单击“添加”，然后按“更新”\



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Under Sw 400

Interface Vlan 198
Vrf forwarding IoT
Ip add 10.4.198.2 255.255.255.0
Interface Vlan 199
Vrf forwarding Guest
Ip add 10.4.199.2 255.255.255.0

Interface Vlan 200
Vrf forwarding Employees
Ip add 10.4.200.2 255.255.255.0

Router bgp 65004
Address-family ipv4 vrf IoT
Neighbor 10.4.198.1 remote-as 65004
Neighbor 10.4.198.1 activate
Network 10.4.1.0 mask 255.255.255.0

Address-family ipv4 vrf Guest
Neighbor 10.4.199.1 remote-as 65004
Neighbor 10.4.199.1 activate
Network 10.4.2.0 mask 255.255.255.0

Address-family ipv4 vrf Employees
Neighbor 10.4.200.1 remote-as 65004
Neighbor 10.4.200.1 activate
Network 10.4.3.0 mask 255.255.255.0

Add Network statement of the IP pools that was created in DNAC in question 2.1
Repeat this configuration for switch 501 and 502

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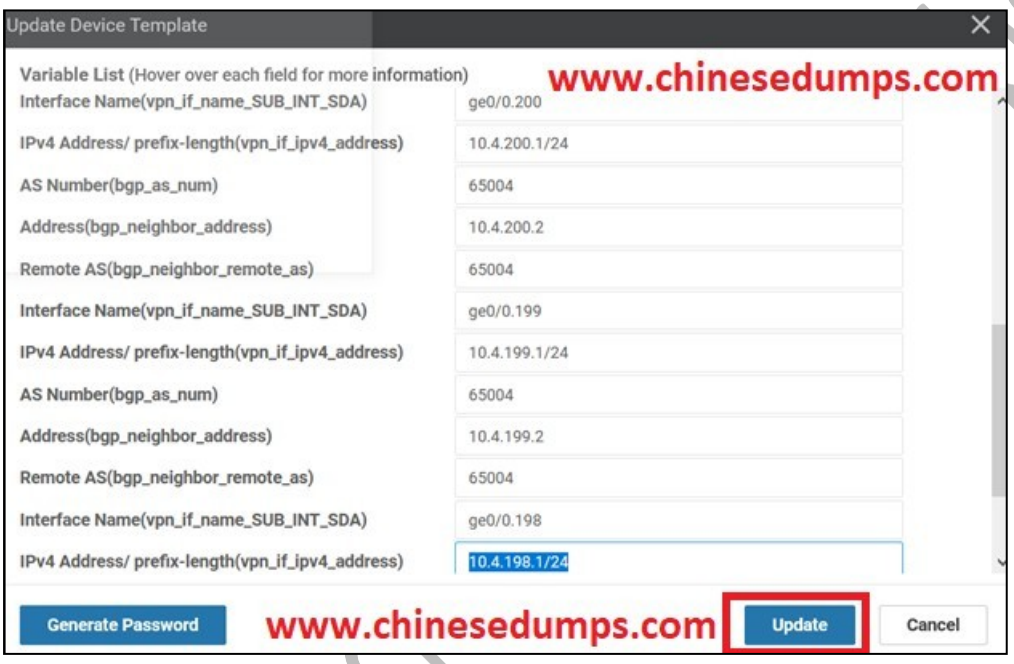
单击右侧的三个点以添加设计值<...>对于branch1 Router vEdge 40，将BGP AS编号设置为65004，将vEdge 51和52 AS设置为65005

将光标放在三个点上<...>以验证接口。



对视频51和52重复上述步骤。

然后单击更新和下一步，并将配置推送到设备。

步骤4：

*配置SDA边界开关Sw 400、Sw 501和Sw 502。

*DNA中心应为员工VN、访客VN和IoTVN生成VRF。

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2.4: Configuring SD-WAN VPN Route Leaking

To allow the traditional parts of the FABD2 network to communication with the employees and IoT VPNs/VNs, configure route leaking in SD-WAN according to these requirements:

- Prefixes in the IoT VPN 198 must be imported into the existing SDA Underlay VPN 999 and tagged with the tag value of 198
- Prefixes in the Employees VPN 200 must be imported into the existing SDA underlay VPN 999 and tagged with the tag value of 200
- Prefixes in the SDA underlay VPN 999 advertised form the DC that are within the 10.4.0.0/15 range must be rejected. Other prefixes in the SDA underlay VPN 999 advertise from DC must be accepted and also imported into IoT VPN 198 and Employees VPN 200
- Redistribution from OMP into OSPF on Branches #1 and #2 in VPN 999 must exclude vRoutes tagged with values 198 or 200.
- Place host41 into Employees VN. Place host51 into IoT VN. Make sure both hosts receive their IP settings from DHCP.
- Ensure that the IoT and Employees VPNs on Branches #1 and #2 have reachability to Branches #3 and #4. It is allowed to modify the VPN 999 OMP settings to accomplish this requirement.

3 Points

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400瑞士法郎以下

接口Vlan 198 Vrf
转发物联网

接口Vlan 199
转发访客
Ip地址为10.4.199.2 255.255.255.0

接口Vlan 200
Vrf转发员工
Ip地址：2000年4月10日255.255.255.0

路由器bgp 65004
地址家族ipv4 vrf物联网
邻居10.4.198.1远程为65004
邻居10.4.198.1激活
网络10.4.1.0掩码255.255.255.0

地址家庭ipv4 vrf来宾邻居
10.4.199.1远程为65004
邻居10.4.199.1激活
网络10.4.2.0掩码255.255.255.0

地址家庭ipv4 vrf员工邻居
10.4.200.1远程为65004
邻居10.4.200.1激活
网络10.4.3.0掩码255.255.255.0

添加问题2.1中在DNAC中创建的IP池的Network语句对交换机501和502重复此配置

Ip地址：1998年4月10日255.255.0

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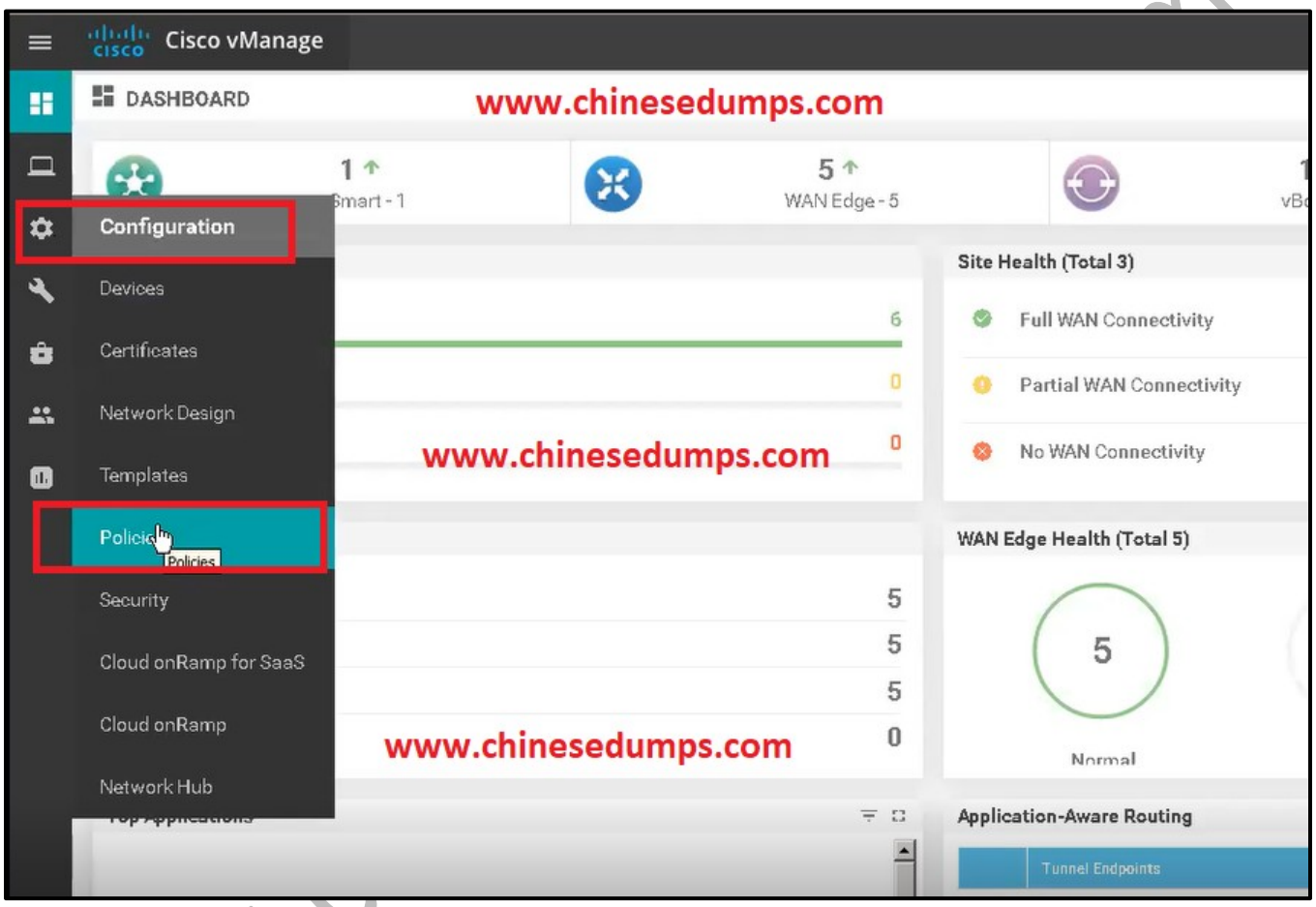
Solution:

Step 1:

In vManage GUI go to **Configuration** tab and select **Policies**



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2.4： 配置SD-WAN VPN路由泄漏

为了使FABD2网络的传统部分能够与员工和物联网VPN/VN通信，请根据以下要求配置SD-WAN中的路由泄漏：

- IoT VPN 198中的前缀必须导入到现有的SDA Underlay VPN 999中，并标记为标签值198
- Employees VPN 200中的前缀必须导入到现有的SDA底层VPN999中，并标记为标签值200
- 必须拒绝从DC通告的SDA底层VPN 999中10.4.0.0/15范围内的前缀。必须接受来自DC的SDA底层VPN 999广告中的其他前缀，并将其导入IoT VPN 198和Employees VPN 200
- 在VPN 999的分支#1和#2上，从OMP到OSPF的重新分发必须排除标记为值198或200的vRoutes。
- 将host41放入Employees VN。将host51放入IoT VN。确保两台主机都从DHCP接收IP设置。
- 确保分支#1和#2上的物联网和员工VPN可以访问分支#3和#4。允许修改VPN 999 OMP设置以满足此要求。

3分

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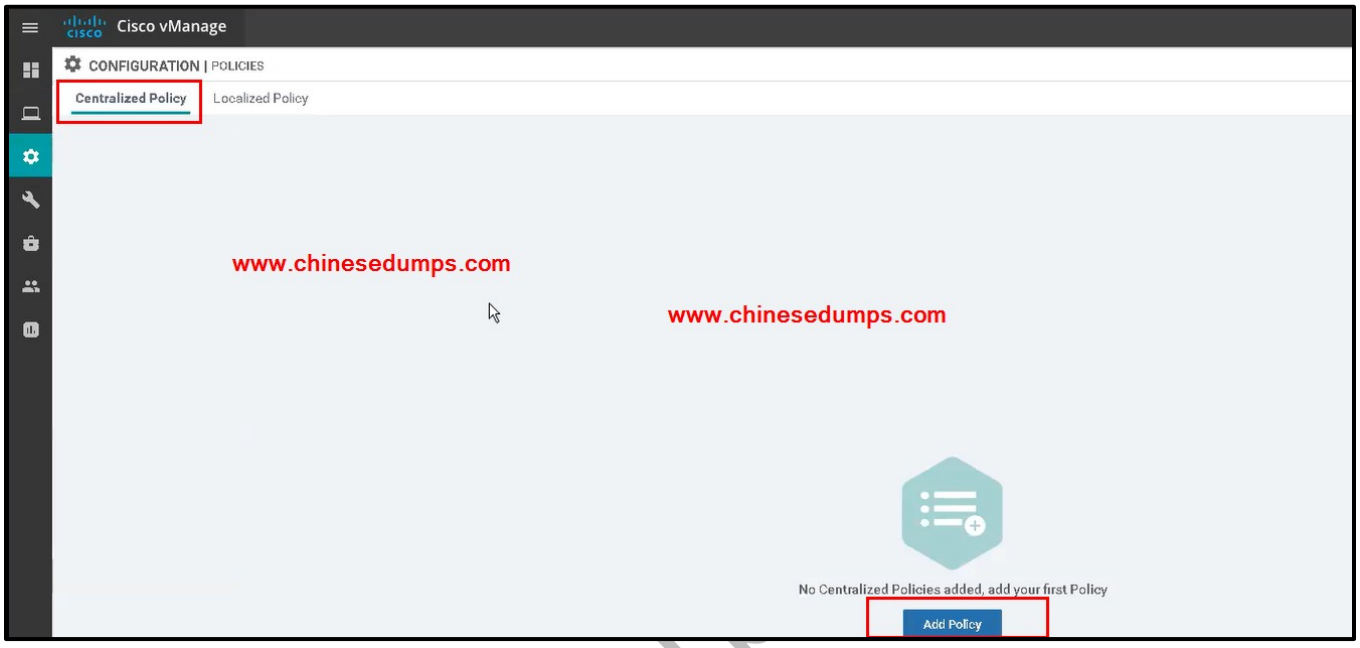
Step 2:

Under **Policies** Select **Centralized policy** and click on **add**



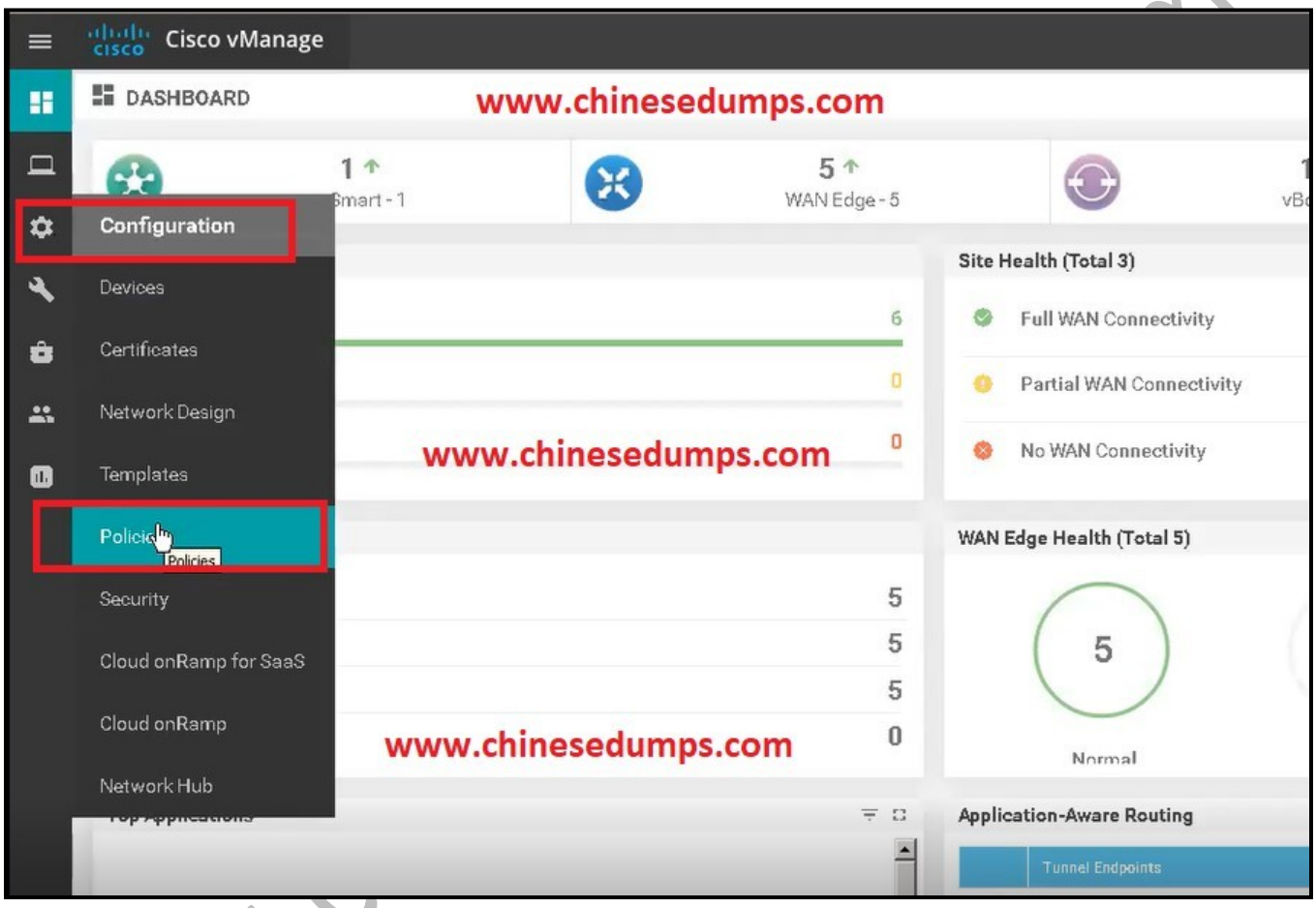
Step 3:

Now under centralized Policy check in the site tab which sites are pre-configured then hit **NEXT** at the bottom of the page

解决方案：

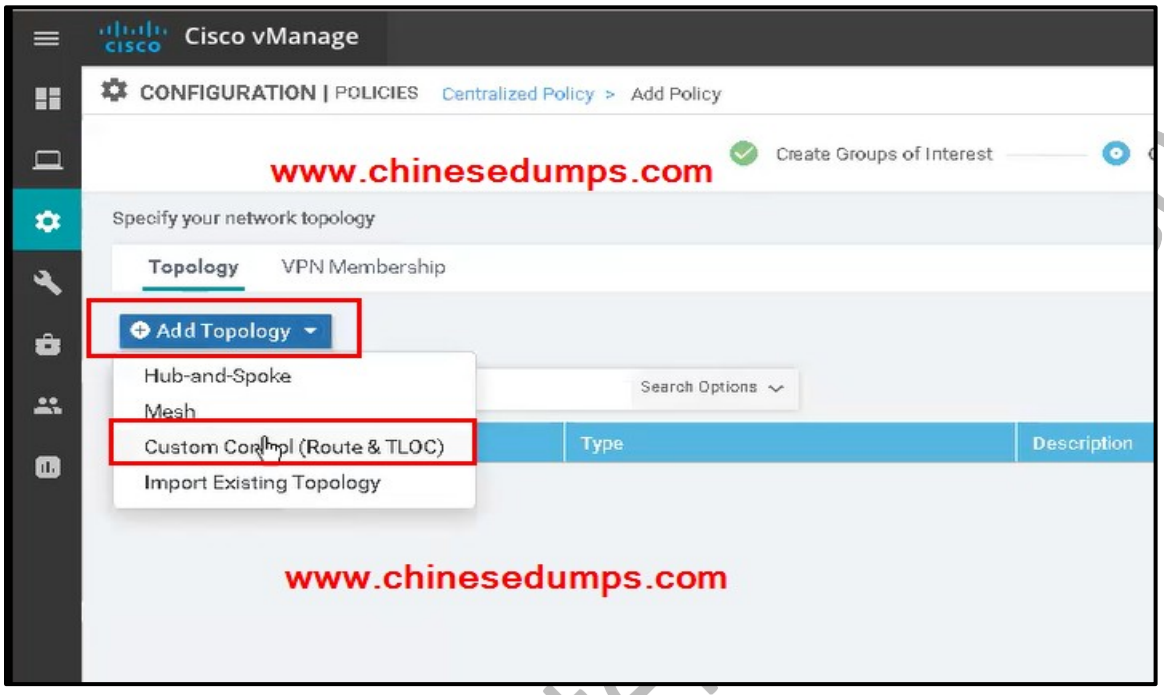
第一步：

在vManage GUI中，转到配置选项卡并选择策略



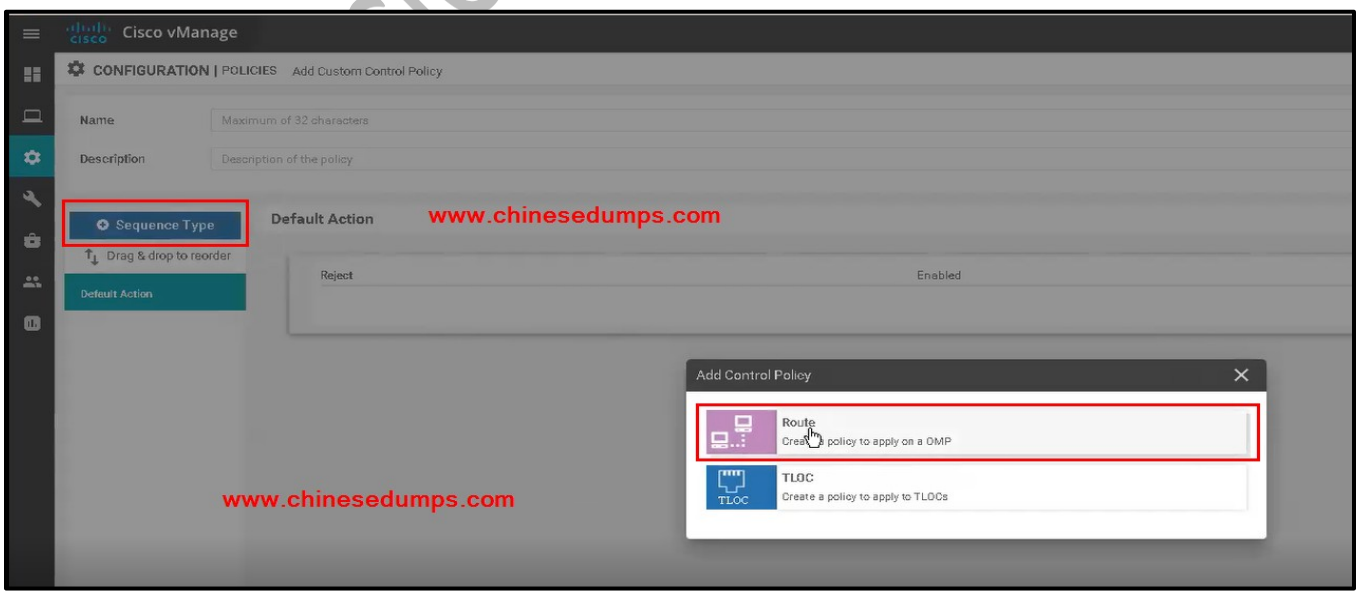
Step 5:

Click on Add **Topology** and select **Custom Control**



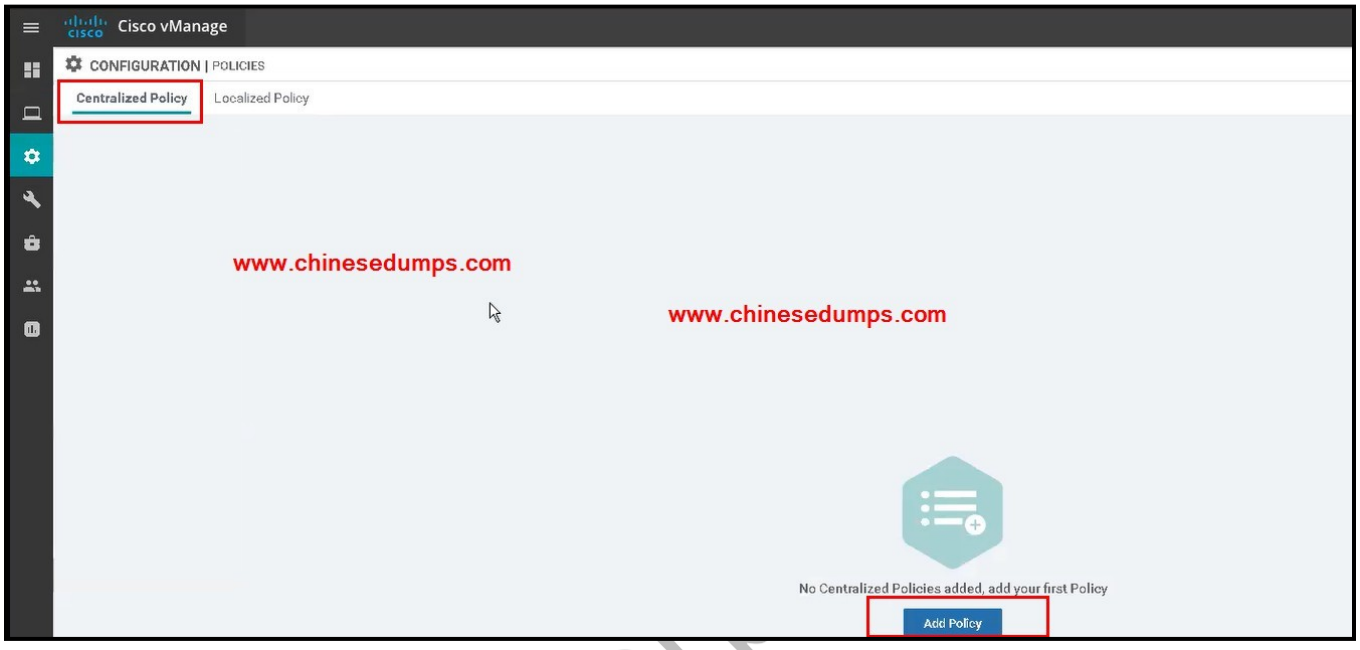
Step 6:

Add a **Sequence Type** and add a **Route Control Policy**



第二步：

在“策略”下选择“集中式策略”，然后单击“添加”

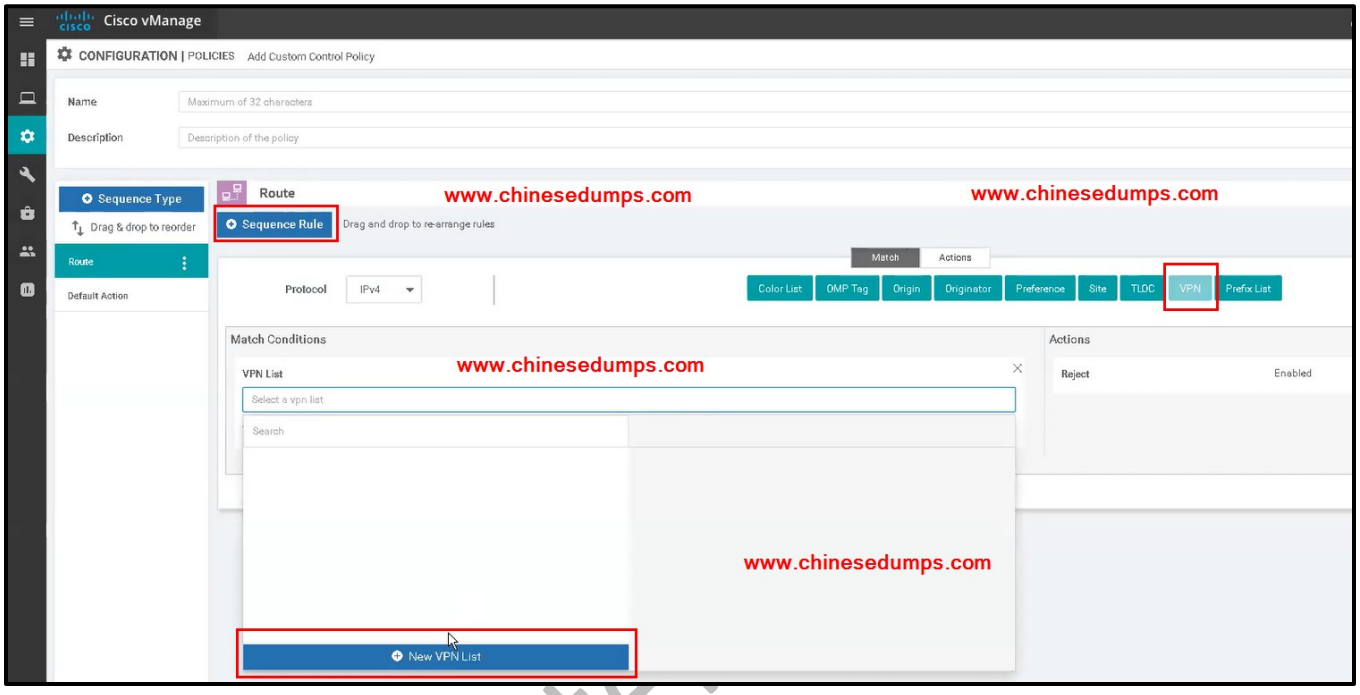


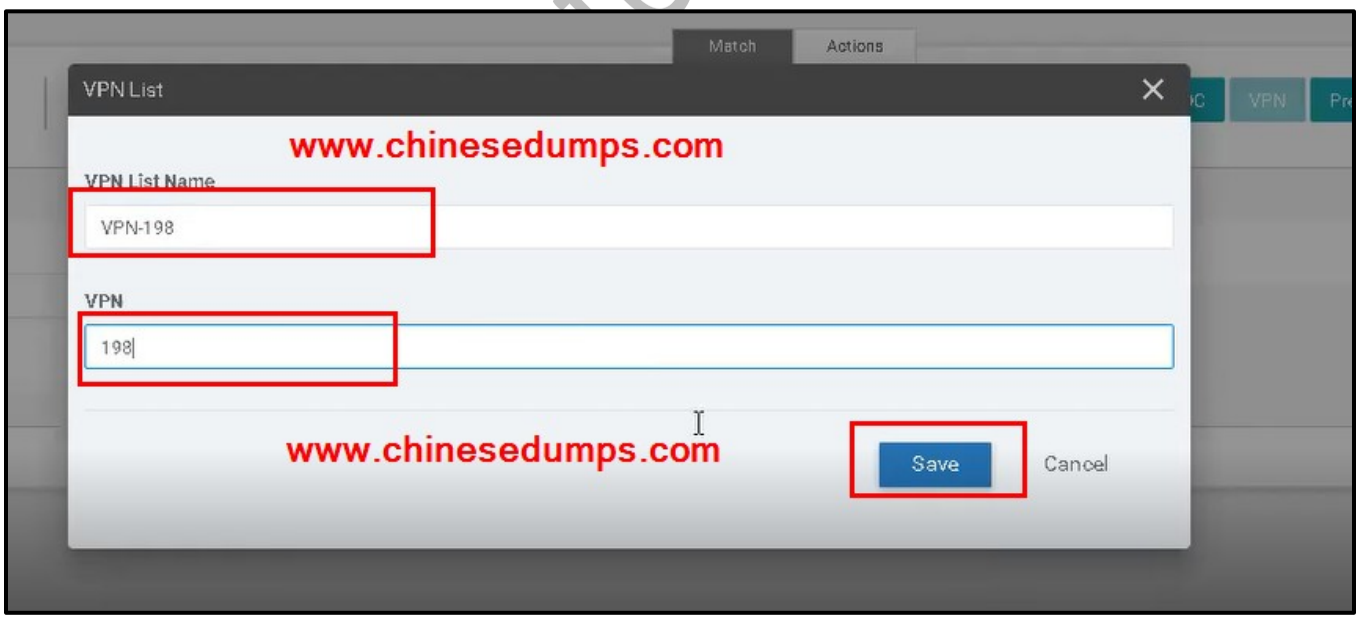
步骤3：

现在，在集中式策略下，在站点选项卡中检查哪些站点已预先配置，然后点击页面底部的下一步

Step 7:

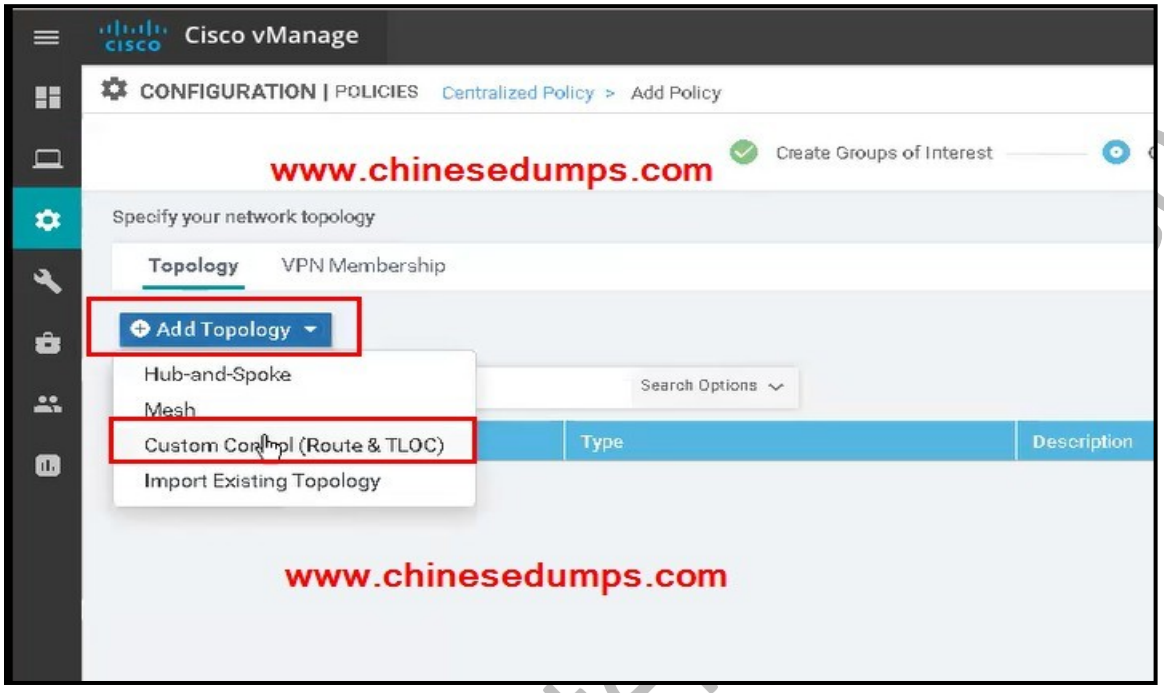
Select VPN list under **Match Conditions** and add a new VPN list





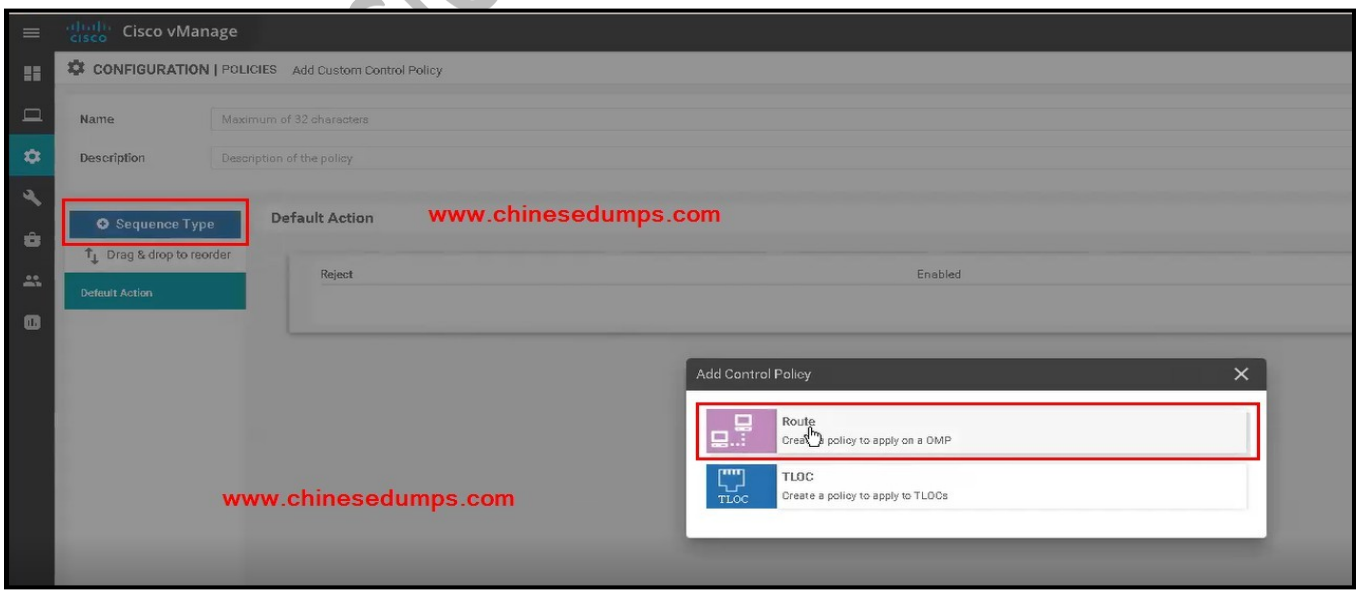
步骤5：

单击添加拓扑并选择自定义控件



步骤6：

添加序列类型并添加路由控制策略



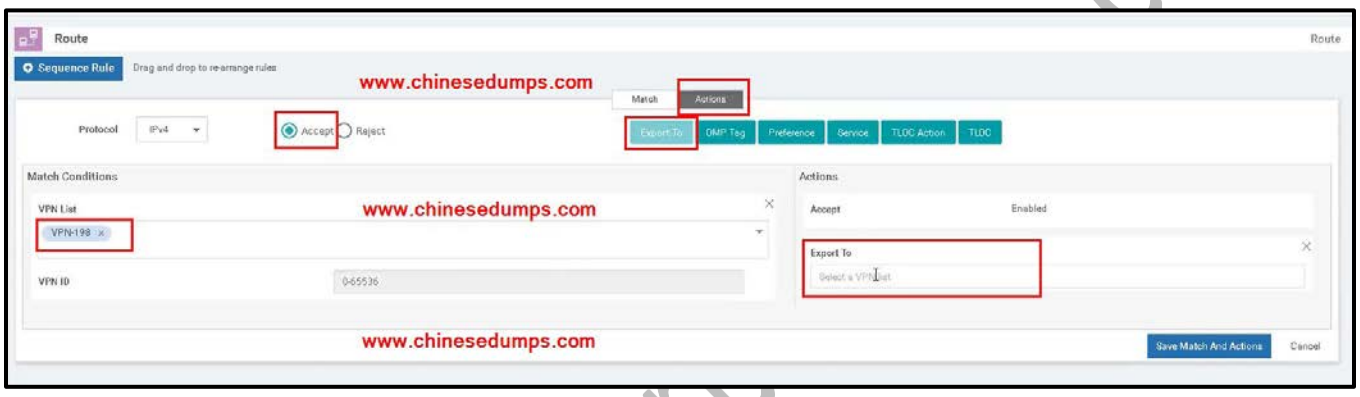
Step 8:

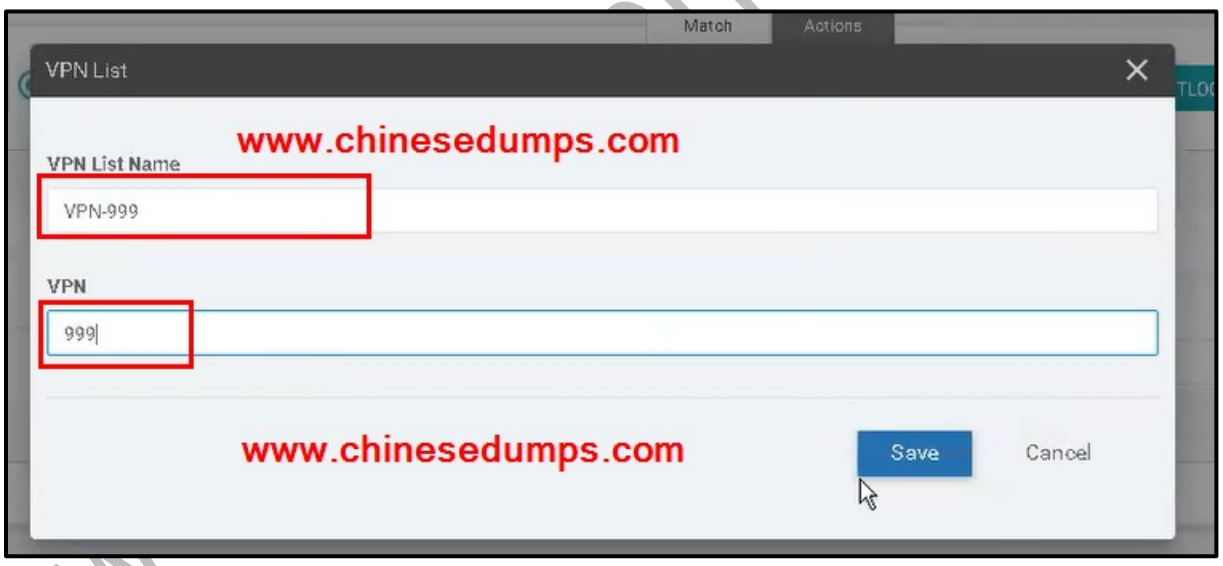
Select the **New VPN list** created

Go to **Actions** Tab

Select **Accept**

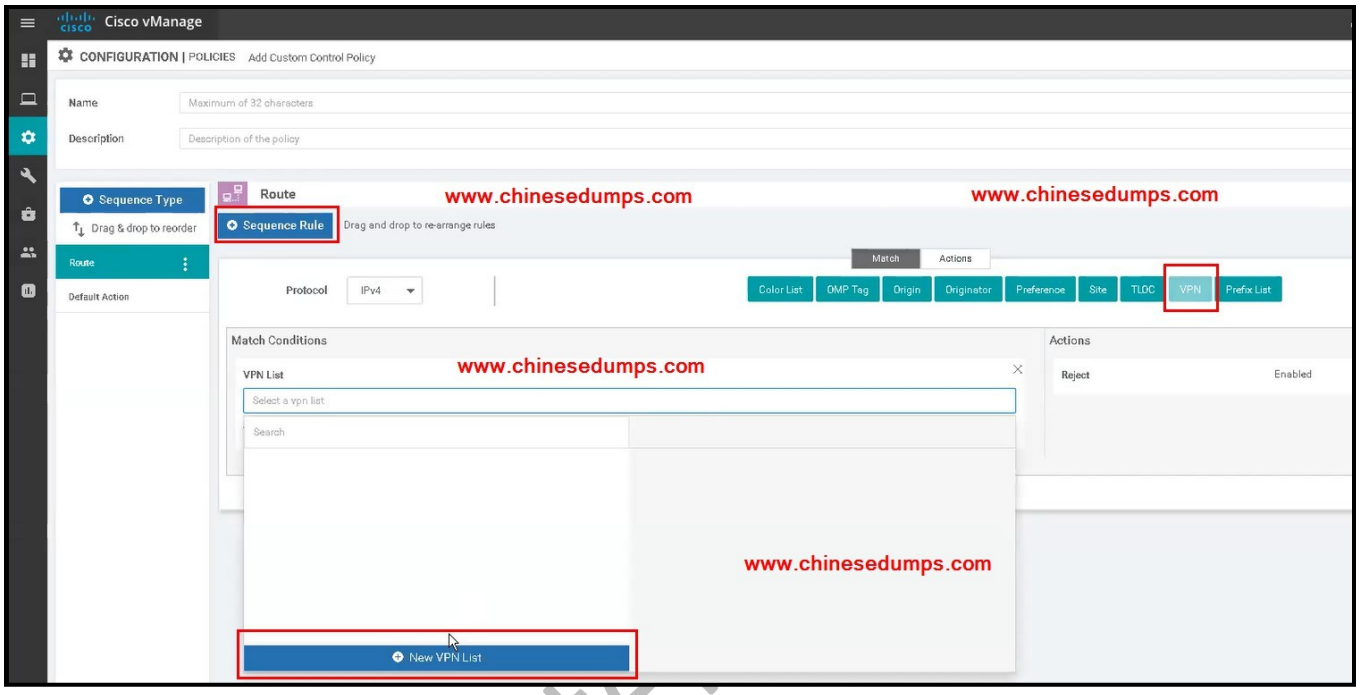
Select **Export** to and create a New VPN list VPN 999 (If not present already)

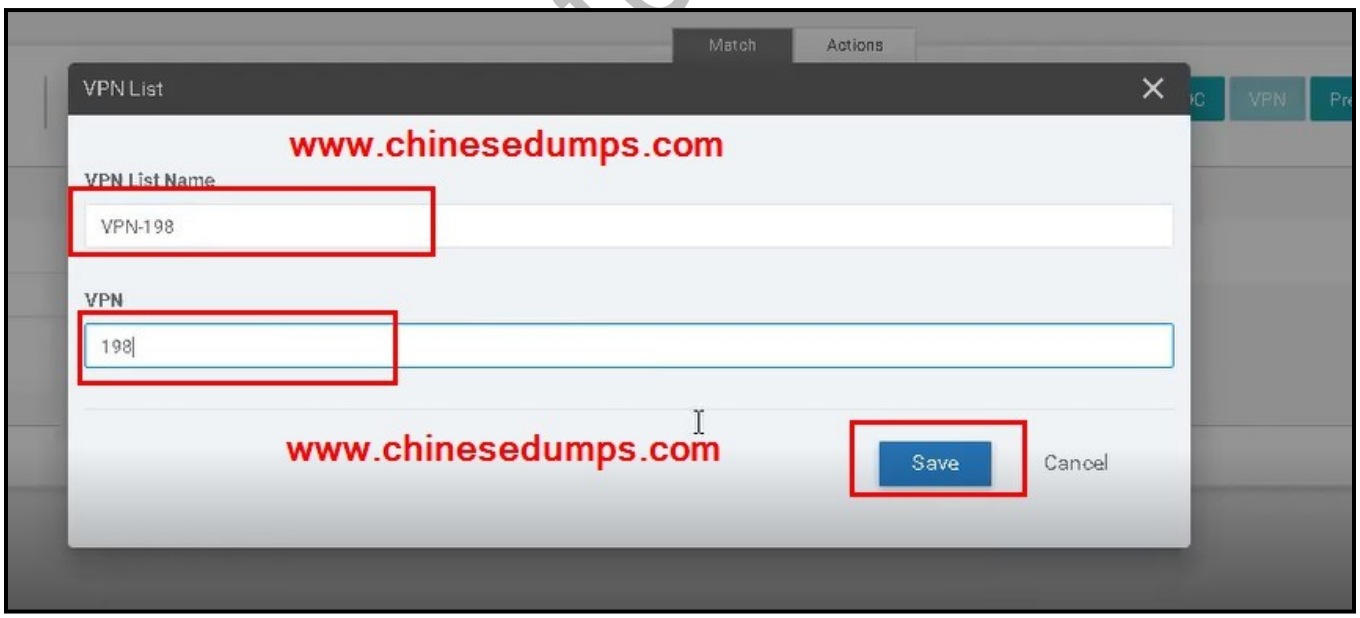




步骤7：

在匹配条件下选择VPN列表，并添加新的VPN列表





Step 9:

Select the VPN 999 in Export to

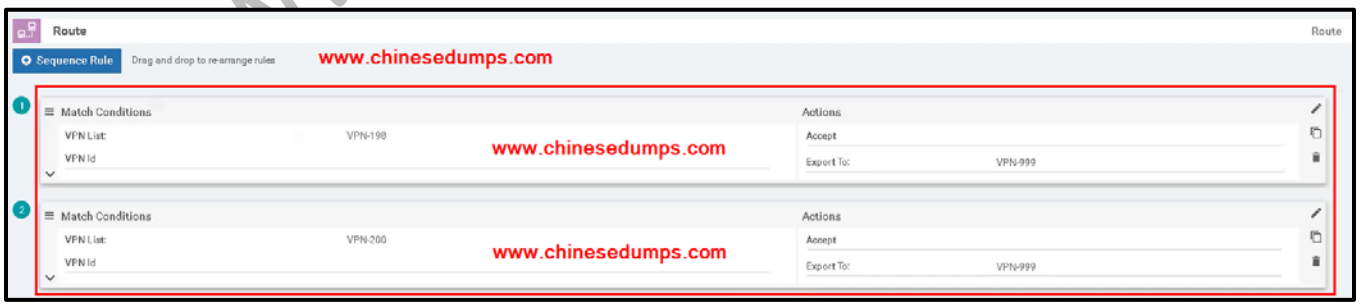
Click on **OMP Tag**

Add **OMP Tag as 198** and **Save Match and Actions**

Step 10:

Repeat step 7, Step 8, Step 9 for VPN 200

Just in OMP Tag select 200



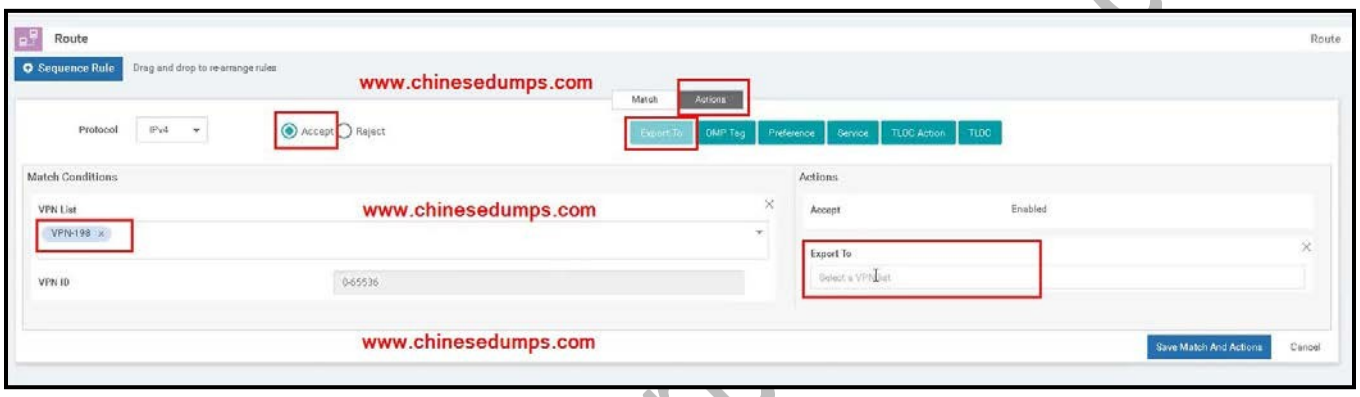
步骤8：

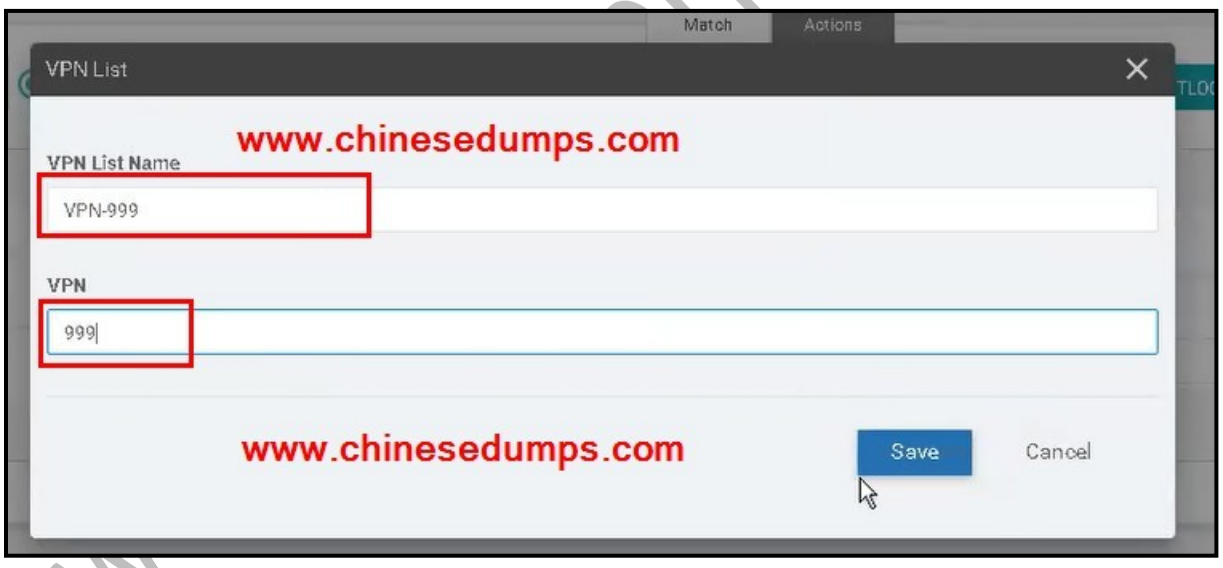
选择已创建的新VPN列表“转到

操作”选项卡

选择接受

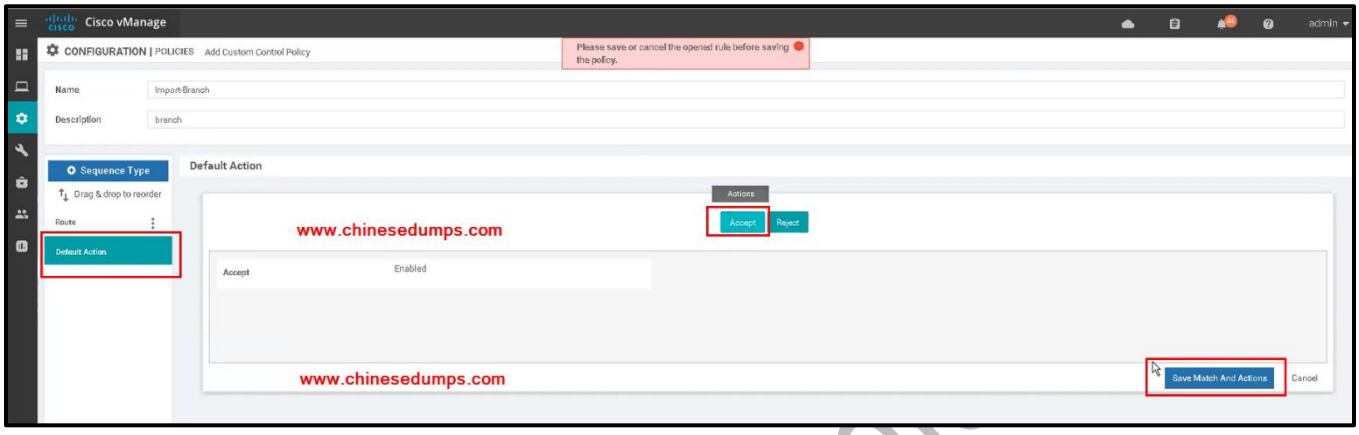
选择导出到并创建新的VPN列表VPN 999（如果尚未存在）





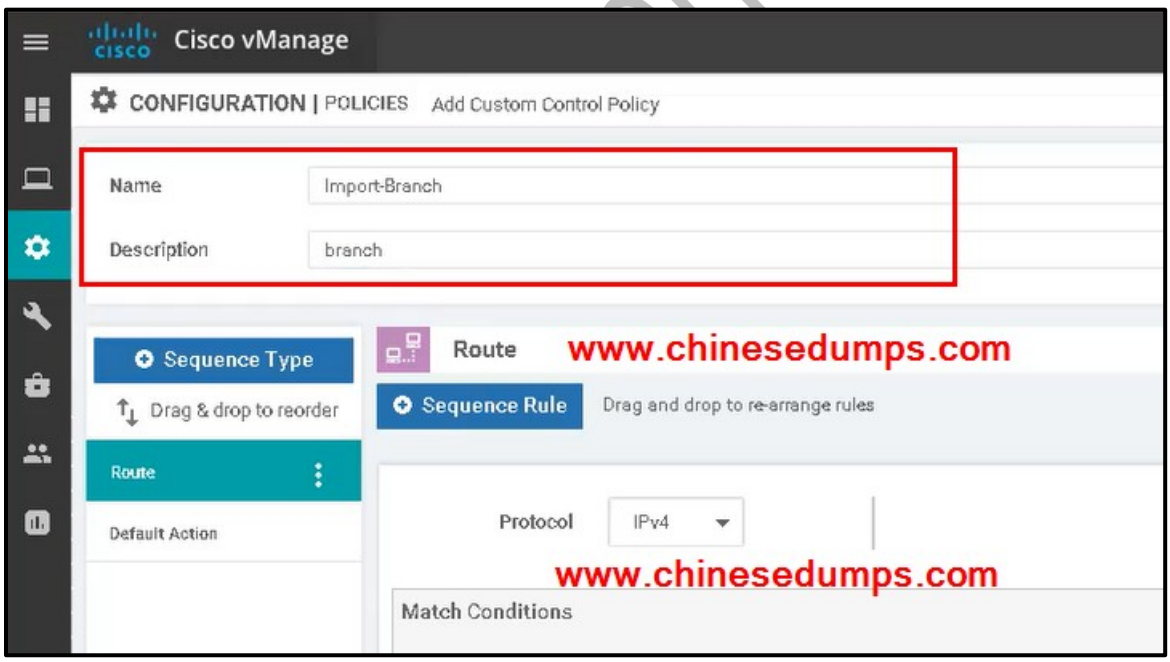
Step 11:

In Default Actions select Accept



Step 12:

Enter **Name** and **Description** and **Save Control Policy**

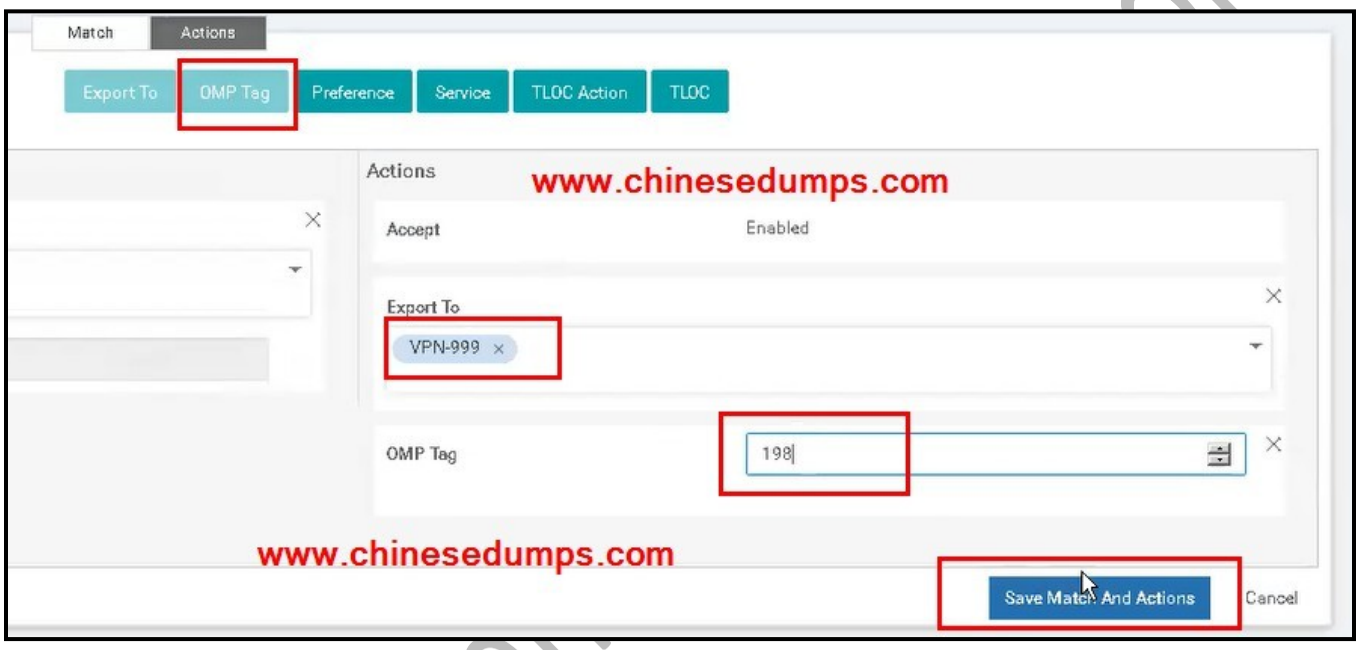


步骤9：

在导出中选择VPN 999，点击

OMP标签

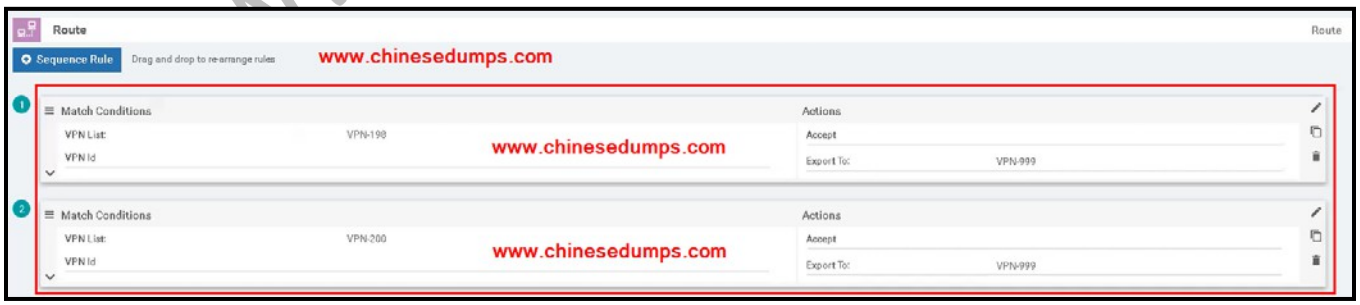
将OMP标记添加为198并保存匹配和操作



步骤10：

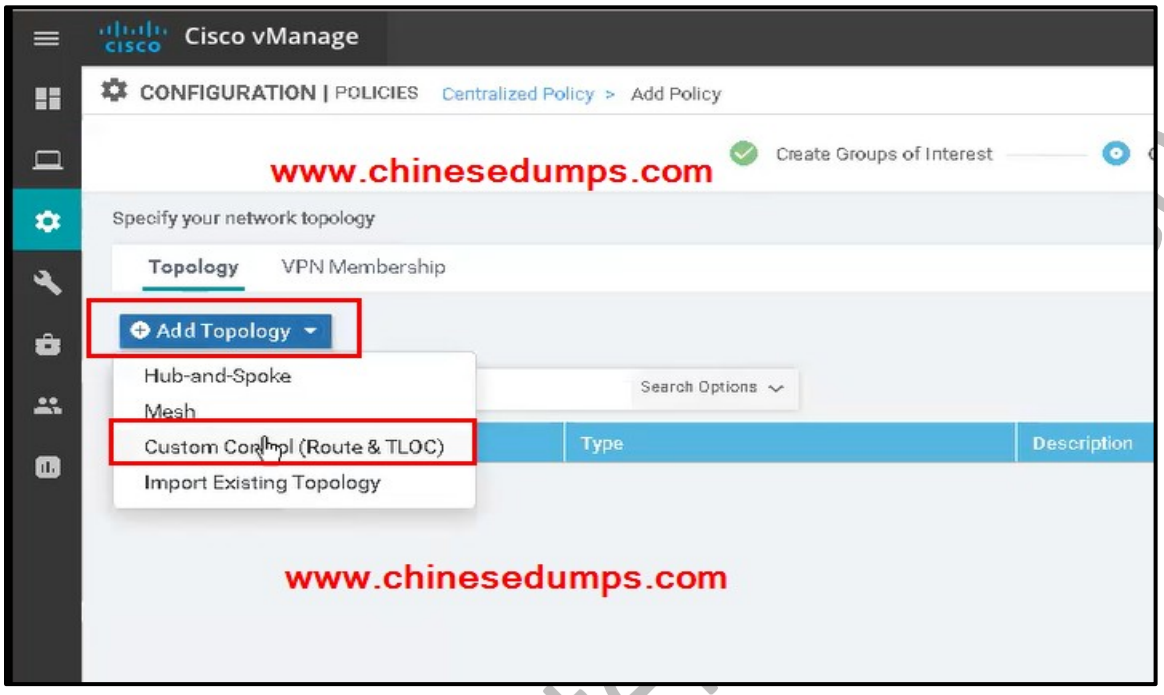
对VPN 200重复步骤7、步骤8、步骤9。

只需在OMP标签中选择200



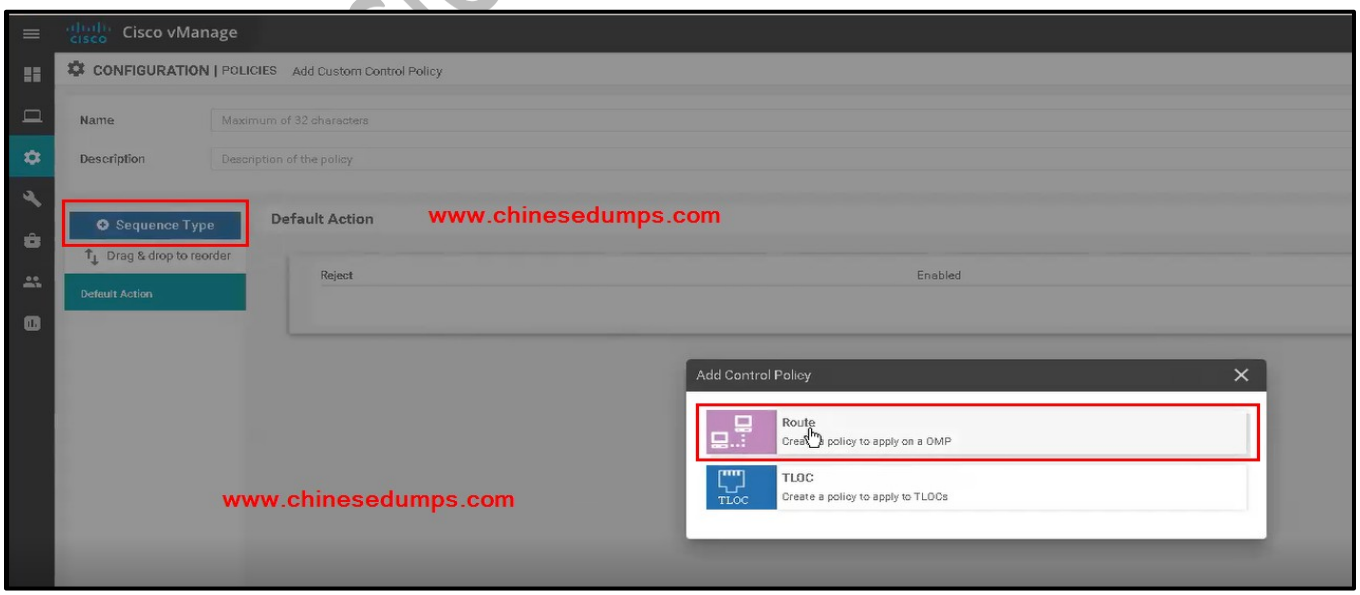
Step 13:

Click on Add **Topology** and select **Custom Control** to add another Policy



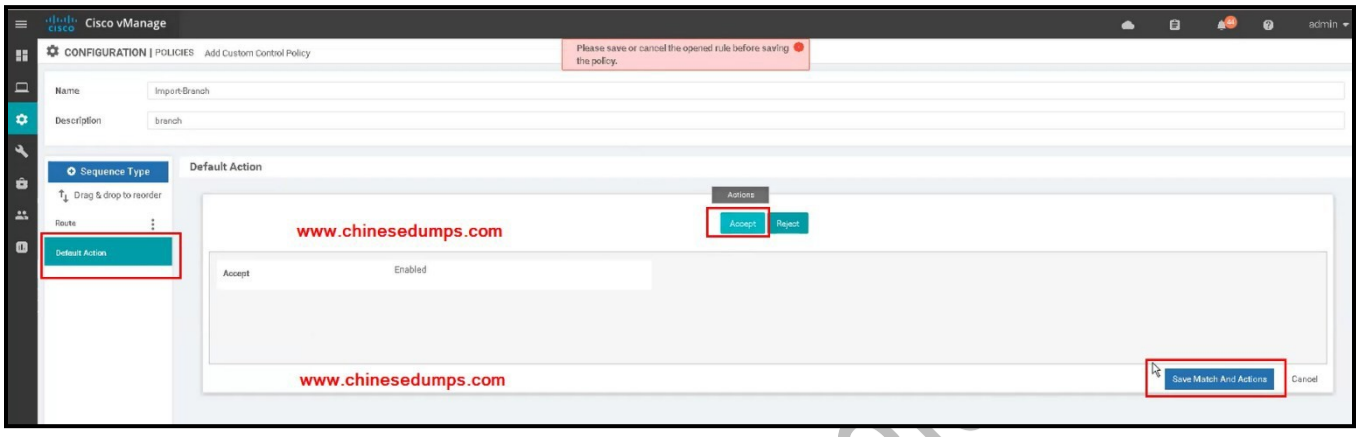
Step 14:

Add a **Sequence Type** and add a **Route Control Policy**



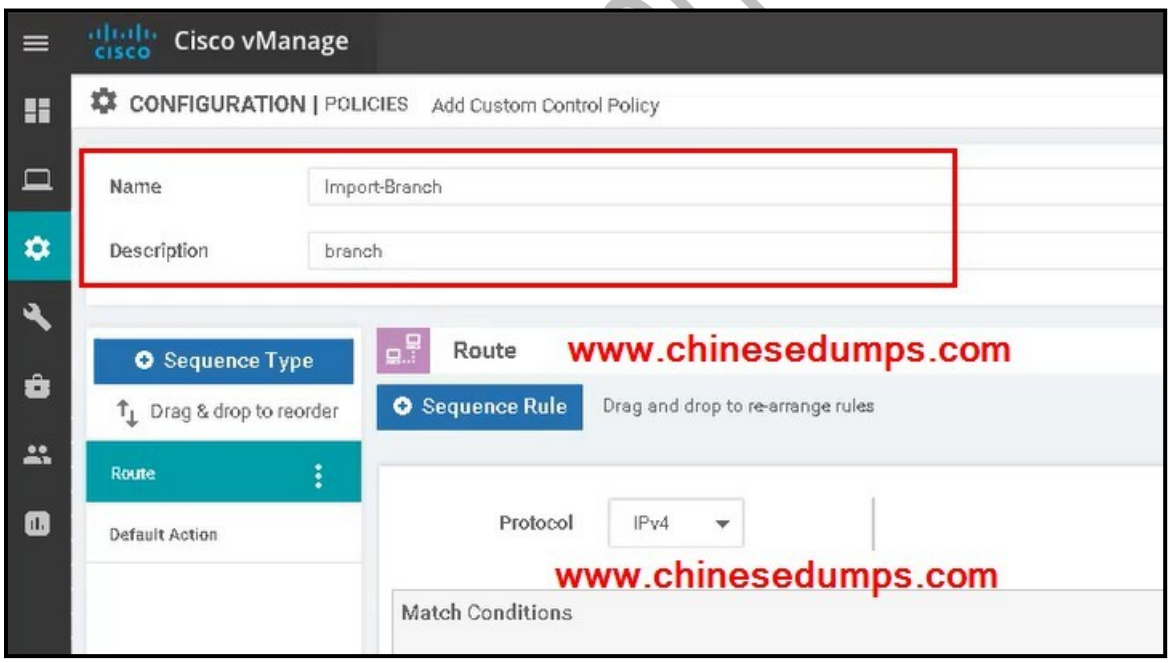
步骤11：

在默认操作中选择接受



步骤12：

输入名称和描述并保存控制策略

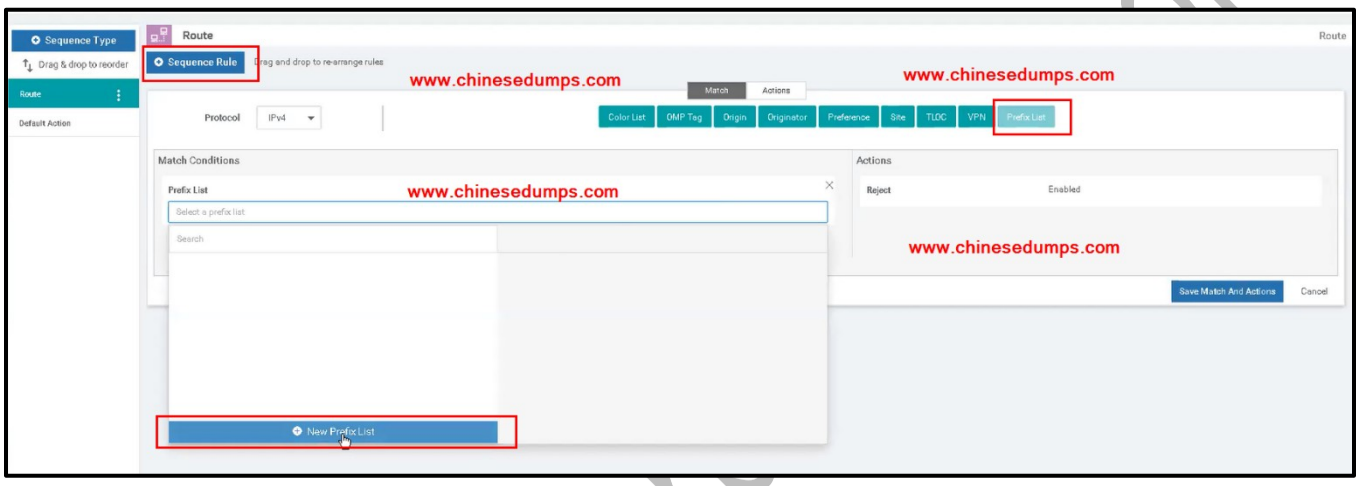


Step 15:

Add **New Sequence**

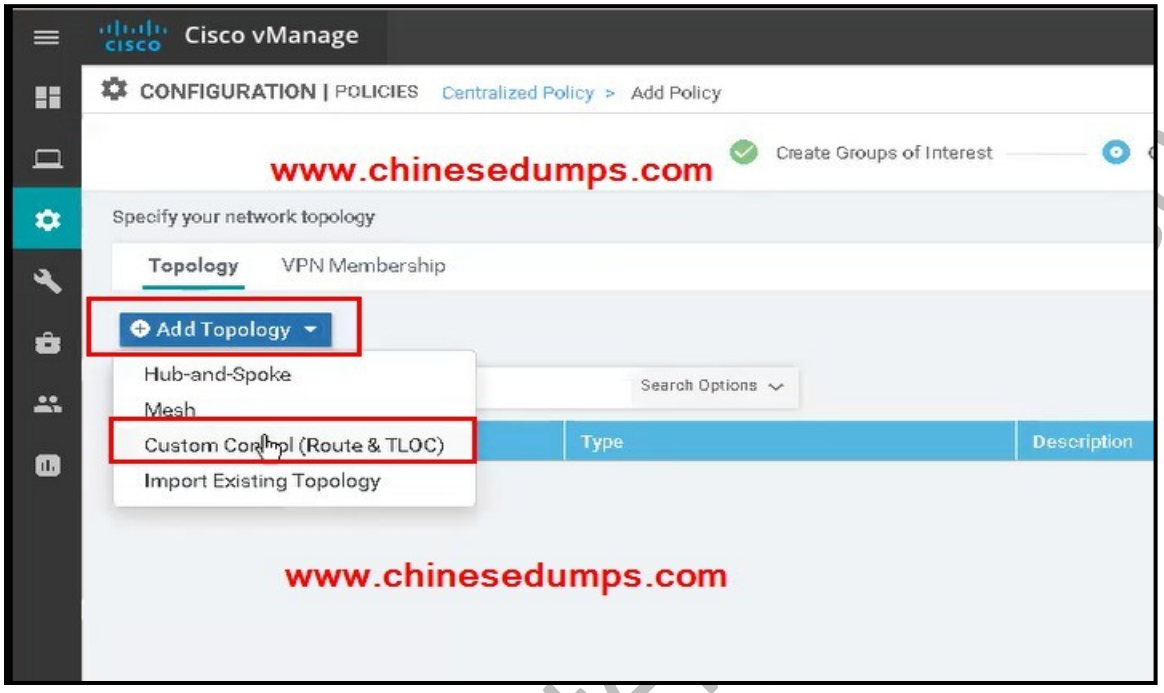
Select **Prefix-list**

Add **New-Prefix-list**



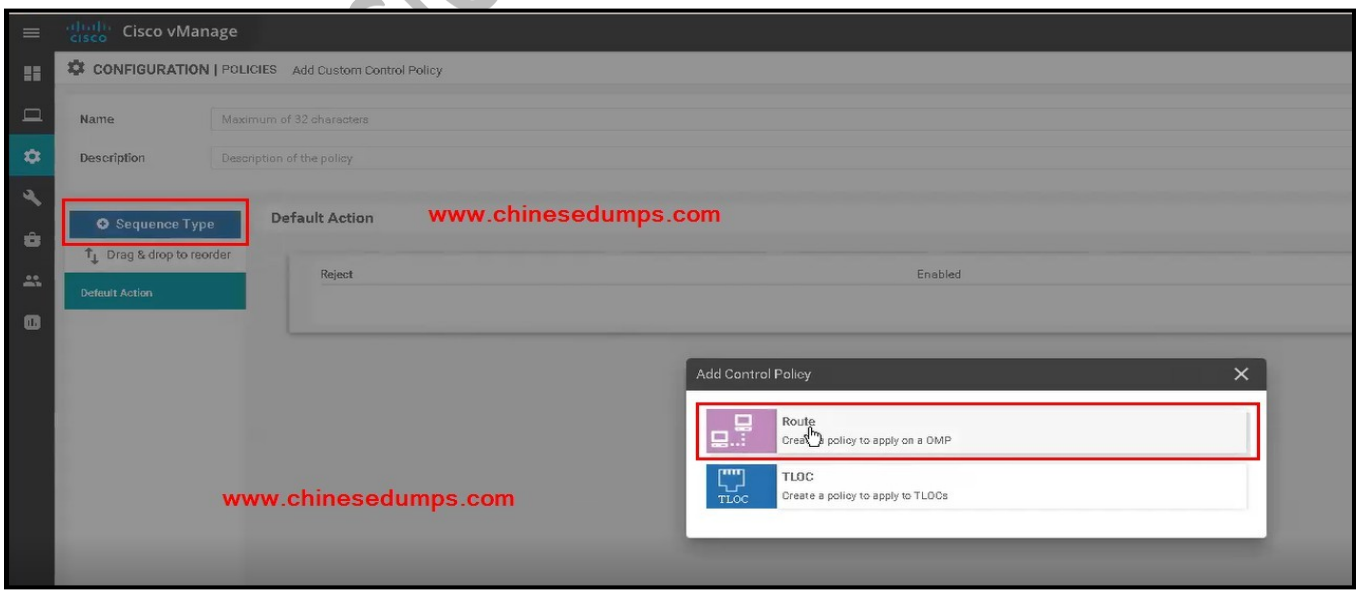
步骤13：

单击添加拓扑，然后选择自定义控件以添加另一个策略

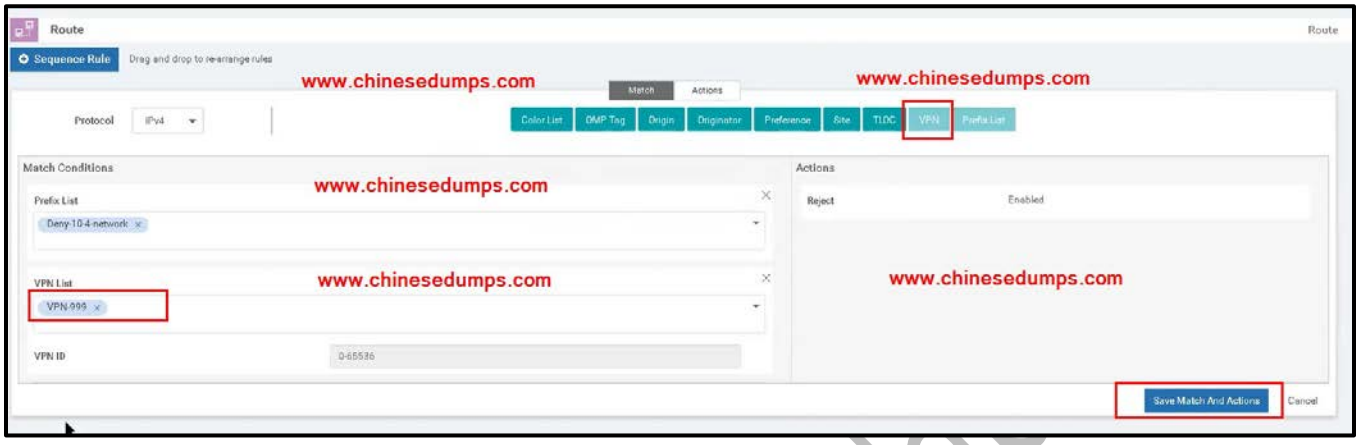


步骤14：

添加序列类型并添加路由控制策略



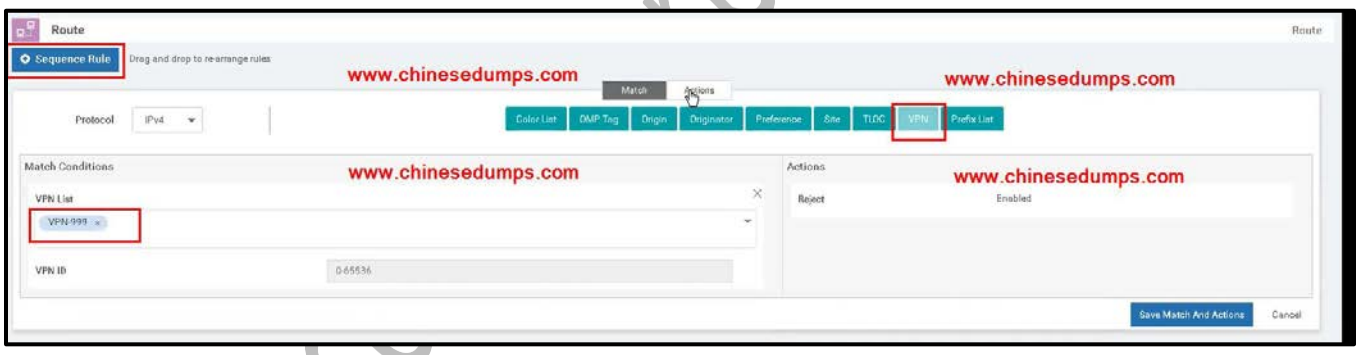
Click on VPN and select VPN 999



Step 16:

Add another Sequence Rule for VPN 999

Select VPN 999 from VPN list



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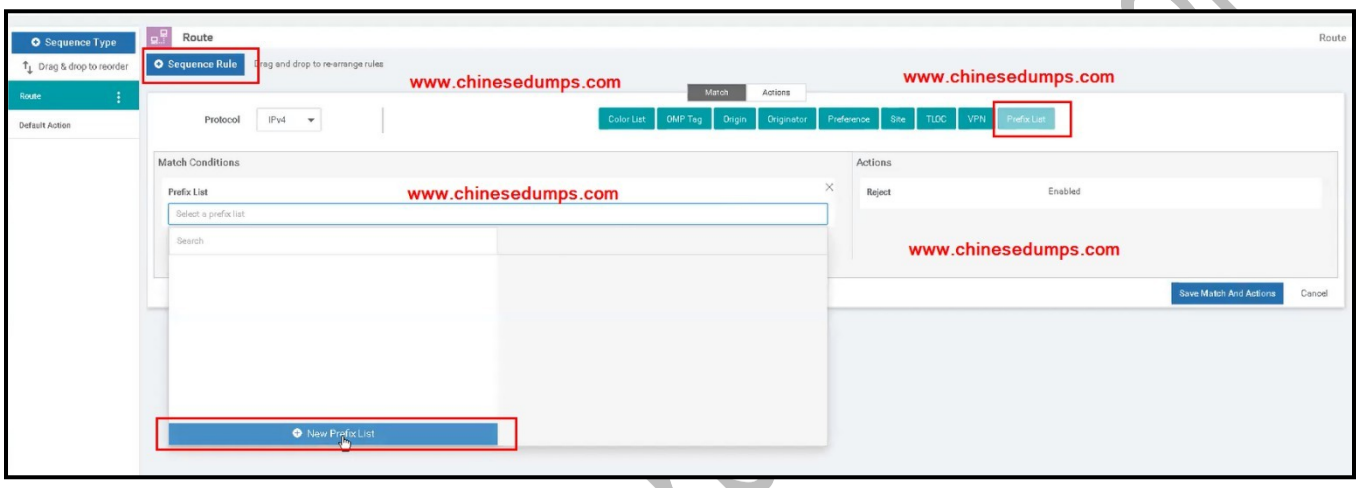
步骤15：

添加新序列

选择前缀列表

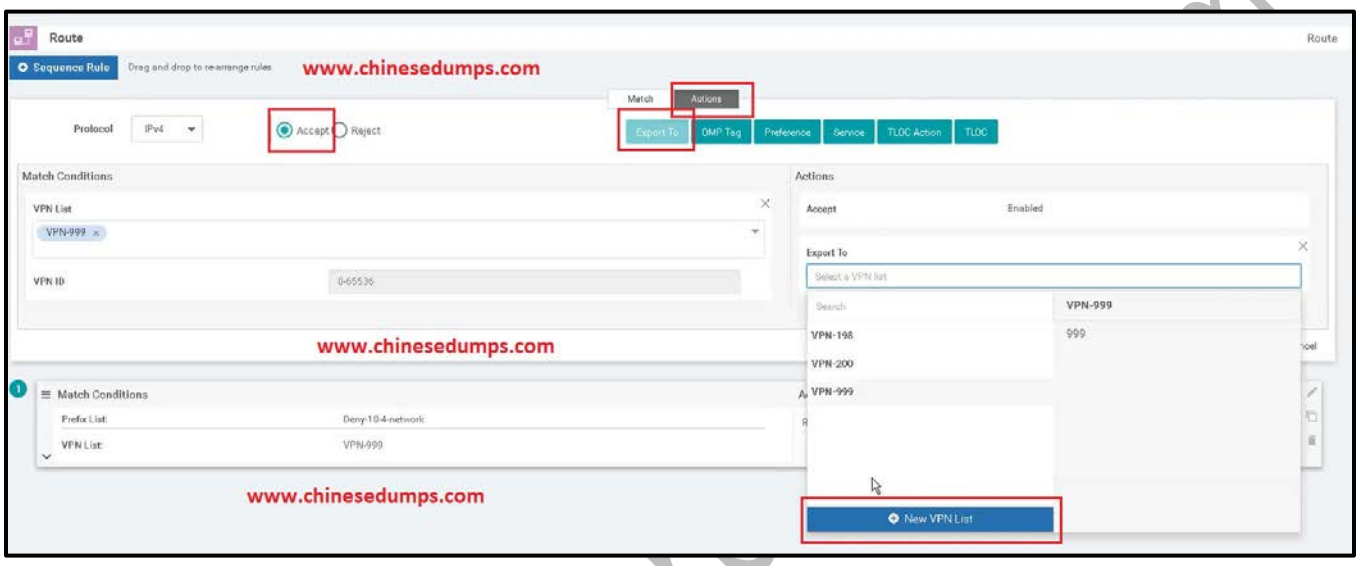
添加新前缀列表

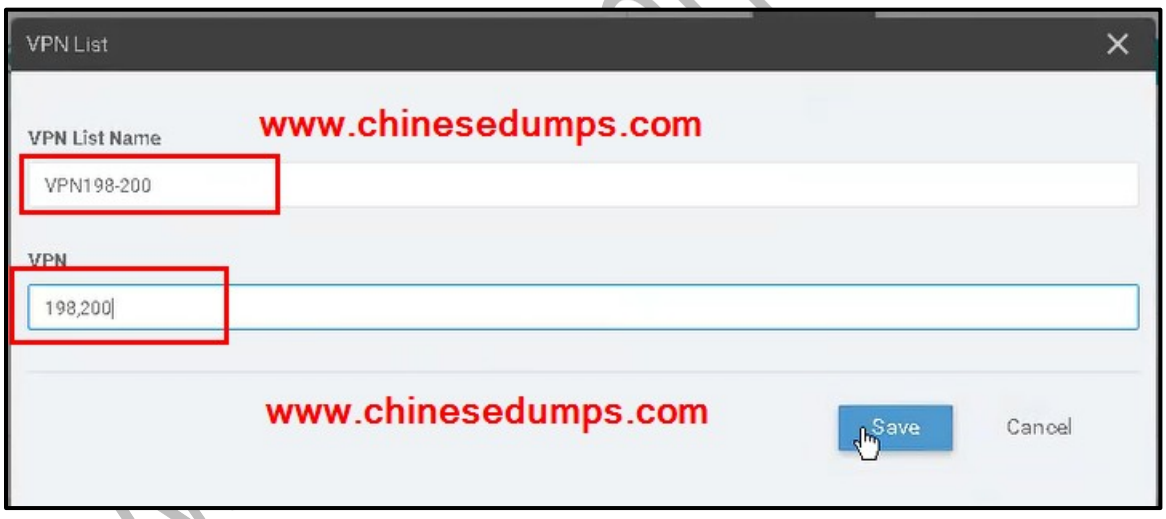


Go to Actions

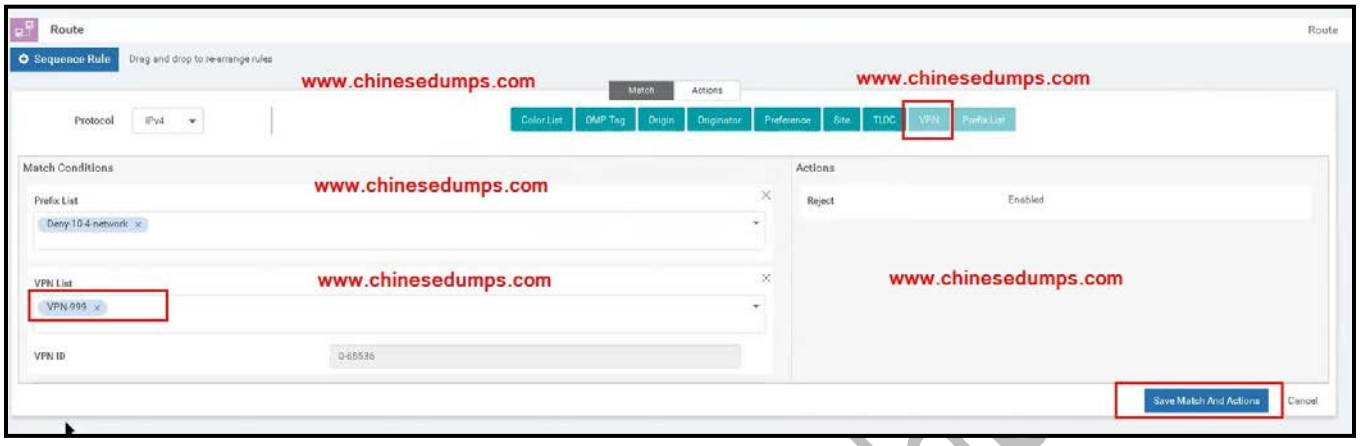
Select Accept

Select Export To and add a NEW VPN LIST





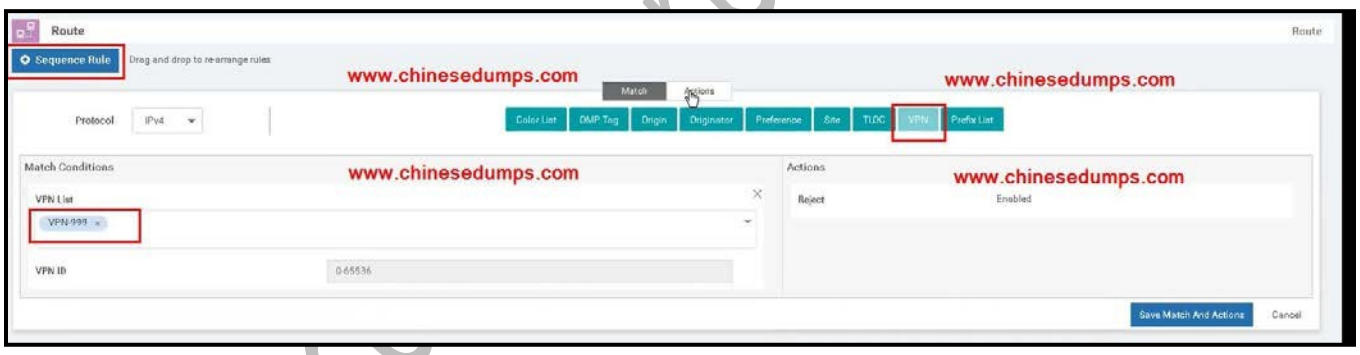
点击VPN并选择VPN 999



步骤16：

为VPN 999添加另一个序列规则从VPN列

表中选择VPN 999



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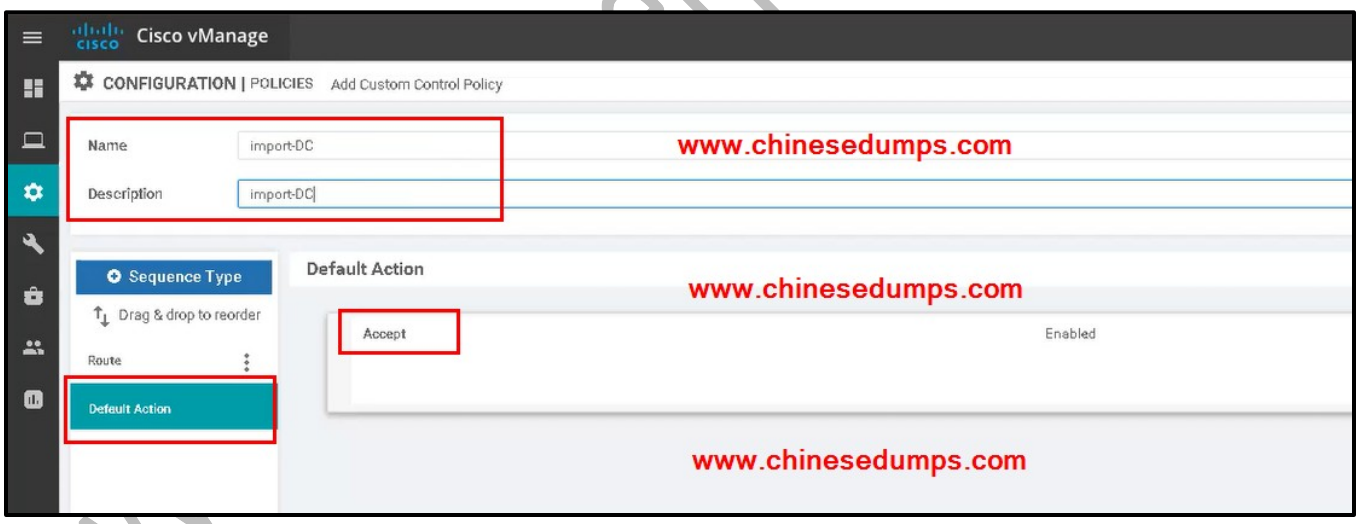
Select VPN list 198-200 under Export To and Save Match and Actions

Step 17:

In Default Actions select Accept

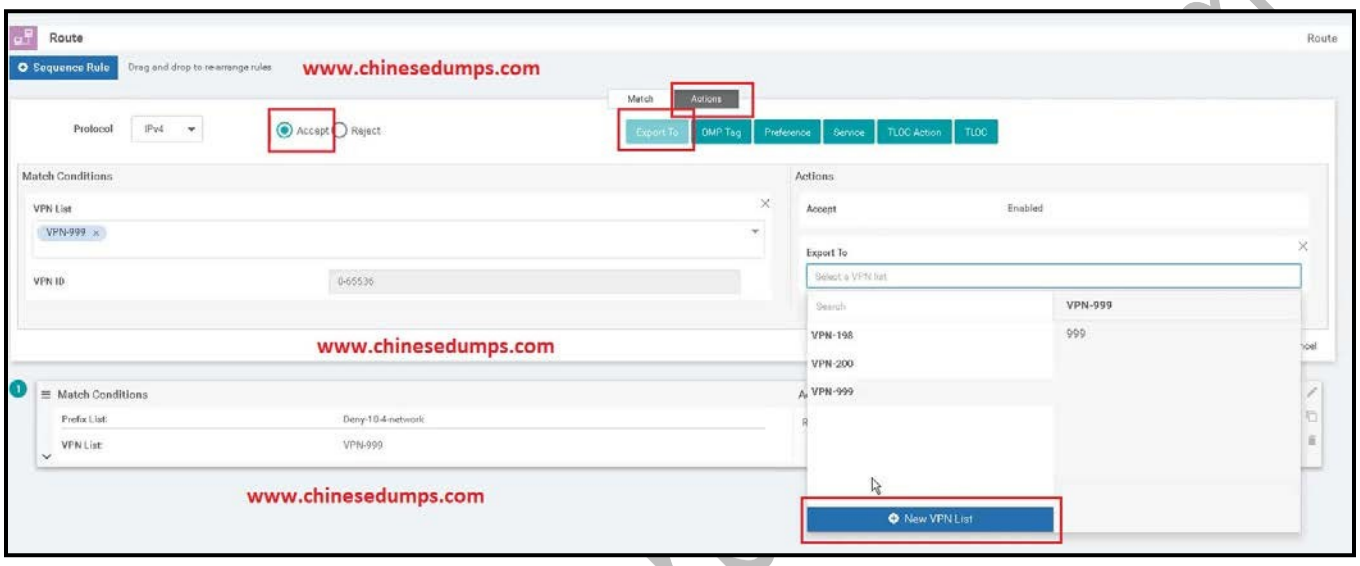


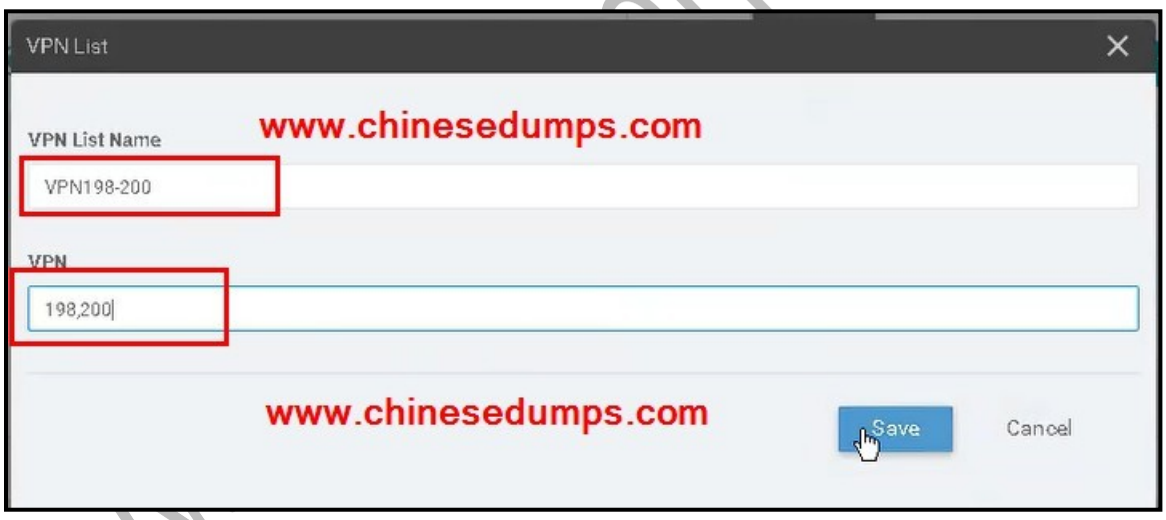
Also Give Name and Description

转到操作选择

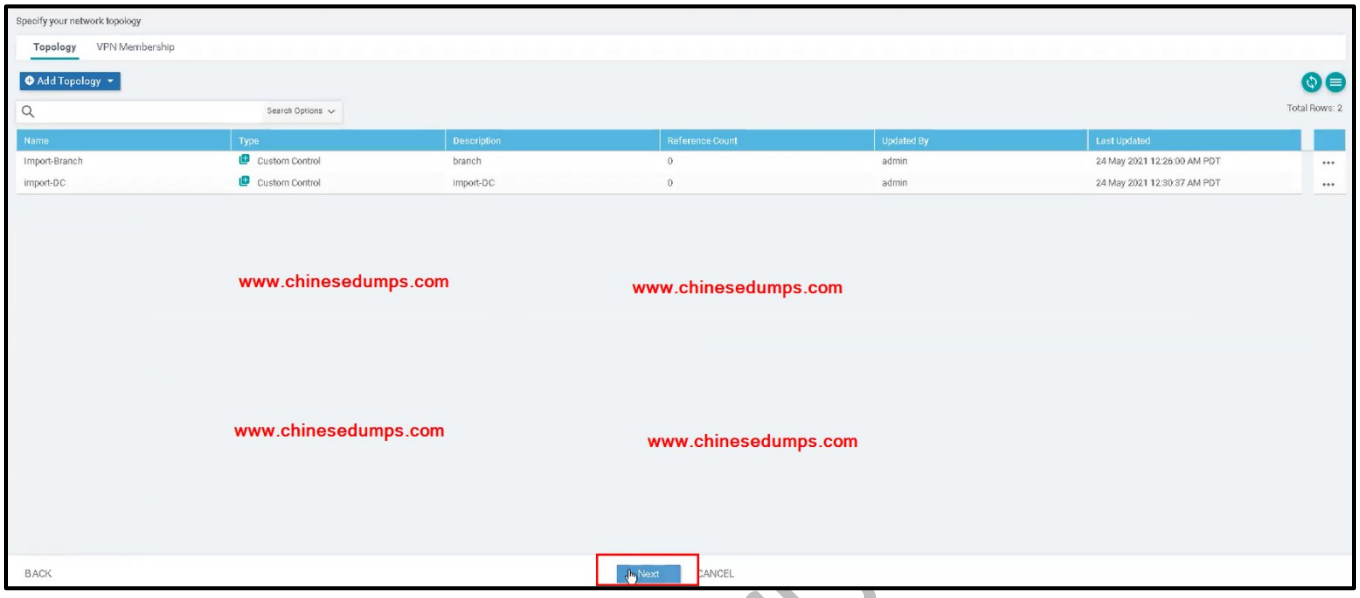
接受

选择导出到并添加新的VPN列表





After adding Hit next 2 times



Step 18:

Under Add Policy

Policy Name

Description

Under **Import-DC**

In **New Site List** in **Outbound List** Select **Branch 1** and **Branch 2**

In **New Site List** in **Inbound List** Select **DC**

Under **Import-Branch**

In **New Site List** in **Inbound List** Select **Branch 1** and **Branch 2**

In **New Site List** in **Outbound List** Select **DC**

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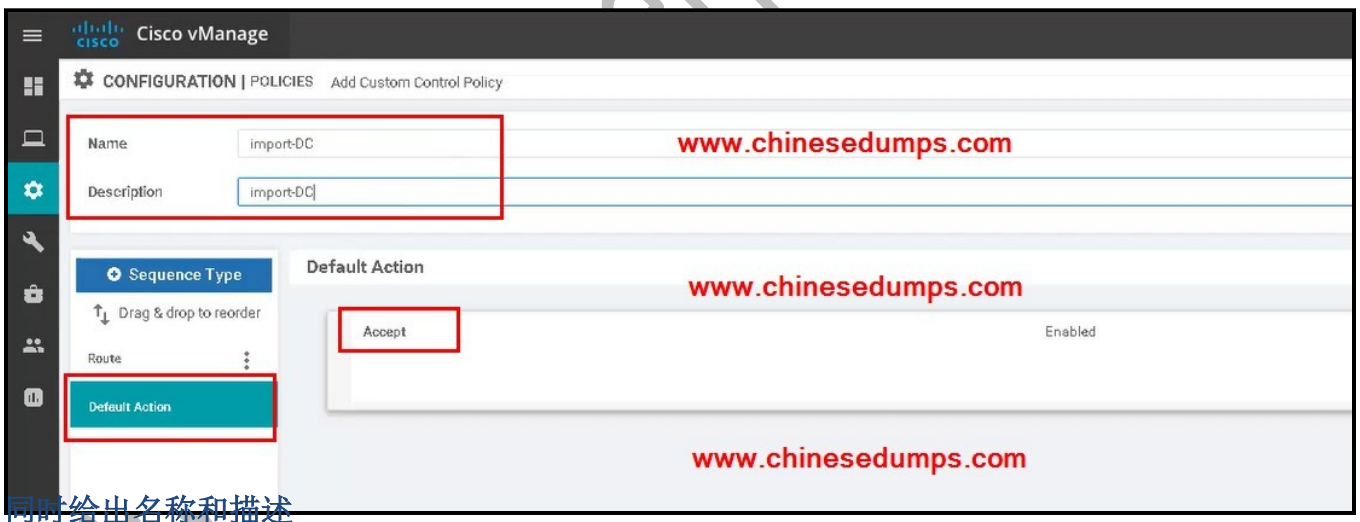
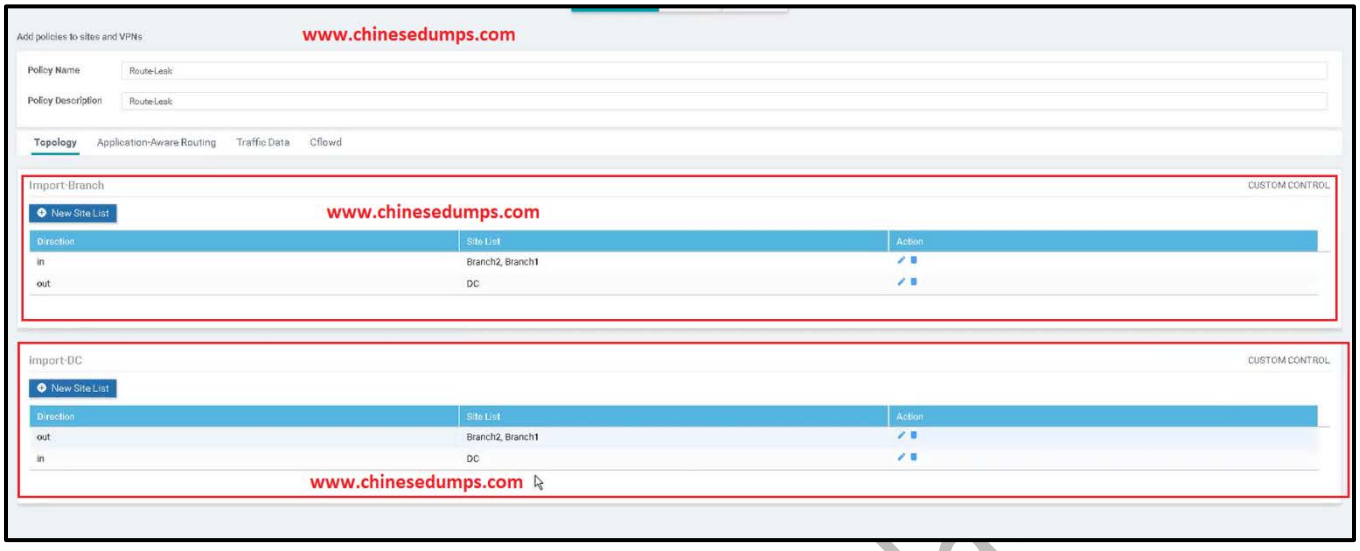
在导出到并保存匹配和操作步骤17下选择VPN列表198-200：

在默认操作中选择接受



同时给出名称和描述

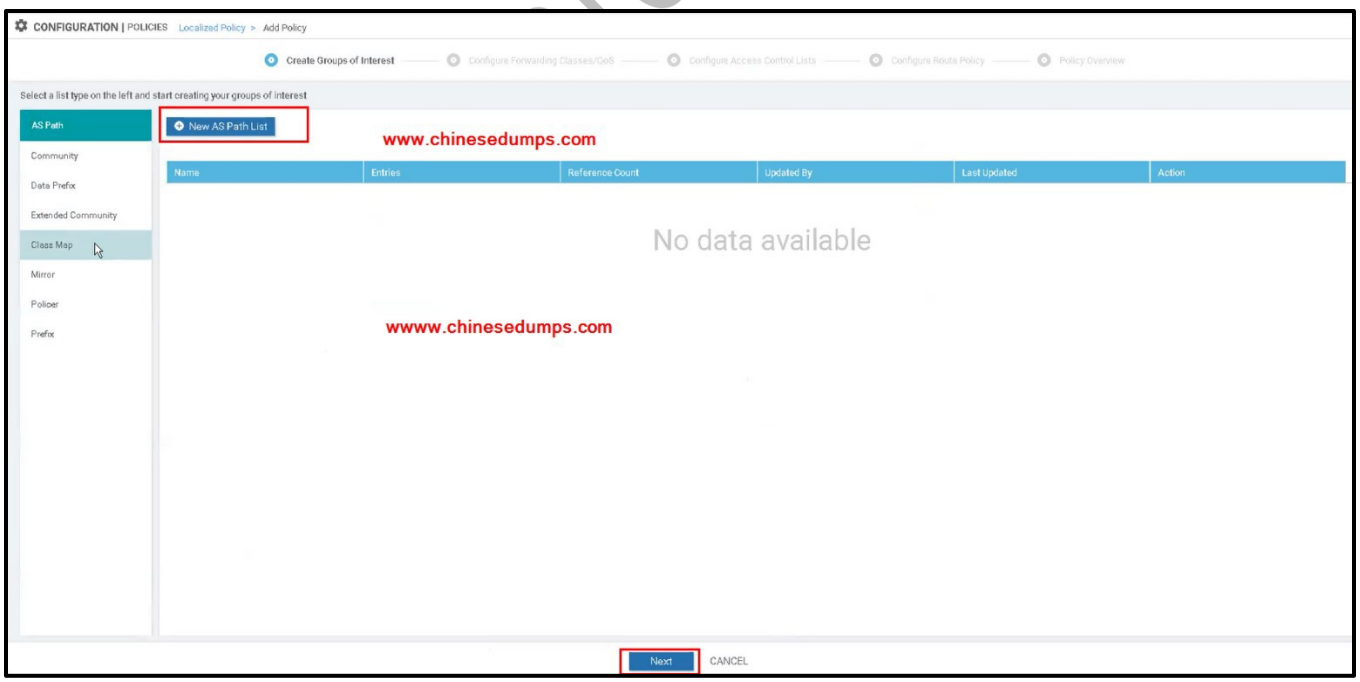


Also Add **New Site-list** as **DC**

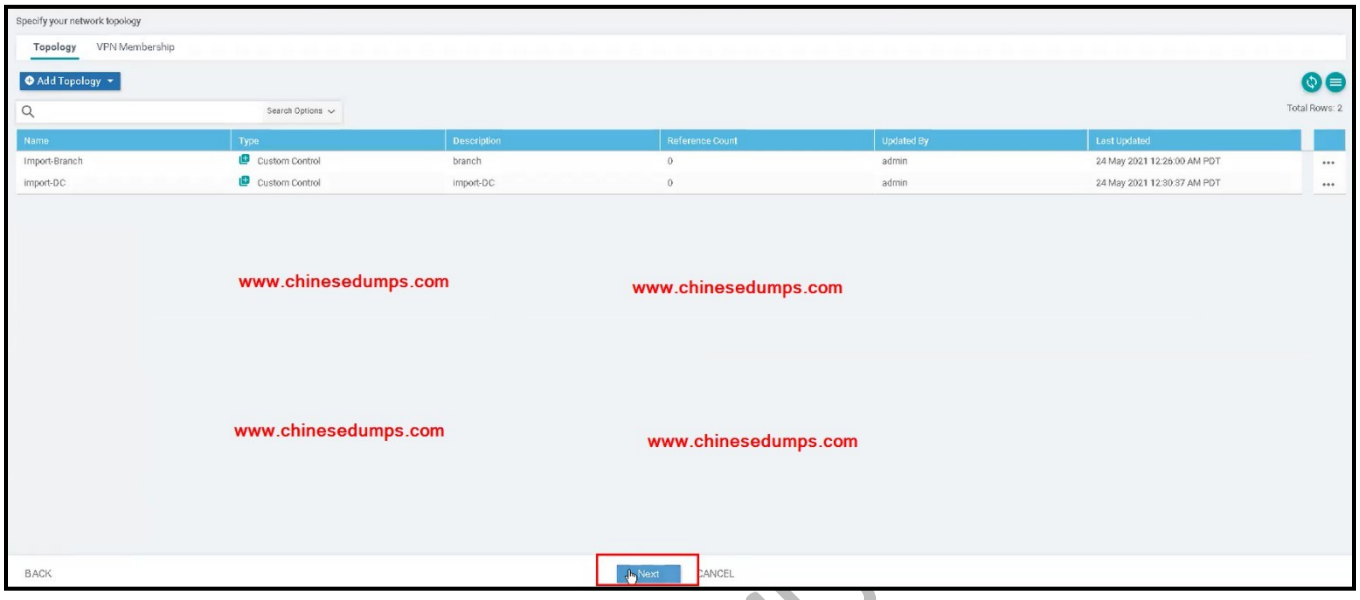
Then Save Policy

Step 19:

Create a Localized Policy



添加Hit next 2次后



步骤18：

在“导入DC”下的

“添加策略名称

描述”下

在出站列表中的新站点列表中，选择分支1和分支2

在入站列表中的新站点列表中选择DC

进口分行下

在入站列表中的新站点列表中，选择分支1和分支2

在出站列表中的新站点列表中选择DC

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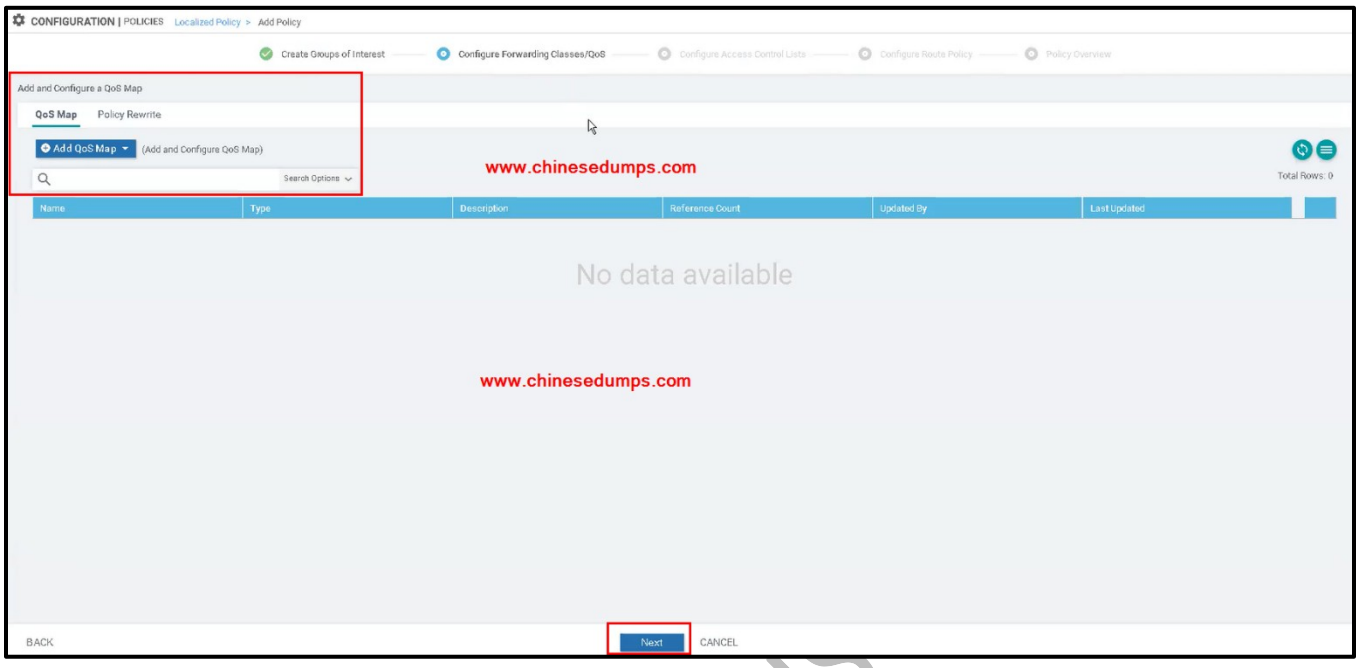

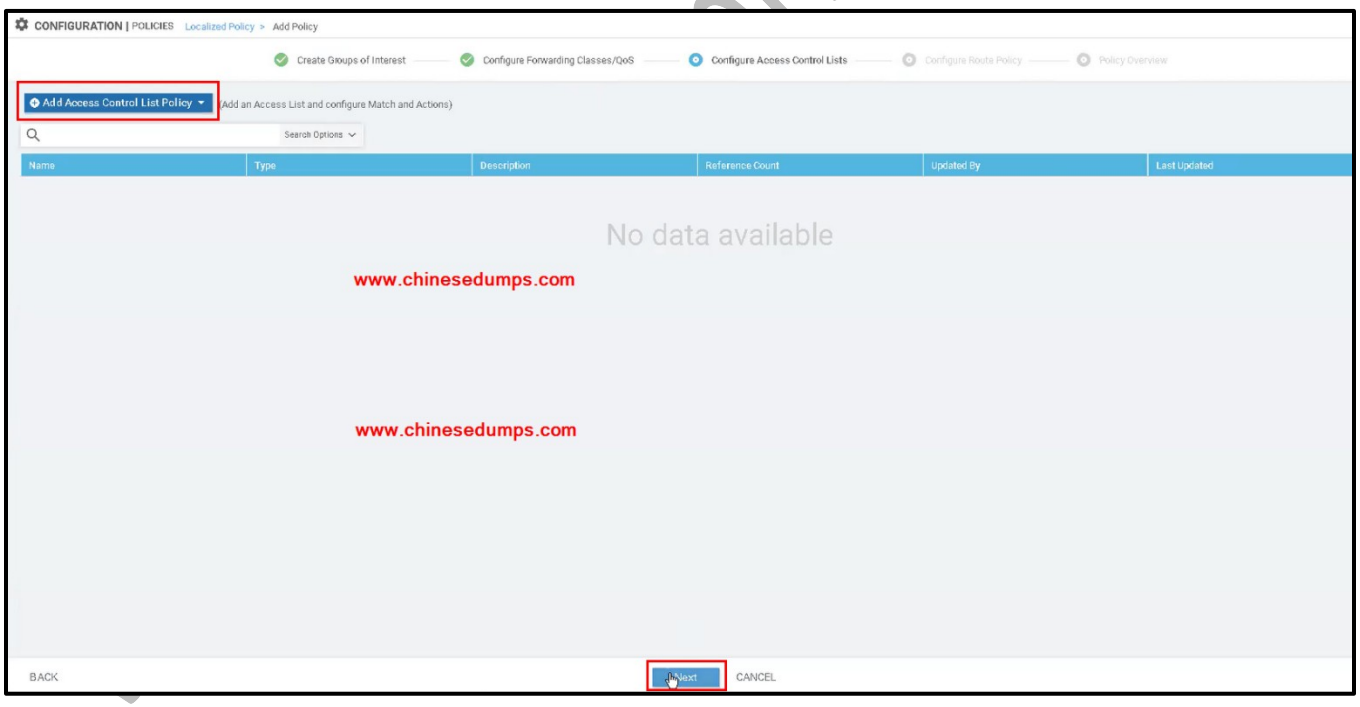

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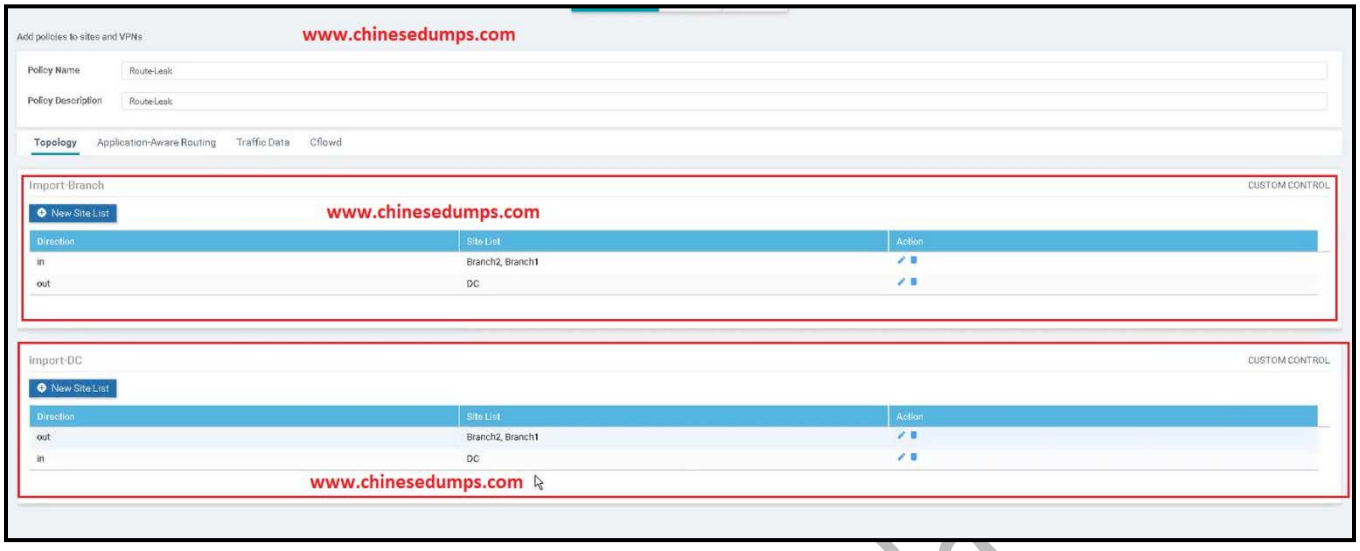
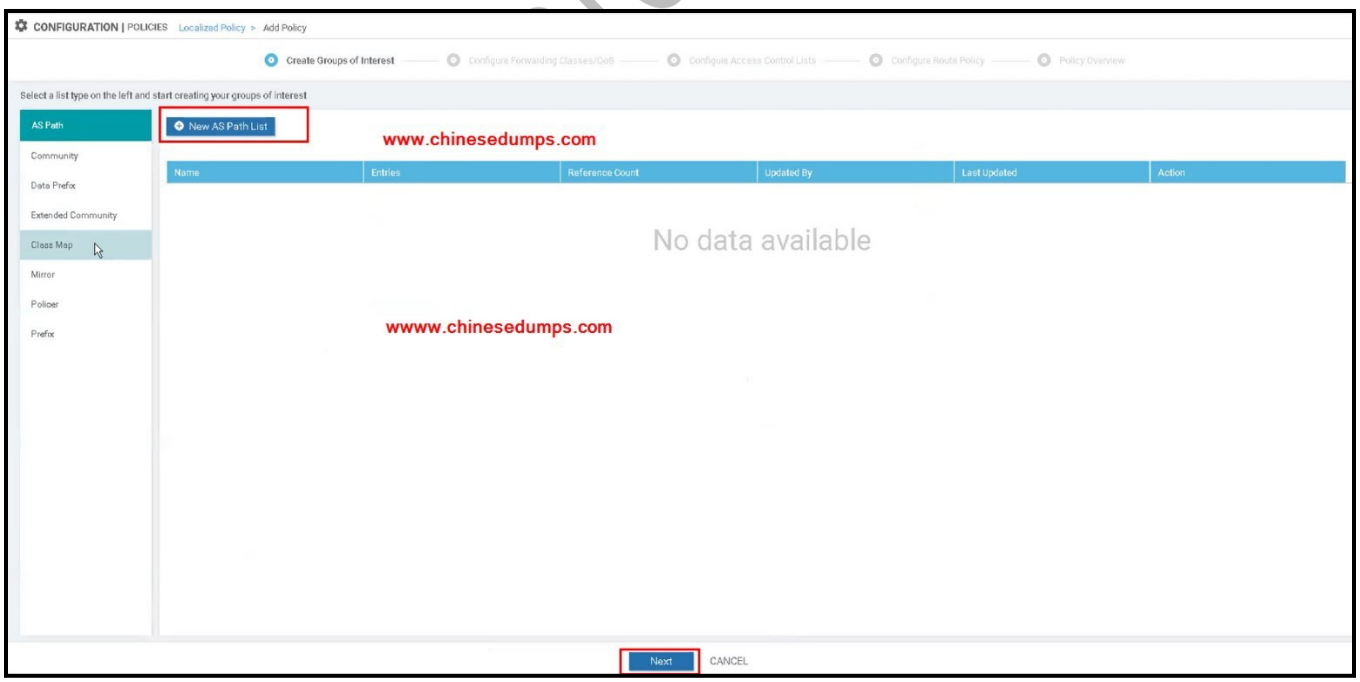


还将新站点列表添加为DC，然

后保存策略

步骤19：

创建本地化策略



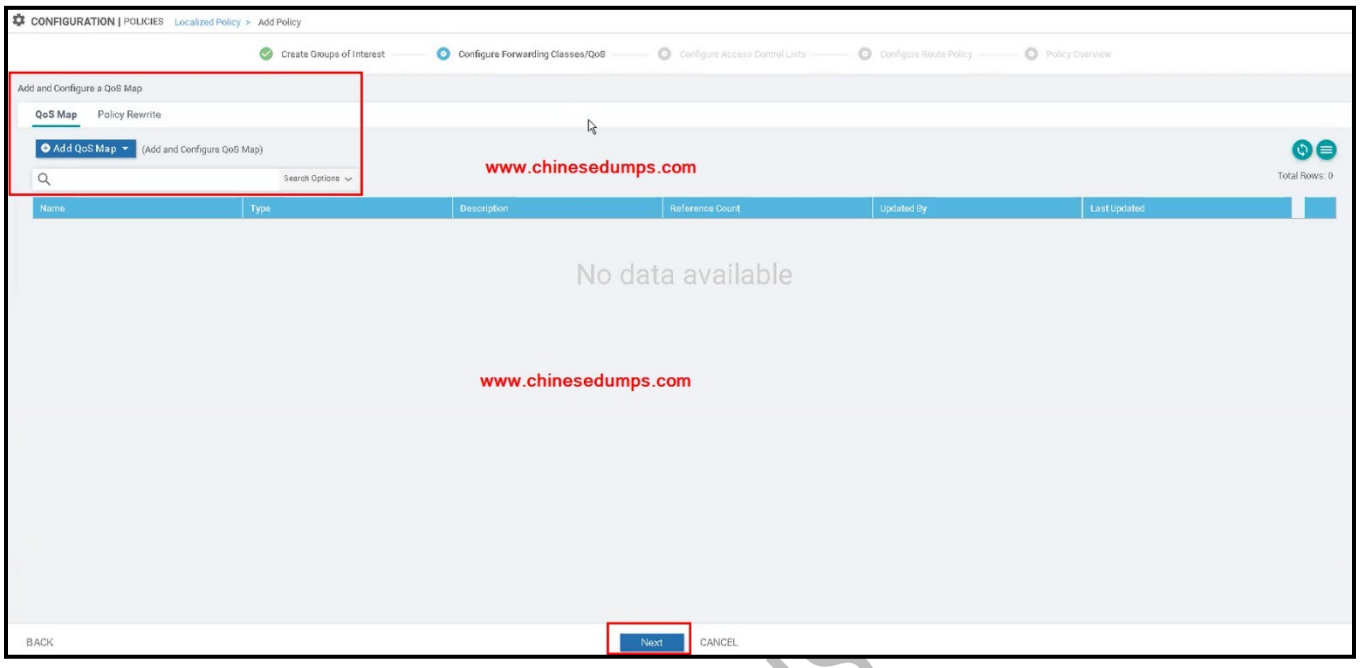
Click on Add Route Policy

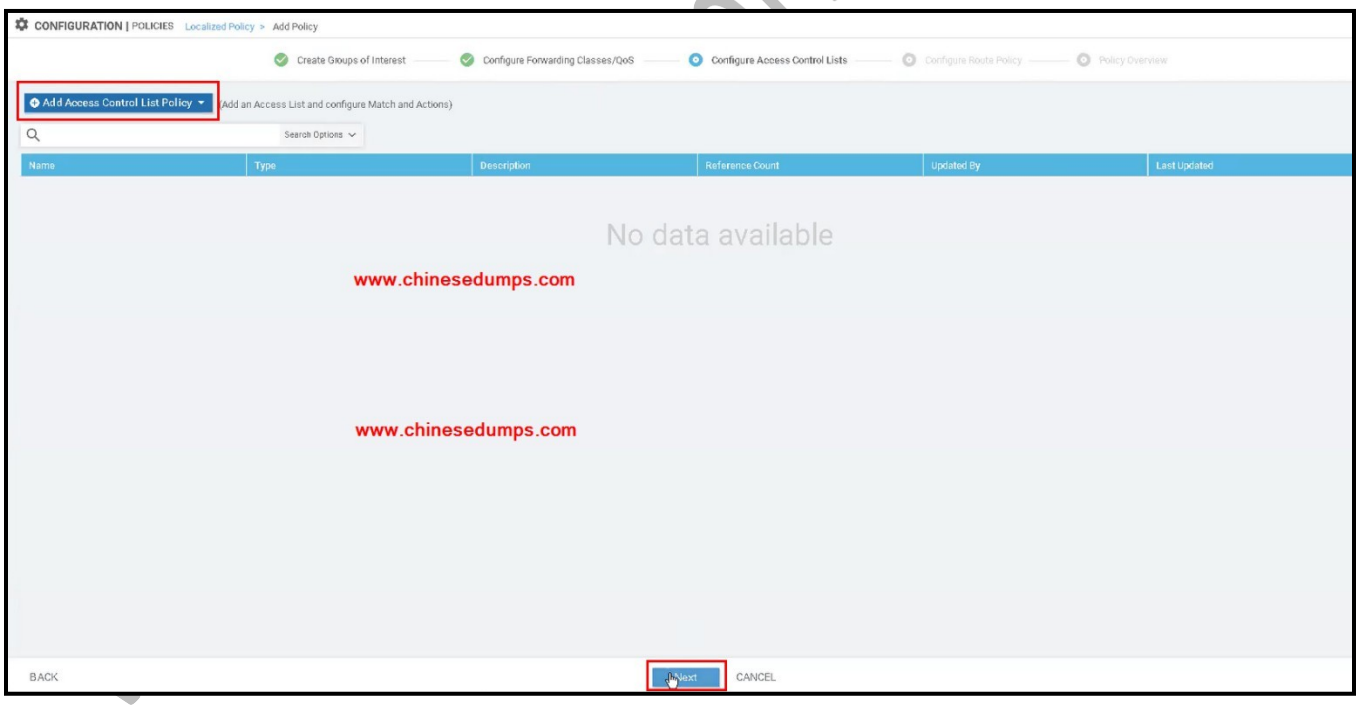
Add Name of the Policy

Add Description

Create Sequence Rule for 198 and 200 as below

In Default Action select Accept





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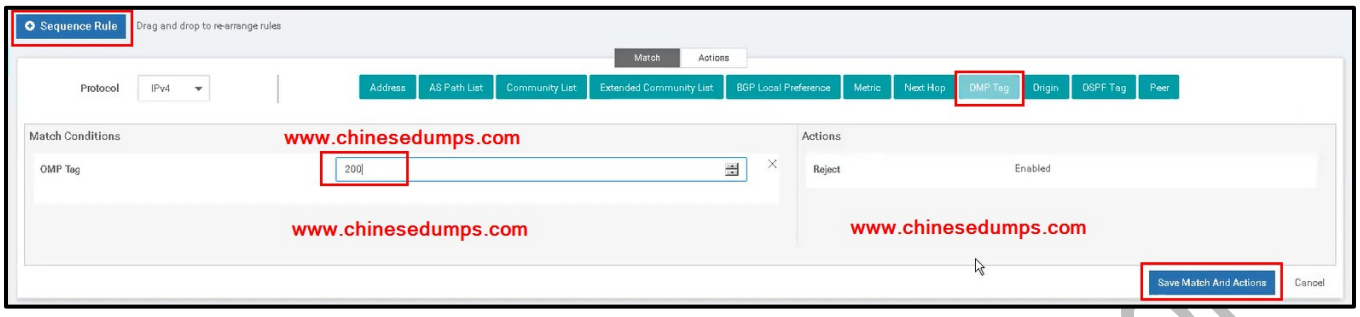
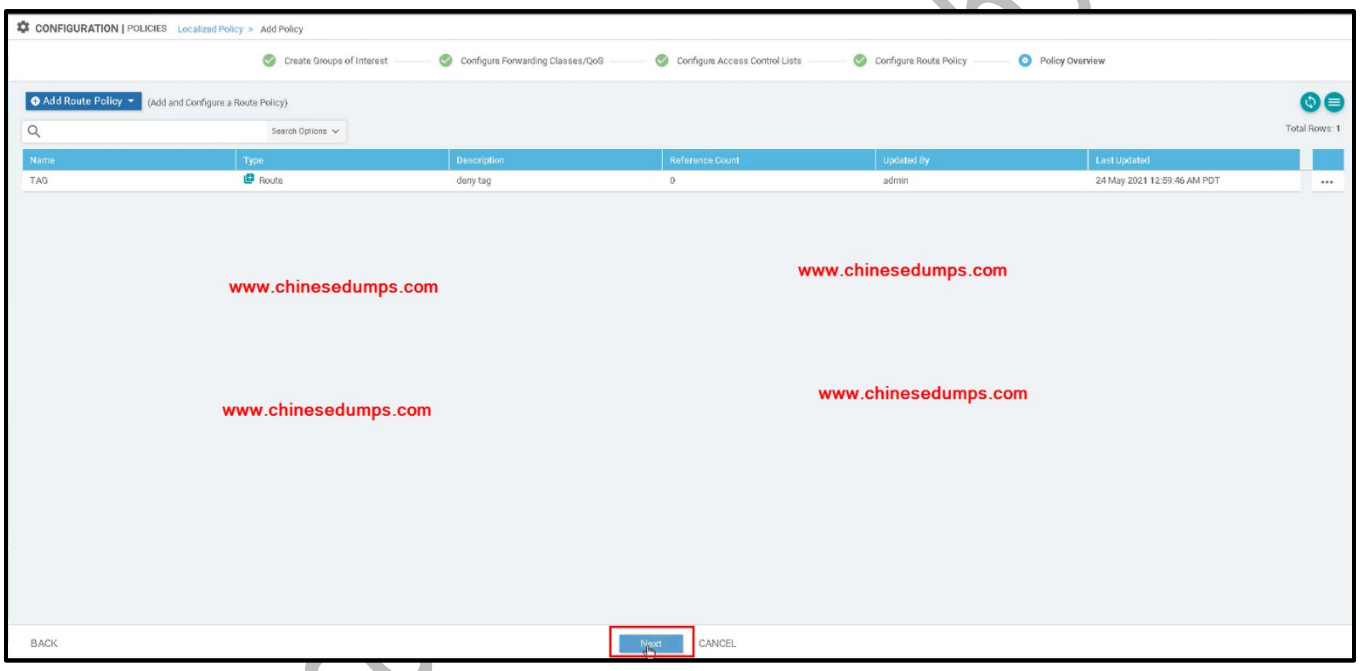


Hit NEXT



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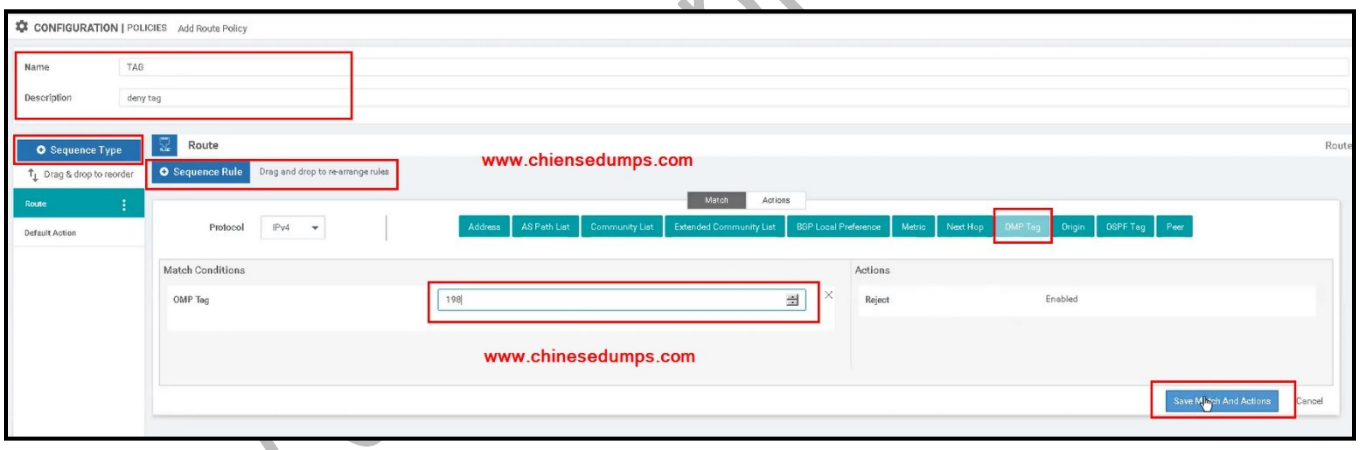
点击添加路线策略

添加策略名称添加说

明

为198和200创建序列规则，如下所示



在默认操作中选择接受

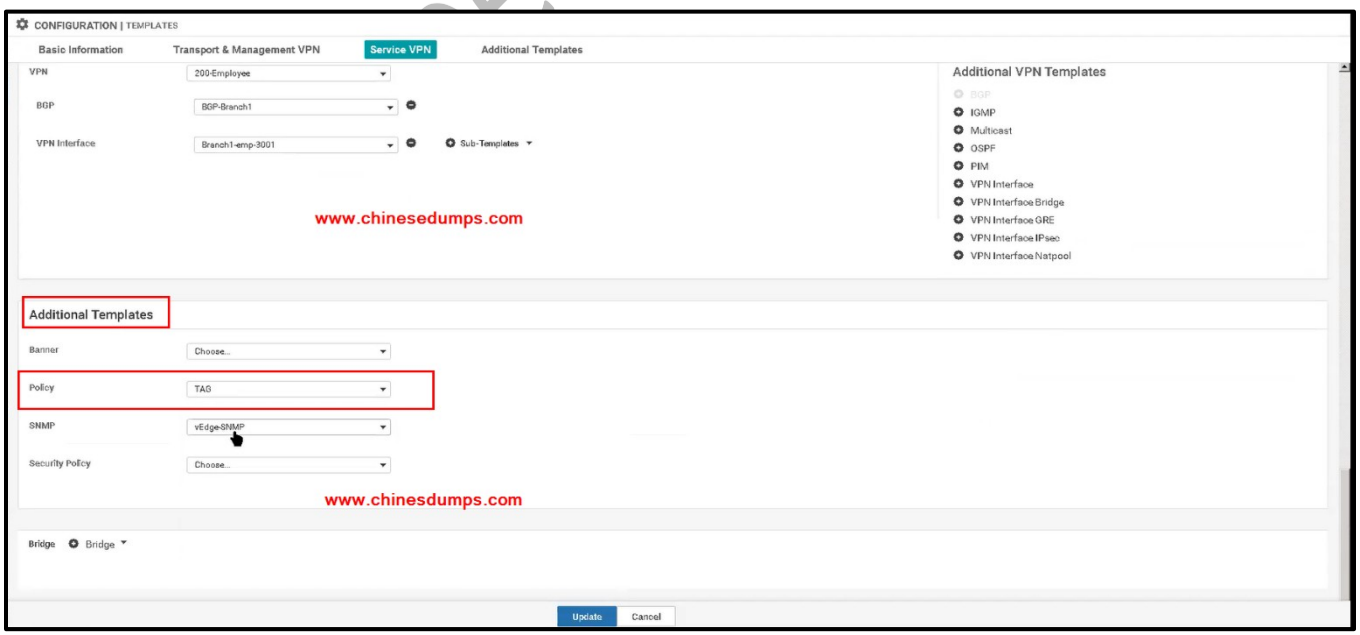
Step 20:

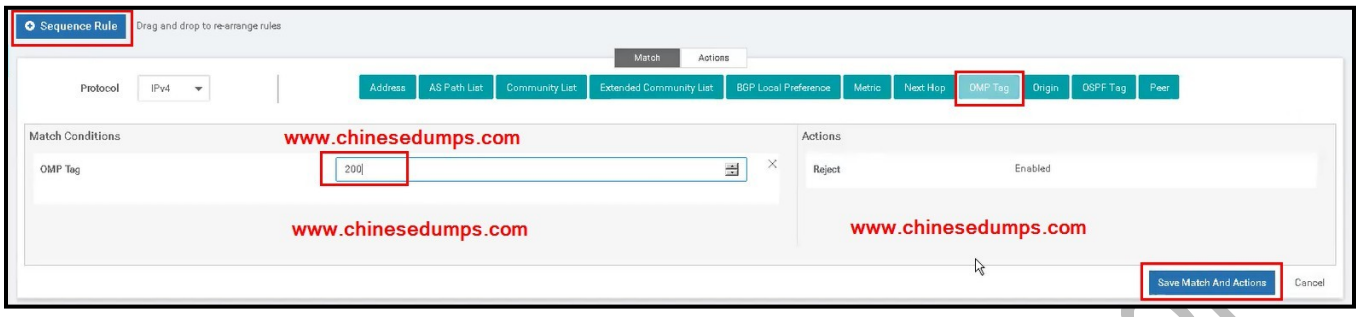
Go Under Device Template

Select Branch 1 Device Template

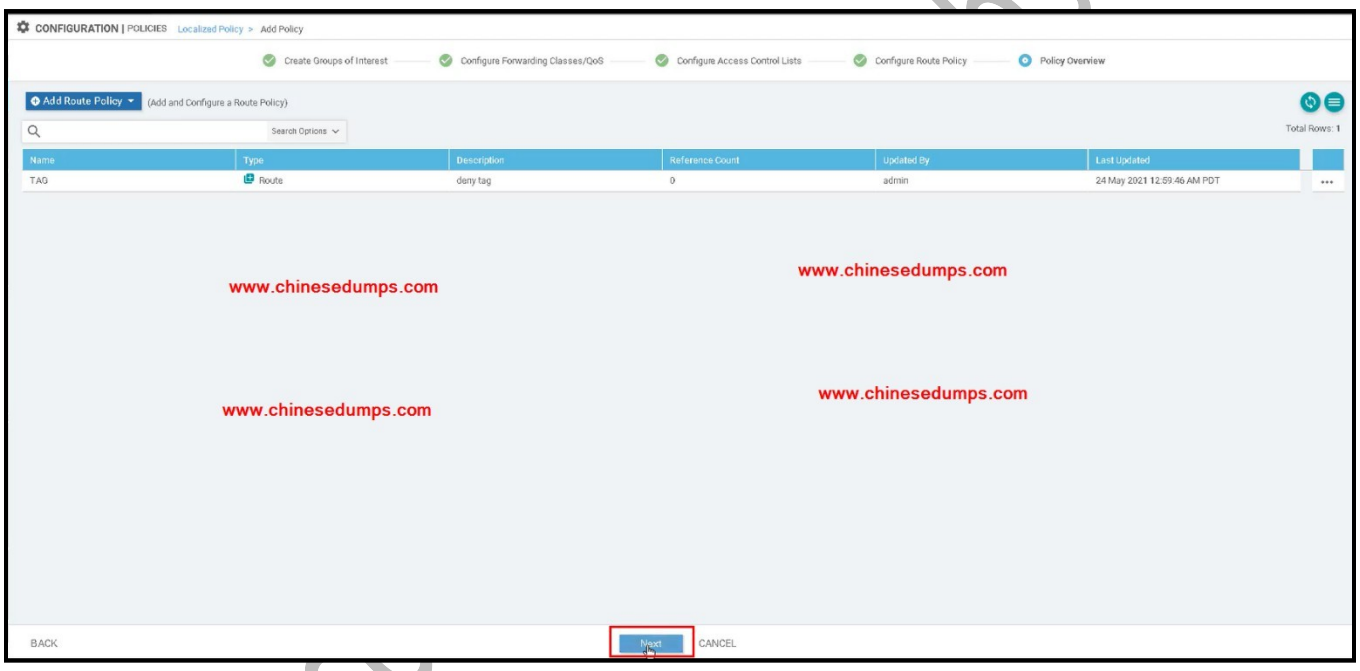
Under Branch 1 Device Template Go to Additional Templates

Add Policy





点击下一步



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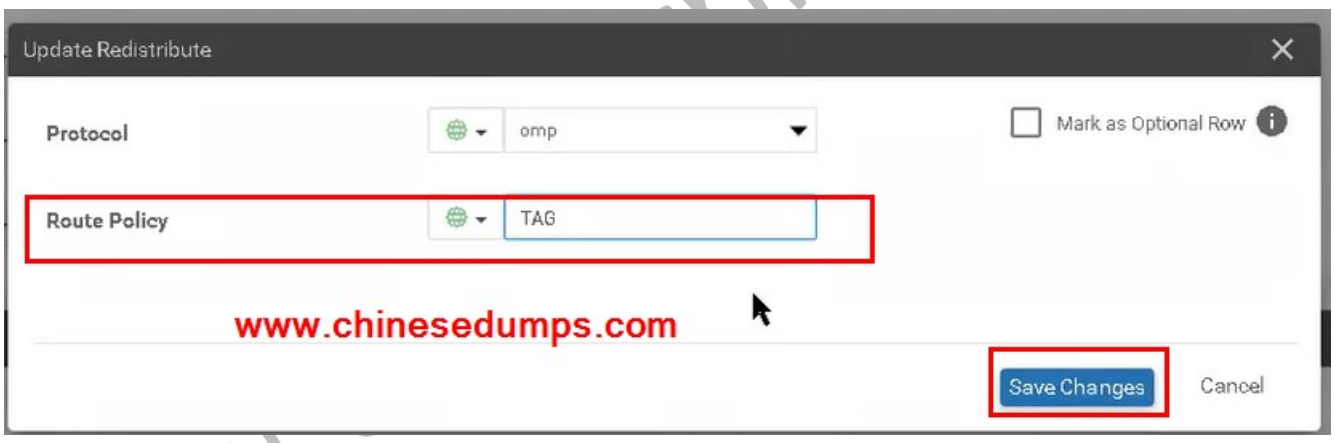
Go to Branch 1 OSPF VPN 999 Feature Template and add the tag under New redistribution



Click on Update

Repeat the Step 20 for Branch 2

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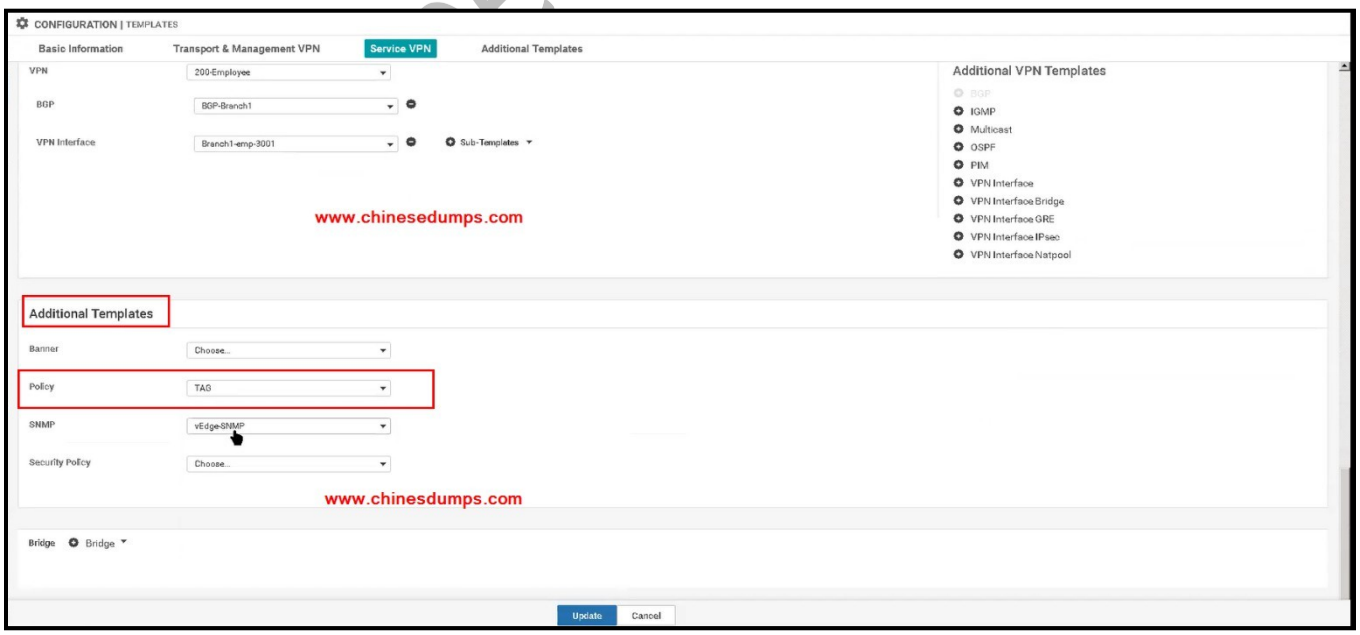
步骤20：

转到设备模板下，选择分支1设

备模板

在分支1设备模板下，转到其他模板添加策略



2.5: Handling Guest Traffic

The guest VN/VPN on Branches #1 and #2 must remain isolated from the rest of the company network. It is only allowed to reach internet through r23 and r24 in the DC. Enable internet connectivity for the Guest VPN according to these requirements:

- On vedge21 and vedge22, place the ge0/2 interfaces into the Guest VPN 199.
- On r23 and r24, create a new VRF named Guest using the RD of 65002:199, and place the gi4 interfaces into this VRF.
- Assign addresses to these interfaces:
    - R23 gi4: 10.2.123.1/24
    - R24 gi4: 10.2.224.1/24
    - Vedge21 ge0/2: 10.2.123.2/24
    - Vedge22 ge0/2: 10.2.224.2/24
- Peer r23 and vedge21 in the Guest VRF/VPN using iBGP.
- Peer r24 and vedge22 in the Guest VRF/VPN using iBGP.
- Ensure that r23 and r24 learn the routes in the Guest VRF/VPN over iBGP.
- On r23 and r24, configure a static default route in the Guest VRF and point it to the ISP's IP address 200.99.23.1 or 200.99.24.1 as appropriate. Advertise this default route in iBGP to vedge21 and vedge22.
- On r23 and r24, configure PAT to allow the Guest VPN to access internet by translating it to the router address on the link toward the ISP. Reuse the NAT ACL already created on the router. Do not use NAT pools.

Configure r23 as the DHCP server for Guest VPN according to these requirements:

- Create Loopback1 interface on r23 associated with the Guest VRF and having the IP address 10.2.255.211/32
- Advertise this prefix in BGP toward vedge21.
- Create DHCP pool named br1_guest for branch #1 Guest subnet.
- Create DHCP Pool names br2_guest for branch #2 Guest subnet.
- Explicitly associate both DHCP pools with the VRF guest.
- In each subnet, assign addresses from .101 up to .254 inclusively, and the appropriate gateway to clients.
- Associate host42 and host52 with the guest VN in DNAC, and make sure that both hosts receive the appropriate address.
- Make sure that host42 and host52 can ping 8.8.8.8 in the ISP cloud

4 Points

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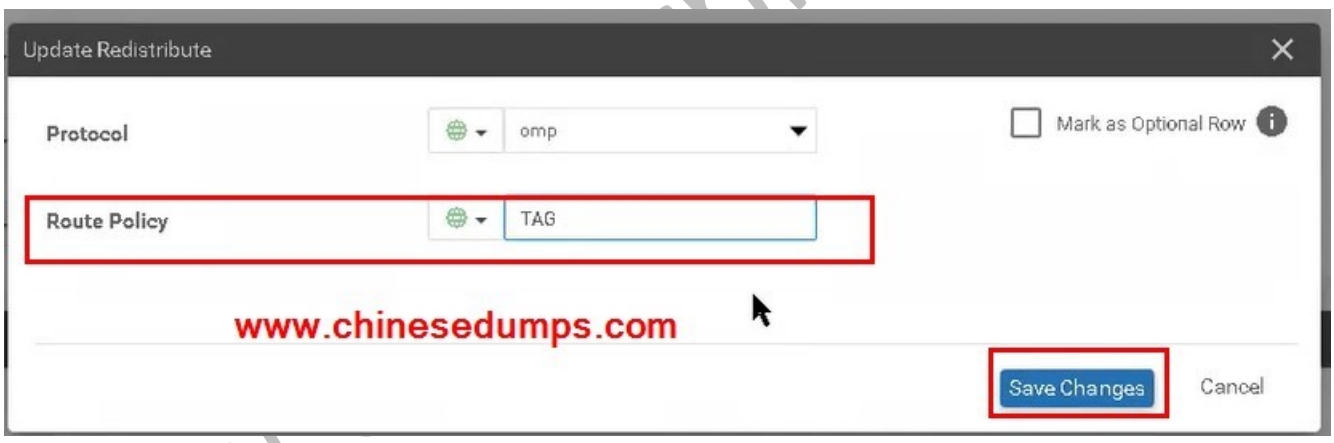
转到分支1 OSPF VPN 999功能模板，并在新重新分发下添加标签



点击更新

对分支2重复步骤20

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Solution:

r23

```
router bgp 65002
neighbor 200.99.23.1 remote 19999

vrf def Guest
rd 65002:199
address-fam ipv4
exit

int gi 4
vrf for Guest
ip add 10.2.123.1 255.255.255.0
ip nat inside

int gi 1
ip nat out
exit

ip access-list stand NAT
permit 10.0.0.0 0.255.255.255
exit

router bgp 65002
address-fam ipv4 vrf Guest
neighbor 10.2.123.2 remote 65002
network 10.2.255.211 mask 255.255.255.255
net 0.0.0.0 mask 0.0.0.0

ip nat inside list NAT int gig 1 vrf Guest overload

ip route vrf Guest 0.0.0.0 0.0.0.0 200.99.23.1 global

int loo1
vrf for Guest
ip add 10.2.255.211
exit
```

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2.5： 处理客流量

分支#1和#2上的访客VN/VPN必须与公司网络的其他部分保持隔离。它只允许通过DC中的r23和r24访问互联网。根据以下要求为Guest VPN启用互联网连接：

- 在vedge21和vedge22上，将ge0/2接口插入Guest VPN 199。
- 在r23和r24上，使用65002:199的RD创建一个名为Guest的新VRF，并将gi4接口放入此VRF中。
- 为这些接口分配地址：
    - R23日期：2014年12月10日
    - R24日期：2014年2月22日
    - 吠陀21日0/2:12.123.2/24
    - 吠陀22日0/2:10.224.2/24
- 使用iBGP的访客VRF/VPN中的对等端r23和vedge21。
- 使用iBGP的访客VRF/VPN中的对等端r24和vedge22。
- 确保r23和r24通过iBGP学习Guest VRF/VPN中的路由。
- 在r23和r24上，在Guest VRF中配置一个静态默认路由，并将其指向ISP的IP地址200.99.23.1或200.99.24.1（视情况而定）。在iBGP中向vedge21和vedge22通告此默认路由。
- 在r23和r24上，配置PAT以允许Guest VPN通过将其转换为指向ISP的链路上的路由器地址来访问互联网。重用路由器上已创建的NAT ACL。不要使用NAT池。

根据以下要求将r23配置为Guest VPN的DHCP服务器：

- 在r23上创建与Guest VRF关联的Loopback1接口，IP地址为10.2.255.211/32
- 在BGP中向vedge21通告此前缀。
- 为分支#1 guest子网创建名为br1_guest的DHCP池。
- 为分支#2 guest子网创建DHCP池名称br2_guest。
- 显式地将两个DHCP池与VRF客户机相关联。
- 在每个子网中，为客户端分配从.101到.254的地址以及相应的网关。
- 将host42和host52与DNAC中的访客VN相关联，并确保两个主机都收到了适当的地址。
- 确保主机42和主机52可以在ISP云中ping 8.8.8.8

4分

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```
ip dhcp class branch1
exit

ip dhcp pool br1_guest
network 10.4.199.0 /24
default-router 10.4.199.1
class branch1
address range 10.4.199.101 10.4.199.254

ip dhcp class branch2
exit

ip dhcp pool br2_guest
network 10.5.199.0 /24
default-router 10.5.199.1
class branch2
address range 10.5.199.101 10.5.199.254
ip dhcp use connected vrf
```

r24

```
vrf def Guest
rd 65002:199
address-fam ipv4
exit

int gi 4
vrf for Guest
ip add 10.2.224.1 255.255.255.0
ip nat inside

int gi 1
ip nat out

ip access-list stand  NAT
permit 10.0.0.0 0.255.255.255
exit
```

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解决方案：

r23
路由器bgp 65002
邻居200.99.23.1远程19999

访客街
65002:199
地址fam ipv4出
口

整数gi 4
vrf为客人
ip地址10.2.123.1 255.255.255.0
内部ip nat

整数gi 1
ip nat出口

ip访问列表支持NAT许可
10.0.0.0 0.255.255.255
退出

路由器bgp 65002
地址fam ipv4 vrf访客邻居
10.2.123.2远程65002
网络10.2.255.211掩码255.255.255
net 0.0.0.0掩码0.0.0.0

ip nat内列表nat int gig 1 vrf访客过载

ip路由vrf访客0.0.0.0 0.0.0.0 200.99.23.1全

局整数1
vrf为客人
ip地址10.2.255.211
退出

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router bgp 65002
address-fam ipv4 vrf Guest
neighbor 10.2.224.2 remote 65002
network 0.0.0.0 mask 0.0.0.0

ip nat inside list NAT int gig 1 vrf Guest overload
ip route vrf Guest 0.0.0.0 0.0.0.0 200.99.23.1 global

ip dhcp use connected vrf

Step 1:

In DC Device Templates

Under Service VPN add VPN 199 (Guest)

Under VPN 199 add Interface Template and BGP Template

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ip-dhcp类分支1出口

ip dhcp池br1_客户
网络10.4.199.0/24
默认路由器10.4.199.1类
分支1
地址范围10.4.199.101 10.4.199.254

ip-dhcp类branch2出
口

ip dhcp池br2_客户
网络10.5.199.0/24
默认路由器10.5.199.1类
分支2
地址范围10.5.199.101 10.5.199.254
ip dhcp使用连接的vrf

r24

访客街
65002:199
地址fam ipv4出
口

整数gi 4
vrf为客人
ip地址10.2.224.1 255.255.255.0
内部ip nat

整数gi 1
ip nat输出

ip访问列表支持NAT许可
10.0.0.0 0.255.255.255
退出

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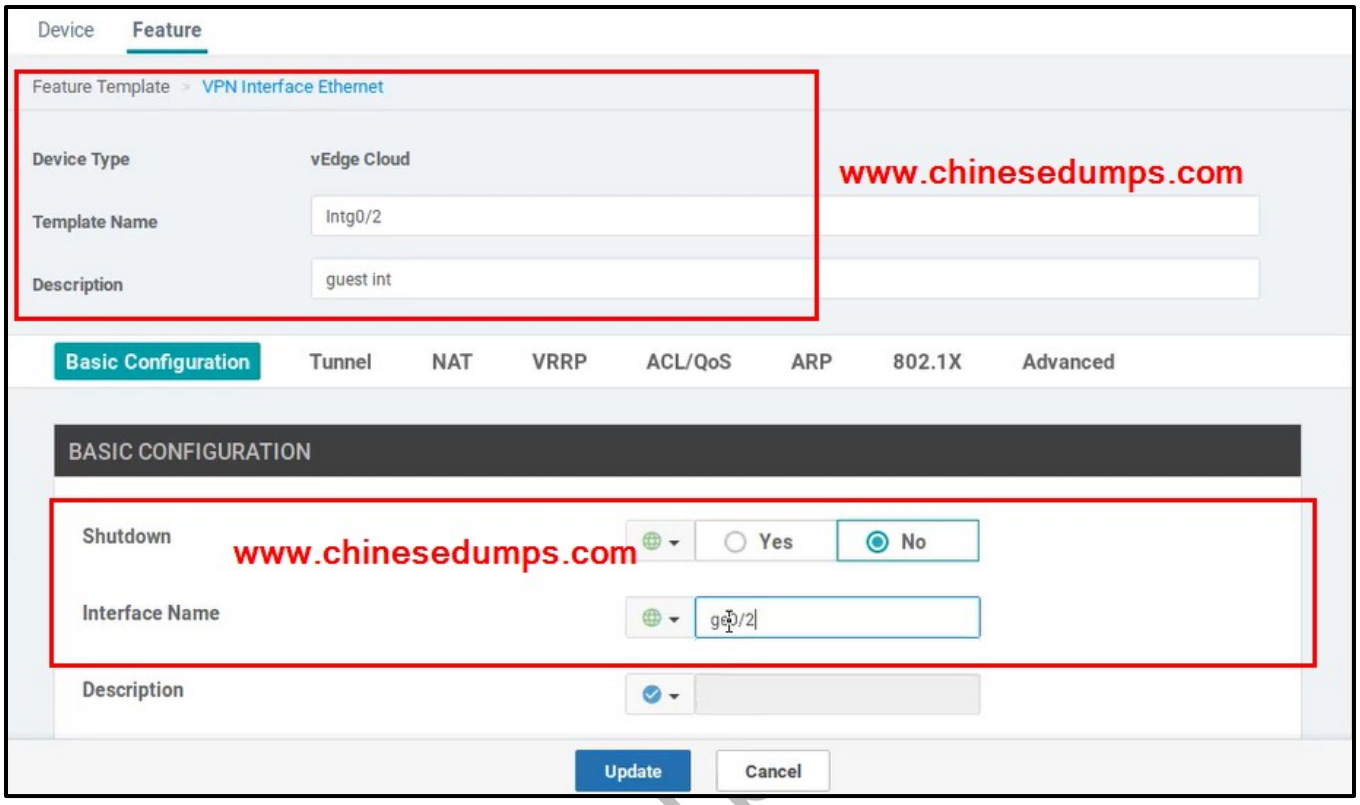
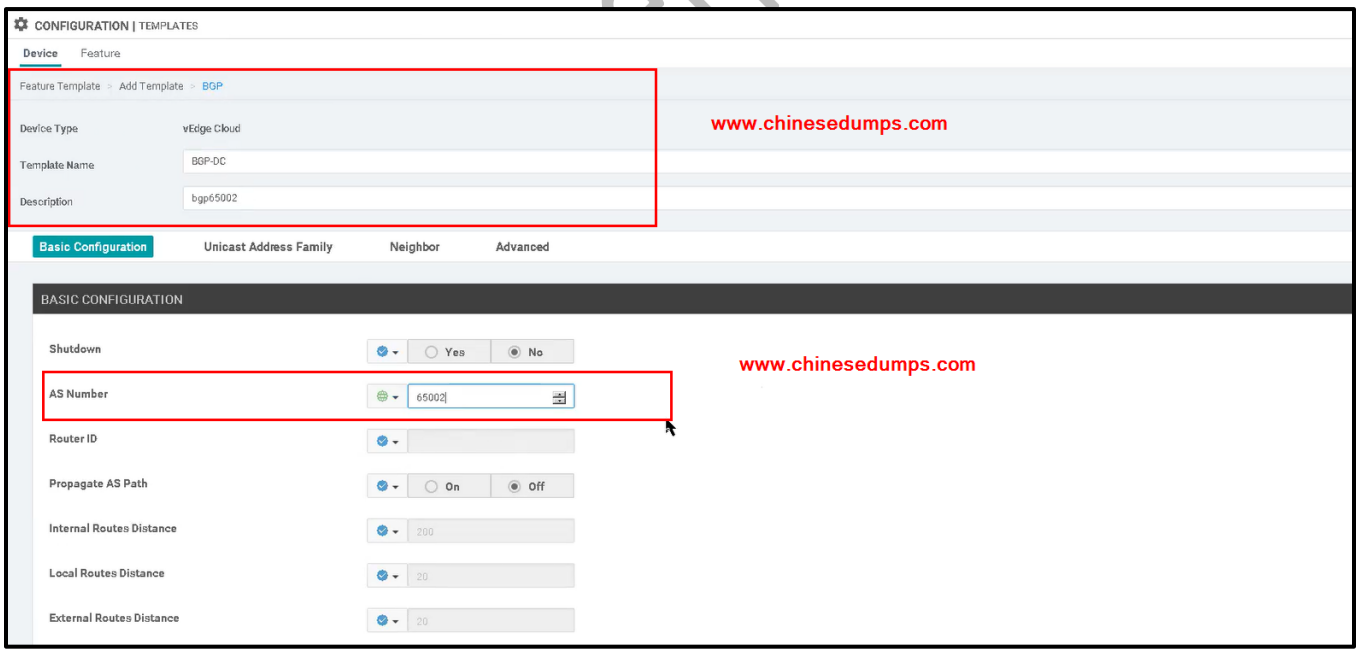

Device
Feature
Feature Template > VPN Interface Ethernet
Device Type
vEdge Cloud
Template Name
Intg0/2
Description
guest int
www.chinesedumps.com
Basic Configuration
Tunnel
NAT
VRRP
ACL/QoS
ARP
802.1X
Advanced
BASIC CONFIGURATION
Shutdown
Yes
No
Interface Name
ge0/2
Description
Update
Cancel


CONFIGURATION | TEMPLATES
Device
Feature
Feature Template > Add Template > BGP
Device Type
vEdge Cloud
Template Name
BGP-DC
Description
bgp65002
www.chinesedumps.com
Basic Configuration
Unicast Address Family
Neighbor
Advanced
BASIC CONFIGURATION
Shutdown
Yes
No
AS Number
65002
Router ID
Propagate AS Path
On
Off
Internal Routes Distance
200
Local Routes Distance
20
External Routes Distance
20
www.chinesedumps.com

路由器bgp 65002
地址fam ipv4 vrf访客邻居
10.2.224.2远程65002
网络0.0.0.0掩码0.0.0.0

ip nat内列表nat int gig 1 vrf访客过载
ip路由vrf访客0.0.0.0 0.0.0.0 200.99.23.1全

局ip dhcp使用连接的vrf

第一步：

在DC设备模板中

在服务VPN下添加VPN 199（访客）

在VPN 199下添加接口模板和BGP模板

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Click **SAVE**

Click Update in DC Device Template

Step 2:

Go Under DNAC

Go to Provision > Fabric Default LAN Fabric

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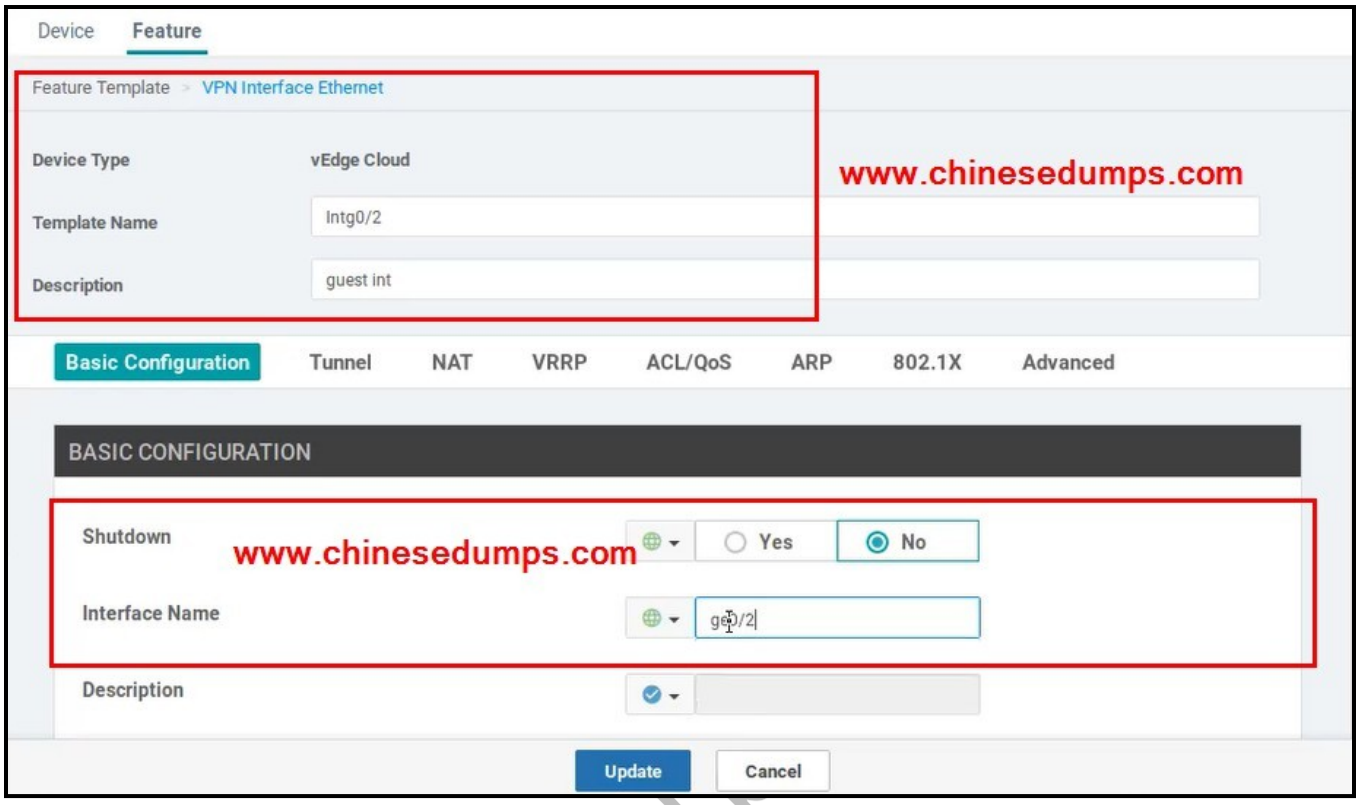
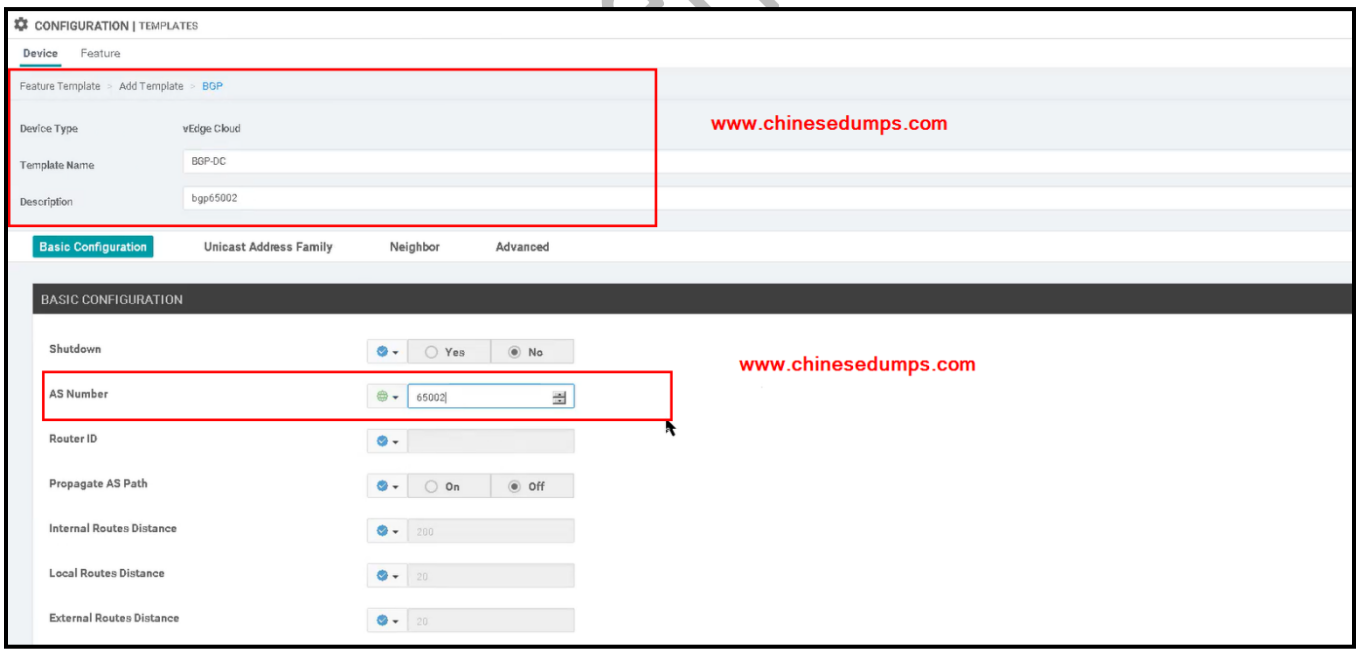

Device
Feature
Feature Template > VPN Interface Ethernet
Device Type
vEdge Cloud
Template Name
Intg0/2
Description
guest int
www.chinesedumps.com
Basic Configuration
Tunnel
NAT
VRRP
ACL/QoS
ARP
802.1X
Advanced
BASIC CONFIGURATION
Shutdown
Yes
No
Interface Name
ge0/2
Description
Update
Cancel


CONFIGURATION | TEMPLATES
Device
Feature
Feature Template > Add Template > BGP
Device Type
vEdge Cloud
Template Name
BGP-DC
Description
bgp65002
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Basic Configuration
Unicast Address Family
Neighbor
Advanced
BASIC CONFIGURATION
Shutdown
Yes
No
AS Number
65002
Router ID
Propagate AS Path
On
Off
Internal Routes Distance
200
Local Routes Distance
20
External Routes Distance
20
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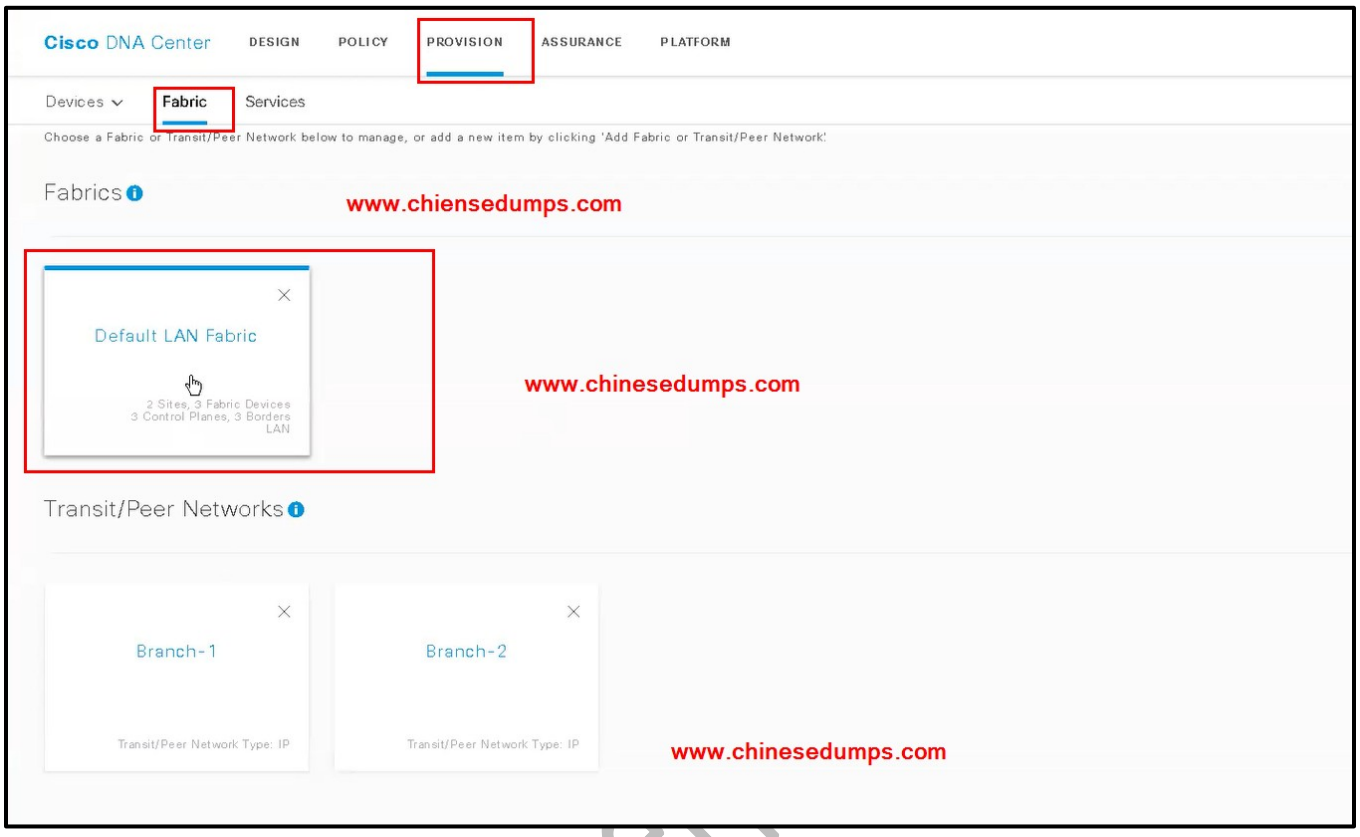

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单击保存

在DC设备模板中单击“更新”

第二步：

转到DNAC下

转到配置>结构默认LAN结构

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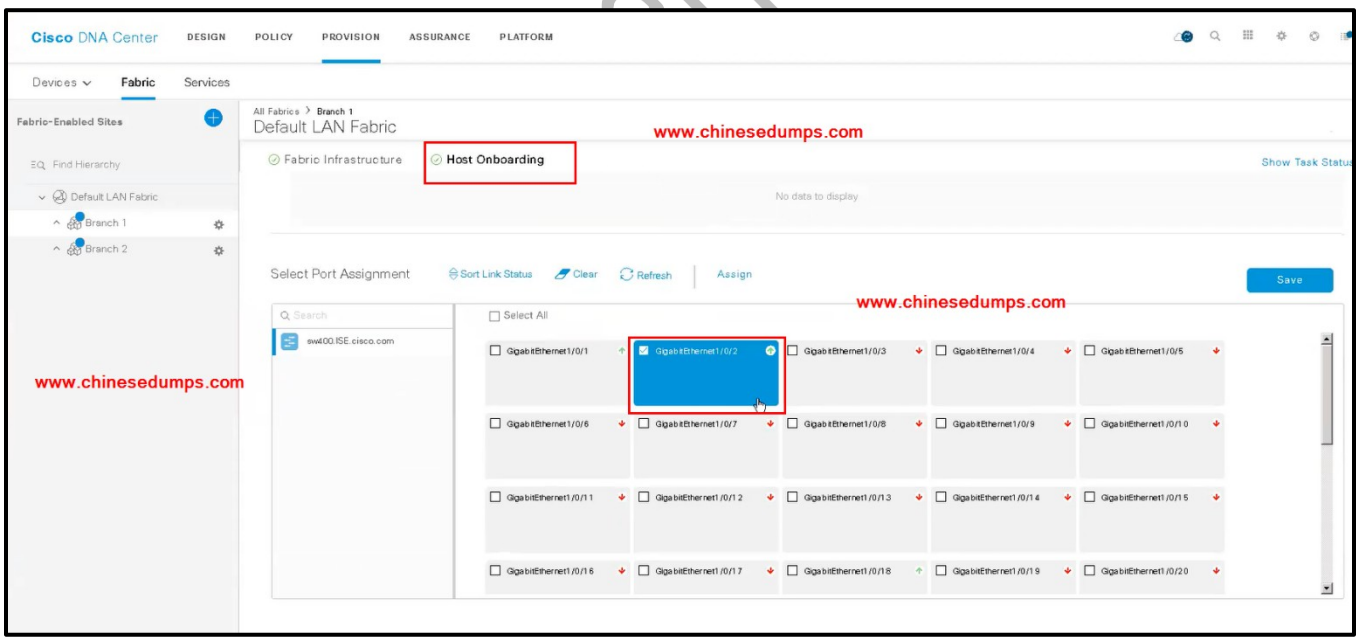
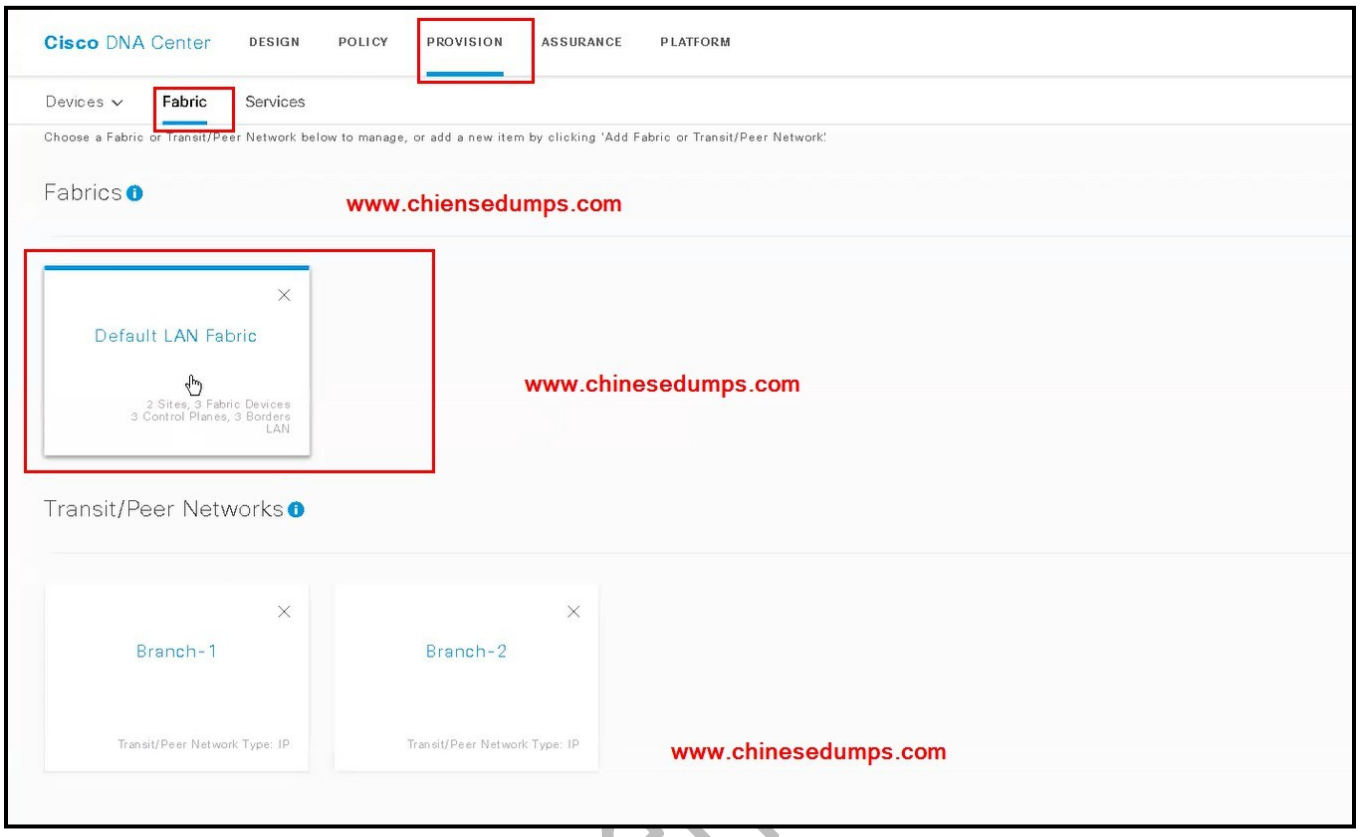
Select Branch 1

Go to Host Onboarding





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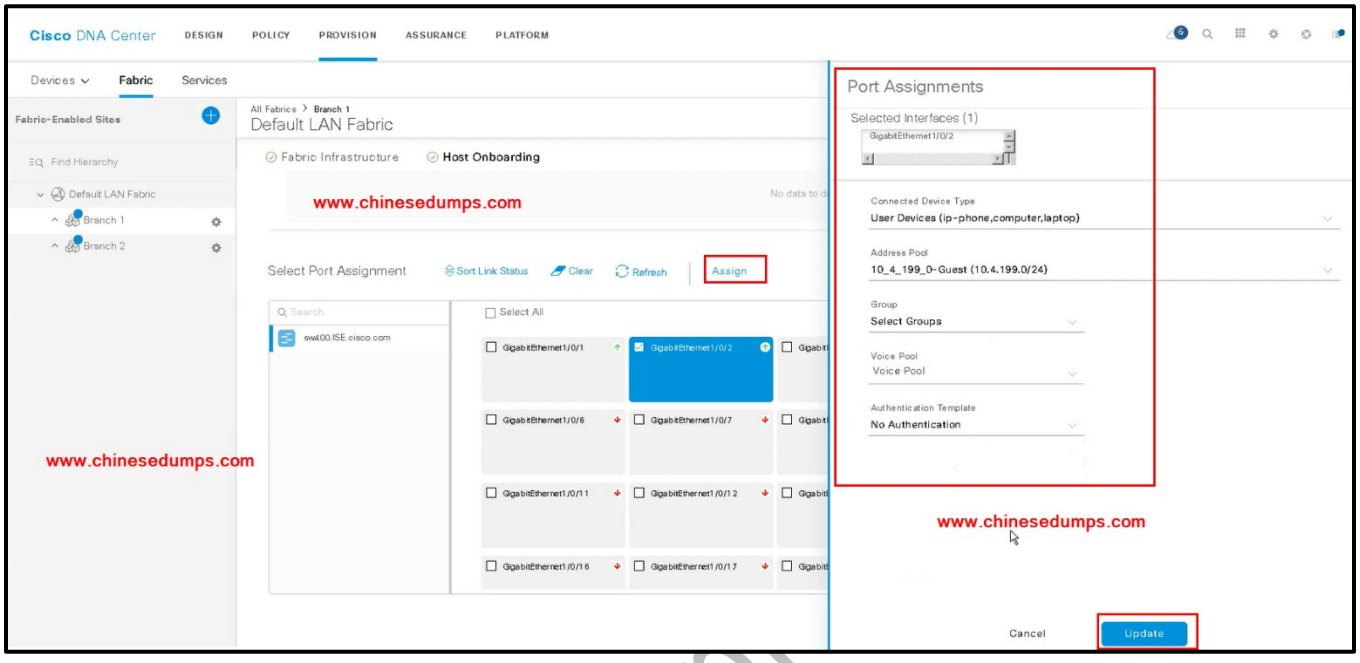
Select the details as per below screenshot



Then SAVE and Apply

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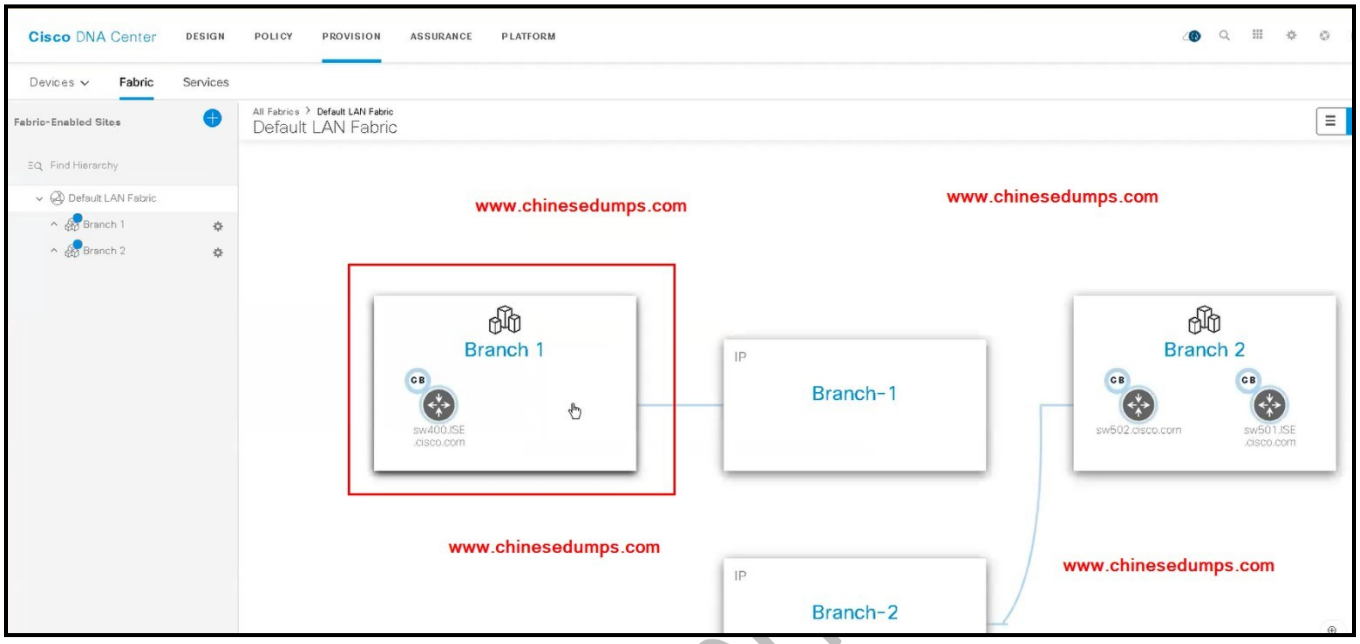
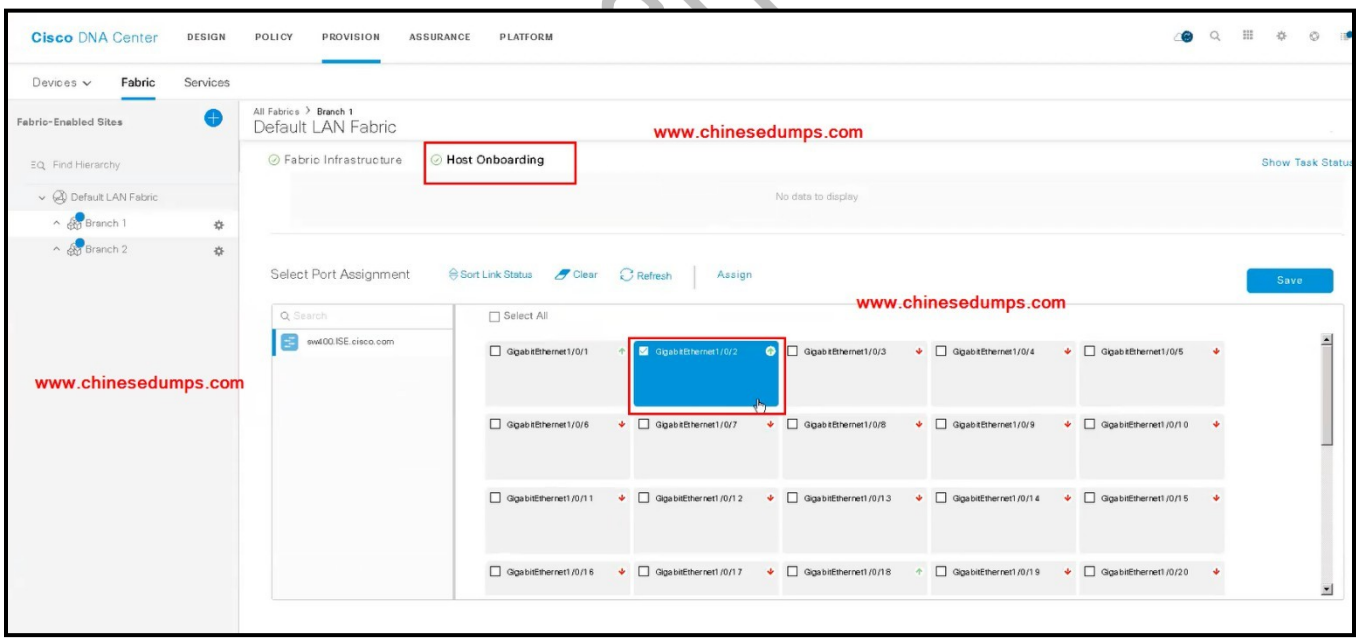
选择分支1



前往主机入职培训



2.6: Support for silent Hosts in Branch #2

This item consists of multiple questions. You may need to scroll down to be able to see all questions.
In future, Branch #2 will be equipped with IP-based IoT endpoints operating in speak-when-spoken-to mode, also called silent hosts. Which of the following SDA features enables a working connectivity with these IoT endpoints?

- o Native Multicast
- o Endpoint Mobility
- o Layer 2 Flooding
- o Layer 2 Extension

In the statement below, select one of the options from the drop-down list to complete the sentence and form a correct statement.

For SDA to support silent hosts, _____Select Option_____ in the underlay as a prerequisite.

Options:

- o IP multicast routing with PIM-SM must be enabled
- o No additional capability aside from unicast IP connectivity is required
- o IS-IS must be used as a routing protocol
- o DHCP Snooping must be enabled

3 Points

Answer: Layer 2 Flooding
Answer: IP Multicast routing with PIM-SM must be enabled

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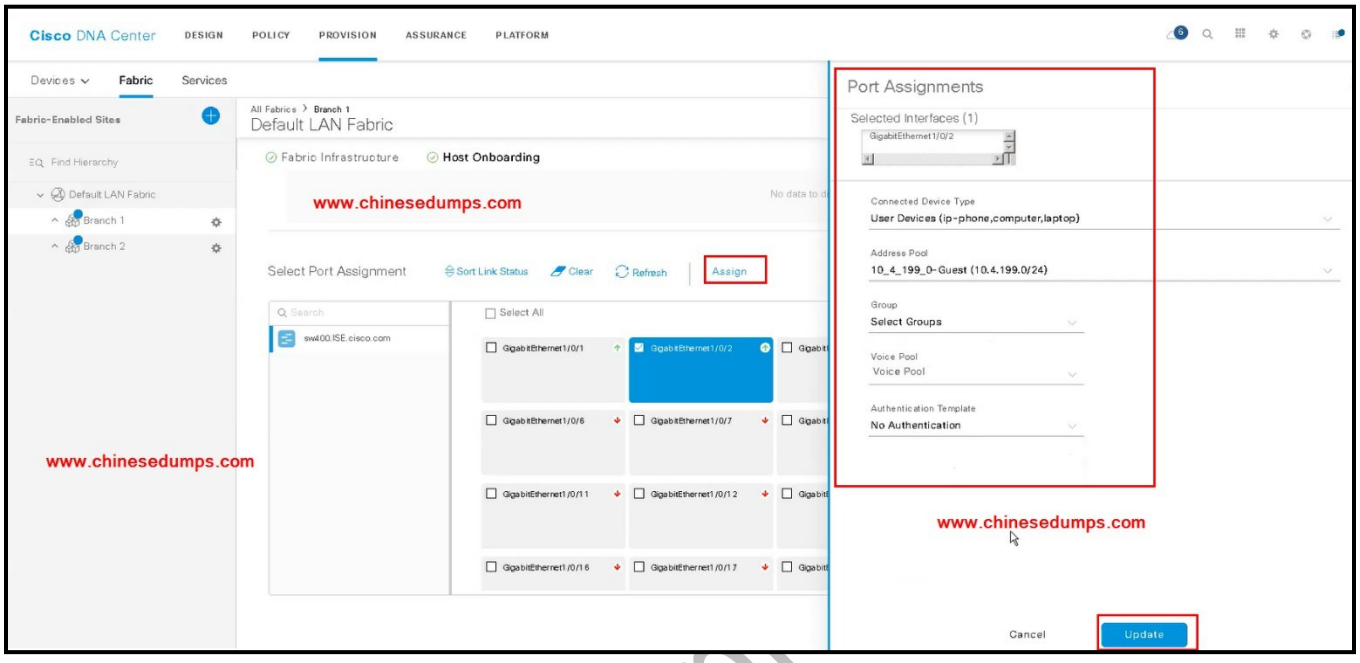
根据下面的屏幕截图选择详细信息



然后保存并应用

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3.1: Enabling CLI access to r30

There is no direct console access provided to the router r30. Moreover, r30 does not accept any remote connections because its VTY lines are configured with transport input none. Using RESTCONF, enable remote access to r30 for all remote access protocols, according to these requirements:

- You can use host31 to access router r30 using IP address 10.3.11.1
- You can use any method of accessing the RESTCONF API on r30 from host31, including curl, Python, or Postman
- You must change the input transport protocol on all configurable VTY lines
- The input transport protocol value setting must be changed from none to all

Important parameters:

- Username/password for HTTP authentication
  - admin/admin
- URL
  - https://10.3.11.1:443/restconf/data/Cisco-IOS-XE-native/line/vty
- HTTP method to retrieve the configuration
  - GET
- HTTP method to modify the configuration
  - PATCH
- HTTP headers
  - Content-Type: application/yang-data+json
  - Accept: application/yang-data+json
- Recommended curl switches
  - –i,-k,-X,-H,-u,-d

2 Points

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2.6 ： 支持分支#2中的静默主机

此项目由多个问题组成。您可能需要向下滚动才能看到所有问题。
未来，2号分支将配备基于IP的物联网端点，这些端点在通话模式下运行，也称为静音主机。以下哪种SDA功能支持与以下设备的工作连接
这些物联网端点？

- 本地多播
- 端点移动性
- 第2层洪水
- 第2层扩展

在下面的语句中，从下拉列表中选择一个选项来完成句子并形成正确的语句。

为了使SDA支持静默主机，_____选择Option作为先决条件。

选项：
- 必须启用带有PIM-SM的IP多播路由
- 除了单播IP连接外，不需要其他功能
- IS-IS必须用作路由协议
- 必须启用DHCP侦听

3分

答案：第二层洪水
答：必须启用带有PIM-SM的IP多播路由

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Solution:

On Chinesedumps.com-r30:

Chinesedumps.com-r30#conf t
Chinesedumps.com-r30(config)# ip http server
Chinesedumps.com-r30(config)# ip http authentication local
Chinesedumps.com-r30(config)# ip http secure-server

Chinesedumps.com-r30(config)# username admin privi 15 secret admin

Chinesedumps.com-r30(config)# restconf

Chinesedumps.com-r30(config)#interface GigabitEthernet2
Chinesedumps.com-r30(config-if)#no shutdown
Chinesedumps.com-r30(config-if)#ip address 10.3.11.1 255.255.255.0
Chinesedumps.com-r30(config-if)#negotiation auto

Step 1:

Open Linux machine host31 and check reachability for r30 , also check restconf reachability

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3.1：启用对r30的CLI访问

路由器r30没有提供直接的控制台访问。此外，r30不接受任何远程连接，因为其VTY线路配置了无传输输入。根据以下要求，使用RESTCONF为所有远程访问协议启用远程访问r30：

- 您可以使用主机31通过IP地址10.3.11.1访问路由器r30
- 您可以使用任何方法从host31访问r30上的RESTCONF API，包括curl、Python或Postman
- 您必须更改所有可配置VTY线路上的输入传输协议
- 输入传输协议值设置必须从none更改为all重要参数：

- HTTP身份验证的用户名/密码

    - 管理员/管理
- 统一资源 https://10.3.11.1:443/restconf/data/Cisco-IOS XE原生/
- 用于检索配置的HTTP方法
    - 获取
- 修改配置的HTTP方法
    - 补丁
    - http头
    - 内容类型：应用程序/yang数据+json
    - 接受：应用程序/yang数据+json
    - 推荐的卷曲开关

o-i、-k、-X、-H、-u、-d

2分

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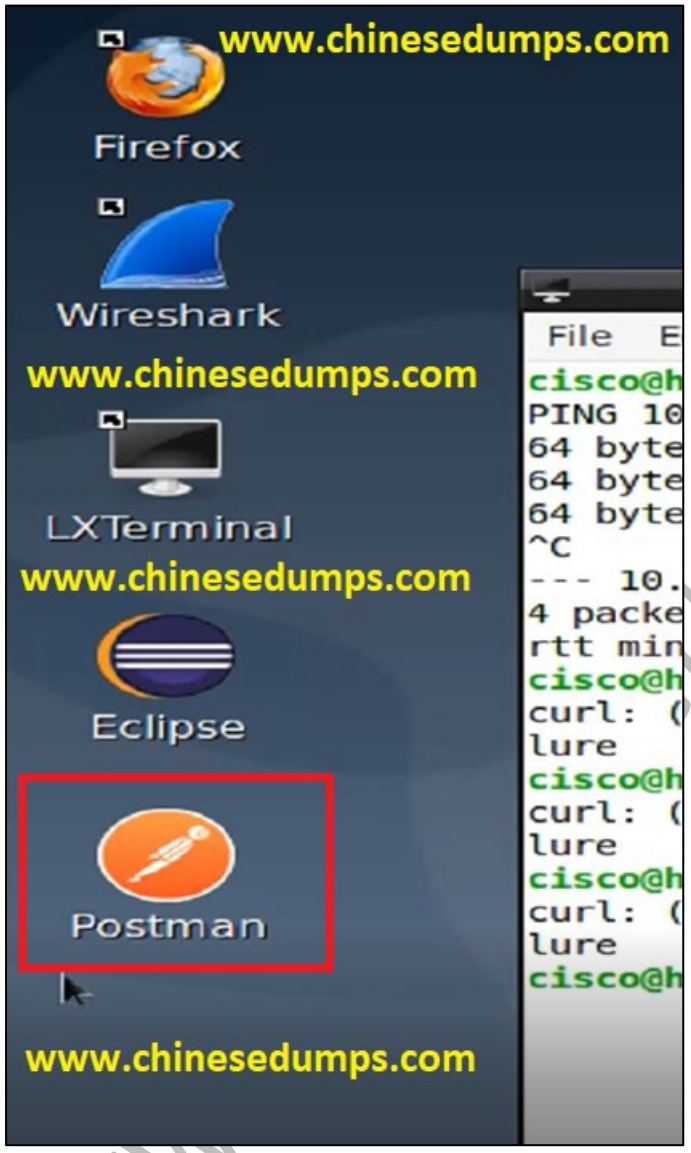
Step 2:

From linux Machine open Postman



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解决方案：

在Chinese edumps.com-r30上：

Chineseedumps.com-r30#conf t
Chineseedumps.com-r30（配置）#ip http
服务器
Chinese edumps.com-r30（配置）#ip http认证本地
Chinese edumps.com-r30（设置）#ip http安全服务器

Chinese edumps.com-r30（配置）#用户名admin privi 15

secretadmin Chinese edumps.com-r30（设置）#restconf

Chinese edumps.com-r30（配置）#接口千兆以太网2
Chinese edumps.com-r30（如果配置）#不关机
Chinese edumps.com-r30（配置if）#ip地址10.3.11.1
255.255.255.0 Chinese edumps.com-r30（设置if）#自动协商

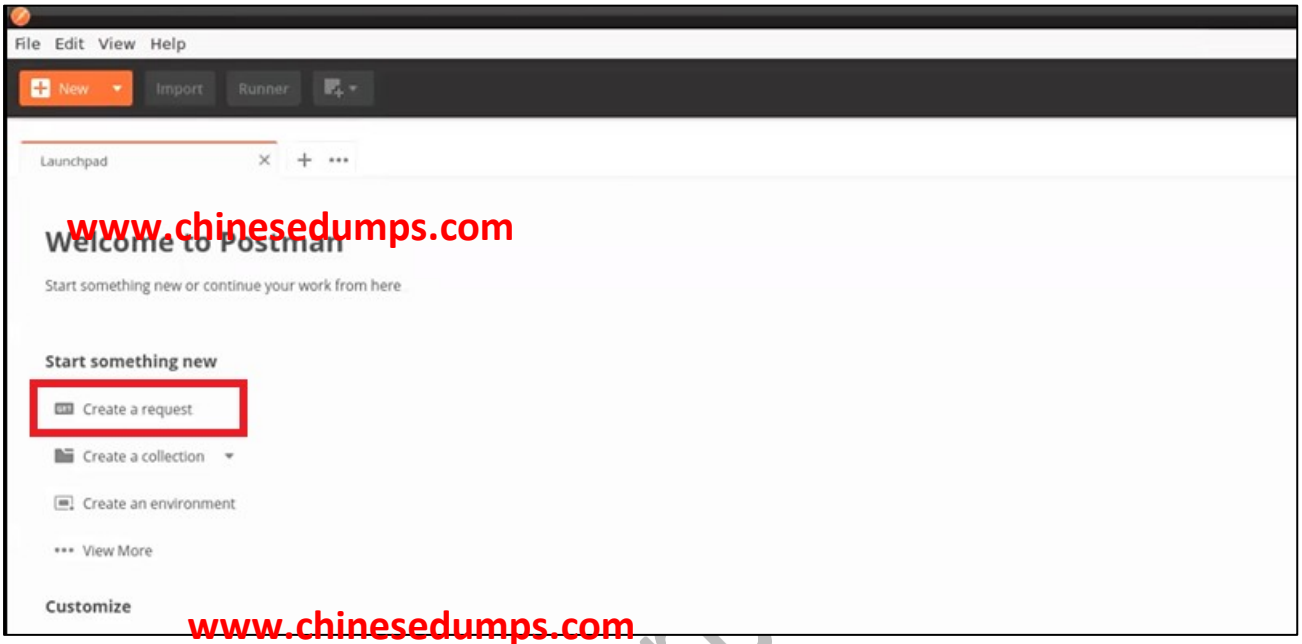
第一步：

打开Linux主机31，检查r30的可达性，并检查restconf的可达性

terpriselabs.com

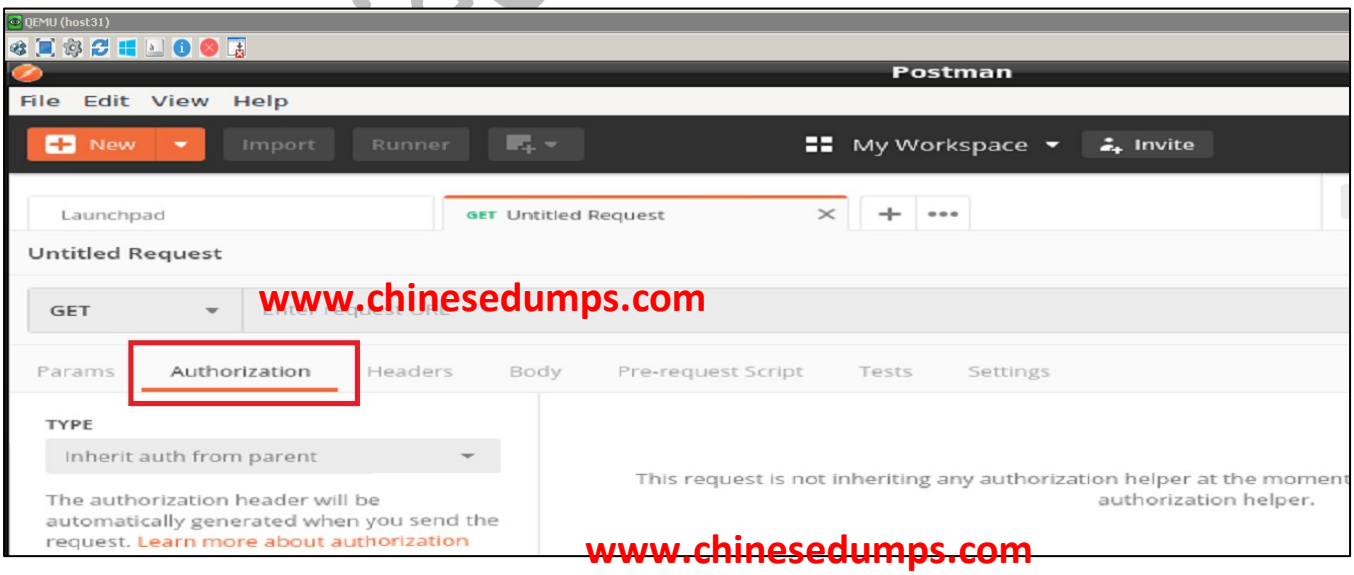
Step 3:

In postman select **Create a request**



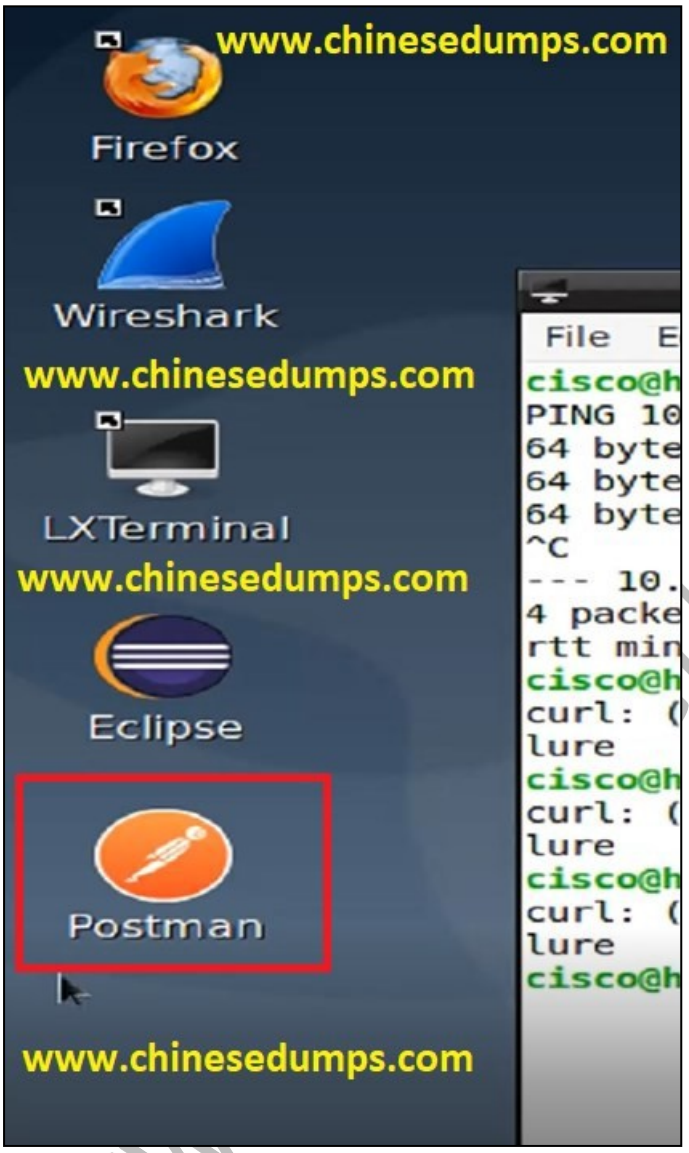
Step 4:

Go under **Authorization** tab



第二步：

从linux机器打开Postman



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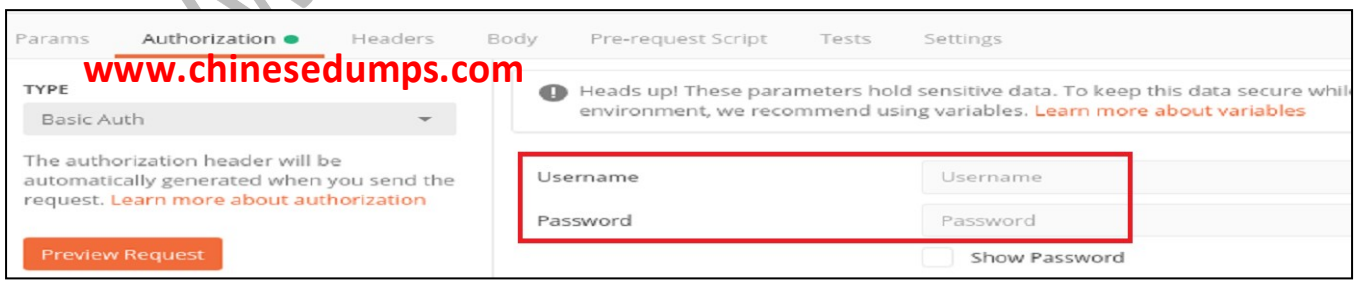
Step 5:

Under **Type** select **Basic Auth**

Step 6:

Insert Username/Pasword : **admin/admin**



步骤3：

在邮递员界面，选择“创建请求”

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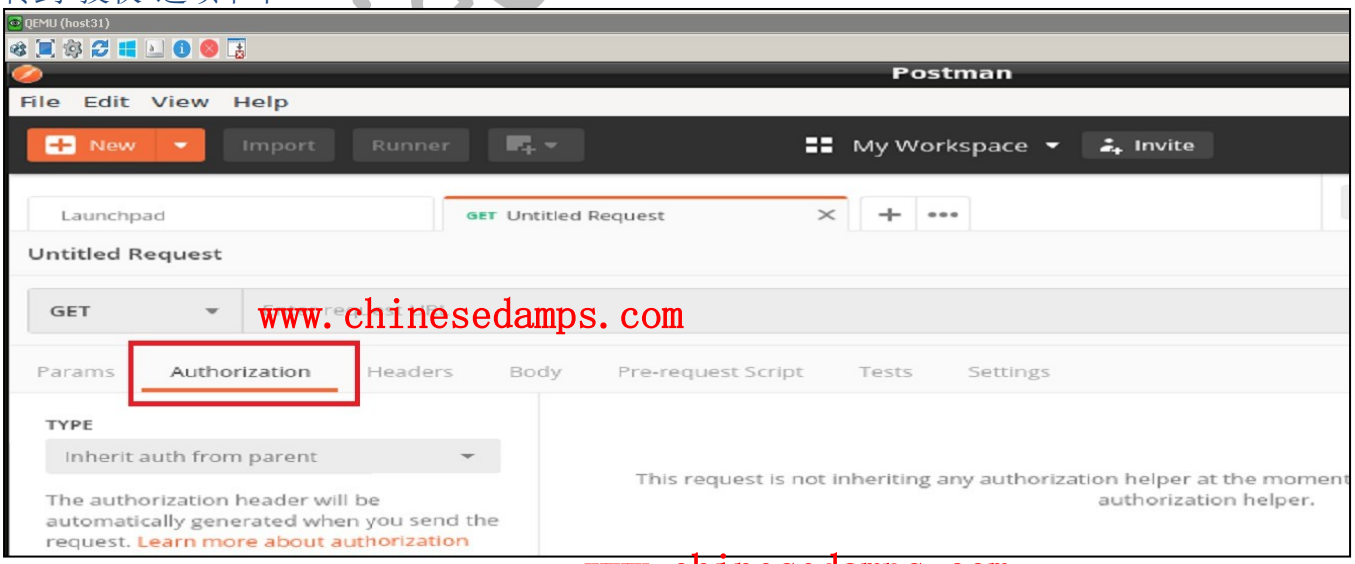
步骤4：

转到“授权”选项卡下



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Step 7:

Select **Setting** and then **Default setting**

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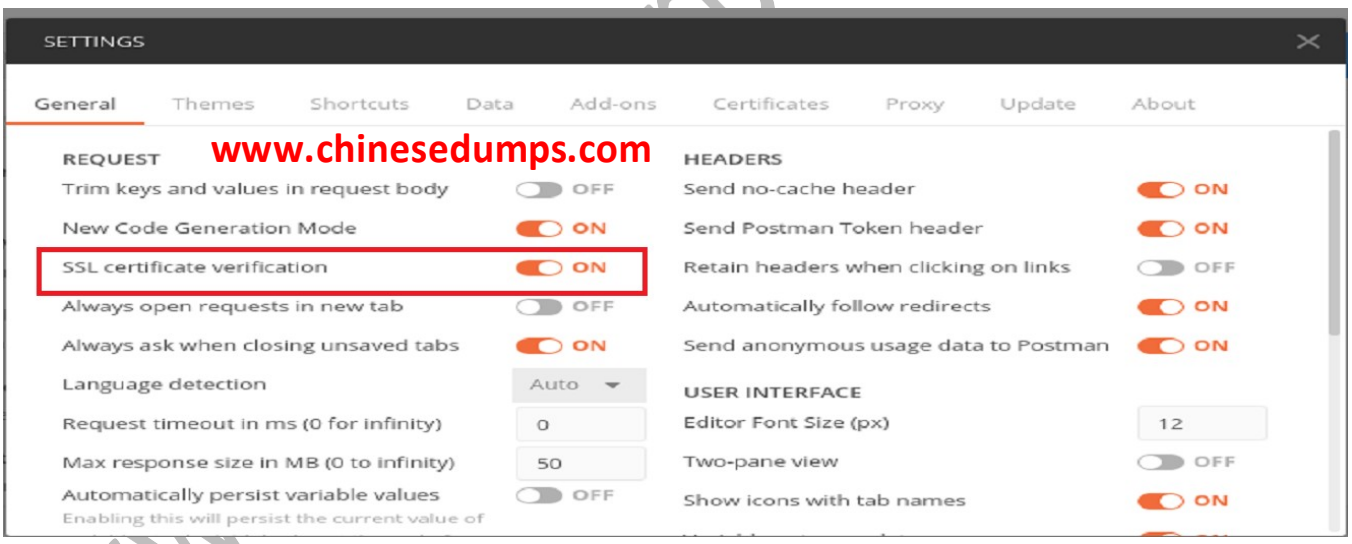
Step 8:

Turn off the **SSL Certificate verification**



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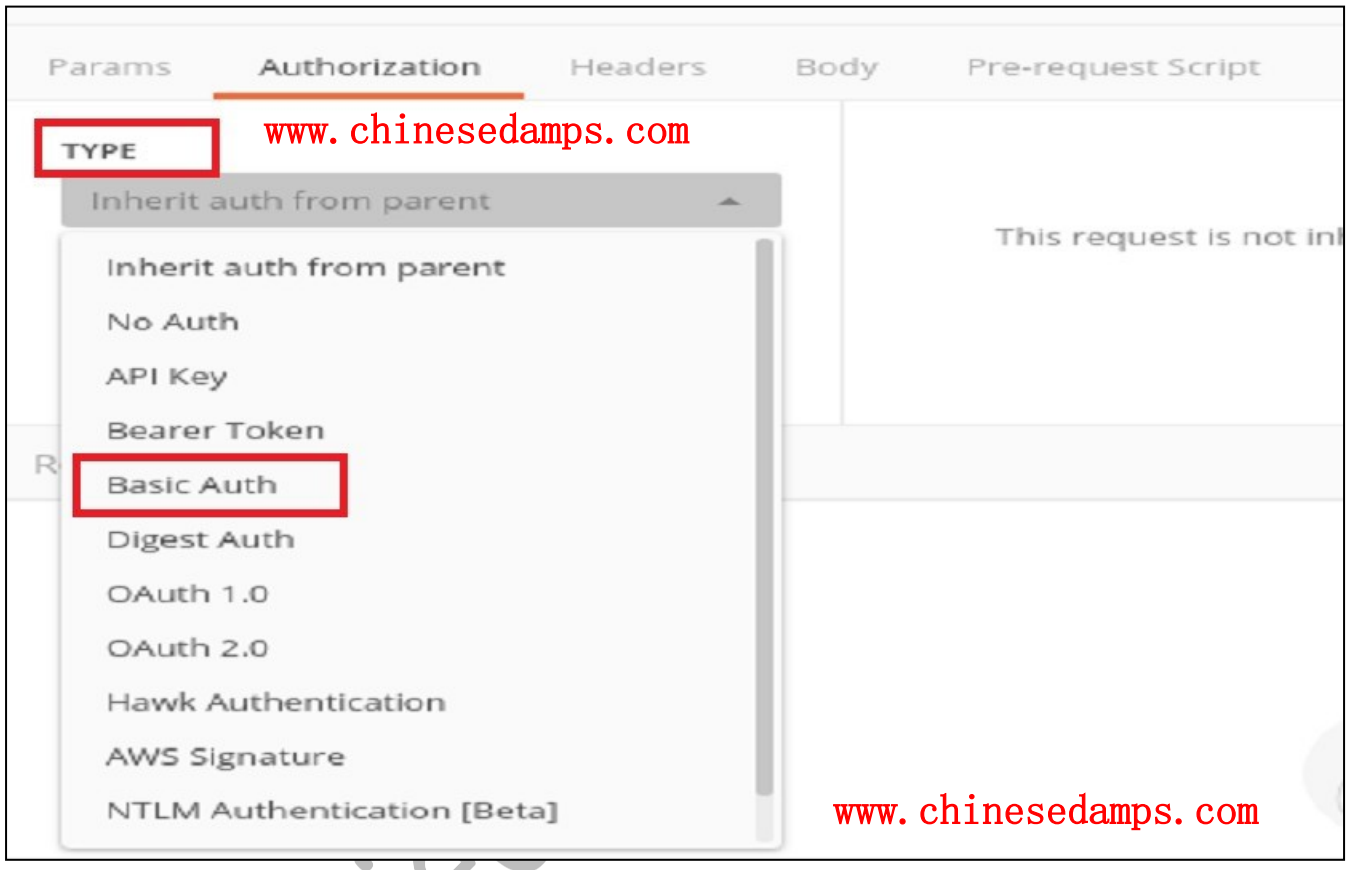
NOTE: This step 8 is sometimes not mandatory to perform in LAB

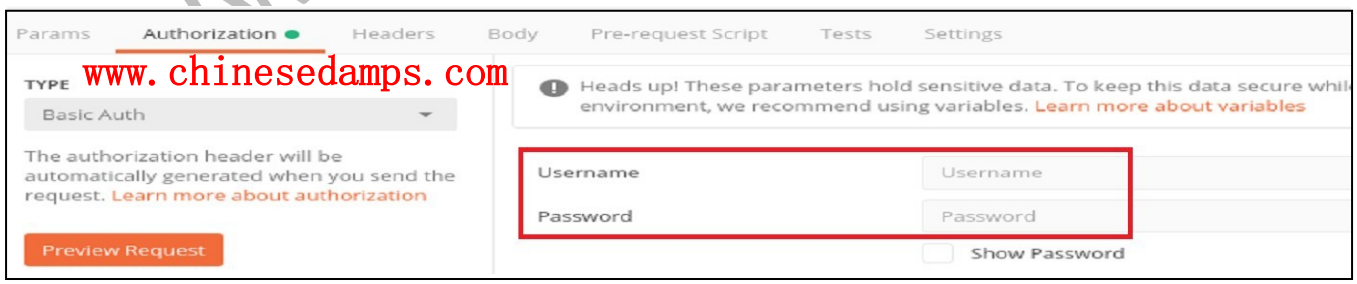
步骤5：

在“类型”下选择“基本身份验证”



步骤6：

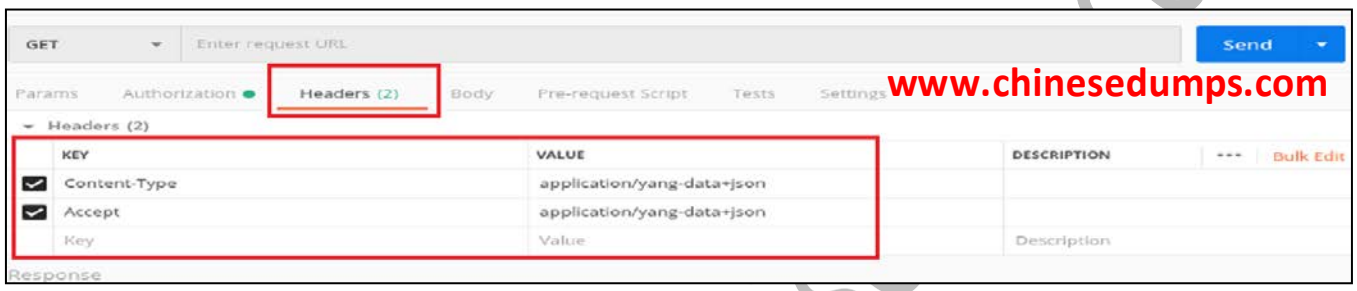
插入用户名/密码：admin/admin



Step 9:

Select **Headers** tab and create to header

- HTTP headers
  - Content-Type: application/yang-data+json
  - Accept: application/yang-data+json



Step 10:

Enter the **URL** under **GET** and hit **SEND**

- URL
  - https://10.3.11.1:443/restconf/data/Cisco-IOS-XE-native:native/line/vty

步骤7：

选择设置，然后选择默认设置

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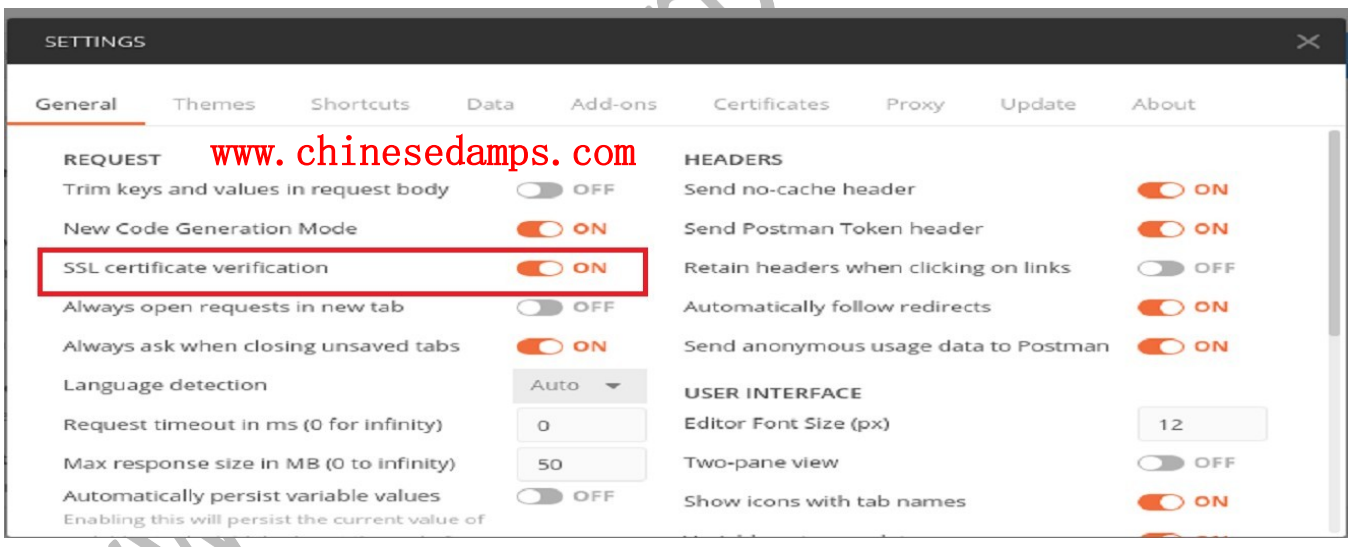
步骤8：

关闭SSL证书验证



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注：此步骤8有时不是必须在实验室中执行的

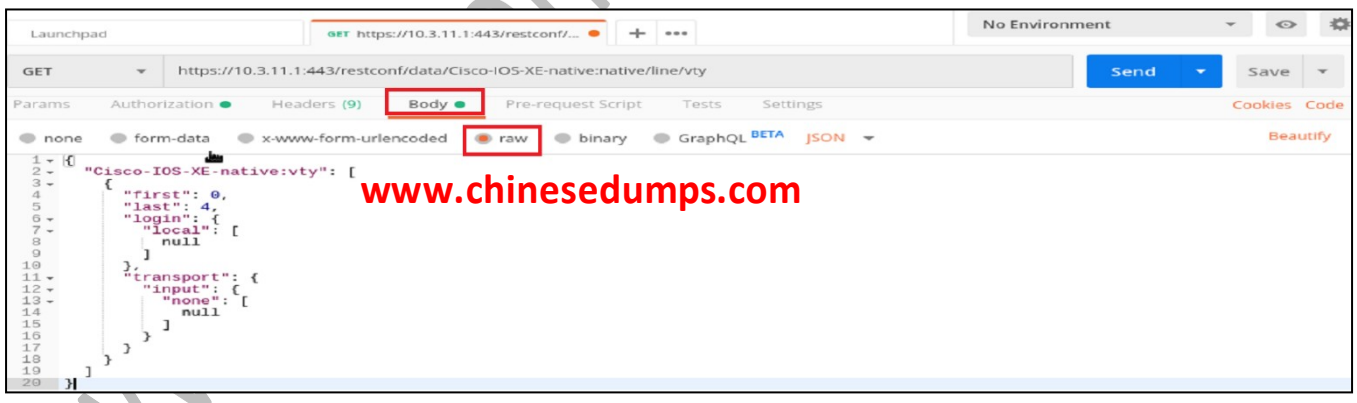
Step 11:

The above step 10 will generate an output in **Body** tab

Copy the generated output

Step 12:

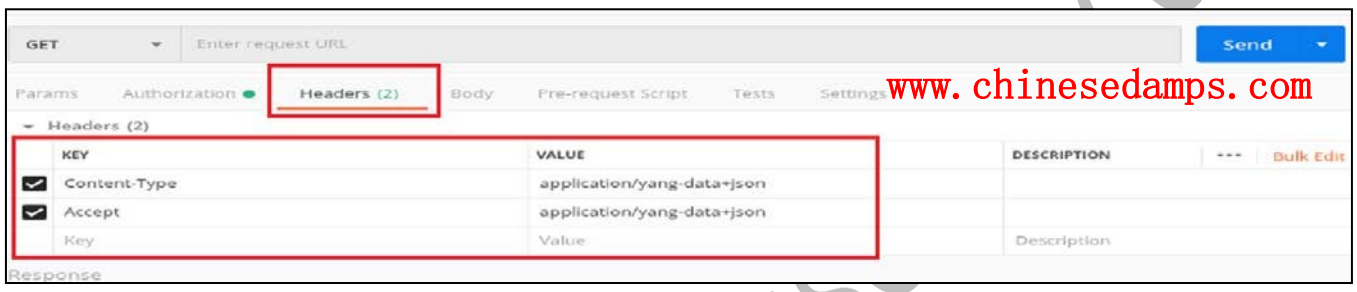
Paste the generated output in **Body** > **RAW**



步骤9：

选择“标题”选项卡并创建标题

- http头
  - 内容类型：应用程序/yang数据+json
  - 接受：应用程序/yang数据+json



步骤10：

在GET下输入URL，然后点击SEND

- 统一资源定位符
  - https://10.3.11.1:443/restconf/data/Cisco-IOS XE本机：本机

Step 13:

Change the **transport input** option from **“none”** to **“all”**

Then **Patch** the output by selecting **Patch** from the list

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步骤11：

上述步骤10将在“正文”选项卡中生成输出。复制生

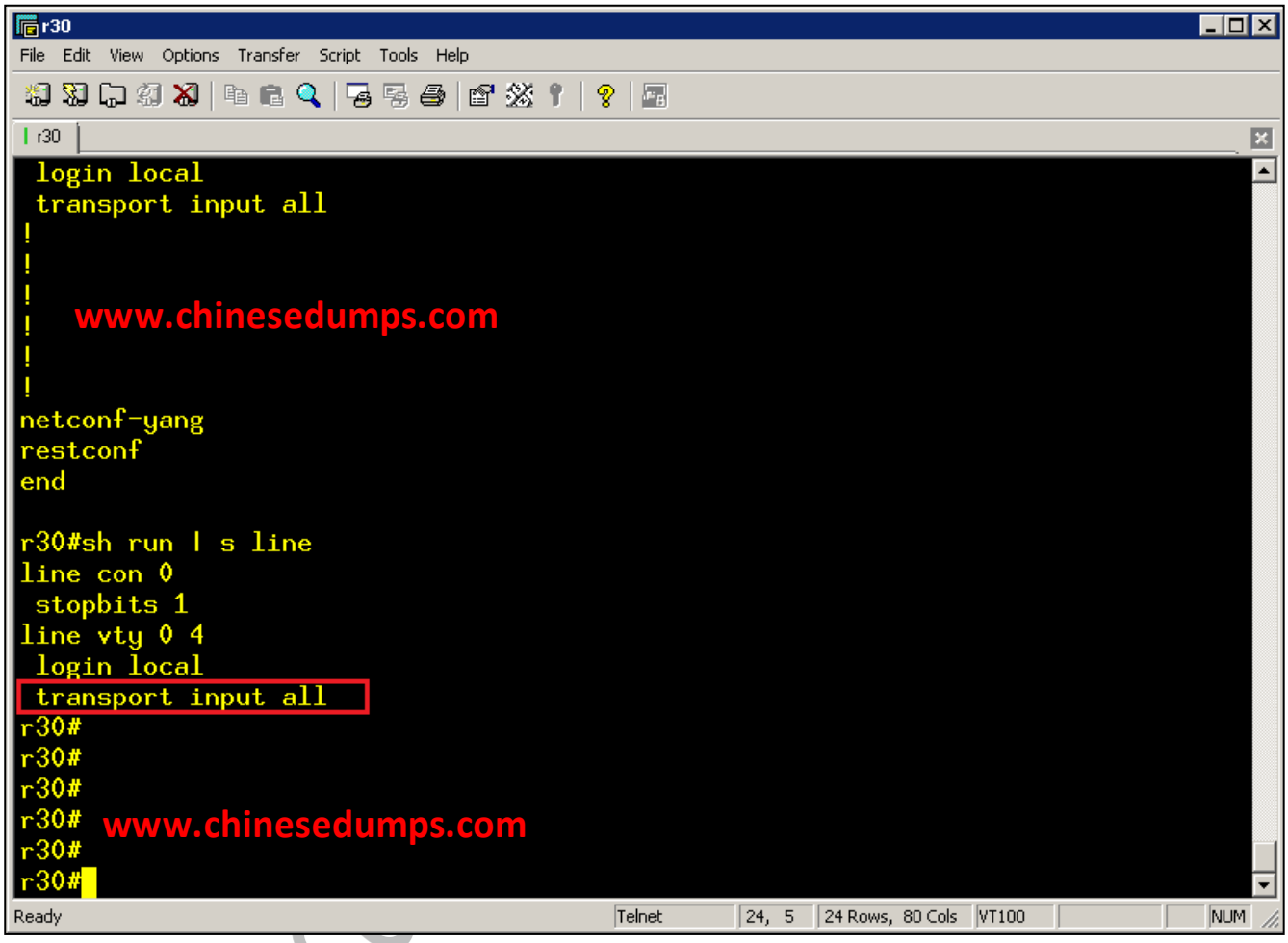
成的输出

步骤12：

将生成的输出粘贴到Body>RAW中

Verification:

Go to **CLI of r30** and run command **show run | s line**



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步骤13：

将运输输入选项从“无”更改为“所有”

然后通过从列表中选择Patch来修补输出

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3.2 : Using Guest shell and python on r30

On r30, enable guestshell and create a python script named ribdump.py in the guestshell according to these requirements:

- If an additional IP network is necessary to start guestshell, you are allowed to use addresses from the range 192.168.255.0/24. This range must not be advertised in any routing protocol.
- The python script must be saved under the name ribdump.py in the home directory of the guestshell user.
- The purpose of the script is to display the complete contents of all routing tables in non-default VRFs created on the router.
- The script must execute the show ip route vrf… or show ipv6 route vrf… command for every non default VRF created on the router, depending on what address families are enabled in that VRF.
- The script must determine the list of created VRFs and enabled address families dynamically every time it is run using, for example, show vrf brief | include ipv
- The script must not attempt to display the VRF routing table for an address family that is not enabled in the VRF.
- It must be possible to run the script using the guestshell run python ribdump.py command from privileged EXEC mode.

3 Points

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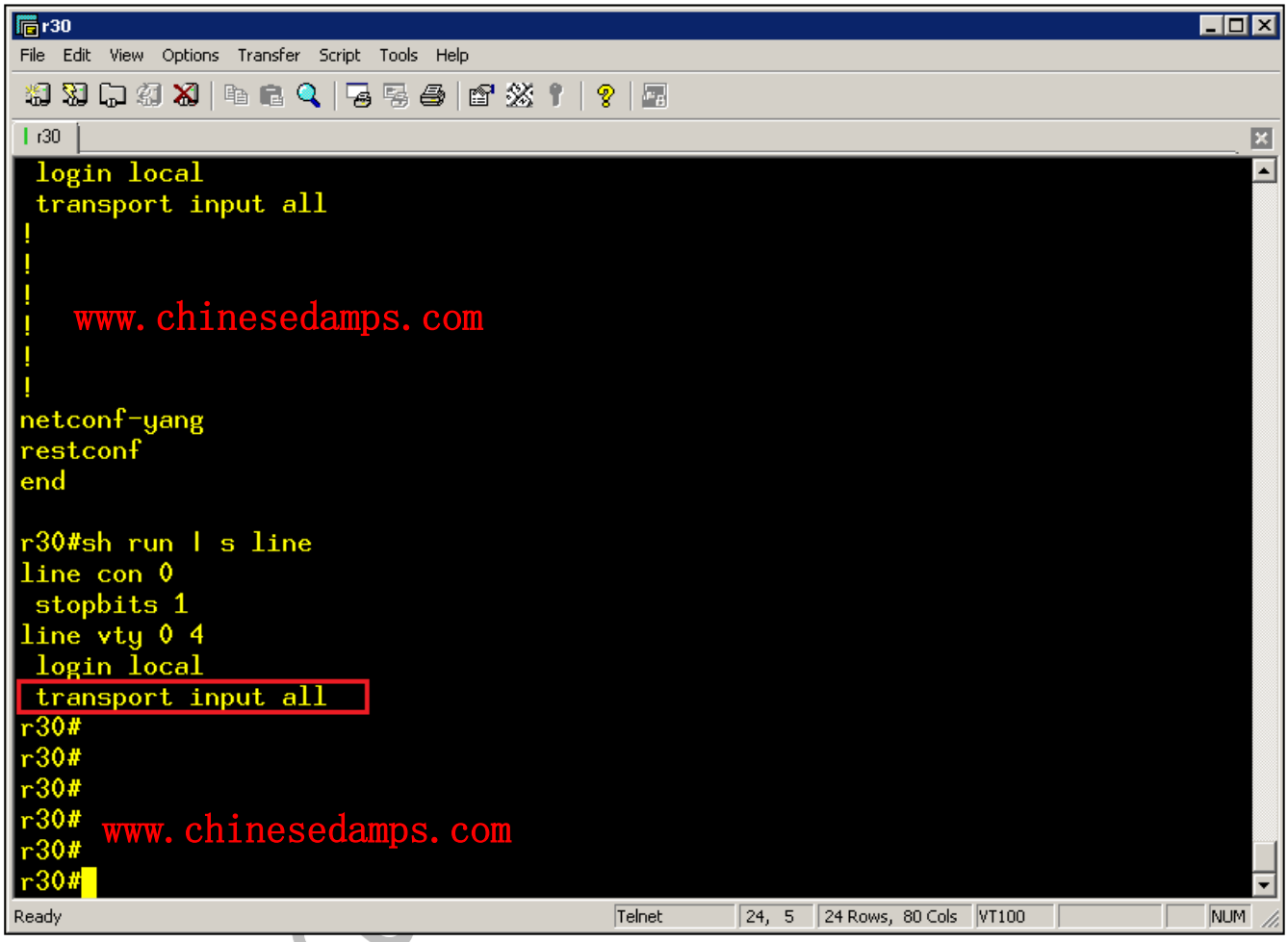
验证：

转到r30的CLI并运行命令show run|s line



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Solution:

- Need to enable Guestshell which needs IOX service to run
- Intent of script
- Execute command "show vrf brief | include ipv" and store output in a variable named vrf
- Run for loop for each vrf in variable vrf
- Check if ipv4 and ipv6 address family is in the vrf
- And execute the "show ipvx route vrf x" command per vrf, keep the output in variable and print the variable

Chinesedumps.com-r30

```
r30#conf t
r30(config)#iox

r30(config)#interface VirtualPortGroup0
r30(config-if)#ip address 192.168.255.1 255.255.255.0
r30(config)#exit

r30(config)#app-hosting appid guestshell
r30(config)#app-vnic gateway1 virtualportgroup 0 guest-interface 0
r30(config)#guest-ipaddress 192.168.255.2 netmask 255.255.255.0
r30(config)#app-default-gateway 192.168.255.1 guest-interface 0
r30(config)#name-server0 8.8.8.8
r30(config)#end
```

***Block this prefix to get advertised out in case connected redistribute or network statement is advertising this subnet**
***create a prefix list to deny 192.168.255./24 and allow everything else**
***apply on bgp neighbor**

```
r30#guestshell enable
```

***wait untili guestshell is enabled**

```
r30#guestshell run bash
```

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3.2 ：在r30上使用Guest shell和python

在r30上，启用guestshell，并根据以下要求在guestshell中创建一个名为ribdump.py的python脚本：

- 如果启动guestshell需要额外的IP网络，则允许您使用192.168.255.0/24范围内的地址。不得在任何路由协议中公布此范围。
- python脚本必须以ribdump.py的名称保存在guestshell用户的主目录中。
- 该脚本的目的是显示在路由器上创建的默认VRF之外的所有路由表的完整内容。
- 对于在路由器上创建的所有非默认vrf，脚本必须执行show ip route vrf…或show ipv6 route vrf…命令，具体取决于该vrf中启用的地址族。
- 每次使用show vrf brief | include ipv运行脚本时，该脚本必须动态地确定创建的vrf和启用的地址系列的列表
- 脚本不得尝试显示VRF中未启用的地址族的VRF路由表。
- 必须能够在特权EXEC模式下使用guestshell run python ribdump.py命令运行脚本。

3分

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***you will get its Linux BASH shell (most of linux bash shell commands willrun)**
***create a file named rib dump using vi editor (better learn the shortcut/commands to use vieditor)**

vi ribdump.py

The above command will edit the ribdump.pyfile

To insert into the editor hit " i "

Paste the below Script in the file and save

Script : Be mindful of indentation used else script willfail

```
import sys
import cli

vrf = cli.execute('show vrf brief | include ipv')
for line in vrf.splitlines():
 ipv4 = False
 ipv6 = False
 vrfString = line.split()
 if(vrfString[2] == "ipv4"):
  ipv4 = True
 elif(vrfString[2] == "ipv6"):
  ipv6 = True
 elif(vrfString[2] == "ipv4,ipv6"):
  ipv4 = True
  ipv6 = True
 if(ipv4 == True):
  vrfv4 = cli.execute('show ip route vrf ' + vrfString[0])
  print(vrfv4)
 if(ipv6 == True):
  vrfv6 = cli.execute('show ipv6 route vrf ' + vrfString[0])
  print(vrfv6)
```

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解决方案：

- 需要启用需要IOX服务才能运行的Guestshell
- 脚本意图
- 执行命令“show vrf short | include ipv”并将输出存储在名为vrf的变量中
- 为变量vrf中的每个vrf运行for循环
- 检查ipv4和ipv6地址族是否在vrf中
- 并根据vrf执行“show ipvx route vrf x”命令，将输出保存在变量中并打印变量

中文教育网-r30

r30#conf t
r30（配置）
#iox

r30（配置）#接口VirtualPortGroup0
r30（如果配置）#ip地址192.168.255.1 255.255.255.0 r30（
配置）#退出

r30（config）#应用程序托管appid guestshell
r30（配置）#app-vnic网关1虚拟端口组0访客界面0
r30（配置）#访客IP地址192.168.255.2网络掩码255.255.255.0
r30（config）#应用默认网关192.168.255.1访客界面0
r30（config）#名称服务器0
8.8.8.8 r30（配置）#结束

***阻止此前缀以进行播发，以防连接的重新分发或网络声明正在播发此子网**
***创建前缀列表以拒绝192.168.255./24并允许一切**
***应用于bgp邻居**

r30#guestshell启用

***等待guestshell启用**

r30#guestshell运行bash

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To exit the editor :

Press ESC then :wq

Then run the file with the below command

guestshell run python ribdump.py

guestshell ls

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*您将得到它的LinuxBASH shell（大多数linuxbashshell命令都会运行）
*使用vi编辑器创建一个名为rib-dump的文件（更好地学习使用vieditor的快捷方式/命令）

```
vi ribdump.py
```

上面的命令将编辑ribdump.pyfile

要插入编辑器，请点击“i”

将以下脚本粘贴到文件中并保存

脚本：注意使用缩进，否则脚本将失败

```
导入sys
导入cli

vrf=cli.execute（“显示vrf简报|包括
ipv”）
 ipv4=假
 ipv6=假
 vrfString=line.split（）
 if（vrfString[2]==“ipv4”
）：
  ipv4=真
 elif（
vrfString[2]==“ipv6”）：
 ipv4=真
 ipv6=真
 如果（ipv4==True）：
  vrfv4=cli.execute（’显示ip路由
vrf’+vrfString[0]）
  打印（vrfv4）
 如果（ipv6==True）：
  vrfv6=cli.execute（’显示ipv6路由
```

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Alternate Solution:

Chinesedumps.com-r30

```
conf t
iox

interface VirtualPortGroup0
ip address 192.168.255.1 255.255.255.0
ip nat inside
exit
ip nat inside source list GS_NAT_ACL interface GigabitEthernet1 overload

ip access-list standard GS_NAT_ACL
permit 192.168.0.0 0.0.255.255

app-hosting appid guestshell
app-vnic gateway1 virtualportgroup 0 guest-interface 0
guest-ipaddress 192.168.255.2 netmask 255.255.255.0
app-default-gateway 192.168.255.1 guest-interface 0
name-server0 8.8.8.8
end

int gi1.100
ip nat outside

guestshell enable
guestshell run bash

vi ripdump.py
```

The above command will edit the ribdump.pyfile

To insert into the editor hit " i "

Paste the below Script in the file and save

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要退出编辑器，请执行以下操作：

按ESC键，然后：wq

然后使用以下命令运行该文件

guestshell运行python ribdump.py

guestshell ls

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Script : Be mindful of indentation used else script willfail

```
import sys
import cli

vrf = cli.execute('show vrf brief | include ipv')
for line in vrf.splitlines():
 ipv4 = False
 ipv6 = False
 vrfString = line.split()
 if(vrfString[2] == "ipv4"):
  ipv4 = True
 elif(vrfString[2] == "ipv6"):
  ipv6 = True
 elif(vrfString[2] == "ipv4,ipv6"):
  ipv4 = True
  ipv6 = True
 if(ipv4 == True):
  vrfv4 = cli.execute('show ip route vrf ' + vrfString[0])
  print(vrfv4)
 if(ipv6 == True):
  vrfv6 = cli.execute('show ipv6 route vrf ' + vrfString[0])
  print(vrfv6)
```

To exit the editor :

Press ESC then :wq

Then run the file with the below command

guestshell run python ribdump.py

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替代解决方案：

Chinese edumps.com-

r30会议室
iox

接口VirtualPortGroup0
ip地址192.168.255.1 255.255.255.0
ip-nat内部出
口
ip nat内源列表GS_nat_ACL接口千兆以太网1过载

ip访问列表标准GS_NAT_ACL许可
192.168.0.0 0.0.255.255

应用程序托管appid guestshell
app-vnic网关1虚拟端口组0访客界面0
访客IP地址192.168.255.2网络掩码255.255.255.0
应用默认网关192.168.255.1访客界面0
名称服务器0 8.8.8.8
结束

整数gi1.100
外部ip nat

guestshell启用
guestshell运行bash

vi ripdump.py

上面的命令将编辑ribdump.pyfile要插入编辑器

，请点击“i”

将以下脚本粘贴到文件中并保存

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3.3 : Automated Configuration Backup Script

This item consists of multiple questions. You may need to scroll down to be able to see all questions. You are tasked with writing a python script to back up the configuration of a number of IOS-XE devices through RESTCONF, and store the configurations in text files. The starting section of this script has already been written and contains the following lines:

```
#!/usr/bin/python3

import requests

Credentials = * ("192.168.1.1", "admin", "s3cr3t"),
                ("192.168.1.2", "netadmin", "0th3rs3cr3t") +

Headers = , "Content-Type" : "application/yang-data+Jason",
            "Accept" : "application/yang-data+Jason" -
```

This script needs to be completed by dragging the individual's command lines below into their correct order to allow the script to correctly accomplish its purpose. Indicate the ends of the **for** block and of the **while** block by properly placing the **"- -End of for**" and **"- - End of while**" symbols.

Drag the lines into their correct order to complete the script as required. Make sure to also properly place the **"- - End of for**" and **"- - End of with**" symbols to indicate the end of the respective blocks in code.

-- End of with	Command
Response= requests.get(URL,auth=(Login,Password),headers=Headers,verify=False)	Command
For IP,Login,Password in Credentials:	Command
URL=f"https://,IP-:443/restconf/data/Cisco-IOS-XE-native:native"	Command
File.write(Response.text)	Command
With open(f",IP-.conf","w") as File:	Command
-- End of for	Command

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脚本：注意使用缩进，否则脚本将失败

```
导入sys
导入cli

vrf=cli.execute（“显示vrf简报|包括
ipv”）
 ipv4=假
 ipv6=假
 vrfString=line.split（）
 if（vrfString[2]==“ipv4”
）：
  ipv4=真
 elif（
vrfString[2]==“ipv6”）：
 ipv4=真
 ipv6=真
 如果（ipv4==True）：
  vrfv4=cli.execute（’显示ip路由
vrf’+vrfString[0]）
 打印（vrfv4）
 如果（ipv6==True）：
  vrfv6=cli.execute（’显示ipv6路由
vrf’+vrfString[0]）
 打印（vrfv6）
```

要退出编辑器，请执行以下操作：

按ESC键，然后：wq

然后使用以下命令运行该文件

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There are plans to extend the script to display a list of known IOS-XE devices by their IP addresses and allow the administrator to select which devices to backup. Aside from other necessary changes in the script, which of the following storage options for the credentials would allow for the most straight forward implementation?

- Credentials = * ("192.168.1.1", “admin”, “s3cr3t”),
  (“192.168.1.2”, “netadmin”, “oth3rs3cr3t”) +
- Credentials = , “192.168.1.1” : (“admin”, “s3cr3t”),
  “192.168.1.2”: (“netadmin”, “ oth3rs3cr3t”) -
- Credentials = “192.168.1.1,admin,s3cr3t,” \
  “192.168.1.2,netadmin,oth3rs3cr3t”
- Credentials = * “192.168.1.1, admin, s3cr3t”,
  “192.168.1.2, netadmin, oth3rs3cr3t” +

2 Points

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3.3 ： 自动配置备份脚本

此项目由多个问题组成。您可能需要向下滚动才能看到所有问题。您的任务是编写一个python脚本，通过RESTCONF备份多个IOS-XE设备的配置，并将配置存储在文本文件中。本脚本的起始部分有
已编写，包含以下行：

#! /usr/bin/python3导

入请求

凭据=*（“192.168.1.1”、“admin”、“s3cr3t”），
（“192.168.1.2”、“netadmin”、“0th3rs3cr3t”）+

Headers=，“内容类型”：“应用程序/yang数据+Jason”，“接受”：
“申请程序/yangdata+Jason”-

此脚本需要通过将下面的个人命令行拖动到正确的顺序来完成，以使脚本正确完成其目的。通过正确放置“for结束”和“while结束”符号来指示for块和while块的结束。

根据需要将线条拖动到正确的顺序以完成脚本。请确保正确放置“结束于”和“结束于带”符号，以表示相应代码块的结束。

--结束with	命令
响应=请求.get（URL，auth=（登录，密码），headers=headers，verify=False）	命令
对于IP、登录、凭据中的密码：	命令
URL=f“https://，IP-：443/restconf/data/Cisco IOS XE原生：原生”	命令
File.write（Response.text）	命令
将open（f“，IP-.conf”，“w”）作为文件：	命令
--for结束	命令

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Solution:

```
for IP,Login,Password in credentials:
URL=f"https://{ip}:443/restconf/data/cisco-IOS-XE-native:native"
   Response=request.get(URL,auth=(login,Password),headers=Headers,verify=false)

with open(f"{ip}.conf,"w") as file:
   file.write(reponse.text)let
end of with
end of for
```

Answer: B

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计划扩展该脚本，按IP地址显示已知IOS-XE设备的列表，并允许管理员选择要备份的设备。除了脚本中的其他必要更改外，以下哪种凭据存储选项允许最直接的实现？

o凭据=*（“192.168.1.1”、“admin”、“s3cr3t”），
（“192.168.1.2”、“netadmin”、
“oth3rs3cr3t”）+
o凭据=，“192.168.1.1”：（“管理员”，“s3cr3t”），
“192.168.1.2”：（“网络管理员”、
“oth3rs3cr3t”）-
o 凭据=“192.168.1.1，管理员，s3cr3t，”\
192.168.1.2，网络管理，oth3rs3cr3t

2分

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解决方案：

对于IP、登录、凭据中的密码：
URL=f“https://{IP}:443/restconf/data/cisco IOS XE
native:native”
Response=request.get（URL，auth=（登录名，密码），headers=headers，verify=false）

将open（f“{ip}.conf，”w“）
作为文件：
file.write（
response.text）以结尾

答案：B

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